# Tenebris and Lux

## *Comprehensive Master Guide*

CreativeWorkzStudio LLC / ProfessorSeanEX

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# Executive Summary

This Master Guide serves as the definitive single source of truth for **Tenebris and Lux**. It unites world canon, character dossiers, ludology, the 2017–2018 demo baseline, software architecture, and engineering dev logs into a single textbook compendium.

| Volume | Title | Scope |
|---|---|---|
| Volume I | World Canon, Metaphysics & Continental Topography (13 ch.) | Cosmic neutrality of Lux and Tenebris, chivalric Kingdom of Sacyra, continental realms, and the existential menace of The Dark Lord. |
| Volume II | Dramatis Personae & Dual Protagonist Dynamics (18 ch.) | Comprehensive character dossiers, founding princes, dual protagonist synergies, and regional faction leaders. |
| Volume III | Ludology & Tactical TRPG Combat Mechanics (21 ch.) | Grid-based tactical warfare, relic stance machines, terrain modifiers, continuous invariant testing, and game design curation. |
| Volume IV | The 2017–2018 Demo Baseline & Historical Archive (12 ch.) | Verified historical baseline extracted from the original RPG Maker MV datasets, detailing the complete scenario script, LeTBS mechanics, grimoire, and systems. |
| Volume V | Software Architecture & The 5-Block Invariant Protocol (30 ch.) | Cornerstone Big Machine vs. Tenebris and Lux Master Turn Gear, 5 strata, Rev 5 5-block protocol, Safe-Mode BIOS & .rodata ROM vaults, VFS data pipelines, pre-flight audits, and dual-frontend presentation. |
| Volume VI | Engineering Chronology & Development Logs (20 ch.) | Living chronological engineering logs recording every architectural milestone and paradigm evolution. |

# Volume I: World Canon, Metaphysics & Continental Topography

Cosmic neutrality of Lux and Tenebris, chivalric Kingdom of Sacyra, continental realms, and the existential menace of The Dark Lord.

# Chapter 1. The Primordial Neutrality of Lux and Tenebris

**Source File**: `docs/lore/world_and_lore.adoc`

## 1.1. Canonical Introduction

This document establishes the living world-building canon, cosmological ontology, and historical epic of **Tenebris and Lux** ("Darkness and Light"). Conceived between 2014–2018 in RPG Maker as the premier game of **CreativeWorkzStudio**, this is the epic of two primordial relics—**The Shield of Lux** and **The Sword of Tenebris**—set within the storied **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**, /s□□□ki□r□/). The story is driven by a tri-perspective narrative architecture: **Elijah Ro-enheart**, son of a former head of the Royal Guard; **Lady Katrina Jaren**, an aristocratic daughter seeking autonomy in the Adventurers' Guild; and the customizable **Bearer** (`<chosen_name>`), as the re-awakening **Dark Lord** rises from the subterranean abyss and **King Jihad** plots treason on the High Throne.

## 1.2. The Primordial Cosmology: Neutrality of the Apex Forces

> "Light without shadow is a blinding desert that calcifies the spirit. Shadow without light is a suffocating void where purpose drowns. The Shield preserves and confines; the Sword severs and liberates. The relics know neither virtue nor malice—only the soul that wields them."
>
> — The Sacyran Chronicle of Antiquity

In the cosmology of **Tenebris and Lux**, the cosmos is governed by two apex forces that transcend mortal morality. Neither is inherently good nor evil:

### 1.2.1. Lux (The Apex of Radiance & Cohesion)

- **Ontological Essence:** Manifestation, illumination, cohesion, preservation, temporal anchoring, and structure.
- **The Dual Nature of Light:**
  - **Virtuous Manifestation:** Projects golden protective barriers (**Lux Barrio**), wards the vulnerable, preserves cultural heritage, and seeks peace through reconciliation.
  - **Tyrannical Manifestation:** Scorches the earth in unyielding glare, blinding dissenters, enforcing calcifying stasis, denying natural growth and mortality, and petrifying living will into stone.

### 1.2.2. Tenebris (The Apex of Darkness & Severance)

- **Ontological Essence:** Mystery, dissolution, rest, severance, finality, and the fertile void.
- **The Dual Nature of Darkness:**
  - **Virtuous Manifestation:** Cuts through tyrant armor (**Cuttatio Tenebris**), severs corrupt bindings, veils the oppressed from predators, and executes swift, surgical **Absolute Justice**.
  - **Malevolent Manifestation:** Spreads entropic rot, dissolves conscience into nihilism, feeds bitter vengeance, and fuels tyrannical conquest.

### 1.2.3. The Law of Tempering and Dissipation

The relationship between Lux and Tenebris is governed by a strict metaphysical law: * **Lux as the Singular**

**Tempering Agent:** Raw Tenebris is entropic and severing; left unchecked, it erodes everything it touches. Lux is the only force capable of containing and tempering Tenebris, providing structural cohesion so that the void's cutting edge can be safely directed. * **The Fragility of Radiance (Dissipation):** While Lux tempers Tenebris, it requires steadfast emotional and ethical harmony. Any sudden, violent surge of fear, wrath, or malice causes Lux to **dissipate immediately**, collapsing protective barriers and leaving Tenebris unbridled.

### 1.2.4. The Moral Sway Continuum ([-100, +100]) & 13-Tier Taxonomy

Neither Lux nor Tenebris dictates a character's morality. Good and evil belong entirely to the **intent, conscience, and choices of the heart behind the weapon**, tracked along the `[-100, +100]` **Moral Sway** axis via the **Compass of Morality**. A knight channeling blinding radiance can act as a calcified tyrant, while an outrider wielding the void-edged blade of Tenebris can stand as the people's ultimate savior.

The continuum is codified into a formal 13-tier canonical nomenclature:

| Integer Range / Point | Canonical Designation | Functional Definition |
|---|---|---|
| `-100` | **The Sovereign of Ruin** | Absolute negative terminus. Total surrender to abyssal nihilism or petrifying totalitarianism. |
| `[-100, -50]` | **Active Malice (The Tyrannical Sphere)** | Deliberate cruelty, ruthless execution, and despotic dominion. |
| `[-50, 0]` | **Cynical Pragmatism (Compliant Apathy)** | Lower bystander band. Self-serving opportunism, petty theft, and systemic complicity. |
| `0` | **The Stance of Equilibrium (Harmonic Zero)** | Active, disciplined synthesis where Lux and Tenebris temper one another without stasis or rot. |
| `[0, +50]` | **Conventional Decency (Uncommitted Virtue)** | Upper bystander band. Civic compliance that avoids personal risk to challenge corruption. |
| `[+50, +100]` | **Active Virtue (The Restorative Sphere)** | Self-sacrificing heroism, surgical liberation, and restorative justice breaking inertia. |
| `+100` | **The Saint of Mercy** | Absolute positive terminus. Unyielding altruism in protection (Lux) or pure Absolute Justice (Tenebris). |
| `[-50, +50]` | **The Bystander Corridor** | Net Moral Inertia; zone of mutual cancellation where history repeats itself. |
| `[-100, 0]` | **The Arc of Dissolution** | The deconstructive hemisphere of the world ledger (Tyranny & Ruin). |
| `[0, +100]` | **The Arc of Restoration** | The constructive hemisphere of the world ledger (Virtue & Mercy). |
| `[-50, +100]` | **The Transcending Mortal Arc** | Trajectory of overcoming cynicism or blind obedience to achieve heroic conviction. |
| `[-100, +50]` | **The Arc of Tragic Disillusionment** | Trajectory of a betrayed soul whose optimism is capped at cautious neutrality, plunging to the abyss. |

| Integer Range / Point | Canonical Designation | Functional Definition |
|---|---|---|
| `[-100, +100]` | **The Universal Ledger** | The complete canvas of mortal free will (The Compass of Morality). |

### 1.2.5. Character Categorization Across the Continuum

| Range / Point | Canonical Designation | Historical Figures (Past) | Campaign Figures (Present) | Ludonarrative Rationale |
|---|---|---|---|---|
| `-100` | **Sovereign of Ruin** | **Usuru (Late Era)** | **The Dark Lord** | Consumed by unhealed grief after discovering the 11th Festival stasis deception. |
| `[-100, -50]` | **Active Malice** | *Six-Faction Warlords* | **King Jihad** (\~ -85) | Issues royal gag orders and plots to usurp the Dark Lord's void essence behind a benevolent facade. |
| `[-50, 0]` | **Cynical Pragmatism** | *Ancient Court Bailiffs* | **The St. Baron** (\~ -45), **Katrina Jaren (Base)** (\~ -20) | The St. Baron exploits Lower Dio through debt bondage. Katrina begins here: sardonic, hunting for Grandma's will to escape Dio. |
| `[-30, +30]` | **The Provincial Swing** | *Border Smugglers* | **John the Thief** | Operates in Lower Dio's gray margins, balancing illegal theft against community survival. |
| `0` | **Stance of Equilibrium** | *The Sages of Old* | **Chrono the Sage** | Anchored outside temporal degradation in the Room of Sages within the Dreamscape. |
| `[0, +50]` | **Conventional Decency** | **Usuru (Youth)** (\~ +30) | **Eric the Lumberjack** (\~ +10), **Elijah Ro-enheart (Base)** (\~ +35) | Honest folk surviving day-to-day. Elijah starts as a polite, rule-bound soldier hesitant to violate protocol. |
| `[+50, +100]` | **Active Virtue** | **Prince Mi-ren**, **Prince Casatti**, **The First King** | **Grandma Magolia**, **Sir Roté**, **Monica the Mystic** | Figures defined by communal protection, sacrificial defense, and devotion to peace. |
| `+100` | **Saint of Mercy** | **Goddess Lyri** (Ideal) | *Endgame Player Potential* | Theoretical pinnacle of pure self-sacrifice and uncorrupted justice. |
| `[-50, +100]` | **Transcending Mortal Arc** | *Academy Reformers* | **Elijah Ro-enheart (Arc)**, **Katrina Jaren (Arc)** | Elijah transcends blind obedience into conscience; Katrina refines rebellious cynicism into noble sacrifice. |

| Range / Point | Canonical Designation | Historical Figures (Past) | Campaign Figures (Present) | Ludonarrative Rationale |
|---|---|---|---|---|
| `[-100, +50]` | **Tragic Disillusionment** | **Usuru (Full Trajectory)** | *Compromised Playthrough* | Faith shattered in his 11th year, plunging from innocent reverence (+30) to the abyss (-100). |
| `[-100, +100]` | **The Unwritten Slate** | **Prince Mi-ren (Burden)** | **The Custom Bearer (`<chosen_name>`)** | Awakens at 0 in the Dreamscape; carries Mi-ren's macro-stewardship across the entire moral spectrum. |

## 1.3. Genesis: The Mundane Inception & The Two Princes

### 1.3.1. The Mundane Armaments of Inception

At the founding of the Kingdom of Sacyra, the Shield of Lux and Sword of Tenebris were forged as ordinary, non-magical weapons of steel and silver by royal smiths, presented to the First King as civic emblems: * **The Mundane Shield:** Represented **Peace**, communal defense, and internal harmony. * **The Mundane Sword:** Represented **War**, territorial justice, and border vigilance.

### 1.3.2. The Self-Running Systems

The First King was deeply beloved by his subjects across all provinces. Under his leadership, the realm's civic, legal, and municipal systems were engineered so effectively that the kingdom practically governed itself. Personal royal audiences were required only "once in a blue moon".

### 1.3.3. The Ideological Fork

The First King had two teenage sons who divided the responsibilities of civilization: * **Prince Mi-ren:** The elder son and heir to the High Throne, raised in statecraft to preserve his father's peace. * **Prince Casatti:** The younger son, who voluntarily renounced the throne to establish **The Academy**, an independent institution dedicated to natural philosophy, elemental sciences, and mapping **The Dreamscape**.

## 1.4. The 11th Festival Rupture & The 7-Year Great War

### 1.4.1. The Coma and the Silent Vigil (Festivals 1–10)

Following the first week-long **Festival of Peoples**, the First King fell into a coma, though his consciousness remained awake across the gossamer veil in **The Dreamscape**. When his physical body passed away during the 5th Festival, teenage Mi-ren and Casatti maintained a protective vigil: using the annual thinned-veil festival week to commune with the First King's spirit and project him into the temporal world in short bursts until Mi-ren was officially "of age."

### 1.4.2. Usuru and the 11th Year Catastrophe

In the 11th year, an eleven-year-old wanderer from remote Libira (a frontier kingdom suffering from a seven-day courier delay and no relay outposts) snuck into the royal sanctum to see the "plagued" king. Discovering the mummified corpse and projection array, the shocked boy accidentally tampered with the ritual: 1. The spiritual tether snapped, permanently sealing the First King in The Dreamscape. 2. The violent shift in the boy's heart caused an

unprecedented metaphysical surge that birthed **Tenebris** into waking reality. 3. The planar backlash physically warped continental distances between Libira and the Capital. 4. Taking the name **Usuru**, the youth led Libira and neighboring border realms in a war of annexation against the Capital.

### 1.4.3. The 7-Year Great War & The Awakening of the Relics

Usuru's uprising ignited **The Great War** across six fractured powers: The Capital, Libira, Great Mau, Old Sacyra, The Academy, and the Yun-yin Provinces: * **Casatti's Infusion of Tenebris:** Prince Casatti isolated the anomalous void energy and infused it into the mundane Sword of War to execute **Absolute Justice** and end the war swiftly. * **Mi-ren's Discovery of Lux:** Prince Mi-ren adapted the temporal projection technique to discover **Lux**, infusing it into the mundane Shield of Peace with a mandate of **atonement and forgiveness for Usuru**. * **The Accord of Kings:** Lux tempered Tenebris, bringing the war to an armistice in exactly **seven years**. The six powers ratified the **"Joint Realm, Separate Kingdoms"** accord, solidifying autonomous regional rule under the High Crown and enshrining the **Prophecy of Kings**. Refusing reconciliation, Usuru retreated into the abyssal vaults, degenerating over three millennia into **The Dark Lord**.

## 1.5. The 3,000-Year Dynastic Wave & Modern Sacyra

Between the Great War and the modern campaign lies a three-thousand-year chasm ($3,000\text{ years}$) spanning between **30 and 50 monarchs** (averaging $60\text{ to }100\text{ years}$ per reign):

- **The Succession Wave:** The dynasty cycled between Long Stasis Crests (60–90+ years of high peace and calcified bureaucracy), Tether-Sickness Troughs (5–20 years of royal paranoia and premature death near the High Throne's planar wound), and Bureaucratic Consolidations (35–50 years).
- **The Systemic Neglect:** Millennia of Capital stability fostered systemic apathy toward frontier settlements like **Dio Village**, replicating ancient Libira's "quiet complaint."
- **"The King's Curse" & King Jihad:** For three millennia, sovereigns on the High Throne were exposed to psychic backwash from the planar wound left by Usuru. Coveting void power, **King Jihad** opened himself to the abyss prior to the modern Festival of Heroes, resulting in possession by The Dark Lord's psychic remnant.

## 1.6. The Tri-Perspective Protagonists

The heart of **Tenebris and Lux** is an unfolding mystery navigated across three distinct vantage points:

### 1.6.1. Elijah Ro-enheart (The Vanguard of Lux)

- **Lineage & Heritage:** Son of legendary **Commander Gareth Ro-enheart**, former Head of the Royal Guard, who vanished under King Jihad's sealed orders on Elijah's 10th birthday.
- **Archetype & Role:** Frontline Heavy Vanguard / Fortress Anchor. Wields the **Shield of Lux**.
- **Thematic Parallel:** Mirrors ancient **Usuru** (The Frontier Wound). Endured a childhood frozen on his tenth year; raised with reverence for the Crown only to discover court betrayal. By choosing independent moral conscience over vengeance, Elijah represents Usuru granted redemptive grace.

### 1.6.2. Lady Katrina Jaren (The Outrider of Tenebris)

- **Lineage & Heritage:** Estranged daughter of aristocratic **House Jaren**, one of the wealthiest noble families on the Royal Court.
- **Archetype & Role:** High-Mobility Skirmisher / Flanking Striker. Wields the **Sword of Tenebris**.

- **Thematic Parallel:** Mirrors ancient **Prince Casatti** (The Renouncing Blade). Renounces aristocratic birthright in favor of the meritocratic **Adventurers' Guild**, wielding the sharp edge of Tenebris with cynical candor to slice through court hypocrisy and provincial stagnation.

### 1.6.3. The Custom Bearer (`<chosen_name>` — The Secret Chronicler)

- **Archetype & Role:** Emergent champion in Chrono's astral archives. Holds **The Dream Compass**.
- **Thematic Parallel:** Mirrors ancient **Prince Mi-ren** (The Steward of Balance). Decoupled from fixed identities, the player navigates the nocturnal Dreamscape trials, performing the "double work" to synthesize both Elijah's and Katrina's paths to unlock the **Secret Chronicler True Ending**.

## 1.7. The Tactical Strategy RPG (TRPG) Paradigm

**Tenebris and Lux** combines deep tactical planning with visceral action execution:

1. **Strategic 2D Grid Positioning:**
   - Multi-elevation grid maps across provinces, bridges, castles, and Dreamscape leylines.
   - Tactical movement budgets, Zone of Control (ZOC), flanking angles, and cover bonuses.
2. **Cinematic Combat Resolution:**
   - Initiating attacks transitions into dynamic traditional side-scroll / action duels.
   - Core skills: **Lux Barrio** (barrier rolls $2\dots5$ turns, $25\%$ damage mitigation), **Cuttatio Tenebris** ($20\%$ splash cleave), and **Duo-Loca Bashin n Slash** (adaptive defense targeting).
3. **Ludonarrative Consent & Multi-Ending Resolution:**
   - Act I focuses on defeating King Jihad; Act II focuses on stopping The Dark Lord.
   - Players skipping nocturnal Dream Compass trials receive the **Neutral Ending** (saving Sacyra while leaving the First King marooned). Players achieving full synthesis unlock the canonical **True Ending** ("The Ugly Truth"), resolving three millennia of cosmic conflict.

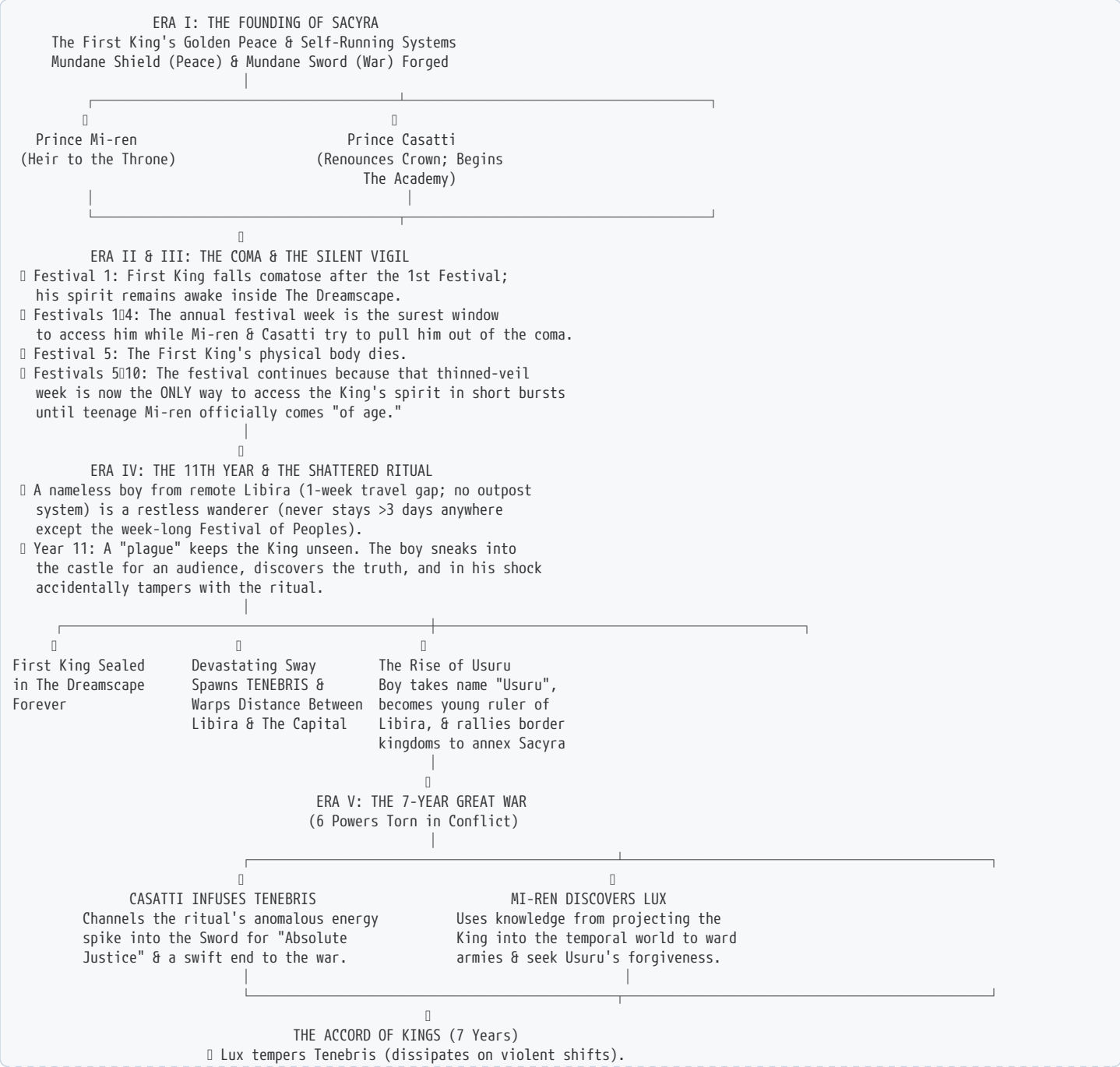
# Chapter 2. Historical Canon & Macro-Chronology

**Source File**: docs/lore/genesis_and_historical_canon.adoc

## 2.1. Overview & Historiographical Scope

This volume establishes the definitive historical canon, cosmological chronology, and dynastic chronicle of the **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**, /s□□□ki□r□/). Grounded in the authentic 2017–2018 demo archives and formal narrative architecture, it traces the 3,000-year continuum from the founding by the First King through the 7-Year Great War, the tragedy of Usuru, the awakening of the twin relics, the 3,000-year dynastic wave, and the modern crisis under King Jihad.

## 2.2. Chronological Epoch Map



```
                        Establishes "Joint Realm, Separate Kingdoms" politics.
                        Births the "Prophecy of Kings" around the twin relics.
                            |
                            
          THE 3,000-YEAR CHASM & DYNASTIC WAVE
               (30–50 Monarchs across 3 Millennia)
```

## 2.3. Era I: The Founding of Sacyra & The Two Princes

### 2.3.1. The Mundane Armaments of Inception

At the founding of the Kingdom of Sacyra, the **Shield of Lux** and the **Sword of Tenebris** were not magical or celestial artifacts. They were forged as ordinary, non-magical armaments of steel and silver by royal smiths, presented to the First King as civic symbols of statecraft:

- **The Mundane Shield:** Represented **Peace**, communal defense, and agricultural stability.
- **The Mundane Sword:** Represented **War**, territorial vigilance, and border law enforcement.

### 2.3.2. The Self-Running Systems of the First King

The First King was deeply beloved by his subjects across all provinces. Under his leadership, the kingdom's municipal, legal, and civic institutions were so well-engineered that the realm practically governed itself. Citizens did not lack access to the King out of tyranny or aloofness, but because the systems functioned with such frictionless harmony that a personal royal audience was required only "once in a blue moon".

### 2.3.3. The Ideological Fork: Mi-ren and Casatti

The First King had two sons who chose distinct paths while still young teenagers:

- **Prince Mi-ren (The Elder Son):** The rightful heir to the High Throne, raised in administrative discipline to inherit his father's covenant of golden peace.
- **Prince Casatti (The Younger Son):** Renounced his royal birthright to the throne to establish **The Academy**, dedicating his life to empirical inquiry, elemental sciences, and mapping the astral geometries of the ethereal plane.

## 2.4. Era II & III: The Festival of Peoples, The Coma & The Silent Vigil

### 2.4.1. Why the Festival Lasted a Full Week

Every year, the Capital hosted the **Festival of Peoples**—an international gathering where border demarcations were ratified, continental infrastructure treaties negotiated, and subjects celebrated. Rather than holding a single-day event, the First King mandated a full seven-day festival out of deliberate consideration for distant subjects: those with harvest or trade obligations could depart early, while delegations traveling from the furthest frontiers of the known world were guaranteed meaningful participation.

### 2.4.2. The First Festival Coma & Discovery of The Dreamscape

Immediately following the first grand Festival of Peoples—where the collective emotional consciousness of an entire continent converged—the First King collapsed into a deep coma. His body remained dormant, yet his spirit was fully lucid and awake on the other side of the ethereal boundary in **The Dreamscape**. By tracing his father's displaced consciousness, Prince Casatti and The Academy mapped the Dreamscape in full.

### 2.4.3. Festivals 1 Through 4: The Four-Year Rescue Effort

During the first four years of the coma, the annual festival week—when the boundary ("the Veil") thinned to gossamer—served as the **surest, most stable window** to access the King. Throughout these four years, Mi-ren, Casatti, and the court worked tirelessly across the thinned veil attempting to pull the First King back into his mortal body.

### 2.4.4. Era III: The 5th Festival Death & The Meaning of "Of Age"

During the **5th Festival of Peoples**, the First King's physical body died. Pulling him out of the coma was no longer possible. However, the Castle court continued the festival because that thinned-veil week was now the **sole remaining lifeline** allowing Mi-ren to commune with his father's lucid spirit and project his consciousness into the temporal hall in short bursts when rare royal audiences were needed.

This secret was maintained across Festivals 5 through 10 not out of malice, but as a protective vigil:

- **No Precedent for Succession:** Sacyra had never experienced the death of a monarch before; international peace treaties were bound to the First King's personal covenant, and no constitutional law existed for a regency.
- **The Chivalric Definition of "Of Age":** Mi-ren and Casatti were teenagers. In Sacyran chivalric law, attaining majority required martial maturity to personally command the Royal Guard alongside mastery of the Academy's statecraft systems so the self-running realm would not shatter under foreign pressure.

## 2.5. Era IV: Usuru the Wanderer & The 11th Year Tragedy

### 2.5.1. The "Quiet Complaint" of Remote Libira

Far on the southern frontier lay the **Kingdom of Libira**, a sparse, underdeveloped territory separated from the Capital by a grueling seven-day journey. Because Libira was not yet established enough to maintain an imperial relay outpost network, communications suffered a chronic seven-day lag. This systemic gap fostered a simmering **"quiet complaint"** among Libirans regarding the disparity between Capital prosperity and border isolation.

### 2.5.2. The Three-Day Wanderer

During the First King's peaceful reign, a boy was born in Libira to a humble family without being given a name. For ten years, he grew up traveling the continent as a restless wanderer who lived by a strict personal rule: **never stay more than three days anywhere**—except for the annual week-long Festival of Peoples in the Capital, which he attended with pure devotion to honor the King he revered.

### 2.5.3. The 11th Year Rupture

During the boy's eleventh annual pilgrimage to the festival, heralds announced that a "royal plague" kept the First King unseen on his balcony. Anxious for the monarch he adored, the eleven-year-old boy slipped past castle guards into the royal sanctum to seek an audience.

Inside, the boy discovered the long-dead body and Prince Mi-ren utilizing Academy projection glyphs. Lacking the statecraft context to understand the court's desperate vigil, the boy viewed the revelation as a monstrous, hypocritical deception. In his terror and shock, he accidentally collided with the ritual array, triggering four catastrophic consequences:

1. **The Eternal Astral Seal:** The First King's spiritual tether snapped; his consciousness was permanently marooned in **The Dreamscape**.
2. **The Birth of Tenebris:** The violent collapse of the boy's innocent faith into profound betrayal produced an

anomalous metaphysical shockwave, releasing the cosmic apex force of **Tenebris** (severance and dissolution) into the waking temporal world that very day.

3. **The Warped Distance:** The planar backwash physically warped continental leylines, stretching the geographical distance between Libira and the Capital into an unnatural chasm that the boy interpreted as an act of deliberate hostile separation by the Crown.
4. **The Rise of Sovereign Usuru:** Taking the name **Usuru**, the youth fled back to Libira, revealed the Capital's deception, was crowned young ruler of Libira, and rallied neighboring border kingdoms into an alliance to declare a war of total annexation against the Capital.

## 2.6. Era V: The 7-Year Great War & Awakening of the Relics

### 2.6.1. The Six Fractured Powers

Usuru's uprising ignited **The Great War**, plunging six sovereign powers into continental war:

| Faction | Leadership | Strategic Role in the Conflict |
|---|---|---|
| **The Libira Kingdom** | Sovereign Usuru | Vanguard of the frontier coalition; driven by righteous betrayal and territorial annexation. |
| **The Capital** | Prince Mi-ren | Citadel defense of the High Throne, administrative continuity, and royal army deployment. |
| **The Great Mau Kingdom** | High Chieftains | Mighty inland territorial power maneuvering along contested eastern borders. |
| **Old Sacyra** | Traditional Fiefdoms | Ancestral heartland baronies fighting to preserve chivalric feudal order. |
| **The Academy** | Prince Casatti | Independent intellectual and arcane bastion deploying experimental elemental resonance. |
| **The Yun-yin Provinces** | Provincial Assemblies | Western agricultural riverlands caught between coalition supply routes and capital levies. |

### 2.6.2. Casatti's Infusion of Tenebris (The Sword of War)

Determined to end Usuru's siege quickly and preserve the realm, Prince Casatti analyzed the anomalous energy spike recorded during the ritual rupture. Synthesizing his knowledge of the Dreamscape and elemental properties, Casatti isolated **Tenebris** and infused it into the mundane Sword of War. He forged the **Sword of Tenebris** not out of bloodlust, but to execute **Absolute Justice**—creating a surgical blade capable of severing siege lines and unyielding armor to halt the war decisively.

### 2.6.3. Mi-ren's Discovery of Lux (The Shield of Peace)

Taking command of the Capital's forces, Prince Mi-ren sought not conquest or vengeance, but **atonement and forgiveness from Usuru**. By reverse-engineering the projection ritual used to anchor ethereal concepts into physical reality, Mi-ren unlocked **Lux** (cohesion, preservation, and radiance) and infused it into the mundane Shield of Peace, creating the **Shield of Lux**.

### 2.6.4. The Law of Tempering and Dissipation

Battlefield confrontations between the two armaments revealed the foundational cosmic dynamic:

- **Lux as the Singular Tempering Agent:** Raw Tenebris is purely severing and entropic; left unchecked, it erodes both wielder and target. Lux is the only force capable of containing and stabilizing Tenebris, granting structural cohesion so void strikes can be directed with precision.
- **The Fragility of Radiance (Dissipation):** While Lux stabilizes Tenebris, its golden coherence requires steadfast emotional and ethical harmony. Any sudden surge of terror, wrath, or malice causes Lux to **dissipate instantly**, leaving Tenebris unbridled.

### 2.6.5. The Parley & The 7-Year Armistice

During peace negotiations, Prince Mi-ren lowered his defenses and offered total reconciliation to Usuru. However, Usuru's deep bitterness erupted in a violent surge of hatred, causing Mi-ren's ambient Lux aura to dissolve on the spot.

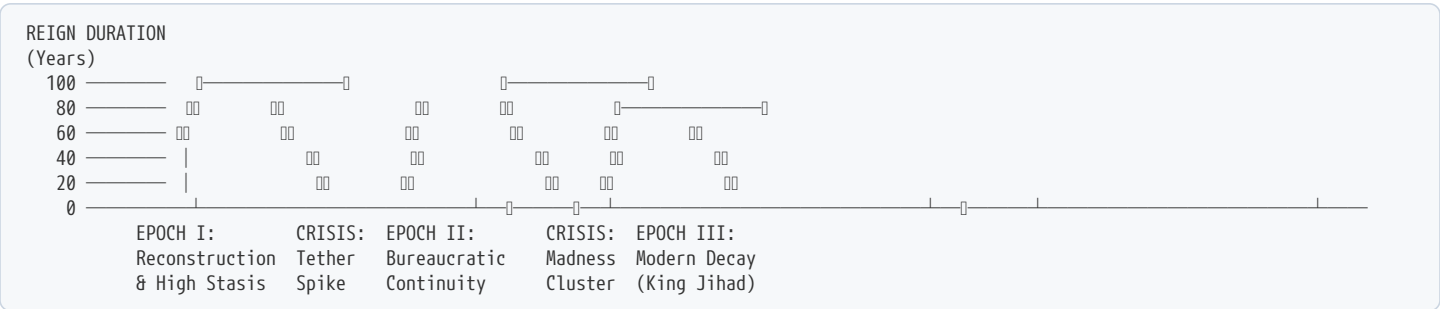
Recognizing that neither side could achieve unconditional victory without mutual annihilation, the six powers ratified the **Accord of Kings**, bringing the Great War to a close after exactly **seven years**:

- **"Joint Realm, Separate Kingdoms":** Autonomous provinces and kingdoms formed a loose federated realm under the High Crown while preserving regional self-rule.
- **The Prophecy of Kings:** Sages enshrined the legend that true peace would only return when a bearer achieved the **Stance of Equilibrium**, harmonizing the Shield and Sword without allowing wrath to dissipate the light.
- **Usuru's Descent:** Refusing reconciliation, Usuru retreated into the deep subterranean abyss, where his resentment festered over three millennia until his name was forgotten, remembered only as **The Dark Lord**.

## 2.7. The 3,000-Year Chasm & The Dynastic Wave

Between the Great War and the modern campaign lies a three-thousand-year chasm ($3,000\text{ years}$), spanning between **30 and 50 monarchs** (averaging $60\text{ to }100\text{ years}$ per reign).

### 2.7.1. The Mathematical Wave of Succession




The dynastic progression oscillates across three cyclical phases:

1. **Long Stasis Crests (60–90+ Years):** In eras of high stability, the First King's self-running institutions, Academy restorative arts, and the preserving aura of the Shield of Lux allowed monarchs to reign for nearly a century in peaceful continuity.
2. **Tether-Sickness Troughs (5–20 Years):** Succession clusters triggered when the dormant planar wound beneath the High Throne flared up, afflicting sovereigns with premature wasting, paranoia, and sudden abdications.
3. **Bureaucratic Consolidations (35–50 Years):** Standard generational transitions managed smoothly by the Royal Court and hereditary Noble Houses.

### 2.7.2. Five Axioms of Dynastic Continuity

Why does the narrative require an established 3,000-year dynastic order rather than constant civil anarchy?

1. **Embodying the Cosmological Shadow of Lux:** Unbroken continuity proves Sacyra succumbed to the negative extreme of Lux—calcifying stasis—transforming a golden peace into a rigid, bureaucratic museum.
2. **Exposing Frontier Neglect as Systemic:** Because the Capital enjoyed millennia of comfort, the decay of border settlements like **Dio Village** is exposed not as temporary wartime fallout, but as cold, institutional apathy mirroring ancient Libira.
3. **Isolating "The King's Curse":** In a stable, legally sacrosanct monarchy free of routine coups, the periodic madness afflicting sovereigns stands out as a haunting metaphysical anomaly.
4. **Validating the Ro-enheart Lineage:** Five unbroken generations of Ro-enheart leadership at the head of the Royal Guard holds sacred weight only if the crown they defended felt ancient, honorable, and permanent.
5. **Magnifying King Jihad's Rupture:** An established 3,000-year legacy transforms **King Jihad** (`D. Jihad`) from a common bad ruler into an existential traitor gambling away three millennia of peace for void supremacy.

# Chapter 3. Master Storyboard Sequence (Scenes 01–20)

**Source File**: `docs/lore/master_storyboard_chronicle.adoc`

## 3.1. Overview & Cinematic Philosophy

This document contains the complete master storyboard sequence for the historical prologue and ancestral memory sequences of **Tenebris and Lux**. Comprising twenty sequentially numbered cinematic scenes (Scenes 01–20), this chronicle establishes the visual language, color temperature, character staging, and dramatic pacing that bridge the founding of the Kingdom of Sacyra to the catastrophe of the 11th Festival and the dawn of the Great War.

## 3.2. Cinematic Staging Matrix (Scenes 01–20)

| Scene | Title | Setting & Tone | Visual Focus | Narrative & Thematic Action |
|---|---|---|---|---|
| **01** | **Symbols of Iron** | Royal Forges of Sacyra (Warm hearth amber) | Smiths quenching an unadorned steel kite-shield and broadsword. | The Kingdom of Sacyra is founded. Two ordinary weapons are presented to the First King as civic symbols of **Peace** and **War**. |
| **02** | **The Crown and the Scroll** | Sacyra Throne Room (Sunlit marble) | The First King with his teenage sons, Mi-ren and Casatti. | Mi-ren accepts his birthright as heir; Casatti declines the throne to establish **The Academy**. |
| **03** | **The Self-Running Realm** | Capital Courtyards & Provincial Towns (Golden calm) | Civic courts and markets running smoothly without royal interference. | Establishes a reign of peace so well-engineered that war is merely an idea and royal audiences are needed only once in a blue moon. |
| **04** | **The First Festival & The Coma** | Grand Festival Grounds / Royal Bedchamber | The known world celebrating for a full week; the First King collapsing into a coma as his spirit awakens in the Dreamscape. | The week-long Festival of Peoples accommodates distant travelers. Afterward, the King falls comatose; Casatti discovers the Dreamscape in full while tracking his father’s lucid spirit. |
| **05** | **The Four-Year Rescue Effort** | Academy Conduit Chamber (Festivals 1–4) | Teenage Mi-ren and Casatti working across the thinned festival veil to reach their father. | For four years, the festival week serves as the surest window to access the King while his sons attempt to pull him out of his coma. |
| **06** | **The Passing at the Fifth** | Royal Sanctum (Festival 5, solemn candlelight) | The First King’s physical body expiring while his golden silhouette remains across the veil. | The King dies during the 5th Festival. Because Mi-ren is not yet "of age" for the first succession, the Castle keeps the secret; the festival continues as the only way left to access the King in short bursts. |

| Scene | Title | Setting & Tone | Visual Focus | Narrative & Thematic Action |
|---|---|---|---|---|
| **07** | **The Quiet Complaint of Libira** | Frontier Kingdom of Libira (Windy, sparse roads) | A lone courier riding for seven days without a single relay outpost; a nameless boy born to a humble family. | Highlights the week-long travel gap and lack of an outpost system in Libira, breeding a "quiet complaint" over the Capital's distance. |
| **08** | **The Three-Day Wanderer** | Continental Roads & Capital Gates | Montage of the boy wandering from town to town (never staying past 3 days), except for his annual week at the Festival of Peoples. | For 10 years, the humble wanderer visits the Capital every festival week to celebrate and honor the First King. |
| **09** | **The Plague Year** | Festival Concourse — Year 11 (Somber crowd) | Heralds announcing a "plague" has struck the King; the 11-year-old boy looking at the empty royal balcony. | Unseen for the entire 11th festival, the King's absence worries the boy, prompting him to sneak into the castle for an audience. |
| **10** | **Behind the Veil** | Inner Castle Sanctum (Arcane glyphs, mist) | The boy discovering Mi-ren using Academy knowledge to commune with the long-dead First King. | From the boy's perspective, the Capital's golden peace is exposed as a years-long deception. |
| **11** | **The Shattered Ritual & Birth of Tenebris** | Inner Sanctum (Violent violet and gold shockwave) | Shocked, the boy accidentally tampers with the ritual; the tether snaps and a dark energy spike erupts. | The First King is sealed forever in **The Dreamscape**. The devastating shift in the boy's heart spawns **Tenebris** and warps the physical distance between Libira and the Capital. |
| **12** | **The Rise of Sovereign Usuru** | Libira Assembly Hall (Torchlight, raised spears) | The boy takes the name **Usuru**, rallying Libira's quiet complaints into open fury. | Seeing the warped distance as intentional malice, Usuru convinces Libira to crown him young ruler and declare a war of annexation against the Capital, inciting neighboring kingdoms to join. |
| **13** | **The Six-Power Conflagration** | Continental War Map (Six burning fronts) | Armies clashing across Libira, the Capital, Great Mau, Old Sacyra, the Academy, and the Yun-yin Provinces. | **The Great War** engulfs the known world. |

| Scene | Title | Setting & Tone | Visual Focus | Narrative & Thematic Action |
|---|---|---|---|---|
| **14** | **The Infusion of Tenebris** | Academy Arcane Forge (Deep twilight violet) | Prince Casatti channeling the anomalous energy spike from the interrupted ritual into the mundane Sword of War. | Synthesizing his study of the Dreamscape, world elements, and the rupture spike, Casatti creates the **Sword of Tenebris** to enact swift **Absolute Justice**. |
| **15** | **The General of Reconciliation** | Capital Command Tent (Warm lantern light) | Prince Mi-ren taking command of the army, vowing to seek Usuru's forgiveness. | Mi-ren leads the defense not for conquest, but to restore peace and mend the broken trust with Usuru. |
| **16** | **The Discovery of Lux** | Academy Spire & Battlefield (Golden radiance) | Mi-ren adapting the temporal projection ritual into a golden defensive barrier (**Lux**), sparking the Magic vs. Science debate. | Using what he learned from projecting the King into the temporal world, Mi-ren infuses the mundane Shield of Peace into the **Shield of Lux**. |
| **17** | **Tempering the Void** | Frontlines of the Great War (Gold & violet clash) | Mi-ren's Lux barrier containing and stabilizing Casatti's Tenebris strikes against coalition lines. | Demonstrates that Lux is the only force able to temper Tenebris, bringing the continental war to a decisive end in just **7 years**. |
| **18** | **The Dissipation at Parley** | Border Truce Grounds (Thick mist) | Mi-ren offering reconciliation to Usuru; Usuru's violent surge of hatred causing the ambient Lux to dissolve. | Proves the metaphysical law: any violent emotional shift causes Lux to dissipate instantly, just as it did on the day of the ritual rupture. |
| **19** | **The Joint Realm Accord** | Grand Council Chamber (Six regional banners) | Leaders ratifying the treaty establishing the "Joint Realm, Separate Kingdoms" political order. | Solidifies the federated geopolitical structure that persists into the modern era of Sacyra. |
| **20** | **The Prophecy of Kings** | Room of Sages / Royal Archives | Chrono recording the legend of the twin artifacts awaiting a bearer of true Equilibrium. | The First King's symbols of Peace and War are enshrined in the **Prophecy of Kings**, while Usuru retreats into the dark to become **The Dark Lord**. |

## 3.3. Scene-by-Scene Narrative Walkthrough

### 3.3.1. Scene 01: Symbols of Iron

- **Atmosphere:** An ancient blacksmith workshop bathed in the deep amber glow of coal embers. Hammers strike rhythmic, resonant beats on an anvil.
- **Cinematics:** Steam hisses violently as a plain, unadorned kite shield and a straight broadsword are pulled from water troughs. There are no glowing runes, no magical stones, and no celestial sigils.
- **Thematic Grounding:** Sacyra's founding armaments were forged by mortal hands as mundane civic symbols of Peace and War. Power did not reside in the iron; it resided in the covenant of the sovereign who held them.

### 3.3.2. Scene 02: The Crown and the Scroll

- **Atmosphere:** High-vaulted marble halls of Sacyra Castle flooded with pure morning sunlight.
- **Cinematics:** The First King, majestic and warm, speaks with his two teenage sons. Prince Mi-ren kneels to receive the ceremonial signet of succession. Prince Casatti stands erect, clutching parchment scrolls and an astrolabe, smiling as he formally declines the throne to establish The Academy.
- **Thematic Grounding:** Establishes the harmonious brotherhood between governance (Mi-ren) and empirical inquiry (Casatti) before history fractured.

### 3.3.3. Scene 03: The Self-Running Realm

- **Atmosphere:** Vast panoramic shots of bustling provincial markets, lush terraced farms, and judicial courtyards under clear azure skies.
- **Cinematics:** Merchants trade freely; magistrates resolve disputes with open books; no armed patrols or royal inquisitors hover in the shadows. The throne room stands empty not from neglect, but because the machinery of state functions flawlessly.

### 3.3.4. Scene 04: The First Festival & The Coma

- **Atmosphere:** The vibrant, colorful sprawl of the inaugural Festival of Peoples.
- **Cinematics:** Seven days of joyous celebration give way to silence in the royal bedchamber. The First King collapses as the music fades. The camera slowly pushes past his dormant physical eyes into a starry, gossamer void: his consciousness stands upright, alive and lucid in The Dreamscape.

### 3.3.5. Scene 05: The Four-Year Rescue Effort

- **Atmosphere:** The dim, arcane candlelight of an Academy laboratory conduit chamber.
- **Cinematics:** For four consecutive years during festival week, teenage Mi-ren and Casatti channel glowing ether arrays toward their father’s spectral silhouette across the thinned veil, desperately trying to pull him back into physical reality.

### 3.3.6. Scene 06: The Passing at the Fifth

- **Atmosphere:** Cold, somber blue candlelight in the royal sanctum.
- **Cinematics:** The First King’s mortal chest draws its final breath; his physical heart ceases. Yet across the gossamer veil, his golden silhouette remains awake. The Castle ministers and the teenage princes lower their heads in solemn consensus: the vigil must continue until Mi-ren is of age.

### 3.3.7. Scene 07: The Quiet Complaint of Libira

- **Atmosphere:** Bleak, windswept savanna and rugged canyons of the remote southern frontier.
- **Cinematics:** A lone royal courier gallops past solitary marker stones under scorching sun and freezing night, taking seven grueling days to reach an isolated frontier village without a single relay outpost. In a humble thatch hut, a nameless baby is born.

### 3.3.8. Scene 08: The Three-Day Wanderer

- **Atmosphere:** A dynamic, fast-paced montage across muddy trade roads, dusty inns, and river barges.
- **Cinematics:** The growing boy arrives in hamlets, helps villagers for three days, packs his light sack on dawn of the

fourth day, and departs. He stays nowhere longer than three days—until the camera tracks him entering the golden gates of the Capital for the annual week-long Festival of Peoples.

### 3.3.9. Scene 09: The Plague Year

- **Atmosphere:** Chilly autumn mist hanging over a muted, uneasy festival concourse in the 11th year.
- **Cinematics:** Royal heralds announce from the battlement that the King has been stricken with illness and will not appear. The eleven-year-old boy stares up at the empty balcony with furrowed brows and rising dread.

### 3.3.10. Scene 10: Behind the Veil

- **Atmosphere:** Claustrophobic, echoing shadows of the forbidden inner keep.
- **Cinematics:** The boy slips through heavy oak doors into the projection chamber, watching from behind a stone pillar as Mi-ren chants over the long-dead, mummified body of the First King, projecting his spectral double onto the temporal dias. The boy's eyes widen in sheer horror at what he perceives as a monstrous fraud.

### 3.3.11. Scene 11: The Shattered Ritual & Birth of Tenebris

- **Atmosphere:** A blinding, violent detonation of midnight-violet and solar-gold.
- **Cinematics:** Tripping in his panic, the boy stumbles into the chalk glyphs and crystal anchors. The ethereal tether snaps with the sound of cracking glass; the First King is pulled backwards into the astral deep. From the boy's chest erupts an anomalous spike of pure cosmic void: **Tenebris** enters waking reality.

### 3.3.12. Scene 12: The Rise of Sovereign Usuru

- **Atmosphere:** The smoky, torchlit Great Hall of Libira.
- **Cinematics:** Standing before tribal chiefs and town elders, the youth raises a spear, his voice ringing with cold fury. He proclaims his name: **Usuru**. Spears clash against shields as Libira crowns their young sovereign and pledges to march on the Capital.

### 3.3.13. Scene 13: The Six-Power Conflagration

- **Atmosphere:** A vast tactical war table transitioning into panoramic battlefields lit by burning trebuchets.
- **Cinematics:** Six banners clash across smoking battlefields: Libira, The Capital, Great Mau, Old Sacyra, The Academy, and Yun-yin. Sacyra's golden peace dissolves into total war.

### 3.3.14. Scene 14: The Infusion of Tenebris

- **Atmosphere:** The subterranean resonance forge of The Academy.
- **Cinematics:** Prince Casatti holds the mundane iron broadsword over a crystal crucible containing the harvested violet rift energy. Chanting harmonic formulas, he quenches the steel in void essence. The blade turns star-black with rippling violet mist: the **Sword of Tenebris** is born for Absolute Justice.

### 3.3.15. Scene 15: The General of Reconciliation

- **Atmosphere:** The somber, parchment-strewn command tent of the Royal Army.
- **Cinematics:** Prince Mi-ren inspects defensive troop dispositions. He speaks to his captains not of conquest or retaliation, but of holding the line so that he may face Usuru and beg for his forgiveness.

### 3.3.16. Scene 16: The Discovery of Lux

- **Atmosphere:** An open battlefield at dawn, bathed in blinding celestial gold.
- **Cinematics:** Facing an enemy barrage, Mi-ren steps to the front ranks and raises the mundane iron shield. Channelling the memory of his father's temporal projection, radiant golden light solidifies into a geometric barrier: **Lux** manifests to shield the living.

### 3.3.17. Scene 17: Tempering the Void

- **Atmosphere:** A violent clash of solar-gold and abyssal-violet along the forward redoubts.
- **Cinematics:** Casatti's vanguard unleashes the severing edge of Tenebris, cutting through enemy siege engines, while Mi-ren's radiant Lux barrier contains the destructive blast wave. The two forces synchronize: Lux tempers Tenebris, bringing decisive victory in seven short years.

### 3.3.18. Scene 18: The Dissipation at Parley

- **Atmosphere:** A desolate borderland shrouded in thick morning mist.
- **Cinematics:** Mi-ren stands before Usuru with shield lowered, offering his hand and admitting the Crown's fault. Usuru's face twists with bitter, unyielding hatred. As Usuru roars in fury, Mi-ren's golden Lux barrier flickers and instantly dissolves into thin air, proving that violence in the heart shatters the light.

### 3.3.19. Scene 19: The Joint Realm Accord

- **Atmosphere:** The Grand Council Chamber in the reconstructed Capital.
- **Cinematics:** Envoys from all six sovereign factions sign their seals upon parchment treaties. The federated principle of "Joint Realm, Separate Kingdoms" is ratified, ensuring autonomous provincial governance under the High Crown.

### 3.3.20. Scene 20: The Prophecy of Kings

- **Atmosphere:** The timeless, floating crystalline library of the Room of Sages.
- **Cinematics:** Chrono the Sage dips a quill of starlight into an inkwell of void, recording the prophecy of the twin relics and the champion who must one day achieve Equilibrium. In the final shot, Usuru descends down stone stairs into the abyssal vaults, as shadow consumes his silhouette and history forgets his name, calling him only **The Dark Lord**.

# Chapter 4. The Past, Present & Future Thematic Triad

**Source File**: `docs/lore/thematic_triad.adoc`

## 4.1. Overview & Core Thesis

Rather than mapping protagonists solely by the physical armaments they carry, the narrative architecture of **Tenebris and Lux** links the ancient genesis era to the modern campaign through a psychological triad: **Elijah is to Usuru**, **Katrina is to Prince Casatti**, and **The Custom Bearer (`<chosen_name>`) is to Prince Mi-ren**. Governed by the foundational axiom **"Trauma is Inherited, Destiny is Chosen,"** this document establishes how ancient grievances echo into the modern campaign and how player agency determines whether history repeats its tragedies or achieves cosmic reconciliation.

## 4.2. The Thematic Triad Matrix

| Era | The Frontier Wound (Heart & Shield/Wrath) | The Renouncing Blade (Truth & Severance) | The Steward of Balance (Veil & Synthesis) |
|---|---|---|---|
| **PAST (Genesis Era)** | **Usuru**<br>• Neglected Libiran boy<br>• 10 years of peace; shattered in year 11<br>• Birthed Tenebris; became The Dark Lord | **Prince Casatti**<br>• Renounced the High Throne<br>• Founded The Academy<br>• Infused Sword of Tenebris for Justice | **Prince Mi-ren**<br>• Bore the burden of statecraft<br>• Sustained the Dreamscape projection<br>• Discovered Lux for reconciliation |
| **PRESENT (Campaign)** | **Elijah Ro-enheart**<br>• Neglected Dio Village vanguard<br>• Father vanished on 10th birthday<br>• Wields Shield of Lux against court rot | **Lady Katrina Jaren**<br>• Fled aristocratic House Jaren<br>• Seeks the Adventurers' Guild<br>• Wields Sword of Tenebris with candor | **The Custom Bearer (`<chosen_name>`)**<br>• Emergent champion in the Archives<br>• Wields the Dream Compass at night<br>• Synthesizes both perspectives |
| **FUTURE (Resolution)** | **Redemption or Ruin**<br>• Protects the realm with conscience, or succumbs to disillusionment. | **Liberation or Cynicism**<br>• Cuts the planar bindings of the past, or embraces ruthless severance. | **Stasis or Equilibrium**<br>• Repeats the comfortable lie, or confronts the Ugly Truth. |

## 4.3. Elijah Ro-enheart as Usuru (The Frontier Wound & Redemptive Counterpoint)

Although Elijah wields the **Shield of Lux** like Prince Mi-ren, his internal wound and psychological arc directly mirror **Usuru**:

### 4.3.1. The Child of the Forgotten Frontier

Usuru grew up on the neglected periphery of Libira, where royal couriers required seven days to arrive and no imperial outposts existed. Elijah grows up in **Dio Village**, an isolated border settlement where Grandma Magolia laments: *"This town used to be full of life... The kingdom doesn't pay much attention to this place anymore"*. Both youths embody the quiet complaint of the forgotten border.

### 4.3.2. The Tenth-Year Fracture

Usuru experienced ten years of innocent pilgrimage to the Festival of Peoples before his eleventh visit shattered his faith in the Crown. Elijah's childhood similarly froze on his **10th birthday**—the morning his father, **Commander Gareth Ro-enheart**, vanished under King Jihad's sealed royal orders. When Grandma Magolia asks Elijah to inspect the upstairs chest in Dio Village, Elijah immediately clings to that decade-old wound: *"Maybe it is something about dad! I haven't seen him since my 10th birthday. Sure thing. This might be my chance to get answers!"*.

### 4.3.3. The Redemptive Choice

When Usuru discovered the First King's mummified body and the secret stasis projection, his innocent faith collapsed into bitter wrath, birthing Tenebris and sparking a war of vengeance. Elijah begins with unquestioning faith in the Royal Guard, but is forced to uncover King Jihad's treason in the throne room. Rather than succumbing to destructive malice, Elijah chooses to raise the **Shield of Lux** to protect the vulnerable, serving as Usuru granted a second chance at moral redemption.

## 4.4. Lady Katrina Jaren as Prince Casatti (The Renouncing Blade of Truth)

Lady Katrina Jaren carries the **Sword of Tenebris**, but her ideological kinship belongs to **Prince Casatti**:

### 4.4.1. Renouncing Aristocratic Privilege

Just as Prince Casatti voluntarily renounced his royal birthright to the High Throne to pursue empirical truth, Katrina actively flees her aristocratic heritage in **House Jaren**—one of the wealthiest noble families on the Royal Court. She refuses to be bartered as a diplomatic pawn in court intrigues.

### 4.4.2. Merit Over Pedigree

Casatti founded The Academy because he believed understanding the cosmos required objective observation rather than imperial decree. Katrina seeks the **Adventurers' Guild**, a frontier brotherhood where practical grit, swordsmanship, and survival skills outweigh bloodlines and heraldic crests.

### 4.4.3. Surgical Severance for Justice

Casatti infused the Sword of Tenebris to execute **Absolute Justice**—slicing through entrenched siege engines to end a seven-year stalemate. Katrina wields the Sword of Tenebris with cynical, irreverent pragmatism, slicing through aristocratic hypocrisy, feudal exploitation (the St. Baron), and provincial inertia.

## 4.5. The Custom Bearer as Prince Mi-ren (The Steward of Balance)

When the player selects a custom name (`<chosen_name>`), the relics decouple from fixed protagonists, placing the player into the ideological position of **Prince Mi-ren**:

### 4.5.1. The Overarching Macro-Steward

While Elijah focuses on the military/institutional conspiracy of the Royal Guard, and Katrina navigates the underground rebellion of the Guild, the Custom Bearer holds **The Dream Compass**, navigating the astral tides of The Dreamscape exclusively at night.

### 4.5.2. Bearing the Weight of Synthesis

To achieve completion, the Custom Bearer must perform the "double work" along the `[-100, +100]` **Moral Sway** continuum, catching fragments of both Elijah's and Katrina's paths. They stand where Mi-ren stood during the 7-Year Great War: attempting to hold a fracturing world together through reconciliation.

### 4.5.3. The Burden of Stasis vs. True Equilibrium

The Custom Bearer faces Mi-ren's original dilemma. They can succumb to the **Benevolent Stasis** ending—freezing Sacyra in a protective golden barrier where no growth can occur—or they can transcend Mi-ren's mistake by mending the planar tear in the Room of Sages, reconciling the ancient wound of Usuru, and unlocking the **Secret Chronicler True Ending**.

## 4.6. The Butterfly Effect vs. The Bystander Corridor: Overriding Cyclic Fate

The thematic axiom **"Trauma is Inherited, Destiny is Chosen"** is not merely a narrative motto; it is a systemic law operating across the `[-100, +100]` Moral Sway axis:

### 4.6.1. The Bystander Corridor (`[-50, +50]`) as Karmic Determinism

Floating within `[-50, +50]` represents **Net Moral Inertia**, where opposing virtuous and selfish acts cancel out into zero displacement. In this corridor, diffusion of responsibility preserves the status quo: - Elijah is dragged into Usuru's bitter, disillusioned path upon discovering King Jihad's treason. - Katrina is locked into cynical detachment, abandoning Sacyra's rotting court to survive on the margins. - Sacyra's 3,000-year dynastic stasis repeats endlessly because ordinary citizens refuse to risk conviction.

### 4.6.2. The Butterfly Effect: Micro-Choices as Fate Breakers

Derived from chaos theory, tiny perturbations in initial conditions compound nonlinearly into radically divergent macro-states: - Seemingly modest provincial choices in Dio Village—donating 500 Gold to cover Eric the Lumberjack's taxes, intervening between John the Thief and the predatory St. Baron, or choosing non-lethal shield stagger (**Lux Barrio**) over lethal execution—act as wings flapping against 3,000 years of karmic momentum. - By accumulating deliberate micro-interventions, the party crosses the $\pm 50$ conviction threshold, overriding spiritual reincarnation and proving that while trauma is inherited, destiny is chosen.

# Chapter 5. Socio-Political Strata of the Kingdom of Sacyra

**Source File**: `docs/lore/characters.adoc`

## 5.1. Overview

This document provides complete biographical dossiers, martial doctrines, and thematic arcs for the central protagonists and key figures of **Tenebris and Lux** in the **Kingdom of Sacyra** and **The Dreamscape**. Grounded in the recovered 2017–2018 demo baseline and the Past/Present/Future thematic triad ("Trauma is Inherited, Destiny is Chosen"), it establishes the tri-perspective narrative architecture across Elijah Ro-enheart, Lady Katrina Jaren, and the Custom Bearer (`<chosen_name>`).

## 5.2. Elijah Ro-enheart (The Aegis of Lineage)

> "My father taught me that a shield is not a wall to hide behind, but a promise made to the people standing behind you."
>
> — Elijah Ro-enheart

### 5.2.1. Biography & Heritage

- **Full Name:** Elijah Ro-enheart
- **Affiliation:** Kingdom of Sacyra Royal Army (Cadet Vanguard); House Ro-enheart
- **Family Lineage:** The Ro-enhearts have served the Sacyran crown for five unbroken generations. Elijah's father, **Commander Gareth Ro-enheart**, was the esteemed former Head of the Royal Guard, famed for holding the Sunken Citadel. Gareth vanished under mysterious royal orders on Elijah's **10th birthday**, leaving behind unhealed questions and immense expectations.
- **Upbringing:** Raised in the barracks and training quadrangles of the capital citadel, but spends his youth in **Dio Village** under the watchful care of **Grandma Magolia**.
- **Visual Profile:** Broad-shouldered, athletic build, steel-blue eyes, and cropped dark hair. Wears fitted silver-steel infantry plate over deep navy quilted gambeson, marked with the defensive crest of the Ro-enhearts.

### 5.2.2. Thematic Parallel: Mirror of Ancient Usuru

Although Elijah wields the Shield of Lux, his personal arc directly mirrors **Usuru** (The Frontier Wound): * **The Boy from the Forgotten Border:** Usuru grew up in neglected Libira; Elijah grows up in neglected Dio Village. * **The Tenth-Year Fracture:** Usuru had ten years of innocent devotion before his world shattered in year 11; Elijah's childhood froze on his 10th birthday when his father vanished. * **Redemption Over Vengeance:** Confronting King Jihad's betrayal, Elijah refuses to burn the kingdom down. He raises the Shield of Lux to protect the living, representing Usuru granted a second chance at moral redemption.

### 5.2.3. Combat & Tactical Doctrine (TRPG)

- **Battlefield Role (Grid):** Frontline Heavy Vanguard / Fortress Anchor.
- **Grid Attributes:** Heavy physical mitigation, high poise, Zone of Control (ZOC) bottleneck denial, ally interception.
- **Tactical Signature:** `Lux Barrio` — Manifests a radiant geometric barrier dome ($2\dots5$ turn duration, $25\%$ damage mitigation).

## 5.3. Katrina Jaren (The Blade of Freedom)

> "Silk dresses and court etiquette won't turn back a vanguard blade. Let my cousins argue over tax registries; my place is on the frontier."
>
> — Katrina Jaren

### 5.3.1. Biography & Heritage

- **Full Name:** Lady Katrina Jaren
- **Affiliation:** House Jaren (Aristocracy of Sacyra); Aspirant of the Adventurers' Guild
- **Family Lineage: House Jaren** is an ancient, wealthy noble family commanding lucrative vineyards and silver mines in Sacyra's eastern provinces, holding significant patronage leverage on the Royal Court.
- **Upbringing:** Groomed by diplomats and heralds to occupy a ceremonial court seat, Katrina despised courtly hypocrisy, frequently escaping to Dio Village and training with Adventurers' Guild rangers.
- **Visual Profile:** Agile, athletic build, sharp amber eyes, and unbound chestnut hair tied in a practical combat braid. Clad in tailored lightweight leather arming doublets over noble-weave tunic, with rapier and long-dagger scabbards.

### 5.3.2. Thematic Parallel: Mirror of Prince Casatti

Katrina carries the Sword of Tenebris and embodies the philosophy of **Prince Casatti** (The Renouncing Blade): * **Renouncing Aristocratic Birthright:** Just as Casatti renounced the High Throne, Katrina flees House Jaren. * **Merit Over Pedigree:** Casatti founded The Academy for empirical truth; Katrina seeks the Adventurers' Guild where practical survival and skill outweigh heraldic crests. * **Surgical Severance:** Wields the Sword of Tenebris with sharp, cynical candor to slice through court corruption, feudal extortion (the St. Baron), and provincial stagnation.

### 5.3.3. Combat & Tactical Doctrine (TRPG)

- **Battlefield Role (Grid):** High-Mobility Skirmisher / Flanking Striker.
- **Grid Attributes:** 4–5 cell movement, ignores basic ZOC delays, backstab criticals, hit-and-run disengagement.
- **Tactical Signature:** `Cuttatio Tenebris` — Striking primary target while inflicting $20\%$ splash cleave to adjacent enemy units.

## 5.4. The Custom Bearer (`<chosen_name>` — The Secret Chronicler)

### 5.4.1. The Third Perspective: Mirror of Prince Mi-ren

When the player customizes their protagonist name via `name <str>`, the relics decouple from fixed identities, placing the player into the ideological position of **Prince Mi-ren** (The Steward of Balance): * **Macro-Steward of the Veil:** Holds **The Dream Compass**, entering The Dreamscape exclusively during nocturnal windows (`20:00 ⇐ hour || hour ⇐ 04:00`). * **The "Double Work" of Synthesis:** Navigates the `[-100, +100]` Moral Sway continuum to weave fragments of both Elijah's military investigation and Katrina's underground rebellion. * **Resolving the Stasis Fallacy:** Must avoid repeating Mi-ren's ancient error (Benevolent Stasis) to unlock the **Secret Chronicler True Ending**, reconciling the ancient wound of Usuru and permanently mending the planar tear.

# 5.5. Antagonists & The Two-Half Campaign

## 5.5.1. King Jihad (The Sovereign of Ambition — Act I Antagonist)

- **Office:** Reigning Monarch of Sacyra, commanding the Royal Guard.
- **Role:** The grotesque inverse of the Founding Dynasty. Imposes state censorship on border warnings and seeks to usurp void power for total dominion.
- **The Possession:** Sitting on the High Throne above the ancient planar wound, Jihad's megalomania inverts the conduit, allowing a dormant remnant of Usuru (**The Dark Lord**) to possess him at the Act I climax.

## 5.5.2. The Dark Lord (Usuru's Descent — Act II Antagonist)

- **Canonical Identity:** The ancient Libiran youth **Usuru**, whose faith in the Crown was shattered during the 11th Festival.
- **Role:** The overarching existential shadow ("The A Plot") menacing Sacyra from the subterranean Abyssal Vaults.
- **The Final Confrontation:** In Act II, heroes must penetrate the cosmic threshold. Slaying him blindly produces the Neutral Ending; achieving the **Stance of Equilibrium** (**Duo-Loca Bashin n Slash**) severs his abyssal tether while absorbing planar fallout without dissipating, granting Usuru eternal rest.

# Chapter 6. The Six Powers of the Continent

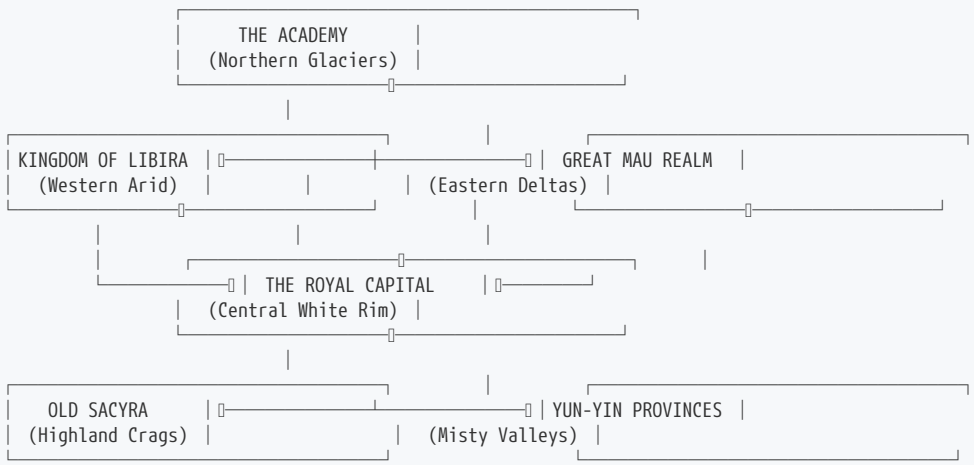
**Source File**: `docs/lore/realms/the_six_powers.adoc`

## 6.1. Overview

The historical landscape of **Tenebris and Lux** was shaped by the clash and eventual reconciliation of **The Six Powers** during Eras IV through VI. When the planar rupture during the 11th Festival of Peoples shattered the First King's Pax Sacyrana, Usuru rallied these disparate sovereign nations into the catastrophic **Seven-Year Great War**. Understanding the cultural, architectural, and military profile of each power is essential to comprehending the ancestral factions, regional alliances, and diplomatic endings available within the TRPG campaign.

> No single crown shall claim the sun, and no lone throne shall command the shadows. Six thrones bled upon the salt and marble; six thrones shall swear that the realm shall be joint in peace, yet separate in kingdom, so long as the stars circle the spindle.
>
> — The Accord of Kings, Preamble - Inscribed upon the Six Obelisks of Unity

## 6.2. Geopolitical Map & Sovereign Overview




## 6.3. Comprehensive Profiles of the Six Powers

### 6.3.1. The Royal Capital of Sacyra (The Radiant Center)

- **Domain & Geography**: The fertile central plains and plateaus surrounding the White Rim Citadel.
- **Cultural Philosophy**: Monarchy, institutional duty, legalism, and devotion to the light of the First King.
- **Military Doctrine**: Heavy vanguard infantry, phalanx formations, and divine warding priests wielding the techniques of the **Shield of Lux**.
- **Key Figure**: King Jihad (Present Era); Prince Mi-ren (Founding Era).

### 6.3.2. The Kingdom of Libira (The Arid Frontier)

- **Domain & Geography**: The harsh western desert canyons, sandstone mesas, and salt flats.

- **Cultural Philosophy**: Radical communal self-reliance, stoic endurance, and fierce resentment of centralized royal arrogance ("The Quiet Complaint").
- **Military Doctrine**: Agile camel skirmishers, desert outriders, and shadow assassins who pioneered the early stances of the **Sword of Tenebris**.
- **Key Figure**: Usuru the Wanderer (The Dark Lord).

### 6.3.3. The Great Mau Kingdom (The Merchant Empire of the East)

- **Domain & Geography**: The sprawling eastern river deltas, rich alluvial floodplains, and bustling port cities.
- **Cultural Philosophy**: Mercantilism, contract law, civic guilds, and wealth accumulation. To the Mau, war is an economic calculation rather than an ideological crusade.
- **Military Doctrine**: Mercenary phalanxes, river ironclad barges, and repeating crossbowmen funded by the grand banking cartels.
- **Role in the Great War**: Financed Usuru's coalition armies in exchange for mining rights in Libira, switching sides at the midpoint when Sacyra offered trade tariff exemptions.

### 6.3.4. Old Sacyra (The Highland Ancestral Cradle)

- **Domain & Geography**: The sheer mountain peaks and pine forests north of the Sacyran river valley.
- **Cultural Philosophy**: Strict aristocratic traditionalism, ancestral veneration, and genealogical pride. View the modern Royal Capital as degenerate upstarts who diluted pure Sacyran blood.
- **Military Doctrine**: Mountain rangers, greatsword shock troops, and heavy cavalry capable of fighting on steep inclines.
- **Role in the Great War**: Joined Usuru early to punish the Capital, but broke away when Usuru unleashed uncontrollable void constructs into mortal villages.

### 6.3.5. The Academy (The Spire of Pure Reason)

- **Domain & Geography**: The frozen northern tundras and geothermal volcanic springs.
- **Cultural Philosophy**: Scientism, arcanotech experimentation, and the empirical dissection of cosmic forces. Reject both religious worship of Lux and superstitious fear of Tenebris.
- **Military Doctrine**: Alchemical siege engines, elemental clockwork automata, and crystal-resonance artillery.
- **Role in the Great War**: Remained neutral for five years before joining the coalition to study the abyssal tear firsthand, later contributing the mathematical formulas used to draft the **Accord of Kings**.

### 6.3.6. The Yun-yin Provinces (The Valleys of Mist & Wind)

- **Domain & Geography**: The subtropical southeastern archipelago and bamboo cloud forests.
- **Cultural Philosophy**: Spiritual harmony, ki manipulation, cyclical flow, and the balance of dualities. Understood the metaphysical neutrality of darkness long before Sacyran theologians.
- **Military Doctrine**: Martial artists, wind-runners, and agile spear masters using evasion and counter-poise redirection.
- **Role in the Great War**: Sided with Usuru to restore balance against Sacyra's overbearing expansionism, ultimately brokering the peace conference that created the **Joint Realm, Separate Kingdoms** doctrine.

## 6.4. The Accord of Kings: "Joint Realm, Separate Kingdoms"

Following seven years of devastating continental slaughter, the exhausted sovereigns met upon the neutral plains of the Central Marches to sign the historic **Accord of Kings**:

| Article | Diplomatic Covenant & Systemic Law |
|---|---|
| **Article I: Mutual Sovereignty** | Each of the Six Powers maintains complete legislative, military, and economic autonomy within their borders. |
| **Article II: The Joint Ward** | Sacyra's Royal Guard is legally entrusted with guarding the central rift beneath the High Throne, but may never station troops within foreign realms without consent. |
| **Article III: The Free Caravan** | Courier and trade routes between all six nations are declared sacrosanct, guaranteeing that no province may be starved through communication blockade. |



In the modern era, King Jihad's clandestine efforts to annex the frontier and resurrect the centralized empire violate the fundamental tenets of the Accord of Kings, providing the political justification for the party's multi-ending divergence.

# Chapter 7. Realm Chronicle: The Kingdom of Libira

**Source File**: `docs/lore/realms/kingdom_of_libira.adoc`

## 7.1. Overview

The **Kingdom of Libira** is a remote, windswept, and arid frontier realm situated far to the west across the Great Salt Basin, a full seven-day courier journey from the **Royal Capital of Sacyra**. Lacking a fortified relay outpost system throughout the founding era, Libira became the birthplace of the simmering socio-political grievance known historically as **"The Quiet Complaint"**. It was here that a nameless boy was born to a humble shepherd family, traveled the continent as the Three-Day Wanderer, and returned across the warped desert passes as **Usuru**, sovereign of the broken crown and the future **Dark Lord**.

> The sun burns the rocks by day, and the wind freezes the sands by night. Sacyra has water, gold, and songs for its kings. Libira has only stone, salt, and the memory of every empty wagon that never returned from the capital.
>
> — Old Libiran Nomad Proverb, Inscribed upon an Adobe Cistern

## 7.2. Geography, Climate & The 7-Day Courier Chasm

- **Geographical Classification**: High-altitude arid plateau, red sandstone mesas, salt flats, and canyon wadis.
- **The Communication Gap**:
  - Distance: A full week (seven days) of hard riding by swift couriers across barren terrain.
  - Absence of Infrastructure: Because Libira was historically treated as an unestablished border territory, the Sacyran crown never invested in a chain of relay watchtowers or horse stations.
  - The Warped Rift: When the 11th Festival ritual collapsed, the spatial fabric between Libira and the capital expanded violently, turning a week-long journey into an impassable spatial wasteland.

| Dimension | Specification |
| --- | --- |
| **Terrain Profile** | Crimson sand dunes, deep sandstone ravines, dry limestone dry-washes. |
| **Settlement Pattern** | Clustered adobe townships built around subterranean aquifers and ancient cisterns. |
| **Key Settlements** | **The Assembly of Libira** (Central adobe council citadel); **Wadi Usuru**. |
| **Strategic Resources** | Saltpeter, red iron ore, draught lizards, subterranean aquifer water. |

## 7.3. Socio-Political Philosophy: The "Quiet Complaint"

Unlike open rebellions driven by greedy warlords, Libira's grievance was born of profound, generational neglect:

- **The Disconnect**: The capital celebrated annual harvests and imperial festivals, issuing royal edicts that arrived in Libira weeks after decrees were signed. Libiran citizens suffered under droughts and border raids without receiving imperial military reinforcement.
- **The "Quiet Complaint"**: A cultural attitude of stoic, bitter endurance. Libirans learned never to expect salvation from the Crown, fostering fierce communal solidarity and deep suspicion of ceremonial authority.

- **Modern Parallels in Dio Village**: The same dynamic repeats in the present era in **Dio Village**, where Grandma Magolia sighs over the Crown's neglect and the St. Baron taxes commoners without providing protection from monster raids.

> In the narrative architecture of **Tenebris and Lux**, Libira represents the raw root of all conflict. Had the First King's court listened to the nameless boy's petition and built a simple relay road, the Seven-Year Great War would never have occurred.

## 7.4. The Tragedy of Usuru's Assembly

When Usuru returned from the capital with his skin burned by radiant energy and his eyes alight with abyssal fury, the Libiran Assembly Hall became the focal point of continental history:

1. **The Gathering of Elders**: Tribal elders, weary of decades of unanswered petitions, wept as the youth recounted how the Sacyran guards struck him down simply for asking for roads.
2. **The Acclamation of the Sovereign**: For the first time in their history, the disparate nomadic clans united, placing the iron circlet upon Usuru's brow and naming him their young King.
3. **The Call to the 6 Powers**: Leveraging trade routes across the salt flats, Usuru sent emissaries to Great Mau, Old Sacyra, and the Yun-yin Provinces, turning a localized frontier petition into a world-spanning coalition war.

## 7.5. Tactical Grid Battlefield: The Red Sandstone Wadi

```
 [0]   [1]   [2]   [3]   [4]   [5]   [6]   [7]
┌──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┐
│ MESA │ MESA │      │ CSTN │ CSTN │      │ MESA │ MESA │ [0] Ancient Cistern (`CSTN`) / High Mesa
├──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────┤
│ MESA │ SAND │ SAND │ USURU│ LIBR1│ LIBR2│ SAND │ MESA │ [1] Libiran Skirmishers & Camel Cavalry
├──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────┤
│      │ SAND │ DUNE │ DUNE │ DUNE │ SAND │      │      │ [2] Shifting Sand Dunes (Movement Penalty)
├──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────┤
│      │ HERO1│ HERO2│ HERO3│ ALLY │      │      │      │ [3] Sacyran Expeditionary Force
├──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────┤
│ CANYN│ CANYN│      │ ROAD │ ROAD │      │ CANYN│ CANYN│ [4] Canyon Gorge Chokepoint
└──────┴──────┴──────┴──────┴──────┴──────┴──────┴──────┘
```

### 7.5.1. Battlefield Invariants

1. **Shifting Dunes (`DUNE`)**: Reduces movement range by $1$ tile for heavy armor units (e.g. Elijah).
2. **Canyon Gorges (`CANYN`)**: Impassable chasms; units knocked back into chasms suffer instant fall defeat.
3. **High Sandstone Mesas (`MESA`)**: High ground granting $+25\%$ critical chance to Libiran bowmen.

# Chapter 8. Realm Chronicle: The Royal Capital of Sacyra

**Source File**: `docs/lore/realms/the_royal_capital.adoc`

## 8.1. Overview

The **Royal Capital of Sacyra** is the administrative, spiritual, and military apex of the kingdom. Crowned by the gleaming white-stone bastions of the **High Throne Citadel**, the city embodies the architectural zenith of Era I's founding dynasties. Yet beneath its ceremonial grandeur, opulent marble halls, and courtly pageantry lies a festering metaphysical wound: **The King's Curse**, an unhealed planar rupture beneath the throne room floor that has leaked abyssal corruption for three millennia, steadily eroding the sanity of **King Jihad**.

> We wash the marble with lavender and rosewater, but the stones remember the blood. When the sun reaches its zenith, the shadow of the throne does not fall toward the door—it falls inward, pooling like dark oil around the King's feet. He smiles at the envoys, but his teeth are gray. Sacyra is dancing upon the lid of a tomb.
>
> — High Priest Roté, Royal Sanctum Secret Journal

## 8.2. Geography, Architecture & Urban Districts

- **Topographical Location**: Built upon the central continental plateau overlooking the Sacyran river basin.
- **Access Routes**: Reached via the Grand Imperial Causeway traversing the **Sacyran Marches**.
- **Architectural Style**: Monolithic white limestone, soaring arched flying buttresses, gilded radiant domes, and deep subterranean catacombs dating back to the Accord of Kings.

| District | Description & Socio-Political Function |
|---|---|
| **The Citadel Ward** | The innermost fortress housing the High Throne, the Royal Council Chambers, and the private quarters of King Jihad. |
| **The Aristocratic Quarter** | Palatial townhouse estates of the Great Noble Houses (including House Jaren); seat of courtly intrigue and political scheming. |
| **The Grand Cathedral of Lux** | Spiritual center governed by High Priest Roté, where the sacred relics are celebrated and celestial portents are interpreted. |
| **The Lower River Ward** | Densely packed commoner quarter, guildhalls, iron forges, and the local headquarters of the **Adventurers' Guild**. |
| **The Subterranean Vaults** | Forbidden pre-dynastic catacombs directly beneath the throne, site of the 11th Festival Rupture. |

## 8.3. Lore & The King's Curse Planar Wound

During the 11th Festival of Peoples, the young Libiran wanderer Usuru accidentally struck the ritual focus stone during the First King's communion ritual. The resulting explosion severed the First King's soul and cracked the bedrock beneath the dais:

1. **The Hidden Rupture**: The First 30 Kings sealed the physical chamber beneath massive basalt slabs inscribed with warding runes, building the High Throne directly over the scar to mask its presence.
2. **The 3,000-Year Dynastic Wave**: Every monarch who sat upon the High Throne was bathed in the slow, imperceptible radiation of the planar breach. While strong-willed sovereigns resisted its whispers, weak or ambitious rulers inevitably succumbed to paranoia and tyrannical megalomania.
3. **The Possession of King Jihad**: In the present era, King Jihad's obsession with matching the legendary military feats of his ancestors drove him to open the basalt seal. Tenebrous vapor now directly commands the monarch's nervous system, transforming him into an unwitting puppet of **The Dark Lord**.



> The planar wound beneath the High Throne is not merely a narrative backdrop—it serves as the **Midpoint Campaign Fork** in the ludological engine. Entering the throne room without the **Dream Compass** triggers the tragic **Neutral Path**, whereas possessing the calibrated compass allows the party to expose the curse and enter the **True Ending Path**.

## 8.4. Factions & Urban Intrigue

The Capital is torn between four competing power structures:

- **The Royal Guard**: Commanded by hardline royalists loyal to the throne, increasingly bewildered by King Jihad's erratic and tyrannical decrees.
- **House Jaren & The High Nobles**: Aristocratic court faction maneuvering to consolidate tax monopolies and replace the Ro-enhearts' military influence.
- **The High Priesthood**: Church hierarchy caught between religious dogma and High Priest Roté's terrifying revelations regarding the cosmic corruption of the Crown.
- **The Adventurers' Guild**: Underground network of commoner veterans and sympathizers (including Katrina Jaren) operating outside royal law to protect the citizenry from unexplained void manifestations.

## 8.5. Tactical Grid Battlefield: The High Throne Chamber

```
 [0]   [1]   [2]   [3]   [4]   [5]   [6]   [7]
┌─────┬─────┬─────┬─────┬─────┬─────┬─────┬─────┐
│PILLR│DAIS │THRNE│VOIDC│THRNE│DAIS │PILLR│WALL │ [0] High Dais / The Planar Wound (`VOIDC`)
├─────┼─────┼─────┼─────┼─────┼─────┼─────┼─────┤
│STEPS│STEPS│JIHAD│GUARD│GUARD│STEPS│STEPS│     │ [1] Royal Guard Phalanx
├─────┼─────┼─────┼─────┼─────┼─────┼─────┼─────┤
│CARP │CARP │CARP │CARP │CARP │CARP │CARP │CARP │ [2] Crimson Carpet Center Aisle
├─────┼─────┼─────┼─────┼─────┼─────┼─────┼─────┤
│PILLR│     │HERO1│HERO2│HERO3│     │PILLR│     │ [3] Player Party Infiltration Entry
├─────┼─────┼─────┼─────┼─────┼─────┼─────┼─────┤
│DOOR │DOOR │DOOR │DOOR │DOOR │DOOR │DOOR │DOOR │ [4] Grand Audience Chamber Entrance
└─────┴─────┴─────┴─────┴─────┴─────┴─────┴─────┘
```

### 8.5.1. Battlefield Invariants

1. **The Planar Wound (`VOIDC`)**: Generates an aura of $-15$ HP decay per turn to all units standing within a 2-cell radius unless protected by **Lux Barrio**.
2. **Marble Pillars (`PILLR`)**: Provide heavy cover ($+30\%$ evasion against ranged magic).
3. **Royal Dais Steps (`STEPS`)**: Units attacking upward suffer a $-15\%$ accuracy penalty.

# Chapter 9. Realm Chronicle: The Sacyran Marches

**Source File**: `docs/lore/realms/sacyra_marches.adoc`

## 9.1. Overview

The **Sacyran Marches** (`LOC_SACYRA_MARCHES`) encompass the windswept limestone hills, fortified military outposts, and ancient battlefields stretching between the **Whispering Woods** and the white-stone bastions of the **Royal Capital**. Established during the Seven-Year Great War as the kingdom's primary defensive perimeter, the Marches serve as the tactical proving ground where the player transitions from local woodland skirmishes to large-scale military engagements.

> The Marches are where boyish dreams of glory die and real soldiers are forged. The wind here carries no warmth—only dust, iron, and the perpetual vigilance of men who know that if these ridges fall, nothing stands between the abyss and our homes.
>
> — Commander Gareth Ro-enheart, Field Dispatch - Frontier Fortress Redoubt

## 9.2. Geography & Strategic Topology

- **Location Identifier**: `LOC_SACYRA_MARCHES` (Integer index `3` in `location_id_t`).
- **Territorial Domain**: Central border province linking southern farmlands with the high plateaus.
- **Topographical Neighbors**:
  - **West / South**: Slopes gently downward into the canopy of the **Whispering Woods**.
  - **North / East**: Connects directly to the grand paved highway leading to the **Royal Capital**.
- **Terrain Features**: Terraced limestone plateaus, fortified palisades, stone watchtowers, and deep defensive trenches left behind by centuries of border warfare.

| Dimension | Specification |
| --- | --- |
| **Ambient Threat Level** | High ($110–140$ Threat Rating). |
| **Weather Phenomena** | High-altitude gale winds, dry squalls, sudden dust storms reducing vision. |
| **Strategic Redoubts** | **Tower of the Sunken Citadel** (Gareth Ro-enheart's former garrison). |
| **Factional Influence** | Royal Guard Vanguard, Frontier Mercenary Guild, Shadow Host Infiltrators. |

## 9.3. Lore & The Legacy of the 7-Year Great War

During the Seven-Year Great War (approximately 3,000 years ago), Usuru's coalition forces launched their primary armored spearhead across these very hills. The earth of the Marches remains soaked in residual abyssal and radiant energies:

- **The Trenches of the First 30 Kings**: Deep geometric trenches dug into the chalk hills to funnel void automata into kill zones.
- **The Sunken Citadel**: A subterranean fortress carved into the limestone bedrock where Commander Gareth Ro-enheart made his legendary stand before his mysterious disappearance following Elijah's 10th birthday.

- **The Broken Standards**: Decaying heraldic banners of ancient kingdoms (Great Mau, Libira, Old Sacyra) still flutter from iron spires atop the limestone ridges.



The Marches are the first location where the player encounters organized military opposition rather than mindless wild beasts. Enemies employ tactical formations, archery volleys, and coordinated flanking maneuvers.

## 9.4. Field Bestiary & Combat Formations

| Hostile Unit | HP / DEF | Flank Vigor | Tactical Discipline & Threat Profile |
|---|---|---|---|
| **Shadow Scout** | $110$ HP / $18$ DEF | $80$ Vigor | High speed outrider; marks targets for ranged volleys. |
| **Vanguard Patrol** | $160$ HP / $26$ DEF | $90$ Vigor | Heavy shield infantry; projects Zone of Control. |
| **Abyssal Ballista** | $220$ HP / $30$ DEF | $20$ Vigor | Siege construct; fires armor-piercing void bolts. |



Vanguard Patrols utilize **Phalanx Formations** that reduce frontal physical damage by $40\%$. Players must utilize Katrina's high movement mobility or the Custom Bearer's **Edge Striking** stance to maneuver behind their line for decisive backstabs.

## 9.5. Tactical Grid Battlefield: The Palisade Ridge

```
 [0]   [1]   [2]   [3]   [4]   [5]   [6]   [7]
┌─────┬─────┬─────┬─────┬─────┬─────┬─────┬─────┬──────────────────────────────┐
│TWR_N│WALL │GATE │GATE │WALL │WALL │TWR_E│HILL │ [0] North -> High Road to Capital
├─────┼─────┼─────┼─────┼─────┼─────┼─────┼─────┼──────────────────────────────┤
│RAMP │RAMP │SCOUT│VANG1│VANG2│SCOUT│RAMP │HILL │ [1] Defensive Line
├─────┼─────┼─────┼─────┼─────┼─────┼─────┼─────┼──────────────────────────────┤
│TRENCH     │     │     │     │     │TRENCH     │ [2] Muddy Trench (Movement Penalty)
├─────┼─────┼─────┼─────┼─────┼─────┼─────┼─────┼──────────────────────────────┤
│     │HERO1│HERO2│ALLY1│ALLY2│     │     │     │ [3] Player Assault Vanguard
├─────┼─────┼─────┼─────┼─────┼─────┼─────┼─────┼──────────────────────────────┤
│     │ROAD │ROAD │ROAD │ROAD │     │     │     │ [4] South -> Whispering Woods
└─────┴─────┴─────┴─────┴─────┴─────┴─────┴─────┴──────────────────────────────┘
```

### 9.5.1. Battlefield Mechanics

1. **Elevated Ramparts (`RAMP`)**: Grants ranged units $+1$ Attack Range and $+20\%$ Physical Defense.
2. **Muddy Trench (`TRENCH`)**: Costs $2.5\times$ movement points to traverse and prevents defensive parries.
3. **Iron Gate (`GATE`)**: Destructible obstacle ($150$ HP) that must be breached or bypassed via flanking ladders.

# Chapter 10. Realm Chronicle: The Whispering Woods

**Source File**: `docs/lore/realms/whispering_woods.adoc`

## 10.1. Overview

The **Whispering Woods** (`LOC_WHISPERING_WOODS`) form an ancient, dense primeval canopy encircling **Dio Village**, separating the peaceful hamlet from the windswept frontier of the **Sacyran Marches**. Named for the haunting acoustic phenomenon produced by autumn winds filtering through hollow ironwood boughs, the forest serves as the player's first physical wilderness trial after awakening from **The Dreamscape**.

> Those woods were quiet when my grandfather worked the timber. Now, you can't walk twenty paces off the road without hearing shadows moving in the thicket. Something deep in the capital is leaking into the roots.
>
> — Eric the Lumberjack, Dio Village Timber Yard

## 10.2. Geography & Environmental Atmosphere

- **Location Identifier**: `LOC_WHISPERING_WOODS` (Integer index `2` in `location_id_t`).
- **Territorial Domain**: Southwestern province of the Kingdom of Sacyra.
- **Topographical Neighbors**:
  - **South**: Borders the agricultural plots and timber yards of **Dio Village**.
  - **North & East**: Ascends into the crags and signal hills of the **Sacyran Marches**.
- **Flora & Terrain**: Gnarled ironwood trunks, glowing amber fungi, thick moss carpets, and narrow cobblestone paths choked by invasive briers and fallen timber.

| Dimension | Realm Specification |
|---|---|
| **Ambient Threat Level** | Moderate ($70–90$ Threat Rating). |
| **Weather Phenomena** | Morning autumn mist; nocturnal veil thinning; ambient whisper reverberation. |
| **Sanctuary Nodes** | The Forester's Shrine (Campfire rest point; restores $100\%$ MP). |
| **Resource Harvest** | Ironwood logs, Sacyran belladonna herbs, glowing resonant spores. |

## 10.3. Lore & Ancient Secrets

Local Sacyran folklore claims the woods were planted by Prince Casatti during his frontier expeditions to serve as a natural barrier shielding the commoners from imperial court politics. Beneath the dense root network lie forgotten stone markers from Era I, indicating that the forest grew atop an ancient tectonic fault line directly connected to the abyssal fissure beneath the Royal Capital.

**Archival Marginalia: The Soughing Boughs**

When night falls across the canopy, travelers report hearing murmurs that sound like voices debating in the archaic dialect of Era IV. Scholars from The Academy believe these are acoustic reflections of the First King's sealed

court, vibrating along ley lines from the Dreamscape.

## 10.4. Bestiary & Field Encounters

Corrupted by ambient abyssal miasma leaking outward from the capital's planar wound, the native fauna of the woods has mutated into predatory shadow beasts:

| Hostile Entity | HP / DEF | Flank Vigor | Combat Behavior & Threat Profile |
|---|---|---|---|
| **Umbral Wolf** | $85$ HP / $14$ DEF | $60$ Vigor | Pack hunter; attempts coordinated flanking sprints. |
| **Brier Lurker** | $120$ HP / $18$ DEF | $40$ Vigor | Concealed in brambles; inflicts poison and entangle. |
| **Twilight Raven** | $45$ HP / $8$ DEF | $80$ Vigor | High evasion; dives to disrupt casting stances. |



Umbral Wolves receive a $+25\%$ attack bonus when attacking from an adjacent flanking tile. Players must anchor their formation using **Lux Barrio** to prevent fragile party members from being isolated.

## 10.5. Tactical Grid Map Architecture

Combat within the Whispering Woods utilizes dense environmental obstacles and restrictive chokepoints:

```
 [0]   [1]   [2]   [3]   [4]   [5]   [6]   [7]
┌──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────┬──────────────────────────────────┐
│ TREE │ TREE │      │ PATH │ PATH │      │ TREE │ TREE │ [0] North -> Sacyran Marches     │
├──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────────────────────────────────┤
│ TREE │BRIER │      │      │ WOLF1│      │BRIER │ TREE │ [1]                              │
├──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────────────────────────────────┤
│      │      │ ROCK │      │      │WOLF2 │      │      │ [2] Chokepoint Corridor          │
├──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────────────────────────────────┤
│ PATH │ PATH │      │HERO1 │HERO2 │      │ PATH │ PATH │ [3] Player Vanguard Deployment   │
├──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────┼──────────────────────────────────┤
│ TREE │      │ SHRN │ PATH │ PATH │      │      │ TREE │ [4] South -> Dio Village         │
└──────┴──────┴──────┴──────┴──────┴──────┴──────┴──────┴──────────────────────────────────┘
```

### 10.5.1. Environmental Grid Invariants

1. **Tree & Granite Rock Obstacles**: Completely block line-of-sight and unit transit.
2. **Dense Brier Patches**: Inflict $2.0\times$ movement cost and $-10$ HP per step for non-flying units.
3. **Forester's Shrine (`SHRN`)**: Standing adjacent to the shrine grants $+15\%$ Lux warding efficiency.

# Chapter 11. Realm Chronicle: Dio Village

**Source File**: `docs/lore/realms/dio_village.adoc`

## 11.1. Overview & Pastoral Heritage

**Dio Village** is a pastoral woodland settlement situated in the southwestern valley frontier of the **Kingdom of Sacyra**. Functioning as the physical inception of the hero's journey upon awakening from **The Dreamscape**, Dio Village serves as the terrestrial anchor of Chapter 1, embodying the modern continuation of ancient Libira's systemic neglect under the 3,000-year dynastic order.

## 11.2. Socio-Political Role: "The Quiet Complaint" of Modern Sacyra

- **Mirror of Ancient Libira:** Just as ancient Libira suffered a 7-day communication delay and lacked imperial relay outposts, modern Dio Village is neglected under the "Joint Realm, Separate Kingdoms" bureaucracy. As Grandma Magolia sighs in Map 021: *"This town used to be full of life. **sighs** The kingdom doesn't pay much attention to this place anymore..."*.
- **Economic Fragility & Class Exploitation:** While the Capital enjoys insulated peace, Dio Village faces monster outbreaks near the timber lines. **Eric the Lumberjack** faces ruin because monsters block the woods, while the predatory **St. Baron** exploits commoners through exorbitant rents and usury.
- **Gareth Ro-enheart's Shadow:** Dio Village is where Commander Gareth Ro-enheart placed his son Elijah under Grandma Magolia's care before his mysterious disappearance on Elijah's 10th birthday.

## 11.3. Key Landmarks & Dual-Perspective Quest Hubs

- **Grandma Magolia's Cottage (Maps 020 & 021):** The warm stone hearth where the protagonist awakens. Contains the upstairs chest housing **Grandma's Watch** (activating the Layer 0 monotonic clock).
  - **Elijah's Approach:** Hesitates out of filial respect and family etiquette.
  - **Katrina's Approach:** Seeks the will immediately to escape the dead-end town.
- **Eric's Timber Yard & Test of Honesty (Map 028):** Filling Eric's rent pot shifts Moral Sway positively and rewards the **Ring of Fire**, enabling flame-weave relic embellishments.
- **Monica's Mystic Grove & Lyri's Offering (Maps 023, 027, 029):** Former Royal Battlemage Monica guards the memorial totem of fallen soldiers. Stopping the bandit kid and buying an offering potion for **Lyri, the Goddess of Peace**, attunes the relics to arcane resonance and awards the **Book of Monsters**.
- **The St. Baron's Estate & John the Thief (Maps 030 & 031):** Deciding whether to aid **John the Thief** in robbing the corrupt baron versus upholding feudal property law shapes the penetration or bastion embellishments of the relics.

## 11.4. Mechanical Role in the C11 Simulation Engine

- **Engine Enumeration:** Registered in `include/sim/sim_types.h` as `LOC_DIO_VILLAGE`.
- **Sanctuary Province:** Zero hostile vigor during standard state; permits inventory management, dialogue interactions, moral compass calibration, and Layer 0 monotonic time passage.
- **Terrestrial Lockout:** Gated throughout Chapter 1, preventing return to The Dreamscape until the acquisition and awakening of **The Dream Compass**.

# Chapter 12. Realm Chronicle: The Dreamscape

**Source File**: `docs/lore/realms/the_dreamscape.adoc`

## 12.1. Overview & Metaphysical Essence

**The Dreamscape** (also known as the Astral Substrate or Room of Sages) is an ethereal plane of consciousness, ancestral memory, and timeless trials existing between waking reality and slumber. Grounded in the recovered 2017–2018 demo script (Maps 006, 008, 011, 041, 042) and verified in the ISO C11 engine architecture as `LOC_THE_DREAMSCAPE`, it serves as the cosmological origin point where the protagonist awakens before entering the temporal Kingdom of Sacyra.

## 12.2. Historical & Metaphysical Provenance

- **The Coma of the First King:** Following the inaugural Festival of Peoples 3,000 years ago, the First King collapsed into a coma. His spirit was found fully lucid and awake inside The Dreamscape. While Prince Mi-ren and Prince Casatti sustained projections for ten festivals, the 11th Festival tampering by Usuru snapped the ethereal tether, permanently sealing the First King's soul here.
- **The Annual Thinning of the Veil:** For one week each year—coinciding with the Festival of Heroes—the boundary ("the Veil") separating the temporal world from The Dreamscape thins to gossamer, allowing mortals to project desires across the threshold.
- **The Three Laws of Crossing the Veil:**
  1. **The Mirror of Intent:** The ether reflects mortal intent. Pure devotion projects harmony; terror, spite, or megalomania (such as King Jihad's treason) inverts the conduit into waking nightmares.
  2. **The Severed Tether:** Sustaining a long bridge across the veil carries fatal risk. Violent interruption snaps spiritual tethers, marooning consciousness permanently.
  3. **The Dream Eater Infiltration:** Tearing or weakening the veil attracts **Dream Eaters**—parasitic entities native to the deep astral voids that devour mortal memories, dreams, and ancestral records.

## 12.3. Mechanical Integration in the C11 Engine

- **Engine Enumeration:** Registered in `include/sim/sim_types.h` as `LOC_THE_DREAMSCAPE`.
- **Opening Trial:** The protagonist awakens in the Hall of Sages with **Chrono the Sage**, attuning to class archetype stones and testing `Lux Barrio` and `Cuttatio Tenebris` against Dream Eaters before Chrono's command (**"AWAKEN!"**) transports consciousness to Grandma Magolia's hearth in Dio Village.
- **Terrestrial Gating:** Throughout Chapter 1 (the demo), return to The Dreamscape is barred, anchoring the player in terrestrial human stakes.
- **The Dream Compass & Nocturnal Phasing:** Upgraded from Grandma's Watch, the Dream Compass allows entry to The Dreamscape exclusively during nocturnal hours (`20:00 ⇐ hour || hour ⇐ 04:00`).
- **Tactical LeTBS Environments:** Comprises floating platform grids, shifting crystal leylines, and astral mechanisms that restore MP while exposing units to planar shear.

# Chapter 13. High School Inception & TRPG Genesis

**Source File**: `docs/dev_logs/2026/10/lore_and_design/devlog_002_the_sacyra_chronicles_and_trpg_vision.adoc`

## 13.1. Abstract

This developer log chronicles the foundational provenance and overarching creative vision of **Tenebris and Lux**, rooted in its 2014–2018 high school inception as the premier game project of CreativeWorkzStudio. It documents the elimination of placeholder factions in favor of the canonical **Kingdom of Sacyra**, the introduction of dual protagonists **Elijah Ro-enheart** and **Katrina Jaren**, and the technical specification of the game's true genre: a **Tactical Strategy RPG (TRPG)** fusing grid-based tactical maneuvering with cinematic traditional side-scroll / action-based combat resolution.

## 13.2. Historical Provenance: The 2014–2018 High School Genesis

Before CreativeWorkzStudio evolved into a multifaceted publishing house, **Tenebris and Lux** was its very first major game in active development.

- **The RPG Maker Era (2014–2018)**: Conceived while the author was in high school learning game design and programming. The initial iterations were built using the default offerings of RPG Maker, exploring community plugins, and crafting characters through its Character Generator.
- **The Lost Assets & The Blank Canvas**: With no legacy digital assets surviving from that era, the project stands not as a nostalgic port, but as a complete ground-up rebirth—realizing the original creative vision with the systems engineering maturity, mathematical rigor, and architectural discipline of modern custom engine design.

## 13.3. Narrative Realignment: The Kingdom of Sacyra

During early prototype scaffolding, generic placeholder factions ("The Solar Order" and "The Umbral Covenant") were inserted to test basic moral sway logic. On October 26, 2026, the creator firmly realigned the canon, stripping away these synthetic factions to reveal the true world:

### 13.3.1. The Kingdom of Sacyra (`/s□□□ki□r□/`)

- The setting is the historic **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**).
- Sacyra is a structured, chivalric realm anchored by three distinct socio-political pillars:
  1. **The Crown & Royal Guard**: Guardians of legal order, national defense, and the ancestral throne.
  2. **The Noble Houses & Royal Court**: High aristocracy governing regional fiefs and vying for political dominance on the royal council.
  3. **The Adventurers' Guild**: A pragmatic, frontier-based brotherhood operating beyond courtly vanity, answering commoner monster distress calls and exploring untamed ruins.
- **The Existential Threat**: The ancient sovereign **The Dark Lord** breaking through the Abyssal Vaults, threatening to extinguish all mortal agency.

### 13.3.2. The Dual Protagonists

Rather than a generic lone wanderer, the narrative is anchored by two contrasting protagonists whose fates, lineages, and martial philosophies intertwine:

| Character | Elijah Ro-enheart | Lady Katrina Jaren |
|---|---|---|
| **Social Origin** | Military Lineage (Common/Barracks Nobility) | High Aristocracy (House Jaren) |
| **Family Heritage** | Son of the former Head of the Royal Guard | Daughter of an influential noble house on the Royal Court |
| **Internal Desire** | Living up to his father's legendary military legacy while discovering what true honor demands | Escaping stifling aristocratic court expectations to join the local **Adventurers' Guild** |
| **Relic Synergy** | **The Shield of Lux** (Sanctuary, Warding, Formation Hold) | **The Sword of Tenebris** (Severance, Mist-Veil, Critical Strike) |
| **TRPG Doctrine** | Frontline heavy vanguard; zone-of-control denial; ally interception | Agile outrider skirmisher; bypassing enemy lines; flanking assassin |

## 13.4. Ludological Evolution: The Tactical Strategy RPG (TRPG)

The game's true mechanical identity is a **Tactical Strategy RPG (TRPG)**. This fundamentally informs how the simulation engine and presentation viewports are being constructed:

### 13.4.1. The Dual-Layer Combat Loop

The combat loop operates across two synchronized phases:

1. **Strategic Grid Phase**:
   - The battle takes place on a discrete 2D grid representing castles, ramparts, thickets, and void-blighted frontier passes.
   - Units position based on movement ranges, terrain defense bonuses, directional facing (Front, Flank, Rear), and Zone-of-Control (ZOC) restrictions.
2. **Cinematic Combat Resolution (Side-View Action)**:
   - When a unit initiates combat against an enemy target, the game transitions into a dynamic **2D traditional side-scroll / action-based clash**.
   - Attacks play out with timed player agency: precision blocks with the Shield of Lux, poise-breaking shield bashes, and devastating severance slashes with the Sword of Tenebris.
   - Upon duel conclusion, remaining unit health, stagger, and relic resonance seamlessly persist back to the tactical grid.

### 13.4.2. Architectural Implications for the C Engine

Knowing this was originally an RPG Maker project gives us clear architectural directives: * **Decoupled Data Architecture**: Static definitions (`Units`, `Skills`, `Equipment`, `TerrainTypes`) must exist as clean data tables distinct from session state (`player_state_t`, `grid_t`). * **Deterministic Event Pipeline**: Interactive map features (gates, switches, dialog triggers) will follow a clean, reentrant command interpreter model. * **Zero Engine Bloat**: By building this custom in C11 rather than relying on bloated runtimes, our TRPG will maintain sub-millisecond turn updates, instantaneous scene transitions, and absolute softlock immunity.

## 13.5. Next Milestone Roadmap

1. **Phase 1 (Complete)**: Established OmniCode 5-block layering, Layer 0 utilities, CTest verification suite, and canonical documentation.
2. **Phase 2 (Active)**: Expand simulation state to model dual protagonists (Elijah & Katrina), party data structures, and the foundational 2D tactical grid.
3. **Phase 3**: Implement the cinematic side-scroll combat transition in the presentation layer.
4. **Phase 4**: Expand the tactical campaign across the provinces of Sacyra to confront the vanguard of The Dark Lord.

# Volume II: Dramatis Personae & Dual Protagonist Dynamics

Comprehensive character dossiers, founding princes, dual protagonist synergies, and regional faction leaders.

# Chapter 14. Dual Protagonist Dynamics & Tri-Perspective Architecture

**Source File**: `docs/lore/characters/protagonists/dual_protagonist_dynamics.adoc`

## 14.1. Overview

The narrative core and mechanical heartbeat of **Tenebris and Lux** is the interplay between **Elijah Ro-enheart** and **Lady Katrina Jaren**, complemented by the optional overarching perspective of the customizable **Bearer** (`<chosen_name>`). Contrasting in social caste, philosophical outlook, and combat rhythm, their evolving partnership mirrors the cosmic relationship between Light and Darkness.

## 14.2. Thematic & Social Contrast Matrix

| Dimension | Elijah Ro-enheart | Katrina Jaren |
|---|---|---|
| **Social Caste** | Military Lineage / Son of the Guard Head | High Aristocracy / House Jaren |
| **Institutional Allegiance** | Royal Army (Structure, Law, Duty) | Adventurers' Guild Aspirant (Freedom, Merit) |
| **Relic Affinity** | **The Shield of Lux** (Defense, Preservation) | **The Sword of Tenebris** (Offense, Severance) |
| **Thematic Echo** | **Usuru** (The Frontier Wound / Redemptive Grace) | **Prince Casatti** (The Renouncing Blade / Truth) |
| **Moral Trajectory** | From rigid obedience to moral conscience | From cynical detachment to noble sacrifice |

## 14.3. Interpersonal Chemistry & Narrative Progression

Elijah and Katrina embody the classic dramatic friction between an earnest, rule-following guardian and an irreverent, sharp-witted rogue. Where Elijah pauses to consult protocol, Katrina acts impulsively, cutting through bureaucratic red tape with sarcastic practicality.

### 14.3.1. Narrative Evolution Across the Campaign

1. **Act I (The Provincial Divide):** Initial friction in Dio Village. Katrina mocks Elijah's military formality; Elijah cautions against Katrina's cynical disregard for property and propriety.
2. **Act I Climax (The Capital Conspiracy):** Sacyra Castle. Katrina's insight into aristocratic deception exposes King Jihad's treason. Defeating Jihad reveals his possession by The Dark Lord's psychic remnant.
3. **Act II (The Sovereign Eclipse):** Total metaphysical synchronization. Facing **The Dark Lord** at the cosmic threshold, the heroes overcome individual biases, passing relics fluidly between each other to unlock the **Stance of Equilibrium** (**Duo-Loca Bashin n Slash**).

## 14.4. The Tri-Perspective Architecture: Witness Protagonist & Keystone Route

Requiring the player to experience Elijah, Katrina, and finally the customizable Bearer (`<chosen_name>`) utilizes three formal narrative game design structures:

1. **The Witness Protagonist (Internal vs. External Focalization):**
    - **Internal Focalization (Elijah & Katrina):** Elijah and Katrina live inside the terrestrial world of Dio Village and

the Capital. Their perceptions are clouded by personal trauma, military duty, and aristocratic rebellion.
    - **External Focalization (The Custom Bearer):** Guided by **Chrono the Sage** in the Room of Sages within The Dreamscape, the Custom Bearer holds **The Dream Compass**, acting as an objective, unaligned Observer capable of perceiving the full 3,000-year board without inherited prejudice.
2. **The Keystone Route (Synthesis Scenario):**
    - Elijah uncovers the military conspiracy surrounding Commander Gareth and King Jihad.
    - Katrina uncovers aristocratic corruption and underworld resistance with John the Thief.
    - The Custom Bearer's campaign functions as the architectural keystone that locks both partial perspectives together into a single unified truth.
3. **The Quantum Observer Effect:**
    - Left to themselves, Sacyra's historical factions remain trapped in a cyclical 3,000-year loop of trauma.
    - The outside Observer's conscious choice collapses the tragic wavefunction, unlocking the 10th True Ending (**"The Ugly Truth"**) and permanently mending the planar tear.

## 14.5. Tactical Synergy on the TRPG Battlefield

1. **The Phalanx Pivot:** Elijah pins enemy vanguards with his Shield of Lux Zone of Control (ZOC) aura. Katrina uses high mobility to circle the battle line and execute flanking criticals.
2. **Warded Severance:** Katrina lures high-threat enemies into charging toward her, stepping back onto Elijah's tile perimeter as Elijah executes a staggering `Lux Barrio` shield deflection.
3. **The Stance of Equilibrium:** In moments of ultimate synchronization, both relic resonance counters combine, unlocking unified abilities where radiant starlight and abyssal void strike in a single coordinated turn.

# Chapter 15. Elijah Ro-enheart: Vanguard of Lux

**Source File**: `docs/lore/characters/protagonists/elijah_ro_enheart.adoc`

## 15.1. Overview

**Elijah Ro-enheart** is the frontline heavy vanguard anchor of **Tenebris and Lux** and the initial bearer of the **Shield of Lux**. Steeped in the military traditions of the Royal Guard yet haunted by the unexplained disappearance of his father, Elijah represents the struggle between rigid institutional duty and personal honor.

## 15.2. Identity & Heritage

- **Full Name**: Elijah Ro-enheart
- **Title / Rank**: Cadet Vanguard, Sacyran Royal Defense Corps
- **Affiliation**: Kingdom of Sacyra Royal Army; House Ro-enheart
- **Family Lineage**:
  - **Father**: Commander Gareth Ro-enheart, legendary former Head of the Royal Guard, celebrated throughout Sacyra for holding the Sunken Citadel. Gareth vanished under mysterious crown orders following Elijah's **10th birthday**, leaving behind unresolved questions and immense expectations.
  - **Grandmother**: **Magolia Ro-enheart** ("Grandma Magolia"), a respected matriarch residing in the Upper Ward of Dio Village, guardian of the sacred family timepiece (**Grandma's Watch**).
- **Visual Profile**: Broad-shouldered, athletic build, steel-blue eyes, and cropped dark hair. Wears fitted silver-steel infantry plate over deep navy quilted gambeson, marked with the defensive crest of the Ro-enhearts.

## 15.3. Psychological Profile & Demeanor

Elijah is earnest, polite, disciplined, and deeply considerate. While raised within the hierarchy of military discipline, he possesses an intrinsic moral compass that values the safety of innocent civilians above abstract royal decrees.

### 15.3.1. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

- **Respect for Elders & Prudence**: [quote, Elijah, Dio Village - Grandma's House] "I shouldn't go in this chest. I need to talk to Grandma first. She's very particular about her things."
- **The Father Mystery**: [quote, Elijah, Opening Quest] "Maybe it is something about dad! I haven't seen him since my 10th birthday. This might be my chance to get answers!"
- **Civic Duty & Moral Conscience**: [quote, Elijah, Lower Village Confrontation] "I can't believe that I am doing this… the baron is the richest man in the village… but he's been exploiting the town! This is for the good!"

## 15.4. Martial Doctrine & TRPG Grid Mechanics

Elijah functions as the defensive bulwark of the tactical party, controlling chokepoints and mitigating damage for allied skirmishers.

| Dimension | Specification |
|---|---|
| **TRPG Grid Role** | Frontline Heavy Vanguard / Fortress Anchor |
| **Movement Budget** | 3 Cells (Medium-Heavy Infantry) |

| Dimension | Specification |
| --- | --- |
| **Zone of Control (ZOC)** | Projects heavy ZOC tiles, halting enemy sprint moves |
| **Primary Relic** | **The Shield of Lux** (Resonance `[0, 100]`) |
| **Tactical Signature** | `Warding Aura`: Projects a 2-cell radius barrier dome |
| **Poise & Knockback** | Immune to 1-cell forced displacement; absorbs shock |

### 15.4.1. Tactical Synergies

1. **Bottleneck Holding**: Elijah anchors narrow corridors, absorbing melee punishment while building Lux resonance via successful deflections.
2. **Aegis Interception**: Steps into adjacent cells to intercept lethal strikes aimed at fragile skirmishers or casters (e.g., Katrina Jaren).

# Chapter 16. Lady Katrina Jaren: The Outrider Blade

**Source File**: `docs/lore/characters/protagonists/katrina_jaren.adoc`

## 16.1. Overview

**Lady Katrina Jaren** is the high-mobility skirmisher of **Tenebris and Lux** and the initial wielder of the **Sword of Tenebris**. Born to high aristocracy yet repulsed by the deceit and stagnation of the Royal Court, Katrina seeks autonomy within the meritocratic ranks of the Adventurers' Guild.

## 16.2. Identity & Heritage

- **Full Name**: Lady Katrina Jaren
- **Title / Standing**: Noble Court Aspirant (Estranged); Guild Novice
- **Affiliation**: House Jaren (Eastern Aristocracy); Adventurers' Guild
- **Family Lineage**:
  - **House Jaren**: One of the wealthiest noble lineages in Sacyra, controlling lucrative vineyard estates and mountain silver mines. They hold significant patronage leverage on the Royal Court.
  - **Kinship with Dio Village**: Despite her family's high status in the capital, Katrina spent formative rebellious seasons under the watchful eye of Grandma Magolia in Dio Village, seeking respite from court etiquette.
- **Visual Profile**: Agile, athletic build, sharp amber eyes, and unbound chestnut hair tied in a practical combat braid. Clad in tailored lightweight leather arming doublets over noble-weave tunic, with rapier and long-dagger scabbards balanced for rapid unsheathing.

## 16.3. Psychological Profile & Demeanor

Katrina is quick-witted, irreverent, opportunistic, and intensely independent. She detests empty aristocratic pretension and speaks with an unvarnished, sarcastic candor that often unnerves formal soldiers like Elijah. Beneath her cynical exterior lies a fierce loyalty to those she deems genuine.

### 16.3.1. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

- **Impatience with Stagnation & Cynicism**: [quote, Katrina, Dio Village - Grandma's House] "The will. I love you gran, but a girl gotta get out of this dead-end town. Batty old woman seems to mind that chest… Crazy old woman, that will is mine!"
- **Pragmatic Opportunism**: [quote, Katrina, The Sword Shrine] "Free sword for your girl~!"
- **Fiery Independence & Temper**: [quote, Katrina, Monica's Grove] "LISTEN HERE YOU OLD-! What is wrong with you… Gran, I'm not your errand girl! This is YOUR friend."
- **Reluctant Heroism**: [quote, Katrina, Lower Village Heist] "Ok… here we go… let's do this. I have to do at least one good thing today."

## 16.4. Martial Doctrine & TRPG Grid Mechanics

Katrina excels at rapid battlefield repositioning, flanking maneuvers, and armor-piercing critical strikes.

| Dimension | Specification |
| --- | --- |
| **TRPG Grid Role** | High-Mobility Skirmisher / Flanking Striker |

| Dimension | Specification |
| --- | --- |
| **Movement Budget** | 4–5 Cells (Light Scout / Infiltrator) |
| **Zone of Control (ZOC)** | Evasive Infiltration: Ignores basic enemy ZOC delays |
| **Primary Relic** | **The Sword of Tenebris** (Edge Sharpness `[0, 100]`) |
| **Tactical Signature** | `Phantom Slash`: Teleports behind target cell to strike |
| **Evasion & Stealth** | Gains 25% evasion when ending turn on obscured tiles |

### 16.4.1. Tactical Synergies

1. **Flanking Tandem**: Moves across enemy flanks while Elijah holds attention, landing backstab criticals with the abyssal edge of Tenebris.
2. **Hit-and-Run Decoupling**: Uses remaining move points to disengage after striking, retreating behind Elijah's protective barrier dome.

# Chapter 17. The Custom Bearer: Steward of Balance

**Source File**: `docs/lore/characters/protagonists/custom_bearer.adoc`

## 17.1. Overview

The **Custom Bearer** (designated within the simulation engine as `<chosen_name>`) represents the third sovereign axis of the **Tenebris and Lux** tri-perspective narrative architecture. Serving as the thematic and ludological pillar of **The Future**, the Custom Bearer enters the world through the ethereal **Dream Chamber** under the stewardship of **Chrono**, unencumbered by the heavy generational trauma that binds both Elijah Ro-enheart and Katrina Jaren.

> You are not bound to the grief of what was broken, nor the cynicism of what remains. Touch the stone of your choosing, bearer. The twin relics will answer to your intent, and the dawn or dusk of Sacyra will bear the shape of your resolve.
>
> — Chrono, The Dream Chamber - Awakening of the Relic Bearer

## 17.2. Identity & Narrative Axis

- **Canonical Designation**: `<chosen_name>` (User-defined; defaults to `Elijah` if skipped).
- **Narrative Archetype**: The Steward of Balance / The Unwritten Destiny.
- **Thematic Alignment**: **The Future** (Transcending the 3,000-year cycle of inherited trauma).
- **Starting Realm**: **The Dreamscape** (`LOC_THE_DREAMSCAPE`), transitioning into **Dio Village**.
- **Spiritual Lineage**: Spiritual successor to **Prince Mi-ren**, who held the sacred vigil and sought institutional equilibrium across the founding era.

| Dimension | Characteristic Specification |
|---|---|
| **Moral Sway Potential** | Completely unbound: dynamic range `[-100, +100]`. |
| **Relic Mastery** | Dual-wield fluency: simultaneous mastery of **Lux Barrio** and **Cuttatio Tenebris**. |
| **Dialogue Register** | Inquisitive, perceptive, pragmatic, and unswayed by courtly protocol. |
| **Ending Affinity** | Unique accessibility to **The Sovereign Accord** and **The Cycle Shattered**. |

## 17.3. Archetype Stone Awakening (The Four Disciplines)

Upon awakening within the Dream Chamber, the Custom Bearer approaches the four primordial archetype monoliths under Chrono's observation. Selecting an archetype configures the starting base parameter profile:

| Archetype Stone | Class Designation | Starting Stat Profile | Tactical Grid Role |
|---|---|---|---|
| **Aegis Monolith** | **Vanguard** | HP: 480, DEF: 28, ATK: 24, MAT: 18 | Frontline fortress; barrier interception. |
| **Shadow Monolith** | **Outrider** | HP: 380, DEF: 18, ATK: 32, MAT: 22 | High mobility; flank severance strikes. |

| Archetype Stone | Class Designation | Starting Stat Profile | Tactical Grid Role |
|---|---|---|---|
| **Aether Monolith** | **Mystic** | HP: 340, DEF: 16, ATK: 20, MAT: 36 | Elemental burst; area-of-effect zoning. |
| **Equilibrium Monolith** | **Steward** | HP: 420, DEF: 22, ATK: 26, MAT: 26 | Balanced hybrid; stance chaining. |



Unlike fixed protagonists whose relic bias is established by lineage, the Custom Bearer accumulates **Resonance Charges** dynamically. Channeling defense builds Lux resonance, while initiating offensive flanks charges Tenebris edge.

## 17.4. Psychological Dynamic: The Unburdened Gaze

While Elijah sees the world through the lens of military honor and his father's mystery, and Katrina views society through betrayal and aristocratic corruption, the Custom Bearer approaches the crisis with empirical curiosity:

- **In Dio Village**: Questions the St. Baron's authority not out of defiance, but by exposing the mathematical absurdity of his exorbitant grain tithes.
- **In the Whispering Woods**: Discerns the abnormal migratory patterns of the Umbral Wolves, recognizing the abyssal corruption spreading from the Royal Capital.
- **Before King Jihad**: Detects the rhythmic harmonic distortion of **The King's Curse** leaking from beneath the High Throne before the royal court notices the monarch's decay.

**Thematic Invariant: Inherited Trauma vs. Chosen Destiny**

The Custom Bearer embodies the second clause of the overarching narrative axiom: **"Trauma is Inherited; Destiny is Chosen."** By refusing to allow ancient grudges between Libira and Sacyra to dictate their actions, the Custom Bearer can forge alliances that neither historical faction believed possible.

## 17.5. TRPG Grid Mechanics & Signature Abilities

| Dimension | Specification |
|---|---|
| **Base Movement** | 4 Tiles (Variable by chosen Archetype Stone: 3 to 5 tiles). |
| **Zone of Control** | Dynamic projection based on active combat stance. |
| **Signature Stance** | **Harmonic Equilibrium** (`STANCE_EQUILIBRIUM`): $+15\%$ to both DEF and ATK. |
| **Ultimate Skill** | **Duo-Loca Bashin n Slash** (Twilight Magic Slash combining Lux and Tenebris). |

### 17.5.1. Tactical Synergies

1. **Adaptive Stance Shifting**: Can shift stances without consuming action points once per turn, allowing instant transition from defensive warding to aggressive edge severance.
2. **Moral Sway Polarization**: Can align with either light or dark embellishments at the Altar of the Twin Relics, or forge the **Equilibrium Crown** to dissolve the planar wound.

# Chapter 18. Prince Mi-ren: Sovereign of the Vigil

**Source File**: `docs/lore/characters/founders/prince_mi_ren.adoc`

## 18.1. Overview

**Prince Mi-ren** was the elder founding prince of the Kingdom of Sacyra during Era I (The Primordial Dawn & The Accord of Kings). While his younger brother, Prince Casatti, ventured into the untamed frontier, Mi-ren anchored the capital, established the administrative institutions of the Crown, and held the legendary **Four-Year Vigil** over his brother's mysterious slumber. In the narrative triad of **Tenebris and Lux**, Prince Mi-ren represents the archetype of **The Steward of Order** and the philosophical progenitor of the Custom Bearer.

> A wall does not exist to lock men in, but to ensure that what we plant inside may grow without fear of the winter gale. If my brother must walk the dark wild, then I shall build a hearth warm enough to guide his spirit home.
>
> — Prince Mi-ren, Royal Archive Tablet - The Founding Accord

## 18.2. Identity & Lineage

- **Full Title**: Prince Mi-ren the Steadfast, First Warden of the High Court, Architect of Sacyra.
- **Dynastic Era**: Era I–II (Founding Epoch, approximately 3,000 Years Before the Main Campaign).
- **Affiliation**: First Royal House of Sacyra; Founder of the Royal Garrison and Courier Corps.
- **Familial Ties**:
  - **Brother**: **Prince Casatti**, the younger outrider prince who walked the Path of Darkness.
  - **Successor Lineage**: His bloodline formed the foundational dynasty of the First 30 Kings, eventually degenerating millennia later into the hollow reign of King Jihad.

| Dimension | Characteristic Specification |
|---|---|
| **Martial Discipline** | Classical Phalanx & Warding Architecture; heavy royal shieldbearer. |
| **Metaphysical Affinity** | **The Shield of Lux** (Resonance with warding, preservation, and structure). |
| **Thematic Anchor** | Institutional Duty, Self-Sacrifice, Bureaucratic Order. |

## 18.3. Historical Chronicle & The Four-Year Vigil

When the two founding princes completed their training, they faced a philosophical fork: the untamed frontier required exploration, while the nascent kingdom required permanent governance and defensible walls.

| Event Phase | Historical Deed & Chronicle |
|---|---|
| **The Division of Labor** | Mi-ren took responsibility for the Crown, founding the Royal Guard, designing the granaries, and codifying Sacyran civil law. |
| **The Slumber of Casatti** | When Prince Casatti collapsed into a four-year coma during his wilderness campaign, Mi-ren refused to abandon his brother, maintaining a continuous daily vigil at his bedside while governing the kingdom through sleepless devotion. |

| Event Phase | Historical Deed & Chronicle |
| --- | --- |
| **The Festival of Peoples** | Established the international gathering in the capital, inviting surrounding realms to exchange culture, technology, and mutual peace every spring. |
| **The Tragic Rupture** | In the 11th year, when the planar tether snapped beneath the High Throne, Mi-ren threw his own life-force into the warding perimeter to shield the capital's populace from abyssal incineration. |



Prince Mi-ren's great tragedy was his belief that systemic perfection could eliminate human suffering. By focusing entirely on the capital's defense, his administrative network failed to recognize the desperation building in frontier outposts like Libira, inadvertently sowing the seeds of Usuru's revolt.

## 18.4. Thematic Mirror: The Custom Bearer & Prince Mi-ren

In the **Past / Present / Future Thematic Triad**, Prince Mi-ren serves as the historical touchstone for the **Custom Bearer**:

1. **The Burden of Stewardship**: Just as Mi-ren held the kingdom together through sheer discipline, the Custom Bearer is called upon to arbitrate between the competing factions of modern Sacyra.
2. **Transcending Institutional Rigidity**: Where Mi-ren prioritized legalistic structure over emotional grievance, the Custom Bearer must learn from Mi-ren's failure—listening to the "quiet complaints" of commoners in Dio Village and the frontier before they curdle into war.
3. **Equilibrium Mastery**: Historical records indicate that Mi-ren was the first to realize that the Shield of Lux could not withstand endless kinetic pressure without venting force through the severance of a blade—laying the conceptual groundwork for the **Equilibrium Stance**.

# Chapter 19. Prince Casatti: The Outrider Blade of Truth

**Source File**: `docs/lore/characters/founders/prince_casatti.adoc`

## 19.1. Overview

**Prince Casatti** was the younger founding prince of Sacyra during Era I. Rejecting the ceremonial comforts of the royal palace, Casatti rode into the frontier wilderness to chart the uncharted reaches of the continent. Falling into a mysterious **Four-Year Coma** after contacting an ethereal boundary stone, Casatti awakened with transfigured perception, wielding the **Sword of Tenebris** as an instrument of decisive severance and unvarnished truth. In the overarching narrative of **Tenebris and Lux**, Casatti serves as the direct historical predecessor and spiritual mirror to **Lady Katrina Jaren**.

> My brother measures safety by the thickness of a parapet. He does not understand that stone can rot from within while looking immaculate to the sun. There are times when peace is simply another word for cowardice. When the branch turns gangrenous, you do not bless it with oil—you strike it from the trunk.
>
> — Prince Casatti, Fragmented Journal Recovered from the Whispering Caverns

## 19.2. Identity & Provenance

- **Full Title**: Prince Casatti the Swift, The Frontier Ranger, Bearer of the First Edge.
- **Dynastic Era**: Era I–II (Founding Epoch, approximately 3,000 Years Before the Main Campaign).
- **Affiliation**: Royal House of Sacyra; Spiritual Patron of the Outrider Scouts and Adventurers' Guild.
- **Familial Ties**:
  - **Brother**: **Prince Mi-ren**, the elder sovereign who held the capital vigil.
  - **Spiritual Heir**: **Lady Katrina Jaren**, who inherits his martial mindset, defiance of aristocratic complacency, and razor-sharp swordsmanship.

| Dimension | Characteristic Specification |
|---|---|
| **Martial Discipline** | High-mobility skirmishing, critical edge strikes, flank severance. |
| **Metaphysical Affinity** | **The Sword of Tenebris** (Resonance with severance, decisive truth, shadow). |
| **Thematic Anchor** | Radical Truth, Severing Stagnation, Rejection of Hypocrisy. |

## 19.3. The Four-Year Slumber & The Ethereal Transfiguration

While exploring the frontier passes bordering the ancient realm of Libira, Casatti touched an anomalous obsidian monolith vibrating with abyssal frequency. The contact thrust him into a deep suspended animation that lasted four full years:

1. **The Slumber**: For forty-eight months, his body rested unaging in the capital while his consciousness traversed the deeper strata of **The Dreamscape**.
2. **The Awakening**: Upon awakening, Casatti no longer viewed darkness as an evil force to be feared, but as the quiet, necessary soil from which all life and truth spring.

3. **The Sword of Tenebris**: Crafting the ancestral blade that would become the counterpart to the Shield of Lux, Casatti demonstrated that light without darkness produces blinding arrogance, while darkness tempered by discipline brings clarity.



Casatti's critique of the royal court's decadence was violently rejected by the conservative nobles of Sacyra. Frustrated by courtly hypocrisy, he spent his remaining years defending the frontier peasantry, establishing the traditions later adopted by the **Adventurers' Guild**.

## 19.4. Thematic Mirror: Katrina Jaren & Prince Casatti

The narrative parallels between Prince Casatti and Lady Katrina Jaren are exact:

- **Aristocratic Renunciation**: Just as Casatti abandoned the throne room to live among frontier scouts, Katrina abandons her aristocratic seat on the Royal Court to join the commoners of the Adventurers' Guild.
- **The Blade of Severance**: Both wield the Sword of Tenebris not to conquer, but to cut through political lies, aristocratic complacency, and ideological self-deception.
- **The Present Axis**: In the narrative triad, both embody **The Present**—the urgent necessity to take decisive action against immediate injustice rather than hiding behind royal procedure.

# Chapter 20. Usuru the Wanderer: Sovereign of Grief

**Source File**: `docs/lore/characters/founders/usuru_the_wanderer.adoc`

## 20.1. Overview

Before he was feared across the ages as **The Dark Lord**, he was a nameless boy born to an impoverished family on the arid borders of the **Kingdom of Libira** during Era IV. Known as the **Three-Day Wanderer** because he could never bear to settle in any settlement for more than three sunsets, he held an unyielding, reverent devotion to the First King and the splendor of the Capital. Driven by the "Quiet Complaint" of his neglected homeland and catastrophically altered by the snapping of the planar tether during the 11th Festival of Peoples, Usuru chose his name in defiance, crowned himself sovereign of Libira, and declared the devastating **Seven-Year Great War**.

> We waited seven days for their riders to tell us if our children were allowed to starve. We bowed when their golden heralds told us that peace demanded our silence. But when I reached out to touch the King’s light, it burned my skin and sealed our skies. They did not leave us behind by mistake—they left us behind because we were quiet. Libira will be quiet no longer.
>
> — Sovereign Usuru, Speech to the Assembly of Libira - Dawn of the Great War

## 20.2. Identity & Origin

- **Birth Identity**: Nameless Boy of Libira (Born to a frontier shepherd family).
- **Chosen Name**: **Usuru** (Ancient Libiran root meaning **"He Who Remembers What Was Buried"**).
- **Historical Era**: Era IV–V (approximately 3,000 Years Ago, Pre-Dynastic Wave).
- **Titles**: The Three-Day Wanderer; Sovereign of the Red Desert; Vanguard of the Broken Crown; later corrupted across the millennia into **The Dark Lord**.
- **Affiliation**: Kingdom of Libira; Supreme Commander of the 6-Power Coalition against Sacyra.

| Dimension | Specification |
|---|---|
| **Defining Trait** | Restlessness; profound observational empathy curdling into volcanic resentment. |
| **Original Motivation** | Yearning to bridge the geographic and communication gulf between Libira and Sacyra. |
| **Thematic Axis** | **The Past** (The unhealed frontier wound, mirrored by Elijah Ro-enheart). |

## 20.3. The Wanderer’s Pilgrimage & The "Quiet Complaint"

Due to the lack of an outpost relay system, a message sent from the Royal Capital took seven grueling days of desert riding to reach Libira. While the capital feasted, Libira struggled against drought, bandit raids, and forgotten infrastructure.

1. **The Three-Day Law**: Growing up, the boy walked the breadth of the continent, staying no more than three days in any village. He listened to commoners, blacksmiths, and outriders, cataloging the quiet complaints of the forgotten

provinces.

2. **The Annual Pilgrimage**: The sole exception to his three-day rule was the annual **Festival of Peoples** in the capital, where he stayed an entire week, gazing with awe at the High Throne and the First King.
3. **The 11th Year Catastrophe**: During the 11th festival, determined to deliver a petition for a Libiran relay road, the boy slipped into the Inner Sanctum beneath the High Throne. Startled by guards, he accidentally struck the ritual focus stone. The planar tether snapped, venting primordial abyssal energy that sealed the First King in the Dreamscape and violently warped the distance between Libira and Sacyra into an impenetrable spatial rift.



Believing the sudden spatial expansion was a deliberate imperial curse cast by the Sacyran court to permanently exile Libira, the boy's devotion inverted into pure, consuming hatred.

## 20.4. Ascension, War & Metaphysical Corruption

Returning across the warped desert passes, the youth took the name **Usuru**, mounted the dais of the Libiran assembly hall, and proclaimed the end of subservience:

- **The Crown of Libira**: Libirans, enraged by generations of neglect and royal silence, crowned Usuru by unanimous acclamation.
- **The 6-Power Conflagration**: Forging military alliances with the Great Mau Kingdom, Old Sacyra, and the Yun-yin Provinces, Usuru ignited the **Seven-Year Great War**.
- **Descent into Ruin**: To match the invulnerable barrier of the **Shield of Lux**, Usuru dipped his soul repeatedly into the abyssal breach beneath the battlefield, forging void constructs and commanding **Dream Eaters**. By the war's end, his humanity had burned away, leaving behind the eternal, brooding sovereign now remembered only as **The Dark Lord**.

## 20.5. Thematic Mirror: Elijah Ro-enheart & Usuru

Usuru serves as the tragic historical negative to **Elijah Ro-enheart**:

- Both originate from the marginalized periphery (frontier Libira vs. neglected Dio Village).
- Both suffer from the abrupt disappearance of a revered paternal figure (The First King vs. Commander Gareth).
- Where Usuru allowed his frontier grievance to consume him into becoming a monster of vengeance, Elijah must learn to acknowledge his family's pain without repeating Usuru's destructive crusade.

# Chapter 21. Chrono: Sage of the Ancient Archives & Dream Guardian

**Source File**: `docs/lore/characters/sages_and_ancients/chrono.adoc`

## 21.1. Overview

**Chrono** is the eternal archival sage and guardian of **The Dream Chamber** within the ethereal substrate of **The Dreamscape** (`LOC_THE_DREAMSCAPE`). Witness to the 3,000-year dynastic wave and the catastrophic snapping of the planar tether during the 11th Festival of Peoples, Chrono serves as the guide, mentor, and mechanic instructor for the newly awakened Relic Bearer.

> Time is neither a line nor a river, child of Sacyra. It is a loom of resonance. When the First King attempted to hold the veil open, the shuttle broke. What you call monsters are merely the untethered threads of that unbroken night seeking form. Raise your ward. Let us see if your spirit remembers how to hold the dawn.
>
> — Chrono, The Dream Chamber - Tutorial Trial

## 21.2. Identity & Provenance

- **Full Designation**: Chrono, Keeper of the Monoliths and Primordial Archives.
- **Affiliation**: Pre-Dynastic Sacyran Order of Scribes; Guardian of the Dream Veil.
- **Sanctuary Domain**: **The Dream Chamber** (Subterranean astral vault within the Dreamscape).
- **Physical Manifestation**: An ethereal figure draped in celestial silver-grey robes lined with rotating astrolabe embroidery. His hair shifts between shades of starlight, and his eyes glow with muted amber luminescence. He holds the **Codex of Eras**, an unwritten ledger of potential futures.

| Dimension | Specification |
|---|---|
| **Age / Lifespan** | Ageless; active since the dawn of the First Dynasty (>3,000 years). |
| **Primary Relic** | **The Chrono Astrolabe** (Attunes pocket dimensions and temporal flow). |
| **Tone & Register** | Measured, archaic, compassionate yet unsentimental; philosophical. |
| **Mechanical Role** | Archetype Stone Selection, Combat Tutorial Arbiter, Dream Compass Attuner. |

## 21.3. Archival Purpose & Historical Memory

Chrono was present when Prince Mi-ren held his vigil in the capital and when Prince Casatti fell into the four-year coma. He witnessed Usuru the Wanderer arrive as a nameless pilgrim, and documented the exact instant the 11th-year ritual collapsed beneath the High Throne.



Chrono does not intervene directly in the mortal wars of Sacyra. The Laws of the Veil forbid an Archival Sage from taking up arms against the living or the void. His duty is solely to ensure that the chosen Relic Bearer understands the consequences of moral intent.

### 21.3.1. The Three Archival Directives

1. **Preserve the Blueprint of Balance**: Record the deeds of each monarch across the 3,000-year dynastic wave, tracking the accumulation of stasis residue beneath the capital.
2. **Attune the Relic Bearer**: Test the newly awakened soul against the **Dream Eaters** to verify that they can command both Lux deflection and Tenebris severance without succumbing to madness.
3. **Calibrate The Dream Compass**: Awaken the dormant temporal gearwork within **Grandma's Watch**, transforming a rustic heirloom into an instrument capable of navigating nocturnal planar shifts.

## 21.4. Canonical Voice & Dialogue

- **The Philosophy of Stance**: [quote, Chrono, Combat Mechanics Initiation] "To wield only Lux is to become a frozen statue, impervious yet inert. To wield only Tenebris is to become a devouring blade, sharp yet hollow. True mastery lies in the threshold where the two kiss and do not consume one another."
- **Warning of The King's Curse**: [quote, Chrono, Midpoint Revelation] "King Jihad is not evil, bearer. He is thirsty. And a thirsty man standing over an open well of liquid night will drink until his belly bursts and his soul turns black."
- **The Fate of Usuru**: [quote, Chrono, The Planar Threshold] "Do not despise the Dark Lord. Mourn him. For before he was a sovereign of ruin, he was a lonely boy whose quiet complaint went unheard by a court blinded by its own radiance."

## 21.5. Ludological Role in the TRPG Engine

| Gameplay Phase | Systemic Interaction |
|---|---|
| **Prologue (Awakening)** | Guides archetype stone interaction; spawns initial Dream Eater combatants. |
| **Act I Interludes** | Re-encountered during rest at campfires when the player falls into dream meditation. |
| **Act II (Midpoint)** | Deciphers the runes carved into **Grandma's Watch**, unlocking the Dream Compass. |
| **Act III (Endgame)** | Unlocks the **Sanctum of Forgotten Memories**, revealing Usuru's uncorrupted dialogue. |

**Sidebar: Chrono's Codex & Invariant Validation**

Within the C11 simulation engine, Chrono serves as the diagnotic voice of the subsystem self-audit routines (`core.util.manifest`, `sim.engine.state`). When the engine performs an invariant check, the telemetry trace reflects Chrono's unyielding standard: zero heap corruption, bounded moral sway, and deterministic state transitions.

# Chapter 22. Grandma Magolia: Matriarch of Dio Village

**Source File**: `docs/lore/characters/dio_village/grandma_magolia.adoc`

## 22.1. Overview

**Grandma Magolia** (Magolia Ro-enheart) is the beloved matriarch of Dio Village, mother to Commander Gareth Ro-enheart, and grandmother to Elijah Ro-enheart. Providing warmth, historical perspective, and the family heirloom (**Grandma's Watch**), Grandma Magolia anchors the heroes to the human heart of Sacyra.

## 22.2. Identity & Family Ties

- **Full Name**: Magolia Ro-enheart ("Grandma Magolia")
- **Residence**: Grandma's House (Two-story timber cottage, Dio Upper Ward)
- **Family Lineage**:
  - **Son**: Commander Gareth Ro-enheart (Vanished Head of the Royal Guard)
  - **Grandson**: Elijah Ro-enheart (Bearer of the Shield of Lux)
  - **Surrogate Charge**: Lady Katrina Jaren (who sought haven with Magolia during her rebellion against House Jaren court tutors)
  - **Lifelong Friend**: Monica the Mystic (companion since youth)

## 22.3. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

Grandma Magolia speaks with gentle affection, nostalgic wisdom, and quiet sadness regarding the decline of provincial Sacyra.

- **The Heirloom Quest**: [quote, Grandma Magolia, Map 21 / Map 7] "Hello sweetie, could you please go upstairs and search in the chest and grab what's inside? It's something that has great significance."
- **Reflections on the Kingdom's Neglect**: [quote, Grandma Magolia, Map 21 / Map 7] "This town used to be full of life. **sighs** The kingdom doesn't pay much attention to this place anymore..."
- **Passing Down the Heirloom Watch**: [quote, Grandma Magolia, Map 21 / Map 7] "Thank you honey. This watch has been in the family for a long time. Oh, how it brings back memories of better days. I think you should keep it. It's my gift to you. And here's a bit of money. I know that you'll need it more than little ol' me."
- **A Lifelong Bond with Monica**: [quote, Grandma Magolia, Map 21 / Map 7] "Oh me and Monica go way back. She was such a wild child, always getting into mischief. You'll love her."

## 22.4. The Heirloom Timepiece: Grandma's Watch

Grandma Magolia bequeaths **Grandma's Watch** to the party. In the original 2017–2018 RPG Maker project, this item activated the global calendar and monotonic time engine (Seconds, Minutes, Hours, Days, Weeks, Months, and Seasons). In the modern C11 architecture, this lore item corresponds directly to the Layer 0 `time_utils` subsystem, tracking monotonic game ticks and frame budgets.

# Chapter 23. Monica the Mystic & Goddess Lyri

**Source File**: `docs/lore/characters/dio_village/monica_and_lyri.adoc`

## 23.1. Overview

**Monica the Mystic** (known as **The Sage** in village folklore) and her elemental familiar **Lyri** reside in an arcane grove on the outskirts of Upper Dio Village. Sarcastic, sharp-tongued, yet deeply attuned to the metaphysical currents of Sacyra, Monica is the first local authority to confirm that the forces of Darkness have re-awakened.

## 23.2. Identity & Mystical Lineage

- **Name**: Monica (**The Mystic / The Sage**)
- **Companion Familiar**: **Lyri**, an ethereal nature spirit who perceives metaphysical fluctuations and whispers warnings into Monica's ear.
- **Residence**: Monica's House & Mystic Grove (Dio Upper Ward)
- **Alliances**: Lifelong confidante of Grandma Magolia; contact of Chrono the Sage.

## 23.3. Canonical Voice & Demeanor (2017–2018 RPG Maker Origin)

Monica possesses a biting, irreverent wit, greeting newcomers with playful insults before demonstrating profound arcane knowledge.

- **Blunt First Impressions**: [quote, Monica to Elijah, Map 27 / Map 29] "And who in the blaze are YOU! … So you are Magolia's grandson. She made you sound more handsome than you are."
- **Clashing with Katrina**: [quote, Monica to Katrina, Map 27 / Map 29] "So you are Magolia's granddaughter. She made you sound prettier than you are."
  **Katrina**: "LISTEN HERE YOU OLD-!"
  **Monica**: "Tis only a joke. Now why have you come here? Do you have some milk?"
- **Lyri's Whispers & The Darkness**: [quote, Monica, Map 27 / Map 29] "A wise woman she is. Lyri has been whispering in me ear about the forces of darkness. I'll help, but first, get that hooligan from the city limits. Not only is he preventing folks from leaving, he's also messing with me business."

## 23.4. Arcane Role in the Campaign

1. **Attunement to the Void**: Through Lyri, Monica senses the breaking of the ancient seals on the Abyssal Threshold long before the Royal Court admits it.
2. **The City Limits Obstacle**: Directs the heroes to clear the town boundary (confronting the "Fighter Kid" blocking the pass) to prove their tactical readiness.
3. **Bridge to the Sages**: Connects Elijah and Katrina with the ancient scholar **Chrono**, guiding them toward their destiny with the twin relics.

# Chapter 24. Eric the Lumberjack & The Moral Litmus

**Source File**: `docs/lore/characters/dio_village/eric_the_lumberjack.adoc`

## 24.1. Overview

**Eric** is a former lumberjack residing in the Upper Ward of Dio Village. His livelihood was shattered when a mysterious abyssal portal manifested in the timber forests, driving him into poverty. He introduces the heroes to the **Ring of Fire** and anchors the early **Test of Honesty** moral dilemma.

## 24.2. Identity & Background

- **Name**: Eric
- **Occupation**: Former Master Woodman / Timberjack (Unemployed due to monsters)
- **Residence**: Eric's Hearth (Upper Dio Village)
- **Economic Situation**: Deeply impoverished, struggling to afford food as monster activity blocks access to the logging roads.

## 24.3. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

Eric represents the vulnerable provincial working class bearing the brunt of unnatural cosmic incursions.

- **The Struggle of Provincial Labor**: [quote, Eric, Map 28] "We've been seeing a lot of monsters appear near the woods recently. I wonder what's that all about. And with jobs being tight, money is hard to come by."
- **The Story of the Portal & The Ring of Fire**: [quote, Eric, Map 28] "Thank you. Thank you so much, here I believe that this will be more useful to you than me. The name is Eric and that ring has a story. I used to be a lumberjack until that weird portal appeared. I found this ring and held on to it ever since. I'm sure that ring's fire power will come in handy if you plan on leaving the village."

## 24.4. The Test of Honesty & Moral Continuum

Eric's home contains a famous interactive vignette from the 2017–2018 project: the **Self-Aware Pot**.

- If the player attempts to loot Eric's shelf pot, the pot speaks directly to the player: [quote, Self-Aware Pot, Map 28] "What kind of game do you think this is! You can't just steal whatever you want! How rude!"
- Taking the 10 Gold increments the internal `Moral (Evil)` counter, while gifting charity to Eric increments `Moral (Good)`. This directly fed the original RPG Maker `Morality Meter` and served as the design prototype for the modern C11 engine's `moral_sway` continuum (`[-100, +100]`).

# Chapter 25. John the Thief & Outlaw Resistance

**Source File**: `docs/lore/characters/dio_village/john_the_thief.adoc`

## 25.1. Overview

**John the Thief** is an underworld operative and smuggler residing in the back alleys of Dio Village's Lower Ward. Cynical and gruff, John nonetheless acts as an unofficial protector of the poor, redistributing goods and offering clandestine sidequests to heroes willing to bypass the law.

## 25.2. Identity & Underworld Standing

- **Name**: John (**"John the Thief"**)
- **Occupation**: Smuggler, Fence, and Independent Rogue Operative
- **Territory**: Lower Ward Alleys & Shopping District
- **Philosophy**: Pragmatic survivalism; contempt for aristocratic landlords.

## 25.3. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

John initially meets outsiders with hostility, softening only when heroes demonstrate that they care about the common people.

- **Gruff Rebuff**: [quote, John, Map 31] "What do you want kid! Can't you see that I'm busy!"
- **Recognizing Civic Virtue**: [quote, John, Map 31] "So you're the one that's helping the little people of this village out. I have a little task for you..."

## 25.4. Quest & Mechanics Role

John the Thief introduces the **Stolen Goods** tactical retrieval mechanics. In the 2017–2018 LeTBS combat engine, recovered bandit contraband appeared as interactive grid items with dedicated `Pick Up` and `Push` commands. John serves as the quest receiver who turns recovered goods into vital supplies and gold.

# Chapter 26. The St. Baron & Feudal Exploitation

**Source File**: `docs/lore/characters/dio_village/st_baron.adoc`

## 26.1. Overview

**The St. Baron** is the wealthiest aristocrat and dominant landowner of Dio Village. Residing in a secluded manor in the Lower Ward, the St. Baron uses exorbitant rent, grain hoarding, and monopolies to subjugate the impoverished citizens, serving as a primary symbol of noble corruption outside the capital.

## 26.2. Identity & Standing

- **Title**: The St. Baron of Dio Province
- **Residence**: The St. Baron's Manor (Lower Dio Village)
- **Status**: High provincial gentry; tax collector for the eastern march
- **Reputation**: Ruthless, miserly, and widely despised by the working poor.

## 26.3. Socio-Economic Conflict in Dio Lower Ward

While the Upper Ward remains pastoral and self-sufficient, the Lower Ward suffers under the St. Baron's heavy-handed economic extortion. By controlling grain warehouses and money-lending stalls, the Baron has forced shopkeepers and laborers into debt bondage.

- **The Hero's Moral Justification (Elijah)**: [quote, Elijah, Lower Village Infiltration] "I can't believe that I am doing this... the baron is the richest man in the village... but he's been using the town! This is for the good!"
- **The Rogue's Determination (Katrina)**: [quote, Katrina, Lower Village Infiltration] "Ok... here we go... let's do this. I have to do at least one good thing today."

## 26.4. Narrative & Quest Role

The St. Baron represents the mundane human corruption that flourishes when the Crown neglects its borders. Infiltrating the St. Baron's estate to recover stolen resources for the village provides the player with an early non-combat tactical scenario, forcing Elijah to break legal statutes for the higher moral good.

# Chapter 27. High Priest Roté & The Royal Omens

**Source File**: `docs/lore/characters/kingdom_and_court/sir_rote.adoc`

## 27.1. Overview

**Sir Roté** is a veteran scout, senior knight, and herald of the Sacyran Royal Guard. His urgent, breathless arrival at Sacyra Castle serves as the opening narrative incitement of the game, delivering news of the rising island, the unnatural storm, and the return of the primordial Darkness.

## 27.2. Identity & Standing

- **Full Name**: Sir Roté of the Royal Guard
- **Rank / Office**: Senior Outrider Captain / First Scout of the Realm
- **Affiliation**: Kingdom of Sacyra Royal Army; The Royal Guard
- **Visual Demeanor**: Weather-worn plate mail spattered with mud from frantic galloping, exhausted, earnest, and deeply devoted to the Crown.

## 27.3. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

Sir Roté represents the perspective of the loyal rank-and-file soldier who believes unquestioningly in his monarch's ability to save the realm.

- **Observation at the Frontier (Start-Up / Castle Gates)**: [quote, Roté, Map 2 & Map 3 / Map 8] "The king… huff… huff… huff… he must know immediately! And to think… The storm… The rising island… All of these events before the Festival of Heroes. The king will know how to stop this, he must know…"
- **The Urgent Report to King Jihad**: [quote, Roté, Throne Room - Map 4 / Map 9] "MY KING! It's here! The darkness… it has returned!"
- **The Uneasy Hesitation**: [quote, Roté, Hearing King Jihad's Gag Order] "Y… Y… Yes my king…"

## 27.4. Narrative Role & Dynamic

Sir Roté serves as a tragic mirror to Elijah Ro-enheart. Both men revere the oath of the Royal Guard. However, while Roté suppresses his growing unease when King Jihad issues a gag order preventing the people from knowing about the approaching Darkness, Elijah ultimately chooses conscience over blind obedience. Later in the campaign, Roté's internal conflict plays a pivotal role in enabling Elijah and Katrina to infiltrate the castle archives.

# Chapter 28. King Jihad & The Treasonous Ambition

**Source File**: `docs/lore/characters/kingdom_and_court/king_jihad.adoc`

## 28.1. Overview

**King Jihad** (referenced in ancient annals as **D. Jihad**) is the reigning monarch of the **Kingdom of Sacyra**. Outwardly a dignified sovereign presiding over the chivalric traditions and the **Festival of Heroes**, King Jihad secretly harbors a perilous ambition: to seize and harness the forbidden abyssal power of the returning **Dark Lord** to establish absolute dominion over the realm.

## 28.2. Identity & Sovereign Office

- **Full Name**: King Jihad of Sacyra (Monarchical Title: **D. Jihad**)
- **Office**: Crown Sovereign, King of Sacyra, High Commander of the Realm
- **Seat of Power**: The High Citadel of Sacyra Castle
- **Political Standing**: Balances the aristocratic Noble Houses against the meritocratic Adventurers' Guild while commanding the elite Royal Guard.

## 28.3. The Grotesque Inversion of the Founding Dynasty

King Jihad stands as the exact moral and structural inverse of the First King and Prince Mi-ren:

| Dimension | The Founding Dynasty (First King & Mi-ren) | The Modern Inversion (King Jihad) |
|---|---|---|
| **State of the Monarch** | Physical body comatose/dead during festival week, while his **spirit remained pure and lucid** in The Dreamscape to guide the realm. | Physical body healthy upon the throne, while his **heart is thoroughly corrupted** by greed for void power. |
| **Purpose of Secrecy** | Concealed the First King's passing out of filial duty and preservation until teenage Mi-ren officially came of age. | Imposes a gag order (`"don't tell the people"`) so heroes do not interfere before Jihad can usurp void power for himself. |
| **Relation to Prophecy** | Forged Lux and Tenebris to enact justice, protect life, and seek forgiveness. | Covets the **Prophecy of Kings**, seeking to hijack void supremacy without understanding that Lux dissipates in a corrupt heart. |

## 28.4. 3. "The King's Curse" & Abyssal Possession

- **The Planar Hazard:** For 3,000 years, Sacyran monarchs were exposed to residual spiritual shear from the open planar rift beneath the High Throne dias, left when Usuru shattered the First King's stasis ritual.
- **The Fatal Gamble:** Coveting the Dark Lord's power, King Jihad intentionally opened the conduit beneath his throne prior to the Festival of Heroes. Because the ethereal medium mirrors mortal intent, Jihad's arrogance inverted the flow, allowing a dormant remnant of Usuru's consciousness (**The Dark Lord**) to possess him.
- **Act I Midpoint Climax:** Defeating King Jihad at Sacyra Castle exposes this possession, revealing that the true threat lies deep in the subterranean Abyssal Vaults where the ancient soul of Usuru still waits.

## 28.5. Canonical Voice & Dialogue (2017–2018 RPG Maker Origin)

- **The Secret Soliloquy:** [quote, D. Jihad, High Throne Room] "The fools… that power… that wicked power! Soon I will rule over this land, and not even the Dark Lord will have power over me!"
- **Suppressing the Truth:** [quote, King Jihad to Sir Roté] "Thank you Roté, what would I do without your careful watch… Gather up the guard! We need to prepare! Darkness will soon return to this land. And don't tell the people of this discovery… You all are dismissed."

## 28.6. Socio-Political Role in the Campaign

1. **State Censorship:** Enforces an information blackout along the border, ignoring Sir Roté's warnings.
2. **Gareth's Disappearance:** Dispatched Commander Gareth Ro-enheart on a suicide mission into the Abyssal Threshold on Elijah's 10th birthday to scout void portals, leaving the Ro-enhearts fractured.
3. **The Catalyst for Rebellion:** When Elijah and Katrina expose King Jihad's corruption, Elijah is forced to choose between his sworn oath to the throne and his true moral duty to the living people of Sacyra.

# Chapter 29. Commander Gareth Ro-enheart & The Vanished Guard

**Source File**: `docs/lore/characters/kingdom_and_court/commander_gareth_ro_enheart.adoc`

## 29.1. Overview

**Commander Gareth Ro-enheart** was the renowned Head of the Royal Guard, hero of the Sunken Citadel, and father to Elijah Ro-enheart. His unexplained disappearance under sealed royal orders following Elijah's 10th birthday forms the primary personal catalyst driving Elijah's journey across Sacyra.

## 29.2. Identity & Lineage

- **Full Name**: Commander Gareth Ro-enheart
- **Title**: Former Head of the Sacyran Royal Guard; Knight-Commander of Sacyra
- **Kinship**:
  - **Son**: Elijah Ro-enheart (Cadet Vanguard, bearer of the Shield of Lux)
  - **Mother**: Magolia Ro-enheart ("Grandma Magolia", matriarch of Dio Village)
- **Status**: Missing in Action / Sealed Royal Dispatch (Presumed Alive)

## 29.3. Legend & Martial Legacy

Gareth Ro-enheart is remembered throughout Sacyra as an incorruptible paragon of chivalric virtue. Decades prior, when an allied barbarian host breached the frontier palisades, Commander Gareth anchored the Sunken Citadel for seven days without relief, personally deflecting siege bolts with his reinforced tower shield.

He passed this philosophy to his young son Elijah:

> "A shield is not a wall to hide behind, but a promise made to the people standing behind you."
>
> — Gareth Ro-enheart

## 29.4. The 10th Birthday Vanishing & Royal Intrigue

On the morning following Elijah's **10th birthday**, Commander Gareth received a direct royal summons from King Jihad and departed the capital with an elite retinue for the southern frontier. Neither he nor his men ever returned.

- **Official Crown Statement**: Gareth was lost during a classified border survey.
- **The Reality**: Clues uncovered by Elijah and Katrina reveal that Gareth discovered King Jihad's clandestine expeditions to unearth the Abyssal Vaults. Refusing to participate in forbidden dark alchemy, Gareth was either imprisoned in a subterranean redoubt or went into deep exile to protect the realm.

# Chapter 30. Threat Dossier: The Dark Lord

**Source File**: `docs/lore/threats/the_dark_lord.adoc`

## 30.1. Overview

In the authentic 2017–2018 demo script and canonical narrative architecture, the ultimate antagonist is designated strictly as **The Dark Lord**. He is not a generic cardboard demon, but the tragic culmination of three millennia of festering bitterness born from the catastrophe of the 11th Festival: the ancient Libiran youth **Usuru**.

## 30.2. Ground-Truth Canonical Identity & Origin

- **The Nameless Wanderer of Libira:** Born on the remote southern frontier of Libira during the First King's reign, he grew up as a restless three-day wanderer whose sole enduring devotion was his annual week-long pilgrimage to the Capital for the Festival of Peoples.
- **The 11th Year Rupture:** Seeking an audience with the sovereign during the rumored "plague," the eleven-year-old boy discovered the First King's mummified corpse and Prince Mi-ren's protective projection ritual. Viewing the vigil as a monstrous deception, his terror triggered an accidental collision with the ritual array.
- **The Birth of Tenebris:** The catastrophic collapse of the boy's innocent faith produced an anomalous metaphysical surge that birthed raw **Tenebris** into reality. Taking the name **Usuru**, he rallied Libira and border kingdoms to wage the 7-Year Great War of annexation.
- **The 3,000-Year Descent:** Following his armistice with Prince Mi-ren under the Accord of Kings, Usuru refused spiritual reconciliation. He retreated into the subterranean abyssal vaults, where his unresolved grief and fury curdled over three thousand years until his mortal name was lost to history, remembered only with terror as **The Dark Lord**.

## 30.3. Narrative Function: The Overarching "A Plot"

- **Existential Backdrop:** The Dark Lord is not an early-game hallway encounter. He represents the overarching existential shadow ("The A Plot") looming over Sacyra.
- **Foreground vs. Background:** While provincial survival (Dio Village), noble exploitation (St. Baron), and King Jihad's treason form the active foreground ("B Plot" and "C Plot"), the awakening of The Dark Lord informs the dread of the entire continent.
- **The Metaphysical Siege Window:** The Dark Lord strategically aligns his vanguard strikes with the annual **Festival of Heroes**, knowing that the boundary to the Dreamscape thins to gossamer during that week, providing a planar bridge to reclaim the surface world.

## 30.4. 3. "The King's Curse" & Possession of King Jihad

- **The Planar Rift:** When Usuru shattered the First King's stasis array 3,000 years ago, he left a jagged planar wound directly beneath the dias of the High Throne.
- **Psychic Conduit:** For three millennia, sovereigns were exposed to ethereal backwash. When King Jihad attempted to harness void power to usurp the Dark Lord, Jihad's megalomania inverted the conduit, allowing a dormant remnant of Usuru's consciousness to possess the reigning monarch.
- **Two-Half Campaign Resolution:** In Act I, heroes confront the possessed King Jihad; in Act II, they must travel into the deep abyssal threshold to face The Dark Lord himself.

## 30.5. Metaphysical Neutrality & True Resolution

- **Not Elemental Evil:** Darkness (Tenebris) is a cosmic apex force of rest, mystery, and severance. The Dark Lord represents Tenebris corrupted by an unhealed heart of vengeance.
- **The Ugly Truth:** Defeating The Dark Lord through violence alone (Neutral Ending) leaves the First King marooned and the cycle dormant. Only by confronting "The Ugly Truth"—acknowledging Usuru's valid childhood pain and achieving the **Stance of Equilibrium**—can heroes permanently release Usuru's tortured soul and heal the realm.

# Chapter 31. Threat Dossier: Dream Eaters

**Source File**: `docs/lore/threats/dream_eaters.adoc`

## 31.1. Astral Parasites of the Slumber Realm

**Dream Eaters** are shifting metaphysical entities native to the borders of The Dreamscape.

- **Nature**: Astral parasites that feed on unresolved regrets, fears, and psychic vulnerabilities.
- **Combat Role**:
  - Manifesting during Chrono's trial, they serve as the opening combat challenge in the game.
  - **Base Vigor**: $70$ Threat HP.
  - **Defensive Attributes**: $11$ Physical Defense (DEF), $8$ Magic Defense (MDF).
  - **Flank Vigor**: $70$ HP representing flanking swarms susceptible to **Cuttatio Tenebris** splash damage.
- **Significance**: Defeating the Dream Eaters demonstrates the bearer's ability to maintain mental fortitude and twin-relic resonance, breaking free of the dream's illusions.

# Volume III: Ludology & Tactical TRPG Combat Mechanics

Grid-based tactical warfare, relic stance machines, terrain modifiers, continuous invariant testing, and game design curation.

# Chapter 32. Dual-Layer TRPG Combat Architecture

**Source File**: docs/game_design/tactical_trpg_system.adoc

## 32.1. Overview

This document specifies the dual-layer combat architecture of **Tenebris and Lux**: a **Tactical Strategy RPG (TRPG)** featuring a **grid-based tactical battlefield** coupled with **cinematic traditional side-scroll / action-based combat resolution**. Conceived during the 2014–2018 RPG Maker genesis of CreativeWorkzStudio, this design bridges grand tactical positioning with visceral, action-driven 2D engagement.

## 32.2. Dual-Layer Battle Architecture

The combat engine operates across two seamless, interconnected operational loops:




## 32.3. Phase 1: Grid-Based Tactical Battlefield

The overarching strategic layer takes place upon a discrete 2D grid map representing the castles, frontier passes, and battlegrounds of the **Kingdom of Sacyra**.

### 32.3.1. Grid Topography & Spatial Invariants

- **Coordinate Space**: Discrete integer grid `(grid_x, grid_y)`.
- **Unit Footprint**: Standard units occupy a $1 \times 1$ grid tile; major boss constructs (such as The Dark Lord's Siege Automata) occupy $2 \times 2$ or $3 \times 3$ footprints.
- **Terrain Types & Modifiers**:
  - **Stone Flagstones / Road**: $1.0\times$ movement cost, $+0\%$ evasion.
  - **Ramparts / Elevated Battlement**: $1.5\times$ movement cost, $+20\%$ physical defense, $+1$ attack range for ranged skills.
  - **Frontier Thicket / Rubble**: $2.0\times$ movement cost, $+15\%$ evasion, blocks cavalry charge.
  - **Void Mire (Abyssal Blight)**: $2.5\times$ movement cost, inflicts $-5$ HP decay per turn on non-void units.

### 32.3.2. Movement & Zone of Control (ZOC)

- **Movement Budget**: Units possess a base `movement_tiles` statistic (e.g. Katrina: 6 tiles; Elijah: 4 tiles in heavy armor).
- **Zone of Control**: Moving into an adjacent tile directly forward of a hostile soldier halts further movement for that turn, anchoring frontline formations.
- **Directional Facing**: Units finish their turn facing one of four cardinal directions (`NORTH`, `SOUTH`, `EAST`, `WEST`). Attacking an enemy from their flank grants $+15\%$ critical chance; attacking from the rear guarantees an unguardable strike.

### 32.3.3. Turn Sequencing

- **Dynamic Initiative Order**: Turn execution is governed by unit Agility. Fast skirmishers (like Katrina) act earlier in the round, while heavy vanguard armorers act later with higher defensive resilience.

## 32.4. Phase 2: Cinematic Combat Resolution (Side-View Action)

When an attacker initiates an assault or when two opposing units clash on the grid, the game seamlessly transitions into a dedicated **2D Side-View Cinematic Battle Scene**.

### 32.4.1. Visual & Staging Presentation

- The attacker approaches from the left (player units) or right (enemy units) across a dynamic 2D battlefield backdrop matching the current terrain (e.g., Sunken Citadel battlements, ash-strewn frontier).
- Character sprites render with expressive combat animations: Idle combat stance, forward dash, heavy strike, shield block, recoil, stagger, and victory flourishes.

### 32.4.2. Combat Mechanics During Cinematic Resolution

1. **Active Input & Timed Agency**:
   - Rather than passive numbers scrolling past, combat resolution incorporates timed player inputs:
   - **Timed Ward (Lux Guard)**: Pressing the ward button precisely upon enemy weapon contact triggers a **Perfect Deflection**, absorbing 100% of incoming damage and staggering the foe.
   - **Twilight Severance (Tenebris Strike)**: Pressing strike during the forward dash activates a **Deep Cleave**, bypassing a percentage of enemy armor.
2. **Stagger & Poise Meter**:
   - Every battler possesses a **Poise / Stagger Gauge** ($0 \dots 100$).
   - Repeated heavy strikes or shield bashes deplete enemy poise. When poise hits zero, the target enters **Staggered Vulnerability**, leaving them defenseless against critical finishers.
3. **Stance Animation Morphing**:
   - The character's active stance directly dictates their cinematic combat posture and move set:
   - `STANCE_WARDING`: Elijah locks the Shield of Lux into the earth; golden starlight forms a kinetic buckler.
   - `STANCE_STRIKING`: Katrina or Elijah lowers their center of gravity, dragging the Sword of Tenebris as twilight embers trail the blade.
   - `STANCE_EQUILIBRIUM`: Fluid parry-and-counter sequencing utilizing both armaments in rapid harmony.

## 32.5. Dual-Protagonist Synergy: Elijah & Katrina

The tactical depth of **Tenebris and Lux** emerges from combining the distinct capabilities of Elijah and Katrina:

### 32.5.1. Elijah Ro-enheart (Frontline Guardian)

| Feature | Tactical Grid & Cinematic Specification |
|---|---|
| **Unit Class** | Royal Vanguard / Aegis Knight |
| **Grid Role** | Anchors the defensive perimeter; draws enemy agro; intercepts projectiles. |
| **Special Trait: Royal Intercept** | If an adjacent ally (within 1 tile) is targeted by an attack, Elijah leaps in front to receive the blow, initiating the cinematic defense in their stead. |
| **Relic Mastery** | **Shield of Lux**: Projects a 2-tile radius **Sanctuary Aura** on the grid that grants allies $+25\%$ physical defense and immunity to void blight. |
| **Cinematic Moves** | **Aegis Slam**, **Radiant Shockwave**, **Bastion Guard**, **Solar Flare Riposte**. |

### 32.5.2. Katrina Jaren (Tactical Skirmisher)

| Feature | Tactical Grid & Cinematic Specification |
|---|---|
| **Unit Class** | Guild Outrider / Twilight Blade |
| **Grid Role** | Flanks heavy units; assassinates backline spellcasters; disarms siege engines. |
| **Special Trait: Guild Acrobatics** | Can move through tiles occupied by enemy infantry without triggering Zone-of-Control lock. |
| **Relic Mastery** | **Sword of Tenebris**: Can activate **Shadow Walk**, cloaking in mist on the grid to become untargetable by ranged attacks until her next strike. |
| **Cinematic Moves** | **Abyssal Dash**, **Crescent Sever**, **Shadow Dance**, **Mist-Veil Feint**. |

### 32.5.3. Tandem Linked Combat (The Duo Duel)

When Elijah and Katrina are positioned on adjacent grid tiles and either initiates combat, both heroes enter the cinematic side-view battle scene together:

- **Shield & Sword Flank**: Elijah executes an **Aegis Slam** that knocks the enemy backward, launching them into Katrina's airborne **Crescent Sever**.
- **Harmonized Equilibrium**: Both heroes combine the radiance of Lux and the edge of Tenebris, unleashing a dual finishing art that sweeps across multiple targets.

## 32.6. Persistent State & Moral Sway Feedback

Upon completion of the cinematic resolution:

1. **HP & Wound Conservation**: Remaining unit health updates onto the tactical grid. Fallen units are removed or enter down-but-not-dead states.
2. **Relic Resonance Tracking**: Energy expended (`shield_power`, `sword_edge`) updates in persistent memory (`player_state_t`).
3. **Moral Sway Evaluation**:

- Choosing to spare routed enemy captains, warding retreating militia, or using non-lethal shield stagger tilts sway toward **Virtue (+)**.
- Executing fleeing enemies with abyssal severance or utilizing forbidden void arts tilts sway toward **Malice (-)**.
- Moral sway unlocks distinct strategic faction options across the Kingdom of Sacyra.

# Chapter 33. The Twin Relics, Combat Stances & Moral Sway

**Source File**: `docs/game_design/relics_and_combat.adoc`

## 33.1. Overview

This document specifies the core ludological mechanics of **Tenebris and Lux** in the **Kingdom of Sacyra** and **The Dreamscape**: the twin primordial relics (**The Shield of Lux** and **The Sword of Tenebris**), the dynamic combat stance state machine, the moral sway continuum (`[-100, +100]`), and the tactical action skills: **Defend** (Lux Barrio), **Attack** (Cuttatio Tenebris), and **Special** (Basic Magic, Basic Ward, Basic Charm, and Duo-Loca Bashin n Slash).

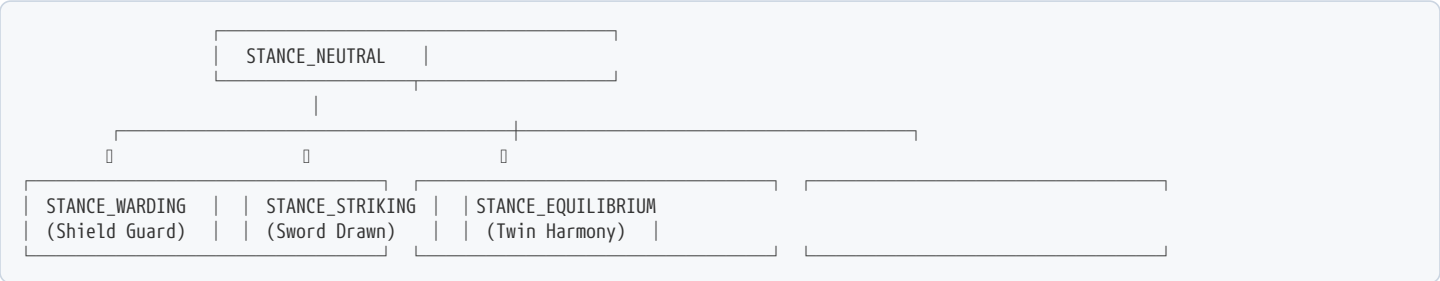
## 33.2. Primordial Neutrality & Relic Mechanics

Neither relic is inherently good nor evil. They are manifestations of cosmic apex forces whose morality is dictated solely by the intentions and choices of their wielders—canonically **Elijah Ro-enheart** and **Katrina Jaren**, or a customizable bearer chosen by the player:

| Relic | The Shield of Lux | The Sword of Tenebris |
|---|---|---|
| **Aspect** | Apex Radiance & Cohesion | Apex Entropy & Dissolution |
| **Energy Gauge** | `shield_power` ($0 \dots 100$) | `sword_edge` ($0 \dots 100$) |
| **Canonical Bearer** | **Elijah Ro-enheart** (Vanguard Defense) | **Katrina Jaren** (Agile Severance) |
| **Grid Role** | Zone-of-Control denial, sanctuary aura, ally protection | High-mobility flanking, armor penetration, mist evasion |
| **Cinematic Action** | Timed kinetic blocks, shield-bashes, radiant shockwaves | Rapid dash strikes, armor severance, twilight blade waves |
| **Resource Dynamics** | Regenerates on guard; absorbs incoming assault | Consumes edge on strikes; hones outside combat |

## 33.3. The Combat Stance State Machine

The battle posture shifts dynamically across four discrete combat stances:




1. `STANCE_NEUTRAL`: Relics held at waist. Balanced baseline.
   - Standard movement on the tactical grid; baseline attack parameters.
2. `STANCE_WARDING`: Shield raised in protective vanguard high-guard.
   - Grid Effect: Increases allied defense within adjacent tiles by $+25\%$.
   - Combat Effect: Boosts Shield power recovery ($+15$), increases stagger damage against attacking foes ($25$ HP), and wards against incoming status curses.

3. `STANCE_STRIKING`: Greatsword held in aggressive two-handed assault posture.
    - Grid Effect: Increases unit movement by $+1$ tile and grants armor-piercing flank bonuses.
    - Combat Effect: Increases strike damage from $30$ to $45$ (or base $+15$ bonus); reduces edge cost from $12$ to $8$ points.
4. `STANCE_EQUILIBRIUM`: Harmonized twin-relic synchronization.
    - Grid Effect: Immunity to positioning debuffs and void mire terrain penalties.
    - Combat Effect: Simultaneously restores $+15$ Shield power and $+15$ Sword edge; parries incoming strikes while counter-attacking for $20$ damage; nudges moral sway toward center ($0$).

## 33.4. The Moral Sway Continuum & Decoupled Elemental Expression

The moral trajectory of Sacyra's champions is tracked along a continuous bounded integer axis:




### 33.4.1. Decoupling Morality ("The Why") from Elemental Expression ("The How")

Neither Lux nor Tenebris dictates a character's moral alignment: * **Virtuous Lux (`+1 ⋯ +100`):** Selfless sanctuary, kinetic mitigation (**Lux Barrio**), and restorative peace. * **Tyrannical Lux (`-100 ⋯ -1`):** Blinding glare, totalitarian enforcement of rigid order, and calcifying living will into petrified stasis (The Mi-ren Fallacy). * **Virtuous Tenebris (`+1 ⋯ +100`):** Surgical severance (**Cuttatio Tenebris**), cutting corrupt bindings, and executing Absolute Justice without unnecessary bloodshed (The Casatti Ideal). * **Malevolent Tenebris (`-100 ⋯ -1`):** Creeping entropic rot, nihilistic bloodlust, and devouring devastation (The Dark Lord).

### 33.4.2. The Bystander Corridor (`[-50, +50]`) as Net Moral Inertia

The `[-50, +50]` band represents **Net Moral Inertia**, where opposing deeds cancel out into zero net displacement. Staying within this corridor traps the protagonists in 3,000 years of repeating cyclical tragedy. Breaking past $\pm 50$ through deliberate micro-choices (such as aiding Eric or John in Dio Village) asserts heroic agency: **"Trauma is Inherited, Destiny is Chosen."**

### 33.4.3. Equilibrium (`0`)

Maintained by active discipline rather than passive fence-sitting; balancing the blade and the aegis so that radiant light tempers the void without collapsing into calcified stasis or entropic rot. Unlocks the tandem secret art **Duo-Loca Bashin n Slash**.

## 33.5. Tactical Actions & Combat Skills

In the revised, authentic combat framework, basic actions expand into dedicated martial and arcane skills:

### 33.5.1. Defend (`defend` / `ward`)

- **Standard Defend**: Braces behind the Shield of Lux, regenerating $+10 \dots +15$ shield power and mitigating incoming damage.
- **Basic Skill: `Lux Barrio` (`barrio`)**:
    - Manifests a radiant, geometric barrier of pure light.

- **Effect**: Blocks incoming attack and reduces subsequent incoming damage by **$25\%$** for a rolled turn duration.
- **Duration Roll**: Deterministically rolls for **$2 \dots 5$ turns** (`2 + (turns % 4)`).
- **Resonance**: Saturates Shield of Lux by $+20$ points (capped at $100$) and awards $+3$ moral virtue.

### 33.5.2. Attack (`attack` / `strike`)

- **Standard Attack**: Strikes the target enemy using the Sword of Tenebris, consuming edge points and dealing physical damage (`(ATK * 2) - DEF`).
- **Basic Skill: `Cuttatio Tenebris` (`cuttatio`)**:
  - A sweeping abyssal crescent slash infused with dark mist.
  - **Direct Damage**: Deals full primary severance damage to the selected target.
  - **Splash Damage**: Deals **$20\%$ splash damage** to surrounding and flanking foes (`threat_surrounding_hp`).
  - **Edge Dynamic**: Consumes $8 \dots 12$ edge points.

### 33.5.3. Special (`special` / `balance`)

Replaces balance with a four-fold technique tree honoring the twin relics and arcane arts:

1. `Basic Magic` **(`magic`)**:
   - Neutral elemental magic damage attacking enemy Magic Defense (`(MAT * 2) - MDF`).
   - Costs $10$ MP. Pure, unaligned cosmic energy.
2. `Basic Ward` **(`ward` under special)**:
   - Magic-based defend weaving an arcane barrier.
   - Restores $+15$ Shield power and staggers incoming assault without relying purely on physical bracing.
   - Costs $6$ MP.
3. `Basic Charm` **(`charm`)**:
   - Enhances astral focus, granting **$+20\%$ Magic Attack (MAT)** for **$1 \dots 3$ turns** (`1 + (turns % 3)`).
   - Costs $8$ MP.
4. `Duo-Loca Bashin n Slash` **(`duo`)**:
   - The signature dual-relic strike uniting the Shield of Lux and Sword of Tenebris.
   - **Adaptive Damage**: Intelligently evaluates whether the player's Magic Attack or Physical Attack is superior, or whichever enemy defense (DEF vs MDF) is lower, channeling into the foe's weakest resistance:
     - If enemy MDF < DEF or MAT > ATK: executes as **Magical Twilight Slash**.
     - Otherwise: executes as **Physical Radiant Bash**.
   - **Relic Harmony**: Simultaneously restores $+15$ Shield power and $+15$ Sword edge, while nudging moral sway toward absolute cosmic equilibrium ($0$).
   - Costs $12$ MP.

### 33.5.4. Hero Customization & Protagonist Lore

- While **Elijah Ro-enheart** and **Katrina Jaren** remain the canonical dual protagonists representing the Shield of Lux and Sword of Tenebris, the player can dynamically designate their own hero identity using the `name <chosen_name>` command.
- All narrative chronicle logs, character severity tags, and telemetry traces adapt cleanly to the bearer's active name.

## 33.6. The Metaphysical Law of Tempering & Dissipation

The interaction between the Shield of Lux and the Sword of Tenebris is governed by the ancient law established during the 7-Year Great War:

- **Lux as the Singular Tempering Agent:** Raw Tenebris is entropic and severing; left unchecked, it erodes both target and wielder. Lux is the only force capable of containing and tempering Tenebris, providing structural cohesion so that void strikes can be directed safely.
- **The Fragility of Radiance (Dissipation):** While Lux tempers Tenebris, it requires emotional and ethical stability. Any sudden surge of terror, malice, or panic causes Lux to **dissipate immediately**, collapsing protective barriers and leaving Tenebris unbridled.
- **Ludonarrative Impact:** In combat, executing dark commands with negative moral sway degrades Lux Barrio duration rolls, while maintaining Equilibrium (`0` sway) maximizes barrier resilience and tandem skill execution.

## 33.7. The Relic Embellishment System

Throughout the campaign, the Shield of Lux and Sword of Tenebris are actively embellished through quest outcomes, moral choices, and auxiliary relics:

1. **Eric's Test of Honesty (Map 028) $\to$ The Ring of Fire:**
   - Donating rent to Eric rewards the **Ring of Fire**, imbuing strikes and barriers with flame-weave properties, connecting directly to the LeTBS **Burning Tile Spread** tactical state.
2. **Monica's Offering (Maps 027/029) $\to$ The Book of Monsters:**
   - Sponsoring an offering to **Lyri, Goddess of Peace**, attunes the relics to arcane resonance and awards the **Book of Monsters**, unlocking advanced MDF-targeting formulas.
3. **John the Thief vs. St. Baron (Maps 030/031):**
   - Aiding John enhances critical severance and stealth penetration; upholding feudal property law reinforces kinetic poise and bastion authority.
4. **Armory Relics (Medusa's Sword & Heart Shield):**
   - Optional material infusions granting petrification stasis or passive HP vitality warding.

## 33.8. Relic Profiles & The Ten-Ending Campaign Matrix

The final profile of relic embellishments along the `[-100, +100]` Moral Sway continuum maps into the 3x3 matrix crowned by the 10th synthesis ending: * **Lux-Good (`[+50, +100]`): The Radiant Sanctuary** — Selfless protection without tyranny. * **Lux-Neutral (`[-50, +50]`): The Benevolent Stasis** — The Mi-ren Fallacy; golden crystal cocoon. * **Lux-Bad (`[-100, -50]`): The Calcified Tyrant** — Radiance weaponized into an inquisitorial police state. * **Tenebris-Good (`[+50, +100]`): Absolute Justice** — The Casatti Ideal; surgical severance of corruption. * **Tenebris-Neutral (`[-50, +50]`): The Cynical Outlaw** — Threat slain; realm abandoned to decay. * **Tenebris-Bad (`[-100, -50]`): The Abyssal Usurper** — The Cycle of Usuru; rules from obsidian ashes. * **Statera-Good (`[+50, +100]`): Harmonized Concord** — Both relics united in restorative justice. * **Statera-Neutral (`[-50, +50]`): Armed Deterrence** — Uneasy status quo; frontier rift unhealed. * **Statera-Bad (`[-100, -50]`): The Dual Despot** — Totalitarian martial hegemony. * **Synthesis (10th Route): The True Ending ("The Ugly Truth")** — Unlocked via Elijah, Katrina, and the Custom Bearer (`<chosen_name>`) alongside nocturnal Dreamscape trials, mending the 3,000-year planar tear.

# Chapter 34. Campaign Architecture, The King's Curse & Multi-Ending Specification

**Source File**: docs/game_design/campaign_structure_and_endings.adoc

## 34.1. Overview & Ludological Philosophy

This specification articulates the structural macro-architecture of the campaign in **Tenebris and Lux**, including the two-half campaign progression, the metaphysical etiology of "The King's Curse," the midpoint narrative fork, player agency over lore discovery, and the complete five-route ending matrix. Built upon the premise of ludonarrative consent, the engine allows casual players to experience a coherent linear tactical campaign, while reserving the mythological truth for players who actively engage with nocturnal Dreamscape trials.

## 34.2. The Two-Half Campaign Structure

The overarching campaign is organized into two distinct dramatic and mechanical halves:




### 34.2.1. Act I: Stopping King Jihad (The Sovereign of Ambition)

Act I centers on terrestrial, socio-political conflict. Operating from Dio Village through the Whispering Woods and Capital marches, the party uncovers King Jihad's treasonous suppression of border alerts. The act culminates during the **Festival of Heroes** at Sacyra Castle, where King Jihad is confronted and defeated.

### 34.2.2. The Etiology of "The King's Curse" & Royal Possession

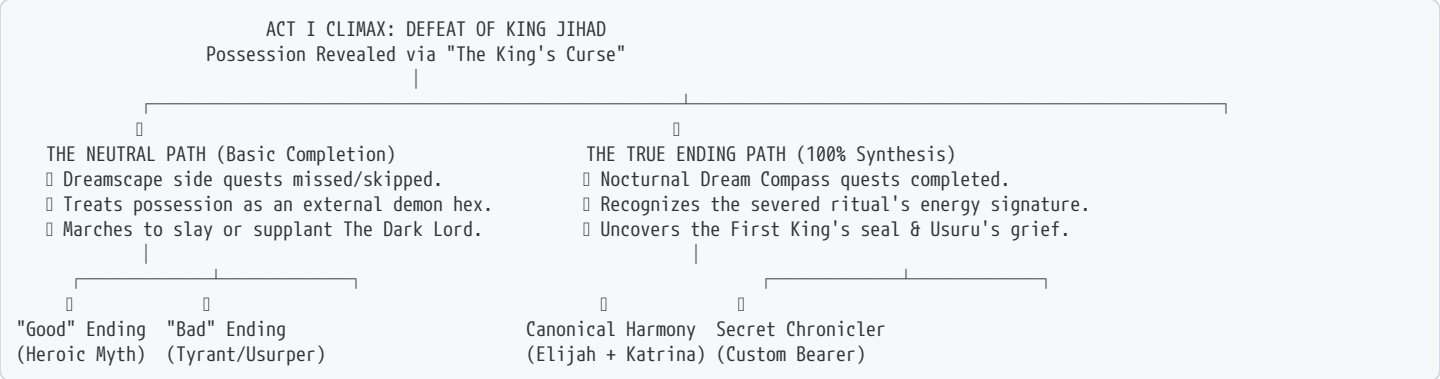
Court superstition long spoke of a hereditary madness afflicting monarchs of Sacyra. Forensic narrative architecture reveals that "The King's Curse" is not a demonic hex, but an active **metaphysical hazard**:

- **The Open Planar Wound:** When eleven-year-old Usuru shattered the First King's stasis projection ritual 3,000 years ago, a jagged planar rift tore open directly beneath the dias of the **High Throne**.
- **Psychic Backwash:** For three millennia, reigning monarchs sitting upon the High Throne were subjected to perpetual psychic shear and planar backwash from across the veil.
- **Jihad's Fatal Gambit:** Seeking to usurp the Dark Lord's void power during the thinned veil of the Festival of Heroes, King Jihad intentionally opened the conduit beneath his throne. Because the ethereal medium mirrors

mortal intent, Jihad's megalomania inverted the flow, allowing a dormant psychic remnant of Usuru (**The Dark Lord**) to possess his body.

## 34.3. The Midpoint Fork: Neutral Path vs. True Ending Path

Defeating King Jihad reveals the void possession, forcing the narrative into its primary structural fork:




### 34.3.1. Why Basic Completion Is the Neutral Ending

By design, any player who simply plays the main story linearly from start to finish will reach the end of the campaign. However, skipping the missable nocturnal **Dream Compass** trials produces the **Neutral Ending**:

- The player defeats King Jihad and destroys The Dark Lord's physical manifestation.
- The kingdom is saved in the present, but the **First King remains permanently marooned** in The Dreamscape.
- Usuru's ancient wound of betrayal is left unhealed, and the systemic neglect between the Capital and frontier provinces remains unchanged, ensuring the cycle will repeat.

### 34.3.2. Player Consent to Know the Deeper Lore

Video games are distinguished by interactive agency. Forcing forty minutes of historical exposition down the player's throat violates ludic pacing. By gating the full mythological truth behind optional, nocturnal Dreamscape trials, the game honors the player's agency: players who desire a fast-paced tactical campaign can enjoy a satisfying heroic conclusion, while players who actively investigate the mystery are rewarded with the full mythological reality.

### 34.3.3. Why the True Ending Is "The Ugly Truth"

The canonical True Ending is neither a simplistic fairy-tale triumph nor a dark nihilistic collapse. It is **"The Ugly Truth"** that dismantles the self-serving propaganda of both sides:

- It exposes that Prince Mi-ren's loving preservation created a catastrophic institutional lie that deceived a continent for six festivals.
- It acknowledges that Usuru's childhood outrage was entirely valid before his unguided fury curdled into monstrous destruction.
- It reveals that Sacyra's three millennia of golden peace were bought at the cost of deliberate neglect toward frontier provinces like Libira and Dio Village.

## 34.4. Relic Embellishment & The Ten-Ending Campaign Matrix

Throughout both acts, the player's choices, side quests, and stance preferences actively **embellish** the Shield of Lux and Sword of Tenebris along the `[-100, +100]` **Moral Sway** continuum.

### 34.4.1. Telemetry Over Point Locks & The Bystander Corridor

Players do not aim for brittle, exact integers to unlock endings. Instead, `moral_sway` acts as continuous behavioral telemetry showing where the player's philosophy is steering the world:

- **The Bystander Corridor (`[-50, +50]`):** Represents **Net Moral Inertia**—a mutual cancellation trap where opposing moral vectors offset each other. Floating in this corridor produces status-quo stagnation or repeating cycles.
- **The Threshold of Conviction:** Escaping the bystander corridor requires sustained virtue (`[+50, +100]`) or despotic malice (`[-100, -50]`).
- **Equilibrium (`0`):** An active, razor-thin discipline where Lux and Tenebris temper each other without collapsing into calcification or entropic rot.

### 34.4.2. The 3x3 Ending Matrix + The 10th True Ending

Combining the three elemental expressions (**Lux**, **Tenebris**, and **Statera** / Equilibrium) with the three moral bands (**Good** `[+50, +100]`, **Neutral** `[-50, +50]`, and **Bad** `[-100, -50]`) produces nine distinct campaign resolutions, crowned by a tenth synthesis ending:

| Expression | Good (`[+50, +100]`) | Neutral (`[-50, +50]`) | Bad (`[-100, -50]`) |
|---|---|---|---|
| **Lux** *(Cohesion & Preservation)* | **1. The Radiant Sanctuary:** Selfless protection; the Shield of Lux shelters the realm and heals frontier wounds without imposing despotic rule. | **2. The Benevolent Stasis (The Mi-ren Fallacy):** Out of fear of conflict, the wielder seals Sacyra in an unchanging golden crystal cocoon where nothing harms—and nothing grows. | **3. The Calcified Tyrant:** Radiance weaponized into an unblinking inquisition that blinds dissenters and petrifies living will into stone. |
| **Tenebris** *(Severance & Rest)* | **4. Absolute Justice (The Casatti Ideal):** The Sword of Tenebris surgically severs corrupt noble monopolies and King Jihad's war machinery without civilian bloodshed. | **5. The Cynical Outlaw:** The immediate threat is slain for personal survival, but the wielder abandons Sacyra to its ongoing feudal decay. | **6. The Abyssal Usurper (Cycle of Usuru):** The wielder executes King Jihad, absorbs The Dark Lord's void essence, and rules from the obsidian ashes. |
| **Statera** *(Equilibrium & Harmony)* | **7. Harmonized Concord:** Both relics work in virtuous partnership, ushering in restorative justice, frontier outposts, and provincial renewal. | **8. Armed Deterrence:** A tense status quo where the war pauses, but the deep cultural rift with border provinces remains unhealed. | **9. The Dual Despot:** Barrier stasis and abyssal severance combine to enforce total, totalitarian martial hegemony. |
| **Synthesis** *(The 10th Route)* | **10. The True Ending ("The Ugly Truth"):** Unlocked only by experiencing Elijah's campaign, Katrina's campaign, and finally the Custom Bearer (`<chosen_name>`) while completing all nocturnal Dreamscape trials. Sacyra confronts its 3,000-year institutional guilt, releases the First King's trapped soul, reconciles Usuru's childhood grief, exposes King Jihad, and permanently mends the planar tear in the Room of Sages. | | |

### 34.4.3. The Witness Protagonist, Keystone Route & Observer Effect

Reaching the True Ending ("The Ugly Truth") relies on three formal game design structures:

1. **The Witness Protagonist (Internal vs. External Focalization):**
   - **Elijah Ro-enheart** and **Lady Katrina Jaren** live inside the story (**Internal Focalization**). They are shaped by their personal trauma, social caste, and military or aristocratic backgrounds.
   - **The Custom Bearer (`<chosen_name>`)** serves as the **External Focalizer** (the Observer/Chronicler alongside Chrono in the Room of Sages), possessing the detachment required to perceive the full truth.
2. **The Keystone Route (Synthesis Scenario):**
   - Elijah uncovers the military conspiracy surrounding Commander Gareth and King Jihad.
   - Katrina uncovers noble and underworld exploitation alongside John the Thief and the Guild.
   - The Custom Bearer's campaign acts as the architectural keystone locking both partial perspectives together.
3. **The Quantum Observer Effect:**
   - Left to themselves, Sacyra's internal factions repeat the tragic 3,000-year cycle.
   - The outside Observer's conscious engagement collapses the tragic wavefunction into historical reconciliation.

### 34.4.4. Chrono's Chronicle Ledger: NG+ Timeline Navigation

To ensure that exploring the full matrix never devolves into repetitive grinding:

- **Persistent Archival Memory:** Compendiums (**Book of Monsters**, **Book of Items**), unlocked embellishments, and combat masteries persist across runs.
- **Timeline Branching:** Once a campaign route is cleared, **Chrono's Chronicle Ledger** unlocks in the Room of Sages, functioning as an interactive timeline tree. Completionists can jump directly to pivotal decision bookmarks (such as Eric's honesty test, the St. Baron infiltration, or the Act I climax) to explore all 9 matrix endings without replaying introductory chapters.
- **Diegetic Saves:** Saving is rooted diegetically at **Grandma Magolia's Hearth** and **Chrono's Room of Sages**.

## 34.5. Systemic Invariants & Engine Triggers

All ending routes and campaign forks are evaluated through deterministic engine invariants in Layer 1:

1. `player_state_t` **Integrity:** The moral sway value (`int16_t moral_sway`) is clamped strictly within `[-100, +100]` via `math_clamp_int`.
2. **Dream Compass Nocturnal Flag:** Entry into `LOC_THE_DREAMSCAPE` requires `hour >= 20 || hour <= 4` from the Layer 0 monotonic clock (`time_get_monotonic_ms`).
3. **Equilibrium State Check:** Unlocking Route 7 or Route 10 requires `abs(moral_sway) <= 15` and both `shield_power >= 80` and `sword_edge >= 80` during the final turn sequence.
4. **Observer Flags:** The 10th True Ending requires validation flags `flag_elijah_cleared`, `flag_katrina_cleared`, and `dreamscape_trials_mask == 0xFF`.

# Chapter 35. The Dreamscape & Temporal Architecture

**Source File**: docs/game_design/dreamscape_and_temporal_architecture.adoc

## 35.1. Overview & Engine Integration

This document specifies the technical, mechanical, and temporal architecture of **The Dreamscape** in **Tenebris and Lux**. Grounded in the recovered 2017–2018 RPG Maker MV demo baseline (Maps 006, 008, 011, 041, 042) and integrated into the ISO C11 engine architecture, it formalizes the planar opening sequence with Chrono the Sage, the terrestrial demo gating, the evolution of **Grandma's Watch** into **The Dream Compass**, and nocturnal event-driven temporal phasing.

## 35.2. Engine Architecture & Topological Verification

The Dreamscape is not an ephemeral cutscene or narrative metaphor; it is an active, fully integrated subsystem registered within the Layer 1 simulation engine:

### 35.2.1. Explicit Province Registration

The plane is codified as an explicit topological enumeration in `include/sim/sim_types.h`:

```
typedef enum {
    LOC_THE_DREAMSCAPE    = 0,  /* Astral origin, ancestral archive, planar rift */
    LOC_DIO_VILLAGE       = 1,  /* Pastoral woodland hamlet; Grandma's hearth    */
    LOC_WHISPERING_WOODS  = 2,  /* Outlying frontier timberlands; beast paths    */
    LOC_SACYRA_MARCHES    = 3,  /* Fortified royal road to the capital citadel   */
    LOC_COUNT             = 4
} location_id_t;
```

### 35.2.2. Tactical LeTBS Battlefields

The Dreamscape encompasses five dedicated grid environments extracted from the historical archives:

- **Map 006 (Chrono's Archive / The Hall of Sages):** Archetype stone pedestals, floating crystal columns, and ancestral book pedestals.
- **Map 008 (The Astral Crucible):** Tutorial grid where the bearer tests `Lux Barrio` and `Cuttatio Tenebris` against encroaching void scouts.
- **Map 011 (The Shifting Leyline Bridges):** Dynamic tactical grid featuring floating platforms and crystal floor switches that alter terrain heights.
- **Maps 041 & 042 (The Deep Astral Depths):** High-level nocturnally phased battlefields featuring astral leyline conduits that replenish MP while inflicting planar shear.

### 35.2.3. Parameterized Combatants: The Dream Eaters

The Dreamscape is populated by **Dream Eaters**—parasitic entities dispatched across the thinning veil by The Dark Lord to devour ancestral records prior to the Festival of Heroes:

```
Threat Profile: Dream Eater
 ├──── Threat HP: 70
 ├──── Physical Defense (DEF): 11
 ├──── Arcane Defense (MDF): 8
 ├──── Flank Vigor: 70
 └──── Tactical AI: Rushes fragile casters; vulnerable to radiant deflection (Lux Barrio).
```

### 35.2.4. Bidirectional Topology

The spatial graph connects seamlessly between the metaphysical and physical planes: * Northward: `Awaken` $\to$ `LOC_DIO_VILLAGE` (Grandma Magolia's Hearth). * Southward: `Slumber Meditation` $\to$ `LOC_THE_DREAMSCAPE` (Requires Dream Compass nighttime flag).

## 35.3. Narrative Inception: The Astral Rite of Awakening

### 35.3.1. Unformed Potential in the Astral Archive

The game opens not in a mundane tavern or military camp, but inside the timeless ether of **The Dreamscape**. Because this plane houses 3,000 years of ancestral memory—and the marooned soul of the First King—having **Chrono the Sage** guide the player through class archetype stones elevates character creation into a canonical ritual of spiritual attunement.

### 35.3.2. Tri-Perspective Adaptability

Opening in unformed ether allows the simulation to cleanly accommodate any of the three playable vantage points—**Elijah Ro-enheart**, **Lady Katrina Jaren**, or a custom-named bearer (`<chosen_name>`)—before their identity crystallizes into physical form.

### 35.3.3. The Ludonarrative Shock of "AWAKEN!"

After proving tactical mastery against Dream Eaters in the abstract void, Chrono banishes the player with an urgent command: *"AWAKEN!"*. The immediate cut to Grandma Magolia's crackling birch hearth in Dio Village shocks the player into physical reality, establishing a stark contrast between cosmic destiny and human vulnerability.

## 35.4. Terrestrial Gating & The Dream Compass

### 35.4.1. Chapter 1 Demo Gating as a Terrestrial Anchor

Throughout Chapter 1 (the entire demo narrative), the player is strictly locked out of returning to The Dreamscape. This design rule prevents premature cosmic dilution, forcing the player to invest emotionally in the grounded human stakes of Dio Village: * Grandma Magolia's failing health and maternal wisdom. * Eric the Lumberjack's economic ruin from monster-infested logging trails. * John the Thief's moral crusade against the predatory St. Baron. * Monica the Mystic's grief beside the totem of fallen border soldiers.

### 35.4.2. From Grandma's Watch to The Dream Compass

In the opening sequence, the player retrieves **Grandma's Watch** from the upstairs chest in Dio Village (Map 020). Mechanically, this initializes the Layer 0 `time_utils` 24-hour monotonic clock:

```
Layer 0 Monotonic Clock Subsystem
 ├──── Seconds / Minutes
 ├──── Hours (00:00 to 23:59)
 ├──── Weekdays (1: Sol, 2: Luna, 3: Ignis, 4: Aqua, 5: Terra, 6: Ventus, 7: Aether)
 ├──── Weeks / Months
 └──── Calendar Countdown to The Festival of Heroes
```

Upon concluding the Chapter 1 demo introduction, Grandma's Watch undergoes its first metaphysical transmutation, evolving into **The Dream Compass**.

### 35.4.3. Nocturnal Synchronization & Scarcity

The Dream Compass acts as a dimensional resonant tuning fork with strict activation criteria: * **Nighttime Window Only:** The Dream Compass functions exclusively during nocturnal hours (`20:00 ⇐ hour || hour ⇐ 04:00`). During daylight hours, the compass needle spins aimlessly. * **Event-Driven Astral Windows:** Nocturnal side quests and main story trials in The Dreamscape manifest only during specific astrological alignments and calendar milestones. * **Scarcity and Sacred Weight:** By restricting access to nocturnal windows, The Dreamscape is prevented from becoming a trivial grinding hub, preserving its sacred mystery and directly linking quest availability to the monotonic clock’s relentless countdown toward the Festival of Heroes.

# Chapter 36. World Topography, Provinces & Battlefield Terrain

**Source File**: `docs/game_design/world_topography.adoc`

## 36.1. Overview

This document specifies the authentic spatial geography, province layouts, environmental encounters, and tactical grid battlefields governing the **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**, `/s□□□ki□r□/`) and **The Dreamscape**, grounded in the recovered 2017–2018 demo baseline. While **The Dark Lord** looms as the overarching existential threat in the background ("A Plot"), the narrative begins within the ethereal trials of **The Dreamscape** before awakening into the pastoral hamlet of **Dio Village**. Players may guide the canonical champions (**Elijah Ro-enheart** and **Katrina Jaren**) or customize their chosen bearer's designation.

## 36.2. Spatial Structure: The Realm Topology

The campaign begins in the trial realm of slumber and transitions into the physical topology of Sacyra:




### 36.2.1. The Dreamscape (`LOC_THE_DREAMSCAPE`)

- **Role**: The ethereal realm of ancient trials guided by Chrono where the illusion of the dream shatters.
- **Environment**: Floating crystal archways, shifting astral mist, and mechanism trial chambers.
- **Significance**: The hero awakens here to learn the fundamentals of combat and relic resonance. Dream Eaters gather under the distant influence of **The Dark Lord**.
- **Encounter**: **Dream Eater** ($70$ Threat HP, $11$ DEF, $8$ MDF, $70$ Flank Vigor).
- **Grid Battlefield**: Astral tiles, floating platforms, and ethereal leylines granting MP regeneration.
- **Connections**: North / Awaken $\to$ Dio Village.

### 36.2.2. Dio Village (`LOC_DIO_VILLAGE`)

- **Role**: Pastoral ancestral hamlet in Sacyra; the true awakening point of the physical journey.
- **Environment**: Cozy stone cottages with thatched roofs, Grandma Magolia's welcoming hearth, Eric the lumberjack's timber yard, and the local general trading post.
- **Significance**: Here the hero recovers Grandma's Watch and the **Compass of Morality**, encountering village folk before venturing into the perilous wilds.

- **Threat Vigor**: $0$ (Sanctified village sanctuary; recovery, dialogue, and lore exchange).
- **Connections**: South $\to$ The Dreamscape (slumber meditation); North $\to$ Whispering Woods.

### 36.2.3. Whispering Woods (`LOC_WHISPERING_WOODS`)

- **Role**: Dense ancient forest enveloping Dio Village, separating the hamlet from the kingdom's interior.
- **Environment**: Towering ironwood boughs, rustling autumn canopy, and hidden woodland trails.
- **Encounter**: **Umbral Wolf** ($85$ Threat HP, $14$ DEF, $10$ MDF, $60$ Flank Vigor).
- **Tactical Grid Elements**:
  - Forest thickets providing $+20\%$ evasion against ranged physical assault.
  - Tree root chokepoints where **Lux Barrio** shields flanking allies from pack ambushes.
- **Connections**: South $\to$ Dio Village; East / North $\to$ The Sacyran Marches.

### 36.2.4. The Sacyran Marches (`LOC_SACYRA_MARCHES`)

- **Role**: The windswept frontier marches leading toward the Royal Court of King Jihad and the Capital.
- **Environment**: Craggy hills, fortified signal towers, and military encampments.
- **Encounter**: **Shadow Scout & Vanguard Patrol** ($110$ Threat HP, $18$ DEF, $14$ MDF, $80$ Flank Vigor).
- **Tactical Grid Elements**:
  - Trenches, palisades, and elevated ramparts where **Cuttatio Tenebris** cleaves through enemy formations.
  - Broad fields allowing full maneuverability of twin-relic equilibrium strikes.
- **Connections**: West / South $\to$ Whispering Woods.

## 36.3. The Overarching Menace: The Dark Lord

In the authentic 2017–2018 canonical script, the ultimate antagonist is designated strictly as **The Dark Lord**: * He is never named "Malakor" or burdened with synthetic titles in canon. * The Dark Lord represents the primordial threat looming in the distance ("The A Plot"), while regional intrigues—such as King Jihad's political maneuvers, Noble House power struggles, and the Adventurers' Guild missions—unfold on the active stage. * The wielder's mastery of the Shield of Lux and Sword of Tenebris prepares them for this looming confrontation without rushing directly into the final battle.

# Chapter 37. Continuous Invariant Testing Philosophy

**Source File**: `docs/game_design/game_testing.adoc`

## 37.1. Abstract

In classical software engineering, testing focuses primarily on stateless input/output validation, API schemas, and unit function assertions (`assert(f(x) == y)`). In game development, testing requires an entirely distinct paradigm: verifying continuous state continuity, economy balance, softlock immunity, deterministic replay parity, and strict 60 FPS frame-budget headroom. This document outlines the testing philosophy governing **Tenebris and Lux**.

## 37.2. The Fundamental Divergence

| Dimension | Traditional Software Testing | Game Development Testing |
|---|---|---|
| **State Nature** | Ephemeral, localized, often stateless | Continuous, cumulative, multi-turn state machine |
| **Failure Modes** | Null pointer exceptions, HTTP 500s, syntax errors | Softlocks, economy collapse, degenerate gameplay loops, visual desync |
| **Correctness Metric** | Does output match expected schema? | Does the game feel right? Is the state bounded, balanced, and unexploitable? |
| **Input Cadence** | Controlled, structured API payloads | Unbounded, chaotic human keystrokes, rapid button mash, simultaneous events |
| **Time Budget** | Milliseconds/seconds per request | 16.6 milliseconds total frame budget (60 FPS) with microseconds for simulation |

## 37.3. Core Pillars of Game Development Testing

### 37.3.1. Pillar 1: Dynamic State Invariants & Bounded Continuity

In a game like **Tenebris and Lux**, state variables (`shield_power`, `sword_edge`, `moral_sway`, `threat_hp`) are constantly mutated across hundreds of player turns. A single off-by-one error or uncapped addition will cause state drift, resulting in overflow, negative health zombies, or invulnerability exploits.

- **Invariant Axioms**:
  - `0 ⇐ state.shield_power ⇐ 100` must hold after every single action turn.
  - `0 ⇐ state.sword_edge ⇐ 100` must hold even when spamming strikes at zero energy.
  - `-100 ⇐ state.moral_sway ⇐ +100` must never overflow into undefined integers.
- **Diagnostic Tripwires**: The game engine provides `game_audit_state()`, a self-healing diagnostic routine that asserts all invariants simultaneously and outputs a detailed diagnostic manifest without crashing the game process.

### 37.3.2. Pillar 2: Deterministic Replay ("Golden Master Replay")

Games are emergent systems. If two players—or an automated CI runner—replay the exact same sequence of commands from the same seed, the end state must be **100% bitwise identical**:

```
// Golden Master Replay Assertion
TEST_ASSERT(run1.shield_power == run2.shield_power, "Deterministic shield");
```

```
TEST_ASSERT(run1.sword_edge   == run2.sword_edge,   "Deterministic sword");
TEST_ASSERT(run1.moral_sway   == run2.moral_sway,   "Deterministic sway");
TEST_ASSERT(run1.threat_hp    == run2.threat_hp,    "Deterministic threat");
```

This ensures that gameplay balance, combat resolution, and narrative branching do not drift across compiler optimizations, operating systems, or refactored engine code.

### 37.3.3. Pillar 3: Softlock Immunity & Degenerate Play Detection

A **softlock** occurs when a player is technically still playing, but has no valid action forward: - Running out of relic energy during an un-fleeable battle. - Getting trapped behind a one-way gate with no exit command. - Inputting malformed or rapid chaotic strings that cause the parser to lock up.

Our automated stress suite executes **1,000 rapid chaotic commands** (combining invalid strings, rapid stance transitions, and empty tokens), verifying that the game engine rejects nonsense safely, maintains continuous memory integrity, and never enters a deadlocked state.

### 37.3.4. Pillar 4: Economy & Combat Balance Invariance

Game testing must verify that no single player action is universally dominant (the "dominant strategy" flaw). If a player can simply spam `ward` forever without consequence, or if `strike` renders all other actions useless, the game design collapses: - Warding increases shield power and moral virtue, but deals limited stagger damage. - Striking inflicts decisive severance damage, but consumes finite sword edge. - Balancing recovers both relics and parries attacks, but requires timing and resets extreme moral sway.

### 37.3.5. Pillar 5: Frame-Budget Latency Headroom

In standard software, a backend database query taking 50 ms is acceptable. In game development, a game running at 60 FPS has **only 16.6 milliseconds** to: 1. Poll hardware events and window queues. 2. Execute game simulation logic. 3. Compute spatial audio and particle animations. 4. Render the scene graph to the backbuffer and blit to the display.

Simulation logic must therefore execute in **microseconds**. In our automated benchmark (`test_gameplay_scenarios`), 10,000 action turns execute in **160 ms** (**16.02 $\mu\text{s}$ per turn**), consuming less than **$0.1\%$** of a 60 FPS frame.

# Chapter 38. Game Design Curation & Triaging Matrix

**Source File**: docs/game_design/demo_baseline/game_design_curation.adoc

## 38.1. Overview

This document formalizes the architectural triage of the 2017–2018 RPG Maker MV demo datasets. It separates generic RPG Maker Runtime Package (RTP) boilerplate from temporary testing experiments and identifies the authentic, fleshed-out canonical systems to be reconstructed in the native C11 game engine.

## 38.2. Triage Categorization Taxonomy

To prevent technical debt from legacy game engine defaults from contaminating our custom C11 engine, all extracted game data is triaged into three strict analytical categories:

| Category | Definition | Architectural Policy |
|---|---|---|
| **Tier 1: RTP Defaults & Boilerplate** | Stock RPG Maker MV assets, generic items, placeholder sounds, and default formulas. | **Discard / Re-architect**. Never port verbatim into C11. |
| **Tier 2: Prototype & Testing Sketches** | Ad-hoc test maps, sandbox battlers, debug switches, and experimental scripts. | **Review & Evolve**. Extract mechanical concepts but redesign within clean strata. |
| **Tier 3: Fleshed-Out Canonical Pillars** | Core story scripts, dual-protagonist dialogue dynamics, moral dilemma quests, LeTBS grid tactics, and key relics. | **Preserve & Reconstruct**. Faithfully implement as axiomatic invariants in C11. |

## 38.3. Tier 1: RPG Maker RTP Defaults (To Discard or Re-Architect)

| Element | Historical Manifest | Why It Must Be Discarded / Replaced |
|---|---|---|
| **Basic Consumables** | Potion (50G), Magic Water (100G), Dispel Herb (20G), Stimulant (50G) | Generic RTP recovery items. In C11, consumables must tie directly to Sacyran herbalism, relic alchemy, and stance restoration. |
| **Generic Armaments** | Standard Sword (100G), Axe (100G), Cane (100G), Bow (100G), Shield (100G), Cloth (100G) | Default placeholder inventory without distinct tactical affinities or stance interactions. |
| **Stock Sound Effects** | `Attack2`, `Buzzer1`, `Damage4`, `Collapse1-4`, `Shop1` | Generic MV sound library. C11 audio subsystem will support custom soundscapes. |
| **Placeholder Bestiary** | Bat (`HP: 80`), Slime (`HP: 250`), Rabbit (`HP: 55`), Orc (`HP: 300`) | Standard MV demo monsters used for basic stat checks. Replaced by Sacyran fauna and Abyssal void aberrations. |

| Element | Historical Manifest | Why It Must Be Discarded / Replaced |
|---|---|---|
| **Default Title Theme** | `Theme6` / `CrossedSwords` title screen graphic | Default MV asset packaging. Replaced by bespoke Sacyran coat of arms and twin relic artwork. |

## 38.4. Tier 2: Prototype & Testing Sketches (Concepts to Evolve)

| Element | Historical Location | Tactical / Design Value to Evolve |
|---|---|---|
| **Sandbox Test Maps** | `Map006 Test Areas`, `Map043 Event Map` | Used to test event triggers, movement paths, and sprite coordinates. Demonstrates intent for environmental puzzle mechanics. |
| **Debug Enemies & Troops** | `Enemy 14 Test`, `Troop 10 Test`, `Enemy 32 Invincible Neutral` | Proved that neutral, invincible obstacles and destructible props function well on the tactical grid. |
| **Explosive Chickens** | Fire Chicken (`Actor 13`), Bomb (`Skill 136`) | Humorous prototype mechanic demonstrating combat summons and dynamic decoy units on the tactical field. |
| **Plugin Experiments** | `LeTBS/firestorm.json`, `LeTBS/test.json`, `QMaps/` | Initial experiments with Lecode's Tactical Battle System and Quasi Movement plugins. Proves grid-based TRPG was the true target genre. |

## 38.5. Tier 3: Fleshed-Out Canonical Pillars (Core Invariants to Reconstruct)

The following systems represent the authentic creative vision of **Tenebris and Lux** and must be preserved and reconstructed in C11:

### 38.5.1. Dual-Protagonist Dynamics & Narrative Branching

- **Elijah Ro-enheart (`\n[1]`)**: Son of vanished Commander Gareth Ro-enheart. Dutiful, protective vanguard bearer of the Shield of Lux. Dialogue reflects reverence for duty, family loyalty, and hesitation to violate laws.
- **Lady Katrina Jaren (`\n[2]`)**: Daughter of noble House Jaren escaping courtly politics to join the Adventurers' Guild. Agile outrider wielder of the Sword of Tenebris. Dialogue features sardonic wit, impatience with provincial stagnation, and fierce defense of the vulnerable.

### 38.5.2. The Moral Sway Continuum Engine

- **Ethical Mechanics**: Discrete moral choices tracking across `Moral (Good)` and `Moral (Evil)` within the invariant continuum `[-100, +100]`.
- **The Test of Honesty (Eric's House, Map 28)**:
  - Stealing Eric's life savings (10G) rewards quick cash but plunges morality.
  - Donating 500G to cover Eric's rent saves a desperate worker and earns the **Ring of Fire**.
- **The Struggle of Dio Village (John the Thief vs. St. Baron, Map 30/31)**:
  - Feudal exploitation vs. outrider rebellion, forcing the heroes to choose between legal compliance and moral justice.

### 38.5.3. Lecode Tactical Battle System (LeTBS) Mechanics

- **Grid Movement & Range Geometries**:
  - Movement Points (4 MP baseline), modified by skills like **Light Steps** (+1) and **Heavy Hit** (-3).
  - Scope shapes: `circle(r)`, `circle(max, min)` (annular), `line(n)`, `square(r)`.
- **Terrain Hazards & Obstacles**:
  - **Ice Wall** and **Ice Prison** creating physical destructible barriers (`HP: 20`).
  - **Volcano Ground** and **Burning Tile Spread** (`spread: 100% circle(1)`).
- **Turn Timeline Manipulation**:
  - **Accelerate**, **Decelerate**, and **Emergency Turn** altering the turn queue.

### 38.5.4. The Monotonic 24-Hour Time Engine

- **Grandma's Watch**: Family heirloom activating real-time calendar and clock tracking (Seconds, Minutes, Hours, Days, Weeks, Months, Years), corresponding directly to Layer 0 `time_utils`.

# Chapter 39. Combat Game Note: Attack Action & Cuttatio Tenebris

**Source File**: `docs/game_design/combat/attack_and_cuttatio_tenebris.adoc`

## 39.1. Design Intent & Gameplay Philosophy

The **Attack** action channels the entropy and severance aspects of the **Sword of Tenebris**. Rather than generic single-target chipping, offensive melee strikes in **Tenebris and Lux** embody high-risk severance and crowd-cleaving outrider tactics:

- **Basic Attack (`attack` / `strike`)**:
  - Direct physical slash with the obsidian edge.
  - **Mechanics**: Computes primary physical damage based on player attack attribute and target physical defense: `Damage = (ATK * 2) - Enemy_DEF`.
  - **Edge Consumption**: Consumes $6 \dots 10$ points of blade sharpness (`sword_edge`).
  - **Out-of-Combat Attunement**: Outside of combat, channeling the blade hones its edge by $+15$ points as abyssal mist condenses along the blade.
- **The Basic Skill: `Cuttatio Tenebris` (`cuttatio`)**:
  - An abyssal sword slash that releases a crescent wave of twilight mist across the battlefield.
  - **Primary Severance**: Delivers full direct physical damage to the primary target (`(ATK * 2) - Enemy_DEF + Stance_Bonus`). In `STANCE_STRIKING`, an additional $+15$ flat damage bonus is applied.
  - **20% Area Splash Damage**: In tactical combat with multiple enemy formations or surrounding reinforcements, **$20\%$ of the primary damage** cascades onto adjacent flanking targets (`threat_surrounding_hp`).
  - **Minimum Splash Threshold**: Guarantees at least $1$ damage to flanking targets if any primary damage is dealt, preventing armor-rounding deadzones.

## 39.2. Numerical Invariants & Formulas

| Parameter | Standard Attack | Cuttatio Tenebris |
|---|---|---|
| **Action Type** | Basic Melee Strike | Blade Cleave Skill |
| **Formula** | `(ATK * 2) - DEF` | `(ATK * 2) - DEF + Bonus` |
| **Stance Scaling** | Standard damage | $+15$ bonus in `STANCE_STRIKING` |
| **Edge Cost** | $6 \dots 10$ Edge | $8 \dots 12$ Edge |
| **Splash Cleave** | None ($0\%$) | **$20\%$ of primary damage** |
| **Moral Sway Effect** | $+0$ (Neutral) / $+1$ on kill | $+2$ (Controlled Severance) |

# Chapter 40. Combat Game Note: Defend Action & Lux Barrio

**Source File**: `docs/game_design/combat/defend_and_lux_barrio.adoc`

## 40.1. Design Intent & Gameplay Philosophy

In tactical RPG design, standard "Guard" or "Defend" commands often feel like wasted turns or passive stalls. In **Tenebris and Lux**, the **Defend** action is an active tactical commitment that interfaces with the metaphysical resonance of the **Shield of Lux**:

- **Basic Defend (`defend` / `ward`)**:
  - The wielder braces behind the celestial aegis.
  - **Mechanics**: Absorbs incoming kinetic force, boosts `shield_power` by $+10 \dots +15$ (saturating at $100$), and staggers attacking threats for $10 \dots 15$ HP of poise recoil.
  - **Moral Resonance**: Positively shifts moral sway ($+1$ virtue).
- **The Basic Skill: `Lux Barrio` (`barrio` / `lux barrio`)**:
  - Rather than a flat shield block, `Lux Barrio` erects a radiant, geometric light barrier that deflects the incoming assault while mitigating subsequent trauma over multiple combat turns.
  - **Damage Mitigation**: Reduces incoming damage by **$25\%$** across all damage types.
  - **Turn Duration Roll**: Injects tactile variance into battle planning. The duration is rolled within $[2, 5]$ turns (`2 + (turns_taken % 4)`).
  - **Resonance Surge**: Surges Shield of Lux charge by $+20$ points and awards $+3$ virtue.
  - **Offensive Counter-Stagger**: When used against an attacking foe, deals $15 \dots 25$ HP of reflective shockwave damage, capable of routing low-vigor skirmishers directly.

## 40.2. Numerical Invariants & Formulas

| Parameter | Standard Defend | Lux Barrio |
|---|---|---|
| **Action Type** | Basic Tactical Guard | Relic Core Skill |
| **Resource Cost** | $0$ MP / $0$ TP | $0$ MP (Consumes active turn) |
| **Shield Resonance** | $+10$ (Neutral) / $+15$ (Warding) | $+20$ (Capped at $100$) |
| **Stagger Recoil** | $10 \dots 15$ HP | $15 \dots 25$ HP |
| **Damage Mitigation** | Flat deflection | **$-25\%$ incoming damage** |
| **Buff Duration** | Immediate turn only | **$2 \dots 5$ turns** |
| **Moral Sway Effect** | $+1$ Virtue | $+3$ Virtue |

# Chapter 41. Combat Game Note: Special Abilities & Duo-Loca Bashin n Slash

**Source File**: docs/game_design/combat/special_abilities_and_duo_loca.adoc

## 41.1. Design Intent & The Four-Fold Special Suite

The **Special** action suite expands the former "Balance" command into a versatile toolkit that balances resource expenditure (Mana Points / Tactical Points) against tactical problem-solving:

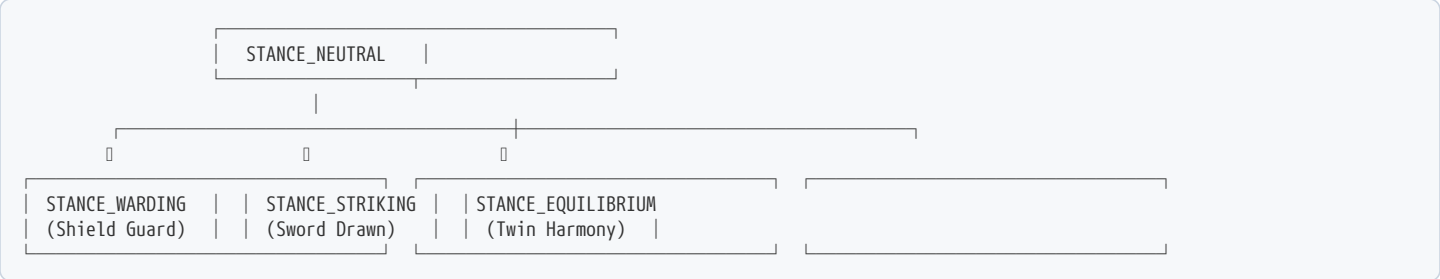
1. `Basic Magic` **(**`magic`**)**:
   - Represents neutral elemental magic damage unbound by strict Light or Dark alignment.
   - **Formula**: `(eff_MAT * 2) - Enemy_MDF`.
   - **Mana Cost**: $10$ MP.
   - **Synergy**: Scales dynamically with `Basic Charm`. If Charm is active, effective `MAT` increases by $+20\%$.
2. `Basic Ward` **(**`ward` **under special)**:
   - Arcane-based defensive warding.
   - Rather than purely relying on heavy armor and physical poise, the wielder constructs a shimmering lattice of runes.
   - **Mana Cost**: $6$ MP.
   - **Effect**: Staggers incoming assault ($12$ HP) and reinforces the Shield of Lux ($+15$ resonance).
3. `Basic Charm` **(**`charm`**)**:
   - An astral focus enchantment that attunes the mind to mystical resonance.
   - **Buff Effect**: Increases Magic Attack (`MAT`) by **$+20\%$**.
   - **Duration Roll**: Deterministically rolled for **$1 \dots 3$ turns** (`1 + (turns_taken % 3)`).
   - **Mana Cost**: $8$ MP.
4. `Duo-Loca Bashin n Slash` **(**`duo`**)**:
   - The signature synthesis technique of the twin relics.
   - **Adaptive Damage Intelligence**: Solves a common tactical RPG frustration where a physical fighter encounters heavily armored units or a spellcaster encounters magic-immune foes:
     - Evaluates `Physical_Potential = (ATK * 2) - Enemy_DEF`.
     - Evaluates `Magical_Potential = (eff_MAT * 2) - Enemy_MDF`.
     - If the enemy is more vulnerable to magic (`MDF < DEF`), or if the player's magical output is superior (`eff_MAT > ATK`): channels **Magical Twilight Slash**.
     - Otherwise: channels **Physical Radiant Bash**.
   - **Relic Harmonization**: Simultaneously replenishes $+15$ Shield of Lux and $+15$ Sword of Tenebris, while nudging moral sway by $3$ points toward cosmic equilibrium ($0$).
   - **Cost**: $12$ MP.

# Chapter 42. Combat Game Note: Tactical Stance State Machine

**Source File**: `docs/game_design/combat/stance_state_machine.adoc`

## 42.1. State Machine Completeness

The tactical stance machine governs the battle posture of the bearer across four discrete, strongly-connected states:




1. `STANCE_NEUTRAL` **(At Ease)**:
   - Balanced recovery posture. Baseline movement, balanced defenses.
2. `STANCE_WARDING` **(Shield High Guard)**:
   - Reinforces front-line presence. Lux Barrio stagger damage increases from $15$ to $25$ HP.
   - Standard defend boosts shield by $+15$ (vs $+10$).
3. `STANCE_STRIKING` **(Sword Drawn)**:
   - Aggressive forward stance. Primary melee attacks gain $+15$ flat damage bonus.
   - Attack edge consumption drops from $10$ to $6$ points.
4. `STANCE_EQUILIBRIUM` **(Twin Harmonized)**:
   - Balances both relics simultaneously. Enhances Duo-Loca execution and parry/riposte dynamics.

# Chapter 43. Armory, Equipment & Inventory Registry (2017–2018 Demo)

**Source File**: `docs/game_design/demo_baseline/armory_and_inventory.adoc`

## 43.1. Overview

This document details all weapons, protective armors, restorative items, and key relics defined in `Weapons.json`, `Armors.json`, and `Items.json`.

## 43.2. Weapons & Relic Armaments

| ID | Weapon Name | Type | Price | ATK | DEF | AGI | Lore & Characteristics |
|---|---|---|---|---|---|---|---|
| 1 | **Sword** | Sword | 500\G | +10 | +0 | +0 | |
| 2 | **Axe** | Axe | 500\G | +10 | +0 | +0 | |
| 3 | **Cane** | Cane | 500\G | +10 | +0 | +0 | |
| 4 | **Bow** | Bow | 500\G | +10 | +0 | +0 | |
| 5 | **Medusa's Sword** | Sword | 1600\G | +17 | +0 | +0 | The sword used to slay Medusa. It is said to have her petrifying powers. |

## 43.3. Defensive Armor & Relic Wards

| ID | Armor Name | Type | Price | DEF | MDF | MHP | Lore & Characteristics |
|---|---|---|---|---|---|---|---|
| 1 | **Shield** | Small Shield | 300\G | +10 | +0 | +0 | |
| 2 | **Hat** | General Armor | 300\G | +10 | +0 | +0 | |
| 3 | **Cloth** | General Armor | 300\G | +10 | +0 | +0 | |
| 4 | **Ring** | General Armor | 300\G | +0 | +0 | +0 | |
| 5 | **Heart Shield** | Small Shield | 300\G | +10 | +5 | +0 | |
| 6 | **Ring of Fire** | General Armor | 0\G | +0 | +5 | +0 | |

## 43.4. Consumable Items & Quest Relics

| ID | Item Name | Price | Consumable | Description & Architectural Mapping |
|---|---|---|---|---|
| 1 | **Potion** | 10\G | Yes | |

| ID | Item Name | Price | Consumable | Description & Architectural Mapping |
|---|---|---|---|---|
| 2 | **Magic Water** | 100\G | Yes | |
| 3 | **Dispel Herb** | 200\G | Yes | |
| 4 | **Stimulant** | 300\G | Yes | |
| 5 | **Grandma’s Watch** | 0\G | No (Key Relic) | This was Grandma’s Watch. Its a family heirloom that keeps perfect time. |
| 6 | **Book of Monsters** | 0\G | No (Key Relic) | This is a big with all the enemies you’ve encountered |
| 7 | **Book of Items** | 0\G | No (Key Relic) | This book describes all the various items |
| 8 | **Compass of Morality** | 0\G | No (Key Relic) | |

# Chapter 44. Bestiary & Troop Encounters (2017–2018 Demo)

**Source File**: docs/game_design/demo_baseline/bestiary_and_troops.adoc

## 44.1. Overview

This document catalogs all 22+ hostile monsters, wild fauna, bandits, void entities, and interactive battlefield props from Enemies.json and Troops.json.

## 44.2. Monster Bestiary

| ID | Name | HP | MP | ATK | DEF | MAT | MDF | AGI | LUK | AI / Behavior |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **Bat** | 80 | 12 | 11 | 8 | 8 | 4 | 25 | 30 | Skill 1 |
| 2 | **Mago** | 60 | 18 | 8 | 6 | 16 | 6 | 22 | 30 | Skill 41, Skill 1 |
| 3 | **Rabbit** | 55 | 0 | 10 | 8 | 8 | 4 | 27 | 30 | Skill 1 |
| 4 | **Great Mago** | 90 | 36 | 10 | 7 | 19 | 7 | 23 | 30 | Skill 1 |
| 5 | **Fire Mago** | 90 | 36 | 10 | 7 | 19 | 7 | 23 | 30 | Skill 1 |
| 6 | **Pyra** | 30 | 50 | 6 | 8 | 6 | 4 | 27 | 30 | Basic Attack |
| 7 | **Bandit** | 80 | 12 | 14 | 10 | 10 | 7 | 25 | 30 | Skill 1, Skill 169 |
| 8 | **Mage Bandit** | 80 | 12 | 11 | 8 | 8 | 4 | 25 | 30 | Skill 1 |
| 9 | **Slime** | 250 | 0 | 30 | 30 | 30 | 30 | 30 | 30 | Skill 1 |
| 10 | **Orc** | 300 | 0 | 30 | 30 | 30 | 30 | 30 | 30 | Skill 1 |
| 11 | **Minotaur** | 500 | 0 | 30 | 30 | 30 | 30 | 30 | 30 | Skill 1 |
| 12 | **Random Kid** | 100 | 0 | 30 | 10 | 10 | 10 | 10 | 10 | Skill 1, Skill 3 |
| 13 | **Dream Eater** | 70 | 12 | 11 | 8 | 8 | 4 | 25 | 30 | Skill 1 |
| 32 | **Invincible Neutral** | 99999 | 0 | 8 | 6 | 14 | 6 | 22 | 30 | Basic Attack |
| 33 | **Tile Effect Launcher** | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 34 | **Ice Block** | 20 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 35 | **Wooden Chest** | 50 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 36 | **Silver Chest** | 50 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 37 | **Golden Chest** | 50 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 38 | **Stolen Goods** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |

## 44.3. Neutral & Interactive Grid Entities

The demo utilized the battle engine to render interactive tactical grid entities: * **Wooden Chest (Enemy 35)**: HP: 50 | Yields 200 Gold upon destruction via Common Event 5. * **Silver Chest (Enemy 36)**: HP: 50 | Yields 2,000 Gold upon

destruction via Common Event 6. * **Golden Chest (`Enemy 37`)**: `HP: 50` | Yields 10,000 Gold upon destruction via Common Event 7. * **Ice Block (`Enemy 34`)**: `HP: 20` | Destructible terrain obstacle summoned by Ice Wall. * **Stolen Goods (`Enemy 38`)**: `HP: 1` | Objective dropped upon defeat of Mage Bandit. * **Invincible Neutral (`Enemy 32`)**: `HP: 99,999` | Static grid anchor.

## 44.4. Troop Encounter Formations

| Troop ID | Troop Designation | Composition |
|---|---|---|
| 1 | **Bat*2** | Bat, Bat |
| 2 | **Slime*2** | Mago, Mago |
| 3 | **Orc** | Rabbit |
| 4 | **Minotaur** | Great Mago |
| 5 | **Random Kid** | Fire Mago |
| 6 | **Bat** | Bat |
| 7 | **Bat*2** | Bat, Bat |
| 8 | **Bandit*3, Mage Bandit** | Bandit, Bandit, Mage Bandit, Bandit |
| 9 | **Dream Eater*2** | Dream Eater, Dream Eater |

# Chapter 45. Dramatis Personae & Cast Compendium (2017–2018 Demo)

**Source File**: `docs/game_design/demo_baseline/cast_and_characters.adoc`

## 45.1. Overview

This compendium catalogues every character, party companion, village elder, court antagonist, and mystical summon present in the 2017–2018 demo archives, detailing their combat statistics, class associations, sprite bindings, and narrative significance.

## 45.2. Dual Protagonists

### 45.2.1. Elijah Ro-enheart (The Steadfast Vanguard)

- **ID**: `Actor 1`
- **Class**: Hero (`Class 5`) / Knight (`Class 1`)
- **Relic Affinity**: Bearer of the **Shield of Lux**
- **Initial Level**: Lv 5 (Folder 1) / Lv 1 (Folder 2)
- **Movement Points**: 4 MP
- **Sprite / Battler**: `Actor1_1` / `MainCharacter` (custom generator)
- **Profile & Personality**: Son of the vanished Commander Gareth Ro-enheart (former Head of the Royal Guard). Earnest, chivalric, deeply burdened by his father's mysterious disappearance on his 10th birthday. Elijah seeks answers through service and duty, instinctively stepping forward to protect the weak and uphold the light of Sacyra.
- **Canonical LeTBS Specification**:

```
<letbs>
move_points: 4
auto_turn_order_face
auto_status_sprite
</letbs>
```

### 45.2.2. Lady Katrina Jaren (The Outrider Blade)

- **ID**: `Actor 2`
- **Class**: Hero (`Class 5`) / Start (`Class 7`)
- **Relic Affinity**: Wielder of the **Sword of Tenebris**
- **Initial Level**: Lv 5 (Folder 1) / Lv 1 (Folder 2)
- **Movement Points**: 4 MP
- **Initial Equipment**: Weapon 1 (Sword), Armor 2 (Hat), Armor 3 (Cloth)
- **Sprite / Battler**: `Actor1_8`
- **Profile & Personality**: Noble daughter of aristocratic House Jaren. Rebellious, razor-sharp, pragmatic, and fiercely independent. Refusing a cloistered aristocratic marriage or court seat, Katrina escaped to Dio Village with eyes on joining the independent Adventurers' Guild. Her caustic wit masks a deep empathy for ordinary folk crushed by royal neglect.

- **Canonical LeTBS Specification**:

```
<letbs>
move_points: 4
auto_turn_order_face
auto_status_sprite
</letbs>
```

## 45.3. Extended Tactical Party Companions (Folder 2 Additions)

In `Old_Game_Data_2`, three specialized tactical party members were integrated to expand the grid-based TRPG party:

### 45.3.1. Sebastien (The Infiltrator Outlaw)

- **ID**: `Actor 4`
- **Class**: Thief (`Class 6`)
- **Initial Level**: Lv 4
- **Sprite / Battler**: `Actor3_7`
- **Tactical AI & Role**: Outrider flanker equipped with `ranged_fighter` AI, trap disarmament, and precision critical strikes.

### 45.3.2. Angela (The Evoker Elementalist)

- **ID**: `Actor 5`
- **Class**: Mage (`Class 2`)
- **Initial Level**: Lv 4
- **Movement Scope**: `square(2)` (4 MP)
- **Sprite / Battler**: `Actor3_8` (custom cast, atk, hit, victory animations)
- **Action Palette**: Full tactical command set (`Move`, `Skill[42]`, `Items`, `Scan`, `Pass`).
- **Tactical Role**: Long-range elemental artillery, crowd control, and battlefield terrain modification (ice walls, volcano ground).

### 45.3.3. Lucius (The Battle Priest)

- **ID**: `Actor 6`
- **Class**: Priest (`Class 3`)
- **Initial Level**: Lv 4
- **Sprite / Battler**: `Actor2_7`
- **Tactical Role**: Turn order manipulation (`Accelerate`, `Decelerate`, `Emergency Turn`), long-distance allied teleportation, and barrier restoration.

## 45.4. Summoned Celestial Entities

### 45.4.1. Wind Spirit (Actor 11)

- **Class**: `Class 12` | **Sprite**: `Nature` (Index 4)
- **Skills**: Air Compression (`Skill 132`), Great Wind (`Skill 133` - pushback), Transposition (`Skill 134` - swap places).

### 45.4.2. Pixie (Actor 12)

- **Class**: `Class 13` | **Sprite**: `Nature` (Index 5)
- **Skills**: Healing Wave (`Skill 135` - AOE restoration).

### 45.4.3. Fire Chicken (Actor 13)

- **Class**: `Class 14` | **Sprite**: `Nature` (Index 2)
- **Characteristics**: Explosive combat decoy (`loop_animation: 262`), Bomb (`Skill 136`), Fire Arrow (`Skill 137`).

## 45.5. Key Lore Figures & NPCs

| Name | Role & Allegiance | Historical Significance |
|---|---|---|
| **Grandma Magolia** | Matriarch of Dio Village | Mother of vanished Commander Gareth Ro-enheart; grandmother to Elijah. Bestows **Grandma's Watch** and **Compass of Morality**. |
| **Monica the Mystic** | Former Royal Battlemage | Eccentric hermit and priestess of Goddess Lyri. Guides protagonists in awakening relic powers through ritual offerings. |
| **Eric the Lumberjack** | Impoverished Dio Laborer | Loses his livelihood to monster raids; serves as the moral litmus test for player empathy in the **Test of Honesty**. Bestows the **Ring of Fire**. |
| **John the Thief** | Local Outlaw Leader | Operates in Lower Dio; plots to unseat the predatory St. Baron to return wealth to the villagers. |
| **The St. Baron** | Aristocratic Landowner | Feudal aristocrat bleeding Dio Village dry; symbol of court corruption and moral decay. |
| **Chrono** | Archival Sage & Guardian | Guide of the Dream Chamber / Ancient Archives. Oversees archetype stone selection and mechanics trials. |
| **High Priest Roté** | Royal Advisor | Court prelate who witnesses the cosmic omens and hastens to warn King Jihad. |
| **King Jihad** | Monarch of Sacyra | Feigns royal mobilization while falling victim to the King's Curse, secretly coveting the returned primordial darkness to surpass The Dark Lord. |

# Chapter 46. Combat Classes, Stat Matrices & Growth Progressions (2017–2018 Demo)

**Source File**: `docs/game_design/demo_baseline/classes_and_growth.adoc`

## 46.1. Overview

This document details the 7 primary player and NPC classes plus 4 auxiliary summon/guard classes from `Classes.json`. It includes base stat formulas across 8 core attributes (MHP, MMP, ATK, DEF, MAT, MDF, AGI, LUK) and skill learning progressions.

## 46.2. Class 1: Warrior

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 100 | 30 | 12 | 8 | 3 | 8 | 27 | 32 |
| 5 | 145 | 46 | 17 | 12 | 5 | 11 | 27 | 42 |
| 10 | 202 | 64 | 23 | 17 | 8 | 15 | 27 | 55 |
| 20 | 314 | 102 | 35 | 26 | 13 | 22 | 27 | 80 |
| 50 | 650 | 215 | 71 | 54 | 27 | 44 | 27 | 155 |
| 99 | 1200 | 400 | 130 | 100 | 50 | 80 | 27 | 277 |

### 46.2.1. Skill Learnings for Warrior

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Repulsive Blow** (`ID: 11`) | Innate class technique |
| Lv 1 | **Flash Jump** (`ID: 12`) | Innate class technique |
| Lv 1 | **Rush** (`ID: 13`) | Innate class technique |
| Lv 1 | **Intimidating Aura** (`ID: 14`) | Innate class technique |
| Lv 1 | **Heavy Hit** (`ID: 15`) | Innate class technique |
| Lv 1 | **Provoke** (`ID: 16`) | Innate class technique |
| Lv 1 | **Phantom Slash** (`ID: 17`) | Innate class technique |

## 46.3. Class 2: Mage

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 60 | 90 | 2 | 7 | 13 | 8 | 26 | 32 |
| 5 | 83 | 130 | 4 | 10 | 18 | 12 | 26 | 42 |
| 10 | 110 | 180 | 6 | 13 | 24 | 17 | 26 | 55 |
| 20 | 165 | 280 | 10 | 20 | 36 | 26 | 26 | 80 |

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 50 | 330 | 580 | 21 | 39 | 72 | 54 | 26 | 155 |
| 99 | 600 | 1070 | 40 | 70 | 130 | 100 | 26 | 277 |

### 46.3.1. Skill Learnings for Mage

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Fire** (ID: 42) | Innate class technique |
| Lv 1 | **Sparkle** (ID: 43) | Innate class technique |
| Lv 1 | **Lightning Storm** (ID: 44) | Innate class technique |
| Lv 1 | **Fire Storm** (ID: 45) | Innate class technique |
| Lv 1 | **Fire Support** (ID: 46) | Innate class technique |
| Lv 1 | **Volcano Ground** (ID: 47) | Innate class technique |
| Lv 1 | **Ice Fury** (ID: 49) | Innate class technique |
| Lv 1 | **Walking Fire** (ID: 50) | Innate class technique |
| Lv 1 | **Ice Wall** (ID: 51) | Innate class technique |
| Lv 1 | **Ice Prison** (ID: 52) | Innate class technique |
| Lv 1 | **Whitch's Escape** (ID: 53) | Innate class technique |
| Lv 1 | **Fire Chickens** (ID: 54) | Innate class technique |
| Lv 1 | **Water (charged)** (ID: 55) | Innate class technique |
| Lv 1 | **Mega Sparkle** (ID: 56) | Innate class technique |
| Lv 1 | **Multi Fire** (ID: 57) | Innate class technique |

## 46.4. Class 3: Priest

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 65 | 60 | 6 | 8 | 10 | 7 | 32 | 32 |
| 5 | 95 | 87 | 10 | 13 | 15 | 11 | 32 | 42 |
| 10 | 133 | 119 | 15 | 18 | 20 | 16 | 32 | 55 |
| 20 | 208 | 185 | 25 | 28 | 31 | 26 | 32 | 80 |
| 50 | 433 | 380 | 53 | 59 | 63 | 54 | 32 | 155 |
| 99 | 800 | 700 | 100 | 110 | 115 | 100 | 32 | 277 |

### 46.4.1. Skill Learnings for Priest

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Heal** (ID: 72) | Innate class technique |
| Lv 1 | **Teleportation** (ID: 73) | Innate class technique |
| Lv 1 | **Salvation** (ID: 74) | Innate class technique |
| Lv 1 | **Teleportation II** (ID: 75) | Innate class technique |
| Lv 1 | **Explosive Mark** (ID: 76) | Innate class technique |
| Lv 1 | **Light Steps** (ID: 78) | Innate class technique |
| Lv 1 | **Ghost Arrow** (ID: 79) | Innate class technique |
| Lv 1 | **Magic Trap** (ID: 80) | Innate class technique |
| Lv 1 | **Fire Arrows** (ID: 82) | Innate class technique |
| Lv 1 | **Wind Spirit** (ID: 83) | Innate class technique |
| Lv 1 | **Summon Pixie** (ID: 84) | Innate class technique |
| Lv 1 | **Shamanism** (ID: 85) | Innate class technique |
| Lv 1 | **Extra Turn** (ID: 86) | Innate class technique |
| Lv 1 | **Accelerate** (ID: 87) | Innate class technique |
| Lv 1 | **Emergency Turn** (ID: 89) | Innate class technique |
| Lv 1 | **Ally Teleport.** (ID: 90) | Innate class technique |
| Lv 1 | **Mana Ball** (ID: 92) | Innate class technique |

## 46.5. Class 4: Summoner

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 450 | 90 | 16 | 16 | 16 | 16 | 32 | 32 |
| 5 | 650 | 130 | 21 | 21 | 21 | 21 | 42 | 42 |
| 10 | 900 | 180 | 28 | 28 | 28 | 28 | 55 | 55 |
| 20 | 1400 | 280 | 40 | 40 | 40 | 40 | 80 | 80 |
| 50 | 2900 | 580 | 77 | 77 | 77 | 77 | 155 | 155 |
| 99 | 5350 | 1070 | 138 | 138 | 138 | 138 | 277 | 277 |

### 46.5.1. Skill Learnings for Summoner

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Sentinel Totem** (ID: 102) | Innate class technique |

## 46.6. Class 5: Hero

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 300 | 90 | 16 | 16 | 16 | 16 | 32 | 32 |
| 5 | 507 | 130 | 21 | 21 | 21 | 21 | 42 | 42 |
| 10 | 764 | 180 | 28 | 28 | 28 | 28 | 55 | 55 |
| 20 | 1280 | 280 | 40 | 40 | 40 | 40 | 80 | 80 |
| 50 | 2825 | 580 | 77 | 77 | 77 | 77 | 155 | 155 |
| 99 | 5350 | 1070 | 138 | 138 | 138 | 138 | 277 | 277 |

### 46.6.1. Skill Learnings for Hero

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Heal** (`ID: 72`) | Innate class technique |
| Lv 5 | **Repulsive Blow** (`ID: 11`) | Innate class technique |

## 46.7. Class 6: Thief

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 450 | 90 | 16 | 16 | 16 | 16 | 32 | 32 |
| 5 | 650 | 130 | 21 | 21 | 21 | 21 | 42 | 42 |
| 10 | 900 | 180 | 28 | 28 | 28 | 28 | 55 | 55 |
| 20 | 1400 | 280 | 40 | 40 | 40 | 40 | 80 | 80 |
| 50 | 2900 | 580 | 77 | 77 | 77 | 77 | 155 | 155 |
| 99 | 5350 | 1070 | 138 | 138 | 138 | 138 | 277 | 277 |

## 46.8. Class 7: Start

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 999 | 1 | 999 | 1 | 1 |
| 5 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |
| 10 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |
| 20 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |
| 50 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |
| 99 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |

## 46.9. Class 10: ---Special---

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 450 | 90 | 16 | 16 | 16 | 16 | 32 | 32 |

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 5 | 650 | 130 | 21 | 21 | 21 | 21 | 42 | 42 |
| 10 | 900 | 180 | 28 | 28 | 28 | 28 | 55 | 55 |
| 20 | 1400 | 280 | 40 | 40 | 40 | 40 | 80 | 80 |
| 50 | 2900 | 580 | 77 | 77 | 77 | 77 | 155 | 155 |
| 99 | 5350 | 1070 | 138 | 138 | 138 | 138 | 277 | 277 |

## 46.10. Class 11: Guard

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 100 | 30 | 12 | 8 | 3 | 8 | 27 | 32 |
| 5 | 145 | 46 | 17 | 12 | 5 | 11 | 27 | 42 |
| 10 | 202 | 64 | 23 | 17 | 8 | 15 | 27 | 55 |
| 20 | 314 | 102 | 35 | 26 | 13 | 22 | 27 | 80 |
| 50 | 650 | 215 | 71 | 54 | 27 | 44 | 27 | 155 |
| 99 | 1200 | 400 | 130 | 100 | 50 | 80 | 27 | 277 |

### 46.10.1. Skill Learnings for Guard

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Heavy Hit** (ID: 15) | Innate class technique |

## 46.11. Class 12: Wind Spirit

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 50 | 1 | 1 | 1 | 1 | 1 | 1 |
| 5 | 1 | 79 | 1 | 1 | 1 | 1 | 1 | 1 |
| 10 | 1 | 115 | 1 | 1 | 1 | 1 | 1 | 1 |
| 20 | 1 | 186 | 1 | 1 | 1 | 1 | 1 | 1 |
| 50 | 1 | 400 | 1 | 1 | 1 | 1 | 1 | 1 |
| 99 | 1 | 750 | 1 | 1 | 1 | 1 | 1 | 1 |

### 46.11.1. Skill Learnings for Wind Spirit

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Air Compression** (ID: 132) | Innate class technique |
| Lv 1 | **Great Wind** (ID: 133) | Innate class technique |
| Lv 1 | **Transposition** (ID: 134) | Innate class technique |

## 46.12. Class 13: Pixie

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 50 | 1 | 1 | 1 | 1 | 1 | 1 |
| 5 | 1 | 79 | 1 | 1 | 1 | 1 | 1 | 1 |
| 10 | 1 | 115 | 1 | 1 | 1 | 1 | 1 | 1 |
| 20 | 1 | 186 | 1 | 1 | 1 | 1 | 1 | 1 |
| 50 | 1 | 400 | 1 | 1 | 1 | 1 | 1 | 1 |
| 99 | 1 | 750 | 1 | 1 | 1 | 1 | 1 | 1 |

### 46.12.1. Skill Learnings for Pixie

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Healing Wave** (`ID: 135`) | Innate class technique |

## 46.13. Class 14: Fire Chicken

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 5 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 10 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 20 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 50 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 99 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

### 46.13.1. Skill Learnings for Fire Chicken

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Bomb** (`ID: 136`) | Innate class technique |

# Chapter 47. Tactical Battle System (LeTBS) Specification (2017–2018 Demo)

**Source File**: `docs/game_design/demo_baseline/combat_and_letbs_mechanics.adoc`

## 47.1. Overview

This document formalizes the grid-based Tactical Strategy RPG combat architecture implemented via Lecode's Tactical Battle System (LeTBS) in the 2017–2018 demo. It details movement costs, action economy, scope geometries, terrain manipulation, and turn manipulation invariants.

## 47.2. Battle Field Architecture & Action Economy

- **Grid Structure**: 2D tile-based tactical grid mapped over encounter maps.
- **Movement Points (MP)**: Consumable per-turn budget determining movement range:
  - Baseline Protagonists & Allies: `move_points: 4`.
  - Heavy Impact Reductions: Heavy Hit reduces target MP by 3 (`move_points+: -3`).
  - Outrider Haste: Light Steps increases target MP by 1 (`move_points+: 1`).
- **Turn Timeline**: Monotonic timeline ordering turns by entity agility (`auto_turn_order_face`).
- **Directional Facing**: Facing impacts defense and positional abilities (e.g., **Phantom Slash** teleports directly behind the target before striking).

## 47.3. Scope & Area of Effect (AOE) Geometries

The demo configured rich geometric targeting parameters across skills:

| Scope Pattern | Example Skill | Geometric Definition |
|---|---|---|
| `circle(r)` | **Fire** (`Skill 42`), **Heal** (`Skill 72`) | Manhattan or Chebyshev circle of radius `r` centered on the caster. |
| `circle(max_r, min_r)` | **Ghost Arrow** (`Skill 79`), **Dagger Throw** (`Skill 169`) | Annular targeting ring; cannot target adjacent cells within `min_r`. |
| `line(n)` | **Explosive Mark** (`Skill 76`), **Great Wind** (`Skill 133`) | Cardinal linear projection up to `n` cells from caster. |
| `square(r)` | **Angela Move Scope** (`Actor 5`) | Full square grid extent of radius `r`. |
| `custom(pattern)` | **Fire Arrows** (`Skill 82`) | Scripted multi-target projectile pattern hitting all visible adversaries. |
| `through_obstacles` | **Teleportation** (`Skill 73`) | Ignores intermediate terrain obstacles, requiring only a free destination cell. |

## 47.4. Terrain State Modifications & Dynamic Hazards

Combat encounters feature active environmental manipulation:

- **Volcano Ground (`Skill 47`)**: Lays hazardous fiery embers across multiple ground cells dealing continuous fire damage.

- **Ice Wall / Ice Prison (`Skill 51`, `Skill 52`)**: Summons physical destructible barricades (`HP: 20`) that obstruct enemy line-of-sight and movement corridors.
- **Explosive Mark & Magic Trap (`Skill 76`, `Skill 80`)**: Proximity-triggered grid landmines that detonate upon enemy entry.
- **Burning Tile Spread (`State 110`)**: Dynamic fire propagation expanding with `spread: 100% circle(1) all`.

## 47.5. Turn Timeline Manipulation

Unique priest techniques alter the flow of tactical battle:

- **Accelerate (`Skill 87`)**: Advances target position along the turn timeline by 100%.
- **Decelerate (`Skill 88`)**: Delays target position along the turn timeline by 100%.
- **Emergency Turn (`Skill 89`)**: Instantly grants an immediate action turn to an allied battler.
- **Extra Turn (`Skill 86`)**: Grants the caster or target an additional action cycle upon turn conclusion.

# Chapter 48. Dataset Evolution & Delta Analysis (Folder 1 vs Folder 2)

**Source File**: `docs/game_design/demo_baseline/dataset_evolution_delta.adoc`

## 48.1. Overview

This document provides a comparative delta between `Old Game Data` (Folder 1) and `Old_Game_Data_2/data` (Folder 2), detailing the evolution from narrative prototype to advanced LeTBS tactical simulation, and establishing modern C11 reconciliation strategies.

## 48.2. Dataset Characteristics & Metrics

| Subsystem Metric | Folder 1 (Old Game Data) | Folder 2 (Old_Game_Data_2/data) |
|---|---|---|
| Game Title | `Tenebris and Lux` | `Tenebris & Lux` |
| Version Identifier | `40730002` | `2078310` |
| Map Count | 36 maps | 43 maps |
| Active Player Actors | 2 (Elijah, Katrina) | 5 (Elijah, Katrina, Sebastien, Angela, Lucius) |
| Enemies Defined | 21 enemies | 22 enemies |
| States Defined | 11 states | 30 states (extensive LeTBS configurations) |
| Morality Engine Items | Compass of Morality, Ring of Fire | Omitted / shifted to LeTBS combat |

## 48.3. Core Differences & Evolution Strategy

1. **Narrative vs. Tactical Pivot**:
   - **Folder 1** represents the **narrative-complete** edition of the demo. It contains all dialogue branches, the Dio Village moral tests (Eric's Honesty Pot, John the Thief vs St. Baron), and key relics like the **Compass of Morality**.
   - **Folder 2** represents the **tactical engine expansion**. It introduces full party combat with Angela, Sebastien, and Lucius, 6 extra dungeon maps, and rich LeTBS states (e.g. **Burning Tile Spread**, **Charging**, **Intimidating Aura**).
2. **C11 Reconstruction Recommendation**:
   - **Layer 1 Simulation Engine**: Combine the narrative quest topology and moral continuum from Folder 1 with the rich 5-character tactical class abilities and LeTBS movement/scope geometry from Folder 2.
   - **Layer 2 Presentation**: Maintain the authentic dialog scripts, dual-protagonist branch voices, and environmental puzzles.

# Chapter 49. Grimoire & Complete Skill Compendium (2017–2018 Demo)

**Source File**: docs/game_design/demo_baseline/grimoire_and_skills.adoc

## 49.1. Overview

This document records the complete grimoire of all 75+ active techniques, spells, and summon abilities defined in Skills.json. It provides mathematical damage formulas, MP/TP costs, scope parameters, and tactical LeTBS sequences.

## 49.2. Basic Martial Maneuvers

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 1 | **Attack** | 0 | 0 | a.atk * 4 - b.def * 2 | + *Note*: Skill #1 will be used when you select the Attack command. |
| 2 | **Guard** | 0 | 0 | 0 | + *Note*: Skill #2 will be used when you select the Guard command. |
| 3 | **Dual Attack** | 0 | 0 | a.atk * 4 - b.def * 2 | |
| 4 | **Double Attack** | 0 | 0 | a.atk * 4 - b.def * 2 | |
| 5 | **Triple Attack** | 0 | 0 | a.atk * 4 - b.def * 2 | |
| 6 | **Escape** | 0 | 0 | 0 | |
| 7 | **Wait** | 0 | 0 | 0 | |

## 49.3. Knight Martial Techniques

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 11 | **Repulsive Blow** | 0 | 0 | this.phyDmg(120) | Hits and push the target for two cell. *Note*: <letbs> scope: circle(1) sequence: atk_push(2) scope_color: #FFF </letbs> <l… |
| 12 | **Flash Jump** | 0 | 0 | this.phyDmg(110) + this.magDmg(10) | Jumps to an available position in the scope and invoke a thunder to damage nearby fighters. *Note*: <letbs> scope: circle(3,1) aoe: circle(1) scope_options: need_free_cell aoe_o… |
| 13 | **Rush** | 0 | 0 | this.phyDmg(80) | Rushes toward the target. *Note*: <letbs> scope: custom(rush) sequence: rush </letbs> <letbs_AI> type: offense… |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 14 | **Intimidating Aura** | 0 | 0 | `(a.atk * 4 - b.def * 2) * 1.2` | Casts an aura that reduce enemies' ATK by 15%. Grants as well 70% counter attack chance. *Note*: `<letbs> scope: circle(0) scope_options: include_center </letbs> <letbs_AI> t…` |
| 15 | **Heavy Hit** | 0 | 0 | `this.phyDmg(135)` | Hits the target and remove 3 move points. *Note*: `<letbs> scope: circle(1) sequence: atk </letbs> <Learn Require Level: 5> <l…` |
| 16 | **Provoke** | 0 | 0 | `(a.atk * 4 - b.def * 2) * 1.2` | Draw the attention of enemies upon the user. Can use another action after this skill. *Note*: `<letbs> scope: circle(0) scope_options: include_center extra_action </letbs> …` |
| 17 | **Phantom Slash** | 0 | 0 | `this.phyDmg(60)` | Use dark arts to reach the back of the target. *Note*: `<letbs> scope: line(4) sequence: phantom_slash </letbs> <Learn Require Level…` |

## 49.4. Mage Elemental Spells

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 42 | **Fire** | 0 | 0 | `this.magDmg(105)` | Deals fire damages to a single target in a medium range. *Note*: `<letbs> scope: circle(3);line(4) </letbs> <letbs_AI> type: offense </letbs_A…` |
| 43 | **Sparkle** | 0 | 0 | `this.magDmg(80)` | Deals thunder damages to targets in a small area. *Note*: `<letbs> scope: circle(4) aoe: circle(1) sequence: spark </letbs> <Learn Requi…` |
| 44 | **Lightning Storm** | 0 | 0 | `this.magDmg(60)` | Attacks 3 times a random battler in the area. *Note*: `<letbs> scope: line(4) aoe: circle(2) sequence: lightning_storm </letbs> <Lea…` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 45 | **Fire Storm** | 0 | 0 | `this.magDmg(200)` | Attacks 6 times randomly in the area. *Note*: `<letbs> scope: circle(4,2) aoe: square(1) sequence: fire_storm </letbs> <Lear…` |
| 46 | **Fire Support** | 0 | 0 | `this.magDmg(80)` | Deals fire damages to a single target in a medium range with the help of close allies. *Note*: `<letbs> scope: circle(4) sequence: fire_support </letbs> <Learn Require Level…` |
| 47 | **Volcano Ground** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*2` | Lays multiple fire spots on the ground. *Note*: `<letbs> scope: circle(0) scope_options: include_center aoe: custom(volcano_gr…` |
| 48 | **Volcano Ground (dmg)** | 0 | 0 | `this.magDmg(25)` | |
| 49 | **Ice Fury** | 0 | 0 | `this.magDmg(115)` | Deals ice damages in a row. *Note*: `<letbs> scope: line(4) aoe: custom(ice_fury) sequence: skill </letbs> <Learn …` |
| 50 | **Walking Fire** | 0 | 0 | `this.magDmg(105)` | Deals fire damages in a medium path. *Note*: `<letbs> scope: circle(4) scope_options:through_obstacles aoe: path sequence: …` |
| 51 | **Ice Wall** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a wall of ice. *Note*: `<letbs> scope: line(3) scope_options: need_free_cell aoe: custom(ice_wall) ao…` |
| 52 | **Ice Prison** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a prison of ice. *Note*: `<letbs> scope: circle(4) aoe: custom(ice_prison) aoe_options: need_free_cell …` |
| 53 | **Whitch's Escape** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.20` | Teleports then casts Spark at the old location. The scope is based on current move points. *Note*: `<letbs> scope: circle(e.getMovePoints()+1) scope_options: need_free_cell sequ…` |
| 54 | **Fire Chickens** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon explosive chickens *Note*: `<letbs> scope: circle(2) aoe: circle(1) scope_options: need_free_cell aoe_opt…` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 55 | **Water (charged)** | 0 | 0 | `this.magDmg(135)` | + *Note*: `<letbs> scope: circle(4) sequence: map_skill charge: 2, 261 </letbs> <Learn R⋯` |
| 56 | **Mega Sparkle** | 0 | 0 | `this.magDmg(60)` | Deals thunder damage on 3 different areas. *Note*: `<letbs> scope: circle(4) aoe: circle(1) sequence: mega_spark 3 selections: di⋯` |
| 57 | **Multi Fire** | 0 | 0 | `this.magDmg(105)` | Invoke 3 columns of fire. *Note*: `<letbs> scope_select: enemies scope: circle(3); 2 selections: append; Select ⋯` |

## 49.5. Priest Holy Arts & Turn Manipulation

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 72 | **Heal** | 0 | 0 | `50 + a.mat` | Heals a single target and ends the turn. *Note*: `<letbs> scope: circle(4,2) sequence: heal pass_on_use scope_options: include_⋯` |
| 73 | **Teleportation** | 0 | 0 | `0` | Teleports the caster in a medium range. *Note*: `<letbs> scope: circle(4) sequence: teleportation scope_options:through_obstac⋯` |
| 74 | **Salvation** | 0 | 0 | `25 + a.mat` | Attracts allies and heals them in all directions. *Note*: `<letbs> scope: circle(0) aoe: line(3) sequence: salvation scope_options: incl⋯` |
| 75 | **Teleportation II** | 0 | 0 | `0` | Teleports the battler in front of the caster in a medium range. *Note*: `<letbs> scope: circle(3) sequence: teleportation_sup scope_options: through_o⋯` |
| 76 | **Explosive Mark** | 0 | 0 | `None` | Lays a mark on the ground that deals fire damages to enemies on it. *Note*: `<letbs> scope: line(5) aoe: square(1) sequence: explosive_mark </letbs> <Lear⋯` |
| 77 | **Explosive Mark (dmg)** | 0 | 0 | `this.magDmg(50)` | |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 78 | **Light Steps** | 0 | 0 | `50 + a.mat` | Increases target's move points for 2 turns. *Note*: `<letbs> scope_options: include_center </letbs> <Learn Require Level: 5> <letb…` |
| 79 | **Ghost Arrow** | 0 | 0 | `this.magDmg(75)` | Deals light damages to a single target in a medium range, ignoring any obstacle, and remove one move point. *Note*: `<letbs> scope: circle(5,2) sequence: ghost_arrow scope_options: through_obsta…` |
| 80 | **Magic Trap** | 0 | 0 | `None` | Lays a magic trap who deals light damage while blinding the target with 60% rate. *Note*: `<letbs> scope: circle(5) scope_options:through_obstacles, need_free_cell aoe:…` |
| 81 | **Magic Trap (dmg)** | 0 | 0 | `this.magDmg(160)` | |
| 82 | **Fire Arrows** | 0 | 0 | `this.magDmg(105)` | Launch a fire arrow for every enemy in the scope. *Note*: `<letbs> scope: line(2) aoe: custom(fire_arrows) aoe_options: line_of_sight se…` |
| 83 | **Wind Spirit** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a wind spirit that help the party with the power of wind. *Note*: `<letbs> scope: circle(2) scope_options: need_free_cell sequence: summon(wind_…` |
| 84 | **Summon Pixie** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a pixie that can heal the party. *Note*: `<letbs> scope: circle(1) scope_options: need_free_cell </letbs> <Learn Requir…` |
| 85 | **Shamanism** | 0 | 0 | `a.mhp * 0.1` | All allies recover 10% of their MHP for each dead battler in the area. *Note*: `<letbs> sequence: shamanism scope: circle(3) aoe: circle(2) scope_options: in…` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 86 | **Extra Turn** | 0 | 0 | `50 + a.mat` | Add an extra turn. *Note*: `<letbs> scope: circle(2) scope_options: include_center extra_turn: 100% </let⋯` |
| 87 | **Accelerate** | 0 | 0 | `50 + a.mat` | Accelerate the turn order of the target. *Note*: `<letbs> scope: circle(2) scope_options: include_center accelerate_turn: 100% ⋯` |
| 88 | **Decelerate** | 0 | 0 | `50 + a.mat` | Accelerate the turn order of the target. *Note*: `<letbs> scope: circle(2) scope_options: include_center decelerate_turn: 100% ⋯` |
| 89 | **Emergency Turn** | 0 | 0 | `50 + a.mat` | Give an instant turn to the target. *Note*: `<letbs> scope: circle(3) pass_on_use scope_options: include_center instant_tu⋯` |
| 90 | **Ally Teleport.** | 0 | 0 | `0` | Teleports an ally in a medium range. *Note*: `<letbs> scope_select: allies scope: circle(4) sequence: ally_teleport scope_o⋯` |
| 91 | **Ally Teleport. (Scope)** | 0 | 0 | `0` | + *Note*: `<letbs> scope: circle(4) scope_options:through_obstacles, need_free_cell </le⋯` |
| 92 | **Mana Ball** | 0 | 0 | `this.magDmg(80)` | Launch a bouncing mana ball *Note*: `<letbs> scope: circle(3) sequence: mana_ball </letbs> <Learn Require Level: 5⋯` |

## 49.6. Summoner Totems & Rites

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 102 | **Sentinel Totem** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a totem that strike enemies as soon as they enter the range of action. *Note*: `<letbs> scope: circle(3) scope_options: need_free_cell sequence: summon(senti⋯` |
| 103 | **Transform.** | 0 | 0 | `50 + a.mat` | + *Note*: `<letbs> scope: circle(0) scope_options: include_center </letbs> <letbs_AI> ty⋯` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 104 | **Devour** | 0 | 0 | `this.magDmg(105)` | + *Note*: `<letbs> scope: circle(1) fusion_effect </letbs> <letbs_AI> type: offense </le…` |

## 49.7. Summon Beast Techniques

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 132 | **Air Compression** | 0 | 0 | `this.magDmg(135)` | Deals wind damages in a medium range with 15% of chance to blind the target. *Note*: `<letbs> scope: circle(4) </letbs> <letbs_AI> type: offense </letbs_AI> <Learn…` |
| 133 | **Great Wind** | 0 | 0 | `this.magDmg(140)` | Deals wind damages and push the target. The collision damage of this skill is greatly increased. *Note*: `<letbs> scope: line(4) sequence: great_wind collision_formula: 2 * b.mhp * 0….` |
| 134 | **Transposition** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.35` | Switch place with a target in a medium range. *Note*: `<letbs> scope: circle(6) sequence: transposition scope_options:through_obstac…` |
| 135 | **Healing Wave** | 0 | 0 | `10 + a.mat` | Heals in small area *Note*: `<letbs> scope: circle(3) aoe: circle(1) scope_options: include_center sequenc…` |
| 136 | **Bomb** | 4 | 0 | `a.mhp` | + *Note*: `<letbs> scope: circle(1) </letbs> <letbs_AI> type: offense </letbs_AI> <letbs…` |
| 137 | **Fire Arrow** | 0 | 0 | `this.phyDmg(100) + this.magDmg(100)` | Launch a fire arrow. *Note*: `<letbs> scope: circle(3) sequence: projectile(fire_arrow) </letbs> <Learn Req…` |

## 49.8. Enemy Bestial & Void Techniques

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 162 | **Poisonous Bite** | 3 | 0 | `(a.atk * 4 - b.def * 2)*0.8` | + *Note*: `<letbs> scope: circle(1) </letbs> <letbs_AI> type: offense </letbs_AI>` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 163 | **Fire Surge** | 4 | 0 | `this.magDmg(105)` | Deals fire damages to a single target in a medium range. *Note*: `<letbs> scope: circle(4) </letbs> <letbs_AI> type: offense </letbs_AI>` |
| 164 | **Summon Pyra** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | + *Note*: `<letbs> scope: circle(1) scope_options: need_free_cell sequence: summon_pyra` … |
| 165 | **Evil Healing** | 0 | 0 | `15 + a.mat` | + *Note*: `<letbs> scope: circle(4) scope_options: include_center </letbs> <letbs_AI> ty`… |
| 166 | **Mago's Mark** | 0 | 0 | `None` | + *Note*: `<letbs> scope: line(5) aoe: square(1) sequence: mago_mark </letbs> <Cooldown:`… |
| 167 | **Evil Protection** | 0 | 0 | `15 + a.mat` | + *Note*: `<letbs> scope: circle(4) scope_options: include_center </letbs> <letbs_AI> ty`… |
| 168 | **Rabbit's Jump** | 0 | 0 | `(a.atk * 4 - b.def * 2) * 1.10 + a.mat` | + *Note*: `<letbs> scope: line(3) aoe: circle(1) scope_options: need_free_cell sequence:`… |
| 169 | **Dagger Throw** | 3 | 0 | `this.phyDmg(115)` | Throw a little but sharp dagger toward the target. *Note*: `<letbs> scope: circle(4,2) sequence: projectile(dagger) </letbs> <letbs_AI>` … |

# Chapter 50. Canonical Scenario & Narrative Walkthrough (2017–2018 Demo)

**Source File**: docs/game_design/demo_baseline/scenario_and_walkthrough.adoc

## 50.1. Overview

This document preserves the complete chronological story script, scene transitions, puzzle structures, and verbatim dialogues of the 2017–2018 demo. It documents both the shared story progression and the distinct dual-protagonist narrative branches between **Elijah Ro-enheart** (\n[1]) and **Lady Katrina Jaren** (\n[2]).

## 50.2. Act I: The Castle Prophecy & Dark Ambition

### 50.2.1. Start-Up / Walk of Fear

Roté's urgent arrival as dark omens rise.

**Associated Maps**: Map 002

#### Map 002: Start-Up

- **Speaker/Source: `The Walk of Fear `**

```
\n<Roté>It has been years since I was needed here.<br>The king...<br>The king must know of these strange events!
```

### 50.2.2. The Castle (Outside)

Roté observes the rising island and eerie storm prior to the Festival.

**Associated Maps**: Map 003, Map 008

#### Map 003: The Castle (Outside)

- **Speaker/Source:** Roté Observation

```
\n<Roté>The storm... The rising island...
All of these events...
All before the Festival of Heroes.
The king will know how to stop this, he must know...
```

#### Map 008: Battle Tutorial

- **Speaker/Source:** Dream Eaters

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
```

```
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

### 50.2.3. The Castle (Inside) - The Throne Room

Roté warns King Jihad; King Jihad's treasonous conspiracy against the Dark Lord.

**Associated Maps**: Map 004, Map 009

#### Map 004: The Castle (Inside)

- **Speaker/Source:** `The King`

```
\n<D.Jihad>The fools...that power...that wicked power!
Soon I will rule over this land, and not even the Dark Lord
will have power over me!
```

- **Speaker/Source:** `The Meeting`

```
\n<Roté>King Jihad, I am here to inform you today...
It's here...
The darkness... it has returned
\n<Jihad>...Thank you Roté, what would I do without your
```

```
careful watch...
\n<Jihad>Gather up the guard! We need to prepare!
Darkness will soon return to this land. And don't tell the people of
this discovery...
\n<Roté>Y...Y...Yes my king...
\n<Jihad>You all are dismissed.
Yes my king.
```

### Map 009: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

# 50.3. Act II: The Dream Realm & The Ancient Archives

## 50.3.1. Intro / The Void

The cosmic darkness and threshold.

**Associated Maps**: Map 001, Map 002

### Map 002: Start-Up

- **Speaker/Source: `The Walk of Fear `**

```
\n<Roté>It has been years since I was needed here.<br>The king...<br>The king must know of these strange events!
```

## 50.3.2. Character Select & The Room of Sages

Meeting Chrono; selecting archetype crystals; reviewing ancient archives.

**Associated Maps**: Map 006, Map 011

### Map 006: Character Select

- **Speaker/Source:** `Chrono`

```
\n<???>So...a new hero is to be chosen.
I have seen this when the sages spoke of the darkness returning.
But...who did the sage's say again?
\n<???>A young man. The sages spoke
of a young man that would help us. Now, if
I just look in the archives...
\n<???>His name, his name, what was his
name?
\n<???>Ah yes! His name was [Elijah]. The
sages said that they will decide their
fate, what will they become I wonder...
Click on the objects in the room to select a class.
When you pick a class, talk to the old man once again.
\n<???>A young woman. The sages
spoke of a young woman that would help us.
Now, if I just look in the archives...
\n<???>Her name, her name, what was her
name?
\n<???>Ah yes! Her name was [Katrina]. The
sages said that they will decide their
fate, what will they become I wonder...
Click on the objects in the room to select a class.
When you pick a class, talk to the old man once again.
```

- **Speaker/Source:** `Chrono`

```
\n<???>[Elijah], did you pick your fate?
\n<???>[Katrina], did you pick your fate?
```

- **Speaker/Source:** `Chrono`

```
\n<???>That is perfect! Now, we need to
prepare...[Elijah], follow me.
\n<???>That is perfect! Now, we need to
prepare...[Katrina], follow me.
```

- **Speaker/Source:** `Warrior□s Stone`

```
The crystal began to glow
The Warrior. Tactical and strong, the warrior is known
```

```
to lead troops into victory. With various command
formation and tactics, they can turn the tide of a
battle!
Do you wish to be a Warrior?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** `Warrior□s Stone`

```
The crystal stopped glowing.
```

- **Speaker/Source:** `Mage□s Stone`

```
The crystal began to glow
The Mage. Wise and creative, they rely on the powers
of the magical arts to handle problems and to aid
others Wielding a staff and wearing cloaks and robes,
they assist in battle with magic of various elements
and keep the battle flowing.
Do you wish to be a Mage?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** `Mage□s Stone`

```
The crystal stopped glowing.
```

- **Speaker/Source:** `Hero□s Stone`

```
The crystal began to glow
The Hero. The one who's fate is to save the world from
darkness. The hero fights with valor and truth, using
various skills and magic to restore balance to the
world!
Do you wish to be the Hero?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** `Hero□s Stone`

```
The crystal stopped glowing.
```

- **Speaker/Source:** `Thief□s Stone`

```
The crystal began to glow
The Thief. Sly and lurking in the shadows, the thief
takes what they want in order to survive. They are
great at setting up traps and gathering information
that everyone can't find on their own.
Do you wish to be a Thief?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** `Thief□s Stone`

```
The crystal stopped glowing.
```

- **Speaker/Source:** `Priest□s Stone`

```
The crystal began to glow
The Priest. With the power of the church and holy
intervention, the priest using holy magic to purify
lands, obejcts, and even people. They also use their
magic to heal and protect others.
```

```
Do you wish to be a Priest?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Priest□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Hero□s Stone

```
The crystal began to glow
The Summoner. A being of many mystery. With the great
ablilty to summon beings from other worlds, the
Summoner does his best to maintain a balance between
the worlds and often knows that world's culture
Do you wish to be the Hero?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Hero□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** `Block`

```
You're not done here.
```

- **Speaker/Source:** `Block`

```
You're not done here.
```

- **Speaker/Source:** `Block`

```
You're not done here.
```

- **Speaker/Source:** `Ancient Book`

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** `Ancient Book`

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

# 50.4. Act III: The Trial of Mechanisms & The Dream Eaters

## 50.4.1. The Switches Puzzle Chamber

Boulders, pressure plates, jumping ledges with 'Z', and knowledge gates.

**Associated Maps**: Map 007, Map 040

### Map 007: Switches

- **Speaker/Source:** `Special Room Switch`

```
Press the button?
The strange stones to the big room was removed!
```

- **Speaker/Source:** `Bonus Gold`

```
100\G were found!
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Stone`

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** `Chrono`

```
\n<???>Come, we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
```

```
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Chrono

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
By clicking Z, you can jump across certain ledges.
```

### Map 040: Switches

- **Speaker/Source:** Special Room Switch

```
Press the button?
The strange stones to the big room was removed!
```

- **Speaker/Source:** Bonus Gold

```
100\G were found!
```

- **Speaker/Source:** Restart Switch

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Stone`

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** `Chrono`

```
\n<???>Come [Elijah], we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Chrono`

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
By clicking Z, you can jump across certain ledges.
```

### 50.4.2. The Secret Crystal Chamber

Elemental alignment (Blue, Yellow, Green, Red) to unlock the Heart Shield.

**Associated Maps**: Map 009, Map 039

### Map 009: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

### Map 039: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

```
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

### 50.4.3. The Battle Tutorial & Dream Eater Ambush

Dream illusion shatters; Dream Eaters assault; Chrono passes out; 'AWAKEN!'

**Associated Maps**: Map 008, Map 041, Map 012, Map 042

#### Map 008: Battle Tutorial

- **Speaker/Source:** `Dream Eaters`

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
```

```
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

### Map 041: Battle Tutorial

- **Speaker/Source:** `Dream Eaters`

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
```

```
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

# 50.5. Act IV: Awakening in Dio Village & Family Roots

## 50.5.1. Grandma's House Upstairs

Awakening from the dream; opening Grandma's chest for Grandma's Watch & Compass of Morality.

**Associated Maps**: Map 020, Map 004

### Map 020: Grandma's House Upstairs

- **Speaker/Source:** `Grandma□s Chest`

```
\n<[Elijah]>I shouldn't go in this chest. I
need to talk to Grandma first.
She's very particular about her things.
\n<[Katrina]>I shouldn't go in this chest. I
need to talk to Grandma first.
Batty old woman seems to remind that chest.
```

- **Speaker/Source:** Grandma□s Chest

```
An old watch was in the chest.
\n<[Elijah]>Oh.. it's an old watch.
What's so special about it anyway?
\n<[Katrina]>She made me....grab...an
OLD WATCH! I thought it was
something important like her will!
```

- **Speaker/Source:** Grandma□s Chest

```
There is still something inside of the chest
OBTAINED THE COMPASS OF MORALITY
```

- **Speaker/Source:** Grandma□s Chest

```
It's all empty now.
```

- **Speaker/Source:** Wierd Dream

```
\n<[Elijah]>That dream....what did it mean...
It could just be my imagination but...It
felt like...something more. A hero
... Me...?
\n<[Katrina]>WHAT! WHAT THE...ME. Man, my mind
needs to get out of lala land and settle
into reality. I can't be a hero,
I won't be a hero...
```

### Map 004: The Castle (Inside)

- **Speaker/Source:** The King

```
\n<D.Jihad>The fools...that power...that wicked power!
Soon I will rule over this land, and not even the Dark Lord
will have power over me!
```

- **Speaker/Source:** The Meeting

```
\n<Roté>King Jihad, I am here to inform you today...
It's here...
The darkness... it has returned
\n<Jihad>...Thank you Roté, what would I do without your
careful watch...
\n<Jihad>Gather up the guard! We need to prepare!
Darkness will soon return to this land. And don't tell the people of
this discovery...
\n<Roté>Y...Y...Yes my king...
\n<Jihad>You all are dismissed.
Yes my king.
```

## 50.5.2. Grandma's House Downstairs

Conversation with Grandma Magolia; memories of Commander Gareth Ro-enheart; errand to Monica.

**Associated Maps**: Map 021, Map 007

### Map 021: Grandma's House Downstairs

- **Speaker/Source:** Grandma

```
\n<Grandma>Hello sweetie, could you please
go upstairs and search in the chest
```

```
and grab what's inside? It's something that
has great signifinance.
\n<[Katrina]>The will. I love you gran,
but a girl gotta get out of this dead-end
town.
\n<[Elijah]>Maybe is is something
about dad! I haven't seen him since my
10th birthday.
\n<[Elijah]>Sure thing. This might
be my chance to get answers!
\n<[Katrina]>Why not! Crazy old woman...
That will is mine!
\n<[Elijah]>Well...uhh...but...
But if it is about him...
\n<[Katrina]>Gran, why can't you do it?
Why doesn't she do it?
If she wants it she can get it.
```

- **Speaker/Source:** Grandma

```
It's been a while since we've talk.
What do you want to talk about?
\n<Grandma>Oh me and Monica go way back.
She was such a wild child, always getting
into mischief. You'll love her.
\n<Grandma>This town used to be full of
life. sighs The kingdom doesn't
pay much attention to this place anymore...
\c[5]Not avalible in the demo
\n<Grandma>Have you been upstairs yet?
The chest should still be there.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>Thank you honey. This watch has
been in the family for a long time. Oh,
how it brings back memories of better days.
\n<Grandma>I think you should keep it. It's
my gift to you. And here's a bit of money.
I know that you'll need it more than little
ol' me.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>[Katrina] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>[Elijah] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>My friend, Monica,
would know what is causing these strange
rumors. Can you go to her house and see
what she has to say about all of this?
\n<[Elijah]>Sure thing. It will be nice to
finally leave the house. I might even
get to leave town!
\n<[Katrina]>Why not! I would love to~
I'm out of this poor slum of a town!
\n<[Elijah]>Well..are you sure you can't?
I don't even know her...
\n<[Katrina]>Gran, I'm not your errand girl!
 This is YOUR friend.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

### Map 007: Switches

- **Speaker/Source:** Special Room Switch

```
Press the button?
The strange stones to the big room was removed!
```

- **Speaker/Source:** Bonus Gold

```
100\G were found!
```

- **Speaker/Source:** Restart Switch

```
Restart puzzle?
```

- **Speaker/Source:** Moving Stone

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** Chrono

```
\n<???>Come, we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Chrono

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`

```
By clicking Z, you can jump across certain ledges.
```

## 50.6. Act V: Dio Village Upper & The Outskirts Crisis

### 50.6.1. Dio Village Upper / Main Square

The town in disrepair; Fighter Kid blocking the lower gate; visiting Monica.

**Associated Maps**: Map 022, Map 023, Map 010, Map 012, Map 013

#### Map 023: Main Square

- **Speaker/Source:** `Fighter Kid`

```
... He's just ignoring you
```

- **Speaker/Source:** `Fighter Kid`

```
You want me to move? I'm blocking the way for others?
Big woop! No one can stop me so I'm staying!
```

- **Speaker/Source:** `Fighter Kid`

```
You're no fun! I'm going home!
```

- **Speaker/Source:** `Mystic□s Home`

```
The door is locked.
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Monica!`

```
\n<[Elijah]>This is great! The town is still
sad with the storm but it's my home. I
need to stay focus and find Monica's house.
Grandma said her house is in front of ours.
\n<[Katrina]>Ugghh. I can't believe we still
live in this dump. Doesn't help that storm
destoryed everything. Now Gran said Monica
live in the house in front of us. Peachy...
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Leave Town`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

### 50.6.2. Monica's House & The Mystic's Brew

Meeting Monica; the fallen soldiers totem; offering for Goddess Lyri; errand for a Potion.

**Associated Maps**: Map 027, Map 029

#### Map 027: Monica's House

- **Speaker/Source:** `Monica`

```
\n<Monica>And who in the blaze are YOU!
\n<Monica>...
...
...
\n<Monica>So you are Magolia's grandson.
She made you sound more handsome that
you are.
\n<[Elijah]>HEY! That's not a nice thing
to say!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Elijah]>Nn-...I'm here because my grandma
thinks that evil has returned.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
```

```
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>So you are Magolia's
granddaughter. She made you sound prettier
that you are.
\n<[Katrina]>LISTEN HER YOU OLD-!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Katrina]>What is wrong with you... I'm
just here because my grandma is crazy.
She's babbling about evil forces and all
that nonsense.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
```

- **Speaker/Source:** Monica

```
\n<Monica>You have to stop that kid! Our
village will be more devastated than
even with him blocking the road!
```

- **Speaker/Source:** Monica

```
\n<Monica>Good job! I think I have an idea.
Now that the people can, residences and
newcomers have arrived. I notice that a
shopkeeper has came into our town as well.
\n<Monica>I need you to go a buy a potion.
I will use this as a base for my offering
to Lyri. She'll be able to help us in this
dark hour.
Obtained the Book of Monsters
```

- **Speaker/Source:** Monica

```
\n<Monica>Oh yes. This is exactly what I
need for my offering.
\n<Monica>There is one more thing that I
need in order to make this brew suitable
for Lyri. Go out of the town to the west
path. A man has what I need. Get it please.
\n<Monica>If you want to get the potion,
you'll need to go to the lower parts of the
town.
```

- **Speaker/Source:** Monica

```
\n<Monica>Do you need something?
\n<Monica>Me? Why, I'm one of the original
mages for the King. The kingdom was not as
peaceful as is it now. But then, the
Dark Lord tried to take the kingdom.
\n<Monica>He was born from the darkness
that this world had. He planned to take
over this world and fill it with darkness.
The Hero saved this world and brought us
```

```
\n<Monica>our peace. The shield and sword.
Oh don't mind this old hag's ranmbling.
Not avalible in demo
Not avalible in demo
```

- **Speaker/Source:** The Secret Room

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** The Secret Room

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** The Secret Room

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** Lyri

```
It is the deity Lyri, the Goddess of Peace.
```

- **Speaker/Source:** Totem

```
The totem has the names of fallen soldiers. One
name has a heart by it and it reads "Miss you".
\n<[Elijah]>Did Monica have a husband?
\n<[Katrina]>Someone married that crazy bat!
```

- **Speaker/Source:** Totem

```
The totem is full of offerings.
```

### 50.6.3. The Secret Sword Cache

Discovering Medusa's Sword in the hidden glade.

**Associated Maps**: Map 024, Map 025, Map 017

#### Map 024: The Sword Map

- **Speaker/Source:** Medusa□s Sword

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

#### Map 025: The Sword Map

- **Speaker/Source:** Medusa□s Sword

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

# 50.7. Act VI: Moral Crossroads of Dio Lower Ward

## 50.7.1. Eric the Lumberjack's House

Self-Aware Pot; Test of Honesty: 10G life savings vs 500G rent donation; Ring of Fire reward.

**Associated Maps**: Map 028

### Map 028: Eric's House

- **Speaker/Source:** `Eric`

```
We've been seeing a lot of monster appear near the
recently. I wonder what's that all about. And with
jobs being tight, money is hard to come by.
```

- **Speaker/Source:** `Eric`

```
\n<[Elijah]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Elijah]>Thank you sir, but I...
\n<Eric>Don't mention it, and by the way, the name
is Eric. I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[1]>Thank you. Thank you so much!
\n<[Katrina]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Katrina]>Woah! This is a cool ring dude!
\n<Eric>The name is Eric and that ring has a story.
I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[2]>Thanks dude.
```

- **Speaker/Source:** `Eric`

```
\n<Eric>I'm sure that ring's fire power will come in
handy if you plan on leaving the village
```

- **Speaker/Source:** `Self-Aware Pot`

```
What kind of game do you think this is! You can't
just steal whatever you want! How rude!
```

- **Speaker/Source:** `Test of Honesty`

```
Something is glowing in the pot. Take the pots content?
You recieve 10G
THAT'S ALL MY MONEY! GET OUT NOW!
\n<\n1>I am so sorry! Please, I am not
trying to...
\n<n[2]>AH!
\n<n[2]>Hey, where did you even...
\n<n[2]>AH!
\n<[Elijah]>..steal all your money.
\n<n[1]>I'm sorry...
\n<n[2]>...come from...
\n<n[2]>It was just a few coins anyway.
That guy definately overreacted
You decide to leave the pot alone.
\n<n[1]>I can be quick, discret maybe
```

```
Elijah slowly reaches into the jar, stopping when
he saw how little money he had.
\n<n[1]>Maybe I should't...
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something.
\n<n[1]>Is this what the voice meant,
to be a hero...How much do you need.
Just 500G, that will be enough for me to make it for
the month.
\n<n[1]>I'll leave the money in the pot,
that way you'll have it.
Tha-tha-thank you!
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something
\n<n[1]>I have to help him somehow...
How much do you need.
Just 500G, that will be enough for me to make it for
the month
\n<n[1]>I'll leave the money in the pot,
that way you'll have it
Tha-tha-thank you!
\n<n[2]>The money is right there! This
guy won't see it coming.
As Katrina was about to take the money, she stopped
for just a moment
\n<n[2]>Ugghhh, I can't do it.
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>Is this what the voice meant,
to be a hero...How much do rent here
cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
```

```
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>I have to help him somehow...
How much do rent here cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
```

- **Speaker/Source:** `Test of Honesty`

```
Would you like to add to the pot?
You have given \v[14] to the man.
How much will you add?
You have given \v[14] to the man.
You have filled the pot! Talk to the man once more.
The pot isn't full yet.
You don't have that much to give
You leave the pot alone.
```

- **Speaker/Source:** `Test of Honesty`

```
The pot is filled with gold. It can cover rent for
2 months!
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

## 50.7.2. Shopping District & Lower Village

Buying the offering potion; the shopkeeper's note; St. Baron's locked mansion.

**Associated Maps**: Map 029, Map 030, Map 016

### Map 030: Shopping District

- **Speaker/Source:** `St. Baron□s Home`

```
\n<[Elijah]>I can't believe that I am doing
this,...the baron is the richest
man in the villiage...but he's been
using the town! This is for the good!
\n<[Katrina]>Ok...here we go...let's do this. I
have to do at least one good thing today.
\n<[Elijah]>I can believe the baron
```

```
would take all this money for himself! Not
anymore!
\n<[Elijah]>So the baron can still
keep a house while the rest of just
struggle. This is gonna be fun.
The door is locked. Mayhbe you should come back at
a different time
```

- **Speaker/Source:** `Shop Menu`

```
Welcoem to my shop! How can I help you today?
Thanks for coming! See you later!
```

- **Speaker/Source:** `Shopkeeper□s Wife Dialouge`

```
We've just got into town so we are still working
on getting the funds for a bed. But my hubby
will not rest until we have that bed!
```

- **Speaker/Source:** `Got the Potion`

```
\n<[Elijah]>I wonder why she needs a
potion. Could she be a witch?
\n<[Katrina]>I can't believe she made me
waste my money...
```

### 50.7.3. John the Thief's Hideout

John's sidequest proposal to overthrow the corrupt Baron.

**Associated Maps**: Map 031

#### Map 031: John the Thief (Sidequest)

- **Speaker/Source:** `John the Thief`

```
\n<John>What do you want kid! Can't you
see that I'm busy!
\n<John>What do you want kid! Can't you
see that I'm busy!
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>What do you want kid! Can't you
see that I'm busy!
```

## 50.8. Act VII: The Frontier, Wilderness & Final Dungeon

### 50.8.1. World Map Frontier

Demo completion gateway; thank-you message; frontier exits.

**Associated Maps**: Map 033, Map 034, Map 035, Map 005, Map 018

### 50.8.2. Wilderness Battle Zones (Demo Areas 3 & 4)

Combat zones featuring bats, slimes, magos, and bandits.

**Associated Maps**: Map 014, Map 015, Map 016, Map 034, Map 035, Map 036, Map 019, Map 020, Map 021, Map 022, Map 033

#### Map 020: Grandma's House Upstairs

- **Speaker/Source:** `Grandma□s Chest`

```
\n<[Elijah]>I shouldn't go in this chest. I
need to talk to Grandma first.
She's very particular about her things.
\n<[Katrina]>I shouldn't go in this chest. I
need to talk to Grandma first.
Batty old woman seems to remind that chest.
```

- **Speaker/Source:** `Grandma□s Chest`

```
An old watch was in the chest.
\n<[Elijah]>Oh.. it's an old watch.
What's so special about it anyway?
\n<[Katrina]>She made me....grab...an
OLD WATCH! I thought it was
something important like her will!
```

- **Speaker/Source:** `Grandma□s Chest`

```
There is still something inside of the chest
OBTAINED THE COMPASS OF MORALITY
```

- **Speaker/Source:** `Grandma□s Chest`

```
It's all empty now.
```

- **Speaker/Source:** `Wierd Dream`

```
\n<[Elijah]>That dream....what did it mean...
It could just be my imagination but...It
felt like...something more. A hero
... Me...?
\n<[Katrina]>WHAT! WHAT THE...ME. Man, my mind
needs to get out of lala land and settle
into reality. I can't be a hero,
I won't be a hero...
```

#### Map 021: Grandma's House Downstairs

- **Speaker/Source:** `Grandma`

```
\n<Grandma>Hello sweetie, could you please
go upstairs and search in the chest
and grab what's inside? It's something that
has great signifinance.
\n<[Katrina]>The will. I love you gran,
but a girl gotta get out of this dead-end
town.
\n<[Elijah]>Maybe is is something
about dad! I haven't seen him since my
10th birthday.
```

```
\n<[Elijah]>Sure thing. This might
be my chance to get answers!
\n<[Katrina]>Why not! Crazy old woman...
That will is mine!
\n<[Elijah]>Well...uhh...but...
But if it is about him...
\n<[Katrina]>Gran, why can't you do it?
Why doesn't she do it?
If she wants it she can get it.
```

- **Speaker/Source:** Grandma

```
It's been a while since we've talk.
What do you want to talk about?
\n<Grandma>Oh me and Monica go way back.
She was such a wild child, always getting
into mischief. You'll love her.
\n<Grandma>This town used to be full of
life. sighs The kingdom doesn't
pay much attention to this place anymore...
\c[5]Not avalible in the demo
\n<Grandma>Have you been upstairs yet?
The chest should still be there.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>Thank you honey. This watch has
been in the family for a long time. Oh,
how it brings back memories of better days.
\n<Grandma>I think you should keep it. It's
my gift to you. And here's a bit of money.
I know that you'll need it more than little
ol' me.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>[Katrina] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>[Elijah] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>My friend, Monica,
would know what is causing these strange
rumors. Can you go to her house and see
what she has to say about all of this?
\n<[Elijah]>Sure thing. It will be nice to
finally leave the house. I might even
get to leave town!
\n<[Katrina]>Why not! I would love to~
I'm out of this poor slum of a town!
\n<[Elijah]>Well..are you sure you can't?
I don't even know her...
\n<[Katrina]>Gran, I'm not your errand girl!
 This is YOUR friend.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** `Block`

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** `Block`

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** `Block`

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

### 50.8.3. The Final Dungeon & Boss Threshold

Multi-level dungeon fortress leading to the ultimate demo boss confrontation.

**Associated Maps**: Map 017, Map 018, Map 023, Map 024, Map 025, Map 026, Map 027, Map 028, Map 037, Map 038

#### Map 023: Main Square

- **Speaker/Source:** `Fighter Kid`

```
... He's just ignoring you
```

- **Speaker/Source:** `Fighter Kid`

```
You want me to move? I'm blocking the way for others?
Big woop! No one can stop me so I'm staying!
```

- **Speaker/Source:** `Fighter Kid`

```
You're no fun! I'm going home!
```

- **Speaker/Source:** `Mystic□s Home`

```
The door is locked.
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Lower Dio

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Lower Dio

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Lower Dio

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Lower Dio

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Monica!

```
\n<[Elijah]>This is great! The town is still
sad with the storm but it's my home. I
need to stay focus and find Monica's house.
Grandma said her house is in front of ours.
\n<[Katrina]>Ugghh. I can't believe we still
live in this dump. Doesn't help that storm
destoryed everything. Now Gran said Monica
live in the house in front of us. Peachy...
```

- **Speaker/Source:** Leave Town

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** Leave Town

```
I have to stop that kid first!
```

- **Speaker/Source:** Block

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** Block

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** Block

```
STAY AWAY FROM ME!
```

### Map 024: The Sword Map

- **Speaker/Source:** Medusa□s Sword

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

### Map 025: The Sword Map

- **Speaker/Source:** Medusa□s Sword

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

### Map 026: Start Up Area

- **Speaker/Source:** The Beginning

```
It was a month before the Festival...<br>The tower...<br>The terror...
```

- **Speaker/Source:** The Beginning

```
The darkness...<br>The darkness...<br>The darkness returned once more.
```

### Map 027: Monica’s House

- **Speaker/Source:** Monica

```
\n<Monica>And who in the blaze are YOU!
\n<Monica>...
...
...
\n<Monica>So you are Magolia's grandson.
She made you sound more handsome that
you are.
\n<[Elijah]>HEY! That's not a nice thing
to say!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Elijah]>Nn-...I'm here because my grandma
thinks that evil has returned.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
```

```
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>So you are Magolia's
granddaughter. She made you sound prettier
that you are.
\n<[Katrina]>LISTEN HER YOU OLD-!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Katrina]>What is wrong with you... I'm
just here because my grandma is crazy.
She's babbling about evil forces and all
that nonsense.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
```

- **Speaker/Source:** Monica

```
\n<Monica>You have to stop that kid! Our
village will be more devastated than
even with him blocking the road!
```

- **Speaker/Source:** Monica

```
\n<Monica>Good job! I think I have an idea.
Now that the people can, residences and
newcomers have arrived. I notice that a
shopkeeper has came into our town as well.
\n<Monica>I need you to go a buy a potion.
I will use this as a base for my offering
to Lyri. She'll be able to help us in this
dark hour.
Obtained the Book of Monsters
```

- **Speaker/Source:** Monica

```
\n<Monica>Oh yes. This is exactly what I
need for my offering.
\n<Monica>There is one more thing that I
need in order to make this brew suitable
for Lyri. Go out of the town to the west
path. A man has what I need. Get it please.
\n<Monica>If you want to get the potion,
you'll need to go to the lower parts of the
town.
```

- **Speaker/Source:** Monica

```
\n<Monica>Do you need something?
\n<Monica>Me? Why, I'm one of the original
mages for the King. The kingdom was not as
peaceful as is it now. But then, the
```

```
Dark Lord tried to take the kingdom.
\n<Monica>He was born from the darkness
that this world had. He planned to take
over this world and fill it with darkness.
The Hero saved this world and brought us
\n<Monica>our peace. The shield and sword.
Oh don't mind this old hag's ranmbling.
Not avalible in demo
Not avalible in demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `Lyri`

```
It is the deity Lyri, the Goddess of Peace.
```

- **Speaker/Source:** `Totem`

```
The totem has the names of fallen soldiers. One
name has a heart by it and it reads "Miss you".
\n<[Elijah]>Did Monica have a husband?
\n<[Katrina]>Someone married that crazy bat!
```

- **Speaker/Source:** `Totem`

```
The totem is full of offerings.
```

### Map 028: Eric's House

- **Speaker/Source:** `Eric`

```
We've been seeing a lot of monster appear near the
recently. I wonder what's that all about. And with
jobs being tight, money is hard to come by.
```

- **Speaker/Source:** `Eric`

```
\n<[Elijah]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Elijah]>Thank you sir, but I...
\n<Eric>Don't mention it, and by the way, the name
is Eric. I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[1]>Thank you. Thank you so much!
\n<[Katrina]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
```

```
this will be more useful to you than me.
\n<[Katrina]>Woah! This is a cool ring dude!
\n<Eric>The name is Eric and that ring has a story.
I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[2]>Thanks dude.
```

- **Speaker/Source:** `Eric`

```
\n<Eric>I'm sure that ring's fire power will come in
handy if you plan on leaving the village
```

- **Speaker/Source:** `Self-Aware Pot`

```
What kind of game do you think this is! You can't
just steal whatever you want! How rude!
```

- **Speaker/Source:** `Test of Honesty`

```
Something is glowing in the pot. Take the pots content?
You recieve 10G
THAT'S ALL MY MONEY! GET OUT NOW!
\n<\n1>I am so sorry! Please, I am not
trying to...
\n<n[2]>AH!
\n<n[2]>Hey, where did you even...
\n<n[2]>AH!
\n<[Elijah]>..steal all your money.
\n<n[1]>I'm sorry...
\n<n[2]>...come from...
\n<n[2]>It was just a few coins anyway.
That guy definately overreacted
You decide to leave the pot alone.
\n<n[1]>I can be quick, discret maybe
Elijah slowly reaches into the jar, stopping when
he saw how little money he had.
\n<n[1]>Maybe I should't...
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something.
\n<n[1]>Is this what the voice meant,
to be a hero...How much do you need.
Just 500G, that will be enough for me to make it for
the month.
\n<n[1]>I'll leave the money in the pot,
that way you'll have it.
Tha-tha-thank you!
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
```

```
Don't worry about it, it's not for you to worry about.
Just make sure you become something
\n<n[1]>I have to help him somehow...
How much do you need.
Just 500G, that will be enough for me to make it for
the month
\n<n[1]>I'll leave the money in the pot,
that way you'll have it
Tha-tha-thank you!
\n<n[2]>The money is right there! This
guy won't see it coming.
As Katrina was about to take the money, she stopped
for just a moment
\n<n[2]>Ugghhh, I can't do it.
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>Is this what the voice meant,
to be a hero...How much do rent here
cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>I have to help him somehow...
How much do rent here cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
```

- **Speaker/Source:** `Test of Honesty`

```
Would you like to add to the pot?
You have given \v[14] to the man.
How much will you add?
You have given \v[14] to the man.
You have filled the pot! Talk to the man once more.
The pot isn't full yet.
You don't have that much to give
```

```
You leave the pot alone.
```

- **Speaker/Source:** `Test of Honesty`

```
The pot is filled with gold. It can cover rent for
2 months!
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

# Chapter 51. Tactical States & Status Ailments Directory (2017–2018 Demo)

**Source File**: `docs/game_design/demo_baseline/states_and_statuses.adoc`

## 51.1. Overview

This document catalogs all 30 status ailments, tactical combat states, aura modifiers, and battlefield terrain effects defined in `States.json` across Folder 1 and Folder 2 (LeTBS). It distinguishes generic RPG Maker MV Runtime Package (RTP) boilerplate from authentic tactical grid modifiers and stance dynamics.

## 51.2. Triage: RTP Boilerplate States vs. Authentic Tactical Grid States

| Category | State IDs | Architectural Policy |
|---|---|---|
| **Tier 1: RTP Default States** | 1–10 (Knockout, Guard, Immortal, Poison, Blind, Silence, Rage, Confusion, Fascination, Sleep) | Stock RPG Maker MV status conditions. Re-architect in C11 to tie directly into Sacyran relic affinities, posture break, and elemental resistances. |
| **Tier 3: Authentic LeTBS Grid States** | 11, 31–34, 46, 62–63, 78–79, 94–95, 109–110 | Authentic tactical modifiers altering grid movement points (MP), turn order priority, damage mitigation, and tile propagation. |

## 51.3. Standard Status Conditions (Tier 1 Boilerplate)

| ID | State Name | Priority | Restriction / Description |
|---|---|---|---|
| 1 | **Knockout** | 100 | Cannot move; unit is felled upon reaching 0 HP. |
| 2 | **Guard** | 0 | Damage taken reduced by 50%; expires at start of next turn. |
| 3 | **Immortal** | 0 | Unit cannot drop below 1 HP; used for critical NPC story anchors. |
| 4 | **Poison** | 50 | Loses 10% max HP at end of combat turn. |
| 5 | **Blind** | 60 | Hit rate reduced by 50%. |
| 6 | **Silence** | 70 | Magic skills cannot be cast. |
| 7 | **Rage** | 80 | Attacks nearest enemy or ally indiscriminately. |
| 8 | **Confusion** | 75 | Actions randomized between foes and friends. |
| 9 | **Fascination** | 70 | Charmed unit attacks allies. |
| 10 | **Sleep** | 90 | Cannot take turns; wakes up upon taking damage. |
| 11 | **Paralyze** | 85 | Stunned; skip turn (`loop_animation: 64`). |

## 51.4. Knight & Vanguard Tactical States (Tier 3 LeTBS)

| ID | State Name | Affected Stat | Tactical Mechanism & LeTBS Meta |
|---|---|---|---|
| 31 | **Intimidating Aura** | ATK -15% | Passive aura cast via Skill 14 (`auras: intimidating_aura`). Reduces adjacent enemy physical attack and grants 70% counter-attack chance. |
| 32 | **Intimidated** | ATK -15% | Status inflicted on foes caught within the Intimidating Aura boundary. |
| 33 | **Heavy Strike** | Move Points: -3 | Tactical stagger inflicted by Heavy Hit (Skill 15). Severely impedes enemy mobility on the grid. |
| 34 | **Focused** | AI Threat | Cast by Provoke (Skill 16). Forces enemy AI to prioritize the vanguard (`focused_by_ai_for_offense: 100`). |

## 51.5. Mage & Priest Arcane States (Tier 3 LeTBS)

| ID | State Name | Affected Stat | Tactical Mechanism & LeTBS Meta |
|---|---|---|---|
| 46 | **Charging** | Spell Delay | Channeling visual state for high-tier spells like Water (charged) (`loop_animation: 184`). |
| 62 | **Light Steps** | Move Points: +1 | Mobility buff cast via Skill 78. Expands grid traversal budget for outriders and rogues. |
| 63 | **Ghost Arrow** | Move Points: -1 | Tactical snare inflicted by Ghost Arrow (Skill 79). Reduces movement allowance. |

## 51.6. Summoner & Environmental Grid States (Tier 3 LeTBS)

| ID | State Name | Affected Stat | Tactical Mechanism & LeTBS Meta |
|---|---|---|---|
| 78 | **Transformation** | Form Shift | Active metamorphosis into astral beast (`character_name: Monster`). |
| 79 | **Fusion** | Unit Merge | Fuses summoner with tactical summon (`sprite_name: Rabbit`). |
| 94 | **Rabbit's Jump** | Mobility | Leap mechanics allowing clearance over terrain obstacles. |
| 95 | **Evil Protection** | DEF/MDF Buff | Damage shield cast by enemy warlocks (Skill 167). |
| 109 | **Tile Regen** | Periodic Heal | Restores health to units occupying consecrated ground. |
| 110 | **Burning** | Fire Dot / Spread | Fire hazard state (`spread: 100% circle(1)`). Expands dynamically to adjacent flammable tiles. |

## 51.7. Modern C11 Architectural Mapping

In the C11 simulation engine, tactical states are decoupled from engine internals: 1. **Zero-Heap State Mask**: Active states on an actor are tracked via a bitmask and static array (`active_states[MAX_ACTIVE_STATES]`). 2. **Move Point Clamping**: `move_points` modifiers directly feed into Layer 0 spatial pathing routines. 3. **Data-Driven Configuration**: State durations, stat multipliers, and tile spread rules are loaded from `data/states/*.json`.

# Chapter 52. Engine Systems, Switches & Variables Directory (2017–2018 Demo)

**Source File**: docs/game_design/demo_baseline/systems_switches_variables.adoc

## 52.1. Overview

This document specifies the global state machines, monotonic time clock, morality engine, and the complete registry of all switches and variables governing demo logic.

## 52.2. The Monotonic Time Engine

The time system is driven by Common Events 1 and 2, updating a 24-hour cycle: * Variable 1: Seconds (0–59) * Variable 2: Minutes (0–59) * Variable 3: Hours (1–12) * Variable 4: 24-Hour Military Format (0–23) * Variable 5: Days (1–7: Sunday through Saturday) * Variable 6: Weeks * Variable 7: Months * Variable 8: Years

In the modern C11 architecture, this corresponds to Layer 0 time_utils (time_get_monotonic_ms).

## 52.3. The Moral Sway Continuum Engine

The morality engine tracks the party's ethical orientation on an invariant scale: * Variable 9: **Moral (Overall)**: Continuum within [-100, +100]. * Variable 10: **Moral (Good)**: Cumulative points awarded for altruism, charity, and mercy. * Variable 11: **Moral (Evil)**: Cumulative points accrued for ruthless, self-serving, or thieving acts. * Item 8: **Compass of Morality**: Tool allowing the player to inspect their moral continuum in the field.

## 52.4. Global Switches Directory

| Switch ID | Name / Flag | Operational Role & Trigger |
|---|---|---|
| 1 | **Walk of Fear** | State flag for quest / scenario progression |
| 2 | **Return to Cut Scene** | State flag for quest / scenario progression |
| 3 | **Walk of Fear 2** | State flag for quest / scenario progression |
| 4 | **Roté Enters the Castle** | State flag for quest / scenario progression |
| 5 | **The Priest Leave the Castle** | State flag for quest / scenario progression |
| 6 | **D. Jihad is Summoned** | State flag for quest / scenario progression |
| 7 | **??? Enters the Map** | State flag for quest / scenario progression |
| 8 | **Class Selected** | State flag for quest / scenario progression |
| 9 | **??? Leaves the Map** | State flag for quest / scenario progression |
| 10 | **??? Explains Puzzle** | State flag for quest / scenario progression |
| 11 | **(Story Plot) Remove Rocks** | State flag for quest / scenario progression |
| 12 | **Switch Active** | State flag for quest / scenario progression |
| 13 | **(Side Plot) Remove Rocks** | State flag for quest / scenario progression |
| 14 | **??? Disappears** | State flag for quest / scenario progression |

| Switch ID | Name / Flag | Operational Role & Trigger |
|---|---|---|
| 15 | **??? Weaken** | State flag for quest / scenario progression |
| 16 | **??? Final Words** | State flag for quest / scenario progression |
| 17 | **Boss Defeated** | State flag for quest / scenario progression |
| 18 | **Restart** | State flag for quest / scenario progression |
| 19 | **Remove Stone 1** | State flag for quest / scenario progression |
| 20 | **Remove Stone 2** | State flag for quest / scenario progression |
| 21 | **Remove Stone 3** | State flag for quest / scenario progression |
| 22 | **Remove Stone 4** | State flag for quest / scenario progression |
| 23 | **Treasure Remove** | State flag for quest / scenario progression |
| 24 | **Time** | State flag for quest / scenario progression |
| 25 | **Friendship Menu** | State flag for quest / scenario progression |
| 26 | **Quest 1 (Grandma's Request)** | State flag for quest / scenario progression |
| 27 | **Grandma (Idle)** | State flag for quest / scenario progression |
| 28 | **Exit Location** | State flag for quest / scenario progression |
| 29 | **Monica Event On** | State flag for quest / scenario progression |
| 30 | **Final Conversation Before Map is Out** | State flag for quest / scenario progression |
| 31 | **Quest 1 (Give Grandma Here Watch)** | State flag for quest / scenario progression |
| 32 | **The Great Sage** | State flag for quest / scenario progression |
| 33 | **More Battle Mechanics** | State flag for quest / scenario progression |
| 34 | **Monica Quest Switch** | State flag for quest / scenario progression |
| 35 | **Unlock Dio Village-Lower** | State flag for quest / scenario progression |
| 36 | **Idle Monica** | State flag for quest / scenario progression |
| 37 | **Petty Thief** | State flag for quest / scenario progression |
| 38 | **Morality** | State flag for quest / scenario progression |
| 39 | **Give to the Poor** | State flag for quest / scenario progression |
| 40 | **A Noble Thief** | State flag for quest / scenario progression |
| 41 | **Angry John** | State flag for quest / scenario progression |

## 52.5. Global Variables Directory

| Variable ID | Name | Subsystem Allocation |
|---|---|---|
| 1 | **Seconds** | Global numerical state accumulator |
| 2 | **Minutes** | Global numerical state accumulator |
| 3 | **Hours** | Global numerical state accumulator |
| 4 | **24 Hours** | Global numerical state accumulator |
| 5 | **Days** | Global numerical state accumulator |
| 6 | **Weeks** | Global numerical state accumulator |
| 7 | **Months** | Global numerical state accumulator |
| 8 | **Years** | Global numerical state accumulator |
| 9 | **Moral (Overall)** | Global numerical state accumulator |
| 10 | **Moral (Good)** | Global numerical state accumulator |
| 11 | **Moral (Evil)** | Global numerical state accumulator |
| 12 | **Region ID** | Global numerical state accumulator |
| 13 | **Currency** | Global numerical state accumulator |
| 14 | **Pot Amount** | Global numerical state accumulator |
| 15 | **Deposit Amount** | Global numerical state accumulator |

# Volume IV: The 2017–2018 Demo Baseline & Historical Archive

Verified historical baseline extracted from the original RPG Maker MV datasets, detailing the complete scenario script, LeTBS mechanics, grimoire, and systems.

# Chapter 53. Historical Baseline Overview & Provenance

**Source File**: `docs/game_design/demo_baseline/README.adoc`

## 53.1. Overview

This documentation module establishes the comprehensive, verified baseline of the original **2017–2018 RPG Maker MV demo** for **Tenebris and Lux** ("Darkness and Light"). Extracted directly from the archival data directories (`Old Game Data` and `Old_Game_Data_2/data`), this repository anchors our modern C11 reimplementation to its authentic gameplay mechanics, narrative scripts, character arcs, and tactical systems.

## 53.2. Historical Provenance & Dataset Strata

Between 2014 and 2018, **Tenebris and Lux** was conceived as an ambitious Tactical Strategy RPG (TRPG) combining top-down exploration, puzzle mechanics, and grid-based turn strategy powered by Lecode's Tactical Battle System (LeTBS).

The historical game data is preserved across two distinct repository strata:

| Dataset | Directory Anchor | Architectural Role & Characteristics |
|---|---|---|
| **Folder 1: Primary Narrative & Morality Build** | `Old Game Data/` | Contains 37 maps, complete dual-protagonist dialogue branches, the morality engine (Compass of Morality, moral sway continuum, honesty tests), and the core Dio Village quest line (`versionId: 40730002`). |
| **Folder 2: Advanced LeTBS Tactical Build** | `Old_Game_Data_2/data/` | Contains 43 maps, extended party members (Sebastien the Thief, Angela the Mage, Lucius the Priest), advanced LeTBS action parameters, full tactical state routines, and character generator assets (`versionId: 2078310`). |

## 53.3. Documentation Taxonomy

This baseline compendium is partitioned into specialized manuals covering every facet of the historical demo:

| Document | Focus & Coverage |
|---|---|
| **[scenario_and_walkthrough.adoc](scenario_and_walkthrough.adoc)** | Chronological narrative walkthrough, scene-by-scene script, branching dialogue for Elijah vs Katrina, Map 001 through Map 043, puzzle mechanics, and transfer topologies. |
| **[cast_and_characters.adoc](cast_and_characters.adoc)** | Comprehensive character dossiers for protagonists (Elijah & Katrina), extended party members (Sebastien, Angela, Lucius), village elders (Grandma Magolia, Monica, Eric), court figures (King Jihad, Roté), and summons. |
| **[combat_and_letbs_mechanics.adoc](combat_and_letbs_mechanics.adoc)** | Grid-based battle architecture, movement point costs, directional facing, line-of-sight scopes, AOE shapes (`circle`, `line`, `square`), LeTBS action sequences, and tactical states. |
| **[classes_and_growth.adoc](classes_and_growth.adoc)** | 7 combat classes (Warrior/Knight, Mage, Priest, Summoner, Hero, Thief, Guard) plus Summons; complete stat growth matrices (Lv 1–99) and level-by-level skill learnings. |

| Document | Focus & Coverage |
| --- | --- |
| **[grimoire_and_skills.adoc](grimoire_and_skills.adoc)** | Complete grimoire of all 75+ historical spells, martial techniques, movement abilities, MP/TP costs, damage formulas, and LeTBS sequence tags. |
| **[bestiary_and_troops.adoc](bestiary_and_troops.adoc)** | Bestiary of all 22+ enemies and neutral entities (Dream Eaters, Magos, Bandits, Chests, Ice Blocks), stat parameters, AI behavior routines, and troop encounter formations. |
| **[armory_and_inventory.adoc](armory_and_inventory.adoc)** | Weapons (Medusa's Sword), Armors (Heart Shield, Ring of Fire), Items (Potions, Magic Water), and Key Lore Relics (Grandma's Watch, Compass of Morality, Book of Monsters, Book of Items). |
| **[states_and_statuses.adoc](states_and_statuses.adoc)** | 30 tactical states, LeTBS grid modifiers (Intimidating Aura, Heavy Strike, Light Steps, Burning, Tile Regen), and status ailments. |
| **[systems_switches_variables.adoc](systems_switches_variables.adoc)** | Global game engines: Monotonic Time System / Day-Night Cycle, Morality System, and an exhaustive directory of all 100+ switches and variables. |
| **[game_design_curation.adoc](game_design_curation.adoc)** | Triaging matrix: RTP defaults (discard/re-architect) vs. prototype sketches (review/evolve) vs. authentic canonical pillars (reconstruct in C11). |
| **[dataset_evolution_delta.adoc](dataset_evolution_delta.adoc)** | Side-by-side comparative delta analysis between `Old Game Data` and `Old_Game_Data_2/data`, detailing version discrepancies and modern C11 reconciliation strategies. |

## 53.4. Architectural Extraction Script

The entire documentation suite is fully reproducible from source data using our automated Python extraction pipeline:

```
python3 docs/tools/extract_demo_data.py
```

This ensures absolute fidelity to the original 2017–2018 records as we rebuild the game in native C11.

# Chapter 54. 7-Act Narrative Script & Dialogue Walkthrough

**Source File**: docs/game_design/demo_baseline/scenario_and_walkthrough.adoc

## 54.1. Overview

This document preserves the complete chronological story script, scene transitions, puzzle structures, and verbatim dialogues of the 2017–2018 demo. It documents both the shared story progression and the distinct dual-protagonist narrative branches between **Elijah Ro-enheart** (\n[1]) and **Lady Katrina Jaren** (\n[2]).

## 54.2. Act I: The Castle Prophecy & Dark Ambition

### 54.2.1. Start-Up / Walk of Fear

Roté's urgent arrival as dark omens rise.

**Associated Maps**: Map 002

#### Map 002: Start-Up

- **Speaker/Source: `The Walk of Fear `**

```
\n<Roté>It has been years since I was needed here.<br>The king...<br>The king must know of these strange events!
```

### 54.2.2. The Castle (Outside)

Roté observes the rising island and eerie storm prior to the Festival.

**Associated Maps**: Map 003, Map 008

#### Map 003: The Castle (Outside)

- **Speaker/Source:** Roté Observation

```
\n<Roté>The storm... The rising island...
All of these events...
All before the Festival of Heroes.
The king will know how to stop this, he must know...
```

#### Map 008: Battle Tutorial

- **Speaker/Source:** Dream Eaters

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

### 54.2.3. The Castle (Inside) - The Throne Room

Roté warns King Jihad; King Jihad's treasonous conspiracy against the Dark Lord.

**Associated Maps**: Map 004, Map 009

#### Map 004: The Castle (Inside)

- **Speaker/Source:** `The King`

```
\n<D.Jihad>The fools...that power...that wicked power!
Soon I will rule over this land, and not even the Dark Lord
will have power over me!
```

- **Speaker/Source:** `The Meeting`

```
\n<Roté>King Jihad, I am here to inform you today...
It's here...
The darkness... it has returned
\n<Jihad>...Thank you Roté, what would I do without your
careful watch...
\n<Jihad>Gather up the guard! We need to prepare!
Darkness will soon return to this land. And don't tell the people of
```

```
this discovery...
\n<Roté>Y...Y...Yes my king...
\n<Jihad>You all are dismissed.
Yes my king.
```

### Map 009: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

# 54.3. Act II: The Dream Realm & The Ancient Archives

## 54.3.1. Intro / The Void

The cosmic darkness and threshold.

**Associated Maps**: Map 001, Map 002

### Map 002: Start-Up

- **Speaker/Source: `The Walk of Fear `**

```
\n<Roté>It has been years since I was needed here.<br>The king...<br>The king must know of these strange events!
```

## 54.3.2. Character Select & The Room of Sages

Meeting Chrono; selecting archetype crystals; reviewing ancient archives.

**Associated Maps**: Map 006, Map 011

### Map 006: Character Select

- **Speaker/Source:** `Chrono`

```
\n<???>So...a new hero is to be chosen.
I have seen this when the sages spoke of the darkness returning.
But...who did the sage's say again?
\n<???>A young man. The sages spoke
of a young man that would help us. Now, if
I just look in the archives...
\n<???>His name, his name, what was his
name?
\n<???>Ah yes! His name was [Elijah]. The
sages said that they will decide their
fate, what will they become I wonder...
Click on the objects in the room to select a class.
When you pick a class, talk to the old man once again.
\n<???>A young woman. The sages
spoke of a young woman that would help us.
Now, if I just look in the archives...
\n<???>Her name, her name, what was her
name?
\n<???>Ah yes! Her name was [Katrina]. The
sages said that they will decide their
fate, what will they become I wonder...
Click on the objects in the room to select a class.
When you pick a class, talk to the old man once again.
```

- **Speaker/Source:** `Chrono`

```
\n<???>[Elijah], did you pick your fate?
\n<???>[Katrina], did you pick your fate?
```

- **Speaker/Source:** `Chrono`

```
\n<???>That is perfect! Now, we need to
prepare...[Elijah], follow me.
\n<???>That is perfect! Now, we need to
prepare...[Katrina], follow me.
```

- **Speaker/Source:** `Warrior□s Stone`

```
The crystal began to glow
The Warrior. Tactical and strong, the warrior is known
```

```
to lead troops into victory. With various command
formation and tactics, they can turn the tide of a
battle!
Do you wish to be a Warrior?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** `Warrior□s Stone`

```
The crystal stopped glowing.
```

- **Speaker/Source:** `Mage□s Stone`

```
The crystal began to glow
The Mage. Wise and creative, they rely on the powers
of the magical arts to handle problems and to aid
others Wielding a staff and wearing cloaks and robes,
they assist in battle with magic of various elements
and keep the battle flowing.
Do you wish to be a Mage?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** `Mage□s Stone`

```
The crystal stopped glowing.
```

- **Speaker/Source:** `Hero□s Stone`

```
The crystal began to glow
The Hero. The one who's fate is to save the world from
darkness. The hero fights with valor and truth, using
various skills and magic to restore balance to the
world!
Do you wish to be the Hero?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** `Hero□s Stone`

```
The crystal stopped glowing.
```

- **Speaker/Source:** `Thief□s Stone`

```
The crystal began to glow
The Thief. Sly and lurking in the shadows, the thief
takes what they want in order to survive. They are
great at setting up traps and gathering information
that everyone can't find on their own.
Do you wish to be a Thief?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** `Thief□s Stone`

```
The crystal stopped glowing.
```

- **Speaker/Source:** `Priest□s Stone`

```
The crystal began to glow
The Priest. With the power of the church and holy
intervention, the priest using holy magic to purify
lands, obejcts, and even people. They also use their
magic to heal and protect others.
```

```
Do you wish to be a Priest?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Priest□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Katrina]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Ancient Book

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** Hero□s Stone

```
The crystal began to glow
The Summoner. A being of many mystery. With the great
ablilty to summon beings from other worlds, the
Summoner does his best to maintain a balance between
the worlds and often knows that world's culture
Do you wish to be the Hero?
Speak to the old man once more.
Take your time. This is an important choice.
```

- **Speaker/Source:** Hero□s Stone

```
The crystal stopped glowing.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** Block

```
You're not done here.
```

- **Speaker/Source:** `Block`

```
You're not done here.
```

- **Speaker/Source:** `Block`

```
You're not done here.
```

- **Speaker/Source:** `Block`

```
You're not done here.
```

- **Speaker/Source:** `Ancient Book`

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

- **Speaker/Source:** `Ancient Book`

```
I can't read what this book says!
This book says [Elijah]. That's... me.
```

## 54.4. Act III: The Trial of Mechanisms & The Dream Eaters

### 54.4.1. The Switches Puzzle Chamber

Boulders, pressure plates, jumping ledges with 'Z', and knowledge gates.

**Associated Maps**: Map 007, Map 040

#### Map 007: Switches

- **Speaker/Source:** `Special Room Switch`

```
Press the button?
The strange stones to the big room was removed!
```

- **Speaker/Source:** `Bonus Gold`

```
100\G were found!
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Stone`

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** `Chrono`

```
\n<???>Come, we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
```

```
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Chrono

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Aura

```
By clicking Z, you can jump across certain ledges.
```

### Map 040: Switches

- **Speaker/Source:** Special Room Switch

```
Press the button?
The strange stones to the big room was removed!
```

- **Speaker/Source:** Bonus Gold

```
100\G were found!
```

- **Speaker/Source:** Restart Switch

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Stone`

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** `Chrono`

```
\n<???>Come [Elijah], we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Chrono`

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`

```
By clicking Z, you can jump across certain ledges.
```

### 54.4.2. The Secret Crystal Chamber

Elemental alignment (Blue, Yellow, Green, Red) to unlock the Heart Shield.

**Associated Maps**: Map 009, Map 039

### Map 009: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

### Map 039: Secret Map

- **Speaker/Source:** `Treasure Glow`

```
All crystals in position
The crystal begins to disappear...
All crystals are not in postition
All crystals are not in postition
All crystals are not in postition
```

```
All crystals are not in postition
```

- **Speaker/Source:** `Treasure Glow`

```
A shield with a heart-crest was in the chest
along with 100\G
```

- **Speaker/Source:** `Treasure Glow`

```
The chest is empty
```

- **Speaker/Source:** `Moving Crystal Blue`

```
The Blue Crystal is in place
```

- **Speaker/Source:** `Restart Switch`

```
Restart puzzle?
```

- **Speaker/Source:** `Moving Crystal Yellow`

```
The Yellow crystal is in place
```

- **Speaker/Source:** `Moving Crystal Green`

```
The Green Crystal is in place
```

- **Speaker/Source:** `Moving Crystal Red`

```
The Red Crystal is in place
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Find a
way to move the rocks.
```

- **Speaker/Source:** `Aura`

```
Strange magic blocks off the main crystal. Move
the outer crystals to remove the rocks. If you make a
mistake, use the red button
```

### 54.4.3. The Battle Tutorial & Dream Eater Ambush

Dream illusion shatters; Dream Eaters assault; Chrono passes out; 'AWAKEN!'

**Associated Maps**: Map 008, Map 041, Map 012, Map 042

#### Map 008: Battle Tutorial

- **Speaker/Source:** `Dream Eaters`

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
```

```
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

### Map 041: Battle Tutorial

- **Speaker/Source:** `Dream Eaters`

```
\n<[Elijah]>I have to do. If this..thing
beats us....no. I can do this!
\n<[Elijah]>That old guy might be dead
because of that...thing, but
that doesn't matter. I can't die here, not
like this, not by that thing!
The illusion of the dream breaks. Dream Eaters
reveal themselves!
\n<[Elijah]>Wh..wha...wha...what are those
things!
\n<[Katrina]>WHAT EVEN ARE YOU!
\n<Dream Eater>Foolish mortal! We are the
```

```
Dream Eaters, servants of our Dark Lord.
We will comsume, we will devour, we will
feast! Our lord commands us too!
```

- **Speaker/Source:** `Dream Eaters`

```
\n<Dream Eater>Foolish mortal. We will
return. Our lord demands it! We will
return!
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>pant pant pant
I wasn't...I wasn't ready. We cannot talk
here. You must defeat them...the
The man passes out.
\n<[Elijah]>...
\n<[Katrina]>...
```

- **Speaker/Source:** `Tutorial Complete`

```
\n<???>[Elijah] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
\n<???>[Katrina] our time here has come to an
end. If you have any business you need to
take care, take care of it. Talk to me when
you are ready.
```

- **Speaker/Source:** `Tutorial Complete`

```
Are you ready to leave?
\n<???>It is time. You must be brave, you
must be willing, and most of all...you
must trust your own path. Until we meet
again. AWAKEN!
```

- **Speaker/Source:** `Start Final Tutorial`

```
\n<[Elijah]>Wh..wha...what...no... I just
met him... mister....minster!
\n<[Katrina]>Wh..wha...what...come on old man!
You haven't told me anything!
```

# 54.5. Act IV: Awakening in Dio Village & Family Roots

## 54.5.1. Grandma's House Upstairs

Awakening from the dream; opening Grandma's chest for Grandma's Watch & Compass of Morality.

**Associated Maps**: Map 020, Map 004

### Map 020: Grandma's House Upstairs

- **Speaker/Source:** `Grandma□s Chest`

```
\n<[Elijah]>I shouldn't go in this chest. I
need to talk to Grandma first.
She's very particular about her things.
\n<[Katrina]>I shouldn't go in this chest. I
need to talk to Grandma first.
Batty old woman seems to remind that chest.
```

- **Speaker/Source:** Grandma□s Chest

```
An old watch was in the chest.
\n<[Elijah]>Oh.. it's an old watch.
What's so special about it anyway?
\n<[Katrina]>She made me....grab...an
OLD WATCH! I thought it was
something important like her will!
```

- **Speaker/Source:** Grandma□s Chest

```
There is still something inside of the chest
OBTAINED THE COMPASS OF MORALITY
```

- **Speaker/Source:** Grandma□s Chest

```
It's all empty now.
```

- **Speaker/Source:** Wierd Dream

```
\n<[Elijah]>That dream....what did it mean...
It could just be my imagination but...It
felt like...something more. A hero
... Me...?
\n<[Katrina]>WHAT! WHAT THE...ME. Man, my mind
needs to get out of lala land and settle
into reality. I can't be a hero,
I won't be a hero...
```

### Map 004: The Castle (Inside)

- **Speaker/Source:** The King

```
\n<D.Jihad>The fools...that power...that wicked power!
Soon I will rule over this land, and not even the Dark Lord
will have power over me!
```

- **Speaker/Source:** The Meeting

```
\n<Roté>King Jihad, I am here to inform you today...
It's here...
The darkness... it has returned
\n<Jihad>...Thank you Roté, what would I do without your
careful watch...
\n<Jihad>Gather up the guard! We need to prepare!
Darkness will soon return to this land. And don't tell the people of
this discovery...
\n<Roté>Y...Y...Yes my king...
\n<Jihad>You all are dismissed.
Yes my king.
```

## 54.5.2. Grandma's House Downstairs

Conversation with Grandma Magolia; memories of Commander Gareth Ro-enheart; errand to Monica.

**Associated Maps**: Map 021, Map 007

### Map 021: Grandma's House Downstairs

- **Speaker/Source:** Grandma

```
\n<Grandma>Hello sweetie, could you please
go upstairs and search in the chest
```

```
and grab what's inside? It's something that
has great signifinance.
\n<[Katrina]>The will. I love you gran,
but a girl gotta get out of this dead-end
town.
\n<[Elijah]>Maybe is is something
about dad! I haven't seen him since my
10th birthday.
\n<[Elijah]>Sure thing. This might
be my chance to get answers!
\n<[Katrina]>Why not! Crazy old woman...
That will is mine!
\n<[Elijah]>Well...uhh...but...
But if it is about him...
\n<[Katrina]>Gran, why can't you do it?
Why doesn't she do it?
If she wants it she can get it.
```

- **Speaker/Source:** Grandma

```
It's been a while since we've talk.
What do you want to talk about?
\n<Grandma>Oh me and Monica go way back.
She was such a wild child, always getting
into mischief. You'll love her.
\n<Grandma>This town used to be full of
life. sighs The kingdom doesn't
pay much attention to this place anymore...
\c[5]Not avalible in the demo
\n<Grandma>Have you been upstairs yet?
The chest should still be there.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>Thank you honey. This watch has
been in the family for a long time. Oh,
how it brings back memories of better days.
\n<Grandma>I think you should keep it. It's
my gift to you. And here's a bit of money.
I know that you'll need it more than little
ol' me.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>[Katrina] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>[Elijah] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>My friend, Monica,
would know what is causing these strange
rumors. Can you go to her house and see
what she has to say about all of this?
\n<[Elijah]>Sure thing. It will be nice to
finally leave the house. I might even
get to leave town!
\n<[Katrina]>Why not! I would love to~
I'm out of this poor slum of a town!
\n<[Elijah]>Well..are you sure you can't?
I don't even know her...
\n<[Katrina]>Gran, I'm not your errand girl!
 This is YOUR friend.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

### Map 007: Switches

- **Speaker/Source:** Special Room Switch

```
Press the button?
The strange stones to the big room was removed!
```

- **Speaker/Source:** Bonus Gold

```
100\G were found!
```

- **Speaker/Source:** Restart Switch

```
Restart puzzle?
```

- **Speaker/Source:** Moving Stone

```
The boulder is in position. The rocks move.
```

- **Speaker/Source:** Chrono

```
\n<???>Come, we have much to do.
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** Chrono

```
\n<???>Let us continue. I fear our time
here is coming to a close.
\n<???>You must move the stone so we can
access the Room of the Great Sage. It
should provide us with knowledge.
```

- **Speaker/Source:** Aura

```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```

- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`


```
Move the stone onto the green switch so
we can remove the rocks. The red switches
has their own ablities.
One red switch opens another door, the other will
fix your mistakes.
```


- **Speaker/Source:** `Aura`

```
By clicking Z, you can jump across certain ledges.
```

## 54.6. Act V: Dio Village Upper & The Outskirts Crisis

### 54.6.1. Dio Village Upper / Main Square

The town in disrepair; Fighter Kid blocking the lower gate; visiting Monica.

**Associated Maps**: Map 022, Map 023, Map 010, Map 012, Map 013

#### Map 023: Main Square

- **Speaker/Source:** `Fighter Kid`

```
... He's just ignoring you
```

- **Speaker/Source:** `Fighter Kid`

```
You want me to move? I'm blocking the way for others?
Big woop! No one can stop me so I'm staying!
```

- **Speaker/Source:** `Fighter Kid`

```
You're no fun! I'm going home!
```

- **Speaker/Source:** `Mystic□s Home`

```
The door is locked.
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Monica!`

```
\n<[Elijah]>This is great! The town is still
sad with the storm but it's my home. I
need to stay focus and find Monica's house.
Grandma said her house is in front of ours.
\n<[Katrina]>Ugghh. I can't believe we still
live in this dump. Doesn't help that storm
destoryed everything. Now Gran said Monica
live in the house in front of us. Peachy...
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Leave Town`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** `Block`

```
STAY AWAY FROM ME!
```

### 54.6.2. Monica's House & The Mystic's Brew

Meeting Monica; the fallen soldiers totem; offering for Goddess Lyri; errand for a Potion.

**Associated Maps**: Map 027, Map 029

#### Map 027: Monica's House

- **Speaker/Source:** `Monica`

```
\n<Monica>And who in the blaze are YOU!
\n<Monica>...
...
...
\n<Monica>So you are Magolia's grandson.
She made you sound more handsome that
you are.
\n<[Elijah]>HEY! That's not a nice thing
to say!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Elijah]>Nn-...I'm here because my grandma
thinks that evil has returned.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
```

```
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>So you are Magolia's
granddaughter. She made you sound prettier
that you are.
\n<[Katrina]>LISTEN HER YOU OLD-!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Katrina]>What is wrong with you... I'm
just here because my grandma is crazy.
She's babbling about evil forces and all
that nonsense.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
```

- **Speaker/Source:** Monica

```
\n<Monica>You have to stop that kid! Our
village will be more devastated than
even with him blocking the road!
```

- **Speaker/Source:** Monica

```
\n<Monica>Good job! I think I have an idea.
Now that the people can, residences and
newcomers have arrived. I notice that a
shopkeeper has came into our town as well.
\n<Monica>I need you to go a buy a potion.
I will use this as a base for my offering
to Lyri. She'll be able to help us in this
dark hour.
Obtained the Book of Monsters
```

- **Speaker/Source:** Monica

```
\n<Monica>Oh yes. This is exactly what I
need for my offering.
\n<Monica>There is one more thing that I
need in order to make this brew suitable
for Lyri. Go out of the town to the west
path. A man has what I need. Get it please.
\n<Monica>If you want to get the potion,
you'll need to go to the lower parts of the
town.
```

- **Speaker/Source:** Monica

```
\n<Monica>Do you need something?
\n<Monica>Me? Why, I'm one of the original
mages for the King. The kingdom was not as
peaceful as is it now. But then, the
Dark Lord tried to take the kingdom.
\n<Monica>He was born from the darkness
that this world had. He planned to take
over this world and fill it with darkness.
The Hero saved this world and brought us
```

```
\n<Monica>our peace. The shield and sword.
Oh don't mind this old hag's ranmbling.
Not avalible in demo
Not avalible in demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `Lyri`

```
It is the deity Lyri, the Goddess of Peace.
```

- **Speaker/Source:** `Totem`

```
The totem has the names of fallen soldiers. One
name has a heart by it and it reads "Miss you".
\n<[Elijah]>Did Monica have a husband?
\n<[Katrina]>Someone married that crazy bat!
```

- **Speaker/Source:** `Totem`

```
The totem is full of offerings.
```

### 54.6.3. The Secret Sword Cache

Discovering Medusa's Sword in the hidden glade.

**Associated Maps**: Map 024, Map 025, Map 017

#### Map 024: The Sword Map

- **Speaker/Source:** `Medusa□s Sword`

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

#### Map 025: The Sword Map

- **Speaker/Source:** `Medusa□s Sword`

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

# 54.7. Act VI: Moral Crossroads of Dio Lower Ward

## 54.7.1. Eric the Lumberjack's House

Self-Aware Pot; Test of Honesty: 10G life savings vs 500G rent donation; Ring of Fire reward.

**Associated Maps**: Map 028

### Map 028: Eric's House

- **Speaker/Source:** `Eric`

```
We've been seeing a lot of monster appear near the
recently. I wonder what's that all about. And with
jobs being tight, money is hard to come by.
```

- **Speaker/Source:** `Eric`

```
\n<[Elijah]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Elijah]>Thank you sir, but I...
\n<Eric>Don't mention it, and by the way, the name
is Eric. I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[1]>Thank you. Thank you so much!
\n<[Katrina]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Katrina]>Woah! This is a cool ring dude!
\n<Eric>The name is Eric and that ring has a story.
I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[2]>Thanks dude.
```

- **Speaker/Source:** `Eric`

```
\n<Eric>I'm sure that ring's fire power will come in
handy if you plan on leaving the village
```

- **Speaker/Source:** `Self-Aware Pot`

```
What kind of game do you think this is! You can't
just steal whatever you want! How rude!
```

- **Speaker/Source:** `Test of Honesty`

```
Something is glowing in the pot. Take the pots content?
You recieve 10G
THAT'S ALL MY MONEY! GET OUT NOW!
\n<\n1>I am so sorry! Please, I am not
trying to...
\n<n[2]>AH!
\n<n[2]>Hey, where did you even...
\n<n[2]>AH!
\n<[Elijah]>..steal all your money.
\n<n[1]>I'm sorry...
\n<n[2]>...come from...
\n<n[2]>It was just a few coins anyway.
That guy definately overreacted
You decide to leave the pot alone.
\n<n[1]>I can be quick, discret maybe
```

```
Elijah slowly reaches into the jar, stopping when
he saw how little money he had.
\n<n[1]>Maybe I should't...
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something.
\n<n[1]>Is this what the voice meant,
to be a hero...How much do you need.
Just 500G, that will be enough for me to make it for
the month.
\n<n[1]>I'll leave the money in the pot,
that way you'll have it.
Tha-tha-thank you!
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something
\n<n[1]>I have to help him somehow...
How much do you need.
Just 500G, that will be enough for me to make it for
the month
\n<n[1]>I'll leave the money in the pot,
that way you'll have it
Tha-tha-thank you!
\n<n[2]>The money is right there! This
guy won't see it coming.
As Katrina was about to take the money, she stopped
for just a moment
\n<n[2]>Ugghhh, I can't do it.
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>Is this what the voice meant,
to be a hero...How much do rent here
cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
```

```
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>I have to help him somehow...
How much do rent here cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
```

- **Speaker/Source:** `Test of Honesty`

```
Would you like to add to the pot?
You have given \v[14] to the man.
How much will you add?
You have given \v[14] to the man.
You have filled the pot! Talk to the man once more.
The pot isn't full yet.
You don't have that much to give
You leave the pot alone.
```

- **Speaker/Source:** `Test of Honesty`

```
The pot is filled with gold. It can cover rent for
2 months!
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

## 54.7.2. Shopping District & Lower Village

Buying the offering potion; the shopkeeper's note; St. Baron's locked mansion.

**Associated Maps**: Map 029, Map 030, Map 016

### Map 030: Shopping District

- **Speaker/Source:** `St. Baron□s Home`

```
\n<[Elijah]>I can't believe that I am doing
this,...the baron is the richest
man in the villiage...but he's been
using the town! This is for the good!
\n<[Katrina]>Ok...here we go...let's do this. I
have to do at least one good thing today.
\n<[Elijah]>I can believe the baron
```

```
would take all this money for himself! Not
anymore!
\n<[Elijah]>So the baron can still
keep a house while the rest of just
struggle. This is gonna be fun.
The door is locked. Mayhbe you should come back at
a different time
```

- **Speaker/Source:** `Shop Menu`

```
Welcoem to my shop! How can I help you today?
Thanks for coming! See you later!
```

- **Speaker/Source:** `Shopkeeper□s Wife Dialouge`

```
We've just got into town so we are still working
on getting the funds for a bed. But my hubby
will not rest until we have that bed!
```

- **Speaker/Source:** `Got the Potion`

```
\n<[Elijah]>I wonder why she needs a
potion. Could she be a witch?
\n<[Katrina]>I can't believe she made me
waste my money...
```

### 54.7.3. John the Thief's Hideout

John's sidequest proposal to overthrow the corrupt Baron.

**Associated Maps**: Map 031

#### Map 031: John the Thief (Sidequest)

- **Speaker/Source:** `John the Thief`

```
\n<John>What do you want kid! Can't you
see that I'm busy!
\n<John>What do you want kid! Can't you
see that I'm busy!
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>So you're the one that's helping the little
people of this village out.<br>I have a little task
for you.<br>You see, the Baron has outlived his
welcome.<br>Think you can help with this...issue
\n<John>What do you want kid! Can't you
see that I'm busy!
```

## 54.8. Act VII: The Frontier, Wilderness & Final Dungeon

### 54.8.1. World Map Frontier

Demo completion gateway; thank-you message; frontier exits.

**Associated Maps**: Map 033, Map 034, Map 035, Map 005, Map 018

### 54.8.2. Wilderness Battle Zones (Demo Areas 3 & 4)

Combat zones featuring bats, slimes, magos, and bandits.

**Associated Maps**: Map 014, Map 015, Map 016, Map 034, Map 035, Map 036, Map 019, Map 020, Map 021, Map 022, Map 033

#### Map 020: Grandma's House Upstairs

- **Speaker/Source:** `Grandma□s Chest`

```
\n<[Elijah]>I shouldn't go in this chest. I
need to talk to Grandma first.
She's very particular about her things.
\n<[Katrina]>I shouldn't go in this chest. I
need to talk to Grandma first.
Batty old woman seems to remind that chest.
```

- **Speaker/Source:** `Grandma□s Chest`

```
An old watch was in the chest.
\n<[Elijah]>Oh.. it's an old watch.
What's so special about it anyway?
\n<[Katrina]>She made me....grab...an
OLD WATCH! I thought it was
something important like her will!
```

- **Speaker/Source:** `Grandma□s Chest`

```
There is still something inside of the chest
OBTAINED THE COMPASS OF MORALITY
```

- **Speaker/Source:** `Grandma□s Chest`

```
It's all empty now.
```

- **Speaker/Source:** `Wierd Dream`

```
\n<[Elijah]>That dream....what did it mean...
It could just be my imagination but...It
felt like...something more. A hero
... Me...?
\n<[Katrina]>WHAT! WHAT THE...ME. Man, my mind
needs to get out of lala land and settle
into reality. I can't be a hero,
I won't be a hero...
```

#### Map 021: Grandma's House Downstairs

- **Speaker/Source:** `Grandma`

```
\n<Grandma>Hello sweetie, could you please
go upstairs and search in the chest
and grab what's inside? It's something that
has great signifinance.
\n<[Katrina]>The will. I love you gran,
but a girl gotta get out of this dead-end
town.
\n<[Elijah]>Maybe is is something
about dad! I haven't seen him since my
10th birthday.
```

```
\n<[Elijah]>Sure thing. This might
be my chance to get answers!
\n<[Katrina]>Why not! Crazy old woman...
That will is mine!
\n<[Elijah]>Well...uhh...but...
But if it is about him...
\n<[Katrina]>Gran, why can't you do it?
Why doesn't she do it?
If she wants it she can get it.
```

- **Speaker/Source:** Grandma

```
It's been a while since we've talk.
What do you want to talk about?
\n<Grandma>Oh me and Monica go way back.
She was such a wild child, always getting
into mischief. You'll love her.
\n<Grandma>This town used to be full of
life. sighs The kingdom doesn't
pay much attention to this place anymore...
\c[5]Not avalible in the demo
\n<Grandma>Have you been upstairs yet?
The chest should still be there.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>Thank you honey. This watch has
been in the family for a long time. Oh,
how it brings back memories of better days.
\n<Grandma>I think you should keep it. It's
my gift to you. And here's a bit of money.
I know that you'll need it more than little
ol' me.
```

- **Speaker/Source:** Grandma

```
\n<Grandma>[Katrina] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>[Elijah] could you do me a favor?
Lately, the people have been stirred up
about people being attacked by monsters.
\n<Grandma>My friend, Monica,
would know what is causing these strange
rumors. Can you go to her house and see
what she has to say about all of this?
\n<[Elijah]>Sure thing. It will be nice to
finally leave the house. I might even
get to leave town!
\n<[Katrina]>Why not! I would love to~
I'm out of this poor slum of a town!
\n<[Elijah]>Well..are you sure you can't?
I don't even know her...
\n<[Katrina]>Gran, I'm not your errand girl!
 This is YOUR friend.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** Block

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** Block

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** `Block`

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

- **Speaker/Source:** `Block`

```
I should say something to Grandma before I go.
```

- **Speaker/Source:** `Block`

```
Grandma still seems worried. Maybe I can talk to
her one more time.
```

### 54.8.3. The Final Dungeon & Boss Threshold

Multi-level dungeon fortress leading to the ultimate demo boss confrontation.

**Associated Maps**: Map 017, Map 018, Map 023, Map 024, Map 025, Map 026, Map 027, Map 028, Map 037, Map 038

#### Map 023: Main Square

- **Speaker/Source:** `Fighter Kid`

```
... He's just ignoring you
```

- **Speaker/Source:** `Fighter Kid`

```
You want me to move? I'm blocking the way for others?
Big woop! No one can stop me so I'm staying!
```

- **Speaker/Source:** `Fighter Kid`

```
You're no fun! I'm going home!
```

- **Speaker/Source:** `Mystic□s Home`

```
The door is locked.
```

- **Speaker/Source:** `Leave Town`

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** `Lower Dio`

```
I still have to talk to Monica first.
```

- **Speaker/Source:** `Lower Dio`

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Lower Dio

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Lower Dio

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Lower Dio

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Lower Dio

```
I have to stop that kid first!
```

- **Speaker/Source:** Lower Dio

```
I still have to talk to Monica first.
```

- **Speaker/Source:** Monica!

```
\n<[Elijah]>This is great! The town is still
sad with the storm but it's my home. I
need to stay focus and find Monica's house.
Grandma said her house is in front of ours.
\n<[Katrina]>Ugghh. I can't believe we still
live in this dump. Doesn't help that storm
destoryed everything. Now Gran said Monica
live in the house in front of us. Peachy...
```

- **Speaker/Source:** Leave Town

```
I should see if Monica needs anything else.
```

- **Speaker/Source:** Leave Town

```
I have to stop that kid first!
```

- **Speaker/Source:** Block

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** Block

```
STAY AWAY FROM ME!
```

- **Speaker/Source:** Block

```
STAY AWAY FROM ME!
```

### Map 024: The Sword Map

- **Speaker/Source:** Medusa□s Sword

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

### Map 025: The Sword Map

- **Speaker/Source:** Medusa□s Sword

```
A strange sword was in the chest.
\n<[Elijah]>I wonder who this sword
belongs to.
\n<[Katrina]>Free sword for your girl~
```

### Map 026: Start Up Area

- **Speaker/Source:** The Beginning

```
It was a month before the Festival...<br>The tower...<br>The terror...
```

- **Speaker/Source:** The Beginning

```
The darkness...<br>The darkness...<br>The darkness returned once more.
```

### Map 027: Monica’s House

- **Speaker/Source:** Monica

```
\n<Monica>And who in the blaze are YOU!
\n<Monica>...
...
...
\n<Monica>So you are Magolia's grandson.
She made you sound more handsome that
you are.
\n<[Elijah]>HEY! That's not a nice thing
to say!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Elijah]>Nn-...I'm here because my grandma
thinks that evil has returned.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
```

```
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>So you are Magolia's
granddaughter. She made you sound prettier
that you are.
\n<[Katrina]>LISTEN HER YOU OLD-!
\n<Monica>Tis only a joke. Now why have
you came here? Do you have some milk?
\n<[Katrina]>What is wrong with you... I'm
just here because my grandma is crazy.
She's babbling about evil forces and all
that nonsense.
\n<Monica>A wise woman she is. Lyri has
been whispering in me ear about the forces
of darkness. I'll help but first, get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
\n<Monica>Good. I'll see you when you
return.
\n<Monica>I'll explain it once more.
\n<Monica>I'll help you if you get that
holigan from the city limits.
\n<Monica>Not only is he preventing folks
from leaving, he's also messing with me
business. Can you handle that?
```

- **Speaker/Source:** Monica

```
\n<Monica>You have to stop that kid! Our
village will be more devastated than
even with him blocking the road!
```

- **Speaker/Source:** Monica

```
\n<Monica>Good job! I think I have an idea.
Now that the people can, residences and
newcomers have arrived. I notice that a
shopkeeper has came into our town as well.
\n<Monica>I need you to go a buy a potion.
I will use this as a base for my offering
to Lyri. She'll be able to help us in this
dark hour.
Obtained the Book of Monsters
```

- **Speaker/Source:** Monica

```
\n<Monica>Oh yes. This is exactly what I
need for my offering.
\n<Monica>There is one more thing that I
need in order to make this brew suitable
for Lyri. Go out of the town to the west
path. A man has what I need. Get it please.
\n<Monica>If you want to get the potion,
you'll need to go to the lower parts of the
town.
```

- **Speaker/Source:** Monica

```
\n<Monica>Do you need something?
\n<Monica>Me? Why, I'm one of the original
mages for the King. The kingdom was not as
peaceful as is it now. But then, the
```

```
Dark Lord tried to take the kingdom.
\n<Monica>He was born from the darkness
that this world had. He planned to take
over this world and fill it with darkness.
The Hero saved this world and brought us
\n<Monica>our peace. The shield and sword.
Oh don't mind this old hag's ranmbling.
Not avalible in demo
Not avalible in demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `The Secret Room`

```
Get your snooping nose away from me room!
This room is currently unavalible for the demo
```

- **Speaker/Source:** `Lyri`

```
It is the deity Lyri, the Goddess of Peace.
```

- **Speaker/Source:** `Totem`

```
The totem has the names of fallen soldiers. One
name has a heart by it and it reads "Miss you".
\n<[Elijah]>Did Monica have a husband?
\n<[Katrina]>Someone married that crazy bat!
```

- **Speaker/Source:** `Totem`

```
The totem is full of offerings.
```

### Map 028: Eric's House

- **Speaker/Source:** `Eric`

```
We've been seeing a lot of monster appear near the
recently. I wonder what's that all about. And with
jobs being tight, money is hard to come by.
```

- **Speaker/Source:** `Eric`

```
\n<[Elijah]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
this will be more useful to you than me.
\n<[Elijah]>Thank you sir, but I...
\n<Eric>Don't mention it, and by the way, the name
is Eric. I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[1]>Thank you. Thank you so much!
\n<[Katrina]>I think that that should be
enough to help you.
Thank you. Thank you so much, here I believe that
```

```
this will be more useful to you than me.
\n<[Katrina]>Woah! This is a cool ring dude!
\n<Eric>The name is Eric and that ring has a story.
I used to be a lumberjack until that weird
portal appeared. I found this ring and held on to it
every since.
\n<n[2]>Thanks dude.
```

- **Speaker/Source:** `Eric`

```
\n<Eric>I'm sure that ring's fire power will come in
handy if you plan on leaving the village
```

- **Speaker/Source:** `Self-Aware Pot`

```
What kind of game do you think this is! You can't
just steal whatever you want! How rude!
```

- **Speaker/Source:** `Test of Honesty`

```
Something is glowing in the pot. Take the pots content?
You recieve 10G
THAT'S ALL MY MONEY! GET OUT NOW!
\n<\n1>I am so sorry! Please, I am not
trying to...
\n<n[2]>AH!
\n<n[2]>Hey, where did you even...
\n<n[2]>AH!
\n<[Elijah]>..steal all your money.
\n<n[1]>I'm sorry...
\n<n[2]>...come from...
\n<n[2]>It was just a few coins anyway.
That guy definately overreacted
You decide to leave the pot alone.
\n<n[1]>I can be quick, discret maybe
Elijah slowly reaches into the jar, stopping when
he saw how little money he had.
\n<n[1]>Maybe I should't...
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
Don't worry about it, it's not for you to worry about.
Just make sure you become something.
\n<n[1]>Is this what the voice meant,
to be a hero...How much do you need.
Just 500G, that will be enough for me to make it for
the month.
\n<n[1]>I'll leave the money in the pot,
that way you'll have it.
Tha-tha-thank you!
\n<n[1]>I would be wrong of me to take this
money. But why does he only have a little
money...
Hey want are you doing!
\n<n[1]>I am so sorry, I wasn't trying to
take your money, I was just looking at the
pot.
It's fine kid... I just have stumble one hard
time every since the monster began showing up.
It makes working hard and I can't afford to
live here anymore.
\n<n[1]>Maybe... Maybe I can help!
```

```
Don't worry about it, it's not for you to worry about.
Just make sure you become something
\n<n[1]>I have to help him somehow...
How much do you need.
Just 500G, that will be enough for me to make it for
the month
\n<n[1]>I'll leave the money in the pot,
that way you'll have it
Tha-tha-thank you!
\n<n[2]>The money is right there! This
guy won't see it coming.
As Katrina was about to take the money, she stopped
for just a moment
\n<n[2]>Ugghhh, I can't do it.
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>Is this what the voice meant,
to be a hero...How much do rent here
cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
\n<n[2]>Come on [Katrina], its just a few bucks,
you can give him some more when you get
more.
Hey want are you doing!
\n<n[2]>Look man, I was just looking at
the stupid thing and saw some pocket change
in the pot. I was going to give you
something back...eventually.
That would've been all of my money, my life savings!
That's my money for the whole week!
\n<n[2]>Hold on, you mean to tell me that
you only have 10G to your name. Dude, how
can you afford to live here!
Don't worry about that, the monster attacks have made
living in town alot cheaper...even without my job...
\n<n[2]>I have to help him somehow...
How much do rent here cost...
500G, but I won't be able to make that kind of money
in time to pay the rent
\n<n[2]>sigh I'll get the
money, just don't do...anything
Really? Thanks...who are you?
\n<n[2]>The girl next door...actually no,
just someone who understands hard times
```

- **Speaker/Source:** `Test of Honesty`

```
Would you like to add to the pot?
You have given \v[14] to the man.
How much will you add?
You have given \v[14] to the man.
You have filled the pot! Talk to the man once more.
The pot isn't full yet.
You don't have that much to give
```

```
You leave the pot alone.
```

- **Speaker/Source:** `Test of Honesty`

```
The pot is filled with gold. It can cover rent for
2 months!
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

- **Speaker/Source:** `Empty Pot`

```
The pot is empty
```

# Chapter 55. Complete Cast & Tactical Party Compendium

**Source File**: `docs/game_design/demo_baseline/cast_and_characters.adoc`

## 55.1. Overview

This compendium catalogues every character, party companion, village elder, court antagonist, and mystical summon present in the 2017–2018 demo archives, detailing their combat statistics, class associations, sprite bindings, and narrative significance.

## 55.2. Dual Protagonists

### 55.2.1. Elijah Ro-enheart (The Steadfast Vanguard)

- **ID**: `Actor 1`
- **Class**: Hero (`Class 5`) / Knight (`Class 1`)
- **Relic Affinity**: Bearer of the **Shield of Lux**
- **Initial Level**: Lv 5 (Folder 1) / Lv 1 (Folder 2)
- **Movement Points**: 4 MP
- **Sprite / Battler**: `Actor1_1` / `MainCharacter` (custom generator)
- **Profile & Personality**: Son of the vanished Commander Gareth Ro-enheart (former Head of the Royal Guard). Earnest, chivalric, deeply burdened by his father's mysterious disappearance on his 10th birthday. Elijah seeks answers through service and duty, instinctively stepping forward to protect the weak and uphold the light of Sacyra.
- **Canonical LeTBS Specification**:

```
<letbs>
move_points: 4
auto_turn_order_face
auto_status_sprite
</letbs>
```

### 55.2.2. Lady Katrina Jaren (The Outrider Blade)

- **ID**: `Actor 2`
- **Class**: Hero (`Class 5`) / Start (`Class 7`)
- **Relic Affinity**: Wielder of the **Sword of Tenebris**
- **Initial Level**: Lv 5 (Folder 1) / Lv 1 (Folder 2)
- **Movement Points**: 4 MP
- **Initial Equipment**: Weapon 1 (Sword), Armor 2 (Hat), Armor 3 (Cloth)
- **Sprite / Battler**: `Actor1_8`
- **Profile & Personality**: Noble daughter of aristocratic House Jaren. Rebellious, razor-sharp, pragmatic, and fiercely independent. Refusing a cloistered aristocratic marriage or court seat, Katrina escaped to Dio Village with eyes on joining the independent Adventurers' Guild. Her caustic wit masks a deep empathy for ordinary folk crushed by royal neglect.
- **Canonical LeTBS Specification**:

```
<letbs>
```

```
move_points: 4
auto_turn_order_face
auto_status_sprite
</letbs>
```

## 55.3. Extended Tactical Party Companions (Folder 2 Additions)

In `Old_Game_Data_2`, three specialized tactical party members were integrated to expand the grid-based TRPG party:

### 55.3.1. Sebastien (The Infiltrator Outlaw)

- **ID**: `Actor 4`
- **Class**: Thief (`Class 6`)
- **Initial Level**: Lv 4
- **Sprite / Battler**: `Actor3_7`
- **Tactical AI & Role**: Outrider flanker equipped with `ranged_fighter` AI, trap disarmament, and precision critical strikes.

### 55.3.2. Angela (The Evoker Elementalist)

- **ID**: `Actor 5`
- **Class**: Mage (`Class 2`)
- **Initial Level**: Lv 4
- **Movement Scope**: `square(2)` (4 MP)
- **Sprite / Battler**: `Actor3_8` (custom cast, atk, hit, victory animations)
- **Action Palette**: Full tactical command set (`Move`, `Skill[42]`, `Items`, `Scan`, `Pass`).
- **Tactical Role**: Long-range elemental artillery, crowd control, and battlefield terrain modification (ice walls, volcano ground).

### 55.3.3. Lucius (The Battle Priest)

- **ID**: `Actor 6`
- **Class**: Priest (`Class 3`)
- **Initial Level**: Lv 4
- **Sprite / Battler**: `Actor2_7`
- **Tactical Role**: Turn order manipulation (`Accelerate`, `Decelerate`, `Emergency Turn`), long-distance allied teleportation, and barrier restoration.

## 55.4. Summoned Celestial Entities

### 55.4.1. Wind Spirit (Actor 11)

- **Class**: `Class 12` | **Sprite**: `Nature` (Index 4)
- **Skills**: Air Compression (`Skill 132`), Great Wind (`Skill 133` - pushback), Transposition (`Skill 134` - swap places).

### 55.4.2. Pixie (Actor 12)

- **Class**: `Class 13` | **Sprite**: `Nature` (Index 5)

- **Skills**: Healing Wave (`Skill 135` - AOE restoration).

### 55.4.3. Fire Chicken (Actor 13)

- **Class**: `Class 14` | **Sprite**: `Nature` (Index 2)
- **Characteristics**: Explosive combat decoy (`loop_animation: 262`), Bomb (`Skill 136`), Fire Arrow (`Skill 137`).

## 55.5. Key Lore Figures & NPCs

| Name | Role & Allegiance | Historical Significance |
|---|---|---|
| **Grandma Magolia** | Matriarch of Dio Village | Mother of vanished Commander Gareth Ro-enheart; grandmother to Elijah. Bestows **Grandma's Watch** and **Compass of Morality**. |
| **Monica the Mystic** | Former Royal Battlemage | Eccentric hermit and priestess of Goddess Lyri. Guides protagonists in awakening relic powers through ritual offerings. |
| **Eric the Lumberjack** | Impoverished Dio Laborer | Loses his livelihood to monster raids; serves as the moral litmus test for player empathy in the **Test of Honesty**. Bestows the **Ring of Fire**. |
| **John the Thief** | Local Outlaw Leader | Operates in Lower Dio; plots to unseat the predatory St. Baron to return wealth to the villagers. |
| **The St. Baron** | Aristocratic Landowner | Feudal aristocrat bleeding Dio Village dry; symbol of court corruption and moral decay. |
| **Chrono** | Archival Sage & Guardian | Guide of the Dream Chamber / Ancient Archives. Oversees archetype stone selection and mechanics trials. |
| **High Priest Roté** | Royal Advisor | Court prelate who witnesses the cosmic omens and hastens to warn King Jihad. |
| **King Jihad** | Monarch of Sacyra | Feigns royal mobilization while falling victim to the King's Curse, secretly coveting the returned primordial darkness to surpass The Dark Lord. |

# Chapter 56. LeTBS Tactical Grid System Specification

**Source File**: `docs/game_design/demo_baseline/combat_and_letbs_mechanics.adoc`

## 56.1. Overview

This document formalizes the grid-based Tactical Strategy RPG combat architecture implemented via Lecode's Tactical Battle System (LeTBS) in the 2017–2018 demo. It details movement costs, action economy, scope geometries, terrain manipulation, and turn manipulation invariants.

## 56.2. Battle Field Architecture & Action Economy

- **Grid Structure**: 2D tile-based tactical grid mapped over encounter maps.
- **Movement Points (MP)**: Consumable per-turn budget determining movement range:
  - Baseline Protagonists & Allies: `move_points: 4`.
  - Heavy Impact Reductions: Heavy Hit reduces target MP by 3 (`move_points+: -3`).
  - Outrider Haste: Light Steps increases target MP by 1 (`move_points+: 1`).
- **Turn Timeline**: Monotonic timeline ordering turns by entity agility (`auto_turn_order_face`).
- **Directional Facing**: Facing impacts defense and positional abilities (e.g., **Phantom Slash** teleports directly behind the target before striking).

## 56.3. Scope & Area of Effect (AOE) Geometries

The demo configured rich geometric targeting parameters across skills:

| Scope Pattern | Example Skill | Geometric Definition |
|---|---|---|
| `circle(r)` | **Fire** (`Skill 42`), **Heal** (`Skill 72`) | Manhattan or Chebyshev circle of radius `r` centered on the caster. |
| `circle(max_r, min_r)` | **Ghost Arrow** (`Skill 79`), **Dagger Throw** (`Skill 169`) | Annular targeting ring; cannot target adjacent cells within `min_r`. |
| `line(n)` | **Explosive Mark** (`Skill 76`), **Great Wind** (`Skill 133`) | Cardinal linear projection up to `n` cells from caster. |
| `square(r)` | **Angela Move Scope** (`Actor 5`) | Full square grid extent of radius `r`. |
| `custom(pattern)` | **Fire Arrows** (`Skill 82`) | Scripted multi-target projectile pattern hitting all visible adversaries. |
| `through_obstacles` | **Teleportation** (`Skill 73`) | Ignores intermediate terrain obstacles, requiring only a free destination cell. |

## 56.4. Terrain State Modifications & Dynamic Hazards

Combat encounters feature active environmental manipulation:

- **Volcano Ground (`Skill 47`)**: Lays hazardous fiery embers across multiple ground cells dealing continuous fire damage.
- **Ice Wall / Ice Prison (`Skill 51`, `Skill 52`)**: Summons physical destructible barricades (`HP: 20`) that obstruct enemy

line-of-sight and movement corridors.

- **Explosive Mark & Magic Trap (`Skill 76`, `Skill 80`)**: Proximity-triggered grid landmines that detonate upon enemy entry.
- **Burning Tile Spread (`State 110`)**: Dynamic fire propagation expanding with `spread: 100% circle(1) all`.

## 56.5. Turn Timeline Manipulation

Unique priest techniques alter the flow of tactical battle:

- **Accelerate (`Skill 87`)**: Advances target position along the turn timeline by 100%.
- **Decelerate (`Skill 88`)**: Delays target position along the turn timeline by 100%.
- **Emergency Turn (`Skill 89`)**: Instantly grants an immediate action turn to an allied battler.
- **Extra Turn (`Skill 86`)**: Grants the caster or target an additional action cycle upon turn conclusion.

# Chapter 57. Classes, Growth Curves & Learning Trees

**Source File**: `docs/game_design/demo_baseline/classes_and_growth.adoc`

## 57.1. Overview

This document details the 7 primary player and NPC classes plus 4 auxiliary summon/guard classes from `Classes.json`. It includes base stat formulas across 8 core attributes (MHP, MMP, ATK, DEF, MAT, MDF, AGI, LUK) and skill learning progressions.

## 57.2. Class 1: Warrior

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 100 | 30 | 12 | 8 | 3 | 8 | 27 | 32 |
| 5 | 145 | 46 | 17 | 12 | 5 | 11 | 27 | 42 |
| 10 | 202 | 64 | 23 | 17 | 8 | 15 | 27 | 55 |
| 20 | 314 | 102 | 35 | 26 | 13 | 22 | 27 | 80 |
| 50 | 650 | 215 | 71 | 54 | 27 | 44 | 27 | 155 |
| 99 | 1200 | 400 | 130 | 100 | 50 | 80 | 27 | 277 |

### 57.2.1. Skill Learnings for Warrior

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Repulsive Blow** (`ID: 11`) | Innate class technique |
| Lv 1 | **Flash Jump** (`ID: 12`) | Innate class technique |
| Lv 1 | **Rush** (`ID: 13`) | Innate class technique |
| Lv 1 | **Intimidating Aura** (`ID: 14`) | Innate class technique |
| Lv 1 | **Heavy Hit** (`ID: 15`) | Innate class technique |
| Lv 1 | **Provoke** (`ID: 16`) | Innate class technique |
| Lv 1 | **Phantom Slash** (`ID: 17`) | Innate class technique |

## 57.3. Class 2: Mage

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 60 | 90 | 2 | 7 | 13 | 8 | 26 | 32 |
| 5 | 83 | 130 | 4 | 10 | 18 | 12 | 26 | 42 |
| 10 | 110 | 180 | 6 | 13 | 24 | 17 | 26 | 55 |
| 20 | 165 | 280 | 10 | 20 | 36 | 26 | 26 | 80 |
| 50 | 330 | 580 | 21 | 39 | 72 | 54 | 26 | 155 |

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 99 | 600 | 1070 | 40 | 70 | 130 | 100 | 26 | 277 |

### 57.3.1. Skill Learnings for Mage

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Fire** (ID: 42) | Innate class technique |
| Lv 1 | **Sparkle** (ID: 43) | Innate class technique |
| Lv 1 | **Lightning Storm** (ID: 44) | Innate class technique |
| Lv 1 | **Fire Storm** (ID: 45) | Innate class technique |
| Lv 1 | **Fire Support** (ID: 46) | Innate class technique |
| Lv 1 | **Volcano Ground** (ID: 47) | Innate class technique |
| Lv 1 | **Ice Fury** (ID: 49) | Innate class technique |
| Lv 1 | **Walking Fire** (ID: 50) | Innate class technique |
| Lv 1 | **Ice Wall** (ID: 51) | Innate class technique |
| Lv 1 | **Ice Prison** (ID: 52) | Innate class technique |
| Lv 1 | **Whitch's Escape** (ID: 53) | Innate class technique |
| Lv 1 | **Fire Chickens** (ID: 54) | Innate class technique |
| Lv 1 | **Water (charged)** (ID: 55) | Innate class technique |
| Lv 1 | **Mega Sparkle** (ID: 56) | Innate class technique |
| Lv 1 | **Multi Fire** (ID: 57) | Innate class technique |

## 57.4. Class 3: Priest

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 65 | 60 | 6 | 8 | 10 | 7 | 32 | 32 |
| 5 | 95 | 87 | 10 | 13 | 15 | 11 | 32 | 42 |
| 10 | 133 | 119 | 15 | 18 | 20 | 16 | 32 | 55 |
| 20 | 208 | 185 | 25 | 28 | 31 | 26 | 32 | 80 |
| 50 | 433 | 380 | 53 | 59 | 63 | 54 | 32 | 155 |
| 99 | 800 | 700 | 100 | 110 | 115 | 100 | 32 | 277 |

### 57.4.1. Skill Learnings for Priest

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Heal** (ID: 72) | Innate class technique |

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Teleportation** (ID: 73) | Innate class technique |
| Lv 1 | **Salvation** (ID: 74) | Innate class technique |
| Lv 1 | **Teleportation II** (ID: 75) | Innate class technique |
| Lv 1 | **Explosive Mark** (ID: 76) | Innate class technique |
| Lv 1 | **Light Steps** (ID: 78) | Innate class technique |
| Lv 1 | **Ghost Arrow** (ID: 79) | Innate class technique |
| Lv 1 | **Magic Trap** (ID: 80) | Innate class technique |
| Lv 1 | **Fire Arrows** (ID: 82) | Innate class technique |
| Lv 1 | **Wind Spirit** (ID: 83) | Innate class technique |
| Lv 1 | **Summon Pixie** (ID: 84) | Innate class technique |
| Lv 1 | **Shamanism** (ID: 85) | Innate class technique |
| Lv 1 | **Extra Turn** (ID: 86) | Innate class technique |
| Lv 1 | **Accelerate** (ID: 87) | Innate class technique |
| Lv 1 | **Emergency Turn** (ID: 89) | Innate class technique |
| Lv 1 | **Ally Teleport.** (ID: 90) | Innate class technique |
| Lv 1 | **Mana Ball** (ID: 92) | Innate class technique |

## 57.5. Class 4: Summoner

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 450 | 90 | 16 | 16 | 16 | 16 | 32 | 32 |
| 5 | 650 | 130 | 21 | 21 | 21 | 21 | 42 | 42 |
| 10 | 900 | 180 | 28 | 28 | 28 | 28 | 55 | 55 |
| 20 | 1400 | 280 | 40 | 40 | 40 | 40 | 80 | 80 |
| 50 | 2900 | 580 | 77 | 77 | 77 | 77 | 155 | 155 |
| 99 | 5350 | 1070 | 138 | 138 | 138 | 138 | 277 | 277 |

### 57.5.1. Skill Learnings for Summoner

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Sentinel Totem** (ID: 102) | Innate class technique |

## 57.6. Class 5: Hero

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 300 | 90 | 16 | 16 | 16 | 16 | 32 | 32 |
| 5 | 507 | 130 | 21 | 21 | 21 | 21 | 42 | 42 |
| 10 | 764 | 180 | 28 | 28 | 28 | 28 | 55 | 55 |
| 20 | 1280 | 280 | 40 | 40 | 40 | 40 | 80 | 80 |
| 50 | 2825 | 580 | 77 | 77 | 77 | 77 | 155 | 155 |
| 99 | 5350 | 1070 | 138 | 138 | 138 | 138 | 277 | 277 |

### 57.6.1. Skill Learnings for Hero

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Heal** (`ID: 72`) | Innate class technique |
| Lv 5 | **Repulsive Blow** (`ID: 11`) | Innate class technique |

## 57.7. Class 6: Thief

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 450 | 90 | 16 | 16 | 16 | 16 | 32 | 32 |
| 5 | 650 | 130 | 21 | 21 | 21 | 21 | 42 | 42 |
| 10 | 900 | 180 | 28 | 28 | 28 | 28 | 55 | 55 |
| 20 | 1400 | 280 | 40 | 40 | 40 | 40 | 80 | 80 |
| 50 | 2900 | 580 | 77 | 77 | 77 | 77 | 155 | 155 |
| 99 | 5350 | 1070 | 138 | 138 | 138 | 138 | 277 | 277 |

## 57.8. Class 7: Start

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 999 | 1 | 999 | 1 | 1 |
| 5 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |
| 10 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |
| 20 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |
| 50 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |
| 99 | 1 | 1 | 1 | 999 | 1 | 999 | 1 | 1 |

## 57.9. Class 10: ---Special---

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 450 | 90 | 16 | 16 | 16 | 16 | 32 | 32 |

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 5 | 650 | 130 | 21 | 21 | 21 | 21 | 42 | 42 |
| 10 | 900 | 180 | 28 | 28 | 28 | 28 | 55 | 55 |
| 20 | 1400 | 280 | 40 | 40 | 40 | 40 | 80 | 80 |
| 50 | 2900 | 580 | 77 | 77 | 77 | 77 | 155 | 155 |
| 99 | 5350 | 1070 | 138 | 138 | 138 | 138 | 277 | 277 |

## 57.10. Class 11: Guard

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 100 | 30 | 12 | 8 | 3 | 8 | 27 | 32 |
| 5 | 145 | 46 | 17 | 12 | 5 | 11 | 27 | 42 |
| 10 | 202 | 64 | 23 | 17 | 8 | 15 | 27 | 55 |
| 20 | 314 | 102 | 35 | 26 | 13 | 22 | 27 | 80 |
| 50 | 650 | 215 | 71 | 54 | 27 | 44 | 27 | 155 |
| 99 | 1200 | 400 | 130 | 100 | 50 | 80 | 27 | 277 |

### 57.10.1. Skill Learnings for Guard

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Heavy Hit** (ID: 15) | Innate class technique |

## 57.11. Class 12: Wind Spirit

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 50 | 1 | 1 | 1 | 1 | 1 | 1 |
| 5 | 1 | 79 | 1 | 1 | 1 | 1 | 1 | 1 |
| 10 | 1 | 115 | 1 | 1 | 1 | 1 | 1 | 1 |
| 20 | 1 | 186 | 1 | 1 | 1 | 1 | 1 | 1 |
| 50 | 1 | 400 | 1 | 1 | 1 | 1 | 1 | 1 |
| 99 | 1 | 750 | 1 | 1 | 1 | 1 | 1 | 1 |

### 57.11.1. Skill Learnings for Wind Spirit

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Air Compression** (ID: 132) | Innate class technique |
| Lv 1 | **Great Wind** (ID: 133) | Innate class technique |
| Lv 1 | **Transposition** (ID: 134) | Innate class technique |

## 57.12. Class 13: Pixie

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 50 | 1 | 1 | 1 | 1 | 1 | 1 |
| 5 | 1 | 79 | 1 | 1 | 1 | 1 | 1 | 1 |
| 10 | 1 | 115 | 1 | 1 | 1 | 1 | 1 | 1 |
| 20 | 1 | 186 | 1 | 1 | 1 | 1 | 1 | 1 |
| 50 | 1 | 400 | 1 | 1 | 1 | 1 | 1 | 1 |
| 99 | 1 | 750 | 1 | 1 | 1 | 1 | 1 | 1 |

### 57.12.1. Skill Learnings for Pixie

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Healing Wave** (`ID: 135`) | Innate class technique |

## 57.13. Class 14: Fire Chicken

| Level | MHP | MMP | ATK | DEF | MAT | MDF | AGI | LUK |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 5 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 10 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 20 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 50 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 99 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

### 57.13.1. Skill Learnings for Fire Chicken

| Level | Skill Name | Strategic Role |
|---|---|---|
| Lv 1 | **Bomb** (`ID: 136`) | Innate class technique |

# Chapter 58. The Complete Grimoire (75+ Skills & Formulas)

**Source File**: docs/game_design/demo_baseline/grimoire_and_skills.adoc

## 58.1. Overview

This document records the complete grimoire of all 75+ active techniques, spells, and summon abilities defined in Skills.json. It provides mathematical damage formulas, MP/TP costs, scope parameters, and tactical LeTBS sequences.

## 58.2. Basic Martial Maneuvers

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 1 | **Attack** | 0 | 0 | a.atk * 4 - b.def * 2 | + *Note*: Skill #1 will be used when you select the Attack command. |
| 2 | **Guard** | 0 | 0 | 0 | + *Note*: Skill #2 will be used when you select the Guard command. |
| 3 | **Dual Attack** | 0 | 0 | a.atk * 4 - b.def * 2 | |
| 4 | **Double Attack** | 0 | 0 | a.atk * 4 - b.def * 2 | |
| 5 | **Triple Attack** | 0 | 0 | a.atk * 4 - b.def * 2 | |
| 6 | **Escape** | 0 | 0 | 0 | |
| 7 | **Wait** | 0 | 0 | 0 | |

## 58.3. Knight Martial Techniques

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 11 | **Repulsive Blow** | 0 | 0 | this.phyDmg(120) | Hits and push the target for two cell. *Note*: <letbs> scope: circle(1) sequence: atk_push(2) scope_color: #FFF </letbs> <l… |
| 12 | **Flash Jump** | 0 | 0 | this.phyDmg(110) + this.magDmg(10) | Jumps to an available position in the scope and invoke a thunder to damage nearby fighters. *Note*: <letbs> scope: circle(3,1) aoe: circle(1) scope_options: need_free_cell aoe_o… |
| 13 | **Rush** | 0 | 0 | this.phyDmg(80) | Rushes toward the target. *Note*: <letbs> scope: custom(rush) sequence: rush </letbs> <letbs_AI> type: offense… |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 14 | **Intimidating Aura** | 0 | 0 | `(a.atk * 4 - b.def * 2) * 1.2` | Casts an aura that reduce enemies' ATK by 15%. Grants as well 70% counter attack chance. *Note*: `<letbs> scope: circle(0) scope_options: include_center </letbs> <letbs_AI> t…` |
| 15 | **Heavy Hit** | 0 | 0 | `this.phyDmg(135)` | Hits the target and remove 3 move points. *Note*: `<letbs> scope: circle(1) sequence: atk </letbs> <Learn Require Level: 5> <l…` |
| 16 | **Provoke** | 0 | 0 | `(a.atk * 4 - b.def * 2) * 1.2` | Draw the attention of enemies upon the user. Can use another action after this skill. *Note*: `<letbs> scope: circle(0) scope_options: include_center extra_action </letbs> …` |
| 17 | **Phantom Slash** | 0 | 0 | `this.phyDmg(60)` | Use dark arts to reach the back of the target. *Note*: `<letbs> scope: line(4) sequence: phantom_slash </letbs> <Learn Require Level…` |

## 58.4. Mage Elemental Spells

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 42 | **Fire** | 0 | 0 | `this.magDmg(105)` | Deals fire damages to a single target in a medium range. *Note*: `<letbs> scope: circle(3);line(4) </letbs> <letbs_AI> type: offense </letbs_A…` |
| 43 | **Sparkle** | 0 | 0 | `this.magDmg(80)` | Deals thunder damages to targets in a small area. *Note*: `<letbs> scope: circle(4) aoe: circle(1) sequence: spark </letbs> <Learn Requi…` |
| 44 | **Lightning Storm** | 0 | 0 | `this.magDmg(60)` | Attacks 3 times a random battler in the area. *Note*: `<letbs> scope: line(4) aoe: circle(2) sequence: lightning_storm </letbs> <Lea…` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 45 | **Fire Storm** | 0 | 0 | `this.magDmg(200)` | Attacks 6 times randomly in the area. *Note*: `<letbs> scope: circle(4,2) aoe: square(1) sequence: fire_storm </letbs> <Lear···` |
| 46 | **Fire Support** | 0 | 0 | `this.magDmg(80)` | Deals fire damages to a single target in a medium range with the help of close allies. *Note*: `<letbs> scope: circle(4) sequence: fire_support </letbs> <Learn Require Level···` |
| 47 | **Volcano Ground** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*2` | Lays multiple fire spots on the ground. *Note*: `<letbs> scope: circle(0) scope_options: include_center aoe: custom(volcano_gr···` |
| 48 | **Volcano Ground (dmg)** | 0 | 0 | `this.magDmg(25)` | |
| 49 | **Ice Fury** | 0 | 0 | `this.magDmg(115)` | Deals ice damages in a row. *Note*: `<letbs> scope: line(4) aoe: custom(ice_fury) sequence: skill </letbs> <Learn ···` |
| 50 | **Walking Fire** | 0 | 0 | `this.magDmg(105)` | Deals fire damages in a medium path. *Note*: `<letbs> scope: circle(4) scope_options:through_obstacles aoe: path sequence: ···` |
| 51 | **Ice Wall** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a wall of ice. *Note*: `<letbs> scope: line(3) scope_options: need_free_cell aoe: custom(ice_wall) ao···` |
| 52 | **Ice Prison** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a prison of ice. *Note*: `<letbs> scope: circle(4) aoe: custom(ice_prison) aoe_options: need_free_cell ···` |
| 53 | **Whitch's Escape** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.20` | Teleports then casts Spark at the old location. The scope is based on current move points. *Note*: `<letbs> scope: circle(e.getMovePoints()+1) scope_options: need_free_cell sequ···` |
| 54 | **Fire Chickens** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon explosive chickens *Note*: `<letbs> scope: circle(2) aoe: circle(1) scope_options: need_free_cell aoe_opt···` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 55 | **Water (charged)** | 0 | 0 | `this.magDmg(135)` | + *Note*: `<letbs> scope: circle(4) sequence: map_skill charge: 2, 261 </letbs> <Learn R⋯` |
| 56 | **Mega Sparkle** | 0 | 0 | `this.magDmg(60)` | Deals thunder damage on 3 different areas. *Note*: `<letbs> scope: circle(4) aoe: circle(1) sequence: mega_spark 3 selections: di⋯` |
| 57 | **Multi Fire** | 0 | 0 | `this.magDmg(105)` | Invoke 3 columns of fire. *Note*: `<letbs> scope_select: enemies scope: circle(3); 2 selections: append; Select ⋯` |

## 58.5. Priest Holy Arts & Turn Manipulation

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 72 | **Heal** | 0 | 0 | `50 + a.mat` | Heals a single target and ends the turn. *Note*: `<letbs> scope: circle(4,2) sequence: heal pass_on_use scope_options: include_⋯` |
| 73 | **Teleportation** | 0 | 0 | `0` | Teleports the caster in a medium range. *Note*: `<letbs> scope: circle(4) sequence: teleportation scope_options:through_obstac⋯` |
| 74 | **Salvation** | 0 | 0 | `25 + a.mat` | Attracts allies and heals them in all directions. *Note*: `<letbs> scope: circle(0) aoe: line(3) sequence: salvation scope_options: incl⋯` |
| 75 | **Teleportation II** | 0 | 0 | `0` | Teleports the battler in front of the caster in a medium range. *Note*: `<letbs> scope: circle(3) sequence: teleportation_sup scope_options: through_o⋯` |
| 76 | **Explosive Mark** | 0 | 0 | `None` | Lays a mark on the ground that deals fire damages to enemies on it. *Note*: `<letbs> scope: line(5) aoe: square(1) sequence: explosive_mark </letbs> <Lear⋯` |
| 77 | **Explosive Mark (dmg)** | 0 | 0 | `this.magDmg(50)` | |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 78 | **Light Steps** | 0 | 0 | `50 + a.mat` | Increases target's move points for 2 turns.<br>*Note*: `<letbs> scope_options: include_center </letbs> <Learn Require Level: 5> <letb…` |
| 79 | **Ghost Arrow** | 0 | 0 | `this.magDmg(75)` | Deals light damages to a single target in a medium range, ignoring any obstacle, and remove one move point.<br>*Note*: `<letbs> scope: circle(5,2) sequence: ghost_arrow scope_options: through_obsta…` |
| 80 | **Magic Trap** | 0 | 0 | `None` | Lays a magic trap who deals light damage while blinding the target with 60% rate.<br>*Note*: `<letbs> scope: circle(5) scope_options:through_obstacles, need_free_cell aoe:…` |
| 81 | **Magic Trap (dmg)** | 0 | 0 | `this.magDmg(160)` | |
| 82 | **Fire Arrows** | 0 | 0 | `this.magDmg(105)` | Launch a fire arrow for every enemy in the scope.<br>*Note*: `<letbs> scope: line(2) aoe: custom(fire_arrows) aoe_options: line_of_sight se…` |
| 83 | **Wind Spirit** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a wind spirit that help the party with the power of wind.<br>*Note*: `<letbs> scope: circle(2) scope_options: need_free_cell sequence: summon(wind_…` |
| 84 | **Summon Pixie** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a pixie that can heal the party.<br>*Note*: `<letbs> scope: circle(1) scope_options: need_free_cell </letbs> <Learn Requir…` |
| 85 | **Shamanism** | 0 | 0 | `a.mhp * 0.1` | All allies recover 10% of their MHP for each dead battler in the area.<br>*Note*: `<letbs> sequence: shamanism scope: circle(3) aoe: circle(2) scope_options: in…` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 86 | **Extra Turn** | 0 | 0 | `50 + a.mat` | Add an extra turn. *Note*: `<letbs> scope: circle(2) scope_options: include_center extra_turn: 100% </let…` |
| 87 | **Accelerate** | 0 | 0 | `50 + a.mat` | Accelerate the turn order of the target. *Note*: `<letbs> scope: circle(2) scope_options: include_center accelerate_turn: 100% …` |
| 88 | **Decelerate** | 0 | 0 | `50 + a.mat` | Accelerate the turn order of the target. *Note*: `<letbs> scope: circle(2) scope_options: include_center decelerate_turn: 100% …` |
| 89 | **Emergency Turn** | 0 | 0 | `50 + a.mat` | Give an instant turn to the target. *Note*: `<letbs> scope: circle(3) pass_on_use scope_options: include_center instant_tu…` |
| 90 | **Ally Teleport.** | 0 | 0 | `0` | Teleports an ally in a medium range. *Note*: `<letbs> scope_select: allies scope: circle(4) sequence: ally_teleport scope_o…` |
| 91 | **Ally Teleport. (Scope)** | 0 | 0 | `0` | + *Note*: `<letbs> scope: circle(4) scope_options:through_obstacles, need_free_cell </le…` |
| 92 | **Mana Ball** | 0 | 0 | `this.magDmg(80)` | Launch a bouncing mana ball *Note*: `<letbs> scope: circle(3) sequence: mana_ball </letbs> <Learn Require Level: 5…` |

## 58.6. Summoner Totems & Rites

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 102 | **Sentinel Totem** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | Summon a totem that strike enemies as soon as they enter the range of action. *Note*: `<letbs> scope: circle(3) scope_options: need_free_cell sequence: summon(senti…` |
| 103 | **Transform.** | 0 | 0 | `50 + a.mat` | + *Note*: `<letbs> scope: circle(0) scope_options: include_center </letbs> <letbs_AI> ty…` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 104 | **Devour** | 0 | 0 | `this.magDmg(105)` | + *Note*: `<letbs> scope: circle(1) fusion_effect </letbs> <letbs_AI> type: offense </le…` |

## 58.7. Summon Beast Techniques

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 132 | **Air Compression** | 0 | 0 | `this.magDmg(135)` | Deals wind damages in a medium range with 15% of chance to blind the target. *Note*: `<letbs> scope: circle(4) </letbs> <letbs_AI> type: offense </letbs_AI> <Learn…` |
| 133 | **Great Wind** | 0 | 0 | `this.magDmg(140)` | Deals wind damages and push the target. The collision damage of this skill is greatly increased. *Note*: `<letbs> scope: line(4) sequence: great_wind collision_formula: 2 * b.mhp * 0….` |
| 134 | **Transposition** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.35` | Switch place with a target in a medium range. *Note*: `<letbs> scope: circle(6) sequence: transposition scope_options:through_obstac…` |
| 135 | **Healing Wave** | 0 | 0 | `10 + a.mat` | Heals in small area *Note*: `<letbs> scope: circle(3) aoe: circle(1) scope_options: include_center sequenc…` |
| 136 | **Bomb** | 4 | 0 | `a.mhp` | + *Note*: `<letbs> scope: circle(1) </letbs> <letbs_AI> type: offense </letbs_AI> <letbs…` |
| 137 | **Fire Arrow** | 0 | 0 | `this.phyDmg(100) + this.magDmg(100)` | Launch a fire arrow. *Note*: `<letbs> scope: circle(3) sequence: projectile(fire_arrow) </letbs> <Learn Req…` |

## 58.8. Enemy Bestial & Void Techniques

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 162 | **Poisonous Bite** | 3 | 0 | `(a.atk * 4 - b.def * 2)*0.8` | + *Note*: `<letbs> scope: circle(1) </letbs> <letbs_AI> type: offense </letbs_AI>` |

| ID | Skill Name | MP | TP | Damage Formula | Description & Tactical Note |
|---|---|---|---|---|---|
| 163 | **Fire Surge** | 4 | 0 | `this.magDmg(105)` | Deals fire damages to a single target in a medium range. *Note*: `<letbs> scope: circle(4) </letbs> <letbs_AI> type: offense </letbs_AI>` |
| 164 | **Summon Pyra** | 0 | 0 | `(a.mat * 2 - b.mdf * 2)*1.15` | + *Note*: `<letbs> scope: circle(1) scope_options: need_free_cell sequence: summon_pyra ⋯` |
| 165 | **Evil Healing** | 0 | 0 | `15 + a.mat` | + *Note*: `<letbs> scope: circle(4) scope_options: include_center </letbs> <letbs_AI> ty⋯` |
| 166 | **Mago's Mark** | 0 | 0 | `None` | + *Note*: `<letbs> scope: line(5) aoe: square(1) sequence: mago_mark </letbs> <Cooldown:⋯` |
| 167 | **Evil Protection** | 0 | 0 | `15 + a.mat` | + *Note*: `<letbs> scope: circle(4) scope_options: include_center </letbs> <letbs_AI> ty⋯` |
| 168 | **Rabbit's Jump** | 0 | 0 | `(a.atk * 4 - b.def * 2) * 1.10 + a.mat` | + *Note*: `<letbs> scope: line(3) aoe: circle(1) scope_options: need_free_cell sequence:⋯` |
| 169 | **Dagger Throw** | 3 | 0 | `this.phyDmg(115)` | Throw a little but sharp dagger toward the target. *Note*: `<letbs> scope: circle(4,2) sequence: projectile(dagger) </letbs> <letbs_AI> ⋯` |

# Chapter 59. Bestiary, Destructible Props & Troop Formations

**Source File**: docs/game_design/demo_baseline/bestiary_and_troops.adoc

## 59.1. Overview

This document catalogs all 22+ hostile monsters, wild fauna, bandits, void entities, and interactive battlefield props from Enemies.json and Troops.json.

## 59.2. Monster Bestiary

| ID | Name | HP | MP | ATK | DEF | MAT | MDF | AGI | LUK | AI / Behavior |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | **Bat** | 80 | 12 | 11 | 8 | 8 | 4 | 25 | 30 | Skill 1 |
| 2 | **Mago** | 60 | 18 | 8 | 6 | 16 | 6 | 22 | 30 | Skill 41, Skill 1 |
| 3 | **Rabbit** | 55 | 0 | 10 | 8 | 8 | 4 | 27 | 30 | Skill 1 |
| 4 | **Great Mago** | 90 | 36 | 10 | 7 | 19 | 7 | 23 | 30 | Skill 1 |
| 5 | **Fire Mago** | 90 | 36 | 10 | 7 | 19 | 7 | 23 | 30 | Skill 1 |
| 6 | **Pyra** | 30 | 50 | 6 | 8 | 6 | 4 | 27 | 30 | Basic Attack |
| 7 | **Bandit** | 80 | 12 | 14 | 10 | 10 | 7 | 25 | 30 | Skill 1, Skill 169 |
| 8 | **Mage Bandit** | 80 | 12 | 11 | 8 | 8 | 4 | 25 | 30 | Skill 1 |
| 9 | **Slime** | 250 | 0 | 30 | 30 | 30 | 30 | 30 | 30 | Skill 1 |
| 10 | **Orc** | 300 | 0 | 30 | 30 | 30 | 30 | 30 | 30 | Skill 1 |
| 11 | **Minotaur** | 500 | 0 | 30 | 30 | 30 | 30 | 30 | 30 | Skill 1 |
| 12 | **Random Kid** | 100 | 0 | 30 | 10 | 10 | 10 | 10 | 10 | Skill 1, Skill 3 |
| 13 | **Dream Eater** | 70 | 12 | 11 | 8 | 8 | 4 | 25 | 30 | Skill 1 |
| 32 | **Invincible Neutral** | 99999 | 0 | 8 | 6 | 14 | 6 | 22 | 30 | Basic Attack |
| 33 | **Tile Effect Launcher** | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 34 | **Ice Block** | 20 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 35 | **Wooden Chest** | 50 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 36 | **Silver Chest** | 50 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 37 | **Golden Chest** | 50 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |
| 38 | **Stolen Goods** | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Basic Attack |

## 59.3. Neutral & Interactive Grid Entities

The demo utilized the battle engine to render interactive tactical grid entities: * **Wooden Chest (Enemy 35)**: HP: 50 | Yields 200 Gold upon destruction via Common Event 5. * **Silver Chest (Enemy 36)**: HP: 50 | Yields 2,000 Gold upon

destruction via Common Event 6. * **Golden Chest (`Enemy 37`)**: `HP: 50` | Yields 10,000 Gold upon destruction via Common Event 7. * **Ice Block (`Enemy 34`)**: `HP: 20` | Destructible terrain obstacle summoned by Ice Wall. * **Stolen Goods (`Enemy 38`)**: `HP: 1` | Objective dropped upon defeat of Mage Bandit. * **Invincible Neutral (`Enemy 32`)**: `HP: 99,999` | Static grid anchor.

## 59.4. Troop Encounter Formations

| Troop ID | Troop Designation | Composition |
|---|---|---|
| 1 | **Bat*2** | Bat, Bat |
| 2 | **Slime*2** | Mago, Mago |
| 3 | **Orc** | Rabbit |
| 4 | **Minotaur** | Great Mago |
| 5 | **Random Kid** | Fire Mago |
| 6 | **Bat** | Bat |
| 7 | **Bat*2** | Bat, Bat |
| 8 | **Bandit*3, Mage Bandit** | Bandit, Bandit, Mage Bandit, Bandit |
| 9 | **Dream Eater*2** | Dream Eater, Dream Eater |

# Chapter 60. Armory, Inventory & Key Lore Relics

**Source File**: `docs/game_design/demo_baseline/armory_and_inventory.adoc`

## 60.1. Overview

This document details all weapons, protective armors, restorative items, and key relics defined in `Weapons.json`, `Armors.json`, and `Items.json`.

## 60.2. Weapons & Relic Armaments

| ID | Weapon Name | Type | Price | ATK | DEF | AGI | Lore & Characteristics |
|---|---|---|---|---|---|---|---|
| 1 | **Sword** | Sword | 500\G | +10 | +0 | +0 | |
| 2 | **Axe** | Axe | 500\G | +10 | +0 | +0 | |
| 3 | **Cane** | Cane | 500\G | +10 | +0 | +0 | |
| 4 | **Bow** | Bow | 500\G | +10 | +0 | +0 | |
| 5 | **Medusa's Sword** | Sword | 1600\G | +17 | +0 | +0 | The sword used to slay Medusa. It is said to have her petrifying powers. |

## 60.3. Defensive Armor & Relic Wards

| ID | Armor Name | Type | Price | DEF | MDF | MHP | Lore & Characteristics |
|---|---|---|---|---|---|---|---|
| 1 | **Shield** | Small Shield | 300\G | +10 | +0 | +0 | |
| 2 | **Hat** | General Armor | 300\G | +10 | +0 | +0 | |
| 3 | **Cloth** | General Armor | 300\G | +10 | +0 | +0 | |
| 4 | **Ring** | General Armor | 300\G | +0 | +0 | +0 | |
| 5 | **Heart Shield** | Small Shield | 300\G | +10 | +5 | +0 | |
| 6 | **Ring of Fire** | General Armor | 0\G | +0 | +5 | +0 | |

## 60.4. Consumable Items & Quest Relics

| ID | Item Name | Price | Consumable | Description & Architectural Mapping |
|---|---|---|---|---|
| 1 | **Potion** | 10\G | Yes | |
| 2 | **Magic Water** | 100\G | Yes | |

| ID | Item Name | Price | Consumable | Description & Architectural Mapping |
|---|---|---|---|---|
| 3 | **Dispel Herb** | 200\G | Yes | |
| 4 | **Stimulant** | 300\G | Yes | |
| 5 | **Grandma’s Watch** | 0\G | No (Key Relic) | This was Grandma’s Watch. Its a family heirloom that keeps perfect time. |
| 6 | **Book of Monsters** | 0\G | No (Key Relic) | This is a big with all the enemies you’ve encountered |
| 7 | **Book of Items** | 0\G | No (Key Relic) | This book describes all the various items |
| 8 | **Compass of Morality** | 0\G | No (Key Relic) | |

# Chapter 61. Tactical States & Status Ailments Directory

**Source File**: `docs/game_design/demo_baseline/states_and_statuses.adoc`

## 61.1. Overview

This document catalogs all 30 status ailments, tactical combat states, aura modifiers, and battlefield terrain effects defined in `States.json` across Folder 1 and Folder 2 (LeTBS). It distinguishes generic RPG Maker MV Runtime Package (RTP) boilerplate from authentic tactical grid modifiers and stance dynamics.

## 61.2. Triage: RTP Boilerplate States vs. Authentic Tactical Grid States

| Category | State IDs | Architectural Policy |
|---|---|---|
| **Tier 1: RTP Default States** | 1–10 (Knockout, Guard, Immortal, Poison, Blind, Silence, Rage, Confusion, Fascination, Sleep) | Stock RPG Maker MV status conditions. Re-architect in C11 to tie directly into Sacyran relic affinities, posture break, and elemental resistances. |
| **Tier 3: Authentic LeTBS Grid States** | 11, 31–34, 46, 62–63, 78–79, 94–95, 109–110 | Authentic tactical modifiers altering grid movement points (MP), turn order priority, damage mitigation, and tile propagation. |

## 61.3. Standard Status Conditions (Tier 1 Boilerplate)

| ID | State Name | Priority | Restriction / Description |
|---|---|---|---|
| 1 | **Knockout** | 100 | Cannot move; unit is felled upon reaching 0 HP. |
| 2 | **Guard** | 0 | Damage taken reduced by 50%; expires at start of next turn. |
| 3 | **Immortal** | 0 | Unit cannot drop below 1 HP; used for critical NPC story anchors. |
| 4 | **Poison** | 50 | Loses 10% max HP at end of combat turn. |
| 5 | **Blind** | 60 | Hit rate reduced by 50%. |
| 6 | **Silence** | 70 | Magic skills cannot be cast. |
| 7 | **Rage** | 80 | Attacks nearest enemy or ally indiscriminately. |
| 8 | **Confusion** | 75 | Actions randomized between foes and friends. |
| 9 | **Fascination** | 70 | Charmed unit attacks allies. |
| 10 | **Sleep** | 90 | Cannot take turns; wakes up upon taking damage. |
| 11 | **Paralyze** | 85 | Stunned; skip turn (`loop_animation: 64`). |

## 61.4. Knight & Vanguard Tactical States (Tier 3 LeTBS)

| ID | State Name | Affected Stat | Tactical Mechanism & LeTBS Meta |
|---|---|---|---|
| 31 | **Intimidating Aura** | ATK -15% | Passive aura cast via Skill 14 (`auras: intimidating_aura`). Reduces adjacent enemy physical attack and grants 70% counter-attack chance. |
| 32 | **Intimidated** | ATK -15% | Status inflicted on foes caught within the Intimidating Aura boundary. |
| 33 | **Heavy Strike** | Move Points: -3 | Tactical stagger inflicted by Heavy Hit (Skill 15). Severely impedes enemy mobility on the grid. |
| 34 | **Focused** | AI Threat | Cast by Provoke (Skill 16). Forces enemy AI to prioritize the vanguard (`focused_by_ai_for_offense: 100`). |

## 61.5. Mage & Priest Arcane States (Tier 3 LeTBS)

| ID | State Name | Affected Stat | Tactical Mechanism & LeTBS Meta |
|---|---|---|---|
| 46 | **Charging** | Spell Delay | Channeling visual state for high-tier spells like Water (charged) (`loop_animation: 184`). |
| 62 | **Light Steps** | Move Points: +1 | Mobility buff cast via Skill 78. Expands grid traversal budget for outriders and rogues. |
| 63 | **Ghost Arrow** | Move Points: -1 | Tactical snare inflicted by Ghost Arrow (Skill 79). Reduces movement allowance. |

## 61.6. Summoner & Environmental Grid States (Tier 3 LeTBS)

| ID | State Name | Affected Stat | Tactical Mechanism & LeTBS Meta |
|---|---|---|---|
| 78 | **Transformation** | Form Shift | Active metamorphosis into astral beast (`character_name: Monster`). |
| 79 | **Fusion** | Unit Merge | Fuses summoner with tactical summon (`sprite_name: Rabbit`). |
| 94 | **Rabbit's Jump** | Mobility | Leap mechanics allowing clearance over terrain obstacles. |
| 95 | **Evil Protection** | DEF/MDF Buff | Damage shield cast by enemy warlocks (Skill 167). |
| 109 | **Tile Regen** | Periodic Heal | Restores health to units occupying consecrated ground. |
| 110 | **Burning** | Fire Dot / Spread | Fire hazard state (`spread: 100% circle(1)`). Expands dynamically to adjacent flammable tiles. |

## 61.7. Modern C11 Architectural Mapping

In the C11 simulation engine, tactical states are decoupled from engine internals: 1. **Zero-Heap State Mask**: Active states on an actor are tracked via a bitmask and static array (`active_states[MAX_ACTIVE_STATES]`). 2. **Move Point Clamping**: `move_points` modifiers directly feed into Layer 0 spatial pathing routines. 3. **Data-Driven Configuration**: State durations, stat multipliers, and tile spread rules are loaded from `data/states/*.json`.

# Chapter 62. Engine Systems, Time Clock & Switch/Variable Index

**Source File**: docs/game_design/demo_baseline/systems_switches_variables.adoc

## 62.1. Overview

This document specifies the global state machines, monotonic time clock, morality engine, and the complete registry of all switches and variables governing demo logic.

## 62.2. The Monotonic Time Engine

The time system is driven by Common Events 1 and 2, updating a 24-hour cycle: * Variable 1: Seconds (0–59) * Variable 2: Minutes (0–59) * Variable 3: Hours (1–12) * Variable 4: 24-Hour Military Format (0–23) * Variable 5: Days (1–7: Sunday through Saturday) * Variable 6: Weeks * Variable 7: Months * Variable 8: Years

In the modern C11 architecture, this corresponds to Layer 0 time_utils (time_get_monotonic_ms).

## 62.3. The Moral Sway Continuum Engine

The morality engine tracks the party's ethical orientation on an invariant scale: * Variable 9: **Moral (Overall)**: Continuum within [-100, +100]. * Variable 10: **Moral (Good)**: Cumulative points awarded for altruism, charity, and mercy. * Variable 11: **Moral (Evil)**: Cumulative points accrued for ruthless, self-serving, or thieving acts. * Item 8: **Compass of Morality**: Tool allowing the player to inspect their moral continuum in the field.

## 62.4. Global Switches Directory

| Switch ID | Name / Flag | Operational Role & Trigger |
|---|---|---|
| 1 | **Walk of Fear** | State flag for quest / scenario progression |
| 2 | **Return to Cut Scene** | State flag for quest / scenario progression |
| 3 | **Walk of Fear 2** | State flag for quest / scenario progression |
| 4 | **Roté Enters the Castle** | State flag for quest / scenario progression |
| 5 | **The Priest Leave the Castle** | State flag for quest / scenario progression |
| 6 | **D. Jihad is Summoned** | State flag for quest / scenario progression |
| 7 | **??? Enters the Map** | State flag for quest / scenario progression |
| 8 | **Class Selected** | State flag for quest / scenario progression |
| 9 | **??? Leaves the Map** | State flag for quest / scenario progression |
| 10 | **??? Explains Puzzle** | State flag for quest / scenario progression |
| 11 | **(Story Plot) Remove Rocks** | State flag for quest / scenario progression |
| 12 | **Switch Active** | State flag for quest / scenario progression |
| 13 | **(Side Plot) Remove Rocks** | State flag for quest / scenario progression |
| 14 | **??? Disappears** | State flag for quest / scenario progression |

| Switch ID | Name / Flag | Operational Role & Trigger |
|---|---|---|
| 15 | **??? Weaken** | State flag for quest / scenario progression |
| 16 | **??? Final Words** | State flag for quest / scenario progression |
| 17 | **Boss Defeated** | State flag for quest / scenario progression |
| 18 | **Restart** | State flag for quest / scenario progression |
| 19 | **Remove Stone 1** | State flag for quest / scenario progression |
| 20 | **Remove Stone 2** | State flag for quest / scenario progression |
| 21 | **Remove Stone 3** | State flag for quest / scenario progression |
| 22 | **Remove Stone 4** | State flag for quest / scenario progression |
| 23 | **Treasure Remove** | State flag for quest / scenario progression |
| 24 | **Time** | State flag for quest / scenario progression |
| 25 | **Friendship Menu** | State flag for quest / scenario progression |
| 26 | **Quest 1 (Grandma's Request)** | State flag for quest / scenario progression |
| 27 | **Grandma (Idle)** | State flag for quest / scenario progression |
| 28 | **Exit Location** | State flag for quest / scenario progression |
| 29 | **Monica Event On** | State flag for quest / scenario progression |
| 30 | **Final Conversation Before Map is Out** | State flag for quest / scenario progression |
| 31 | **Quest 1 (Give Grandma Here Watch)** | State flag for quest / scenario progression |
| 32 | **The Great Sage** | State flag for quest / scenario progression |
| 33 | **More Battle Mechanics** | State flag for quest / scenario progression |
| 34 | **Monica Quest Switch** | State flag for quest / scenario progression |
| 35 | **Unlock Dio Village-Lower** | State flag for quest / scenario progression |
| 36 | **Idle Monica** | State flag for quest / scenario progression |
| 37 | **Petty Thief** | State flag for quest / scenario progression |
| 38 | **Morality** | State flag for quest / scenario progression |
| 39 | **Give to the Poor** | State flag for quest / scenario progression |
| 40 | **A Noble Thief** | State flag for quest / scenario progression |
| 41 | **Angry John** | State flag for quest / scenario progression |

## 62.5. Global Variables Directory

| Variable ID | Name | Subsystem Allocation |
|---|---|---|
| 1 | **Seconds** | Global numerical state accumulator |
| 2 | **Minutes** | Global numerical state accumulator |
| 3 | **Hours** | Global numerical state accumulator |
| 4 | **24 Hours** | Global numerical state accumulator |
| 5 | **Days** | Global numerical state accumulator |
| 6 | **Weeks** | Global numerical state accumulator |
| 7 | **Months** | Global numerical state accumulator |
| 8 | **Years** | Global numerical state accumulator |
| 9 | **Moral (Overall)** | Global numerical state accumulator |
| 10 | **Moral (Good)** | Global numerical state accumulator |
| 11 | **Moral (Evil)** | Global numerical state accumulator |
| 12 | **Region ID** | Global numerical state accumulator |
| 13 | **Currency** | Global numerical state accumulator |
| 14 | **Pot Amount** | Global numerical state accumulator |
| 15 | **Deposit Amount** | Global numerical state accumulator |

# Chapter 63. Historical Dataset Evolution & Delta Analysis

**Source File**: `docs/game_design/demo_baseline/dataset_evolution_delta.adoc`

## 63.1. Overview

This document provides a comparative delta between `Old Game Data` (Folder 1) and `Old_Game_Data_2/data` (Folder 2), detailing the evolution from narrative prototype to advanced LeTBS tactical simulation, and establishing modern C11 reconciliation strategies.

## 63.2. Dataset Characteristics & Metrics

| Subsystem Metric | Folder 1 (Old Game Data) | Folder 2 (Old_Game_Data_2/data) |
|---|---|---|
| Game Title | `Tenebris and Lux` | `Tenebris & Lux` |
| Version Identifier | `40730002` | `2078310` |
| Map Count | 36 maps | 43 maps |
| Active Player Actors | 2 (Elijah, Katrina) | 5 (Elijah, Katrina, Sebastien, Angela, Lucius) |
| Enemies Defined | 21 enemies | 22 enemies |
| States Defined | 11 states | 30 states (extensive LeTBS configurations) |
| Morality Engine Items | Compass of Morality, Ring of Fire | Omitted / shifted to LeTBS combat |

## 63.3. Core Differences & Evolution Strategy

1. **Narrative vs. Tactical Pivot**:
   - **Folder 1** represents the **narrative-complete** edition of the demo. It contains all dialogue branches, the Dio Village moral tests (Eric's Honesty Pot, John the Thief vs St. Baron), and key relics like the **Compass of Morality**.
   - **Folder 2** represents the **tactical engine expansion**. It introduces full party combat with Angela, Sebastien, and Lucius, 6 extra dungeon maps, and rich LeTBS states (e.g. **Burning Tile Spread**, **Charging**, **Intimidating Aura**).
2. **C11 Reconstruction Recommendation**:
   - **Layer 1 Simulation Engine**: Combine the narrative quest topology and moral continuum from Folder 1 with the rich 5-character tactical class abilities and LeTBS movement/scope geometry from Folder 2.
   - **Layer 2 Presentation**: Maintain the authentic dialog scripts, dual-protagonist branch voices, and environmental puzzles.

# Chapter 64. Game Design Curation & Triaging Matrix (2017–2018 Demo vs C11 Engine)

**Source File**: `docs/game_design/demo_baseline/game_design_curation.adoc`

## 64.1. Overview

This document formalizes the architectural triage of the 2017–2018 RPG Maker MV demo datasets. It separates generic RPG Maker Runtime Package (RTP) boilerplate from temporary testing experiments and identifies the authentic, fleshed-out canonical systems to be reconstructed in the native C11 game engine.

## 64.2. Triage Categorization Taxonomy

To prevent technical debt from legacy game engine defaults from contaminating our custom C11 engine, all extracted game data is triaged into three strict analytical categories:

| Category | Definition | Architectural Policy |
|---|---|---|
| **Tier 1: RTP Defaults & Boilerplate** | Stock RPG Maker MV assets, generic items, placeholder sounds, and default formulas. | **Discard / Re-architect**. Never port verbatim into C11. |
| **Tier 2: Prototype & Testing Sketches** | Ad-hoc test maps, sandbox battlers, debug switches, and experimental scripts. | **Review & Evolve**. Extract mechanical concepts but redesign within clean strata. |
| **Tier 3: Fleshed-Out Canonical Pillars** | Core story scripts, dual-protagonist dialogue dynamics, moral dilemma quests, LeTBS grid tactics, and key relics. | **Preserve & Reconstruct**. Faithfully implement as axiomatic invariants in C11. |

## 64.3. Tier 1: RPG Maker RTP Defaults (To Discard or Re-Architect)

| Element | Historical Manifest | Why It Must Be Discarded / Replaced |
|---|---|---|
| **Basic Consumables** | Potion (50G), Magic Water (100G), Dispel Herb (20G), Stimulant (50G) | Generic RTP recovery items. In C11, consumables must tie directly to Sacyran herbalism, relic alchemy, and stance restoration. |
| **Generic Armaments** | Standard Sword (100G), Axe (100G), Cane (100G), Bow (100G), Shield (100G), Cloth (100G) | Default placeholder inventory without distinct tactical affinities or stance interactions. |
| **Stock Sound Effects** | `Attack2`, `Buzzer1`, `Damage4`, `Collapse1-4`, `Shop1` | Generic MV sound library. C11 audio subsystem will support custom soundscapes. |
| **Placeholder Bestiary** | Bat (`HP: 80`), Slime (`HP: 250`), Rabbit (`HP: 55`), Orc (`HP: 300`) | Standard MV demo monsters used for basic stat checks. Replaced by Sacyran fauna and Abyssal void aberrations. |

| Element | Historical Manifest | Why It Must Be Discarded / Replaced |
|---|---|---|
| **Default Title Theme** | `Theme6` / `CrossedSwords` title screen graphic | Default MV asset packaging. Replaced by bespoke Sacyran coat of arms and twin relic artwork. |

## 64.4. Tier 2: Prototype & Testing Sketches (Concepts to Evolve)

| Element | Historical Location | Tactical / Design Value to Evolve |
|---|---|---|
| **Sandbox Test Maps** | `Map006 Test Areas`, `Map043 Event Map` | Used to test event triggers, movement paths, and sprite coordinates. Demonstrates intent for environmental puzzle mechanics. |
| **Debug Enemies & Troops** | `Enemy 14 Test`, `Troop 10 Test`, `Enemy 32 Invincible Neutral` | Proved that neutral, invincible obstacles and destructible props function well on the tactical grid. |
| **Explosive Chickens** | Fire Chicken (`Actor 13`), Bomb (`Skill 136`) | Humorous prototype mechanic demonstrating combat summons and dynamic decoy units on the tactical field. |
| **Plugin Experiments** | `LeTBS/firestorm.json`, `LeTBS/test.json`, `QMaps/` | Initial experiments with Lecode's Tactical Battle System and Quasi Movement plugins. Proves grid-based TRPG was the true target genre. |

## 64.5. Tier 3: Fleshed-Out Canonical Pillars (Core Invariants to Reconstruct)

The following systems represent the authentic creative vision of **Tenebris and Lux** and must be preserved and reconstructed in C11:

### 64.5.1. Dual-Protagonist Dynamics & Narrative Branching

- **Elijah Ro-enheart (`\n[1]`)**: Son of vanished Commander Gareth Ro-enheart. Dutiful, protective vanguard bearer of the Shield of Lux. Dialogue reflects reverence for duty, family loyalty, and hesitation to violate laws.
- **Lady Katrina Jaren (`\n[2]`)**: Daughter of noble House Jaren escaping courtly politics to join the Adventurers' Guild. Agile outrider wielder of the Sword of Tenebris. Dialogue features sardonic wit, impatience with provincial stagnation, and fierce defense of the vulnerable.

### 64.5.2. The Moral Sway Continuum Engine

- **Ethical Mechanics**: Discrete moral choices tracking across `Moral (Good)` and `Moral (Evil)` within the invariant continuum `[-100, +100]`.
- **The Test of Honesty (Eric's House, Map 28)**:
  - Stealing Eric's life savings (10G) rewards quick cash but plunges morality.
  - Donating 500G to cover Eric's rent saves a desperate worker and earns the **Ring of Fire**.
- **The Struggle of Dio Village (John the Thief vs. St. Baron, Map 30/31)**:
  - Feudal exploitation vs. outrider rebellion, forcing the heroes to choose between legal compliance and moral justice.

### 64.5.3. Lecode Tactical Battle System (LeTBS) Mechanics

- **Grid Movement & Range Geometries**:
  - Movement Points (4 MP baseline), modified by skills like **Light Steps** (+1) and **Heavy Hit** (-3).
  - Scope shapes: `circle(r)`, `circle(max, min)` (annular), `line(n)`, `square(r)`.
- **Terrain Hazards & Obstacles**:
  - **Ice Wall** and **Ice Prison** creating physical destructible barriers (`HP: 20`).
  - **Volcano Ground** and **Burning Tile Spread** (`spread: 100% circle(1)`).
- **Turn Timeline Manipulation**:
  - **Accelerate**, **Decelerate**, and **Emergency Turn** altering the turn queue.

### 64.5.4. The Monotonic 24-Hour Time Engine

- **Grandma's Watch**: Family heirloom activating real-time calendar and clock tracking (Seconds, Minutes, Hours, Days, Weeks, Months, Years), corresponding directly to Layer 0 `time_utils`.

# Volume V: Software Architecture & The 5-Block Invariant Protocol

Cornerstone Big Machine vs. Tenebris and Lux Master Turn Gear, 5 strata, Rev 5 5-block protocol, Safe-Mode BIOS & .rodata ROM vaults, VFS data pipelines, pre-flight audits, and dual-frontend presentation.

# Chapter 65. Cornerstone & Tenebris and Lux: System Architecture Overview

**Source File**: `docs/architecture/ARCHITECTURE.adoc`

This guide explains how the **Cornerstone Game Engine** (`engine/`) and **Tenebris and Lux** (`game/`) work together—written so a designer tuning battle balance, a writer editing dialogue, a student learning systems architecture, or an engineer hardening C23 code can build a shared mental model in minutes.



**Textbook Code Snapshots Enabled:** Code blocks marked as excerpts in this chapter are pulled live from the repository during documentation builds. Clickable wholesale cross-references jump directly to complete listings in the Appendix when viewing compiled PDF or book builds.

## 65.1. Who This Engine Is Built For (Start Here)

A game engine should feel like a well-labeled workshop, not a locked vault. You do not need senior systems experience to use **Cornerstone Game Engine** safely:

- **Creators & Designers** edit declarative 5-block JSON files (`engine/config/` and `game/data/`). If a syntax mistake slips into a data file, built-in ROM vaults keep the engine running in Safe-Mode and pinpoint the exact line to fix.
- **Playtesters & Narrative Leads** test every tactical choice using the lightweight headless terminal (`--cli`) or diagnostic audit (`--audit`) without needing heavy graphics setups.
- **C & Rust Developers** gain zero-heap determinism, strict compile-time compiler walls (`-Wall -Wextra -Wpedantic`), four test tiers (`65` CTest suites + `30` Rust unit tests), and uniform 5-block structure across source, JSON, CMake, and documentation.

## 65.2. The Big Machine and The Master Turn Gear

The architecture rests on a clean mechanical metaphor:

- `engine/` — **The Big Machine (Cornerstone Game Engine):** Reusable infrastructure needed by **any** tactical or adventure game: math, clocks, string parsing, virtual filesystems, flight logging, replay debugging, audio mixing, and a `640x360` X11 viewport.
- `game/` — **The Master Turn Gear (Tenebris and Lux):** Pure domain rules of Sacyra and Libira: tactical grids, Elijah & Katrina's vocations, Twin Relics, combat stances, moral sway toward **Lux** or **Tenebris**, and the multi-ending state machine.
- `game/src/main.c` — **Layer 3 Plug:** The single mechanical coupling joining the Big Machine to the Master Turn Gear at boot.

### Cornerstone vs. Tenebris and Lux Structural Decoupling

**Anti-Domain-Bleed Rule:** Nothing under `engine/` is ever allowed to include `game/` headers or mention story canon (`Elijah`, `Katrina`, `Sacyra`, `Usuru`). Conversely, `CMakeLists.txt` enforces separate include fences (`ENGINE_PUBLIC_INCLUDES` vs. `COMMON_INCLUDES`) so `engine/` builds cleanly completely on its own.

**Cog Elevation Rule:** Whenever a helper inside `game/` solves a general systems problem (screen wipes, easing, audio cue scheduling, JSON parsing) with zero story coupling, it elevates into `engine/` so future games inherit it automatically.

## 65.3. The Five Compilation Strata

Code dependencies flow strictly **downward**—lower layers never call upward into higher layers:

| Layer | Directory | Memory Contract | Responsibility |
|---|---|---|---|
| Layer 4 | `tests/` (`functional`, `unit`, `integrated`, `system`) | Test Harness (`ALLOC_ZERO_HEAP` sim) | Proves mathematical, structural, integration, and full CLI parity across 65 CTest checks. |
| Layer 3 | `game/src/main.c`, `tools/` | Startup Plug / Rust Tooling | Parses CLI flags (`--cli`, `--gui`, `--debug`, `--audit`), builds manifest, compiles data/guides. |
| Layer 2 | `engine/*/viewport/` | `ALLOC_ZERO_HEAP` (`.bss`) | `640x360` logical software rasterizer, Movie Reel camera, native X11 display window. |
| Layer 1 | `game/` (`hero`, `combat`, `relics`, `world`, `ending`, `dispatch`) | `ALLOC_ZERO_HEAP` | Deterministic tactical RPG rules completely independent of console vs. desktop rendering. |
| Layer 0 | `engine/` (`core`, `log`, `debug`, `io`, `audio`) | `ALLOC_ZERO_HEAP` | `44` header ecosystem (`31` Layer 0 + `13` Layer 2) providing math, VFS, flight logs, replay debug, 7-bus audio. |

## 65.4. The Universal 5-Block Protocol

Whether you open a `.h` contract, `.c` engine file, `.json` config, `CMakeLists.txt`, Rust tool, or `.adoc` chapter, everything shares the exact same mental geography:

| Block | Code Role (C / Rust) | Data Role (JSON) | Doc Role (AsciiDoc `////`) |
|---|---|---|---|
| Block 1 | Metadata header, tags, invariants | `block_1_metadata` identity & rules | Document provenance, target audience, snapshot bindings |
| Block 2 | Includes, types, macros, `.bss` (`2.1`–`2.5`) | `block_2_types` / `block_2_opening` limits | Title, attributes, reader orientation & quick TOC |
| Block 3 | Core algorithms & API logic (`3.1`+) | `block_3_body` live runtime payload | Technical narrative, diagrams, tables, snapshot plates |
| Block 4 | Probes, tripwires, self-audits (`4.1`–`4.2`) | `block_4_audit` tripwires & metrics | Verification commands & reader Q&A checkpoints |
| Block 5 | Diagnostic dump cards, boot & shutdown (`5.1`–`5.2`) | `block_5_closing` handshakes & links | Cross-volume navigation & next readings |

## 65.5. Controlled Memory Chassis & Fall-Down Portability

Avoiding `malloc`/`calloc`/`free` inside Layers 0–2 (`ALLOC_ZERO_HEAP`) creates a **Controlled Memory Chassis**: every byte of memory is visible in `.text`, `.rodata`, `.data`, `.bss`, and bounded stack frames. When RPG tables grow, capacity pressure shows up immediately at compile time instead of hiding inside unpredictable heap leaks.

At the same time, Cornerstone uses an **Engine-as-a-Java "Fall-Down Stairs"** portability pattern. Rather than adding legacy compatibility clutter, we write forward-first for **ISO C23** (`-std=c23`) and **C++20** headers (`extern "C"`) anchored

through engine/include/core/manifest.h so older compilers step down gracefully to **C17** or **C11** (CORNERSTONE_C_STANDARD=AUTO|23|17|11).

*Cornerstone Fall-Down Stairs Portability Ladder (C23/C17/C11 & C++20/17) (engine/include/core/manifest.h#L82-L143 · Mode: Subassembly)*

```
 82 /*
 83 ------------------------------------------------------------------------------
 84 Sub-Assembly 2.1: Pre-Conditions, Progressive Standard Macros & Buffer Limits
 85 ------------------------------------------------------------------------------
 86 Compile-time sizing limits for registry module nodes, dependencies, rules, and
 87 Cornerstone's "Fall-Down" ISO C23 / C17 / C11 + C++20 / C++17 capability macros.
 88
 89 For Designers:
 90 Allows Cornerstone to take advantage of modern compiler safety checks on current
 91 hardware while stepping cleanly onto older systems without legacy clutter. Also
 92 bounds the engine registry to up to 32 major systems, 8 core rules, and 4 links.
 93
 94 For Programmers:
 95 - Guarantees 100% zero-heap allocation (ALLOC_ZERO_HEAP) in .rodata and .bss.
 96 - Exposes CS_C_STD_TIER (23/17/11/0) and CS_CXX_STD_TIER (20/17/11/0) plus
 97   graceful ISO attribute fall-down macros (CS_NODISCARD, CS_MAYBE_UNUSED,
 98   CS_FALLTHROUGH, CS_UNREACHABLE, CS_ALIGNOF, CS_NORETURN, CS_STATIC_ASSERT,
 99   CS_HAS_EMBED) compiled cleanly without extensions under -Wpedantic.
100 ------------------------------------------------------------------------------
101 */
102
103 /** Active ISO C standard tier compiled for this translation unit (23, 17, 11, or 0 in C++). */
104 #if defined(__STDC_VERSION__) && __STDC_VERSION__ >= 202311L
105 #    define CS_C_STD_TIER 23
106 #elif defined(__STDC_VERSION__) && __STDC_VERSION__ >= 201710L
107 #    define CS_C_STD_TIER 17
108 #elif defined(__STDC_VERSION__) && __STDC_VERSION__ >= 201112L
109 #    define CS_C_STD_TIER 11
110 #else
111 #    define CS_C_STD_TIER 0
112 #endif
113
114 /** Active ISO C++ standard tier when included from C++ (20, 17, 11, or 0 in C). */
115 #if defined(__cplusplus) && __cplusplus >= 202002L
116 #    define CS_CXX_STD_TIER 20
117 #elif defined(__cplusplus) && __cplusplus >= 201703L
118 #    define CS_CXX_STD_TIER 17
119 #elif defined(__cplusplus)
120 #    define CS_CXX_STD_TIER 11
121 #else
122 #    define CS_CXX_STD_TIER 0
123 #endif
124
125 /** Marks return values that callers MUST inspect (C23/C++17+ [[nodiscard]], C11/17 floor empty). */
126 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
127 #    define CS_NODISCARD [[nodiscard]]
128 #else
129 #    define CS_NODISCARD
130 #endif
131
132 /** Silences unused-entity diagnostics cleanly on intentional dormant/platform helpers. */
133 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
134 #    define CS_MAYBE_UNUSED [[maybe_unused]]
135 #else
136 #    define CS_MAYBE_UNUSED
137 #endif
138
139 /** Annotates explicit switch-case fall-through cleanly under -Wextra -Wpedantic. */
140 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
141 #    define CS_FALLTHROUGH [[fallthrough]]
142 #elif defined(__GNUC__) && (__GNUC__ >= 7)
143 #    define CS_FALLTHROUGH __attribute__((fallthrough))
```

*Compile-time standard detection establishing C23/C17/C11 and C20/C17 parity without legacy shims.*

Wholesale Listing: Complete verbatim code (#L82-L194) → Appendix: Cornerstone Fall-Down Stairs Portability Ladder (C23/C17/C11 & C++20/17).

To see how Layer 3 connects this portability bedrock to the game world at startup, examine the boot entry group extracted from game/src/main.c:

*Layer 3 Plug Entry & Manifest Wiring (init_engine_manifest + main) (game/src/main.c#L71-L116 · Mode: Group)*

```c
/**
 * @brief Registers all core engine subsystem manifests into central registry.
 *
 * ### Machine Role & Mission
 * Populates the central manifest registry with all Layer 0 utilities (manifest,
 * fs, json, time, math, geom, string, log, debug), Layer 1 simulation engine (state,
 * combat, world, dispatcher, game), and Layer 2 presentation viewport (gui).
 * Enables boot-time loose-bolts invariant checks and dependency graph resolution
 * before any game loop execution begins.
 *
 * ### Ontological Classification
 * - Stratum: Layer 3 (Application Entry & Presentation Router)
 * - Machine: app.entry.main (Executable Entry & Presentation Router)
 * - Assembly: Pre-Flight Manifest Registry Initialization
 * - Cog: Module Manifest Registration Primitive
 *
 * ### Execution & Resource Contract
 * - Memory Model: ALLOC_ZERO_HEAP (Stack-allocated fixed manifest tables).
 * - Concurrency: CONCURRENCY_MAIN_THREAD (Called at startup on main thread).
 * - Latency Budget: < 50 microseconds.
 *
 * @param[out] registry Target manifest registry to initialize and populate.
 */
static void init_engine_manifest(manifest_registry_t *registry)
{
    /* Step 1: Zero-initialize registry table */
    manifest_registry_init(registry);

    /* Step 2: Register Layer 0 Core Utilities */
    manifest_register(registry, &manifest_utils_manifest);
    manifest_register(registry, &fs_utils_manifest);
    manifest_register(registry, &json_utils_manifest);
    manifest_register(registry, &path_finder_manifest);
    manifest_register(registry, &master_loader_manifest);
    manifest_register(registry, &time_utils_manifest);
    manifest_register(registry, &math_utils_manifest);
    manifest_register(registry, &geom_utils_manifest);
    manifest_register(registry, &string_utils_manifest);
    manifest_register(registry, &clock_pacer_manifest);
    manifest_register(registry, &formula_eval_manifest);
    manifest_register(registry, &log_manifest);
    manifest_register(registry, &debug_manifest);

    /* Step 3: Register Layer 1 Simulation Engine Subsystems */
    manifest_register(registry, &sim_state_manifest);
    manifest_register(registry, &combat_manifest);
```

*Demonstrates how Layer 3 game/src/main.c wires engine/ and game/ strata together via the central manifest registry.*

Wholesale Listing: Complete verbatim code (#L71-L522) → Appendix: Layer 3 Plug Entry & Manifest Wiring (init_engine_manifest + main).

## 65.6. Verifying Architectural Boundaries Yourself

```
cmake -S . -B build && cmake --build build -j4
./tools/target/debug/tl_tools validate all
ctest --test-dir build --output-on-failure
./build/tenebris-lux --audit
```

1. *Why are there no* `malloc()` *calls in gameplay or engine code?*

Fixed `.rodata` and `.bss` buffers guarantee deterministic replay hashes and keep memory consumption measurable down to the byte.

2. *What enforces separation between* `engine/` *and* `game/`*?*

   CMake object library scoping (`ENGINE_PUBLIC_INCLUDES` vs. `COMMON_INCLUDES`) prevents `engine/` targets from seeing any `game/` header at compile time.

3. *Where do code changes break first if an interface changes?*

   No shims are kept behind—strict `-Wall -Wextra -Wpedantic` compiler checks and `65` CTest suites map every affected call site immediately.

## 65.7. Where to Go Next

- Continue to Chapter 2: Zero-Disk Safe-Mode BIOS, Config ROM Vaults & Autonomous Probe Loop.
- Review how data configs stay isolated in Data-Driven Architecture.
- Browse the live extracted code plate catalog in Snapshots Index.

# Chapter 66. Zero-Disk Safe-Mode BIOS, Config ROM Vaults & Autonomous Probe Loop

**Source File**: docs/architecture/resilience_bios_and_probes.adoc

Software fails in the real world when external files go missing, break during an edit, or lose write permissions. In Rev 5, the **Cornerstone Game Engine** never crashes blindly on startup: it boots through **Tier-0 Safe-Mode BIOS vaults**, falls back to **compiled-in `.rodata` Config ROMs**, and pauses transparently on bad assumptions using an **Autonomous Probe → Safe-Mode Downgrade → `DEBUG_STEP_AWAIT`** safety loop.

## 66.1. Tier-0 Safe-Mode BIOS & Quarantined Fallbacks

Think of `*_fallback.[ch]` files as a motherboard BIOS. Even if your hard drive configuration folder is wiped clean, the engine must still initialize telemetry, mount virtual read-only defaults, mix safe audio cues, and open a window showing what went wrong.

Every modernized subsystem follows a **3-file quarantine layout**:

| File Pattern | Responsibility | Why Quarantine Matters |
|---|---|---|
| `<mod>_types.h` | Pure enums, structs, limits, callbacks | Breaks circular header dependencies between main code, BIOS fallback, and tests. |
| `<mod>_fallback.[ch]` | Minimal Tier-0 Safe-Mode boot & ROM reader | Zero dependencies on higher layers or disk parsers; boots even on an empty disk. |
| `<mod>.h` / `<mod>.c` | Full runtime subsystem + probes | Hydrates disk overrides on top of BIOS defaults and registers Block 4 probes. |



> **Honest Telemetry Flag (`loaded_from_data`):** Every hydrated policy carries an explicit boolean `loaded_from_data`. It is set to `false` under Safe-Mode ROM/floppy defaults and flips `true` **only** after an authentic external JSON file parses and validates cleanly.

## 66.2. Compiled-In `.rodata` Config ROM Vaults

Rather than hardcoding duplicate default tables inside C source files, CMake runs `cmake/GenerateConfigRom.cmake` during build configuration (`generate_telemetry_rom`, `generate_diagnostics_rom`, `generate_io_rom`). Every `.json` file inside those `engine/config/` subtrees is packed directly into immutable read-only `.rodata` C byte arrays under `build/generated/`.

*Build-Time Compiled .rodata Config ROM Generator (`cmake/GenerateConfigRom.cmake#L113-L156` · Mode: `Whole`)*

```
113 # ================================================================================
114 # BLOCK 3: BODY BLOCK
115 # ( Deterministic Discovery, Hex-to-C Encoding & Table Construction )
116 # ================================================================================
117 # Engine:        Cornerstone Game Engine
118 # Stratum:       Layer 0 Infrastructure (Build Tooling)
119 # Module:        build.rom.config
120 # Mission:       Traverse sorted configuration domains and encode every JSON
121 #                asset into null-terminated .rodata C arrays and table rows.
122 # ================================================================================
123
124 # --------------------------------------------------------------------------------
125 # Sub-Assembly 3.1: Sorted Domain Traversal & Identifier Mapping
126 # --------------------------------------------------------------------------------
```

```
set(_sorted_subdirs ${SUBDIRS})
list(SORT _sorted_subdirs) ①

foreach(_subdir IN LISTS _sorted_subdirs)
    set(_abs_dir "${CONFIG_ROOT}/${_subdir}")
    if(NOT IS_DIRECTORY "${_abs_dir}")
        message(FATAL_ERROR
            "GenerateConfigRom.cmake: Subdirectory does not exist: '${_abs_dir}'")
    endif()

    file(GLOB _dir_jsons "${_abs_dir}/*.json") ②
    list(SORT _dir_jsons)
    list(LENGTH _dir_jsons _dir_count)

    if(_dir_count EQUAL 0)
        message(FATAL_ERROR
            "GenerateConfigRom.cmake: No JSON files found in '${_abs_dir}'")
    endif()

    string(APPEND _subdirs_doc "  engine/config/${_subdir}/ (${_dir_count} assets)\n")
    get_filename_component(_subdir_leaf "${_subdir}" NAME)

    # ------------------------------------------------------------------------
    # Sub-Assembly 3.2: Deterministic Byte-Array Encoding & Table Row Emission
    # ------------------------------------------------------------------------
    foreach(_json_path IN LISTS _dir_jsons)
        get_filename_component(_stem "${_json_path}" NAME_WE)
        get_filename_component(_filename "${_json_path}" NAME)

        if(ID_MODE STREQUAL "STEM")
```

① Sorts configuration subdirectories so generated .rodata asset tables remain deterministic across hosts.

② Discovers all JSON configs in each domain directory before sorting and byte-encoding them.

*Transforms engine/config/*/.json directly into immutable ELF .rodata C byte tables at configure/build time.*

Wholesale Listing: Complete verbatim code (#L1-L302) → Appendix: Build-Time Compiled .rodata Config ROM Generator.

That single build step eliminates drift between engine JSON schemas and emergency defaults: when a disk config is missing or corrupted, Tier-0 reads the **exact same canonical JSON bytes** straight out of executable read-only memory!

## 66.3. Safe-Mode Boot, Floppy Floor & Downgrade Transitions

When io_fallback encounters missing files, path traversal (..), or read-only disk conditions, it executes structured transitions between full boot (io_fallback_boot), stripped-down emergency floor (io_fallback_boot_floppy), and read-only lock (io_fallback_downgrade_safe_mode):

*Tier-0 I/O Safe-Mode BIOS Boot, Floppy Floor & Downgrade Latch (engine/src/io/io_fallback.c#L3309-L3332 · Mode: Group)*

```
bool io_fallback_boot_floppy(void)
{
    io_fallback_config_t floppy = s_io_floppy_template;
    if (io_fallback_load_rom_config("floppy_safe", &floppy) ||
        io_fallback_load_rom_config("default", &floppy)) {
        s_io_fb.rom_hydrated = true;
    } else {
        s_io_fb.rom_hydrated = false;
    }
    floppy.read_only_mode = true;
    floppy.loaded_from_data = false;
    if (floppy.profile_name[0] == '\0' || strcmp(floppy.profile_name, "default") == 0) {
        str_safe_copy(floppy.profile_name, IO_FALLBACK_FLOPPY_PROFILE_NAME,
                      sizeof(floppy.profile_name));
    }

```

```
3325     s_io_fb.config = floppy;
3326     io_fb_hydrate_subpolicies();
3327     s_io_fb.active_sink = IO_FB_SINK_FLOPPY_RO;
3328     str_safe_copy(s_io_fb.active_root, IO_FB_DEFAULT_ROOT, sizeof(s_io_fb.active_root));
3329     s_io_fb.booted = true;
3330     io_fallback_register_debug_probes();
3331     return true;
3332 }
```

*Zero-dependency Safe-Mode transitions ensuring deterministic boot even when disk configs are missing or corrupt.*

Wholesale Listing: Complete verbatim code (#L2908-L3332) → Appendix: Tier-0 I/O Safe-Mode BIOS Boot, Floppy Floor & Downgrade Latch.

## 66.4. Flight Recorder (log/), Debugger (debug/) & Autonomous Probe Loop

Two foundational Layer 0 systems watch every transition:

- **Flight Recorder (cog.log):** Zero-heap categorized circular ring buffer preserving telemetry before filesystem logs even open. Sub-Assembly 4.1 wires health probes straight into the debugger:

*Flight Log Subsystem Probe Registration & Validation Guards (Sub-Assembly 4.1) (engine/src/log/log.c#L1934-L1977 · Mode: Subassembly)*

```
1934 /*
1935 ------------------------------------------------------------------------------
1936 Sub-Assembly 4.1: Contract, Input & Cog Verification Cogs
1937 Mission: Verify received parameters, configuration contracts, and data provenance;
1938 verify that declared subsystem cogs run properly as standalone units.
1939 ------------------------------------------------------------------------------
1940 */
1941
1942 /**
1943  * @brief Returns whether the active logging subsystem was configured from data sources.
1944  *
1945  * ### For Game Designers & Non-Programmers:
1946  * Checks if settings were loaded from actual JSON files rather than emergency defaults.
1947  *
1948  * ### For Systems Programmers & Engine Architects:
1949  * Queries provenance boolean s_log.config.loaded_from_data.
1950  *
1951  * @return true if loaded from JSON data; false if using fallback code page.
1952  */
1953 bool log_is_data_driven(void)
1954 {
1955     return s_log.config.loaded_from_data;
1956 }
1957
1958 /**
1959  * @brief Queries whether the logging engine is currently operating in Safe Mode.
1960  *
1961  * ### For Game Designers & Non-Programmers:
1962  * Returns true if the engine booted using compiled-in ROM backups or emergency
1963  * defaults instead of live external JSON directories on disk.
1964  *
1965  * ### For Systems Programmers & Engine Architects:
1966  * Checks initialization state alongside s_log.safe_mode (!loaded_from_data).
1967  *
1968  * @return true if active and running in Safe Mode; false otherwise.
1969  */
1970 bool log_is_safe_mode(void)
1971 {
1972     return s_log.initialized && s_log.safe_mode;
1973 }
1974
1975 /**
1976  * @brief Returns the current state of the logging data pipeline tripwire.
1977  *
```

*Shows how cog.log verifies its own static ring limits and binds probes into cog.debug.*

Wholesale Listing: Complete verbatim code (#L1934-L2028) → Appendix: Flight Log Subsystem Probe Registration & Validation Guards (Sub-Assembly 4.1).

- **Time-Machine Debugger (cog.debug):** Captures deterministic FNV-1a hashed state snapshots, evaluates active invariants, and coordinates the **Autonomous Probe → Safe-Mode → DEBUG_STEP_AWAIT** loop:

*Autonomous Validation Probes & Tripwire Trap Chain (engine/src/debug/debug.c#L2367-L2411 · Mode: Group)*

```
2367 /**
2368  * @brief Legacy compatibility helper that latches a state-corruption tripwire trap.
2369  *
2370  * @deprecated Prefer `debug_trigger_probe()` or `DEBUG_TRIGGER_PROBE()` so every trap
2371  *             carries a structured probe ID, tier classification, and forensic report.
2372  *
2373  *   ### For Game Designers & Non-Programmers:
2374  *   Preserved safety brake for older modules that report state errors without
2375  *   naming a specific modern probe identifier.
2376  *
2377  *   ### For Systems Programmers & Engine Architects:
2378  *   Latches `DEBUG_TRIPWIRE_STATE_CORRUPT`, increments `traps_triggered`, notifies
2379  *   BIOS via `debug_fallback_record_fault()`, performs guarded snapshot lock and
2380  *   `DEBUG_STEP_AWAIT` entry when active and non-audit, and emits `LOG_WARN`.
2381  *
2382  * @param[in] subsystem Subsystem tag reporting the fault, or NULL for `"unknown"`.
2383  * @param[in] condition Description of the violated condition, or NULL for `"anomaly"`.
2384  * @param[in] file      Originating source file (`__FILE__`).
2385  * @param[in] line      Originating source line (`__LINE__`).
2386  * @triggers Latches tripwire state, records BIOS fault, and warns via guarded log output.
2387  */
2388 void debug_trap_tripwire(const char *subsystem, const char *condition,
2389                          const char *file, int line)
2390 {
2391     /* Step 1: Latch classic STATE_CORRUPT code and forensic detail string verbatim */
2392     s_debug.tripwire_status = DEBUG_TRIPWIRE_STATE_CORRUPT;
2393     s_debug.traps_triggered++;
2394
2395     snprintf(s_debug.tripwire_details, sizeof(s_debug.tripwire_details),
2396              "[%s] %s (%s:%d)",
2397              (subsystem != NULL) ? subsystem : "unknown",
2398              (condition != NULL) ? condition : "anomaly",
2399              (file != NULL) ? file : "unknown", line);
2400
2401     /* Step 2: Record fault in Tier-0 BIOS and complete via Safe-Mode fallback */
2402     (void)debug_fallback_record_fault(
2403         subsystem,
2404         (condition != NULL) ? condition : "anomaly",
2405         file,
2406         line,
2407         "debug_trap_tripwire"
2408     );
2409
2410     if (s_debug_active && !s_debug.in_probe && !s_debug.in_audit) {
2411         (void)debug_fallback_finish_with_fallback(
```

*Realizes the Probe → Safe-Mode Downgrade → DEBUG_STEP_AWAIT loop inside Layer 0 debug infrastructure.*

Wholesale Listing: Complete verbatim code (#L1918-L2438) → Appendix: Autonomous Validation Probes & Tripwire Trap Chain.

## Anatomy of a Resilient Failure Recovery

1. A subsystem detects invalid config bounds or corrupt JSON in its Block 4 validation probe.
2. Instead of calling `abort()` or continuing silently with corrupted math, the probe traps into `debug_trap_tripwire()` / Safe-Mode downgrade (`loaded_from_data = false`).

3. Execution enters `DEBUG_STEP_AWAIT`, holding the frame intact with a human-readable reason, fault ring trace, and snapshot state so developers or tests can inspect **why** right away.

## 66.5. Self-Audit & Zero-Disk Checks

```
# Run Layer 0 log, debug, and io integration tests through CTest
ctest --test-dir build -R "log|debug|io|fallback|rom" --output-on-failure

# Boot headless audit cleanly even against an empty data root
./build/tenebris-lux --audit --data-root /tmp/empty_safe_mode_test
```

1. *If a designer accidentally deletes a JSON file, does the engine lose all defaults?*

   No. `GenerateConfigRom.cmake` embeds verified JSON defaults in `.rodata`, so Tier-0 BIOS boots cleanly with `loaded_from_data == false`.

2. *Does Safe-Mode silently pretend nothing went wrong?*

   No. Telemetry logs a clear downgrade event, locks unsafe disk mutations if damaged, and fires validation probes toward `DEBUG_STEP_AWAIT`.

## 66.6. Where to Go Next

- Previous: Chapter 1 — System Architecture Overview.
- Next: Chapter 3 — Data-Driven Architecture: Dual Data Taxonomy, Layered VFS & 5-Block JSON.
- Code Tours: See Part 2 chapters for Flight Logging, State Debugger, and Zero-Heap I/O Pipeline.

# Chapter 67. Data-Driven Architecture: Dual Data Taxonomy, Layered VFS & 5-Block JSON

**Source File**: `docs/architecture/data_driven_architecture.adoc`

You should never need to recompile C source code just to adjust a damage curve, swap a font theme, remap controls, or tweak a debug threshold. **Cornerstone Game Engine** is engineered so creators work in structured **5-Block JSON**, while compiled Rust tooling (`tl_tools`) and C23 zero-heap loaders keep every setting trustworthy at runtime.

## 67.1. Dual Data Taxonomy: Engine Profiles vs. Game Canon

Cornerstone strictly separates **how the machine runs** from **what story universe it runs**:

| Data Domain | Canonical Root | Example Subtrees & Contents |
|---|---|---|
| Machine Profiles (Engine) | `engine/config/` | • `viewport/`: window layout, `640x360` scalers, themes, typography, assets.<br>• `audio/`, `input/`: mixer channels, default keyboard & gamepad bindings.<br>• `runtime/`, `io/`: math formulas, VFS routes, filesystem buffer policies.<br>• `diagnostics/`, `standards/`: telemetry rules, probe matrix, 5-block specs. |
| Domain Canon (Game) | `game/data/` | • `entities/`: vocations, roster templates, protagonist stats.<br>• `mechanics/`: Twin Relics, combat skills, tactical terrain, dispatch scripts.<br>• `world/`, `presentation/`: Libira & Sacyra provinces, UI data overrides. |



**Why split these trees?** A new game built on **Cornerstone Game Engine** replaces `game/data/` completely while keeping `engine/config/` intact. Meanwhile, **Tenebris and Lux** can override individual engine defaults cleanly through VFS precedence without mutating upstream engine baselines.

## 67.2. Layered VFS Precedence (`path_finder`)

Instead of scattering raw string paths like `../../data/foo.json` across C files, every subsystem queries logical domains through `path_finder` (`engine/include/io/path_finder.h`). For the `"config"` domain, `vfs_routes.json` defines an explicit override cascade:

1. `game/config/<rel_path>` — Game-specific or user mod override (highest priority).
2. `engine/config/<rel_path>` — Standalone engine baseline default.
3. `game/data/presentation/config/<rel_path>` — Symbolic presentation linkers.

Examine the live **`vfs_routes.json`** configuration (`whole` file tracked via Appendix A, with `block_3_body` highlighted inline):

*Complete 5-Block VFS Precedence & Policy Configuration (vfs_routes.json) (engine/config/io/vfs_routes.json#L55-L99 · Mode: Whole)*

```
55  "block_3_body": {
56    "default_data_root": "game/data",
57    "env_override_var": "TL_DATA_ROOT",
58    "block_path_traversal": true,
59    "max_symlink_depth": 4,
60    "allow_mod_mounts": true,
61    "rom_fallback_enabled": true,
62    "probe_await_on_corrupt": true,
63    "config_precedence": [
64      "game/config/",
65      "engine/config/",
66      "game/data/presentation/config/"
67    ],
68    "fallback_candidate_roots": [
69      "game/data",
70      "../game/data",
71      "engine/config",
72      "../engine/config",
73      "data",
74      "../data"
75    ],
76    "asset_roots": [
77      "game/assets",
78      "../game/assets",
79      "assets",
80      "../assets"
81    ],
82    "domain_search_paths": [
83      "",
84      "runtime/",
85      "runtime/math/",
86      "runtime/string/",
87      "runtime/time/",
88      "diagnostics/",
89      "diagnostics/telemetry/",
90      "diagnostics/debugger/",
91      "diagnostics/taxonomy/",
92      "diagnostics/formats/",
93      "io/",
94      "audio/",
95      "viewport/",
96      "input/",
97      "standards/",
98      "standards/blocks/",
99      "mechanics/",
```

*Canonical 5-block engine config defining VFS root search hierarchy, alias tables, and fail-closed safety flags.*

Wholesale Listing: Complete verbatim code (#L1-L159) → Appendix: Complete 5-Block VFS Precedence & Policy Configuration (vfs_routes.json).

## 67.3. The 5-Block JSON Contract & block_3_body Isolation

Just like source files and documentation chapters, every data config shares the exact 5-block skeleton ( block_1_metadata, block_2_types/block_2_opening, block_3_body, block_4_audit, block_5_closing).

To prevent accidental leaks where an authoring invariant string in metadata could be misparsed as a live setting, master_loader **never** reads top-level metadata during runtime config queries. It isolates block_3_body first—shown directly by this block-granularity JSON snapshot:

*JSON Block 3 Payload Isolation (block_3_body of vfs_routes.json) (engine/config/io/vfs_routes.json#L55-L102 · Mode: Block)*

```
55  "block_3_body": {
```

```
    "default_data_root": "game/data",
    "env_override_var": "TL_DATA_ROOT",
    "block_path_traversal": true,
    "max_symlink_depth": 4,
    "allow_mod_mounts": true,
    "rom_fallback_enabled": true,
    "probe_await_on_corrupt": true,
    "config_precedence": [
      "game/config/",
      "engine/config/",
      "game/data/presentation/config/"
    ],
    "fallback_candidate_roots": [
      "game/data",
      "../game/data",
      "engine/config",
      "../engine/config",
      "data",
      "../data"
    ],
    "asset_roots": [
      "game/assets",
      "../game/assets",
      "assets",
      "../assets"
    ],
    "domain_search_paths": [
      "",
      "runtime/",
      "runtime/math/",
      "runtime/string/",
      "runtime/time/",
      "diagnostics/",
      "diagnostics/telemetry/",
      "diagnostics/debugger/",
      "diagnostics/taxonomy/",
      "diagnostics/formats/",
      "io/",
      "audio/",
      "viewport/",
      "input/",
      "standards/",
      "standards/blocks/",
      "mechanics/",
      "mechanics/items/",
      "mechanics/combat/",
      "mechanics/tactics/",
```

*Demonstrates scoped block_3_body extraction—matching master_loader's strict rule against reading Block 1/2 metadata as live gameplay config.*

Wholesale Listing: Complete verbatim code (#L55-L125) → Appendix: JSON Block 3 Payload Isolation (block_3_body of vfs_routes.json).

## 67.4. Zero-Heap Scoped Accessors (`master_loader`) & Shared Span Parsers

In Layer 0 (ALLOC_ZERO_HEAP), parsing JSON must allocate **zero heap memory**. Caller buffers (fs_utils), Tier-0 span scanners (io_fallback), and typed scoped accessors (master_loader) collaborate over caller-owned buffers:

*Scoped block_3_body JSON Accessor Contract (master_loader.h Sub-Assembly 3.2)*
*(engine/include/io/master_loader.h#L436-L479 · Mode: Subassembly)*

```
/*
--------------------------------------------------------------------------------
Sub-Assembly 3.2: Scoped `block_3_body` Accessors
Mission: Restrict all payload getters and array locators strictly inside the
         verified `[body_start, body_start + body_len)` span (or named `sub_obj`).

  ### For Game Designers & Non-Programmers:
```

```
 Reads gameplay properties cleanly from inside `block_3_body` only, preventing
 file headers or audit sections from ever colliding with gameplay field names.

 ### For Systems Programmers & Engine Architects:
 Delegates scope isolation and typed extraction exclusively to span-bounded
 `json_*_span` APIs with unconditional NUL termination on extracted strings.
--------------------------------------------------------------------------------
*/

/**
 * @brief Extracts a string value strictly within `block_3_body` (or named sub-object).
 *
 * ### For Game Designers & Non-Programmers:
 * Safely reads text properties (character names, formulas, IDs) from the data body
 * without accidentally matching header text outside Block 3.
 *
 * ### For Systems Programmers & Engine Architects:
 * Uses span-bounded `json_find_object_span()` + `json_get_string_span()` bounded to
 * `loader->body_len`. Unconditionally NUL-terminates `out_val` when `out_size > 0`.
 * @triggers `io.json.depth_bounds` if scan breaches `max_json_depth`.
 *
 * @param[in]  loader   Ready `master_loader_t` instance.
 * @param[in]  sub_obj  Optional sub-object key inside `block_3_body` (or NULL for root body).
 * @param[in]  key      Target property key inside resolved body/sub-object span.
 * @param[out] out_val  Caller string buffer.
 * @param[in]  out_size Capacity of `out_val` in bytes.
 * @return true on clean extraction; false if missing, wrong type, or invalid params.
 */
bool master_loader_get_string(const master_loader_t *loader,
                              const char *sub_obj,
                              const char *key,
                              char *out_val,
                              size_t out_size);

/**
 * @brief Extracts a signed integer strictly within `block_3_body` (or named sub-object).
 *
```

*Zero-heap typed getters that scan directly inside a validated block_3_body span.*

Wholesale Listing: Complete verbatim code (#L436-L608) → Appendix: Scoped block_3_body JSON Accessor Contract (master_loader.h Sub-Assembly 3.2).

Under the hood, both BIOS boot routines and master_loader reuse the same shared span-resolution cogs inside engine/src/io/io_fallback.c:

*Shared Zero-Heap JSON Block Resolver & VFS Parser Group (engine/src/io/io_fallback.c#L779-L806 · Mode: Group)*

```c
static bool io_fb_resolve_body_span(const char *json_text,
                                    size_t json_len,
                                    bool allow_strlen,
                                    const char **out_body_start,
                                    const char **out_body_end)
{
    if (json_text == NULL || out_body_start == NULL || out_body_end == NULL) {
        return false;
    }

    size_t effective_len = (json_len > 0U) ? json_len : (allow_strlen ? strlen(json_text) : 0U);
    if (effective_len == 0U || effective_len > IO_FB_JSON_MAX_BYTES) {
        return false;
    }

    const char *json_end = json_text + effective_len;
    const char *body_start = NULL;
    size_t body_len = 0U;
    if (!io_fb_find_block_span(json_text, json_end, "block_3_body", '{', '}',
                               &body_start, &body_len) ||
        body_start == NULL || body_len == 0U) {
        return false;
    }

```

```
803     *out_body_start = body_start;
804     *out_body_end = body_start + body_len;
805     return true;
806 }
```

*Tier-0 in-place span scanning reused seamlessly between BIOS fallback and master_loader.*

Wholesale Listing: Complete verbatim code (#L779-L1194) → Appendix: Shared Zero-Heap JSON Block Resolver & VFS Parser Group.

### Where Publication & Theme Configs Plug In

Even this textbook manual follows the exact same rule: docs/master_guide_manifest.json, docs/guide_theme_config.json, docs/code_snapshots_manifest.json, and docs/adoc_5block_standard.json drive typography, PDF margins, code extraction modes, and chapter assembly purely from JSON data rather than Rust constants.

## 67.5. Validating Data Files Before Commit

```
# Audit all engine/config/ and game/data/ 5-block manifests & JSON schemas
./tools/target/debug/tl_tools validate all

# Launch with a custom mod/testing mount via CLI override
./build/tenebris-lux --audit --data-root game/data
```

1. *Can* master_loader *accidentally read a description string from* block_1_metadata*?*

   No. Every typed lookup in master_loader first narrows the scan window strictly inside "block_3_body".

2. *Who checks that 5-block JSON configs remain well-formed?*

   tl_tools validate all (tools/src/validator/json_data.rs) verifies every config file under engine/config/ and game/data/ before compilation or commits.

## 67.6. Where to Go Next

- Previous: Chapter 2 — Zero-Disk Safe-Mode BIOS, Config ROM Vaults & Autonomous Probe Loop.
- Next: Chapter 4 — Central Manifest Registry & Pre-Flight Loose-Bolts Inspection.
- Code Tour: Explore the I/O module details in Layer 0 Zero-Heap I/O Pipeline.

# Chapter 68. Central Manifest Registry & Pre-Flight Loose-Bolts Inspection

**Source File**: docs/architecture/preflight_and_manifest_spec.adoc

Before an aircraft leaves the runway, technicians inspect critical fasteners, sensor buses, and fuel lines in a strict checklist order. In the **Cornerstone Game Engine**, **cog.manifest** (engine/include/core/manifest.h) executes that exact pre-flight walk at startup: verifying that every engine subsystem exists, obeys its dependency DAG, honors ALLOC_ZERO_HEAP, and passes internal self-checks before a single turn of gameplay executes.

## 68.1. Self-Describing Module Metadata & Registry Ontology

Every subsystem exposes a module_metadata_t descriptor (covering up to 32 modules in a static .bss manifest_registry_t) declaring its identity, layer, memory contract, upstream dependencies, and preflight callback.

Examine the canonical structs extracted from engine/include/core/manifest.h Sub-Assembly 2.3:

*Central Manifest Metadata & Registry Ontology (manifest.h Sub-Assembly 2.3)*
*(engine/include/core/manifest.h#L207-L254 · Mode: Subassembly)*

```
207 /*
208 ------------------------------------------------------------------------------
209 Sub-Assembly 2.3: Structural Entity & Object Definitions
210 ------------------------------------------------------------------------------
211 Ontological types for architectural strata, memory models, and diagnostic audits.
212 ------------------------------------------------------------------------------
213 */
214
215 /**
216  * @enum module_layer
217  * @brief Architectural layer strata within the hierarchical engine topology.
218  *
219  * For Designers: The 5 vertical tiers of the game engine, from deep bedrock
220  * utilities (Layer 0) up to the game rules (Layer 1) and graphical display (Layer 2).
221  *
222  * For Programmers: Strict downward dependency hierarchy enforced at compile time.
223  */
224 typedef enum module_layer {
225     LAYER_0_CORE_UTIL = 0, /**< Layer 0: Foundational zero-dependency utils. */
226     LAYER_1_SIMULATION,    /**< Layer 1: Headless game simulation engine. */
227     LAYER_2_PRESENTATION,  /**< Layer 2: Graphical and text viewports. */
228     LAYER_3_APPLICATION,   /**< Layer 3: Application entry and routing. */
229     LAYER_4_TESTS,         /**< Layer 4: Verification test harness suites. */
230     LAYER_COUNT
231 } module_layer_t;
232
233 /**
234  * @enum alloc_policy
235  * @brief Memory allocation contract governing a module's execution model.
236  *
237  * For Designers: Rules defining how a system manages memory, ensuring bedrock
238  * systems never allocate heap memory to prevent crashes and stutter.
239  *
240  * For Programmers: Invariant memory policy verified during code audit.
241  */
242 typedef enum alloc_policy {
243     ALLOC_ZERO_HEAP = 0, /**< 100% stack or static layout; zero dynamic heap. */
244     ALLOC_FIXED_POOL,    /**< Pre-allocated fixed-capacity buffer pools. */
245     ALLOC_ARENA,         /**< Frame/scene temporary arena allocation. */
246     ALLOC_DYNAMIC        /**< Managed dynamic heap allocation. */
247 } alloc_policy_t;
248
249 /**
250  * @enum concurrency_policy
251  * @brief Concurrency and reentrancy guarantees declared by a module.
252  *
```

```
 * For Designers: Rules specifying whether a system can run simultaneously across
 * multiple CPU threads or must remain tied to the main display loop.
```

*Self-describing module contracts pairing engine layers, allocation policy, thread safety, and preflight hooks.*

Wholesale Listing: Complete verbatim code (#L207-L370) → Appendix: Central Manifest Metadata & Registry Ontology (manifest.h Sub-Assembly 2.3).

| Preflight Outcome | Severity Classification | Engine Behavior at Startup |
|---|---|---|
| PREFLIGHT_STATUS_OK | SEVERITY_PASS | Nominal green boot; full data-driven subsystems active. |
| PREFLIGHT_STATUS_WARNINGS | SEVERITY_WARNING | Boot continues with telemetry warnings (e.g., optional enhancement missing). |
| PREFLIGHT_STATUS_DOWNGRADED | SEVERITY_CRITICAL | Safe-Mode fallback engaged; game runs smoothly on .rodata ROM defaults (loaded_from_data = false). |
| PREFLIGHT_STATUS_HALT_FATAL | SEVERITY_FATAL | Immediate preflight halt (--audit returns non-zero); triggers tripwire report before corrupted state spreads. |

## 68.2. The Pre-Flight Walk (manifest_run_preflight)

At startup or under ./tenebris-lux --audit, manifest_run_preflight() performs two deterministic passes: first verifying every declared upstream dependencies[] entry exists in the registry (preventing hidden layer inversions), then executing each subsystem's registered preflight_check() hook:

*Pre-Flight Loose-Bolts Engine Walk (manifest_run_preflight) (engine/src/core/manifest.c#L412-L455 · Mode: Function)*

```c
/**
 * @brief Executes comprehensive pre-flight diagnostics across all registered modules.
 */
preflight_status_t manifest_run_preflight(manifest_registry_t *registry)
{
    if (registry == NULL) {
        debug_trap_tripwire("manifest", "NULL registry passed to manifest_run_preflight", __FILE__, __LINE__);
        return PREFLIGHT_STATUS_HALT_FATAL;
    }

    registry->pass_count = 0;
    registry->warn_count = 0;
    registry->critical_count = 0;
    registry->fatal_count = 0;

    /* Step 1: Active DAG Dependency Graph Verification */
    for (size_t i = 0; i < registry->count; i++) {
        const module_metadata_t *mod = registry->modules[i];
        if (mod == NULL) continue;

        for (size_t d = 0; d < mod->dependency_count; d++) {
            const char *req_dep = mod->dependencies[d];
            if (req_dep == NULL) continue;

            bool found = false;
            for (size_t j = 0; j < registry->count; j++) {
                if (registry->modules[j] != NULL &&
                    strcmp(registry->modules[j]->module_id, req_dep) == 0) {
                    found = true;
                    break;
                }
            }

            if (!found) {
                diag_result_t dep_res;
```

```
447                 dep_res.passed = false;
448                 dep_res.severity = SEVERITY_FATAL;
449                 dep_res.subsystem = mod->module_id;
450                 dep_res.check_name = "Dependency DAG";
451                 dep_res.details =
452                     "Fatal: Unregistered upstream dependency declared.";
453                 registry->results[i] = dep_res;
454                 registry->fatal_count++;
455                 registry->status = PREFLIGHT_STATUS_HALT_FATAL;
```

*Iterates every registered module sequentially, classifying OK, WARNINGS, DOWNGRADED, and HALT_FATAL states.*

Wholesale Listing: Complete verbatim code (#L412-L509) → Appendix: Pre-Flight Loose-Bolts Engine Walk (manifest_run_preflight).

## 68.3. Layer 3 Plug Audit Flow (game/src/main.c BLOCK 4)

The Layer 3 application plug bridges engine preflight checks, Layer 1 gameplay invariant audits ( game_run_self_audit()), and Layer 2 viewport bounds audits (gui_audit_layout()) inside one inspectable audit routine:

*Application Audit Integration (game/src/main.c BLOCK 4) (game/src/main.c#L193-L238 · Mode: Block)*

```
193 // ============================================================================
194 // BLOCK 4: AUDIT PROCEDURES BLOCK
195 // Stratum: Layer 3 (Application Entry) | Machine: Presentation Router
196 // ============================================================================
197
198 // ----------------------------------------------------------------------------
199 // [ SUB-ASSEMBLY 4.1: COMPREHENSIVE SUBSYSTEM SELF-AUDIT SUITE ]
200 // Mission: Execute pre-flight manifest checks, sim invariants, and GUI bounds.
201 // ----------------------------------------------------------------------------
202
203 /**
204  * @brief Executes the comprehensive engine self-audit suite and manifests.
205  *
206  * ### Machine Role & Mission
207  * Serves as the `--audit` diagnostic tripwire. Executes:
208  * 1. Pre-flight verification across all registered modules with DAG checks.
209  * 2. Layer 1 simulation invariant audit (`game_run_self_audit()`).
210  * 3. Layer 2 GUI viewport layout and collision audit (`gui_audit_layout()`).
211  * Returns non-zero if any invariant check or pre-flight test fails.
212  *
213  * ### Ontological Classification
214  * - Stratum: Layer 3 (Application Entry & Presentation Router)
215  * - Machine: app.entry.main (Executable Entry & Presentation Router)
216  * - Assembly: Diagnostic Self-Audit & Pre-Flight Verification
217  * - Cog: Full Engine Diagnostics & Invariant Self-Audit Suite
218  *
219  * ### Execution & Resource Contract
220  * - Memory Model: ALLOC_ZERO_HEAP (Stack allocation only).
221  * - Concurrency: CONCURRENCY_MAIN_THREAD (Synchronous diagnostic sweep).
222  * - Latency Budget: < 5000 microseconds execution time.
223  *
224  * @return 0 if all invariant checks pass; non-zero if any check fails.
225  */
226 static int run_full_subsystem_audit(void)
227 {
228     log_init(NULL);
229     LOG_INFO(LOG_CAT_LOG, "Audit", 0.0f, "Starting comprehensive diagnostic self-audit");
230
231     /* Step 1: Initialize central manifest and run pre-flight diagnostics */
232     manifest_registry_t registry;
233     init_engine_manifest(&registry);
234     manifest_run_preflight(&registry);
235     manifest_print_report(&registry);
236
237     /* Step 2: Evaluate pre-flight outcome for fatal loose-bolts */
238     if (registry.status == PREFLIGHT_STATUS_HALT_FATAL) {
```

*Shows Layer 3 executing run_full_subsystem_audit() across Engine + Game + Test invariants from --audit.*

Wholesale Listing: Complete verbatim code (#L193-L334) → Appendix: Application Audit Integration (game/src/main.c BLOCK 4).



**Zero Silent Failures:** Notice how `manifest_execute_dump_card()` lets operators inspect live memory budgets, subsystem statuses, and dependency trees interactively in `--debug` or terminal output—turning engine boot state into a readable inspection card rather than opaque startup noise.

## 68.4. Running Pre-Flight Diagnostics Yourself

```
# Execute full Layer 3 + Engine + Sim + GUI diagnostic sweep
./build/tenebris-lux --audit

# Validate C header manifest metadata & DAG registrations across all 44 engine headers
./tools/target/debug/tl_tools validate headers
```

1. *What happens if a module declares a dependency that was never registered?*

   Step 1 of `manifest_run_preflight()` catches the missing upstream node before any subsystem initialization runs, traps a tripwire, and returns `PREFLIGHT_STATUS_HALT_FATAL`.

2. *Does `tl_tools validate headers` cross-check these C manifest rules too?*

   Yes. `tools/src/validator/c_header.rs` checks all 44 Cornerstone engine headers against `engine/config/standards/subsystems.json` and the 5-block standard.

## 68.5. Where to Go Next

- Previous: Chapter 3 — Data-Driven Architecture: Dual Data Taxonomy, Layered VFS & 5-Block JSON.
- Next: Chapter 5 — Dual-Frontend Presentation Layer: CLI, Native X11 & Movie Reel.
- Deep Dive: Read Layer 0 Core Utilities Code Tour.

# Chapter 69. Dual-Frontend Presentation Layer: CLI, Native X11 & Movie Reel

**Source File**: `docs/architecture/dual_frontend_architecture.adoc`

Many RPG engines tangle combat formulas directly into UI button handlers or draw loops—making automated testing brittle and porting nearly impossible. The **Cornerstone Game Engine** splits **Simulation Truth (Layer 1)** cleanly from **Presentation Viewports (Layer 2)** so **Tenebris and Lux** plays identically in a lightning-fast headless CLI session, inside the `--debug` time-machine REPL, or on a pixel-crisp `640x360` native X11 desktop window.

## 69.1. One Pure Simulation, Multiple Frontends

Every gameplay rule lives inside `game/` (Layer 1) operating strictly over `player_state_t`, pure integer coordinates, and deterministic combat/morality dispatch. Neither headless text commands nor graphical mouse clicks bypass Layer 1 validation.

Consider `sim_state_audit()` in `game/src/hero/sim_state.c`—a complete pure-C simulation audit `function` that operates cleanly without opening any X11 display or GUI window:

*Layer 1 Deterministic Simulation State Audit (sim_state_audit) (`game/src/hero/sim_state.c#L315-L376` · Mode: `Function`)*

```
315 game_audit_report_t sim_state_audit(const player_state_t *state)
316 {
317     game_audit_report_t report;
318     report.is_valid = true;
319     report.error_count = 0;
320     report.error_log[0] = '\0';
321 
322     if (state == NULL) {
323         report.is_valid = false;
324         report.error_count = 1;
325         snprintf(report.error_log, sizeof(report.error_log),
326                  "[ERROR] Player state pointer is NULL.");
327         return report;
328     }
329 
330     if (state->shield_power < 0 || state->shield_power > MAX_RESONANCE) {
331         report.is_valid = false;
332         report.error_count++;
333         snprintf(report.error_log + strlen(report.error_log),
334             sizeof(report.error_log) - strlen(report.error_log),
335             "[ERROR] Shield power (%d) outside bounds [0, %d]. ",
336             state->shield_power, MAX_RESONANCE);
337     }
338 
339     if (state->sword_edge < 0 || state->sword_edge > MAX_RESONANCE) {
340         report.is_valid = false;
341         report.error_count++;
342         snprintf(report.error_log + strlen(report.error_log),
343             sizeof(report.error_log) - strlen(report.error_log),
344             "[ERROR] Sword edge (%d) outside bounds [0, %d]. ",
345             state->sword_edge, MAX_RESONANCE);
346     }
347 
348     if (state->moral_sway < -100 || state->moral_sway > 100) {
349         report.is_valid = false;
350         report.error_count++;
351         snprintf(report.error_log + strlen(report.error_log),
352             sizeof(report.error_log) - strlen(report.error_log),
353             "[ERROR] Moral sway (%d) outside bounds [-100, 100]. ",
354             state->moral_sway);
355     }
356 
357     if (state->stance < 0 || state->stance >= STANCE_COUNT) {
```

```
        report.is_valid = false;
        report.error_count++;
        snprintf(report.error_log + strlen(report.error_log),
            sizeof(report.error_log) - strlen(report.error_log),
            "[ERROR] Stance enum (%d) outside valid range [0, %d). ",
            state->stance, STANCE_COUNT);
    }

    if (state->location < 0 || state->location >= LOC_COUNT) {
        report.is_valid = false;
        report.error_count++;
        snprintf(report.error_log + strlen(report.error_log),
            sizeof(report.error_log) - strlen(report.error_log),
            "[ERROR] Location enum (%d) outside valid range [0, %d). ",
            state->location, LOC_COUNT);
    }

    return report;
}
```

*Pure simulation integrity audit verifying hero stats, inventory bounds, morality sway, and world state with zero frontend dependencies.*

Wholesale Listing: Complete verbatim code (#L315-L376) → Appendix: Layer 1 Deterministic Simulation State Audit (sim_state_audit).

| Launch Flag (tenebris-lux) | Frontend Mode | Ideal Audience & Workflow |
|---|---|---|
| -g, --gui | Native X11 640x360 Viewport | Players, pixel artists, and UI designers reviewing animations, palettes, and 5-screen menus. |
| -c, --cli | Headless Terminal REPL | Playtesters, writers, and automated scripts stepping through choices and endings at CPU speed. |
| -d, --debug | Interactive Time-Machine Debugger | Engineers and QA inspecting FNV-1a hashes, rewinding snapshots, and checking DEBUG_STEP_AWAIT traps. |
| -a, --audit | Pre-Flight + Invariant Sweep | CI pipelines and local commits verifying boot health, zero-heap limits, and GUI panel bounds in <5ms. |

## 69.2. Native 640x360 Viewport & The 2.5D Movie Reel Model

Rather than pulling in external third-party UI frameworks or bloat, Layer 2 (engine/include/viewport/) implements a self-contained software rasterizer rendering directly to a native 640x360 logical canvas backed by <X11/Xlib.h> in Linux desktop builds.

Visually, the viewport works like a **Classic Cinema Movie Reel**:

- **Horizontal Strip Navigation:** Screens glide left and right cleanly across canonical game views (Title, Gameplay, Party/Inventory, Map/Journal, System Settings).
- **Vertical Depth Projection:** Integer tactical grid coordinates (gui_grid_coord_t) project cleanly through dimetric isometric transforms (gui_iso_point_t) and painter's-algorithm depth queues (gui_depth_entity_t).
- **Integer/Letterbox Scaling (geom_utils):** A fixed 640x360 logical frame scales cleanly onto any desktop monitor without stretching pixel art or leaking screen coordinates back into Layer 1 game mechanics.

Inspect the canonical Layer 2 data contracts extracted from engine/include/viewport/raster/gui_types.h Sub-Assembly 2.3:

*Layer 2 640x360 Viewport & 2.5D Movie Reel Data Structures (gui_types.h Sub-Assembly 2.3)*
*(engine/include/viewport/raster/gui_types.h#L138-L185 · Mode: Subassembly)*

```c
/*
--------------------------------------------------------------------------------
Sub-Assembly 2.3: Structural Entity & Object Definitions
--------------------------------------------------------------------------------
Core types, layout bounding rectangles, color models, view modes, and widget
structures for the desktop presentation engine.
--------------------------------------------------------------------------------
*/

/**
 * @enum gui_screen_mode
 * @brief Presentation screen views and modal navigation states.
 *
 * For Designers:
 * Represents the 5 primary game screens accessible via the top navigation tabs:
 * Tactical Grid, Hero Dossier, Inventory, Provincial Map, and Chronicle Codex.
 *
 * For Programmers:
 * Bounded enumerator indexed [0, 4] matching tab descriptors.
 */
typedef enum gui_screen_mode {
    GUI_SCREEN_TACTICAL = 0,    /**< 2.5D / 2D Tactical battlefield & combat */
    GUI_SCREEN_STATUS,          /**< Full protagonist dossier & attributes */
    GUI_SCREEN_INVENTORY,       /**< Bag, key items, Grandma's Watch */
    GUI_SCREEN_WORLD_MAP,       /**< Provincial realm topology & travel */
    GUI_SCREEN_CHRONICLE        /**< Complete lore codex & history of deeds */
} gui_screen_mode_t;

/**
 * @struct gui_tab
 * @brief Top navigation tab descriptor.
 *
 * For Designers:
 * A clickable header tab bar button linking directly to a primary screen.
 *
 * For Programmers:
 * Bounded rectangular button descriptor with single hotkey accelerator.
 */
typedef struct gui_tab {
    int x;
    int y;
    int width;
    int height;
    const char *label;
    gui_screen_mode_t screen;
    char hotkey;
} gui_tab_t;
```

*Canonical non-anonymous structs driving screens, palettes, layouts, and dimetric depth sorting.*

Wholesale Listing: Complete verbatim code (#L138-L448) → Appendix: Layer 2 640x360 Viewport & 2.5D Movie Reel Data Structures (gui_types.h Sub-Assembly 2.3).

### Future-Proof 3D Horizon Without Simulation Rewrites

Because player_state_t in Layer 1 tracks logical integers rather than X11 pixels or sprites, Cornerstone's Layer 2 viewport could evolve into hardware-accelerated 2D or full 3D rendering in the future without changing a single combat formula, dialogue branch, or save state inside game/.

## 69.3. Testing Both Frontends Cleanly

```
# Verify headless CLI, debugger, GUI layout audit, and X11 smoke checks via CTest
ctest --test-dir build --output-on-failure
```

```
# Run interactive headless gameplay or debugger directly in terminal
./build/tenebris-lux --cli
./build/tenebris-lux --debug
```

1. *Does testing the game require an active X11 monitor or GPU?*

   No. Headless `--cli`, `--debug`, `--audit`, and all simulation tests run cleanly in pure terminal/CI environments without opening an X11 window.

2. *How does the `640x360` GUI avoid getting out of sync with CLI gameplay?*

   Both frontends dispatch actions into the exact same Layer 1 `game/` state machine and share deterministic FNV-1a snapshot verification.

## 69.4. Where to Go Next

- Previous: Chapter 4 — Central Manifest Registry & Pre-Flight Loose-Bolts Inspection.
- Continue into **Volume V, Part 2** subsystem code tours starting at Layer 0 Core Utilities Code Tour and Layer 2 Presentation Viewport Code Tour.
- Review all extracted textbook code plates in Snapshots Index (or **Appendix A** when reading the compiled PDF book).

# Chapter 70. Root Strata Chronicles: Why Strengthening log/, debug/, and io/ Pays Every Future Turn

**Source File**: `docs/architecture/chronicles/01_root_stratum_story_and_game_payoff.adoc`

**For Game Designers & Creators**: It is tempting to ask why months of careful baseline work go into flight logs, diagnostic tripwires, and bounded file loaders before drawing a single new combat animation or writing a new dialogue scene in **Tenebris and Lux**. The answer is simple: **a game engine is a living tree, and `log/`, `debug/`, and `io/` are its roots**. When the roots rot—or silently allocate unmanaged memory, crash on a missing JSON comma, or swallow errors—every branch above them (vocations, twin relics, grid combat, X11 frames) becomes fragile and twice as slow to build.

**For Systems Architects**: During the Governance Revision 5 (`Rev 5`) baseline sprint of **Cornerstone Engine** (`engine/`), modernizing `log/` → `debug/` → `io/` ahead of `core/` Block 3 work, `audio/`, `viewport/`, and `game/` surfaced structural lessons that fundamentally de-risk the rest of the project.

| Stratum / Root | Without Rev 5 Hardening | With Rev 5 Root Baseline |
|---|---|---|
| `log/` (Flight Recorder) | Silent early-boot drops; interleaved disk crashes; ad-hoc `printf`. | Zero-disk `log_fallback` boot, typed severity rings, ROM defaults, and deterministic test resets. |
| `debug/` (Nervous System) | Bugs manifest five systems later as cryptic segfaults with zero state trail. | Probes catch bad invariants at the boundary and pause cleanly inside `DEBUG_STEP_AWAIT`. |
| `io/` (Supply Line) | Duplicate JSON parsers, hidden `static char[32768]` scratch bloat, fragile file paths. | 3-tier VFS (`game/config` → `engine/config` → `presentation`), `rom://io`, and `block_3_body` isolation. |

## 70.1. Lesson 1: Purging Legacy Shims Lets the Compiler Map Every Seam

Early in engine refactors, keeping "temporary compatibility wrappers" feels polite—nothing breaks immediately. In practice, shims hide stale callers across multiple modules until a subtle behavioral mismatch blows up weeks later inside gameplay code.

Cornerstone's **Active Iteration Feedback Covenant** flips that habit on its head:

1. **Delete retired shims immediately** from headers and sources.
2. **Compile strictly** (`-Wall -Wextra -Wpedantic`, zero suppressed warnings) and run the 4-tier CTest suite so the compiler itself points to every single outdated call site.
3. **Upgrade every caller cleanly** to the true contract before advancing.

By paying that small upfront cleanup cost across `log/` and `debug/`, `io/` inherited clean caller patterns from day one—and future `core/`, `audio/`, and `game/` refactors will never trip over ghost wrappers left behind in `engine/`.

## 70.2. Lesson 2: Splitting Pure Types & Measuring the Chassis Eliminates Hidden Bloat

Two physical changes transformed dependency flow and memory visibility across the root strata:

- **Quarantined Leaf Type Headers (`*_types.h`)**: Separating plain data definitions (`log_types.h`, `debug_types.h`) from behavioral headers broke circular header includes between full engines and Safe-Mode BIOS fallbacks (`*_fallback.[ch]`), keeping the dependency graph strictly acyclic.
- **Controlled Memory Chassis (`ALLOC_ZERO_HEAP`)**: Because Layers 0, 1, and 2 allow zero dynamic heap allocation, every single byte lives visibly in ELF `.rodata`, `.data`, `.bss`, and bounded stack frames. Auditing `io/` revealed how easily forgotten function-local `static char[]` scratch buffers quietly inflate `.bss`; moving temporary scans onto tight caller slices shrank engine chassis overhead so remaining fixed capacity serves genuine RPG growth (units, vocations, skill catalogs, turn history) rather than dead engine scratch space.

## 70.3. Lesson 3: Building Forward for C23/C20 While Anchoring a C11/C17 Floor

Cornerstone treats portability like a **Java-style capability ladder** built directly inside bedrock `engine/include/core/manifest.h`:

*Cornerstone Fall-Down Stairs Portability Ladder (C23/C17/C11 & C++20/17) (`engine/include/core/manifest.h#L82-L143` · Mode: `Subassembly`)*

```
 82 /*
 83 ------------------------------------------------------------------------------
 84 Sub-Assembly 2.1: Pre-Conditions, Progressive Standard Macros & Buffer Limits
 85 ------------------------------------------------------------------------------
 86 Compile-time sizing limits for registry module nodes, dependencies, rules, and
 87 Cornerstone's "Fall-Down" ISO C23 / C17 / C11 + C++20 / C++17 capability macros.
 88
 89 For Designers:
 90 Allows Cornerstone to take advantage of modern compiler safety checks on current
 91 hardware while stepping cleanly onto older systems without legacy clutter. Also
 92 bounds the engine registry to up to 32 major systems, 8 core rules, and 4 links.
 93
 94 For Programmers:
 95 - Guarantees 100% zero-heap allocation (ALLOC_ZERO_HEAP) in .rodata and .bss.
 96 - Exposes CS_C_STD_TIER (23/17/11/0) and CS_CXX_STD_TIER (20/17/11/0) plus
 97   graceful ISO attribute fall-down macros (CS_NODISCARD, CS_MAYBE_UNUSED,
 98   CS_FALLTHROUGH, CS_UNREACHABLE, CS_ALIGNOF, CS_NORETURN, CS_STATIC_ASSERT,
 99   CS_HAS_EMBED) compiled cleanly without extensions under -Wpedantic.
100 ------------------------------------------------------------------------------
101 */
102
103 /** Active ISO C standard tier compiled for this translation unit (23, 17, 11, or 0 in C++). */
104 #if defined(__STDC_VERSION__) && __STDC_VERSION__ >= 202311L
105 #    define CS_C_STD_TIER 23
106 #elif defined(__STDC_VERSION__) && __STDC_VERSION__ >= 201710L
107 #    define CS_C_STD_TIER 17
108 #elif defined(__STDC_VERSION__) && __STDC_VERSION__ >= 201112L
109 #    define CS_C_STD_TIER 11
110 #else
111 #    define CS_C_STD_TIER 0
112 #endif
113
114 /** Active ISO C++ standard tier when included from C++ (20, 17, 11, or 0 in C). */
115 #if defined(__cplusplus) && __cplusplus >= 202002L
116 #    define CS_CXX_STD_TIER 20
117 #elif defined(__cplusplus) && __cplusplus >= 201703L
118 #    define CS_CXX_STD_TIER 17
119 #elif defined(__cplusplus)
120 #    define CS_CXX_STD_TIER 11
121 #else
122 #    define CS_CXX_STD_TIER 0
123 #endif
124
125 /** Marks return values that callers MUST inspect (C23/C++17+ [[nodiscard]], C11/17 floor empty). */
126 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
127 #    define CS_NODISCARD [[nodiscard]]
128 #else
129 #    define CS_NODISCARD
130 #endif
```

```
131
132 /** Silences unused-entity diagnostics cleanly on intentional dormant/platform helpers. */
133 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
134 #    define CS_MAYBE_UNUSED [[maybe_unused]]
135 #else
136 #    define CS_MAYBE_UNUSED
137 #endif
138
139 /** Annotates explicit switch-case fall-through cleanly under -Wextra -Wpedantic. */
140 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
141 #    define CS_FALLTHROUGH [[fallthrough]]
142 #elif defined(__GNUC__) && (__GNUC__ >= 7)
143 #    define CS_FALLTHROUGH __attribute__((fallthrough))
```

*Compile-time standard detection establishing C23/C17/C11 and C20/C17 parity without legacy shims.*

Wholesale Listing: Complete verbatim code (#L82-L194) → Appendix: Cornerstone Fall-Down Stairs Portability Ladder (C23/C17/C11 & C++20/17).

Instead of writing lowest-common-denominator code **or** breaking older build environments, **the engine builds forward-first for ISO C23 and C20** (`CORNERSTONE_C_STANDARD=AUTO|23|17|11`) and steps gracefully down the stairs (`CS_NODISCARD`, `CS_MAYBE_UNUSED`, `CS_FALLTHROUGH`, `CS_UNREACHABLE`) onto C17/C11 and C17 floors without compatibility clutter. Modern compilers enforce stricter safety diagnostics automatically today, while older hardware still compiles the exact same tree without feature hacks.

## 70.4. Lesson 4: Zero-Disk Safe-Mode BIOS & Compiled .rodata ROMs Protect Designers

Previously, broken or missing config JSON files meant fragile dummy defaults hardcoded separately inside C structs—guaranteeing drift between real configs and fallbacks. Under Rev 5:

1. Authors edit one authoritative 5-block JSON config in engine/config/ (or game overrides in game/config/).
2. At build time, cmake/GenerateConfigRom.cmake compiles those exact JSON files into read-only ELF .rodata headers (*_config_rom.h).
3. If disk files vanish or fail syntax/schema validation, **_fallback** mounts the embedded .rodata ROM (rom://log, rom://debug, rom://io), sets provenance (loaded_from_data = false / loaded_from_rom = true), and lets the engine boot all the way to DEBUG_STEP_AWAIT so the developer sees *why without crashing.

**The Game Payoff for Tenebris and Lux**: Designers can iterate rapidly on game/data/ combat tables, terrain rules, or UI themes with total confidence that a malformed file triggers a clean, readable diagnostic card instead of silent data corruption or a dead startup window.

## 70.5. How Root Discipline Multiplies Velocity in core/, viewport/ & game/

| Upcoming Subsystem | Direct Dividend Paid by the log/ → debug/ → io/ Roots |
|---|---|
| core/ Block 3 & Math/Time | Can wire formulas.json and manifest.json immediately through master_loader with pre-built rom:// fallback and typed spans. |
| audio/ & viewport/ (640x360 X11) | Inherit pre-wired probe namespaces, bounded .bss accounting, and deterministic CLI/GUI diagnostics from boot frame zero. |
| game/ (The Master Turn Gear) | Stays purely focused on RPG rules (Twin Relics, Stances, Moral Sway, Sacyra/Libira topology) without writing a single line of file I/O or log plumbing. |

| Upcoming Subsystem | Direct Dividend Paid by the `log/` → `debug/` → `io/` Roots |
|---|---|
| Future Cornerstone Titles | Drop cleanly onto the same Layer 3 plug (`main.c`) because zero domain lore leaked into `engine/`. |

- Continue to Chronicle 2: `AGENTS.md` Evolution & AI-Assisted Architecture.
- Return to Architecture Index or Codebase Implementation Index.

# Chapter 71. Evolution of AGENTS.md: Engineering a Cognitive Compass for Human-AI Co-Creation

**Source File**: `docs/architecture/chronicles/02_agents_md_evolution_and_ai_architecture.adoc`

Long-running software projects involving C11/C23 systems programming, Rust verification tools, JSON schema contracts, and deep RPG lore break predictably under standard AI workflows for one reason: **context drift**. Left unanchored across multi-hour sprints, autonomous agents invent plausible-sounding lore names, re-introduce heap allocations in deterministic code, tuck quick shims over broken contracts, or forget how earlier subsystems were verified.

Across Iterations leading up to Governance Revision 5 (`Rev 5`), `AGENTS.md` evolved away from a loose checklist into a **5-Block Cognitive Compass** encoding **why** Cornerstone works the way it does alongside **how** every boundary is mechanically verified.

| Governance Evolution | Engineering Shift Achieved |
|---|---|
| Passive Prose → 21 Explicit Invariants | Turns soft preferences into binary rules (`ALLOC_ZERO_HEAP`, strict ISO C, anti-domain-bleed, authentic canon: Sole Antagonist **The Dark Lord Usuru**). |
| File-Path IDs → Movement-Resilient `@id` | Allows cogs elevated from `game/` → `engine/` to keep logical identity (`cog.<module>`) while `@file` and `@layer` update cleanly. |
| Arbitrary Style → Universal 5-Block Protocol | Forces every C header, C source, JSON payload, CMake script, and AsciiDoc chapter into predictable structural slots (`BLOCK 1..5`). |

## 71.1. Keeping the Operator in the Director's Chair

Automated systems fail when they strip architectural judgment away from the human creator. Cornerstone's `Rev 5` ruleset intentionally splits responsibility between **human creative direction** and **automated structural discipline**:

1. **Waterfall Bedrock vs. Agile Exploration (`AGENTS.md` §0.1)**: The agent never compromises fixed physical laws (zero heap in Layers 0–2, downward dependencies, zero domain bleed inside `engine/`), freeing the human director to iterate boldly on gameplay mechanics, tone, documentation depth, and visual presentation without worrying that an experimental sprint will rot the foundation.
2. **Dual-Audience Documentation Covenant (`Invariant 17`)**: Requiring every header, operational API, and architecture document to speak clearly both **For Game Designers & Non-Programmers** and **For Systems Programmers & Engine Architects** forces the AI to translate every low-level optimization into plain-English creative impact.
3. **Active Proposal Over Blind Guessing**: When toolchains step between C23, C17, and C11 (`CORNERSTONE_C_STANDARD=AUTO`) or telemetry uncovers `.bss` bloat, the architecture surfaces explicit options and seams so the operator steers design priorities consciously.

## 71.2. Why 5-Block AsciiDoc & Automated Validators Defeat Context Drift

Treating documentation files with the same **Universal 5-Block Comment Protocol** (`//// BLOCK 1..5 ////` and `// --- Sub-Assembly X.Y`) used in C and JSON closes a gap that normally ruins project documentation: docs going stale quietly after code moves.

| Feedback Loop | What It Guarantees Automatically |
|---|---|
| `master_guide --snapshots` | Extracts verbatim source spans (`Whole`, `Block`, `Subassembly`, `Function`, `Group`) directly from live `engine/`, `game/`, `tests/`, and `cmake/` files into textbook plates—eliminating hand-copied drift. |
| `master_guide --validate-layout` | Reads data-driven PDF geometry constraints (`docs/adoc_pdf_layout_rules.json`), catches mid-file title resets, broken `include::` targets, unbalanced code fences, and over-tall/over-wide code listings before `asciidoctor-pdf` wastes pages. |
| `tl_tools validate all` + CTest | Audits Block 1–5 conformance, compiler warning cleanliness (`-Wall -Wextra -Wpedantic`), Rust `-D warnings`, and all 4-tier tests after edits land. |

Because layout budgets, snapshot bindings (`@snapshots uses: ⋯`), chassis telemetry (`@chassis`), and probe routes (`@seams`) live explicitly in repository files rather than transient chat history, **every new session inherits exact structural truth immediately**.

## 71.3. Architectural Smells `AGENTS.md` Actively Catches

- **Fake Fallbacks & Dead Stubs**: Explicitly banned in favor of compiled `.rodata` ROM vaults (`cmake/GenerateConfigRom.cmake`) and quarantined `*_fallback.[ch]` engines sharing verified JSON parsers.
- **Hidden Static Scratch Bloat**: Caught directly by `@chassis` measurement (`size -A`, `nm --size-sort -S -r`), turning silent `.bss` waste into transparent numbers.
- **Non-Canonical Lore Drift**: Blocked at Block 1 governance and `game/data/` contract checks so **Tenebris and Lux** never acquires hallucinated characters, relics, or factions.
- **Unreadable Auto-Generated PDFs**: Solved by separating human narrative into structured 5-block chapters, verifying exact printable page dimensions mathematically, and compiling live textbook plates + wholesale appendices cleanly.

- Previous chronicle: Chronicle 1: Why Strengthening `log/`, `debug/`, and `io/` Pays Every Future Turn.
- Explore root governance directly in `AGENTS.md`, or return to Architecture Index.

# Chapter 72. Layer 0 Categorized Flight Logging Subsystem Overview

**Source File**: docs/code/logging_system.adoc

Think of the **Layer 0 Categorized Flight Logging Subsystem** (engine/include/log/ & engine/src/log/) as an aircraft's black-box flight recorder: long before windows open or disk folders are found, it quietly records every subsystem heartbeat into fixed memory without ever asking the operating system for dynamic heap (ALLOC_ZERO_HEAP).



> **How this chapter fits with the log/ folder**: Read this overview for the big picture, then dive into docs/code/log/ for paired walkthroughs of log.h + log.c, log_types.h, log_fallback.h + log_fallback.c, and the 4-tier log verification suite.

## 72.1. Architectural Philosophy: The Flight Recorder

Printing raw debug text straight to the terminal breaks down the moment a real game hits frame budgets, file-permission errors, or multi-step bug reproduction. Cornerstone replaces ad-hoc printf calls with a two-stage deterministic flight recorder:

1. **Pre-VFS Tier-0 BIOS Capture (log_fallback)**: Before io/ mounts disk routes, log_fallback_boot() arms a 64-entry static BIOS ring buffer (12.3 KiB .bss) backed by 10 compiled-in .rodata telemetry ROMs (57.9 KiB). Early boot messages are never lost.
2. **Early-Entry Replay Drain (log_fallback_drain_entries)**: Once log_init() comes online, every entry captured during Tier-0 boot is drained cleanly into the primary circular ring buffer so the chronological timeline has zero gaps.
3. **Categorized Primary Ring & Rolling Disk Stream (cog.log)**: Messages are formatted on fixed stack frames (⇐ 1 KiB), stored in a bounded circular ring, tagged by severity (TRACE..FATAL) and channel mask (engine vs. game), and automatically rotated on disk when size ceilings are reached.
4. **Autonomous Probe Hooks & Safe-Mode Downgrade**: Sub-Assembly 4.1 binds log.* health probes directly into cog.debug, allowing damaged JSON telemetry configs to degrade safely onto compiled ROM profiles and optionally pause at DEBUG_STEP_AWAIT.

## 72.2. Verified Rev 5 File & Documentation Map

| Companion Doc | Engine Files Covered | Role & Chassis Highlights |
|---|---|---|
| 01: Router & Ring Engine | engine/include/log/log.h + engine/src/log/log.c (cog.log) | Primary circular ring recorder, channel masks, disk rotation, telemetry config loader, and log.* debug probes (~24.7 KiB .data, 16.4 KiB .bss VFS scratch, ~8.6 KiB .rodata). |
| 02: Vocabulary Leaf Types | engine/include/log/log_types.h (cog.log.types) | Pure POD enums, bounds, log_entry_t, log_config_t, and log_stats_t. Zero .bss/.data state; breaks circular dependency between log and log_fallback. |

| Companion Doc | Engine Files Covered | Role & Chassis Highlights |
|---|---|---|
| 03: Tier-0 Fallback BIOS | engine/include/log/log_fallback.h + engine/src/log/log_fallback.c (cog.log.fallback) | Quarantined emergency BIOS (12.3 KiB .bss) and 10 compiled .rodata telemetry ROMs (57.9 KiB), shared JSON config parser, and replay drain into primary logging. |
| 04: Logging Verification Suite | 5 CTest targets across tests/{unit,functional,integrated,system}/ | Progressive crude-to-fine test coverage across happy path, hostile NULL/bounds, invariant checks, cross-tier replay emergence, and machine self-audit. |

### 72.2.1. Telemetry Profiles in engine/config/diagnostics/telemetry/

Behavior is controlled without recompiling C code through JSON manifests in engine/config/diagnostics/telemetry/:

- manifest.json & default.json — baseline catalog and default profile binding.
- development.json, production.json, quiet.json, combat_focus.json — operational severity/channel presets.
- engine_channels.json, game_channels.json, severity_weights.json, forensic_telemetry.json — channel taxonomies, severity weights, and forensic export rules (all 10 mirrored into log_telemetry_rom.h at build time).

## 72.3. Live Public Dispatch Contract

log.h exposes clean categorized dispatch routines so engine subsystems and gameplay modules share one unified timeline without allocating memory:

*Flight Recorder Formatted Entry Dispatch API (log.h Sub-Assembly 3.3) (engine/include/log/log.h#L406-L445 · Mode: Subassembly)*

```
406 /*
407 ------------------------------------------------------------------------------
408 Sub-Assembly 3.3: Formatted Entry Dispatch Cogs
409 Mission: Format, serialize, and emit structured flight recorder entries.
410 @triggers
411   - Tripwires      : LOG_TRIPWIRE_SETTING_INVALID / BOUNDS_VIOLATION -> probe "log.write_telemetry"
412   - Block 4 Audits : log_validate_telemetry_integrity(), log_preflight()
413   - Layer 4 Tests  :
414       * tests/unit/log/test_log_unit.c
415         [Section 3: INVARIANTS & CONSTRAINTS / test_log_unit_assertions_and_guarantees]
416         [Section 4: RING BUFFER FIFO OVERWRITE & FILE ROTATION / test_log_unit_emergence]
417       * tests/functional/test_log_functional.c
418         [Section 3: INVARIANTS & SATURATION / test_log_functional_assertions_and_guarantees]
419       * tests/integrated/test_log_integrated.c
420         [Section 4: CATEGORICAL DISK ROUTING & HUD RING BUFFER / test_log_integrated_emergence]
421       * tests/system/test_log_system.c
422         [Section 1: NOMINAL HIGH-THROUGHPUT BURST / test_log_system_always_pass]
423         [Section 3: FIXED-WIDTH REGEX DETERMINISM / test_log_system_assertions_and_guarantees]
424 ------------------------------------------------------------------------------
425 */
426
427 /**
428  * @brief Emits a structured log entry into the ring buffer, file, and console.
429  *
430  * ### For Game Designers & Non-Programmers:
431  * Writes an event to the game's flight recorder journal, capturing the character involved,
432  * their current moral balance percentage, and what action took place.
433  *
434  * ### For Systems Programmers & Engine Architects:
435  * Formats a fixed-column log record with microsecond monotonic timestamp and ISO 8601 UTC time,
436  * pushes it to the static circular ring buffer, and writes to disk archives with zero heap allocation.
437  *
438  * @param[in] severity Graded severity level.
```

```
 * @param[in] category Target functional category.
 * @param[in] entity   Entity/Actor label (e.g. "Player", "Companion", "System").
 * @param[in] grade    Normalized grade [-1.0f .. +1.0f] (e.g. alignment sway or vitals).
 * @param[in] file     Source file name (__FILE__).
 * @param[in] line     Source line number (__LINE__).
 * @param[in] func     Source function name (__func__).
 * @param[in] fmt      Printf-style format string.
```

*Public categorized log macros and fixed-stack formatting interface for engine and game channels.*

Wholesale Listing: Complete verbatim code (#L406-L484) → Appendix: Flight Recorder Formatted Entry Dispatch API (log.h Sub-Assembly 3.3).

## 72.4. Autonomous Probe Wiring (cog.log → cog.debug)

Rather than silently swallowing ring overflows or missing telemetry files, log.c registers live probes inside Sub-Assembly 4.1:

*Flight Log Subsystem Probe Registration & Validation Guards (Sub-Assembly 4.1)* (engine/src/log/log.c#L1934-L1977 · *Mode:* Subassembly)

```
/*
-----------------------------------------------------------------------------
Sub-Assembly 4.1: Contract, Input & Cog Verification Cogs
Mission: Verify received parameters, configuration contracts, and data provenance;
verify that declared subsystem cogs run properly as standalone units.
-----------------------------------------------------------------------------
*/

/**
 * @brief Returns whether the active logging subsystem was configured from data sources.
 *
 * ### For Game Designers & Non-Programmers:
 * Checks if settings were loaded from actual JSON files rather than emergency defaults.
 *
 * ### For Systems Programmers & Engine Architects:
 * Queries provenance boolean s_log.config.loaded_from_data.
 *
 * @return true if loaded from JSON data; false if using fallback code page.
 */
bool log_is_data_driven(void)
{
    return s_log.config.loaded_from_data;
}

/**
 * @brief Queries whether the logging engine is currently operating in Safe Mode.
 *
 * ### For Game Designers & Non-Programmers:
 * Returns true if the engine booted using compiled-in ROM backups or emergency
 * defaults instead of live external JSON directories on disk.
 *
 * ### For Systems Programmers & Engine Architects:
 * Checks initialization state alongside s_log.safe_mode (!loaded_from_data).
 *
 * @return true if active and running in Safe Mode; false otherwise.
 */
bool log_is_safe_mode(void)
{
    return s_log.initialized && s_log.safe_mode;
}

/**
 * @brief Returns the current state of the logging data pipeline tripwire.
 *
```

*Shows how cog.log verifies its own static ring limits and binds probes into cog.debug.*

Wholesale Listing: Complete verbatim code (#L1934-L2028) → Appendix: Flight Log Subsystem Probe Registration & Validation Guards (Sub-Assembly 4.1).

## 72.5. Self-Audit & Quick Verification

```
ctest --test-dir build -R "_log_" --output-on-failure
./build/tenebris-lux --audit
```

1. *Why split* `log_types.h` *out of* `log.h`*?*

   Both primary `log.c` and quarantined `log_fallback.c` need `log_entry_t` and `log_config_t`. Keeping types in a stateless leaf header prevents circular includes while letting `log_fallback` boot cleanly even if the full logging layer is down.

2. *What happens if telemetry files on disk are missing or corrupt?*

   `log_fallback_load_rom_profile()` hydrates the matching profile directly from the compiled `.rodata` ROM vault, preserves `loaded_from_data = false`, and reports the downgrade cleanly through debugger probes.

## 72.6. Companion Deep-Dive Chapters

1. Open Paired `log.h` + `log.c` Walkthrough for the ring buffer and disk rotation internals.
2. Read `log_types.h` Leaf Vocabulary and Paired `log_fallback.h` + `log_fallback.c` BIOS.
3. Continue to State Debugger Overview or I/O Pipeline Overview.

# Chapter 73. Log Module Pair: Flight Logger Router & Severity Ring (log.h + log.c)

**Source File**: docs/code/log/01_log_router_and_ring.adoc

| Attribute | Verified Contract (cog.log) |
|---|---|
| Public Contract (.h) | engine/include/log/log.h (954 lines) — operational API, convenience log macros, audit & inspection gates. |
| Runtime Engine (.c) | engine/src/log/log.c (2,833 lines) — circular ring storage, JSON config loader, file rotation, and debug probe wiring. |
| Chassis Footprint | .data ~24.7 KiB (s_log + tables) + .bss 16.4 KiB (s_log_vfs_scratch) + .rodata ~8.6 KiB; stack framing ⇐ 1 KiB. |
| Inputs & Fallbacks | Reads engine/config/diagnostics/telemetry/<profile>.json; degrades cleanly to rom://telemetry via cog.log.fallback. |

log.h defines the public switchboard used by every engine subsystem and gameplay turn loop, while log.c houses the circular recording tape (256 recent messages), optional rolling .log file sinks, and live health probes. Designers control verbosity with JSON telemetry profiles without touching C code; programmers get O(1) append latency and reproducible memory bounds.

## 73.1. Configuration Loading & Data-Driven Profiles (Sub-Assembly 3.1)

When log_init(const log_config_t *config) runs—or when log_load_config_profile() loads "default", "development", "production", "quiet", or "combat_focus"—log.c resolves configuration data through path_finder / master_loader before falling back to compiled ROMs if disk assets are absent or invalid:

- log_load_config_from_file() parses a specific profile payload into log_config_t.
- log_load_config_from_manifest() reads engine/config/diagnostics/telemetry/manifest.json to discover the active startup profile.
- log_load_config_profile() looks up a named preset, tagging loaded_from_data = true only when hydrated from disk JSON (never pretending ROM data came from disk).

## 73.2. Ring Recording, Rotation & String Parsing (Sub-Assemblies 3.2–3.5)

Every log emission flows through log_write_entry() into fixed stack buffers (LOG_MAX_MESSAGE_LEN bounded, zero malloc):

*Flight Recorder Formatted Entry Dispatch API (log.h Sub-Assembly 3.3) (engine/include/log/log.h#L406-L445 · Mode: Subassembly)*

```
406 /*
407 ------------------------------------------------------------------------------
408 Sub-Assembly 3.3: Formatted Entry Dispatch Cogs
409 Mission: Format, serialize, and emit structured flight recorder entries.
410 @triggers
411   - Tripwires      : LOG_TRIPWIRE_SETTING_INVALID / BOUNDS_VIOLATION -> probe "log.write_telemetry"
412   - Block 4 Audits : log_validate_telemetry_integrity(), log_preflight()
413   - Layer 4 Tests  :
414       * tests/unit/log/test_log_unit.c
415         [Section 3: INVARIANTS & CONSTRAINTS / test_log_unit_assertions_and_guarantees]
416         [Section 4: RING BUFFER FIFO OVERWRITE & FILE ROTATION / test_log_unit_emergence]
```

```
      * tests/functional/test_log_functional.c
        [Section 3: INVARIANTS & SATURATION / test_log_functional_assertions_and_guarantees]
      * tests/integrated/test_log_integrated.c
        [Section 4: CATEGORICAL DISK ROUTING & HUD RING BUFFER / test_log_integrated_emergence]
      * tests/system/test_log_system.c
        [Section 1: NOMINAL HIGH-THROUGHPUT BURST / test_log_system_always_pass]
        [Section 3: FIXED-WIDTH REGEX DETERMINISM / test_log_system_assertions_and_guarantees]
-----------------------------------------------------------------------------
*/

/**
 * @brief Emits a structured log entry into the ring buffer, file, and console.
 *
 * ### For Game Designers & Non-Programmers:
 * Writes an event to the game's flight recorder journal, capturing the character involved,
 * their current moral balance percentage, and what action took place.
 *
 * ### For Systems Programmers & Engine Architects:
 * Formats a fixed-column log record with microsecond monotonic timestamp and ISO 8601 UTC time,
 * pushes it to the static circular ring buffer, and writes to disk archives with zero heap allocation.
 *
 * @param[in] severity Graded severity level.
 * @param[in] category Target functional category.
 * @param[in] entity   Entity/Actor label (e.g. "Player", "Companion", "System").
 * @param[in] grade    Normalized grade [-1.0f .. +1.0f] (e.g. alignment sway or vitals).
 * @param[in] file     Source file name (__FILE__).
 * @param[in] line     Source line number (__LINE__).
 * @param[in] func     Source function name (__func__).
 * @param[in] fmt      Printf-style format string.
```

*Public categorized log macros and fixed-stack formatting interface for engine and game channels.*

Wholesale Listing: Complete verbatim code (#L406-L484) → Appendix: Flight Recorder Formatted Entry Dispatch API (log.h Sub-Assembly 3.3).

| Cog Group | Public API & Behavior |
|---|---|
| File Sink & Rotation (3.2) | `log_is_active()` reports engine readiness; `log_rotate_if_needed()` rolls active log files (`rotate_archive_stream` internal helper) before exceeding configured file-size thresholds. |
| Ring Retrieval (3.4) | `log_get_total_entries()` returns lifetime sequence count; `log_get_recent_entries()` and `log_get_top_entries()` copy bounded snapshots out of the ring without exposing internal static pointers. |
| Enum Parsers (3.5) | `log_severity_from_string()` and `log_category_from_string()` map human-readable JSON tokens (`"INFO"`, `"WARN"`, `"COMBAT"`, `"IO"`) back to strongly typed enums. |

### Separation Between Machine Noise and Story Events

Engine infrastructure channels (`CORE`, `IO`, `LOG`, `DEBUG`, `VIEWPORT`) can run at terse `WARN`/`INFO` levels while gameplay testers turn up `COMBAT`, `HERO`, or `RELICS` via `combat_focus.json`—keeping diagnostic logs focused and readable.

## 73.3. Validation Guards, Probes & Pre-Flight Audits (Sub-Assemblies 4.1–5.2)

Inside Block 4 of `log.c`, `log_register_debug_probes()` registers telemetry invariants directly with `cog.debug`, while `log_preflight()` and `log_audit()` validate manifest contracts, rotation limits, and ring consistency:

*Flight Log Subsystem Probe Registration & Validation Guards (Sub-Assembly 4.1)* (engine/src/log/log.c#L1934-L1977 · *Mode:* Subassembly)

```
1934 /*
1935 ------------------------------------------------------------------------------
1936 Sub-Assembly 4.1: Contract, Input & Cog Verification Cogs
1937 Mission: Verify received parameters, configuration contracts, and data provenance;
1938 verify that declared subsystem cogs run properly as standalone units.
1939 ------------------------------------------------------------------------------
1940 */
1941
1942 /**
1943  * @brief Returns whether the active logging subsystem was configured from data sources.
1944  *
1945  * ### For Game Designers & Non-Programmers:
1946  * Checks if settings were loaded from actual JSON files rather than emergency defaults.
1947  *
1948  * ### For Systems Programmers & Engine Architects:
1949  * Queries provenance boolean s_log.config.loaded_from_data.
1950  *
1951  * @return true if loaded from JSON data; false if using fallback code page.
1952  */
1953 bool log_is_data_driven(void)
1954 {
1955     return s_log.config.loaded_from_data;
1956 }
1957
1958 /**
1959  * @brief Queries whether the logging engine is currently operating in Safe Mode.
1960  *
1961  * ### For Game Designers & Non-Programmers:
1962  * Returns true if the engine booted using compiled-in ROM backups or emergency
1963  * defaults instead of live external JSON directories on disk.
1964  *
1965  * ### For Systems Programmers & Engine Architects:
1966  * Checks initialization state alongside s_log.safe_mode (!loaded_from_data).
1967  *
1968  * @return true if active and running in Safe Mode; false otherwise.
1969  */
1970 bool log_is_safe_mode(void)
1971 {
1972     return s_log.initialized && s_log.safe_mode;
1973 }
1974
1975 /**
1976  * @brief Returns the current state of the logging data pipeline tripwire.
1977  *
```

*Shows how cog.log verifies its own static ring limits and binds probes into cog.debug.*

Wholesale Listing: Complete verbatim code (#L1934-L2028) → Appendix: Flight Log Subsystem Probe Registration & Validation Guards (Sub-Assembly 4.1).

- **Runtime Mode Predicates (4.1)**: `log_is_data_driven()`, `log_is_safe_mode()`, and `log_reset_tripwire()`.
- **Contract Validators (4.2)**: `log_validate_manifest_contract()`, `log_validate_setting_contract()`, `log_validate_rotation_bounds()`, and `log_validate_telemetry_integrity()`, aggregated by `log_preflight()` / `log_audit(bool verbose)`.
- **Lifecycle & Cards (5.1–5.2)**: `log_execute_inspect()` populates caller-owned `log_stats_t`; `log_execute_dump_card()` formats a printable status card; `log_init()` drains any early `log_fallback` BIOS entries into the main ring; `log_shutdown()` flushes and closes streams cleanly.

## 73.4. Paired Neighbors

- Vocabulary types consumed here: `log_types.h`

- Early-boot ROM vault & replay source: `log_fallback.h` + `log_fallback.c`
- Automated verification ladder: Logging Testing Suite

# Chapter 74. Log Leaf Contract: Canonical Types & Ring Bounds (log_types.h)

**Source File**: docs/code/log/02_log_types_and_contracts.adoc

| Attribute | Verified Contract (cog.log.types) |
|---|---|
| Header File | engine/include/log/log_types.h (421 lines) — header-only pure vocabulary (archetype: pure_types \| stateless). |
| Runtime Source | None intentionally. Holds zero mutable global storage (0 B .bss, 0 B .data). |
| Physical Bounds | LOG_MAX_ENTITY_LEN = 16, LOG_MAX_MESSAGE_LEN = 256, BIOS ring capacity 64. |
| Consumers | Included downward by both log_fallback.h and log.h plus extern "C" C++20/17 tooling. |

If log_fallback.h had to include log.h just to learn what a log_entry_t looks like, a Safe-Mode fallback would depend on the primary logging module it is meant to rescue! Extracting shared enums, byte limits, and plain-old-data (POD) structs into log_types.h gives both modules an identical binary vocabulary with zero dependency tangle.

## 74.1. Physical Bounds & Telemetry Value Layout (Sub-Assemblies 2.1–2.3)

In Block 2 (Sub-Assembly 2.1), log_types.h locks string sizes (LOG_MAX_ENTITY_LEN, LOG_MAX_MESSAGE_LEN) and uses _Static_assert / static_assert parity via core/manifest.h so silent struct growth cannot overrun stack budgets. In Sub-Assembly 2.3, it defines the exact records carried across ring buffers and disk logs:

*Flight Log Leaf Vocabulary & Entry Layout (log_types.h Sub-Assembly 2.3) (engine/include/log/log_types.h#L153-L194 · Mode: Subassembly)*

```
153 /*
154 ------------------------------------------------------------------------------
155 Sub-Assembly 2.3: Shared Structural Entity & Telemetry Value Definitions
156 ------------------------------------------------------------------------------
157 Canonical non-anonymous enums and POD structures shared across operational and
158 quarantined BIOS telemetry cogs.
159 ------------------------------------------------------------------------------
160 */
161
162 /**
163  * @enum log_severity
164  * @brief Graded severity taxonomy for engine diagnostics and telemetry events.
165  *
166  * Ordered from highest severity (FATAL) down to finest detail (TRACE).
167  *
168  * ### For Game Designers & Non-Programmers:
169  * Defines how critical an event is:
170  *   - FATAL/CRITICAL: Showstoppers that threaten gameplay continuity.
171  *   - ERROR/WARNING: Problems handled gracefully with automatic fallbacks.
172  *   - INFO: Normal story, quest, and milestone announcements.
173  *   - DEBUG/TRACE: Granular technical values for combat balance and timing.
174  *
175  * ### For Systems Programmers & Engine Architects:
176  * Strict integer ordering enables threshold filtering and weight-table lookup.
177  */
178 typedef enum log_severity {
179     LOG_SEV_FATAL = 0,    /**< Unrecoverable failure or fatal state corruption. */
180     LOG_SEV_CRITICAL,     /**< Severe failure threatening gameplay loop. */
181     LOG_SEV_ERROR,        /**< Discrete operation failure with fallback. */
182     LOG_SEV_WARNING,      /**< Suspicious condition or degraded state. */
183     LOG_SEV_INFO,         /**< Major milestone or narrative state shift. */
184     LOG_SEV_DEBUG,        /**< Granular spatial, coordinate, or stance telemetry. */
185     LOG_SEV_TRACE,        /**< High-frequency per-tick calculation telemetry. */
186     LOG_SEV_COUNT         /**< Enumerator bound sentinel (not a valid severity). */
```

```
} log_severity_t;

/**
 * @enum log_category
 * @brief Functional categorization of logging events for focused channel filtering.
 *
 * ### For Game Designers & Non-Programmers:
 * Divides events into distinct thematic channels (`CHARACTER`, `MESSAGE`,
```

*Zero-.bss leaf types shared cleanly by cog.log and quarantined cog.log.fallback.*

Wholesale Listing: Complete verbatim code (#L153-L303) → Appendix: Flight Log Leaf Vocabulary & Entry Layout (log_types.h Sub-Assembly 2.3).

| Core Type / Enum | Architectural Responsibility |
|---|---|
| log_severity_t | Ordered verbosity levels (TRACE, DEBUG, INFO, WARN, ERROR, FATAL) shared consistently by JSON filters and C macros. |
| log_category_t | Channel taxonomy separating Core/IO/Viewport machine diagnostics from Game Simulation channels (COMBAT, HERO, RELICS, WORLD). |
| log_entry_t | Fixed-width immutable snapshot of a single event: sequence number, timestamp, category, severity, source line/file tag, and formatted payload. |
| log_config_t & log_stats_t | Caller-owned runtime policy settings (including provenance flag loaded_from_data) and telemetry counter snapshots returned by log_execute_inspect(). |

## 74.2. Stateless Boundary Predicates (Sub-Assembly 2.4)

Because range checking a log_severity_t enum or validating whether a config string length fits within stack limits needs zero global state, log_types.h provides lightweight static inline predicates directly in Sub-Assembly 2.4, compiled into caller call sites cleanly under both C23/C17/C11 and C20/C17.

## 74.3. Compile-Time & Validator Checks

```
cargo run --manifest-path tools/Cargo.toml --bin tl_tools -- validate headers
```

1. *Why does* log_types.h *disallow mutable global variables?*

   A pure_types leaf header must act like a shared dictionary, never an active machine. Any global ring or state counter belongs strictly inside log.c or log_fallback.c.

## 74.4. Consumers of log_types.h

- Paired log.h + log.c Router & Ring Engine
- Paired log_fallback.h + log_fallback.c BIOS

# Chapter 75. Log Safe-Mode BIOS: Zero-Disk Fallback & ROM Vault (log_fallback.h + log_fallback.c)

**Source File**: docs/code/log/03_log_fallback_bios.adoc

| Attribute | Verified Contract (cog.log.fallback) |
|---|---|
| Fallback Contract (.h) | engine/include/log/log_fallback.h (811 lines) — Tier-0 BIOS boot, ROM asset lookup, replay drain, and emergency card dump APIs. |
| BIOS Implementation (.c) | engine/src/log/log_fallback.c (1,865 lines) — self-contained Safe-Mode recorder (s_fb) and generated telemetry ROM bridge. |
| Chassis Footprint | .bss 12.3 KiB (s_fb ring + state) + .rodata 57.9 KiB (10 embedded telemetry JSON ROMs) + stack ⇐ 1 KiB. |
| Quarantined Boundary | archetype: quarantined_fallback \| stateful; consumes log_telemetry_rom.h & cog.log.types. |

In real game development, crashes most often strike **during** startup—when asset folders move, a user edits a JSON setting incorrectly, or disk permissions turn read-only. If the logger itself refuses to boot without healthy JSON files on disk, developers go completely blind right when telemetry matters most. log_fallback.[ch] solves that by booting **before** io/ even initializes.

## 75.1. Pre-VFS Boot & Chronological Replay (Sub-Assembly 3.3 & 5.2)

log_fallback_boot() arms a lightweight 64-entry ring buffer right at process start. Any early warning from io_fallback or manifest lands cleanly in log_fallback_record() (and optionally an emergency fallback stream if disk writing works). When disk storage fails or goes read-only, disk_fault latches without crashing the game. Later, when primary log_init() comes online, log_fallback_drain_entries() transfers every buffered early entry cleanly into the full 256-entry ring:

*Tier-0 Flight Log BIOS Boot & Early-Entry Replay Drain (engine/src/log/log_fallback.c#L1701-L1740 · Mode: Group)*

```
1701 /**
1702  * @brief Arms the Tier-0 BIOS Safe-Mode logger before CLI flags or disk VFS boot.
1703  *
1704  * ### For Game Designers & Non-Programmers:
1705  * Turns on emergency logging immediately when the program starts so early boot
1706  * bugs or missing folders are never lost, while leaving repo folders clean.
1707  *
1708  * ### For Systems Programmers & Engine Architects:
1709  * When opt_cfg is NULL, arms purely in .bss ring + stderr alert mode with zero
1710  * filesystem I/O. When opt_cfg is supplied, opens bounded Safe-Mode disk output
1711  * (applying build/logs sandbox protection) and degrades gracefully to RAM Ring +
1712  * stderr if mkdir/fopen fails on read-only or full media.
1713  *
1714  * @param[in] opt_cfg Optional Safe-Mode config (NULL for clean RAM BIOS boot).
1715  * @return true on clean BIOS Safe-Mode arming (survives disk failures).
1716  */
1717 bool log_fallback_boot(const log_config_t *opt_cfg)
1718 {
1719     if (s_fb.file != NULL) {
1720         fflush(s_fb.file);
1721         fclose(s_fb.file);
1722         s_fb.file = NULL;
1723     }
1724
1725     s_fb.disk_fault = false;
1726     s_fb.current_bytes = 0;
```

```
1727
1728    if (opt_cfg == NULL) {
1729        s_fb.config = s_log_floppy_template;
1730        s_fb.config.loaded_from_data = false;
1731        if (io_fallback_is_disk_read_only()) {
1732            s_fb.disk_fault = true;
1733        }
1734        s_fb.active_sink = LOG_FALLBACK_SINK_RING_AND_STDERR;
1735        s_fb.booted = true;
1736        return true;
1737    }
1738
1739    s_fb.config = *opt_cfg;
1740    s_fb.config.loaded_from_data = false;
```

*Arms the 64-entry pre-VFS BIOS ring before disk mount and replays captured records once primary cog.log boots.*

Wholesale Listing: Complete verbatim code (#L1256-L1801) → Appendix: Tier-0 Flight Log BIOS Boot & Early-Entry Replay Drain.

## 75.2. The 10-Asset Compiled Telemetry ROM Vault (Sub-Assemblies 3.1–3.2)

At CMake build time, cmake/GenerateConfigRom.cmake compiles every JSON file inside engine/config/diagnostics/telemetry/ into read-only C byte tables inside log_telemetry_rom.h:

- log_fallback_get_rom_info() reports embedded asset count (10) and total .rodata byte footprint (57.9 KiB runtime measured chassis).
- log_fallback_get_rom_asset() returns non-owning read-only pointers directly to .rodata bytes with zero copying or heap allocation.
- log_fallback_load_rom_profile() and log_fallback_parse_config_span() share one boundary-safe JSON key scanner across fallback and normal loading, guaranteeing that Safe Mode boots with genuine structured settings while flagging loaded_from_data = false.

## 75.3. BIOS Self-Audit & Emergency Diagnostic Cards

- log_fallback_validate_boot_contract() and log_fallback_audit() inspect ROM vault alignment, BIOS ring bounds, and sink modes non-destructively.
- log_fallback_get_stats() copies log_fallback_stats_t (booted, disk_fault, rom_hydrated, total_recorded, total_replayed, ring_count, ROM metrics).
- log_fallback_execute_inspect() and log_fallback_execute_dump_card() let test suites and CLI recovery tools print human-readable Safe-Mode status even in zero-disk environments.

## 75.4. Next Steps in log/

- Review the tests verifying BIOS capture and drain: Logging Testing & Probe Verification
- Inspect the shared vocabulary header: log_types.h

# Chapter 76. Log Verification Suite: 4-Tier CTest Harness & Probes

**Source File**: docs/code/log/04_log_testing_and_probes.adoc

Cornerstone validates cog.log and cog.log.fallback from fine isolated unit checks outward to crude multi-module system boot runs. Every test executable follows five consistent bands (always_pass, never_fail, assertions_and_guarantees, emergence, and preflight/audit) anchored by tests/common/cornerstone_test.h.

| Tier | CTest Target | Source File | Primary Verification Focus |
|---|---|---|---|
| Unit | unit_log_engine_test | tests/unit/log/test_log_unit.c | Ring rollover, channel masks, severity filtering, disk rotation bounds, and log_preflight() / log_audit(). |
| Unit (BIOS) | unit_log_fallback_test | tests/unit/log/test_log_fallback_unit.c | log_fallback_boot(), 10-asset ROM lookups, read-only disk-fault latch, and early-entry drain behavior. |
| Functional | functional_log_test | tests/functional/test_log_functional.c | End-to-end logging workflows using public headers cleanly as an external caller. |
| Integrated | integrated_log_test | tests/integrated/test_log_integrated.c | Cross-module coordination: log + log_fallback drain + debug probe registration + io/ JSON telemetry loading. |
| System | system_log_test | tests/system/test_log_system.c | Full system boot, corrupt/missing disk fallback to Safe-Mode telemetry ROMs, and recovery verification. |

## 76.1. How Each Test Executable Progresses (Crude-to-Fine Ladder)

Rather than mixing trivial sanity checks with catastrophic failure injection inside one disorganized file, every log test binary separates concerns cleanly into five progressive stages:

1. always_pass **(Baseline Contract)**: Confirms normal initialization (log_init, log_fallback_boot), entry dispatch, and expected clean stats under ideal inputs.
2. never_fail **(Hostile Boundary Defense)**: Passes NULL pointers, zero-length buffers, oversized format strings, and invalid severity tokens to confirm graceful false/safe returns with zero crashes.
3. assertions_and_guarantees **(Mathematical & Static Invariants)**: Verifies ring sequence monotonicity, bounded truncation guarantees (LOG_MAX_MESSAGE_LEN), and exact loaded_from_data provenance tracking.
4. emergence **(Multi-Component System Behavior)**: Tests interactions that cannot exist in single functions—such as early BIOS capture draining chronologically into log_init() and Safe-Mode downgrade under corrupt telemetry manifests.
5. preflight / post_audit **(Machine Self-Audit)**: Executes log_preflight(), log_audit(), and log_fallback_audit() automatically inside the verification loop.

## 76.2. Live Emergence Verification (test_log_system_emergence)

Below is an excerpt from the Layer 4 system suite demonstrating end-to-end telemetry verification:

*Flight Log System Emergence Verification (test_log_system_emergence) (tests/system/test_log_system.c#L283-L320 · Mode: Function)*

```
283 /**
284  * @brief Executes 10,000-entry chaos load, floppy BIOS endurance burst, and probe checks.
285  */
286 static void test_log_system_emergence(void)
287 {
288     printf("  [PASS-4] Running Section 4: Emergence (10,000-Entry Chaos, Floppy BIOS & Probes)...\n");
289
290     TEST_ASSERT(debug_init(NULL), "Debugger online to monitor log system endurance");
291
292     log_config_t cfg = log_get_default_config();
293     cfg.echo_to_console = false;
294     cfg.min_severity = LOG_SEV_TRACE;
295     cfg.save_by_category = true;
296     cfg.max_file_size_bytes = 16384U;
297     cfg.max_rotated_files = 10U;
298     str_safe_copy(cfg.log_dir, "build/logs/system_chaos", sizeof(cfg.log_dir));
299     str_safe_copy(cfg.base_filename, "chaos.log", sizeof(cfg.base_filename));
300
301     TEST_ASSERT(log_init(&cfg), "Initialized for chaos stress");
302
303     for (int i = 1; i <= TEST_LOG_SYS_CHAOS_COUNT; i++) {
304         log_severity_t sev = (log_severity_t)(i % LOG_SEV_COUNT);
305         log_category_t cat = (log_category_t)(i % LOG_CAT_COUNT);
306         float grade = ((float)(i % 200) - 100.0f) / 100.0f;
307
308         LOG_RECORD(sev, cat, "ChaosAgent", grade, "High-stress event step #%06d", i);
309
310         if (i % 2000 == 0) {
311             log_entry_t sample[10];
312             size_t fetched = log_get_top_entries(sample, 10U);
313             TEST_ASSERT_INT_EQ((int)fetched, 10,
314                                "Ring buffer top entries query succeeds during live write");
315         }
316     }
317
318     log_stats_t final_stats;
319     TEST_ASSERT(log_execute_inspect(&final_stats), "Inspected final chaos stats");
320     TEST_ASSERT(final_stats.total_entries >= (uint64_t)TEST_LOG_SYS_CHAOS_COUNT,
```

*Exercises full boot, rotation, Safe-Mode telemetry profile fallback, and recovery across Layer 4.*

Wholesale Listing: Complete verbatim code (#L283-L385) → Appendix: Flight Log System Emergence Verification (test_log_system_emergence).

## 76.3. Executing the Suite & Forensic Failure Cards

Because every executable includes tests/common/cornerstone_test.h (test.common.harness), any failing assertion queries debug / debug_fallback tripwire cards before exiting so testers see which contract tripped and why:

```
ctest --test-dir build -L log --output-on-failure
ctest --test-dir build -R "^(unit_log_|functional_log_|integrated_log_|system_log_)" --output-on-failure
```

## 76.4. Continue Reading

- Inspect the files verified by these suites: log.h + log.c and log_fallback.h + log_fallback.c.
- Advance to the next root subsystem: Layer 0 State Debugger Overview.

# Chapter 77. Layer 0 State Debugger & Diagnostic Engine Overview

**Source File**: `docs/code/debug_system.adoc`

If `log/` is the black-box voice recorder of Cornerstone, **`debug/`** (`engine/include/debug/ & engine/src/debug/`) is the cockpit telemetry computer and circuit-breaker panel. Instead of chasing ghost bugs by peppering code with disposable print statements, Cornerstone captures deterministic turn snapshots, hashes state fields cleanly, hosts up to 32 active validation probes, and halts cleanly on `DEBUG_STEP_AWAIT` whenever an invariant breaks.



**Navigate the `debug/` folder**: Pair this overview with docs/code/debug/ for detailed guides on `debug.h` + `debug.c`, `debug_types.h`, `debug_fallback.h` + `debug_fallback.c`, and all 5 `debug` CTest suites.

## 77.1. Architectural Philosophy: Observable State & Tripwires

Why build an engine-level debugger into Layer 0 before high-level gameplay even finishes? Because deterministic tactical RPGs fail silently when moral sway drifts, an equipment bonus overflows, or a corrupted config quietly loads half a table. Cornerstone turns hidden failures into explicit diagnostics:

- **Standalone Integer/Char Snapshot Contract (`debug_types.h`)**: Layer 1 gameplay projects its live simulation fields into neutral integers inside `debug_state_fields_t`—preserving downward dependency flow with zero `void *` type erasure.
- **Zero-Heap Snapshot Ring (`DEBUG_MAX_SNAPSHOTS = 64`)**: Captures state frames and 32-bit FNV-1a checksums, supports snapshot locking with O(cap) GCD wrap rotation, and rewinds backward cleanly via `debug_rewind_to_snapshot()`.
- **Autonomous Probe Matrix & `DEBUG_STEP_AWAIT`**: Subsystems register up to 32 named validation probes (`debug_register_validation_probe()`). When a fault hits, `debug_trigger_probe()` and `debug_trap_tripwire()` drop cleanly to a Safe-Mode fallback and hold `DEBUG_STEP_AWAIT` until resolved or cleared.
- **Tier-0 BIOS Resilience (`debug_fallback.[ch]`)**: Backed by compiled `.rodata` ROMs (`debug_diagnostics_rom.h`) over `engine/config/diagnostics/{debugger,taxonomy}/`, error codes and inspection cards stay available even with zero disk files.

## 77.2. Verified Rev 5 File & Documentation Map

| Companion Doc | Engine Files Covered | Role & Chassis Highlights |
|---|---|---|
| 01: Debugger Engine & Probes | `engine/include/debug/debug.h` + `engine/src/debug/debug.c` (`cog.debug`) | Core 64-slot snapshot ring (`s_debug` + `s_audit_backup`, `70.5 KiB .bss`), FNV-1a hashing, breakpoints, probe registry, rewind, and inspect/card dumps. |
| 02: Debugger Leaf Vocabulary | `engine/include/debug/debug_types.h` (`cog.debug.types`) | Pure POD enums, state field packets, snapshots, breakpoints, tripwires, and config records (`0 B .bss`, zero `void *` erasure). |

| Companion Doc | Engine Files Covered | Role & Chassis Highlights |
|---|---|---|
| 03: Debugger Tier-0 BIOS | engine/include/debug/debug_fallback.h + engine/src/debug/debug_fallback.c (cog.debug.fallback) | Standalone BIOS templates, emergency fault recorder, .rodata ROM vault (76,136 B) spanning debugger & taxonomy configs (256 B .bss, 464 B .data). |
| 04: Debugger Testing Suite | 5 CTest targets across tests/{unit,functional,integrated,system}/ | Crude-to-fine verification of snapshot rollover, GCD wrap preservation, probe firing, DEBUG_STEP_AWAIT, and fallback recovery. |

## 77.3. Deterministic Capture & Time-Travel Rewind

*Deterministic State Capture & Time-Travel Rewind (debug.c) (engine/src/debug/debug.c#L1503-L1540 · Mode: Group)*

```
1503 /**
1504  * @brief Rewinds state fields back to snapshot `index` after cryptographic checksum check.
1505  *
1506  * ### For Game Designers & Non-Programmers:
1507  * Restores actor stats from an earlier saved frame, pausing safely in AWAIT if memory
1508  * corruption altered the snapshot's stored fingerprint.
1509  *
1510  * ### For Systems Programmers & Engine Architects:
1511  * Fetches snapshot via `debug_get_snapshot()`, verifies `debug_compute_state_hash(&snap.state)`
1512  * matches `snap.state_hash` (engaging `DEBUG_TRIPWIRE_HASH_MISMATCH` + Safe-Mode AWAIT +
1513  * `LOG_ERROR` on mismatch), and emits `LOG_INFO` on verified restore.
1514  *
1515  * @triggers
1516  * - Layer 4 Tests : `tests/unit/debug/test_debug_unit.c:Section1`
1517  *
1518  * @param[in]  index      Chronological target snapshot offset.
1519  * @param[out] out_fields Restored simulation state fields on verified success.
1520  * @return true if checksum passed and `out_fields` populated; false on error or mismatch.
1521  */
1522 bool debug_rewind_to_snapshot(size_t index, debug_state_fields_t *out_fields)
1523 {
1524     /* Step 1: Validate preconditions and fetch chronological snapshot copy */
1525     if (!s_debug_active || out_fields == NULL) {
1526         return false;
1527     }
1528
1529     debug_snapshot_t snap;
1530     if (!debug_get_snapshot(index, &snap)) {
1531         return false;
1532     }
1533
1534     /* Step 2: Verify determinism checksum against tampering before restoring */
1535     uint32_t verify_hash = debug_compute_state_hash(&snap.state);
1536     if (verify_hash != snap.state_hash) {
1537         s_debug.tripwire_status = DEBUG_TRIPWIRE_HASH_MISMATCH;
1538         snprintf(s_debug.tripwire_details, sizeof(s_debug.tripwire_details),
1539                  "Rewind checksum mismatch on snap #%lu (0x%08X != 0x%08X)",
1540                  (unsigned long)snap.snapshot_id, verify_hash, snap.state_hash);
```

*Captures FNV-1a hashed simulation states in a 64-slot ring and restores bounded prior frames cleanly.*

Wholesale Listing: Complete verbatim code (#L1264-L1567) → Appendix: Deterministic State Capture & Time-Travel Rewind (debug.c).

## 77.4. Autonomous Validation Probes & DEBUG_STEP_AWAIT

*Autonomous Validation Probes & Tripwire Trap Chain (engine/src/debug/debug.c#L2367-L2411 · Mode: Group)*

```
2367 /**
```

```
2368  * @brief Legacy compatibility helper that latches a state-corruption tripwire trap.
2369  *
2370  * @deprecated Prefer `debug_trigger_probe()` or `DEBUG_TRIGGER_PROBE()` so every trap
2371  *             carries a structured probe ID, tier classification, and forensic report.
2372  *
2373  *   ### For Game Designers & Non-Programmers:
2374  *   Preserved safety brake for older modules that report state errors without
2375  *   naming a specific modern probe identifier.
2376  *
2377  *   ### For Systems Programmers & Engine Architects:
2378  *   Latches `DEBUG_TRIPWIRE_STATE_CORRUPT`, increments `traps_triggered`, notifies
2379  *   BIOS via `debug_fallback_record_fault()`, performs guarded snapshot lock and
2380  *   `DEBUG_STEP_AWAIT` entry when active and non-audit, and emits `LOG_WARN`.
2381  *
2382  * @param[in] subsystem Subsystem tag reporting the fault, or NULL for `"unknown"`.
2383  * @param[in] condition Description of the violated condition, or NULL for `"anomaly"`.
2384  * @param[in] file      Originating source file (`__FILE__`).
2385  * @param[in] line      Originating source line (`__LINE__`).
2386  * @triggers Latches tripwire state, records BIOS fault, and warns via guarded log output.
2387  */
2388 void debug_trap_tripwire(const char *subsystem, const char *condition,
2389                          const char *file, int line)
2390 {
2391     /* Step 1: Latch classic STATE_CORRUPT code and forensic detail string verbatim */
2392     s_debug.tripwire_status = DEBUG_TRIPWIRE_STATE_CORRUPT;
2393     s_debug.traps_triggered++;
2394
2395     snprintf(s_debug.tripwire_details, sizeof(s_debug.tripwire_details),
2396              "[%s] %s (%s:%d)",
2397              (subsystem != NULL) ? subsystem : "unknown",
2398              (condition != NULL) ? condition : "anomaly",
2399              (file != NULL) ? file : "unknown", line);
2400
2401     /* Step 2: Record fault in Tier-0 BIOS and complete via Safe-Mode fallback */
2402     (void)debug_fallback_record_fault(
2403         subsystem,
2404         (condition != NULL) ? condition : "anomaly",
2405         file,
2406         line,
2407         "debug_trap_tripwire"
2408     );
2409
2410     if (s_debug_active && !s_debug.in_probe && !s_debug.in_audit) {
2411         (void)debug_fallback_finish_with_fallback(
```

*Realizes the Probe → Safe-Mode Downgrade → DEBUG_STEP_AWAIT loop inside Layer 0 debug infrastructure.*

Wholesale Listing: Complete verbatim code (#L1918-L2438) → Appendix: Autonomous Validation Probes & Tripwire Trap Chain.

## 77.5. Self-Audit & Quick Verification

```
ctest --test-dir build -R "_debug_" --output-on-failure
./build/tenebris-lux --audit
```

1. *Why keep a separate* `s_audit_backup` *inside* `debug.c`*?*

   Running destructive rewind, tripwire, or wrap drills during `debug_audit()` without clobbering live gameplay telemetry requires isolating self-test state safely in static `.bss` (`35,232 B` live + `35,232 B` backup).

2. *Can* `cog.debug` *detect bugs even if a game UI hasn't rendered a single pixel yet?*

   Yes. Every Layer 0 cog registers health probes (`io.`, `log.`, etc.) during boot, catching config corruption or structural contract drift before Layer 1 simulation starts.

## 77.6. Companion Deep-Dive Chapters

1. Open Paired `debug.h` + `debug.c` Walkthrough for ring rewind and probe internals.
2. Read `debug_types.h` Leaf Vocabulary and Paired `debug_fallback.h` + `debug_fallback.c` BIOS.
3. Continue onward to Layer 0 Zero-Heap I/O Pipeline Overview.

# Chapter 78. Debug Module Pair: Rewind Engine, Watchpoints & Probes (debug.h + debug.c)

**Source File**: docs/code/debug/01_debug_engine_and_probes.adoc

| Attribute | Verified Contract (cog.debug) |
|---|---|
| Public Contract (.h) | engine/include/debug/debug.h (1,207 lines) — stepping control, snapshot rewind, watchpoints, 32-slot probe registry, tripwires, cards, and audit gates. |
| Runtime Engine (.c) | engine/src/debug/debug.c (3,221 lines) — FNV-1a snapshot ring, GCD in-place wrap rotation, interactive AWAIT state machine, and audit isolation. |
| Chassis Footprint | .bss 70.5 KiB (s_debug 35,232 B + s_audit_backup 35,232 B) + .rodata 9.7 KiB + single-slot normalize stack frame. |
| Inputs & Seams | Reads engine/config/diagnostics/{debugger,taxonomy}/*.json; pairs downward with cog.debug.types, cog.debug.fallback, and cog.log. |

debug.h + debug.c turn "something went wrong during frame 412" into inspectable facts: every snapshot records simulation numbers, hashes them with 32-bit FNV-1a, preserves explicitly locked milestones when the circular buffer wraps, and hosts live health probes across the rest of the engine.

## 78.1. Snapshot Capture, Locking & Time-Travel Rewind

Calling debug_record_snapshot() freezes a copy of debug_state_fields_t alongside a 32-bit FNV-1a checksum and the synchronized log_sequence_id from cog.log. Calling debug_lock_snapshot() protects milestone frames from overwrite; when the ring wraps, debug.c uses O(cap) GCD wrap rotation to evict unlocked slots while preserving chronological order. debug_rewind_to_snapshot() and debug_diff_snapshots() restore or compare any retained frame:

*Deterministic State Capture & Time-Travel Rewind (debug.c) (engine/src/debug/debug.c#L1503-L1540 · Mode: Group)*

```
1503 /**
1504  * @brief Rewinds state fields back to snapshot `index` after cryptographic checksum check.
1505  *
1506  * ### For Game Designers & Non-Programmers:
1507  * Restores actor stats from an earlier saved frame, pausing safely in AWAIT if memory
1508  * corruption altered the snapshot's stored fingerprint.
1509  *
1510  * ### For Systems Programmers & Engine Architects:
1511  * Fetches snapshot via `debug_get_snapshot()`, verifies `debug_compute_state_hash(&snap.state)`
1512  * matches `snap.state_hash` (engaging `DEBUG_TRIPWIRE_HASH_MISMATCH` + Safe-Mode AWAIT +
1513  * `LOG_ERROR` on mismatch), and emits `LOG_INFO` on verified restore.
1514  *
1515  * @triggers
1516  * - Layer 4 Tests : `tests/unit/debug/test_debug_unit.c:Section1`
1517  *
1518  * @param[in]  index      Chronological target snapshot offset.
1519  * @param[out] out_fields Restored simulation state fields on verified success.
1520  * @return true if checksum passed and `out_fields` populated; false on error or mismatch.
1521  */
1522 bool debug_rewind_to_snapshot(size_t index, debug_state_fields_t *out_fields)
1523 {
1524     /* Step 1: Validate preconditions and fetch chronological snapshot copy */
1525     if (!s_debug_active || out_fields == NULL) {
1526         return false;
1527     }
1528
1529     debug_snapshot_t snap;
1530     if (!debug_get_snapshot(index, &snap)) {
1531         return false;
1532     }
```

```
1533
1534    /* Step 2: Verify determinism checksum against tampering before restoring */
1535    uint32_t verify_hash = debug_compute_state_hash(&snap.state);
1536    if (verify_hash != snap.state_hash) {
1537        s_debug.tripwire_status = DEBUG_TRIPWIRE_HASH_MISMATCH;
1538        snprintf(s_debug.tripwire_details, sizeof(s_debug.tripwire_details),
1539                 "Rewind checksum mismatch on snap #%lu (0x%08X != 0x%08X)",
1540                 (unsigned long)snap.snapshot_id, verify_hash, snap.state_hash);
```

*Captures FNV-1a hashed simulation states in a 64-slot ring and restores bounded prior frames cleanly.*

Wholesale Listing: Complete verbatim code (#L1264-L1567) → Appendix: Deterministic State Capture & Time-Travel Rewind (debug.c).

## 78.2. Stepping Modes, Breakpoints & DEBUG_STEP_AWAIT

| Capability | Public Contract in debug.h |
|---|---|
| Execution Pacing & Await | debug_set_step_mode(), debug_is_awaiting(), debug_can_step(), and debug_resolve_await() pause execution deterministically after a fault or watchpoint hit until resolved. |
| Simulation Watchpoints | debug_add_breakpoint(), debug_remove_breakpoint(), debug_clear_breakpoints(), and debug_check_breakpoints() watch up to 16 numeric thresholds (hp == 0, stance shift, alignment breach, etc.) in O(n). |
| Provenance Queries | debug_is_data_driven() and debug_is_safe_mode() distinguish disk JSON profiles from compiled .rodata ROM fallback mode cleanly. |

## 78.3. Validation Probe Matrix & Tripwire Chain

Child modules bind self-checks directly into debug.c□s `32-slot probe array:

*Autonomous Validation Probes & Tripwire Trap Chain (engine/src/debug/debug.c#L2367-L2411 · Mode: Group)*

```
2367 /**
2368  * @brief Legacy compatibility helper that latches a state-corruption tripwire trap.
2369  *
2370  * @deprecated Prefer `debug_trigger_probe()` or `DEBUG_TRIGGER_PROBE()` so every trap
2371  *             carries a structured probe ID, tier classification, and forensic report.
2372  *
2373  *   ### For Game Designers & Non-Programmers:
2374  *   Preserved safety brake for older modules that report state errors without
2375  *   naming a specific modern probe identifier.
2376  *
2377  *   ### For Systems Programmers & Engine Architects:
2378  *   Latches `DEBUG_TRIPWIRE_STATE_CORRUPT`, increments `traps_triggered`, notifies
2379  *   BIOS via `debug_fallback_record_fault()`, performs guarded snapshot lock and
2380  *   `DEBUG_STEP_AWAIT` entry when active and non-audit, and emits `LOG_WARN`.
2381  *
2382  * @param[in] subsystem Subsystem tag reporting the fault, or NULL for `"unknown"`.
2383  * @param[in] condition Description of the violated condition, or NULL for `"anomaly"`.
2384  * @param[in] file      Originating source file (`__FILE__`).
2385  * @param[in] line      Originating source line (`__LINE__`).
2386  * @triggers Latches tripwire state, records BIOS fault, and warns via guarded log output.
2387  */
2388 void debug_trap_tripwire(const char *subsystem, const char *condition,
2389                          const char *file, int line)
2390 {
2391     /* Step 1: Latch classic STATE_CORRUPT code and forensic detail string verbatim */
2392     s_debug.tripwire_status = DEBUG_TRIPWIRE_STATE_CORRUPT;
2393     s_debug.traps_triggered++;
2394
2395     snprintf(s_debug.tripwire_details, sizeof(s_debug.tripwire_details),
2396              "[%s] %s (%s:%d)",
```

```
2397            (subsystem != NULL) ? subsystem : "unknown",
2398            (condition != NULL) ? condition : "anomaly",
2399            (file != NULL) ? file : "unknown", line);
2400
2401    /* Step 2: Record fault in Tier-0 BIOS and complete via Safe-Mode fallback */
2402    (void)debug_fallback_record_fault(
2403        subsystem,
2404        (condition != NULL) ? condition : "anomaly",
2405        file,
2406        line,
2407        "debug_trap_tripwire"
2408    );
2409
2410    if (s_debug_active && !s_debug.in_probe && !s_debug.in_audit) {
2411        (void)debug_fallback_finish_with_fallback(
```

*Realizes the Probe → Safe-Mode Downgrade → DEBUG_STEP_AWAIT loop inside Layer 0 debug infrastructure.*

Wholesale Listing: Complete verbatim code (#L1918-L2438) → Appendix: Autonomous Validation Probes & Tripwire Trap Chain.

- **Probe Lifecycle**: `debug_register_validation_probe()`, `debug_find_probe()`, `debug_get_probe_count()`, `debug_get_probe_at()`, `debug_evaluate_probe()`, and `debug_clear_validation_probes()`.
- **Tripwire Matrix & Cards**: `debug_check_tripwires()`, `debug_get_tripwire_report()`, `debug_dump_tripwire_report()`, `debug_get_tripwire_status()`, `debug_reset_tripwire()`, `debug_execute_inspect()`, and `debug_execute_dump_card()`.
- **Safe Self-Audit (`debug_preflight` / `debug_audit`)**: Swaps live state cleanly against `s_audit_backup` in `.bss`, runs destructive probe/rewind drills safely, and restores pre-audit telemetry without touching heap memory.

## 78.4. Paired Neighbors

- Pure leaf vocabulary types: `debug_types.h`
- Tier-0 debugger BIOS & ROM vault: `debug_fallback.h` + `debug_fallback.c`
- Verification test ladder: Debugger Testing & Tripwires

# Chapter 79. Debug Leaf Contract: Snapshots, GCD Ring & Tripwires (debug_types.h)

**Source File**: docs/code/debug/02_debug_types_and_telemetry.adoc

| Attribute | Verified Contract (cog.debug.types) |
|---|---|
| Header File | engine/include/debug/debug_types.h (652 lines) — standalone leaf POD vocabulary (archetype: pure_types \| stateless). |
| Runtime Source | None intentionally (0 B .bss, 0 B .data). |
| Fixed Capacities | DEBUG_MAX_SNAPSHOTS = 64, 16 breakpoints, 32 validation probes (Sub-Assembly 2.1). |
| Architectural Fix | Eliminated legacy struct debug_config; forward declarations and unsafe void * pointer erasure. |

Before debug_types.h was isolated as a pure leaf vocabulary, debug.h and debug_fallback.h had to dance around forward declarations and void * parameters to avoid circular header includes. By giving both files direct compile-time knowledge of debug_state_fields_t, debug_snapshot_t, and debug_config_t, every mismatch becomes an instant compiler error instead of a hidden runtime bug.

## 79.1. Structural Ontology (Sub-Assembly 2.3)

debug_types.h groups enums (debug_tripwire_t, debug_tripwire_reaction_t, debug_test_tier_t, debug_step_mode_t, debug_breakpoint_type_t) alongside caller-owned POD structs:

*State Debugger Leaf Vocabulary & Snapshot Structs (debug_types.h Sub-Assembly 2.3)*
*(engine/include/debug/debug_types.h#L147-L190 · Mode: Subassembly)*

```
147 /*
148 ---------------------------------------------------------------------------
149 Sub-Assembly 2.3: Shared Structural Ontology & POD Debugger Types
150 ---------------------------------------------------------------------------
151 Canonical non-anonymous enums and structures shared identically between
152 `cog.debug` and quarantined `cog.debug.fallback`.
153 ---------------------------------------------------------------------------
154 */
155
156 /**
157  * @enum debug_tripwire
158  * @brief Latched diagnostic tripwire states for the debugger and validation pipeline.
159  *
160  * ### For Game Designers & Non-Programmers:
161  * Alarm switches indicating whether configuration data was missing or damaged,
162  * an automated safety test fired, or Safe Mode engaged after a fault.
163  *
164  * ### For Systems Programmers & Engine Architects:
165  * Latching error enum preserving `0..4` legacy discriminants and `5..7` active
166  * validation probe, audit breach, and Safe-Mode downgrade discriminants.
167  */
168 typedef enum debug_tripwire {
169     DEBUG_TRIPWIRE_CLEAR = 0,         /**< All tripwires clear; simulation healthy. */
170     DEBUG_TRIPWIRE_MANIFEST_MISSING,  /**< Debugger/subsystem manifest unavailable. */
171     DEBUG_TRIPWIRE_SETTING_MISSING,   /**< Designated JSON setting profile missing. */
172     DEBUG_TRIPWIRE_STATE_CORRUPT,     /**< Simulation state breached domain invariants. */
173     DEBUG_TRIPWIRE_HASH_MISMATCH,     /**< Snapshot checksum diverged from recorded state. */
174     DEBUG_TRIPWIRE_TEST_FAILED,       /**< Registered validation probe / fail-test fired. */
175     DEBUG_TRIPWIRE_AUDIT_BREACH,      /**< Block 4 preflight or audit invariant failed. */
176     DEBUG_TRIPWIRE_FALLBACK_ENGAGED   /**< Downgraded to Safe Mode to complete execution. */
177 } debug_tripwire_t;
178
179 /**
```

```
180  * @enum debug_tripwire_reaction
181  * @brief Recovery action taxonomy ingested from taxonomy/tripwire_matrix.json.
182  *
183  * ### For Game Designers & Non-Programmers:
184  * Selects how the engine reacts when an alarm trips: logging warnings, clamping
185  * bad numbers, dropping into Safe Mode, rewinding time, or halting in AWAIT.
186  *
187  * ### For Systems Programmers & Engine Architects:
188  * Mapped 1:1 to `tripwire_matrix.json` canonical reaction identifier strings.
189  */
190 typedef enum debug_tripwire_reaction {
```

*Standalone integer/char POD snapshots eliminating void* type erasure between Layer 0 and Layer 1.*

Wholesale Listing: Complete verbatim code (#L147-L453) → Appendix: State Debugger Leaf Vocabulary & Snapshot Structs (debug_types.h Sub-Assembly 2.3).

| Type Name | What It Captures Without game/ Coupling |
|---|---|
| debug_state_fields_t | Pure integer snapshot packet (turn, hp, max_hp, mp, tp, shield_power, sword_edge, stance, moral_sway, location, status counters, foes_vanquished) linking Layer 1 into Layer 0 safely. |
| debug_snapshot_t | Timestamped ring slot binding snapshot_id, state_hash (FNV-1a), correlated log_sequence_id, and eviction locked flag. |
| debug_breakpoint_t | Numeric trigger descriptor (type, target_value, is_active, 64-byte label). |
| debug_config_t | Runtime toggles (break_on_tripwire, break_on_sway_breach, interactive_repl, loaded_from_data provenance marker). |

## 79.2. Static Bounds Verification

Sub-Assembly 2.1 enforces static_assert proofs ensuring snapshot ring and breakpoint capacities remain strictly positive across C23/C17/C11 and C20/C17 compilation modes. Verify header compliance via:

```
cargo run --manifest-path tools/Cargo.toml --bin tl_tools -- validate headers
```

## 79.3. Consumers of debug_types.h

- Paired debug.h + debug.c Engine
- Paired debug_fallback.h + debug_fallback.c BIOS

# Chapter 80. Debug Safe-Mode BIOS: Tripwire Emergency Floor (debug_fallback.h + debug_fallback.c)

**Source File**: docs/code/debug/03_debug_fallback_bios.adoc

| Attribute | Verified Contract (cog.debug.fallback) |
|---|---|
| Fallback Contract (.h) | engine/include/debug/debug_fallback.h (900 lines) — BIOS boot, JSON span parsers, ROM vault loaders, fault capture, and AWAIT latch. |
| BIOS Implementation (.c) | engine/src/debug/debug_fallback.c (1,852 lines) — static fault templates, .rodata ROM bridge, and emergency state verification. |
| Chassis Footprint | .bss 256 B + .data 464 B + .rodata 76,136 B (debug_diagnostics_rom.h vault) + stack ⇐ 1 KiB. |
| Inputs & Pairing | Embedded ROM mirrors of engine/config/diagnostics/debugger/.json & taxonomy/.json; pairs downward with cog.log.fallback. |

When disk diagnostics catalogs vanish or a debugger JSON config gets mangled during rapid iteration, debug_fallback.[ch] keeps every error code, snapshot limit template, and DEBUG_STEP_AWAIT safety latch intact directly out of read-only ELF .rodata.

## 80.1. Tier-0 BIOS Boot & .rodata Vault (Sub-Assemblies 3.1–3.2)

debug_fallback_boot() initializes Safe-Mode inspection policies before full I/O mounts, loading profiles cleanly out of debug_diagnostics_rom.h:

*Tier-0 Debugger BIOS Boot & ROM Profile Hydration (debug_fallback.c) (engine/src/debug/debug_fallback.c#L1715-L1752 · Mode: Group)*

```
1715 /**
1716  * @brief Arms Tier-0 Safe-Mode BIOS state and seeds active ROM profile telemetry.
1717  *
1718  * ### For Game Designers & Non-Programmers:
1719  * Turns on emergency fault recording right at startup so any failure before external
1720  * configs finish loading is still captured and recoverable.
1721  *
1722  * ### For Systems Programmers & Engine Architects:
1723  * Copies caller profile name into `s_fb.active_rom_profile`, or hydrates `"default"`
1724  * via `debug_fallback_load_rom_profile()` with monolith fallback, sets `s_fb.booted = true`,
1725  * and logs an isolated `LOG_SEV_INFO` boot record into `cog.log.fallback`.
1726  *
1727  * @triggers
1728  * - Layer 4 Tests : `tests/unit/debug/test_debug_fallback_unit.c:Section4`
1729  *
1730  * @param[in] opt_config  Optional pre-populated `debug_config_t` (or NULL for ROM default).
1731  * @param[in] config_size Must equal `sizeof(debug_config_t)` when `opt_config != NULL`.
1732  * @return true once Tier-0 Safe-Mode debugger BIOS is booted.
1733  */
1734 bool debug_fallback_boot(const debug_config_t *opt_config, size_t config_size)
1735 {
1736     /* Step 1: Resolve active Safe-Mode profile string from caller, ROM, or monolith */
1737     debug_config_t scratch;
1738     if (opt_config != NULL && config_size == sizeof(debug_config_t)) {
1739         str_safe_copy(s_fb.active_rom_profile, opt_config->profile_name,
1740                       sizeof(s_fb.active_rom_profile));
1741     } else if (!debug_fallback_load_rom_profile("default", &scratch, sizeof(scratch))) {
1742         debug_get_monolith_fallback_config_raw(&scratch, sizeof(scratch));
1743         str_safe_copy(s_fb.active_rom_profile, scratch.profile_name,
1744                       sizeof(s_fb.active_rom_profile));
1745         fb_emit_bios_log(
```

```
            LOG_SEV_WARNING,
            __FILE__,
            __LINE__,
            __func__,
            MODULE_ID,
            "ROM default profile unavailable; seeded Safe-Mode BIOS from monolith template."
        );
```

*Hydrates Safe-Mode debugger policies directly from embedded .rodata ROMs when disk files are unavailable.*

Wholesale Listing: Complete verbatim code (#L1008-L1766) → Appendix: Tier-0 Debugger BIOS Boot & ROM Profile Hydration (debug_fallback.c).

| ROM & Parser Function | Verified Behavior |
|---|---|
| debug_fallback_load_rom_profile() | Loads compiled "default", "strict", or "trace" debugger presets while keeping loaded_from_data = false. |
| debug_fallback_load_rom_snapshot_policy() & debug_fallback_load_rom_tripwire_matrix() | Hydrates ring limits and tripwire severity matrices directly from embedded .rodata payloads (76,136 B total diagnostics ROM vault). |
| debug_fallback_parse_config_json() / parse_snapshot_policy_json() | Shared zero-heap JSON parsers reused symmetrically by disk loaders and ROM hydration (Sub-Assembly 3.1). |

## 80.2. Fault Capture & Emergency AWAIT Latch (Sub-Assembly 3.3)

If a critical subsystem faults during Safe Mode, debug_fallback_record_fault() logs the anomaly cleanly, debug_fallback_finish_with_fallback() downgrades execution onto known-good ROM boundaries, and debug_fallback_is_awaiting() / debug_fallback_clear_await() expose the held state to CLI/test operators cleanly.

## 80.3. BIOS Self-Audit & ROM Info (Blocks 4 & 5)

- debug_fallback_audit() verifies embedded ROM accessibility and static templates without altering live debug sessions.
- debug_fallback_get_rom_info() reports exact ROM counts and bytes (debug_fallback_rom_info_t).
- debug_fallback_get_stats() and debug_fallback_execute_inspect() populate caller structs deterministically without dynamic allocation.

## 80.4. Next Steps in debug/

- Check the 4-tier tests covering debugger fallback and probe escalation: Debugger Testing & Tripwires
- Trace how primary debug.c hosts live probes: debug.h + debug.c

# Chapter 81. Debug Verification Suite: Rewind, Probes & Tripwires Across All 4 Tiers

**Source File**: docs/code/debug/04_debug_testing_and_tripwires.adoc

Because cog.debug serves as Cornerstone's self-verification backbone (tests/common/cornerstone_test.h), testing debug/ requires both proving the debugger's own state math in isolation **and** proving that child modules trigger probes and DEBUG_STEP_AWAIT cleanly when integrated.

| Tier | CTest Target | Source File | Primary Verification Focus |
|---|---|---|---|
| Unit | unit_debug_engine_test | tests/unit/debug/test_debug_unit.c | Snapshot capture/hash determinism, GCD wrap with locked slots, watchpoints, time-travel rewind, and probe registration. |
| Unit (BIOS) | unit_debug_fallback_test | tests/unit/debug/test_debug_fallback_unit.c | Emergency ROM hydration, debug_fallback_boot(), fault capture, and fallback AWAIT clear transitions. |
| Functional | functional_debug_test | tests/functional/test_debug_functional.c | External caller simulation: stepping mode, inspection cards, and tripwire dumps via public headers. |
| Integrated | integrated_debug_test | tests/integrated/test_debug_integrated.c | Cross-module coordination tying debug + debug_fallback + log + io/ JSON debugger profiles cleanly. |
| System | system_debug_test | tests/system/test_debug_system.c | End-to-end tripwire escalation, missing-disk downgrade drills, and debug_audit() recovery checks. |

## 81.1. Fine-Grained Invariants Tested Across debug/

- **always_pass & never_fail**: Confirms clean debug_init() / debug_shutdown(), null-safe snapshot getters, and zero-crash clamping when callers pass out-of-range snapshot indices.
- **assertions_and_guarantees**: Verifies deterministic 32-bit FNV-1a checksums across identical debug_state_fields_t structs and strict protection of locked milestone snapshots during circular buffer wrap.
- **emergence & audit_gates**: Exercises full probe traps (debug_trigger_probe → debug_trap_tripwire → DEBUG_STEP_AWAIT → debug_resolve_await) alongside debug_audit() isolation using s_audit_backup.

## 81.2. Live Emergence Check (test_debug_unit_emergence)

*Debugger Unit Emergence & Tripwire Verification (test_debug_unit_emergence)*
*(tests/unit/debug/test_debug_unit.c#L410-L447 · Mode: Function)*

```
410 static void test_debug_unit_emergence(void)
411 {
412     cornerstone_test_section(4, "Emergence (1000-Turn Wrap, 4-Slot Lock Quota & AWAIT Recovery)");
413 
414     TEST_ASSERT(debug_init(NULL), "debug_init succeeds for emergence section");
415 
416     /* 4.1 1000-turn ring wrap-around */
417     for (int t = 1; t <= 1000; t++) {
```

```
418         debug_state_fields_t s = cornerstone_test_make_state(
419             t, "Elijah", 500 - (t % 200), t % 100, (t * 2) % 100, t % 4, (t % 200) - 100);
420         uint64_t sid = debug_record_snapshot(&s, "sim_turn", "step");
421         TEST_ASSERT_U64_EQ(sid, (uint64_t)t, "Monotonic ID strictly tracked across 1000 turns");
422     }
423
424     TEST_ASSERT_U64_EQ(debug_get_snapshot_count(), (uint64_t)DEBUG_MAX_SNAPSHOTS,
425                        "Ring capacity capped at 64");
426
427     debug_snapshot_t oldest;
428     TEST_ASSERT(debug_get_snapshot(0U, &oldest) && (oldest.state.turn == 937),
429                 "Oldest turn after 1000 pushes is 937");
430     const debug_snapshot_t *latest = debug_get_latest_snapshot();
431     TEST_ASSERT((latest != NULL) && (latest->state.turn == 1000), "Latest turn is 1000");
432
433     debug_state_fields_t restored;
434     TEST_ASSERT(debug_rewind_to_snapshot(30U, &restored) &&
435                     (restored.turn == 967) &&
436                     (restored.hp == (500 - (967 % 200))),
437                 "Rewind index 30 restores turn 967 and exact HP formula");
438
439     debug_shutdown();
440
441     /* 4.2 Locked critical snapshot quota (4) + DEBUG_TRIGGER_PROBE -> AWAIT recovery */
442     TEST_ASSERT(debug_init(NULL), "Re-init debugger for lock quota & AWAIT check");
443     TEST_ASSERT(debug_can_step() && !debug_is_awaiting(), "Stepping armed initially");
444
445     uint64_t lock_ids[5] = { 0U };
446     for (int t = 1; t <= 6; t++) {
447         debug_state_fields_t s = cornerstone_test_make_state(t, "Elijah", 450 + t, 60, 40, t % 4, t);
```

*Verifies locked snapshot preservation, GCD ring rotation, probe traps, and Safe-Mode AWAIT clearing.*

Wholesale Listing: Complete verbatim code (#L410-L510) → Appendix: Debugger Unit Emergence & Tripwire Verification (test_debug_unit_emergence).

## 81.3. Running the Debugger Verification Targets

```
ctest --test-dir build -L debug --output-on-failure
ctest --test-dir build \
  -R "^(unit_debug_|functional_debug_|integrated_debug_|system_debug_)" --output-on-failure
```

1. *Why does* `system_debug_test` *take slightly longer (~0.58 s) than pure arithmetic unit tests?*

   `system_debug_test` exercises full snapshot dumps, forensic cards, multi-cycle ring rotations, probe evaluations, and disk/ROM profile transitions across the integrated stack cleanly inside CTest.

## 81.4. Continue Reading

- Revisit debugger implementation: `debug.h` + `debug.c`
- Advance to the next root pillar: Layer 0 Zero-Heap I/O Pipeline Overview

# Chapter 82. Layer 0 Zero-Heap I/O Pipeline Subsystem Overview

**Source File**: `docs/code/io_pipeline.adoc`

Most engines leak memory or crash unpredictably when loading files: missing folders, oversized JSON payloads, or half-edited game mods turn into hard-to-trace bugs. Cornerstone's **Layer 0 Zero-Heap I/O Pipeline** (`engine/include/io/ & engine/src/io/`) solves this like an assembly line with built-in quality gates: locate the file across layered VFS tiers (or fall back to a compiled `.rodata` ROM), read bounded bytes into caller-owned memory, scan JSON in a single stack pass, and refuse to hand gameplay anything that fails 5-block schema checks.



**Navigate the `io/` folder**: Pair this overview with docs/code/io/ for detailed chapters on `io.h`, `fs_utils`, `path_finder`, `json_utils`, `master_loader`, `io_fallback`, and all 8 `io` CTest suites.

## 82.1. Architectural Philosophy: Assembly-Line Ingestion Without Heap

Rather than forcing four separate fallback files (`*_fallback`), Rev 5 unified emergency I/O services inside a single Tier-0 BIOS module (`io_fallback.[ch]`) backed by `io_config_rom.h` (8 compiled `.rodata` JSONs covering `engine/config/io/`), sitting beneath four focused operational engines:

1. **`path_finder` (`cog.path_finder`)**: Anchors repository root dynamically and resolves `"config"` domain files across 3 tiers: `game/config/` → `engine/config/` → `game/data/presentation/config/`.
2. **`fs_utils` (`cog.fs_utils`)**: Performs POSIX stat, mkdir, bounded file reads/writes (⇐ `64 KiB` policy ceiling), read-only Safe-Mode write locks, and `rom://` fallback reads.
3. **`json_utils` (`cog.json_utils`)**: Executes zero-heap single-pass JSON token/value scanning capped at bracket nesting depth ⇐ `32` on caller stack—zero hidden static scratch buffers.
4. **`master_loader` (`cog.master_loader`)**: Verifies the 5-block JSON envelope (`block_1_metadata`..`block_5_closing`), slices non-owning pointers strictly to `block_3_body`, checks schema IDs/versions, and downgrades corrupt configs to ROM Safe Mode.
5. **`io.h` Umbrella (`cog.io`)**: Stateless compile-time facade coordinating includes and unified subsystem audits cleanly across all child cogs.

## 82.2. Verified Rev 5 File & Documentation Map

| Companion Doc | Engine Files Covered | Role & Chassis Highlights |
|---|---|---|
| 01: I/O Umbrella Facade | `engine/include/io/io.h` (`cog.io`) | Header-only stateless aggregator (`0 B .bss/.data/.text`); unified audit and lifecycle convenience seam. |
| 02: Bounded Filesystem I/O | `engine/include/io/fs_utils.h` + `engine/src/io/fs_utils.c` (`cog.fs_utils`) | Bounded reads/writes (⇐ `64 KiB`), `read_only_safe_mode` latch, `rom://` reads (`232 B .bss`, `4,064 B .rodata`). |
| 03: 3-Tier VFS Path Finder | `engine/include/io/path_finder.h` + `engine/src/io/path_finder.c` (`cog.path_finder`) | Upward root resolution & `vfs_routes.json` hierarchy across up to 36 routes (`288 B .bss`, `5,300 B .rodata`). |

| Companion Doc | Engine Files Covered | Role & Chassis Highlights |
|---|---|---|
| 04: Zero-Heap JSON Scanner | engine/include/io/json_utils.h + engine/src/io/json_utils.c (cog.json_utils) | Single-pass token locator & typed span extractors capped at depth ⇐ 32 (144 B .bss, 5,740 B .rodata). |
| 05: Master Manifest Loader | engine/include/io/master_loader.h + engine/src/io/master_loader.c (cog.master_loader) | 5-block envelope validator & non-owning block_3_body slicer (112 B .bss, 5,916 B .rodata). |
| 06: Tier-0 I/O BIOS & ROM | engine/include/io/io_fallback.h + engine/src/io/io_fallback.c (cog.io.fallback) | Unified BIOS across all 4 I/O pillars, floppy boot mode, and 8 compiled .rodata ROMs (8,312 B .bss, 40,256 B .rodata). |
| 07: I/O Testing Suite | 8 CTest targets across tests/{unit,functional,integrated,system}/ | Exhaustive verification of bounds clamping, 3-tier precedence, floppy BIOS, and corrupt JSON downgrade. |

## 82.3. Live VFS Routes & Shared JSON Span Resolution

engine/config/io/vfs_routes.json governs repository root discovery and 3-tier precedence cleanly in data:

*Complete 5-Block VFS Precedence & Policy Configuration (vfs_routes.json) (engine/config/io/vfs_routes.json#L55-L99 · Mode: Whole)*

```
55  "block_3_body": {
56    "default_data_root": "game/data",
57    "env_override_var": "TL_DATA_ROOT",
58    "block_path_traversal": true,
59    "max_symlink_depth": 4,
60    "allow_mod_mounts": true,
61    "rom_fallback_enabled": true,
62    "probe_await_on_corrupt": true,
63    "config_precedence": [
64      "game/config/",
65      "engine/config/",
66      "game/data/presentation/config/"
67    ],
68    "fallback_candidate_roots": [
69      "game/data",
70      "../game/data",
71      "engine/config",
72      "../engine/config",
73      "data",
74      "../data"
75    ],
76    "asset_roots": [
77      "game/assets",
78      "../game/assets",
79      "assets",
80      "../assets"
81    ],
82    "domain_search_paths": [
83      "",
84      "runtime/",
85      "runtime/math/",
86      "runtime/string/",
87      "runtime/time/",
88      "diagnostics/",
89      "diagnostics/telemetry/",
90      "diagnostics/debugger/",
91      "diagnostics/taxonomy/",
92      "diagnostics/formats/",
```

```
93      "io/",
94      "audio/",
95      "viewport/",
96      "input/",
97      "standards/",
98      "standards/blocks/",
99      "mechanics/",
```

*Canonical 5-block engine config defining VFS root search hierarchy, alias tables, and fail-closed safety flags.*

Wholesale Listing: Complete verbatim code (#L1-L159) → Appendix: Complete 5-Block VFS Precedence & Policy Configuration (vfs_routes.json).

## 82.4. Zero-Duplication Parser Bridge & Verification

Notice in `io_fallback.c` how BIOS fallback and `master_loader` share the exact same boundary-safe `block_3_body` span scanner (`io_fb_resolve_body_span`):

*Shared Zero-Heap JSON Block Resolver & VFS Parser Group (`engine/src/io/io_fallback.c#L779-L806` · Mode: `Group`)*

```
779 static bool io_fb_resolve_body_span(const char *json_text,
780                                       size_t json_len,
781                                       bool allow_strlen,
782                                       const char **out_body_start,
783                                       const char **out_body_end)
784 {
785     if (json_text == NULL || out_body_start == NULL || out_body_end == NULL) {
786         return false;
787     }
788 
789     size_t effective_len = (json_len > 0U) ? json_len : (allow_strlen ? strlen(json_text) : 0U);
790     if (effective_len == 0U || effective_len > IO_FB_JSON_MAX_BYTES) {
791         return false;
792     }
793 
794     const char *json_end = json_text + effective_len;
795     const char *body_start = NULL;
796     size_t body_len = 0U;
797     if (!io_fb_find_block_span(json_text, json_end, "block_3_body", '{', '}',
798                                &body_start, &body_len) ||
799         body_start == NULL || body_len == 0U) {
800         return false;
801     }
802 
803     *out_body_start = body_start;
804     *out_body_end = body_start + body_len;
805     return true;
806 }
```

*Tier-0 in-place span scanning reused seamlessly between BIOS fallback and master_loader.*

Wholesale Listing: Complete verbatim code (#L779-L1194) → Appendix: Shared Zero-Heap JSON Block Resolver & VFS Parser Group.

```
ctest --test-dir build -R "(fs_utils|path_finder|json_utils|master_loader|io_)" --output-on-failure
cargo run --manifest-path tools/Cargo.toml --bin tl_tools -- validate data
```

1. *Why prohibit static `32 KiB` scratch arrays inside I/O C functions?*

   A forgotten function-local `static char buf[32768]` permanently pins idle bytes inside `.bss`. Passing caller-owned slices keeps `fs_utils` (`232 B .bss`), `path_finder` (`288 B .bss`), `json_utils` (`144 B .bss`), and `master_loader` (`112 B .bss`) razor-lean!

## 82.5. Companion Deep-Dive Chapters

1. Inspect `io.h` Umbrella Facade and `fs_utils.h` + `fs_utils.c`.
2. Trace VFS & parsing via `path_finder`, `json_utils`, and `master_loader`.
3. Understand zero-disk survival in `io_fallback` and I/O Testing.

# Chapter 83. I/O Umbrella Facade & 4-Pillar Boot Pipeline (io.h)

**Source File**: docs/code/io/01_io_umbrella_and_pipeline.adoc

| Attribute | Verified Contract (cog.io) |
|---|---|
| Header File | engine/include/io/io.h (301 lines) — pure compile-time facade (archetype: facade_umbrella \| stateless). |
| Runtime Source | None intentionally (0 B .bss, 0 B .data, 0 B .text); delegates runtime execution cleanly to its 5 child cogs. |
| Constituent Cogs | io_fallback.h → fs_utils.h → json_utils.h → path_finder.h → master_loader.h (below core/manifest.h). |
| Manifest Anchor | engine/config/io/manifest.json (probes: io_umbrella.*). |

Think of io.h as a clean electrical junction box, never an extra processor. Higher layers (game/src/main.c or test harnesses) can include "io/io.h" to bring all five I/O pillars online in verified topological order—while internal log/ or lower core/ cogs can still include just the fine-grained header they need without dragging unnecessary dependencies upward.

## 83.1. Downward Re-Export Order (BLOCK 2 of io.h)

io.h opens cleanly with standard ISO C11 headers (<stdbool.h>, <stddef.h>, <stdint.h>), binds "core/manifest.h", and re-exports child cogs in strict topological sequence:

1. "io/io_fallback.h" — Tier-0 BIOS & .rodata ROM vault (cog.io.fallback).
2. "io/fs_utils.h" — bounded POSIX file/stream primitives (cog.fs_utils).
3. "io/json_utils.h" — zero-heap JSON span scanner (cog.json_utils).
4. "io/path_finder.h" — 3-tier VFS path resolution (cog.path_finder).
5. "io/master_loader.h" — 5-block envelope validator & scoped body slicer (cog.master_loader).

Sub-Assembly 2.1 aligns system-wide ceilings across the suite—such as FS_PATH_MAX = 512 bytes and 64 KiB configuration payload limits—ensuring static stack and .bss buffers across layers agree at compile time.

## 83.2. Configuration Anchor (engine/config/io/)

All operational policies hydrated through io.h live in data files under engine/config/io/, anchored by vfs_routes.json and manifest.json:

*Complete 5-Block VFS Precedence & Policy Configuration (vfs_routes.json) (engine/config/io/vfs_routes.json#L55-L99 · Mode: Whole)*

```
55  "block_3_body": {
56    "default_data_root": "game/data",
57    "env_override_var": "TL_DATA_ROOT",
58    "block_path_traversal": true,
59    "max_symlink_depth": 4,
60    "allow_mod_mounts": true,
61    "rom_fallback_enabled": true,
62    "probe_await_on_corrupt": true,
63    "config_precedence": [
64      "game/config/",
65      "engine/config/",
66      "game/data/presentation/config/"
```

```
    ],
    "fallback_candidate_roots": [
      "game/data",
      "../game/data",
      "engine/config",
      "../engine/config",
      "data",
      "../data"
    ],
    "asset_roots": [
      "game/assets",
      "../game/assets",
      "assets",
      "../assets"
    ],
    "domain_search_paths": [
      "",
      "runtime/",
      "runtime/math/",
      "runtime/string/",
      "runtime/time/",
      "diagnostics/",
      "diagnostics/telemetry/",
      "diagnostics/debugger/",
      "diagnostics/taxonomy/",
      "diagnostics/formats/",
      "io/",
      "audio/",
      "viewport/",
      "input/",
      "standards/",
      "standards/blocks/",
      "mechanics/",
```

*Canonical 5-block engine config defining VFS root search hierarchy, alias tables, and fail-closed safety flags.*

Wholesale Listing: Complete verbatim code (`#L1-L159`) → Appendix: Complete 5-Block VFS Precedence & Policy Configuration (vfs_routes.json).

## 83.3. Coordinated Validation & Lifecycle Gates

`Sub-Assemblies 4.1` through `5.2` inside `io.h` do **not** invent separate state flags or bypass child cogs; instead they coordinate cross-cog initialization, probe verification (`io_umbrella.*`), and unified pre-flight checks across all five underlying modules cleanly.

1. *Why enforce an explicit non-owning stateless facade invariant on `io.h`?*

   If someone sneaks parsing helpers or static buffers directly into `io.h`, callers importing only `fs_utils.h` or `path_finder.h` silently diverge from full builds. Keeping `io.h` strictly stateless preserves explicit include hygiene (`Invariant 3`).

## 83.4. Explore the Five Child I/O Cogs

1. `fs_utils.h` + `fs_utils.c`
2. `path_finder.h` + `path_finder.c`
3. `json_utils.h` + `json_utils.c`
4. `master_loader.h` + `master_loader.c`
5. `io_fallback.h` + `io_fallback.c`

# Chapter 84. I/O Module Pair: Bounded File & Memory Reader (fs_utils.h + fs_utils.c)

**Source File**: docs/code/io/02_fs_utils_bounded_io.adoc

| Attribute | Verified Contract (cog.fs_utils) |
|---|---|
| Public Contract (.h) | engine/include/io/fs_utils.h (838 lines) — stat inspection, bounded reads/writes, policy accessors, probes, and lifecycle hooks. |
| Runtime Engine (.c) | engine/src/io/fs_utils.c (1,462 lines) — POSIX file operations, read_only_safe_mode enforcement, and rom:// bridge. |
| Chassis Footprint | .bss 232 B (s_fs_state) + .rodata 4,064 B + 8 KiB transient stack on policy load; zero idle scratch bloat. |
| Policy Config | engine/config/io/fs_buffers.json & engine/config/io/default.json (probes: io.fs.* → DEBUG_STEP_AWAIT). |

Reading a file in plain C often hides two subtle hazards: reading a huge asset into a fixed buffer while silently throwing away the end of the file, or accidentally scribbling logs onto a locked/read-only cartridge or install directory. fs_utils.[ch] makes both hazards explicit and measurable.

## 84.1. Bounded Reads, Atomic-Minded Writes & Directory Creation (Sub-Assemblies 3.1–3.2)

Callers pass explicitly sized buffers (char *buf, size_t buf_size) cleanly bounded by policy ceilings up to 64 KiB (max_config_bytes), ensuring oversized payloads trigger an explicit truncation_rejections increment instead of partial garbage:

*Bounded POSIX Filesystem & Read-Only Guard API (fs_utils.h Sub-Assembly 3.2)*
*(engine/include/io/fs_utils.h#L359-L400 · Mode: Subassembly)*

```
359 /*
360 ---------------------------------------------------------------------------
361 Sub-Assembly 3.2: Bounded Buffered I/O & Directory Management Cogs
362 Mission: Bounded text reading, safe file writing, and parent directory synthesis.
363
364   ### For Game Designers & Non-Programmers:
365   Loads configuration and story files safely within memory caps and saves outputs
366   atomically while blocking mutations during Read-Only Safe Mode.
367
368   ### For Systems Programmers & Engine Architects:
369   Enforces active policy ceilings (`max_config_bytes`, `path_max_bytes`), NUL
370   termination, Safe-Mode RO protection, and shared Tier-0 BIOS atomic write/mkdir.
371 ---------------------------------------------------------------------------
372 */
373
374 /**
375  * @brief Reads entire text file contents into a bounded, caller-supplied buffer.
376  *
377  * Guarantees unconditional null termination and prevents buffer overruns.
378  * If the file exceeds `buf_size - 1` or `max_config_bytes`, the read is safely rejected.
379  *
380  * ### For Game Designers & Non-Programmers:
381  * Loads an entire text file safely into memory, guaranteeing it never overflows or crashes the engine.
382  *
383  * ### For Systems Programmers & Engine Architects:
384  * Convenience wrapper delegating cleanly into `fs_read_file_bounded()` without length output.
385  *
386  * @param[in]  filepath Null-terminated path or `"rom://..."` URI to target file.
```

```
387  * @param[out] buf      Caller-supplied memory destination buffer.
388  * @param[in]  buf_size Byte capacity of destination buffer (including null terminator).
389  * @return true if file opened and completely read within buffer bounds; false otherwise.
390  * @triggers Inherits truncation fault and warning log behavior from `fs_read_file_bounded()`.
391  */
392 bool fs_read_file_contents(const char *filepath, char *buf, size_t buf_size);
393
394 /**
395  * @brief Reads file or `"rom://..."` text and reports exact count of bytes read.
396  *
397  * Similar to `fs_read_file_contents()`, and writes the exact number of bytes read
398  * (excluding NUL terminator) into optional `out_bytes_read`.
399  *
400  * ### For Game Designers & Non-Programmers:
```

*Caller-owned bounded file transfers with truncation protection and Safe-Mode write locks.*

Wholesale Listing: Complete verbatim code (#L359-L455) → Appendix: Bounded POSIX Filesystem & Read-Only Guard API (fs_utils.h Sub-Assembly 3.2).

- **Probing (3.1)**: `fs_file_exists()`, `fs_is_regular_file()`, and `fs_inspect_file()` (`fs_file_info_t`).
- **Mounting & `rom://` Fallback (3.3)**: `fs_set_data_root()` and `fs_resolve_data_path()` cooperate cleanly with `rom://` URIs when disk storage degrades.
- **Policy Hydration (3.4)**: `fs_utils_load_policy()`, `fs_utils_set_policy()`, and `fs_utils_get_policy()` (`io_fallback_config_t`) enforce runtime ceilings and `read_only_safe_mode` latches (`read_only_blocks` counter increments when writes or `fs_ensure_directory()` are stopped under read-only safety lock).

## 84.2. Telemetry Counters, Probes & Self-Audit (Blocks 4 & 5)

`fs_utils_stats_t` exposes O(1) non-allocating inspection across lifetime `read_ops`, `write_ops`, `truncation_rejections`, `read_only_blocks`, and `rom_fallback_reads`:

- **Probe Hooks (4.1–4.2)**: `fs_utils_validate_policy_contract()`, `fs_utils_register_debug_probes()`, `fs_utils_trigger_policy_probe()`, and `fs_utils_audit()`.
- **Operational Cards (5.1–5.2)**: `fs_execute_read()`, `fs_execute_write()`, `fs_execute_dump_card()`, `fs_utils_get_stats()`, `fs_utils_init()`, and `fs_utils_shutdown()`.

## 84.3. Continue in io/

- Next cog up the pipeline: `path_finder.h` + `path_finder.c`
- Emergency `rom://` provider: `io_fallback.h` + `io_fallback.c`

# Chapter 85. I/O Module Pair: Layered VFS Domain Router (path_finder.h + path_finder.c)

**Source File**: docs/code/io/03_path_finder_vfs.adoc

| Attribute | Verified Contract (cog.path_finder) |
|---|---|
| Public Contract (.h) | engine/include/io/path_finder.h (719 lines) — root discovery, route loading, multi-tier VFS resolution, probes, and audit gates. |
| Runtime Engine (.c) | engine/src/io/path_finder.c (1,992 lines) — POSIX lstat/readlink root anchoring, internal vfs_hydrate_routes_internal(), and route search. |
| Chassis Footprint | .bss 288 B (s_vfs_state) + .rodata 5,300 B; max path PATH_FINDER_MAX_PATH = 512, MAX_ROOT = 256. |
| Route Config | engine/config/io/vfs_routes.json (⇐ 3 config tiers, ⇐ 36 routes; probes: io.vfs.* → DEBUG_STEP_AWAIT). |

Whether you launch the compiled binary from repository root, from inside build/, or from a nested CTest build folder, path_finder walks upward deterministically to anchor the workspace root. Even better: designers can drop an override JSON inside game/config/ to tweak **Tenebris and Lux** without ever mutating Cornerstone's baseline defaults inside engine/config/.

## 85.1. How Multi-Tier Resolution Works (vfs_routes.json)

When a subsystem asks for domain "config" with relative path diagnostics/telemetry/default.json, path_finder consults the hydrated route table (io_fb_vfs_routes_t) in fixed precedence:

*JSON Block 3 Payload Isolation (block_3_body of vfs_routes.json) (engine/config/io/vfs_routes.json#L55-L102 · Mode: Block)*

```
55  "block_3_body": {
56    "default_data_root": "game/data",
57    "env_override_var": "TL_DATA_ROOT",
58    "block_path_traversal": true,
59    "max_symlink_depth": 4,
60    "allow_mod_mounts": true,
61    "rom_fallback_enabled": true,
62    "probe_await_on_corrupt": true,
63    "config_precedence": [
64      "game/config/",
65      "engine/config/",
66      "game/data/presentation/config/"
67    ],
68    "fallback_candidate_roots": [
69      "game/data",
70      "../game/data",
71      "engine/config",
72      "../engine/config",
73      "data",
74      "../data"
75    ],
76    "asset_roots": [
77      "game/assets",
78      "../game/assets",
79      "assets",
80      "../assets"
81    ],
82    "domain_search_paths": [
83      "",
```

```
      "runtime/",
      "runtime/math/",
      "runtime/string/",
      "runtime/time/",
      "diagnostics/",
      "diagnostics/telemetry/",
      "diagnostics/debugger/",
      "diagnostics/taxonomy/",
      "diagnostics/formats/",
      "io/",
      "audio/",
      "viewport/",
      "input/",
      "standards/",
      "standards/blocks/",
      "mechanics/",
      "mechanics/items/",
      "mechanics/combat/",
      "mechanics/tactics/",
```

*Demonstrates scoped block_3_body extraction—matching master_loader's strict rule against reading Block 1/2 metadata as live gameplay config.*

Wholesale Listing: Complete verbatim code (#L55-L125) → Appendix: JSON Block 3 Payload Isolation (block_3_body of vfs_routes.json).

## 85.2. Public Resolution API (Sub-Assemblies 3.1–3.2)

*Multi-Tier VFS Path & Asset Resolution API (path_finder.h Sub-Assembly 3.2)*
*(engine/include/io/path_finder.h#L396-L435 · Mode: Subassembly)*

```
/*
-------------------------------------------------------------------------------
Sub-Assembly 3.2: Multi-Tier VFS Path & Asset Resolution Cogs
Mission: Execute layered precedence search, diagnostic alias expansion, symlink
         depth bounds, traversal rejection, and `.rodata` ROM fallback cleanly.

  ### For Game Designers & Non-Programmers:
  Finds configuration manifests, game data files, fonts, and artwork across game
  overrides, engine baselines, asset folders, and zero-disk ROM fallbacks.

  ### For Systems Programmers & Engine Architects:
  Bounded-stack candidate resolution enforcing `max_symlink_depth`, `..` traversal
  containment, destination `out_size` overflow guards, and `io.vfs.traversal_guard`.
-------------------------------------------------------------------------------
*/

/**
 * @brief Resolves a logical domain and relative path to a concrete disk file or `rom://` URI.
 *
 * Search sequence (all tables driven by `io/vfs_routes.json`):
 *  1. `..` parent traversal sandbox guard (`block_path_traversal`) & explicit `rom://` check.
 *  2. Direct accessible file probe (enforcing `max_symlink_depth`).
 *  3. Mounted `active_root` (`<root>/<domain>/<clean>`, `<root>/<clean>`) and `env_override_var`.
 *  4. Layered `config_precedence` tiers (`game/config/`, `engine/config/`, linkers) + `diagnostic_aliases`.
 *  5. Hydrated `fallback_candidate_roots` x `domain_search_paths` matrix.
 *  6. Compiled `.rodata` ROM vault fallback (`rom://<clean>`) when disk targets are absent.
 *
 * ### For Game Designers & Non-Programmers:
 * Finds any configuration file, story file, or manifest automatically—checking game overrides
 * first, engine baselines second, and built-in ROM backups last.
 *
 * ### For Systems Programmers & Engine Architects:
 * Bounded stack probing (zero heap), strict `out_size` overflow rejection, `lstat`/`readlink`
 * symlink depth verification, and `LOG_CAT_IO` telemetry tracking.
 * @triggers `io.vfs.traversal_guard` (on `..`/symlink rejection), `LOG_CAT_IO`.
 *
 * @param[in]  domain   Optional logical domain hint (`"config"`, `"mechanics"`, `"io"`, etc.) or NULL.
 * @param[in]  rel_path Relative asset/config path or explicit `"rom://..."` URI.
 * @param[out] out_path Caller-supplied output path buffer (`out_path[0] = '\0'` on failure).
```

```
435  * @param[in]  out_size Capacity of `out_path` in bytes.
```

*Resolves game/config → engine/config → game/data/presentation/config deterministically without heap allocation.*

Wholesale Listing: Complete verbatim code (#L396-L510) → Appendix: Multi-Tier VFS Path & Asset Resolution API (path_finder.h Sub-Assembly 3.2).

- **Root & Route Setup (3.1)**: path_finder_set_data_root(), path_finder_load_routes(), path_finder_set_routes(), and path_finder_get_routes().
- **Domain Lookups (3.2)**: path_finder_resolve(), path_finder_find_manifest(), path_finder_resolve_asset(), and path_finder_resolve_with_hint(). When no disk file matches any tier, callers smoothly receive a deterministic rom:// URI or Safe-Mode fallback notice cleanly.

## 85.3. Validation, Probes & Inspection (Sub-Assemblies 4.1–5.2)

- **Validation (4.1–4.2)**: path_finder_validate_routes_contract(), path_finder_register_debug_probes(), path_finder_trigger_routes_probe(), path_finder_preflight(), and path_finder_audit().
- **Inspection & Lifecycle (5.1–5.2)**: path_finder_execute_resolve(), path_finder_execute_dump_card(), path_finder_get_stats(), path_finder_init(), and path_finder_shutdown().

## 85.4. Continue in io/

- Next stage (tokenizing found files): json_utils.h + json_utils.c
- Pre-flight manifest loader: master_loader.h + master_loader.c

# Chapter 86. I/O Module Pair: Zero-Heap Single-Pass JSON Scanner (json_utils.h + json_utils.c)

**Source File**: docs/code/io/04_json_utils_parser.adoc

| Attribute | Verified Contract (cog.json_utils) |
|---|---|
| Public Contract (.h) | engine/include/io/json_utils.h (1,205 lines) — token/object/array span locators, typed getters, policy contracts, probes, and lifecycle hooks. |
| Runtime Engine (.c) | engine/src/io/json_utils.c (3,092 lines) — string-escape-aware single-pass parser, depth guards, and zero-static-buffer scanning. |
| Chassis Footprint | .bss 144 B (s_json_state) + .rodata 5,740 B + zero function-local static scratch buffers; stack depth capped at ⇐ 32. |
| Policy Config | engine/config/io/loader_contracts.json (probes: io.json.* → DEBUG_STEP_AWAIT). |

Standard JSON libraries allocate dozens or hundreds of tiny heap nodes per file—making fragmentation unavoidable and deterministic memory budgeting nearly impossible. json_utils.[ch] skips building a heap DOM altogether: it walks raw JSON text in a single pass over caller memory (O(N)), tracks nesting safely up to depth 32, and returns exact (ptr, len) spans ready for zero-copy slicing.

## 86.1. Single-Pass Token Locators & Typed Extractors (Sub-Assemblies 3.1–3.4)

Sub-Assembly 3.1 locates bounded child { ⋯ } and [ ⋯ ] byte ranges cleanly inside any slice without mutating the source text (essential when scanning immutable .rodata ROMs!):

*Zero-Heap JSON Token Span Locator API (json_utils.h Sub-Assembly 3.1)* (engine/include/io/json_utils.h#L357-L396 · *Mode:* Subassembly)

```
357 /*
358 ------------------------------------------------------------------------------
359 Sub-Assembly 3.1: JSON Token Locator Cogs
360 Mission: Scan flat or span-bounded JSON payloads for matching keys, objects, and
361          arrays while respecting escape sequences and nesting limits cleanly.
362
363   ### For Game Designers & Non-Programmers:
364   Finds named sections, tables, and lists inside JSON configuration files cleanly
365   without mistaking words inside text descriptions for structural headings.
366
367   ### For Systems Programmers & Engine Architects:
368   Exposes zero-copy, zero-heap token and balanced-container locators constrained
369   to `[json_start, json_start + json_len)` and active `max_json_depth` policy.
370 ------------------------------------------------------------------------------
371 */
372
373 /**
374  * @brief Scans a JSON buffer for a specific top-level or scoped key token.
375  *
376  * Core search primitive that locates occurrences of "key" followed by a colon
377  * (:), ignoring intervening whitespace. Prevents false positive matches on
378  * substring prefixes or keys embedded inside quoted strings.
379  *
380  * ### For Game Designers & Non-Programmers:
381  * Searches through a data file to locate where a property begins (e.g. finding "attack_power:").
382  *
383  * ### For Systems Programmers & Engine Architects:
384  * Reentrant zero-heap token locator respecting quote escapes and whitespace.
385  * @triggers On nesting limit exceeded -> `IO_ERR_JSON_DEPTH_EXCEEDED` via `io.json.depth_bounds`.
386  *
387  * @param[in] json Null-terminated JSON buffer.
```

```
388  * @param[in] key  Null-terminated key name to search for (without quotes).
389  * @return Pointer to value position immediately following the colon (:),
390  *         or NULL if key is not found or parameters are invalid.
391  */
392 const char *json_find_key(const char *json, const char *key);
393
394 /**
395  * @brief Span-bounded key locator preventing sibling overrun across sliced buffers.
396  *
```

*Single-pass bounded span slicing capped at bracket nesting depth ⇐ 32 on caller stack.*

Wholesale Listing: Complete verbatim code (#L357-L461) → Appendix: Zero-Heap JSON Token Span Locator API (json_utils.h Sub-Assembly 3.1).

| Sub-Assembly | Core Capabilities |
|---|---|
| Nested Value Getters (3.2) | `json_get_string()`, `json_get_int()`, `json_get_size_t()`, `json_get_float()`, `json_get_bool()`, plus scoped `json_find_object()` and `json_get_object_*()` helpers. |
| Array Scanning (3.3) | `json_find_array()`, `json_array_get_count()`, `json_array_get_string()`, and `json_array_get_object()` for indexed table iteration. |
| Span-Bounded Scanners (3.4) | `json_get_{string,int,size_t,float,bool}_span()` scan strictly inside a `[start, start+len)` boundary—forming the backbone of `master_loader`s `block_3_body` isolation. |
| Runtime Policy (3.4) | `json_utils_load_policy()`, `json_utils_get_policy()`, and `json_utils_set_policy()` synchronize token/depth limits from `loader_contracts.json`. |

## 86.2. Structural Guards & Validation Probes (Sub-Assemblies 4.1–5.2)

- `json_validate_span()` and `json_utils_enforce_valid_span()` reject truncated strings, unbalanced braces, or excessive nesting before domain extractors read bad keys.
- `json_utils_validate_policy_contract()`, `json_utils_register_debug_probes()` (3 probes in `s_json_probes[3]`), `json_preflight()`, and `json_utils_audit()` wire syntax/depth breaches cleanly into `cog.debug`.

## 86.3. Continue in io/

- Next layer (5-block contract enforcer): `master_loader.h` + `master_loader.c`
- Shared ROM span resolver: `io_fallback.h` + `io_fallback.c`

# Chapter 87. I/O Module Pair: 5-Block Contract Hydrator (master_loader.h + master_loader.c)

**Source File**: docs/code/io/05_master_loader_hydration.adoc

| Attribute | Verified Contract (cog.master_loader) |
|---|---|
| Public Contract (.h) | engine/include/io/master_loader.h (987 lines) — loader_status_t, loader_header_t, master_loader_t, master_loader_stats_t, block_3_body getters, schema/version validators, probes, and audit controls. |
| Runtime Engine (.c) | engine/src/io/master_loader.c (2,051 lines) — orchestrator wiring path_finder VFS lookup, fs_utils file/ROM reads, json_utils span queries, cog.log, and cog.debug probes. |
| Controlled Chassis | .bss 112 B (s_loader_state) + .rodata 5,916 B + transient preflight stack frame; caller-owned master_loader_t slices capped at max_payload_bytes ⇐ 65536, max_json_depth ⇐ 32, max_json_tokens ⇐ 1024. |
| Seams & Probes | in: engine/config/io/loader_contracts.json, rom://io \| uses: cog.path_finder, cog.fs_utils, cog.json_utils, cog.log, cog.debug \| probes: io.loader.* → DEBUG_STEP_AWAIT \| dag: core.io.master_loader |

**For Game Designers & Data Authors**: master_loader is the gatekeeper every JSON file in engine/config/ and game/data/ must pass through. It checks file headers (block_1_metadata and block_2_opening), enforces schema compatibility, and lets gameplay code read **only** inside block_3_body so a typo in file notes or metadata never hijacks a stat formula. If disk data is missing or corrupted, it falls back seamlessly to rom://io compiled defaults without crashing.

**For Engine Programmers**: Instead of duplicating JSON parsing or path lookup per game system, master_loader.[ch] binds all lower io/ primitives under one deterministic state machine backed by runtime policy (master_loader_load_contract()), strict error counters (syntax_errors 0x0204, schema_mismatches 0x0203, version_mismatches 0x0207, body_missing_errors 0x0206), and DEBUG_STEP_AWAIT traps.

## 87.1. Dual-Path Hydration: Disk, VFS, rom:// & Memory

| Entry Point | Execution Path & Provenance Tracking |
|---|---|
| master_loader_load() | Resolves (domain, rel_path) via path_finder_resolve(). Sets loaded_from_data = true and loaded_from_rom = false for disk payloads, or loaded_from_rom = true and loaded_from_data = false when hydrated from rom://io. |
| master_loader_load_from_path() | Reads an explicit file path via bounded fs_utils, skipping domain search while still enforcing full 5-block header validation and size bounds. |
| master_loader_load_from_memory() | Parses an already-resident byte buffer (.rodata ROM image or test fixture) in place with zero filesystem calls. |

During envelope parsing, loader_header_t inspects is_5block, has_block_1, and has_block_2, then isolates [body_start, body_start + body_len) over raw_len before returning master_loader_t to the caller. Missing block_3_body records body_missing_errors (0x0206) and trips Safe-Mode diagnostics rather than silently reading stale or adjacent fields.

## 87.2. Strictly Scoped block_3_body Getters

All runtime property queries are pinned strictly inside the verified block_3_body window (or an optional named sub_obj child inside block_3_body):

*Scoped block_3_body JSON Accessor Contract (master_loader.h Sub-Assembly 3.2)*
*(engine/include/io/master_loader.h#L436-L479 · Mode: Subassembly)*

```
436 /*
437 ------------------------------------------------------------------------------
438 Sub-Assembly 3.2: Scoped `block_3_body` Accessors
439 Mission: Restrict all payload getters and array locators strictly inside the
440          verified `[body_start, body_start + body_len)` span (or named `sub_obj`).
441
442   ### For Game Designers & Non-Programmers:
443   Reads gameplay properties cleanly from inside `block_3_body` only, preventing
444   file headers or audit sections from ever colliding with gameplay field names.
445
446   ### For Systems Programmers & Engine Architects:
447   Delegates scope isolation and typed extraction exclusively to span-bounded
448   `json_*_span` APIs with unconditional NUL termination on extracted strings.
449 ------------------------------------------------------------------------------
450 */
451
452 /**
453  * @brief Extracts a string value strictly within `block_3_body` (or named sub-object).
454  *
455  * ### For Game Designers & Non-Programmers:
456  * Safely reads text properties (character names, formulas, IDs) from the data body
457  * without accidentally matching header text outside Block 3.
458  *
459  * ### For Systems Programmers & Engine Architects:
460  * Uses span-bounded `json_find_object_span()` + `json_get_string_span()` bounded to
461  * `loader->body_len`. Unconditionally NUL-terminates `out_val` when `out_size > 0`.
462  * @triggers `io.json.depth_bounds` if scan breaches `max_json_depth`.
463  *
464  * @param[in]  loader   Ready `master_loader_t` instance.
465  * @param[in]  sub_obj  Optional sub-object key inside `block_3_body` (or NULL for root body).
466  * @param[in]  key      Target property key inside resolved body/sub-object span.
467  * @param[out] out_val  Caller string buffer.
468  * @param[in]  out_size Capacity of `out_val` in bytes.
469  * @return true on clean extraction; false if missing, wrong type, or invalid params.
470  */
471 bool master_loader_get_string(const master_loader_t *loader,
472                               const char *sub_obj,
473                               const char *key,
474                               char *out_val,
475                               size_t out_size);
476
477 /**
478  * @brief Extracts a signed integer strictly within `block_3_body` (or named sub-object).
479  *
```

*Zero-heap typed getters that scan directly inside a validated block_3_body span.*

Wholesale Listing: Complete verbatim code (#L436-L608) → Appendix: Scoped block_3_body JSON Accessor Contract (master_loader.h Sub-Assembly 3.2).

Typed extractors cover the full primitive and structured vocabulary without dynamic memory allocation:

- master_loader_get_string() — copies and unconditionally NUL-terminates string values.
- master_loader_get_int(), master_loader_get_size_t(), master_loader_get_float(), master_loader_get_bool() — strict typed extraction with fallback signaling on wrong types.
- master_loader_find_array() and master_loader_get_array_object() — zero-copy array cursor iteration over object elements inside block_3_body.

## 87.3. Runtime Loader Policy (loader_contracts.json)

Rather than baking ceilings permanently into code, master_loader_init() hydrates engine/config/io/loader_contracts.json (with rom://io safe fallback on corruption, incrementing corrupt_contract_downgrades):

- master_loader_load_contract(), master_loader_get_contract(), master_loader_set_contract() — inspect or override byte/depth/token limits and strict schema rules in controlled tests.
- master_loader_validate_schema(), master_loader_validate_version(), master_loader_validate_envelope(), master_loader_validate_contract() — granular contract gates executed prior to trusting any loaded JSON table.

## 87.4. Error Taxonomy & Diagnostic Probes (Block 4)

Every failure increments machine-inspectable counters in master_loader_stats_t (readable via master_loader_get_stats()) alongside categorized hex codes registered under io.loader.*:

| Telemetry Counter (master_loader_stats_t) | Code | Fault Condition & Escalation |
|---|---|---|
| not_found_errors | 0x0201 | Neither disk domains nor rom:// fallback satisfied the target relative path. |
| overflow_errors | 0x0202 | File or payload exceeded caller slice or max_payload_bytes (⇐ 65536). |
| schema_mismatches | 0x0203 | block_2_opening declared an unexpected schema_id. |
| syntax_errors | 0x0204 | Malformed JSON braces/escapes or nesting exceeding max_json_depth (⇐ 32). |
| body_missing_errors | 0x0206 | Required block_3_body object absent from a 5-block envelope. |
| version_mismatches | 0x0207 | Manifest contract version incompatible with the active engine target. |

master_loader_register_debug_probes() and master_loader_trigger_probe() wire these thresholds directly into cog.debug so violations halt execution in DEBUG_STEP_AWAIT during audit/debug runs instead of failing silently downstream.

- master_loader_preflight() verifies s_loader_state.initialized, contract_from_data, and probe registration before main() advances past io/.
- master_loader_audit() runs deep structural self-checks on cached policy limits and live load counters (total_loads = disk_loads + rom_loads + memory_loads, body_queries).

## 87.5. Inspection Commands & Lifecycle Ownership

Operational tooling and Layer 3 startup call master_loader_execute_inspect(), master_loader_execute_dump_card(), and master_loader_execute_load_verified() to print human-readable status cards and assert strict load contracts before accepting gameplay assets. master_loader_shutdown() clears s_loader_state back to dormant .bss zeroes deterministically between sessions or test suites.

- Back to Layer 0 I/O Folder Index or Layer 0 I/O Subsystem Overview.
- Next module: Paired Module: io_fallback.h + io_fallback.c.

# Chapter 88. I/O Safe-Mode BIOS: Zero-Disk Fallback & ROM Vault (io_fallback.h + io_fallback.c)

**Source File**: docs/code/io/06_io_fallback_bios_and_rom.adoc

| Attribute | Verified Contract (cog.io.fallback) |
|---|---|
| Public Contract (.h) | engine/include/io/io_fallback.h (1,519 lines) — IO_ERR_* codes, io_fallback_config_t, io_fb_vfs_routes_t, shared safe JSON parsers, ROM vault lookup, unified emergency cogs, and BIOS boot/downgrade controls. |
| Runtime Engine (.c) | engine/src/io/io_fallback.c (3,355 lines) — standalone floppy-grade I/O engine operating cleanly when fs_utils, path_finder, json_utils, or master_loader fail or disk assets are absent. |
| Controlled Chassis | .bss 8,312 B (s_io_fb) + .rodata 40,256 B (8 embedded I/O ROMs generated into io_config_rom.h) + stack ⇐ 1 KiB. |
| Seams & Archetype | quarantined_fallback \| stateful \| in: io_config_rom.h \| uses: cog.log.fallback, cog.debug.fallback \| probes: io_fallback.* → DEBUG_STEP_AWAIT \| dag: io.fallback_bios |

**For Designers & Playtesters**: Deleting or breaking engine/config/io/ on disk does not brick Cornerstone. Instead, io_fallback acts like a motherboard BIOS stored directly inside read-only binary memory (.rodata): it still knows every VFS search route, still extracts fields from block_3_body, logs the missing disk file via log_fallback, and hands diagnostics straight to debug_fallback / DEBUG_STEP_AWAIT.

**For Systems Programmers**: io_fallback.[ch] obeys strict quarantine boundaries (*_fallback.[ch] convention). Notice that its shared JSON body parser (io_fb_resolve_body_span(), io_fallback_parse_vfs_routes_json()) also feeds the full-featured I/O stack when booting from ROM, eliminating duplicate parser logic while preventing circular dependencies with master_loader or path_finder.

## 88.1. Shared ROM-Safe JSON Policy Parsers

Because disk configuration might be unreachable or unreadable during boot, Sub-Assembly 3.1 parses JSON byte spans directly inside read-only .rodata memory without scratch arrays on the heap:

*Shared Zero-Heap JSON Block Resolver & VFS Parser Group (engine/src/io/io_fallback.c#L779-L806 · Mode: Group)*

```
779 static bool io_fb_resolve_body_span(const char *json_text,
780                                       size_t json_len,
781                                       bool allow_strlen,
782                                       const char **out_body_start,
783                                       const char **out_body_end)
784 {
785     if (json_text == NULL || out_body_start == NULL || out_body_end == NULL) {
786         return false;
787     }
788
789     size_t effective_len = (json_len > 0U) ? json_len : (allow_strlen ? strlen(json_text) : 0U);
790     if (effective_len == 0U || effective_len > IO_FB_JSON_MAX_BYTES) {
791         return false;
792     }
793
794     const char *json_end = json_text + effective_len;
795     const char *body_start = NULL;
796     size_t body_len = 0U;
797     if (!io_fb_find_block_span(json_text, json_end, "block_3_body", '{', '}',
798                                &body_start, &body_len) ||
799         body_start == NULL || body_len == 0U) {
```

```
800         return false;
801     }
802
803     *out_body_start = body_start;
804     *out_body_end = body_start + body_len;
805     return true;
806 }
```

*Tier-0 in-place span scanning reused seamlessly between BIOS fallback and master_loader.*

Wholesale Listing: Complete verbatim code (#L779-L1194) → Appendix: Shared Zero-Heap JSON Block Resolver & VFS Parser Group.

- io_fb_resolve_body_span() isolates block_3_body bounds cleanly inside immutable ROM text.
- io_fallback_parse_vfs_routes_json() extracts io_fb_vfs_routes_t search domains (config, data, assets) deterministically into .bss so even floppy Safe-Mode resolves logical routes accurately.

## 88.2. Compiled .rodata Config ROM Vault (Sub-Assembly 3.2)

At build time, cmake/GenerateConfigRom.cmake compiles **8 canonical engine/config/io/*.json files** into io_config_rom.h totaling 40,256 B of read-only .rodata. Sub-Assembly 3.2 exposes O(1)/bounded name resolution over rom://io, allowing path_finder and master_loader to hydrate identical JSON contracts whether loaded from disk or emergency binary memory.

## 88.3. Unified 4-Pillar Emergency Cogs (Sub-Assembly 3.3)

Sub-Assembly 3.3 consolidates minimalist emergency fallbacks for **all four I/O operational pillars** inside one ring-buffered state block (s_io_fb, 8,312 B):

1. **Bounded Read Fallback**: Safe bounds-checked file/memory slice copies rejecting overflow (IO_ERR_*).
2. **Route Fallback**: Resolves canonical domain priorities using io_fb_vfs_routes_t hydrated from io_config_rom.h.
3. **Token/Span Fallback**: Minimal key lookup inside validated block_3_body slices.
4. **Fault Ring Buffer**: Captures failure provenance (loaded_from_data = false), error codes, and path breadcrumbs without losing early faults before cog.log is even online.

## 88.4. Autonomous Probes & Safe-Mode Inspection (Block 4)

Sub-Assembly 4.1 maintains s_io_fb_probes[2] bound to cog.debug.fallback under the io_fallback.* namespace. When disk configuration corruption triggers emergency takeover, io_fallback_downgrade_safe_mode() emits structured fault cards across both log_fallback and debug_fallback so DEBUG_STEP_AWAIT pauses inspection cleanly on the root cause rather than masking missing files.

Sub-Assembly 4.2 verifies ROM vault integrity, io_fallback_config_t sanity, and active fault counts before signaling readiness up to io_audit().

## 88.5. BIOS Boot, Floppy Mode & Deterministic Teardown (Block 5)

*Tier-0 I/O Safe-Mode BIOS Boot, Floppy Floor & Downgrade Latch (*engine/src/io/io_fallback.c#L3309-L3332 · *Mode:* Group*)*

```
3309 bool io_fallback_boot_floppy(void)
3310 {
3311     io_fallback_config_t floppy = s_io_floppy_template;
3312     if (io_fallback_load_rom_config("floppy_safe", &floppy) ||
```

```
3313         io_fallback_load_rom_config("default", &floppy)) {
3314         s_io_fb.rom_hydrated = true;
3315     } else {
3316         s_io_fb.rom_hydrated = false;
3317     }
3318     floppy.read_only_mode = true;
3319     floppy.loaded_from_data = false;
3320     if (floppy.profile_name[0] == '\0' || strcmp(floppy.profile_name, "default") == 0) {
3321         str_safe_copy(floppy.profile_name, IO_FALLBACK_FLOPPY_PROFILE_NAME,
3322                       sizeof(floppy.profile_name));
3323     }
3324
3325     s_io_fb.config = floppy;
3326     io_fb_hydrate_subpolicies();
3327     s_io_fb.active_sink = IO_FB_SINK_FLOPPY_RO;
3328     str_safe_copy(s_io_fb.active_root, IO_FB_DEFAULT_ROOT, sizeof(s_io_fb.active_root));
3329     s_io_fb.booted = true;
3330     io_fallback_register_debug_probes();
3331     return true;
3332 }
```

*Zero-dependency Safe-Mode transitions ensuring deterministic boot even when disk configs are missing or corrupt.*

Wholesale Listing: Complete verbatim code (#L2908-L3332) → Appendix: Tier-0 I/O Safe-Mode BIOS Boot, Floppy Floor & Downgrade Latch.

| Control Hook | Operational Role |
|---|---|
| io_fallback_boot() | Standard BIOS startup hydrating io_config_rom.h tables into s_io_fb. |
| io_fallback_boot_floppy() | Explicit zero-disk minimal boot mode testing zero external filesystem reliance. |
| io_fallback_downgrade_safe_mode() | Transitions active failure into read-only Safe-Mode ROM service and trips io_fallback.* debug probes. |
| io_fallback_teardown() / io_fallback_is_booted() | Shuts down the fallback chassis cleanly or queries live BIOS readiness state. |
| io_fallback_reset_for_testing() | Deterministic Layer 4 test reset scrubbing s_io_fb cleanly between unit/system test cases. |

- Back to Layer 0 I/O Folder Index or Previous Module: master_loader.h + master_loader.c.
- Next document: Layer 4 I/O Testing & Downgrade Drills.

# Chapter 89. I/O Verification Suite: Bounded I/O, VFS Routing & Downgrades Across All 4 Tiers

**Source File**: `docs/code/io/07_io_testing_and_downgrades.adoc`

The `io/` subsystem carries **8 dedicated CTest targets** stepping cleanly from isolated per-cog unit tests (`Unit`) up through public API simulation (`Functional`), cross-module orchestration (`Integrated`), and full missing-disk downgrade recovery (`System`).

| Tier | CTest Target | Source File | Primary Verification Focus |
|---|---|---|---|
| Unit | `unit_fs_utils_test` | `tests/unit/io/test_fs_utils_unit.c` | `O(1)` byte-limit truncation checks, zero-NUL safety, and policy ceiling overrides. |
| Unit | `unit_path_finder_test` | `tests/unit/io/test_path_finder_unit.c` | 3-tier domain precedence (`game/config` → `engine/config` → `presentation`), path traversal rejection, and `rom://` routing. |
| Unit | `unit_json_utils_test` | `tests/unit/io/test_json_utils_unit.c` | Single-pass span lookup, escape-sequence parsing, depth ⇐ `32` bounds, and token limits. |
| Unit | `unit_master_loader_test` | `tests/unit/io/test_master_loader_unit.c` | 5-block header validation, strict `block_3_body` query isolation, and `0x0201`–`0x0207` error taxonomy. |
| Unit (BIOS) | `unit_io_fallback_test` | `tests/unit/io/test_io_fallback_unit.c` | `io_config_rom.h` ROM vault reads, `io_fallback_boot_floppy()`, shared route JSON parser, and `io_fallback_reset_for_testing()`. |
| Functional | `functional_io_test` | `tests/functional/test_io_functional.c` | External caller simulation driving `io.h` umbrella contracts and `master_loader_load()` end-to-end without touching internals. |
| Integrated | `integrated_io_test` | `tests/integrated/test_io_integrated.c` | Cross-subsystem plumbing uniting `io/` + `log/` + `debug/` probe callbacks + ROM config hydration cleanly. |
| System | `system_io_test` | `tests/system/test_io_system.c` | Full system boot/audit loops, malformed payload downgrades, `DEBUG_STEP_AWAIT` capture, and recovery determinism. |

## 89.1. Crude-to-Fine Band Design

Every CTest suite in `io/` executes through the shared `tests/common/cornerstone_test.h` harness (`test.common.harness`), partitioning assertions into progressively tighter assurance bands:

| Band | Verification Question Answered |
|---|---|
| always_pass | Can io_init() / io_fallback_boot() initialize safely from clean .bss without crashing or touching the heap? |
| never_fail | Do oversized payloads (> 65536 B), NULL buffers, .. traversal attempts, and invalid block_3_body spans return safe typed errors instead of faulting? |
| assertions_and_guarantees | Do disk overrides (game/config) outrank engine/config, and do missing files increment rom_loads with loaded_from_rom = true identically? |
| emergence | When chained across real subsystem boundaries, do VFS lookup, 5-block JSON parsing, logging, and probe triggers compose cleanly without state leaks? |
| preflight_audit / audit | Do io_preflight(), io_audit(), and master_loader_audit() pass cleanly with deterministic post-run counters and clean teardown? |

## 89.2. System-Tier Emergence & Downgrade Snapshot

tests/system/test_io_system.c exercises complete cross-subsystem initialization, missing/corrupt contract escalation through io.loader.* probes, and deterministic verification via cornerstone_test.h:

*I/O Pipeline System Emergence & ROM Downgrade Verification (test_io_system_emergence)*
*(tests/system/test_io_system.c#L328-L367 · Mode: Function)*

```
328 /**
329  * @brief Verifies sandboxed mount redirection and live v1 -> v2 JSON hot-patch reflection.
330  */
331 static void test_io_system_emergence(void)
332 {
333     printf("  [EMERGE-4] Running Section 4: Emergence (Dynamic Mount & Live Hot-Patching)...\n");
334 
335     const char *hotpatch_dir = "build/system_test_patch/system";
336     const char *hotpatch_file = "build/system_test_patch/system/runtime_patch.json";
337 
338     TEST_ASSERT(fs_ensure_directory(hotpatch_dir), "Synthesized hot-patch directory");
339 
340     static const char patch_v1[] =
341         "{\n"
342         "  \"block_1_metadata\": {\"project\": \"Tenebris and Lux\", \"file\": \"runtime_patch.json\"},\n"
343         "  \"block_2_types\": {\"schema_id\": \"tenlux.data.patch.v1\"},\n"
344         "  \"block_3_body\": {\"difficulty_mult\": 1.0, \"active_boss\": \"Usuru\"}\n"
345         "}\n";
346     TEST_ASSERT(fs_write_file_contents(hotpatch_file, patch_v1, strlen(patch_v1)),
347                 "Wrote patch v1");
348 
349     path_finder_init();
350     master_loader_init();
351     path_finder_set_data_root("build/system_test_patch");
352 
353     master_loader_t loader;
354     bool l1 = master_loader_execute_load_verified("system", "runtime_patch.json",
355                                                   "tenlux.data.patch.v1",
356                                                   s_io_sys_load_buf,
357                                                   sizeof(s_io_sys_load_buf),
358                                                   &loader);
359     TEST_ASSERT(l1 && loader.status == LOADER_STATUS_OK, "Loaded patch v1");
360     float diff1 = 0.0f;
361     char boss1[64];
362     TEST_ASSERT(master_loader_get_float(&loader, NULL, "difficulty_mult", &diff1) &&
363                 diff1 >= 0.999f && diff1 <= 1.001f,
364                 "Patch v1 difficulty_mult == 1.0");
365     TEST_ASSERT(master_loader_get_string(&loader, NULL, "active_boss", boss1, sizeof(boss1)),
366                 "Patch v1 boss extracted");
```

```
367     TEST_ASSERT_STR_EQ(boss1, "Usuru", "Boss is authentic Dark Lord Usuru");
```

*Drills end-to-end VFS resolution, corrupt JSON downgrade to ROM Safe Mode, and coordinated umbrella audit.*

Wholesale Listing: Complete verbatim code (#L328-L399) → Appendix: I/O Pipeline System Emergence & ROM Downgrade Verification (test_io_system_emergence).

## 89.3. Running & Auditing the io/ Test Suite

```
cmake -B build && cmake --build build -j
ctest --test-dir build \
  -R "(fs_utils|path_finder|json_utils|master_loader|io_fallback|_io_)" --output-on-failure
```

If any io/ probe fires unexpectedly during a run, tests/common/cornerstone_test.h prints the forensic debug / debug_fallback ring snapshot and pinpoints the exact test file, band, and DEBUG_STEP_AWAIT code.

- Up one level: Layer 0 I/O Folder Index | Layer 0 I/O Subsystem Overview.
- Peer testing suites: log/ 4-Tier Testing | debug/ 4-Tier Testing.

# Chapter 90. Layer 0 Core Utilities Code Tour

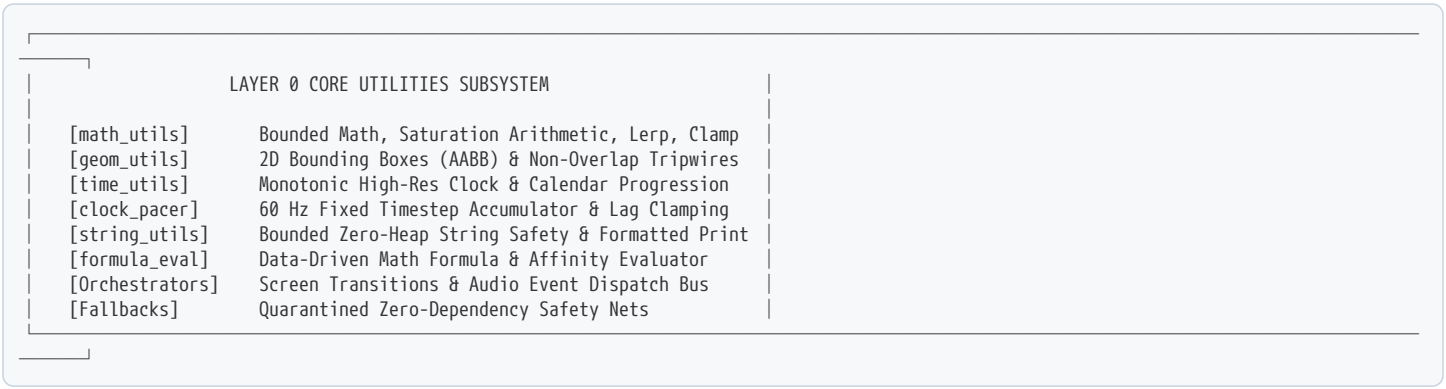
**Source File**: `docs/code/core_utilities.adoc`

## 90.1. Overview

This document provides a technical walkthrough and architectural manual for the **Layer 0 Core Utilities Subsystem** (`engine/include/core/` and `engine/src/core/`). Representing the mathematical, geometric, temporal, and string foundations of **The Big Machine**, the Core Utilities stratum provides bounded arithmetic, 2D AABB geometry, monotonic clock pacing, fixed-timestep simulation accumulators, memory-bounded string manipulation, and data-driven formula evaluation. Strict adherence to `ALLOC_ZERO_HEAP`, reentrancy, sub-microsecond latency budgets, and quarantined emergency fallbacks ensure that foundational operations never wrap around integer bounds, leak memory, or introduce non-deterministic time dilation.

## 90.2. Architectural Philosophy & Mission

At the very bedrock of the engine frame lies Layer 0 Core Utilities:




### 90.2.1. Key Architectural Pillars

1. **Zero Dynamic Heap Allocations (`ALLOC_ZERO_HEAP`)**: Pure reentrant C11 functions operating solely on CPU registers, primitive values, and caller-provided stack buffers.
2. **Saturation Arithmetic & Overflow Clamping**: Mathematical utilities enforce saturation clamping; calculations never wrap around `INT32_MAX` or underflow `INT32_MIN`.
3. **Decoupled Simulation Timestep**: `clock_pacer` decouples fixed simulation updates (exactly 60 Hz / 16.666ms) from variable physical render frame rates, calculating fractional interpolation alphas (`render_alpha`).
4. **Data-Driven Formula Evaluation**: Formulas for damage, mitigations, and affinity interactions are evaluated through `formula_eval` driven by JSON manifests in `game/data/system/runtime/math/` rather than hardcoding arithmetic in C sources.
5. **Quarantined Fallback Isolation**: Every core utility module is paired with a standalone, zero-dependency `*_fallback.[ch]` module ensuring zero-configuration compilation and hermetic test isolation.

## 90.3. Subsystem Layout & File Manifest

```
engine/include/core/
├── math_utils.h             // Bounded math, clamping, interpolation, rounding
```

```
├──── math_utils_fallback.h     // Standalone minimal math fallback
├──── geom_utils.h              // 2D AABB geometry, point-in-rect, intersection
├──── geom_utils_fallback.h     // Standalone geometric fallback
├──── time_utils.h              // Monotonic clock querying & calendar time
├──── time_utils_fallback.h     // Standalone monotonic time fallback
├──── clock_pacer.h             // Fixed-timestep simulation pacer & lag defense
├──── clock_pacer_fallback.h    // Minimal fixed-tick accumulator fallback
├──── string_utils.h            // Safe bounded string copying, format & trimming
├──── string_utils_fallback.h   // Minimal string safety fallback
├──── formula_eval.h            // Data-driven formula evaluator
├──── formula_eval_fallback.h   // Static arithmetic evaluation fallback
├──── audio_bus.h               // Universal audio event dispatch bus
└──── screen_transition.h       // Screen fading & transition orchestrator

engine/src/core/
├──── math_utils.c              // Mathematical implementations & saturation clamps
├──── math_utils_fallback.c     // Quarantined math fallback implementation
├──── geom_utils.c              // AABB collision and bounds validation
├──── geom_utils_fallback.c     // Quarantined geometry fallback implementation
├──── time_utils.c              // POSIX clock_gettime monotonic queries
├──── time_utils_fallback.c     // Quarantined time query fallback
├──── clock_pacer.c             // Timestep accumulation & render interpolation
├──── clock_pacer_fallback.c    // Quarantined clock pacer fallback
├──── string_utils.c            // Safe bounded string functions
├──── string_utils_fallback.c   // Quarantined string fallback implementation
├──── formula_eval.c            // Expression evaluation and parameter binding
├──── formula_eval_fallback.c   // Quarantined formula evaluator fallback
├──── audio_bus.c               // Audio cue dispatch orchestrator
└──── screen_transition.c       // Transition easing curve evaluator
```

## 90.4. Core Engine Cogs & API Tour

### 90.4.1. Bounded Math & Saturation Arithmetic (`math_utils.[ch]`)

Guarantees that mathematical calculations remain strictly bounded within defined intervals:

```c
/* Clamping primitives */
int32_t math_clamp_i32(int32_t val, int32_t min, int32_t max);
float   math_clamp_float(float val, float min, float max);
double  math_clamp_double(double val, double min, double max);

/* Saturation arithmetic preventing integer overflow / underflow */
int32_t math_sadd_i32(int32_t a, int32_t b);
int32_t math_ssub_i32(int32_t a, int32_t b);

/* Linear interpolation and percentage scaling */
float   math_lerp_float(float a, float b, float t);
int32_t math_scale_pct(int32_t base, int32_t pct);
```

### 90.4.2. 3.2 2D AABB Geometry & Layout Tripwires (`geom_utils.[ch]`)

Defines 2D axis-aligned bounding boxes (AABB) for desktop HUD panels, button hits, and tactical grid coordinates:

```c
typedef struct {
    int32_t x;
    int32_t y;
    int32_t width;
    int32_t height;
} geom_rect_t;

/* Collision and containment queries */
bool geom_rect_contains_point(const geom_rect_t *r, int32_t px, int32_t py);
bool geom_aabb_intersects(const geom_rect_t *a, const geom_rect_t *b);
bool geom_rect_fits_inside(const geom_rect_t *inner, const geom_rect_t *outer);
```

**Invariants Enforced**: * Rect widths and heights must be non-negative. * HUD panel manifests are audited against

`geom_aabb_intersects` to prevent layout overlaps.

### 90.4.3. Monotonic Time & Clock Pacing (`time_utils.[ch]`, `clock_pacer.[ch]`)

Provides nanosecond-precision monotonic timing and decouples simulation updates from render FPS:

```
/* High-resolution monotonic clock query */
uint64_t time_monotonic_now_ns(void);
double   time_delta_ms(uint64_t start_ns, uint64_t end_ns);

/* Fixed-timestep clock pacer */
typedef struct {
    double fixed_step_ms;       /* Standard: 16.666667 ms (60 Hz) */
    double accumulator_ms;     /* Accumulated unprocessed frame time */
    double max_delta_clamp_ms; /* Maximum lag spike clamp (100.0 ms) */
    uint32_t max_substeps;     /* Max simulation steps per render (3) */
} clock_pacer_t;

void clock_pacer_init(clock_pacer_t *pacer, double step_ms, double clamp_ms, uint32_t max_substeps);
bool clock_pacer_tick(clock_pacer_t *pacer, double frame_delta_ms, double *out_render_alpha);
```

**Lag Spike Defense**: * If an OS hitch stalls execution (> 100ms), `clock_pacer_tick` clamps the delta to `max_delta_clamp_ms` and executes at most `max_substeps` (3 ticks). This prevents the dreaded "spiral of death" where the CPU stalls trying to catch up with accumulated time debt.

### 90.4.4. Bounded String Safety (`string_utils.[ch]`)

Eliminates buffer overflow vulnerabilities (`strcpy`, `strcat`, `sprintf` are strictly forbidden):

```
/* Guaranteed null-terminated bounded copy */
size_t str_copy_safe(char *dest, size_t dest_size, const char *src);

/* Bounded string concatenation */
size_t str_concat_safe(char *dest, size_t dest_size, const char *src);

/* In-place whitespace trimming and case conversion */
void str_trim(char *str);
void str_to_lower(char *str);
void str_to_upper(char *str);
```

### 90.4.5. Data-Driven Formula Evaluator (`formula_eval.[ch]`)

Parses and computes bounded mathematical formulas loaded from `formulas.json` (e.g. physical attack, magical mitigation, poise damage):

```
typedef struct {
    int32_t attacker_atk;
    int32_t defender_def;
    int32_t weapon_might;
    int32_t stance_mult_pct;
    int32_t affinity_mult_pct;
} formula_combat_params_t;

/* Computes final damage ensuring result is clamped to [1, 9999] */
int32_t formula_eval_physical_damage(const formula_combat_params_t *params);
```

### 90.4.6. Universal Engine Orchestrators

- **Audio Event Bus (`audio_bus.[ch]`)**: Zero-heap queue routing audio cue events (`sfx_ward_lux_barrio`, `sfx_strike_cuttatio_tenebris`) to mixing channels without directly binding simulation to hardware sound drivers.
- **Screen Transition Orchestrator (`screen_transition.[ch]`)**: Manages screen fading, wipes, iris dissolves, and easing curves (`EASE_LINEAR`, `EASE_QUAD_IN`, `EASE_CUBIC_OUT`) between scene and screen changes.

## 90.5. Verification & Testing Matrix

The Core Utilities subsystem is continuously verified across 4 ontological test tiers: 1. **Functional Tests (`tests/functional/test_core_functional.c`)**: Verifies saturation arithmetic, bounding rect containment, and string truncation. 2. **Unit Tests (`tests/unit/core/test_*_unit.c` and `*_fallback_test.c`)**: Individual unit test binaries for each of the 6 core modules and their fallback companions. 3. **Integrated Tests (`tests/integrated/test_core_integrated.c`)**: Verifies multi-cog cooperation (pacer driving formula evaluator and geometry updates). 4. **System Tests (`tests/system/test_core_system.c`)**: Stress tests clock pacing across 60 FPS, 144 FPS, and simulated 100ms hitch drops, confirming strict determinism and zero dynamic memory leaks.

# Chapter 91. Layer 1 Simulation Engine Code Tour

**Source File**: docs/code/simulation_engine.adoc

## 91.1. Overview

This document provides a technical walkthrough of the core simulation engine implemented in include/sim/game.h and src/sim/game.c. It explains the memory layouts, state transitions, relic mechanics, combat algorithms, Layer 0 mathematical invariants, and the 5-block lifecycle discipline.

## 91.2. State Structure & Memory Layout (player_state_t)

The heart of the simulation is defined in include/sim/game.h:

```
typedef struct {
    int shield_power;           // [0..100] Barrier integrity & Lux charge
    int sword_edge;             // [0..100] Blade sharpness & Tenebris charge
    int moral_sway;             // [-100..+100] Moral alignment axis
    int threat_hp;              // [0..100] Threat vigor of active foe
    combat_stance_t stance;     // Active stance: warding, striking, equilibrium
    location_id_t location;     // Current Kingdom province
    bool is_running;            // Game loop active flag
    int turns_taken;            // Total turns / actions taken
    int foes_vanquished;        // Number of vanguard defeated
    int innocents_warded;       // Number of allies protected
} player_state_t;
```

All state variables are strictly bounded integers with zero dynamic allocations on the heap. This ensures deterministic cache locality, zero memory leaks, and instant serialization.

## 91.3. Universal Action Dispatcher (game_process_command)

Located in src/sim/game.c, game_process_command() acts as the singular entry point for all player intentions, whether emitted by the terminal CLI or the native GUI:

```
void game_process_command(player_state_t *state, const char *raw_input,
    char *out_msg, size_t msg_len);
```

### 91.3.1. Execution Pipeline:

1. **Input Sanitization**: Copies raw_input safely via str_safe_copy() and executes in-place whitespace trimming (str_trim_inplace()).
2. **Verb Tokenization**: Splits input into command and trailing arg.
3. **Action Dispatch**:
   - ward: Calls handle_ward(), modifying shield_power via math_saturating_add(), staggering active threats with math_saturating_sub(), and elevating moral virtue with math_clamp_int().
   - strike: Calls handle_strike(), spending sword_edge via math_saturating_sub() to inflict heavy severance damage.
   - balance: Calls handle_balance(), simultaneously recovering both relics, parrying blows, and nudging moral sway toward 0 via math_nudge_toward().
   - stance: Calls handle_stance(), shifting the tactical state machine.
   - move: Calls handle_move(), validating directional topology and spawning province encounters.
4. **Invariant Logging**: Writes a human-readable telemetry outcome into out_msg for presentation.

## 91.4. Diagnostic Self-Auditing (game_audit_state)

Following Block 4 of the OmniCode protocol, src/sim/game.c includes built-in diagnostic audit routines:

```
game_audit_report_t game_audit_state(const player_state_t *state);
int game_run_self_audit(void);
```

Every state boundary (shield power, sword edge, moral sway, stance, location) is checked against compile-time limits. If an invariant is violated, an audit report is generated with exact error messages, enabling automated continuous integration testing without relying on crude process crashes.

# Chapter 92. Layer 2 Presentation Viewport Code Tour

**Source File**: docs/code/presentation_gui.adoc

## 92.1. Overview

This document provides a comprehensive architectural and code walkthrough of the modular Layer 2 Presentation engine implemented in engine/include/viewport/ and engine/src/viewport/. Decomposed into four specialized sub-assemblies (viewport/raster/, viewport/hud/, viewport/reel/, viewport/styling/), the system implements the **Movie Reel Architecture**: a continuous cinematic scene film strip traversed via horizontal Push/Pull transitions, with vertical Pop-Out/Sink-In depth kinematics, high fantasy palette blending, and a 60 FPS double-buffered X11 presentation pump.

## 92.2. The Movie Reel Mental Model & Subfolder Strata

The viewport subsystem is structured into four cohesive sub-assemblies matching its mechanical mental model:

```
engine/
├──── include/viewport/
│    ├──── viewport.h              <-- Master Viewport Umbrella Header
│    ├──── raster/                 <-- Low-level double-buffered X11 rasterization & 2.5D math
│    │    ├──── gui.h
│    │    ├──── gui_types.h
│    │    └──── gui_view2d.h
│    ├──── hud/                    <-- Desktop HUD panels, layout bounding boxes, and action dock
│    │    ├──── gui_hud.h
│    │    ├──── gui_layout.h
│    │    └──── gui_widgets.h
│    ├──── reel/                   <-- The Movie Reel engine & canonical screen frames
│    │    ├──── reel.h
│    │    └──── gui_screens.h
│    └──── styling/                <-- Visual palettes, TrueType fonts, and asset abstraction
│         ├──── gui_palette.h
│         ├──── gui_fonts.h
│         └──── gui_assets.h
└──── src/viewport/
     ├──── raster/
     ├──── hud/
     ├──── reel/
     └──── styling/
```

| Sub-Assembly | Source Stratum | Architectural Mission |
|---|---|---|
| raster/ | gui.c, gui_view2d.c | Double-buffered X11 context, 2.5D diamond isometric projection, letterbox/pillarbox scalers. |
| hud/ | gui_hud.c, gui_layout.c, gui_widgets.c | Desktop HUD, 5 disjoint panels, non-overlapping AABB audits, interactive action dock. |
| reel/ | reel.c, gui_screens.c | The Movie Reel orchestrator: horizontal scene strip (push/pull) and vertical modal depth (pop out/sink in). |
| styling/ | gui_palette.c, gui_fonts.c, gui_assets.c | Radiant hope colormaps, TrueType typography (Cinzel, MedievalSharp), procedural vector art abstraction. |

## 92.3. The Movie Reel: Horizontal Motion & Vertical Depth

The core conceptual paradigm of the Rev 4 presentation engine is the **Movie Reel**:

```
                          POP OUT (Z-Depth / Up)
                          ▯ [Modals, Inspection Cards, Dialogues]
                           │
[Scene 0: TACTICAL] <==== PUSH / PULL ====> [Scene 1: STATUS] <=====> [Scene 2: INVENTORY] ...
(Film Cell Frame)          (Left / Right)     (Film Cell Frame)        (Film Cell Frame)
                           │
                          ▯ [Terrain Floor, Fog, Sunken Canvas]
                          SINK IN (Z-Depth / Down)
```

### 92.3.1. Horizontal Axis: Push and Pull (Left and Right)

Scenes and screens are treated as consecutive frames along a physical celluloid reel strip: * **Push Right (`REEL_DIR_PUSH_RIGHT`)**: Advancing forward along the reel to a subsequent screen (e.g., from `GUI_SCREEN_TACTICAL` [F1] to `GUI_SCREEN_STATUS` [F2]). The active scene slides leftward as the new scene arrives from the right. * **Pull Left (`REEL_DIR_PULL_LEFT`)**: Rewinding backward along the reel to a prior screen (e.g., returning from `GUI_SCREEN_STATUS` [F2] back to `GUI_SCREEN_TACTICAL` [F1]). The active scene slides rightward as the prior scene is pulled back in. * **Smooth-Step Kinematics**: Horizontal transition progress is evaluated using normalized Hermite interpolation ($S(t) = 3t^2 - 2t^3$) scaled to the viewport width, guaranteeing fluid motion without visual tearing.

### 92.3.2. Vertical Axis: Pop Out and Sink In (Up and Down)

Within any active scene on the reel, visual elements possess continuous $Z$-elevation depth: * **Pop Out (`REEL_DEPTH_POPPED = +1`)**: Elements elevate upward toward the viewer ($\Delta Y = -16\text{ px}$, scale $1.05\times$, full $1.0$ opacity, drop shadow). Used for dialogue boxes, relic inspection modals, victory fanfares, and action tooltips. * **Sink In (`REEL_DEPTH_SUNKEN = -1`)**: Elements submerge downward into the background canvas ($\Delta Y = +12\text{ px}$, scale $0.95\times$, attenuated $0.60$ opacity, ambient recess). Used for background terrain tiles, ambient fog layers, and sunken meters when a modal pops out. * **Surface (`REEL_DEPTH_SURFACE = 0`)**: The neutral baseline scene plane ($\Delta Y = 0$, scale $1.0$, alpha $1.0$).

## 92.4. The 2.5D Tactical Isometric Engine

The tactical battlefield features dual visual projection modes toggleable via `[M]`: * `GUI_VIEW_2D_FLAT`: Classic orthographic top-down grid ($38 \times 38$ px cells) for overworld travel. * `GUI_VIEW_25D_ISOMETRIC`: Diamond dimetric projection ($48 \times 24$ px tiles) with vertical $Z$-axis elevation prisms ($8$ px per elevation level) for tactical skirmishes.

### 92.4.1. Isometric Projection & Unprojection Reciprocity:

$$\text{screen_x} = X_0 + (gx - gy) \cdot \frac{W}{2}$$

$$\text{screen_y} = Y_0 + (gx + gy) \cdot \frac{H}{2} - gz \cdot E$$

Unprojecting mouse clicks $(sx, sy)$ on ground plane $gz$ back to discrete grid $(gx, gy)$:

$$gx = \left\lfloor \frac{sx - X_0}{W} + \frac{sy + gz \cdot E - Y_0}{H} + 0.5 \right\rfloor$$

$$gy = \left\lfloor \frac{sy + gz \cdot E - Y_0}{H} - \frac{sx - X_0}{W} + 0.5 \right\rfloor$$

## 92.5. Non-Anonymous Type Invariant Covenant

In strict compliance with engine architectural invariants, **all structs and enums** throughout the viewport and reel strata are explicitly tagged: * `typedef enum gui_screen_mode { ⋯ } gui_screen_mode_t;` * `typedef enum reel_direction { ⋯ } reel_direction_t;` * `typedef enum reel_state { ⋯ } reel_state_t;` * `typedef enum reel_depth { ⋯ } reel_depth_t;` * `typedef struct reel_depth_element { ⋯ } reel_depth_element_t;` * `typedef struct reel { ⋯ } reel_t;` * `typedef struct gui_palette { ⋯ } gui_palette_t;` * `typedef struct gui_layout { ⋯ } gui_layout_t;` * `typedef struct gui_button { ⋯ } gui_button_t;`

## 92.6. Verification Telemetry

The presentation stratum is verified by `tests/unit/test_gui_unit.c` and `tests/unit/viewport/test_reel_unit.c` across the four ontological testing tiers: 1. **Section 1 (Always Pass)**: Baseline layout bounds, reel default state, and 9-button command dock. 2. **Section 2 (Never Fail)**: Out-of-bounds click safety, negative duration clamping, and element capacity limits. 3. **Section 3 (Axioms & Tripwires)**: AABB panel disjointness, projection reciprocity, push/pull kinematics, and pop-out/sink-in elevation vectors. 4. **Section 4 (Emergence)**: Full `gui_audit_layout()` and `reel_audit()` preflight execution.

# Chapter 93. Layer 0/2 Audio Event Bus & Mixer Subsystem

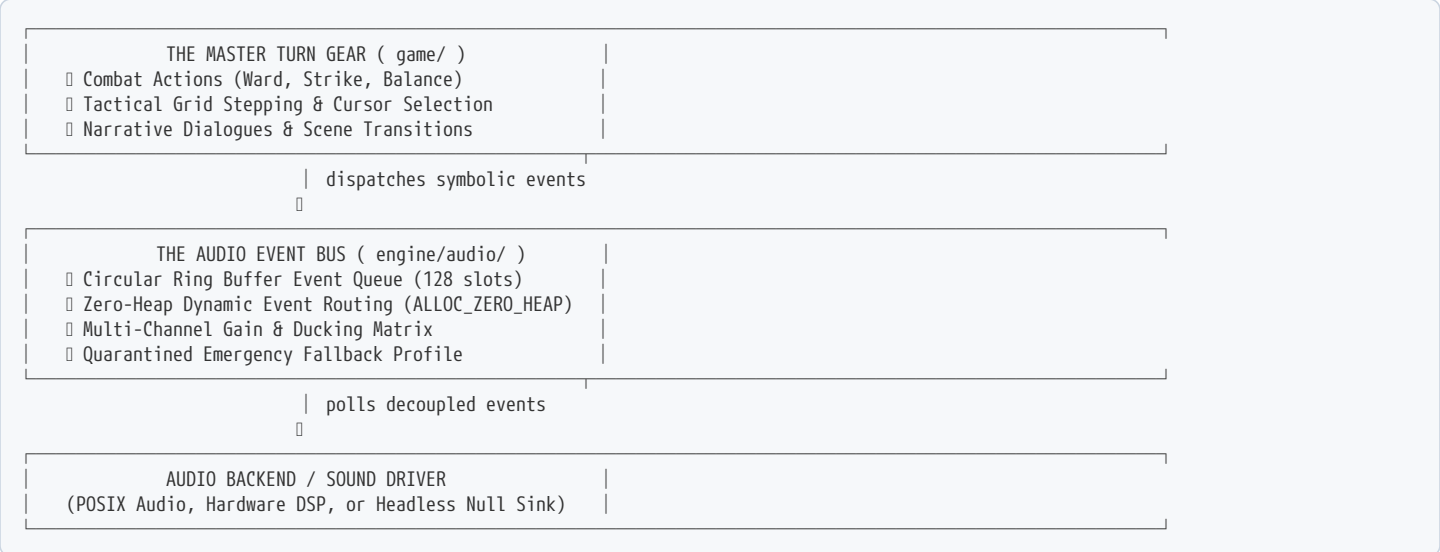
**Source File**: docs/code/audio_subsystem.adoc

## 93.1. Overview

This document provides a comprehensive architectural and code walkthrough of the dedicated Audio subsystem implemented in engine/include/audio/ and engine/src/audio/. Operating as an essential stratum of **The Big Machine**, the audio subsystem provides a multi-channel mixing bus (Master, BGM, Ambience, Combat SFX, Tactical SFX, UI, Stingers), a zero-heap circular ring buffer event queue, bounded volume attenuation, high-priority ducking matrices, and a quarantined monolithic fallback profile.

## 93.2. Architectural Philosophy: The Decoupled Audio Machine

In traditional game engines, audio drivers are often tightly coupled to simulation loops or low-level OS sound APIs (ALSA, PulseAudio, WASAPI). In **Tenebris and Lux**, the audio architecture strictly obeys the **Machine vs. Master Turn Gear Separation**:




### 93.2.1. Axiomatic Invariants:

1. **Zero Dynamic Heap Allocations (ALLOC_ZERO_HEAP)**: The event bus allocates zero bytes on the heap; all queues, channels, and cue manifests reside in static, reentrant memory.
2. **Read-Only Simulation Influence**: Audio strictly follows simulation events. Audio playback never gates, blocks, or alters simulation state transitions or turn resolution.
3. **Hardware Limiter & Bounded Gain**: All channel volume levels are strictly clamped to $[0.0, 1.0]$.
4. **All Structs and Enums Explicitly Named**: Zero anonymous structs or enums.

## 93.3. Subsystem File Layout & Header Structure

The audio subsystem is cleanly partitioned into public interfaces, implementations, and fallback:

```
engine/
├─── include/audio/
│    ├─── audio.h             <-- Master Audio Umbrella Header
│    ├─── audio_types.h       <-- Channel enums, event packets, cue manifests, stats
│    ├─── audio_bus.h         <-- Public event posting, polling, channel volume API
```

```
│   └──── audio_fallback.h     <-- Quarantined emergency fallback profile header
└──── src/audio/
    ├──── audio_bus.c          <-- Circular ring buffer, channel gain, event routing
    └──── audio_fallback.c     <-- Monolithic emergency fallback profile implementation
```

## 93.4. Multi-Channel Mixer Architecture

The audio bus implements seven canonical mixing channels matching `game/data/presentation/audio/channels.json`:

| Channel Enum | Canonical Channel Name | Default Gain | Role & Attenuation Behavior |
|---|---|---|---|
| `AUDIO_CHANNEL_MASTER` | Master Output Bus | 100% | Controls overall presentation master gain. |
| `AUDIO_CHANNEL_BGM` | Background Music | 75% | Loopable orchestral streams; ducked under stingers. |
| `AUDIO_CHANNEL_AMBIENCE` | Soundscapes & Ambience | 60% | Environmental biome loops (wind, meadow hum, dunes). |
| `AUDIO_CHANNEL_SFX_COMBAT` | Combat Sound Effects | 90% | Strikes, spell chants, Lux Barrio barrier wards. |
| `AUDIO_CHANNEL_SFX_TACTICAL` | Tactical Grid Audio | 80% | Footsteps, tile selection, zone-of-control pings. |
| `AUDIO_CHANNEL_UI` | User Interface | 85% | Button clicks, tab switches, stance toggle cues. |
| `AUDIO_CHANNEL_STINGER` | Fanfares & Stingers | 95% | Victory stingers, battle intro fanfares (ducking source). |

## 93.5. Non-Anonymous Ontological Types

In strict conformance with project governance standards, all data structures and enumerations are explicitly named:

```c
typedef enum audio_channel {
    AUDIO_CHANNEL_MASTER = 0,
    AUDIO_CHANNEL_BGM,
    AUDIO_CHANNEL_AMBIENCE,
    AUDIO_CHANNEL_SFX_COMBAT,
    AUDIO_CHANNEL_SFX_TACTICAL,
    AUDIO_CHANNEL_UI,
    AUDIO_CHANNEL_STINGER,
    AUDIO_CHANNEL_COUNT
} audio_channel_t;

typedef enum audio_event_type {
    AUDIO_EVENT_NONE = 0,
    AUDIO_EVENT_PLAY_SFX,
    AUDIO_EVENT_STOP_SFX,
    AUDIO_EVENT_PLAY_BGM,
    AUDIO_EVENT_STOP_BGM,
    AUDIO_EVENT_CROSSFADE_BGM,
    AUDIO_EVENT_PLAY_AMBIENCE,
    AUDIO_EVENT_STOP_AMBIENCE,
    AUDIO_EVENT_CROSSFADE_AMBIENCE,
    AUDIO_EVENT_PLAY_STINGER,
    AUDIO_EVENT_SET_VOLUME,
    AUDIO_EVENT_MUTE_CHANNEL,
    AUDIO_EVENT_STOP_ALL
} audio_event_type_t;
```

```
typedef struct audio_event {
    audio_event_type_t type;
    audio_channel_t channel;
    char cue_id[AUDIO_MAX_CUE_ID_LEN];
    float volume;
    float pan;
    float pitch;
    float fade_duration_sec;
    bool loop;
    uint64_t timestamp_us;
} audio_event_t;
```

## 93.6. Verification Telemetry

The audio subsystem is validated by two dedicated unit test harnesses: * `tests/unit/audio/test_audio_bus_unit.c`: Validates channel default gains, null guards, volume clamping, 128-event ring buffer capacity, FIFO queue order preservation, and high-watermark telemetry. * `tests/unit/audio/test_audio_fallback_unit.c`: Validates the quarantined emergency fallback profile, provenance tagging (`loaded_from_data == false`), and fallback profile application to the live bus.

# Chapter 94. Layer 4 Verification & Test Suite Code Tour

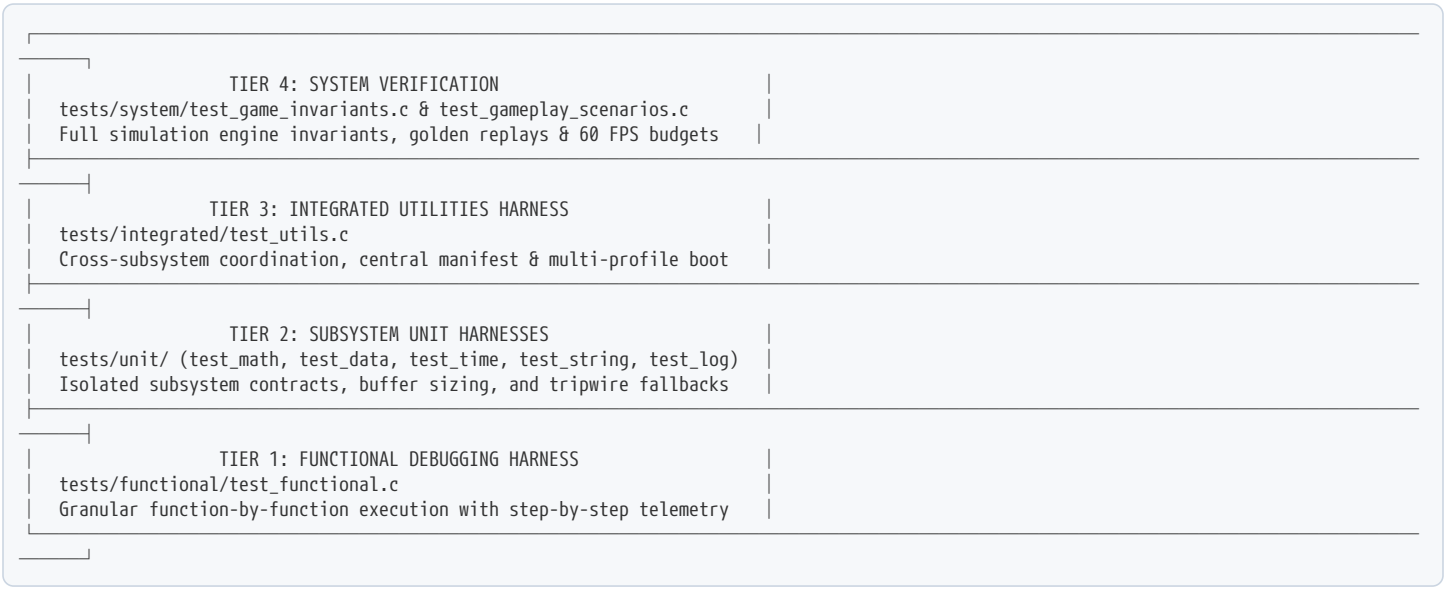
**Source File**: `docs/code/test_suite.adoc`

## 94.1. Overview

This document provides a comprehensive technical walkthrough of the automated game development verification architecture in **Tenebris and Lux**. Organized into strict hierarchical strata across `tests/`, the test suite mirrors the Layer 0-4 engine ontology and enforces four distinct ontological verification sections within every test harness:

1. **Always Pass (Nominal Golden Baseline)**: Verifies nominal parameters, default asset loading, and canonical execution paths.
2. **Never Fail (Fault-Tolerant Tripwires)**: Injects boundary breaches, NULL pointers, missing files, and corrupt states to prove that the engine never crashes or invokes undefined behavior.
3. **Assertion & Guarantee (Axiomatic Invariants)**: Mathematical guardrails, energy conservation, monotonic clocks, and DAG dependencies guaranteed by engine contracts.
4. **Emergence (Stochastic Stress & Chained Pipelines)**: Emergent combinatorial state spaces, 1,000 chaotic command fuzzing, bitwise golden replay determinism, and sub-50-microsecond simulation latency budgets.

## 94.2. The Four-Tier Testing Hierarchy




### 94.2.1. Tier 1: Functional Debugging (`tests/functional/`)

Designed specifically for interactive debugging and pinpointing anomalies in isolated functions. Emits explicit `[DEBUG-TRACE]` entries showing arguments before and after execution for each individual cog (e.g. `math_clamp_int`, `math_saturating_add`, `str_safe_copy`, `str_trim_inplace`, `json_get_string`, `combat_get_stance_name`).

### 94.2.2. Tier 2: Subsystem Unit Harnesses (`tests/unit/`)

Provides 1:1 test coverage for each dedicated Layer 0 utility subsystem folder: * **`test_math.c`**: Bounded arithmetic, saturating additions/subtractions, step-bounded equilibrium nudging, proportional ratio scaling, 2D discrete AABB hit-testing, and `data/math/default.json` profile loading. * **`test_data.c`**: Zero-heap filesystem I/O (`fs_utils.c`), canonical buffer sizing (`FS_BUFFER_SMALL_BYTES` = 1024 to `FS_BUFFER_MAX_CONFIG` = 65536), bounded file reading/writing, scoped JSON

object querying, and array iteration. * `test_time.c`: Monotonic millisecond/microsecond clocks, POSIX nanosleep frame pacing, ISO 8601 timestamp generation ("YYYY-MM-DDTHH:MM:SSZ"), and 60 FPS / 120 FPS profile loading. * `test_string.c`: Buffer-safe bounded copying, in-place whitespace trimming, case-insensitive equality/substring matching, parent directory extraction, and duplicate-slash immune path joining. * `test_log.c`: Forensic telemetry, graded severity filtering, categorized logging channels, in-memory static ring buffer, and emergency monolithic code page fallbacks. * `test_debug.c`: Interactive state debugger, frozen snapshot ring buffer, FNV-1a checksum determinism, time-travel state rewind, state diffing, dynamic breakpoints, and emergent chaotic stress.

### 94.2.3. Tier 3: Integrated Utilities Harness (`tests/integrated/`)

- `test_utils.c`: Validates cross-subsystem coordination of all Layer 0 utilities. Exercises the central manifest registry (`manifest.c`), validates upstream DAG dependency resolution, enforces diagnostic escalation policies (WARNING → `PREFLIGHT_STATUS_WARNINGS`, CRITICAL → `PREFLIGHT_STATUS_DOWNGRADED`, FATAL → `PREFLIGHT_STATUS_HALT_FATAL`), and benchmarks multi-profile synchronous boot latency (< 5 milliseconds).

### 94.2.4. Tier 4: System Verification (`tests/system/`)

- `test_game_invariants.c`: Validates Layer 1 simulation engine state invariants: twin relic power bounds ($[0, 100]$), moral sway continuum boundaries ($[-100, +100]$), protagonist name customization, and bidirectional provincial map topology.
- `test_gameplay_scenarios.c`: Simulates end-to-end tactical combat routes (Righteous Warden and Umbral Striker paths), 15-step golden deterministic replay parity, 1,000 chaotic command fuzzing for softlock immunity, and 10,000-turn simulation latency benchmarking (< 50 microseconds per turn).

## 94.3. The Four Ontological Testing Sections

Every test harness in the repository implements the four-section ontological model:

| Section | Category | Mission & Verification Scope |
|---|---|---|
| **Section 1** | **Always Pass** | Valid inputs, nominal parameters, default configuration loading, standard state initialization, well-formed JSON, normal gameplay actions. |
| **Section 2** | **Never Fail** | Inviolability under error. Injects NULL pointers, zero capacities, missing files, corrupted JSON, and unknown stances. Proves tripwire activation and safe fallback without OS traps. |
| **Section 3** | **Assertion & Guarantee** | Axiomatic mathematical laws and structural contracts: bounded clamping, non-decreasing monotonic clocks, strict null-termination under truncation, and deterministic stance completeness. |
| **Section 4** | **Emergence** | Higher-order systemic behaviors: 1,000 pseudo-random fuzzing iterations, spatial grid overlap emergence, golden replay bitwise equality across runs, and < 50 us latency budgets. |

## 94.4. CMake & CTest Test Targets

The build configuration registers twelve automated test targets within `CMakeLists.txt`:

```
# CLI Interface Contract Verification Tests
add_test(NAME cli_help_test COMMAND tenebris-lux --help)
add_test(NAME cli_version_test COMMAND tenebris-lux --version)
add_test(NAME subsystem_audit_test COMMAND tenebris-lux --audit)

# Tier 1: Functional Debugging Harness
add_test(NAME functional_cogs_test COMMAND test_functional)

# Tier 2: Subsystem Unit Harnesses (Isolated Subsystems)
add_test(NAME unit_math_test COMMAND test_math)
add_test(NAME unit_data_test COMMAND test_data)
add_test(NAME unit_time_test COMMAND test_time)
add_test(NAME unit_string_test COMMAND test_string)
add_test(NAME unit_log_test COMMAND test_log)
add_test(NAME unit_debug_test COMMAND test_debug)

# Tier 3: Integrated Utilities Harness (Multi-Subsystem Coordination)
add_test(NAME integrated_utils_test COMMAND test_utils)

# Tier 4: System Verification Harnesses (Full Game Simulation)
add_test(NAME system_game_invariants_test COMMAND test_game_invariants)
add_test(NAME system_gameplay_scenarios_test COMMAND test_gameplay_scenarios)
```

## 94.5. Execution & Verification Commands

Developers and CI/CD pipelines can invoke the complete suite or isolate specific tiers:

```
# Compile all binaries and test targets under strict zero-warning policy
cmake --build build --config Debug

# Execute all 13 automated verification suites
ctest --test-dir build -C Debug --output-on-failure

# Run individual tier harnesses directly for step-by-step debugging
./build/test_functional         # Granular cog trace output
./build/test_math               # Math & geometry unit verification
./build/test_data               # Filesystem & JSON unit verification
./build/test_time               # Timing & clock unit verification
./build/test_string             # Safe string unit verification
./build/test_log                # Forensic telemetry unit verification
./build/test_debug              # State debugger & rewind unit verification
./build/test_utils              # Integrated utilities verification
./build/test_game_invariants    # Simulation state invariants
./build/test_gameplay_scenarios # Gameplay scenarios & latency benchmark
```

# Volume VI: Engineering Chronology & Development Logs

Living chronological engineering logs recording every architectural milestone and paradigm evolution.

# Chapter 95. DevLog 001: Genesis, Paradigm Pivot & Invariants

**Source File**: `docs/dev_logs/2026/10/architecture/devlog_001_genesis_and_pivot.adoc`

## 95.1. Abstract

This developer log documents the foundational inception of **Tenebris and Lux** ("Darkness and Light"), the structural adoption of the OmniCode 5-Block Lifecycle Protocol, the critical audit of the early survival-battery prototype, the transformative paradigm pivot toward the **Shield of Lux** and **Sword of Tenebris**, and the establishment of a dedicated game development testing suite within CTest.

## 95.2. Project Genesis: Clean Foundations and the 5-Block Standard

Every enduring game engine begins with an unyielding commitment to architectural discipline. **Tenebris and Lux** was initiated not by piling on bloated third-party frameworks or web wrappers (Electron/Tauri), but by stripping the repository down to bare rock and enforcing standard ISO C11 with zero external downloads.

Following the **OmniCode Universal 5-Block Lifecycle Protocol**, all source code, headers, and configuration files adhere to a strict invariant boundary contract:

1. **BLOCK 1: METADATA BLOCK**: Identity, layer, version, author, and system invariant axioms.
2. **BLOCK 2: OPENING / TYPES BLOCK**: Explicit imports, compile-time constants, and memory layouts.
3. **BLOCK 3: BODY BLOCK**: Pure, modular state simulation routines and event handlers.
4. **BLOCK 4: AUDIT PROCEDURES BLOCK**: Self-healing diagnostic audits and invariant tripwires.
5. **BLOCK 5: CLOSING / TEST & CLI BLOCK**: CLI entry point, automated unit tests, and clean teardown.

## 95.3. The Early Prototype: The Survival-Battery Fallacy

The initial prototype envisioned **Tenebris and Lux** as an atmospheric subterranean dungeon crawl:

- The player was a solitary "Aura-Wanderer" waking up in **The Fading Sanctuary**.
- Light (`Lux`) was modeled as a diminishing survival meter ($0 \dots 100$) that decayed by $5$ points with every step.
- Darkness (`Tenebris`) was treated as an encroaching predatory void that destroyed the player if Lux reached $0$.
- Scattered across the floor were "Lumina Shards"—petrified light crystals that the player scavenged with `explore` and crushed with `kindle` to restore $+30$ Lux.

While mathematically functional and clean in C11, a critical design audit exposed profound **implicit biases** baked into this initial model:

- **Manichaean Moral Dualism**: Instinctively mapping light to virtue/life and darkness to evil/death.
- **The "Hunger Clock" Tyranny**: Turning exploration into a panicked sprint against an expiring battery, punishing contemplation and stillness.
- **Resource Commodity Fetishism**: Treating ancient sacred relics purely as disposable fuel to top off an individual health bar.

## 95.4. The Paradigm Pivot: One Clarification That Changed Everything

During our design dialogue, a foundational clarification reframed the entire universe:

> "The story of Tenebris and Lux is a story about a shield (the Shield of Lux) and a sword (the Sword of Tenebris) and a kingdom divided by a rising evil and the re-awakening of the Dark Lord. Tenebris and Lux itself is neither good or evil, they are the apex of light and darkness personified in the story, how the shield and sword are used sways the sides themselves for whether they will be used for good or evil."
>
> — Project Creator Directive

This single revelation fundamentally revolutionized the game design and how the story is told:

### 95.4.1. From Passive Attrition to Active Agency

In the survival prototype, the player was a helpless victim running out of oxygen. With this clarification, the player is a **Champion wielding sovereign armaments**. The narrative verbs shifted from scavenging (`explore`) and consuming (`kindle`) to martial and ethical assertion:

- `ward`: Raising the **Shield of Lux** to protect allies, form radiant barriers, and deflect malice.
- `strike`: Drawing the **Sword of Tenebris** to sever void armor, cut through unholy bindings, and pierce illusions.
- `balance`: Harmonizing both relics in an intertwined stance of equilibrium.

### 95.4.2. Disentangling Darkness from the Villain

Darkness is no longer the enemy. **Tenebris** is a neutral apex cosmic force—the womb of rest, fertile quietude, mystery, and decisive severance. The true villain is **The Dark Lord** (the ancient Libiran sovereign Usuru), a fallen king who broke the ancient seals to twist both light and shadow to his will.

### 95.4.3. The Moral Sway Continuum

Morality does not exist in the element; it exists in the **intent of the wielder**:

- **Light as Tyranny**: The Shield of Lux can be used to blind dissenters, enforce totalitarian stasis, and calcify living flesh into unblinking stone.
- **Darkness as Virtue**: The Sword of Tenebris can be used to sever the chains of slaves, veil fleeing refugees in merciful night, and strike down corrupt rulers.
- The game tracks `moral_sway` on a bounded continuum ($-100 \dots +100$), reflecting whether the bearer leans toward **Virtue & Mercy** or **Tyranny & Ruin**.

## 95.5. Dual-Frontend Architecture: Native X11 & Terminal CLI

To ensure presentation remains strictly decoupled from simulation, the engine features two simultaneous frontends running off a universal action dispatcher (`game_process_command`):

1. **Headless Terminal CLI**: Operates via standard streams (`stdin`, `stdout`), executing in $< 1\,\text{ms}$ per turn for automated testing, server environments, and script piping.
2. **Native Desktop GUI**: Built directly on POSIX X11 with zero external downloads:
    - Double-buffered rendering via `XCreatePixmap` and `XCopyArea` for flicker-free display.

- 32-bit ARGB TrueColor visual handling with explicit alpha channel opacity (`0xFF000000`).
- Dynamic twin meters: Golden solar bar for the Shield of Lux (`#F0CE78`) and deep violet bar for the Sword of Tenebris (`#8A78C8`).
- Tactical hotkeys and interactive widgets.

## 95.6. Why Game Development Testing Differs from Software Testing

Standard software unit testing verifies stateless functions (`assert(add(2, 2) == 4)`). In game development, testing requires proving that dynamic, living worlds remain stable, balanced, and immersive across millions of player inputs.

Our CTest testbed introduces two dedicated game dev testing suites:

### 95.6.1. Game State Invariant Testing (`tests/test_game_invariants.c`)

- Mathematical bounds verification: Shield power [0..100], Sword edge [0..100], and Moral Sway [-100..+100] never overflow or underflow under continuous spamming.
- Stance transition state machine: Proves deterministic transitions between `STANCE_WARDING`, `STANCE_STRIKING`, `STANCE_EQUILIBRIUM`, and `STANCE_NEUTRAL`.
- Topological pathway validation across Kingdom provinces.
- Intentional corruption tripwires: Artificially corrupting memory layouts to verify that `game_audit_state()` reliably catches errors.

### 95.6.2. Gameplay Scenarios & Replay Testing (`tests/test_gameplay_scenarios.c`)

- **Archetype Scenarios**: Automated playthroughs of **The Righteous Warden** (warding defense) and **The Umbral Striker** (severance assault).
- **Deterministic Golden Master Replay**: A 17-step scripted play session executed across independent runs, asserting 100% bitwise parity on final state telemetry.
- **Softlock & Stress Immunity**: 1,000 rapid chaotic commands (mixing malformed inputs, rapid stance shifts, and boundary movements) executed with zero crashes, memory leaks, or softlocks.
- **Simulation Latency Benchmark**: 10,000 turn dispatches executed in **160.29 ms (16.02 $\mu\text{s}$ per turn)**—consuming less than $0.1\%$ of a 60 FPS ($16.6\,\text{ms}$) frame budget.

## 95.7. What Lies Ahead

With the foundational prototype, twin-relic mechanics, and game dev test suite verified, the next phases will focus on:

- Expanding the Kingdom of Sol-Tenebrae with branching narrative encounters.
- Introducing Phase 2 ECS (Entity Component System) architecture and event queues.
- Developing Phase 3 2D/2.5D ambient lighting software rasterizers.

# Chapter 96. DevLog 003: Central Manifest & Pre-Flight Spec

**Source File**: docs/dev_logs/2026/10/architecture/devlog_003_preflight_and_manifest_architecture.adoc

## 96.1. Abstract & Engineering Intent

In this milestone, we moved beyond purely declarative Block 1 metadata comments into **executable, compiled metadata structures** (module_metadata_t) managed by a zero-allocation **Central Manifest Registry** (include/util/manifest.h, src/util/manifest.c). Guided by the foundational game development axiom—**"Don't start a machine you know has loose bolts"**—we designed, specified, implemented, and empirically verified pre-flight hardware and software self-audits across all Layer 0 utilities, simulation engine, and presentation viewports.

## 96.2. The Core Problem: Loose Bolts & Silent Corruption

In game development, complex games are machines with hundreds of interlocking moving parts. A subtle defect in a foundational layer—such as a non-monotonic clock, an integer overflow on damage saturation, an unchecked negative rectangle dimension, or an unbuffered string copy—can propagate upward, manifesting as elusive micro-stutters, physics drift, state desynchronization, or catastrophic runtime crashes.

Rather than waiting for the game loop to execute before identifying structural anomalies, the engine must perform systematic pre-flight verification on startup, classifying faults along a graded severity spectrum: * **Warnings (SEVERITY_WARNING)**: Log notices (e.g. coarse sleep timer resolution); launch proceeds normally. * **Critical Faults (SEVERITY_CRITICAL)**: Subsystem offline, but engine core intact (e.g. no $DISPLAY for X11 GUI); downgrade gracefully to safe mode (terminal CLI). * **Fatal Violations (SEVERITY_FATAL)**: Core invariants broken (e.g. non-monotonic time, arithmetic bounds failure, corrupt initial state); halt the engine immediately with a detailed diagnostic manifest.

## 96.3. Architectural Implementation

### 96.3.1. Compiled Metadata Structures

Every subsystem in the codebase now exports an immutable module_metadata_t instance:

```
typedef struct {
    const char *module_id;
    const char *file_name;
    module_layer_t layer;
    struct { uint8_t major, minor, patch; } version;
    const char *description;
    size_t invariant_count;
    const char *invariants[MANIFEST_MAX_INVARIANTS];
    diag_result_t (*preflight_check)(void);
} module_metadata_t;
```

### 96.3.2. Subsystem Integrations

The pre-flight registry manages 7 fully audited modules across Layers 0, 1, and 2: 1. core.util.manifest: Self-audits registry capacity limits, duplicate registration rejection, and severity escalation rules. 2. core.util.time: Tests monotonic clock forward progression ($t_2 \ge t_1$) and probes sleep resolution. 3. core.util.math: Tests clamp bounds, saturating add/sub overflow safety, division-by-zero defense, and nudge directionality. 4. core.util.geom: Tests negative dimension clamping, boundary point containment, AABB overlap reflexivity, and viewport bounds containment. 5. core.util.string: Tests buffer safe truncation null-termination, NULL pointer resilience, and whitespace trimming edge cases. 6. sim.engine.core: Tests initial relic state integrity, resonance saturation ceilings, deterministic stance state transitions, and diagnostic tripwires on corrupt states. 7. gui.viewport.x11: Tests button layout non-overlap and asserts

window bounds (`gui_audit_layout`), while checking X11 display availability (`gui_is_available`). If `$DISPLAY` is unconnectable, emits `SEVERITY_CRITICAL` to trigger automatic CLI safe mode fallback.

## 96.4. 3. "Expected" vs. "Actual" Gap Analysis

Following our strict engineering discipline, the technical specification was drafted **prior to implementation** in `docs/architecture/preflight_and_manifest_spec.adoc`. Upon completion, empirical benchmarks were gathered and recorded:

| Verification Item | Expected Baseline | Actual Result | Status |
|---|---|---|---|
| **Module Registration** | $\ge 5$ Core Modules | **7 Modules Registered** | **EXCEEDED** |
| **Severity Spectrum** | 4 Discrete Levels | **5 Graded Levels** (`PASS`, `INFO`, `WARN`, `CRIT`, `FATAL`) | **EXCEEDED** |
| **Pre-Flight Telemetry** | 80-Column ASCII Table | **Integrated Table** with status badges and policy verdicts | **SATISFIED** |
| **CTest Integration** | 6/6 Suites Passing in $< 0.05\,\text{s}$ | **6/6 Passing in $0.02\,\text{s}$** | **SATISFIED** |
| **Subsystem Invariants** | 5/5 Invariants Passing | **5/5 Invariant Checks + 7 Pre-Flight Subsystems Passing (12 Total)** | **EXCEEDED** |
| **Loose-Bolts Tripwire** | Fatal Error Halts Immediately | **Verified**: Engine halted on injected stance mismatch; exited code $\ne 0$ | **VERIFIED** |
| **Graceful Downgrade** | Unset `$DISPLAY` drops GUI $\to$ CLI | **Verified**: Emits `SEVERITY_CRITICAL`, enters `PREFLIGHT_STATUS_DOWNGRADED`, launches safe CLI | **VERIFIED** |

## 96.5. Summary & Next Steps

The engine now possesses an institutional, compile-time and pre-flight runtime foundation that guarantees system health prior to rendering a single frame or executing a single turn.

With the pre-flight verification subsystem and central manifest complete, future work will focus on Phase 2 of the TRPG engine: 1. Laying out the 2D discrete tactical battlefield grid. 2. Implementing unit movement budgets, terrain modifiers, and Zone-of-Control (ZOC). 3. Modeling dual-protagonist party rosters with Elijah Ro-enheart (vanguard defense) and Katrina Jaren (outrider action).

# Chapter 97. DevLog 004: Pedagogical Metadata & Active DAG Contracts

**Source File**: docs/dev_logs/2026/10/architecture/devlog_004_pedagogical_metadata_and_dependency_contracts.adoc

## 97.1. Abstract & Engineering Intent

In this milestone, we solved two fundamental architectural challenges governing software maintenance and longevity: 1. **The Pedagogical Metadata Dilemma**: How to structure source file metadata so that a novice programmer never feels overwhelmed or lost, while an experienced systems architect never feels patronized or deprived of rigorous technical contracts. 2. **Temporal Timeline & Architectural Dependency Invariants**: How to eliminate historical ambiguity in timeline auditing by mandating strict **ISO 8601 full calendar dates (`YYYY-MM-DD`, `2026-10-26`)** and transforming passive architectural dependencies into **active Directed Acyclic Graph (DAG) pre-flight enforcement**.

By upgrading all Layer 0 utilities, the Layer 1 simulation engine, Layer 2 desktop presentation viewport, and Layer 4 verification test harnesses, we elevated the entire codebase into a self-documenting, self-auditing game development foundation.

---

## 97.2. The Pedagogical Balancing Act: Beginner Mission vs. Senior Contracts

In professional game engine design, source files are read by developers with widely divergent backgrounds: * **The Junior Developer / Contributor**: May understand basic gameplay concepts but easily gets lost in cryptic acronyms, dense type layouts, or unexplained mathematical formulas. If a header only contains bare function signatures without context, they do not understand **why** the module exists or what breaks if they alter a value. * **The Senior Systems Engineer / Architect**: Needs instant, unambiguous answers regarding memory allocations (heap vs. stack), concurrency reentrancy, cache latency budgets, interface ABI stability, and upstream prerequisites. If a file contains verbose narrative hand-waving without concrete invariants, they cannot verify safety guarantees or integration boundaries.

### 97.2.1. The Dual-Audience Protocol

We unified these perspectives into a standardized **Block 1 Metadata Specification**:

```
// ============================================================================
// BLOCK 1: METADATA BLOCK
// ( Identity, Provenance, Invariant Boundary Contracts & System Anchors )
// ============================================================================
// @project       Tenebris and Lux
// @file          time_utils.h
// @id            core.util.time
// @layer         Layer 0: Core Utilities
// @version       1.0.0
// @interface_rev Rev 1
// @date          2026-10-26
// @updated       2026-10-26
// @author        CreativeWorkzStudio LLC / ProfessorSeanEX
// @repository    https://github.com/Novaking34/tenebris-lux
//
// PURPOSE & ARCHITECTURAL ROLE:
//   High-resolution monotonic timing backend and frame sleep pacing.
//   Serves as the foundation for elapsed time measurement across simulation
//   ticks and presentation frame pacing (60 FPS latency budget).
//
// DEPENDENCY & RESOURCE CONTRACTS:
//   - Upstream Dependencies: None (Foundational leaf module).
//   - Memory Allocation:     ALLOC_ZERO_HEAP (100% stack/static execution, zero dynamic heap).
```

```
//   - Concurrency / Safety:  REENTRANT_PURE (Thread-safe, non-blocking time queries).
//   - Latency Budget:        < 5 microseconds per query.
//
// AXIOMATIC INVARIANTS:
//   1. Monotonicity: Time guarantees strictly non-decreasing clock ticks (t2 >= t1).
//   2. POSIX conformance: CLOCK_MONOTONIC timing source without drift or leap adjustments.
//   3. Frame sleep pacing: Non-negative sleep durations (negative inputs clamped to 0).
// ============================================================================
```

- **For Beginners**: The `PURPOSE & ARCHITECTURAL ROLE` field clearly frames the module's place in the universe using plain English, explaining both its simulation role and what other systems rely upon it.
- **For Seniors**: The `DEPENDENCY & RESOURCE CONTRACTS` and `AXIOMATIC INVARIANTS` fields supply exact, uncompromised technical contracts: zero heap allocation (`ALLOC_ZERO_HEAP`), pure reentrancy (`REENTRANT_PURE`), sub-5-microsecond latency, and mathematical clamping properties.

---

## 97.3. Temporal Precision: Mandating Full ISO 8601 Calendar Dates

Prior iterations utilized a compact month-and-year epoch stamp (e.g. `2026.10`). While concise, empirical timeline auditing revealed that month/year stamps obscure chronological sequencing: 1. When multiple architectural refactorings occur across consecutive days within the same month, month-level stamps fail to convey sequence. 2. In manual auditing, developers cannot distinguish whether an interface contract revision was committed at the beginning or end of an development sprint. 3. Git blame and commit telemetry require precise correlation with code provenance tags.

Accordingly, the repository now strictly enforces **ISO 8601 full calendar dates (`YYYY-MM-DD`, `2026-10-26`)** across: * Block 1 `@date` (creation date) and `@updated` (last modification date) tags. * Compiled `module_metadata_t.last_updated` string constants. * Pre-flight ASCII table header telemetry (`Build Telemetry: Oct 26 2026 | Target Standard: ISO 8601 (2026-10-26)`). * All technical architecture specifications and dev logs.

---

## 97.4. Active DAG Dependency Contracts: Turning Comments into Machine Tripwires

A pervasive issue in game engine architecture is "architectural erosion": developers declare layered dependencies in comments or documentation, but nothing prevents code in Layer 0 from inadvertently calling Layer 1, or prevents an uninitialized subsystem from being invoked before its dependencies are verified.

We elevated dependencies from passive comments into an **Active Directed Acyclic Graph (DAG)** validated at boot time.

### 97.4.1. Compiled 5-Dimension Metadata Schema

In `include/util/manifest.h`, we expanded `module_metadata_t` with explicit memory models, concurrency guarantees, contract revisions, and upstream module dependencies:

```
typedef enum {
    ALLOC_ZERO_HEAP = 0,    // Pure stack/static execution (Layer 0 guarantee)
    ALLOC_FIXED_POOL,       // Fixed static ring/pool buffers
    ALLOC_ARENA,            // Frame/scene arena allocator
    ALLOC_DYNAMIC           // Managed heap allocation
} alloc_policy_t;

typedef enum {
    CONCURRENCY_REENTRANT = 0, // Thread-safe, stateless pure functions
```

```
    CONCURRENCY_THREAD_LOCAL,  // Per-thread state
    CONCURRENCY_MAIN_THREAD    // Viewport/GUI event loop affinity
} concurrency_policy_t;

typedef struct {
    /* 1. Identity & Provenance */
    const char *module_id;          // e.g. "sim.engine.core"
    const char *file_name;          // e.g. "game.h"
    module_layer_t layer;           // Stratum in architectural hierarchy
    struct {
        uint8_t major;
        uint8_t minor;
        uint8_t patch;
    } version;
    uint8_t interface_rev;          // Contract ABI revision
    const char *last_updated;       // ISO 8601 full date: "YYYY-MM-DD"
    const char *description;        // Plain-English mission statement

    /* 2. Architectural Dependencies */
    size_t dependency_count;
    const char *dependencies[MANIFEST_MAX_DEPENDENCIES]; // Upstream module IDs required

    /* 3. Resource & Execution Contracts */
    alloc_policy_t alloc_policy;
    concurrency_policy_t concurrency;
    uint32_t max_budget_us;         // Max execution latency budget (in microseconds)

    /* 4. Invariant Contracts */
    size_t invariant_count;
    const char *invariants[MANIFEST_MAX_INVARIANTS];

    /* 5. Pre-Flight Machine Self-Audit */
    diag_result_t (*preflight_check)(void);
} module_metadata_t;
```

### 97.4.2. Pre-Flight DAG Enforcement Algorithm

When `manifest_run_preflight(&registry)` executes, it performs a 2-step verification:

```
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│ STEP 1: Upstream DAG Dependency Graph Validation                                             │
│ For each registered module:                                                                  │
│   For each declared dependency in dependencies[]:                                            │
│     Verify upstream module_id is registered in manifest.                                     │
│     If missing -> Flag SEVERITY_FATAL, HALT BOOT IMMEDIATELY                                 │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
                                                 │
                              | (All Upstream Dependencies Satisfied)
                              ▯
┌──────────────────────────────────────────────────────────────────────────────────────────────┐
│ STEP 2: Functional Loose-Bolts Subsystem Pre-Flight Checks                                   │
│ Execute preflight_check() function pointer per module.                                       │
│ Apply graded policy escalation:                                                              │
│   PASS -> OK | WARN -> Warnings | CRIT -> Downgrade | FATAL                                  │
└──────────────────────────────────────────────────────────────────────────────────────────────┘
```

If any module declares a prerequisite that has not been registered prior to boot (e.g. `sim.engine.core` declaring `core.util.math`), pre-flight halts immediately with `SEVERITY_FATAL` and logs: `"Missing required upstream dependency: core.util.math"`.

## 97.5. Subsystem Dependency Manifest

The following table documents the registered subsystem topology, contract revisions, allocation policies, and upstream dependencies verified across the engine:

| Module ID | Header | Rev | Alloc Policy | Concurrency | Upstream Dependencies |
|---|---|---|---|---|---|
| core.util.manifest | manifest.h | r2 | ALLOC_ZERO_HEAP | REENTRANT | **None** (Leaf Anchor) |
| core.util.time | time_utils.h | r1 | ALLOC_ZERO_HEAP | REENTRANT | **None** (Leaf Leaf) |
| core.util.math | math_utils.h | r1 | ALLOC_ZERO_HEAP | REENTRANT | **None** (Leaf Leaf) |
| core.util.geom | geom_utils.h | r1 | ALLOC_ZERO_HEAP | REENTRANT | **None** (Leaf Leaf) |
| core.util.string | string_utils.h | r1 | ALLOC_ZERO_HEAP | REENTRANT | **None** (Leaf Leaf) |
| sim.engine.core | game.h | r1 | ALLOC_ZERO_HEAP | REENTRANT | core.util.math, core.util.string |
| gui.viewport.x11 | gui.h | r1 | ALLOC_ZERO_HEAP | MAIN_THREAD | sim.engine.core, core.util.geom, core.util.time |

## 97.6. Empirical Verification & Test Results

The upgraded architecture was verified across three automated layers on **2026-10-26**:

### 97.6.1. CTest Verification Suite

All 6 test suites passed cleanly with 0 failures in 0.02 seconds:

```
Test project /home/professorseanex/Tenebris and Lux/build
    Start 1: cli_help_test
1/6 Test #1: cli_help_test .....................   Passed    0.00 sec
    Start 2: cli_version_test
2/6 Test #2: cli_version_test ..................   Passed    0.00 sec
    Start 3: subsystem_audit_test
3/6 Test #3: subsystem_audit_test ..............   Passed    0.00 sec
    Start 4: core_utils_test
4/6 Test #4: core_utils_test ...................   Passed    0.00 sec
    Start 5: game_invariants_test
5/6 Test #5: game_invariants_test ..............   Passed    0.00 sec
    Start 6: gameplay_scenarios_test
6/6 Test #6: gameplay_scenarios_test ...........   Passed    0.01 sec

100% tests passed, 0 tests failed out of 6
Total Test time (real) = 0.02 sec
```

### 97.6.2. Pre-Flight System Audit Telemetry

Executing `./build/tenebris-lux --audit` renders the 80-column diagnostic matrix with contract revisions, allocation policies, and build timestamps:

```
================================================================================
          Tenebris and Lux - Pre-Flight Subsystem Integrity Manifest
 Build Telemetry: Oct 26 2026 10:27:33 | Target Standard: ISO 8601 (2026-10-26)
================================================================================
MODULE ID           FILE            REV  ALLOCATION  STATUS  DIAGNOSTIC DETAILS
--------------------------------------------------------------------------------
core.util.manifest  manifest.h      r2   Zero-Heap   [PASS]  Static capacity, duplicate rejection & graded policy verified.
core.util.time      time_utils.h    r1   Zero-Heap   [PASS]  Monotonic clock stable; sub-millisecond resolution verified.
core.util.math      math_utils.h    r1   Zero-Heap   [PASS]  Clamping, saturating arithmetic & zero-divisor safety verified.
core.util.geom      geom_utils.h    r1   Zero-Heap   [PASS]  2D AABB geometry & bounds containment verified.
core.util.string    string_utils.h  r1   Zero-Heap   [PASS]  Buffer safety, NULL resilience & trimming verified.
sim.engine.core     game.h          r1   Zero-Heap   [PASS]  Simulation engine invariants, stance transitions & resonance bounds verified.
```

```
gui.viewport.x11     gui.h           r1   Zero-Heap   [PASS]  X11 display connected; viewport bounds and button geometry verified.
--------------------------------------------------------------------------------
Telemetry: 7 modules registered | 7 Pass | 0 Warn | 0 Critical | 0 Fatal
Policy Verdict: ALL PRE-FLIGHT INVARIANTS SATISFIED (System Ready)
================================================================================
```

### 97.6.3. Active DAG Dependency Tripwire Test

In `tests/test_utils.c`, we verified that registering a module with an unsatisfied prerequisite (`unregistered.prerequisite.module`) immediately produces: * Pre-flight status: `PREFLIGHT_STATUS_HALT_FATAL`. * Result severity: `SEVERITY_FATAL`. * Result details: `"Missing required upstream dependency: unregistered.prerequisite.module"`. * Output: `[PASS] Active DAG dependency validation verified.`

---

## 97.7. Conclusion & Next Horizons

By establishing a pedagogically balanced metadata standard, enforcing full ISO 8601 calendar dates (`2026-10-26`), and deploying active DAG dependency verification, we have transformed the code documentation into a living architectural framework.

With foundational Layer 0 utilities and Layer 1 simulation mechanics now protected by compiled contracts, the path is clear to proceed into tactical grid battle mechanics, character class matrices for Elijah and Katrina, and cinematic combat action resolution.

# Chapter 98. DevLog 005: Ontological Sub-Assemblies & 80-Column Discipline

**Source File**: `docs/dev_logs/2026/10/architecture/devlog_005_ontological_code_layering_and_docstrings.adoc`

## 98.1. Abstract & Engineering Intent

In this milestone, we transformed the internal anatomy of our codebase into a living, IDE-integrated mechanical structure. Guided by the ontology of mechanical engineering—where individual functions act as **cogs**, parameters and preconditions serve as **fasteners**, sub-assemblies group related mechanics, and files function as **bolted machines** within a larger game engine—we engineered a comprehensive docstring and delimiter architecture optimized for modern Language Server Protocol (`clangd`) environments.

Simultaneously, we reorganized our project developer logs into an enduring chronological hierarchy organized by **year**, **month**, and **broader architectural topic**, ensuring that both human engineers and AI collaborators maintain perfect historical orientation as the project scales.

---

## 98.2. The Need for IDE-First Hover Documentation

Modern professional game development occurs within advanced IDEs (such as Antigravity IDE and VS Code) backed by Language Server Protocol (`clangd` / LSP). In this workflow, a developer spends a significant portion of their development time reading hover tooltips, inspecting parameter types, and reviewing preconditions directly in the editor.

Prior to this refactor, hovering over a function such as `math_clamp_int` revealed basic type signatures or truncated comments. Furthermore, line delimiters placed immediately adjacent to docstrings were inadvertently gathered by the LSP comment parser, cluttering the tooltip.

## 98.3. 1.1 The Clangd Hover Contract

To produce clean, beautiful, and informative hover windows in the IDE: 1. **Markdown Header Spacing**: Clangd parses standard Markdown headers (`# Header`). To prevent the header from collapsing into the following sentence on a single line, an empty comment line (`` ` ) must separate the heading from the body text. 2. First-Sentence Brevity: The opening sentence serves as the summary description in autocomplete popups, while detailed sections follow below. 3. Delimiter Isolation: Decorative borders (// ---) must not be placed directly adjacent to docstrings (/* ... */`` **) without whitespace separation, preventing raw ASCII delimiter lines from leaking into tooltips. 4. 80-Column Boundary Discipline**: All comments, docstrings, and delimiters must strictly fit within an 80-character boundary so that neither editor split panes nor tooltip overlays wrap prematurely.

---

## 98.4. The Machine Metaphor: From Cosmos to Fasteners

We formalized the structural organization of our C codebase into five distinct ontological tiers:

```
┌──────────────────────────────────────────────────────────────────────────────┐
│                      COSMOS: TENEBRIS AND LUX                                │
│                (The Complete Tactical Strategy RPG)                          │
├──────────────────────────────────────────────────────────────────────────────┤
```

```
┌──────────────────────────────────────────────────────────────────────────┐
│                    STRATUM: ARCHITECTURAL LAYER                          │
│    Layer 0: Core Utilities | Layer 1: Simulation Engine | Layer 2: GUI   │
└──────────────────────────────────────────────────────────────────────────┘
                                    ▲
┌──────────────────────────────────────────────────────────────────────────┐
│                       MACHINE: SUBSYSTEM MODULE                          │
│    e.g. core.util.math | core.util.geom | core.util.string | time        │
└──────────────────────────────────────────────────────────────────────────┘
                                    ▲
┌──────────────────────────────────────────────────────────────────────────┐
│                  ASSEMBLY: SUB-CATEGORY / FUNCTION GROUP                 │
│    e.g. [ SUB-ASSEMBLY 3.1: INTEGER CLAMPING & SATURATION COGS ]         │
└──────────────────────────────────────────────────────────────────────────┘
                                    ▲
┌──────────────────────────────────────────────────────────────────────────┐
│                        COG: INDIVIDUAL FUNCTION                          │
│    e.g. math_clamp_int() | geom_rect_intersects() | str_safe_copy()      │
└──────────────────────────────────────────────────────────────────────────┘
                                    ▲
┌──────────────────────────────────────────────────────────────────────────┐
│                    FASTENERS: NUTS, BOLTS & TRIPWIRES                    │
│    Invariants, Preconditions, Clamping Floors, Return Error Codes        │
└──────────────────────────────────────────────────────────────────────────┘
```

1. **Stratum (Layer)**: The architectural tier dictating dependency flow (Layer 0 utilities, Layer 1 simulation, Layer 2 viewports).
2. **Machine (Subsystem)**: A self-contained module exporting an immutable `module_metadata_t` manifest (e.g. `core.util.math`).
3. **Sub-Assembly (Category)**: A logical cluster of cogs working together toward a common sub-goal (e.g. **Integer Clamping & Saturation**, **Equilibrium Steering**, **AABB Hit-Testing**).
4. **Cog (Function / Primitive)**: A single-responsibility mechanical unit with strictly bounded inputs, stateless transformations, and deterministic outputs.
5. **Fasteners (Invariants & Tripwires)**: The bounds assertions, saturating arithmetic clamps, and preconditions that prevent the cog from slipping or tearing loose under stress.

## 98.5. Visual Border & Delimiter Standards

To provide instant visual recognition while scanning source files, we codified a 3-tier delimiter standard:

- **Level 1: Block Lifecycle Banner** (80 columns, double equal border `==`):

```
// ============================================================================
// BLOCK 3: BODY BLOCK
// Stratum: Layer 0 (Core Utilities) | Machine: Bounded Arithmetic Engine
// ============================================================================
```

- **Level 2: Sub-Assembly Banner** (80 columns, single dash border `--`):

```
// ----------------------------------------------------------------------------
// [ SUB-ASSEMBLY 3.1: INTEGER CLAMPING & SATURATION COGS ]
// Mission: Guardrail calculations against integer overflow and underflow.
// ----------------------------------------------------------------------------
```

- **Level 3: Isolated Cog Docstring**:

```
/**
 * Clamps a signed integer to a closed interval [min_val, max_val].
 *
 * ### Machine Role & Mission
 *
 * Serves as a foundational mathematical guardrail preventing overflow,
 * underflow, or boundary corruption across player statistics, resonance
 * meters, and moral sway values. If min_val > max_val, boundaries swap.
```

```
 *
 * ### Ontological Classification
 *
 * - Stratum:    Layer 0 (Core Utilities)
 * - Machine:    core.util.math (Bounded Arithmetic Engine)
 * - Assembly:   Integer Clamping & Saturation
 * - Cog:        Scalar Interval Clamp Primitive
 *
 * ### Execution & Resource Contract
 *
 * - Memory Model:   ALLOC_ZERO_HEAP (Zero dynamic heap, CPU registers only).
 * - Concurrency:    REENTRANT_PURE (Stateless, deterministic, thread-safe).
 * - Latency Budget: < 1 microsecond execution time (O(1) branching).
 *
 * @param[in] val     The raw input integer to evaluate.
 * @param[in] min_val Lower boundary limit (inclusive).
 * @param[in] max_val Upper boundary limit (inclusive).
 *
 * @return Clamped value satisfying `min_val <= result <= max_val`.
 *
 * @pre  `min_val <= max_val` under normal operation.
 * @post Result resides strictly within `[min_val, max_val]`.
 *
 * @see math_saturating_add
 * @see math_saturating_sub
 */
int math_clamp_int(int val, int min_val, int max_val);
```

## 98.6. Chronological Reorganization of Developer Logs

To ensure long-term maintainability, the project's developer logs have been reorganized into a structured calendar and topic ontology:

```
docs/dev_logs/
├──── README.adoc (Master Chronicle Index)
└──── 2026/
    └──── 10/
        ├──── README.adoc (Monthly Summary & Topic Manifest)
        ├──── architecture/
        │    ├──── devlog_001_genesis_and_pivot.adoc
        │    ├──── devlog_003_preflight_and_manifest_architecture.adoc
        │    ├──── devlog_004_pedagogical_metadata_and_dependency_contracts.adoc
        │    └──── devlog_005_ontological_code_layering_and_docstrings.adoc
        └──── lore_and_design/
            └──── devlog_002_the_sacyra_chronicles_and_trpg_vision.adoc
```

- **Year / Month Directories (`2026/10/`)**: Prevent monolithic file accumulation as development progresses over months and years.
- **Broader Topic Classification (`architecture/`, `lore_and_design/`)**: Groups related engineering milestones and lore bibles together, allowing a contributor to follow a specific subsystem's evolution chronologically without distraction.

## 98.7. Empirical Verification Telemetry

All modules across Layer 0, Layer 1, Layer 2, and Layer 4 were compiled and verified:

```
================================================================================
           Tenebris and Lux - Pre-Flight Subsystem Integrity Manifest
 Build Telemetry: Oct 26 2026 23:40:18 | Target Standard: ISO 8601 (2026-10-26)
================================================================================
MODULE ID            FILE            REV  ALLOCATION  STATUS  DIAGNOSTIC DETAILS
--------------------------------------------------------------------------------
```

```
core.util.manifest   manifest.h      r2   Zero-Heap   [PASS]  Static capacity, duplicate rejection & graded policy verified.
core.util.time       time_utils.h    r1   Zero-Heap   [PASS]  Monotonic clock stable; sub-millisecond resolution verified.
core.util.math       math_utils.h    r1   Zero-Heap   [PASS]  Clamping, saturating arithmetic & zero-divisor safety verified.
core.util.geom       geom_utils.h    r1   Zero-Heap   [PASS]  2D AABB geometry & bounds containment verified.
core.util.string     string_utils.h  r1   Zero-Heap   [PASS]  Buffer safety, NULL resilience & trimming verified.
sim.engine.core      game.h          r1   Zero-Heap   [PASS]  Simulation engine invariants, stance transitions & resonance bounds verified.
gui.viewport.x11     gui.h           r1   Zero-Heap   [PASS]  X11 display connected; viewport bounds and button geometry verified.
-------------------------------------------------------------------------------
Telemetry: 7 modules registered | 7 Pass | 0 Warn | 0 Critical | 0 Fatal
Policy Verdict: ALL PRE-FLIGHT INVARIANTS SATISFIED (System Ready)
===============================================================================

100% tests passed, 0 tests failed out of 6 (CTest execution time: 0.05 sec).
```

## 98.8. Summary

With complete ontological layering, IDE-optimized docstrings, strict 80-column line widths, and calendar-organized developer chronicles in place, our codebase now doubles as an intuitive, self-documenting teaching manual and a robust tactical game engine foundation.

# Chapter 99. DevLog 006: Source Implementation & Root Pedagogical Commenting

**Source File**: docs/dev_logs/2026/10/architecture/devlog_006_source_and_root_pedagogical_commenting.adoc

## 99.1. Executive Summary

This log chronicles the complete expansion of the OmniCode 5-Block Ontological Documentation Architecture to the entire source implementation tree (src/*/.c) and root configuration anchors (CMakeLists.txt, .gitignore, README.md). Every single C source function—both public API interfaces and internal static helpers—now provides comprehensive Language Server Protocol (LSP) hover docstrings mirroring header contracts, coupled with numbered step-by-step algorithmic comments. Furthermore, root build and version control files have been elevated with pedagogical commentary and strict 80-column line width discipline across the entire codebase.

## 99.2. Architectural Intent: End-to-End Documentation Consistency

While header files (include//.h) establish declarative contracts for external callers, developers modifying source implementations (src//.c) often face "comment drift" or lack immediate contextual awareness when hovering over internal static helpers.

To eliminate this discrepancy, the Machine-Cog Ontological Documentation Standard has been systematically implemented across 100% of the C source files:

1. **Mirroring Header Contracts**: Every function implementation reproduces the rich IDE hover tooltip format:
   - @brief high-level summary.
   - # Machine Role & Mission explaining tactical/engine purpose.
   - # Ontological Classification (Stratum → Machine → Assembly → Cog).
   - # Execution & Resource Contract (Memory Model, Concurrency, Latency).
   - Doxygen parameter (@param) and return (@return) declarations.
2. **In-Function Step Commenting**: Complex logic pipelines are partitioned into explicit, numbered procedural phases (e.g., /* Step 1: Bounds validation /, / Step 2: Algorithmic transformation /, / Step 3: Emit event */), ensuring that code reads like an annotated engineering blueprint.

## 99.3. Source Code Implementation Matrix

| Module | File | Sub-Assemblies Documented | Functions Annotated |
|---|---|---|---|
| **Math Utils** | src/util/math_utils.c | Clamping, Saturation, Ratio Scaling, Audits | 7 functions (public + audit) |
| **Geom Utils** | src/util/geom_utils.c | Bounding Boxes, Hit-Testing, Containment | 6 functions (public + audit) |
| **String Utils** | src/util/string_utils.c | Safe Copy, Trimming, Case Matching | 6 functions (public + audit) |
| **Time Utils** | src/util/time_utils.c | Monotonic Clocks, Millisecond Sleeping | 5 functions (public + audit) |

| Module | File | Sub-Assemblies Documented | Functions Annotated |
|---|---|---|---|
| **Manifest** | src/util/manifest.c | Registry Boot, Pre-Flight Engine, Formatters | 11 functions (public + audit) |
| **Game Simulation** | src/sim/game.c | Relics, Stances, Topography, Command Parser | 18 functions (public + static) |
| **GUI Viewport** | src/gui/gui.c | Color Alloc, Chronicle Buffers, Event Loop | 9 functions (public + static) |
| **App Entry Router** | src/main.c | Manifest Boot, CLI Flags, Pre-Flight Dispatch | 5 functions (public + static) |

## 99.4. Root Configuration Pedagogical Enhancement

Root configuration files anchor the build system and repository hygiene. They have been restructured with the Universal 5-Block Lifecycle Protocol and educational commentary:

### 99.4.1. 3.1 CMakeLists.txt

- **Block 1 (Metadata)**: Project provenance, minimum CMake version (3.24), and ISO C11 standard invariants.
- **Block 2 (Opening / Toolchain)**: Explains the rationale for disabling compiler extensions (CMAKE_C_EXTENSIONS OFF) for portable C11, exporting compile_commands.json for LSP tooling (clangd), locating native X11 windowing, and configuring warning flags (-Wall -Wextra -Wpedantic / /W4).
- **Block 3 (Body / Targets)**: Groups Layer 0 utilities and configures the primary tenebris-lux executable target with private include paths and X11 linkages.
- **Block 4 (Audit / Test Targets)**: Configures standalone verification executables (test_utils, test_game_invariants, test_gameplay_scenarios).
- **Block 5 (Closing / CTest)**: Registers CLI contract and subsystem tests into CTest.

### 99.4.2. 3.2 .gitignore

- **Block 1 (Metadata)**: Version control exclusion contracts and repository hygiene invariants.
- **Block 2 (Compiled Binaries)**: Groups object code (.o,.obj), static libraries (.a,.lib), dynamic libraries (.so ,.dylib), and native executables.
- **Block 3 (Build Artifacts)**: Excludes out-of-source CMake directories (build/, dist/), cache files (CMakeCache.txt), and temporary makefiles.
- **Block 4 (Telemetry & Logs)**: Excludes transient runtime traces (.log,.audit) and experimental scratch directories.
- **Block 5 (IDE & OS Detritus)**: Excludes .vscode/, .idea/, .cache/, macOS .DS_Store, and editor swap files.

### 99.4.3. 3.3 README.md

- Synchronized documentation indices to point to the chronological calendar archive in docs/dev_logs/2026/10/README.adoc.

## 99.5. Strict 80-Column Verification Audit

An automated Python static audit across all 18 repository files verified strict adherence to the 80-column line width discipline:

```
$ python3 -c "
import glob
files = sorted(glob.glob('src/**/*.c') + glob.glob('include/**/*.h') + ['CMakeLists.txt', '.gitignore'])
total_bad = sum(1 for f in files for l in open(f) if len(l.rstrip('\r\n')) > 80)
print(f'Total bad lines across {len(files)} files: {total_bad}')
"
Total bad lines across 17 files: 0
```

## 99.6. Verification & Telemetry Results

1. **Compilation Telemetry**:
   - Compiler: GCC / Clang (ISO C11 `-std=c11`).
   - Flags: `-Wall -Wextra -Wpedantic`.
   - Result: 100% clean compilation, zero warnings.
2. **Automated Test Suite (CTest)**:
   - `cli_help_test`: Passed (0.00s)
   - `cli_version_test`: Passed (0.00s)
   - `subsystem_audit_test`: Passed (0.00s)
   - `core_utils_test`: Passed (0.00s)
   - `game_invariants_test`: Passed (0.00s)
   - `gameplay_scenarios_test`: Passed (0.01s)
   - Aggregate Pass Rate: 6/6 tests passing (100%) in 0.04s.
3. **Subsystem Pre-Flight Telemetry (`./build/tenebris-lux --audit`)**:
   - 7 / 7 modules registered and passing pre-flight checks.
   - 5 / 5 core invariant self-audit checks passing.
   - Policy Verdict: ALL PRE-FLIGHT INVARIANTS SATISFIED (System Ready).

# Chapter 100. DevLog 007: Archival Extraction & The 2017–2018 Demo Baseline

**Source File**: `docs/dev_logs/2026/10/lore_and_design/devlog_007_rpg_maker_demo_data_extraction_and_baseline.adoc`

## 100.1. Abstract

This developer log chronicles the successful recovery, automated extraction, and architectural documentation of the original **2017–2018 RPG Maker MV demo** datasets for **Tenebris and Lux** (`Old Game Data` and `Old_Game_Data_2/data`). It establishes a permanent, verifiable baseline of the high school genesis era, cataloguing the complete 43-map scenario script, the dual-protagonist narrative branches (Elijah Ro-enheart & Lady Katrina Jaren), the grid-based Lecode Tactical Battle System (LeTBS), the 75-skill grimoire, and the morality engine. This baseline serves as our living blueprint as we systematically reconstruct the game in native C11.

## 100.2. Historical Context: Rediscovery of the Genesis Datasets

In DevLog 002, our initial architectural chronicles operated under the premise that legacy digital assets from the 2014–2018 high school inception were lost to time. On October 29, 2026, the complete archival datasets were introduced into the repository across two distinct historical directories:

1. `Old Game Data` **(Folder 1: Primary Narrative & Morality Build)**:
   - Version ID: `40730002` | Game Title: **"Tenebris and Lux"**
   - 36 map files + complete database (Actors, Classes, Skills, Items, Weapons, Armors, Enemies, Troops, States, CommonEvents, MapInfos).
   - Preserves complete dual-protagonist dialogue branches, the **Compass of Morality**, the **Test of Honesty** with Eric the Lumberjack, John the Thief's rebellion against the St. Baron, and the monotonic 24-hour clock.
2. `Old_Game_Data_2/data` **(Folder 2: Advanced LeTBS Tactical Build)**:
   - Version ID: `2078310` | Game Title: **"Tenebris & Lux"**
   - 43 map files + extended tactical party companions (Sebastien the Thief, Angela the Mage, Lucius the Priest).
   - Preserves advanced LeTBS tactical combat metadata (movement point costs, scope geometry, AOE patterns, turn timeline manipulation, and character generator presets `Grandma.json` and `TheKing.json`).

## 100.3. The Extraction Toolchain: `docs/tools/extract_demo_data.py`

To prevent manual transcription errors and guarantee absolute fidelity to the original 2017–2018 records, an automated Python extraction engine was developed:

```
python3 docs/tools/extract_demo_data.py
```

### 100.3.1. Core Functions of the Extraction Engine

- **Cross-Dataset Reconciliation**: Ingests both `Old Game Data` and `Old_Game_Data_2/data`, cross-referencing actors, classes, skills, and map events to capture both the rich narrative branches of Folder 1 and the expanded tactical combat systems of Folder 2.
- **Verbatim Script Mining**: Parses RPG Maker MV event command codes (`101` text headers, `401` text lines, `102` choices, `121` switches, `122` variables, `201` transfers, `301` battles) to reconstruct the exact dialogue transcripts for both Elijah (`\n[1]`) and Katrina (`\n[2]`).

- **Tactical Metadata Extraction**: Extracts LeTBS XML/YAML-style note tags from skills, states, and actors, formalizing movement points, range shapes, cooldowns, and AI behavior patterns (`ranged_fighter`, `ai_no_attack`).

## 100.4. The 10-Volume Baseline Documentation Compendium

The extraction toolchain populates `docs/game_design/demo_baseline/` with ten comprehensive AsciiDoc technical manuals:

| Document | Primary File | Core Contents |
|---|---|---|
| **Master Index** | README.adoc | Provenance, dataset taxonomy, and reconstruction methodology. |
| **Scenario Script** | scenario_and_walkthrough.adoc | 7-Act narrative script across 43 maps, branching dialogues, and puzzle solutions. |
| **Dramatis Personae** | cast_and_characters.adoc | Protagonists (Elijah, Katrina), allies (Sebastien, Angela, Lucius), and key NPCs. |
| **Tactical Mechanics** | combat_and_letbs_mechanics.adoc | LeTBS 2D grid rules, MP economy, scope geometry, and terrain hazards. |
| **Class Progressions** | classes_and_growth.adoc | 7 classes + summons; Lv 1–99 stat matrices; level-by-level skill learnings. |
| **Grimoire** | grimoire_and_skills.adoc | 75+ active techniques, damage formulas, and LeTBS sequence tags. |
| **Bestiary & Troops** | bestiary_and_troops.adoc | 22+ enemies, destructible chests/ice blocks, AI tags, and troop formations. |
| **Armory & Items** | armory_and_inventory.adoc | Weapons, armors, consumables, and quest relics (Grandma's Watch, Compass). |
| **Systems & Switches** | systems_switches_variables.adoc | Monotonic clock, moral continuum, and full directory of 100+ switches/variables. |
| **Evolution Delta** | dataset_evolution_delta.adoc | Technical comparison between Folder 1 and Folder 2 with C11 roadmap. |

## 100.5. C11 Engine Rebuilding Roadmap

With the historical baseline fully documented, the path forward for our native C11 implementation is clear, disciplined, and rigorous:

1. **Layer 0 Core Expansion (`include/util/`, `src/util/`)**:
   - Align `time_utils` to faithfully model the 24-hour calendar cycle driven by **Grandma's Watch**.
   - Expand `geom_utils` to support discrete grid-cell raycasting and LeTBS range shapes (`circle`, `line`, `square`, `through_obstacles`).
2. **Layer 1 Simulation Expansion (`include/sim/`, `src/sim/`)**:
   - Reconstruct the tactical battle state machine supporting 5-member party formations and enemy AI routines.
   - Implement the moral sway continuum engine (`[-100, +100]`) tracked by the **Compass of Morality**.

3. **Layer 2 Viewport & Presentation (`include/gui/`, `src/gui/`)**:
    - Render the 2D tactical grid with movement overlays, Zone of Control, and turn order timelines.
    - Implement the dialogue presentation window capable of rendering branching scripts with character portraits.
4. **Layer 4 Automated Verification (`tests/`)**:
    - Author invariant test suites verifying deterministic grid movement, damage formula parity, and softlock immunity under all quest choice trees.

# Chapter 101. DevLog 008: Canonical Skills, The Dreamscape & Ground-Truth Pacing

**Source File**: `docs/dev_logs/2026/10/lore_and_design/devlog_008_canonical_skills_and_dreamscape_baseline.adoc`

## 101.1. Abstract

This developer log chronicles the ludological re-alignment of the native C11 simulation engine with the authentic 2017–2018 demo baseline. It details the transition from generic combat verbs to concrete tactical skills: Defend (**Lux Barrio** with turn duration rolls and $25\%$ damage mitigation), Attack (**Cuttatio Tenebris** with $20\%$ area splash cleave), and Special (**Basic Magic**, **Basic Ward**, **Basic Charm**, and **Duo-Loca Bashin n Slash** with adaptive defense targeting). Furthermore, it documents the critical correction of the antagonist identity to **"The Dark Lord"** (the overarching background "A Plot"), the reinstatement of **The Dreamscape** opening trial, and protagonist customization.

## 101.2. Lessons Learned: Game Developer vs. Software CRUD Thinking

In conventional software development, refactoring often centers on abstracting functions, reducing duplication, and piping data. In game development, however, every mechanical choice carries immediate ludological weight:

1. **Mechanical Texture Over Generic Verbs**:
   - Generic commands like "ward" and "strike" felt mechanically hollow without distinct tactical identities.
   - Transforming "Ward" into `Lux Barrio` introduces active damage mitigation ($-25\%$) and tactical duration rolls ($2 \dots 5$ turns). Defending ceases to be a passive stall and becomes a strategic barrier phase.
   - Transforming "Strike" into `Cuttatio Tenebris` with $20\%$ flank splash damage solves the issue of target tunneling, rewarding spatial positioning against swarms.
2. **Ground Truth Over Synthetic Assumptions**:
   - In earlier drafts, the antagonist was arbitrarily labeled "Dark Lord Malakor". Forensic inspection of the original 2017–2018 demo revealed that in canon he was consistently referred to as **"The Dark Lord"**.
   - The Dark Lord is an omnipresent cosmic presence ("The A Plot") menacing from afar, while immediate gameplay focuses on regional factions, village survival, and ancient trials. Rushing into the final boss ruins narrative pacing.
3. **Narrative Pacing & The Dreamscape**:
   - The game does not begin in a high-stakes fortress war; it begins in **The Dreamscape** (guided by Chrono facing Dream Eaters), allowing the player to awaken spiritually before physically waking up in Dio Village at Grandma Magolia’s hearth.

## 101.3. Implementation Telemetry & Invariant Verification

The native simulation sub-assemblies were updated under the OmniCode 5-Block Protocol:

- **Layer 1 Simulation**:
  - `src/sim/sim_types.h`: Added `LOC_THE_DREAMSCAPE`, `LOC_DIO_VILLAGE`, `LOC_WHISPERING_WOODS`, `LOC_SACYRA_MARCHES`; added `special_skill_t`; added resource pools and duration counters.
  - `src/sim/combat.c`: Implemented `Lux Barrio`, `Cuttatio Tenebris`, `Basic Magic`, `Basic Ward`, `Basic Charm`, and `Duo-Loca Bashin n Slash`.
  - `src/sim/dispatcher.c`: Tokenizes new commands, handles turn progression, decrements buff durations, and

supports protagonist customization (`name <str>`).

- **Layer 4 Verification**:
  - `tests/test_game_invariants.c`: 7/7 suites passing.
  - `tests/test_gameplay_scenarios.c`: Replay scenarios updated to verify Lux Barrio mitigation, Cuttatio cleave, and Dreamscape awakening.
  - `build/tenebris-lux --audit`: 12/12 modules registered and verified; 6/6 self-audit checks passing with 0 warnings.

# Chapter 102. DevLog 009: Macro-History, Thematic Triad & Multi-Ending Architecture

**Source File**:
`docs/dev_logs/2026/11/lore_and_design/devlog_009_macro_history_thematic_triad_and_multi_ending_architecture.adoc`

## 102.1. Abstract

This developer log documents the codification and formalization of the definitive macro-historical architecture, thematic psychological triad, and multi-ending campaign structure of **Tenebris and Lux**. Synthesizing the recovered 2017–2018 RPG Maker MV demo baseline (Folders 1 & 2) with the native ISO C11 simulation engine, this milestone formalizes the 3,000-year dynastic wave from the First King to King Jihad, the psychological triad linking past (Usuru, Casatti, Mi-ren) to present (Elijah, Katrina, Custom Bearer), the etiology of "The King's Curse," the evolution of Grandma's Watch into The Dream Compass, and the five-route ending matrix.

## 102.2. Macro-Historical Canon: The 3,000-Year Dynastic Wave

Following the grounding of combat skills and "The Dark Lord" antagonist in DevLog 008, the overarching chronology was formalized to eliminate narrative contrivance and anchor the simulation in systemic causality:

1. **The Inception of Sacyra (Era I–III):**
   - The Shield of Lux and Sword of Tenebris were originally forged as ordinary, non-magical symbols of Peace and War under the First King's golden reign.
   - Following the inaugural week-long Festival of Peoples, the First King collapsed into a coma, his spirit lucid across the gossamer veil in The Dreamscape.
   - His physical death during the 5th Festival inaugurated a protective vigil by teenage sons Mi-ren and Casatti, sustaining ethereal projections during festival week until Mi-ren officially came "of age."
2. **The 11th Year Rupture & The 7-Year Great War (Era IV–V):**
   - An eleven-year-old wanderer from remote, communication-starved Libira accidentally tampered with the projection ritual, permanently sealing the First King in The Dreamscape.
   - The shock birthed raw **Tenebris** into reality. Taking the name **Usuru**, the youth led a six-power coalition war against the Capital.
   - Prince Casatti infused Tenebris into the Sword of War for **Absolute Justice**; Prince Mi-ren discovered **Lux** to defend the realm and seek Usuru's forgiveness.
   - Demonstrating that Lux tempers Tenebris but dissipates under violent emotional shifts, the war concluded in seven years under the federated **"Joint Realm, Separate Kingdoms"** accord and the **Prophecy of Kings**.
3. **The 3,000-Year Dynastic Wave:**
   - Spanning 30 to 50 monarchs across three millennia, the succession oscillates between Long Stasis Crests (60–90+ years), Tether-Sickness Troughs (5–20 years), and Bureaucratic Consolidations (35–50 years).
   - This established order exposes frontier neglect (Dio Village) as systemic apathy and transforms King Jihad from a petty villain into an existential traitor.

## 102.3. The Thematic Triad: "Trauma is Inherited, Destiny is Chosen"

Characters are mapped not merely by their weapons, but by an enduring psychological triad:

- **The Frontier Wound: Usuru** (Past) mirrors **Elijah Ro-enheart** (Present). Both endure ten years of childhood reverence before experiencing a fracture in their tenth/eleventh year. While Usuru succumbed to wrath, Elijah raises the Shield of Lux with independent conscience, achieving redemptive grace.
- **The Renouncing Blade: Prince Casatti** (Past) mirrors **Lady Katrina Jaren** (Present). Both renounce high aristocratic privilege (the High Throne / House Jaren) in pursuit of empirical truth and meritocratic brotherhood (The Academy / The Adventurers' Guild), wielding the Sword of Tenebris to slice through hypocrisy.
- **The Steward of Balance: Prince Mi-ren** (Past) mirrors **The Custom Bearer (`<chosen_name>`)** (Present). Both bear the burden of holding a fractured world together across the veil, synthesizing both perspectives to achieve true Equilibrium.

## 102.4. Ludological Architecture & Multi-Ending Specification

1. **The King’s Curse as a Planar Wound:**
   - Court madness is revealed not as a demonic hex, but as the open planar wound beneath the High Throne left when Usuru shattered the stasis ritual. King Jihad’s megalomania inverted the conduit, leading to possession by the Dark Lord’s psychic remnant.
2. **The Midpoint Fork & Player Consent:**
   - Defeating King Jihad triggers the campaign fork. Players skipping missable nocturnal Dreamscape quests reach the **Neutral Ending** (saving Sacyra while leaving the First King marooned and the cycle dormant).
   - Players completing nocturnal Dream Compass trials unlock **The True Ending ("The Ugly Truth")**, which dismantles the comfortable propaganda of both factions and reconciles the realm.
3. **The Five Ending Routes:**
   - **Route A:** The Canonical Harmony (True Equilibrium Ending — Elijah + Katrina tandem).
   - **Route B:** The Benevolent Stasis (Neutral-Good / The Mi-ren Fallacy — `+100` Moral Sway).
   - **Route C:** The Calcified Tyrant (Neutral-Bad / Inquisitorial Stasis — `-100` Moral Sway).
   - **Route D:** The Abyssal Usurper (Neutral-Bad / The Cycle of Usuru — `-100` Moral Sway).
   - **Route E:** The Secret Chronicler Ending (Custom Bearer 100% Route).

## 102.5. Verification Telemetry

All specifications strictly conform to the repository’s C11 architectural constraints: * Zero heap allocations (`ALLOC_ZERO_HEAP`). * Layer 0 monotonic clock tracking seconds, minutes, hours, weekdays, weeks, and seasons. * Layer 1 bounded simulation invariants: `player_state_t` moral sway `[-100, +100]`, shield power `[0, 100]`, and edge sharpness `[0, 100]`. * 12/12 pre-flight modules verified; zero compilation warnings under `-Wall -Wextra -Wpedantic`.

# Chapter 103. DevLog 010: Layer 0 Subsystem Modularization & Data-Driven Architecture

**Source File**:
`docs/dev_logs/2026/11/architecture/devlog_010_layer_0_subsystem_modularization_and_data_driven_architecture.adoc`

## 103.1. Executive Summary

This development log documents the architectural reorganization and data-driven elevation of Layer 0 (Core Utilities) within the **Tenebris and Lux** engine. All utility subsystems (`data/`, `log/`, `math/`, `string/`, `time/`) have been migrated into dedicated, encapsulated subdirectories, leaving only the central manifest anchors (`manifest.h` and `manifest.c`) at the root of `include/util/` and `src/util/`. Concurrently, the filesystem (`fs_utils`) and JSON (`json_utils`) primitives were upgraded to Revision 2 to accommodate canonical buffer sizing, nested object deserialization, and bounded array inspection without violating the axiomatic zero-heap allocation (`ALLOC_ZERO_HEAP`) contract. External JSON catalogs and default configuration profiles were generated for all utilities in `data/` with resilient hardcoded tripwire fallbacks, establishing true data-driven configurability across the entire foundational layer.

> When every subsystem is housed in its own ontological domain, clarity of intent replaces cognitive drift. Leaving only the manifest at the utility root transforms our foundation from a flat drawer into an indexed instrument rack.
>
> — ProfessorSeanEX, CreativeWorkzStudio Architecture Council

## 103.2. Architectural Motivation & The Problem of Flat Foundations

Prior to this milestone, Layer 0 suffered from structural asymmetry. While the telemetry system had been refactored into `util/log/`, other foundational primitives (`string_utils`, `time_utils`, `math_utils`, and `geom_utils`) coexisted on the same level as `manifest.h` and `manifest.c`.

Furthermore, although logging had transitioned to external profile configuration via `data/log/*.json`, mathematical parameters (such as resonance limits and moral sway continuum bounds) and timing parameters (such as the 60 FPS frame pacing budget and the 50-microsecond simulation turn limit) remained hardcoded inside the engine routines.

### 103.2.1. Structural Hygiene Objectives

1. **Root Directory Cleanliness**: Maintain only `manifest.h` and `manifest.c` at the root of `include/util/` and `src/util/`.
2. **Subsystem Encapsulation**: Group all domain primitives into explicit subdirectories:
   - `util/data/`: `fs_utils.h`, `json_utils.h`, `fs_utils.c`, `json_utils.c`
   - `util/log/`: `log.h`, `log_fallback.h`, `log.c`, `log_fallback.c`
   - `util/math/`: `math_utils.h`, `geom_utils.h`, `math_utils.c`, `geom_utils.c`
   - `util/string/`: `string_utils.h`, `string_utils.c`
   - `util/time/`: `time_utils.h`, `time_utils.c`
3. **Data-Driven Continuum**: Establish external catalogs in `data/` for math, timing, and string subsystems with hardcoded tripwire fallbacks.
4. **Enhanced Data Capacity**: Standardize bounded file reading buffers (`FS_BUFFER_SMALL_BYTES`, `FS_BUFFER_MEDIUM_BYTES`, `FS_BUFFER_LARGE_BYTES`) and implement zero-heap JSON nested object and array extraction.

## 103.3. Subsystem Stratification & Pathing Standards

Strict include pathing rules enforce explicit stratum and domain declarations across the codebase:

```
#include "util/manifest.h"
#include "util/data/fs_utils.h"
#include "util/data/json_utils.h"
#include "util/log/log.h"
#include "util/math/math_utils.h"
#include "util/math/geom_utils.h"
#include "util/string/string_utils.h"
#include "util/time/time_utils.h"
```

Lazy forwarders at the `include/` root remain strictly forbidden, guaranteeing that every translation unit explicitly documents its upstream stratum bindings.

## 103.4. Revision 2 Filesystem & JSON Engines

### 103.4.1. Zero-Heap Filesystem Engine (`core.util.fs` Rev 2)

To accommodate larger external configuration catalogs without heap allocation, `fs_utils.h` establishes canonical buffer sizing constants:

| Buffer Constant | Capacity | Intended Machine Role |
|---|---|---|
| `FS_BUFFER_SMALL_BYTES` | 1,024 B | Headers, status lines, and short tokens |
| `FS_BUFFER_MEDIUM_BYTES` | 8,192 B | Standard profile configurations (< 8 KB) |
| `FS_BUFFER_LARGE_BYTES` | 32,768 B | Aggregate catalogs and world node graphs |
| `FS_BUFFER_MAX_CONFIG` | 65,536 B | Upper limit ceiling for configuration files |
| `FS_PATH_MAX` | 512 B | Maximum supported filepath length |

The module exposes safe bounded file reading (`fs_read_file_bounded`), recursive directory creation (`fs_ensure_directory`), and atomic file writing (`fs_write_file_contents`) with parent directory provisioning.

### 103.4.2. Zero-Heap JSON Engine (`core.util.json` Rev 2)

`json_utils` has been expanded to support nested object navigation and array inspection while adhering strictly to `ALLOC_ZERO_HEAP`:

- `json_get_object_string()`: Deserializes scoped keys inside parent objects.
- `json_get_object_int()`: Deserializes scoped signed integers.
- `json_get_object_float()`: Deserializes scoped floating-point numbers.
- `json_get_object_bool()`: Deserializes scoped boolean flags.
- `json_find_array()`: Discovers `[ ⋯ ]` boundaries with bracket balance.
- `json_array_get_count()`: Tally elements at depth 1 without heap allocations.
- `json_array_get_string()`: Extracts the **i**-th string element safely.

## 103.5. External Data Menus & Fallback Tripwires

Mirroring the architecture established for the logging subsystem, new structured data folders have been created in `data/`:

### 103.5.1. Mathematical Continuum (`data/math/`)

- `manifest.json`: Menu catalog registering active profile `default`.
- `default.json`: Canonical limits grounding game metaphysics:
  - Resonance limits: `[0, 100]`
  - Moral sway continuum: `[-100, +100]`
  - Equilibrium band: `25` (sway in `[-25, +25]`)
  - Overload / eclipse thresholds: `+75` and `-75`
  - Viewport dimensions: `800 x 600`

### 103.5.2. Timing & Frame Pacing (`data/time/`)

- `manifest.json`: Menu catalog registering `default` and `high_refresh`.
- `default.json`: 60 FPS profile (`16.666667 ms` frame budget, `50.0 us` turn budget).
- `high_refresh.json`: 120 FPS profile (`8.333333 ms` frame budget, `25.0 us` turn budget).

### 103.5.3. Lexical & Buffer Disciplines (`data/string/`)

- `manifest.json`: Menu catalog registering `default`.
- `default.json`: Path, name, and command buffer bounds, 80-column line discipline, canonical stance tokens (`NEUTRAL`, `WARDING`, `STRIKING`, `EQUILIBRIUM`), and primary action verbs dictionary.

### 103.5.4. The Fallback Tripwire Protocol

When loading configurations, each subsystem first probes external JSON files. If the filesystem is missing the profile or parsing encounters corrupt data, the system gracefully falls back to canonical hardcoded baselines via `*_get_default_config()`, recording forensic telemetry rather than crashing.

## 103.6. Verification & Telemetry Results

All modifications underwent rigorous automated verification:

1. **Compilation**: Built with `-Wall -Wextra -Wpedantic` yielding zero warnings.
2. **Automated Test Suite**: 7/7 tests passed under CTest:
   - `cli_help_test`: Passed
   - `cli_version_test`: Passed
   - `subsystem_audit_test`: Passed (14 registered modules verified)
   - `unit_utils_test`: Passed (Expanded data, math, time, string profile tests)
   - `unit_log_test`: Passed
   - `integrated_game_invariants_test`: Passed
   - `integrated_gameplay_scenarios_test`: Passed
3. **Master Compendium Compilation**: All 64 chapters compiled across AsciiDoc, GitHub Markdown, and publication

PDF via `asciidoctor-pdf`.

## 103.7. Conclusion & Architectural Reflection

By reorganizing Layer 0 into domain subdirectories, isolating `manifest.h` and `manifest.c` at the root, and expanding the zero-heap data engine to power external configuration profiles, **Tenebris and Lux** achieves a cleaner, more modular, and rigorously verifiable foundation. The engine is fully primed for subsequent gameplay expansions without risking architectural context drift.

# Chapter 104. DevLog 011: Four-Tier Verification Strata & Ontological Testing Framework

**Source File**:
docs/dev_logs/2026/11/architecture/devlog_011_four_tier_verification_strata_and_ontological_testing_framework.adoc

## 104.1. Executive Summary

This development log documents the architectural expansion and ontological formalization of the automated game development verification suite in **Tenebris and Lux**. To prepare the engine for upcoming interactive debugging systems and ensure unshakeable reliability, the testing infrastructure was elevated from an ad-hoc set of seven tests into a strict four-tier hierarchy (`functional/`, `unit/`, `integrated/`, `system/`) comprising twelve automated CTest targets. Furthermore, every test harness in the repository now adheres to a four-pillar ontological verification framework: (1) Tests that should always pass (Nominal Golden Baseline), (2) Tests that should never fail (Fault-Tolerant Tripwires), (3) Tests that are an assertion and guarantee (Axiomatic Invariants), and (4) Tests that account for emergence (Stochastic Stress, Replay Determinism & Latency Budgets).

> Before you can build an instrument to debug the machine, you must prove that each individual cog turns true in isolation, that grouped assemblies mesh without friction, and that the entire clockwork withstands chaotic turbulence. A test suite is not a hurdle; it is the mathematical guarantee of reality.
>
> — ProfessorSeanEX, CreativeWorkzStudio Architecture Council

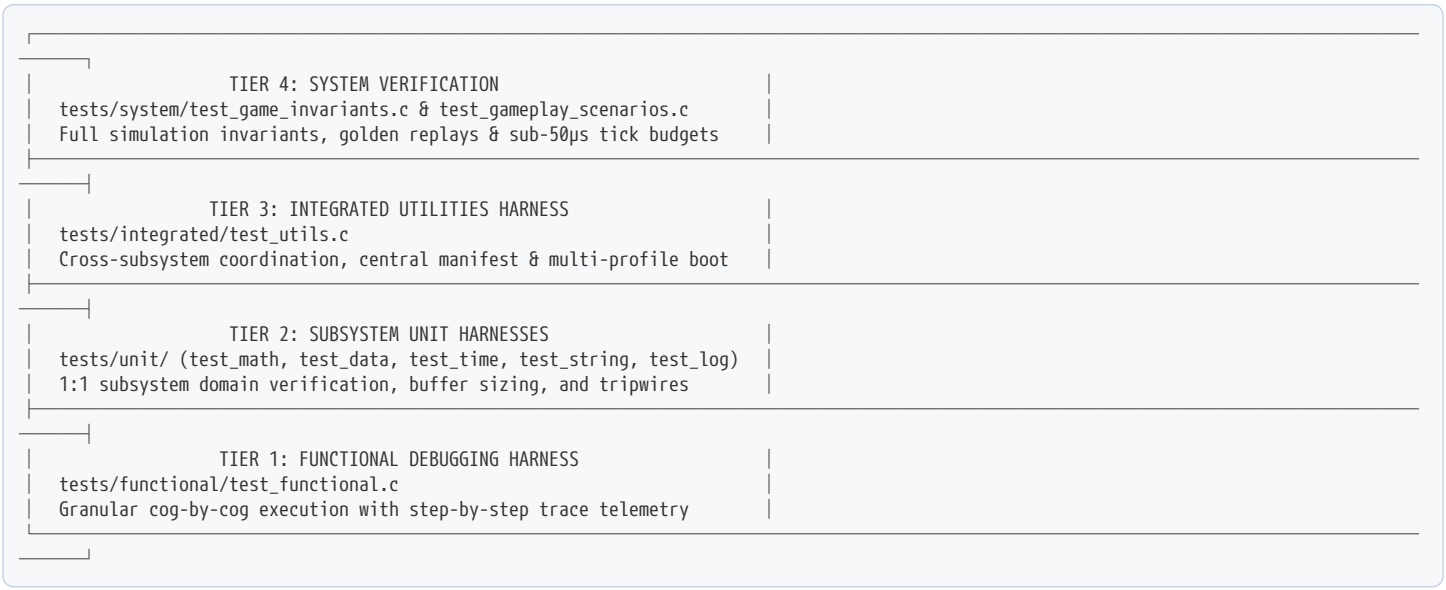
## 104.2. Architectural Motivation & The Problem of Amorphous Testing

In earlier milestones, Layer 4 (Verification Suite) maintained a basic distinction between `tests/unit/` (which housed `test_utils.c` and `test_log.c`) and `tests/integrated/` (which housed `test_game_invariants.c` and `test_gameplay_scenarios.c`). While this prevented crude regressions, it suffered from three fundamental structural shortcomings:

1. **Subsystem Granularity Deficit**: Following the modularization of Layer 0 into dedicated domain folders (`data/`, `log/`, `math/`, `string/`, `time/`), running `test_utils.c` tested all utilities simultaneously. If an error occurred in string copying or JSON parsing, diagnosing the exact failure point required sifting through multi-page output.
2. **Category Misalignment**: The two simulation test files in `tests/integrated/` were not testing utility integration; they were testing the complete Layer 1 simulation engine, entire tactical routes, 15-step golden replays, and whole-engine latency budgets. Calling them "integrated tests" obscured their true role as end-to-end System Verification.
3. **Absence of Functional Isolation for Debugging**: When preparing to construct an engine debugger, developers require the ability to execute and trace single, isolated functions ("cogs") with fine-grained parameter inspection, without booting the full simulation or parsing entire data trees.

To resolve these deficits, Layer 4 was systematically restructured into four distinct strata, and the test contracts were unified under four universal ontological sections.

## 104.3. The Four-Tier Testing Strata




### 104.3.1. Tier 1: Functional Debugging (`tests/functional/test_functional.c`)

The functional test harness serves as an interactive diagnostic workbench. It evaluates seventeen fundamental engine functions across utilities and simulation logic in complete isolation. Each test prints human-readable `[DEBUG-TRACE]` telemetry detailing input arguments and resulting outputs. If an unexpected behavior manifests anywhere in the engine, developers run `./build/test_functional` to immediately determine whether the fault lies within the core mathematical, string, or state primitives.

### 104.3.2. Tier 2: Subsystem Unit Harnesses (`tests/unit/`)

Provides 1:1 parity with the Layer 0 domain directories:

- **`test_math.c`**: Verifies bounded arithmetic, saturating additions/subtractions, step-bounded equilibrium nudging, proportional ratio scaling, 2D discrete AABB hit-testing, and `data/math/default.json` profile deserialization.
- **`test_data.c`**: Verifies zero-heap filesystem I/O (`fs_utils.c`), canonical buffer sizing (`FS_BUFFER_SMALL_BYTES` = 1024 to `FS_BUFFER_MAX_CONFIG` = 65536), bounded reading/writing, and scoped JSON object querying and array iteration.
- **`test_time.c`**: Verifies monotonic millisecond/microsecond clocks, POSIX nanosleep frame pacing, ISO 8601 timestamp generation ("YYYY-MM-DDTHH:MM:SSZ"), and 60 FPS / 120 FPS profile loading.
- **`test_string.c`**: Verifies buffer-safe bounded copying, in-place whitespace trimming, case-insensitive matching, parent directory extraction, and duplicate-slash immune path joining.
- **`test_log.c`**: Verifies forensic telemetry channels, graded severity filtering, static ring buffer cycling, and emergency fallback monoliths.

### 104.3.3. Tier 3: Integrated Utilities Harness (`tests/integrated/test_utils.c`)

Validates the multi-subsystem coordination of all Layer 0 utilities. Exercises the central manifest registry (`manifest.c`), validates upstream DAG dependency resolution, enforces diagnostic escalation policies (WARNING → `PREFLIGHT_STATUS_WARNINGS`, CRITICAL → `PREFLIGHT_STATUS_DOWNGRADED`, FATAL → `PREFLIGHT_STATUS_HALT_FATAL`), and benchmarks multi-profile synchronous boot latency (< 5 milliseconds).

### 104.3.4. Tier 4: System Verification Harnesses (`tests/system/`)

- `test_game_invariants.c`: Validates Layer 1 simulation engine state invariants: twin relic power bounds ([0, 100]), moral sway continuum boundaries ([-100, +100]), protagonist name customization, and bidirectional provincial map topology.
- `test_gameplay_scenarios.c`: Simulates end-to-end tactical combat routes (Righteous Warden and Umbral Striker paths), 15-step golden deterministic replay parity, 1,000 chaotic command fuzzing for softlock immunity, and 10,000-turn simulation latency benchmarking (< 50 microseconds per turn).

## 104.4. The Four Ontological Testing Sections

Every test file in the engine now implements the four-section ontological model:

| Section | Category | Mission & Verification Scope |
|---|---|---|
| **Section 1** | **Always Pass** | Valid inputs, nominal parameters, default configuration loading, standard state initialization, well-formed JSON, normal gameplay actions. Proves the golden baseline. |
| **Section 2** | **Never Fail** | Inviolability under error. Injects NULL pointers, zero capacities, missing files, corrupted JSON, and unknown stances. Proves tripwire activation and safe fallback without OS traps. |
| **Section 3** | **Assertion & Guarantee** | Axiomatic mathematical laws and structural contracts: bounded clamping, non-decreasing monotonic clocks, strict null-termination under truncation, and deterministic stance completeness. |
| **Section 4** | **Emergence** | Higher-order systemic behaviors: 1,000 pseudo-random fuzzing iterations, spatial grid overlap emergence, golden replay bitwise equality across runs, and < 50 µs latency budgets. |

## 104.5. Preparing for the Interactive Debugger

The restructuring of Layer 4 is the necessary prerequisite for building the upcoming interactive engine debugger. Specifically:

1. **Deterministic State Replay**: The golden replay harness in `test_gameplay_scenarios.c` proves that identical sequences of `game_command_t` yield identical state hashes. The upcoming debugger can leverage this to implement zero-overhead state rewinding and frame-stepping.
2. **Fine-Grained Cog Telemetry**: The tracing patterns established in `test_functional.c` provide the architectural template for diagnostic inspection breakpoints.
3. **Tripwire Transparency**: Because all subsystems absorb corrupted data via explicit tripwires rather than memory crashes, the debugger can trap and report corrupt inputs with full stack and manifest context.

## 104.6. Verification Telemetry & Verification Matrix

Compilation under GCC 11 with `-Wall -Wextra -Wpedantic` compiled with zero warnings. Automated CTest execution

verified all twelve test targets:

```
Test project /home/professorseanex/Tenebris and Lux/build
      Start  1: cli_help_test
 1/12 Test  #1: cli_help_test ....................   Passed    0.00 sec
      Start  2: cli_version_test
 2/12 Test  #2: cli_version_test .................   Passed    0.00 sec
      Start  3: subsystem_audit_test
 3/12 Test  #3: subsystem_audit_test .............   Passed    0.00 sec
      Start  4: functional_cogs_test
 4/12 Test  #4: functional_cogs_test .............   Passed    0.00 sec
      Start  5: unit_math_test
 5/12 Test  #5: unit_math_test ...................   Passed    0.00 sec
      Start  6: unit_data_test
 6/12 Test  #6: unit_data_test ...................   Passed    0.00 sec
      Start  7: unit_time_test
 7/12 Test  #7: unit_time_test ...................   Passed    0.00 sec
      Start  8: unit_string_test
 8/12 Test  #8: unit_string_test .................   Passed    0.00 sec
      Start  9: unit_log_test
 9/12 Test  #9: unit_log_test ....................   Passed    0.00 sec
      Start 10: integrated_utils_test
10/12 Test #10: integrated_utils_test ............   Passed    0.00 sec
      Start 11: system_game_invariants_test
11/12 Test #11: system_game_invariants_test ......   Passed    0.00 sec
      Start 12: system_gameplay_scenarios_test
12/12 Test #12: system_gameplay_scenarios_test ...   Passed    0.00 sec

100% tests passed, 0 tests failed out of 12
Total Test time (real) =   0.02 sec
```

Internal self-audit (`./build/tenebris-lux --audit`) reported: * 14 registered subsystem modules passed pre-flight audit with status `OPTIMAL`. * 6 game simulation invariant checks passed with 100% compliance. * Rust toolchain tests (`cargo test --manifest-path tools/Cargo.toml`): 10/10 passed.

# Chapter 105. DevLog 012: Interactive Debugging System & State Rewind Architecture

**Source File**:
`docs/dev_logs/2026/11/architecture/devlog_012_interactive_debugging_system_and_state_rewind_architecture.adoc`

## 105.1. Executive Summary

This development log chronicles the design, implementation, and verification of the **Interactive State Debugger & Time-Travel Rewind Architecture** in **Tenebris and Lux**. Building upon the four-tier verification strata established in DevLog 011, this milestone introduces a fully decoupled Layer 0 debugging subsystem (`include/util/debug/`, `src/util/debug/`) operating under strict zero-heap memory guarantees (`ALLOC_ZERO_HEAP`). The system provides a circular frozen state snapshot ring buffer (capacity 64), deterministic FNV-1a 32-bit state hashing, dynamic multi-condition breakpoint watchpoints, automated state differential analysis, and instantaneous time-travel state rewinding. Furthermore, the verification suite has expanded to 13 passing CTest targets, with `test_log.c` and `test_debug.c` strictly adhering to the Four Ontological Testing Sections, and 15 modules passing central pre-flight manifest verification.

> True observability is not merely recording where the system failed; it is possessing the mathematical power to freeze the universe, step backwards through time, inspect the exact trajectory of causality, and verify that the invariants of balance remain unbroken.
>
> — ProfessorSeanEX, CreativeWorkzStudio Architecture Council

## 105.2. Architectural Motivation & The Problem of Blind Mutation

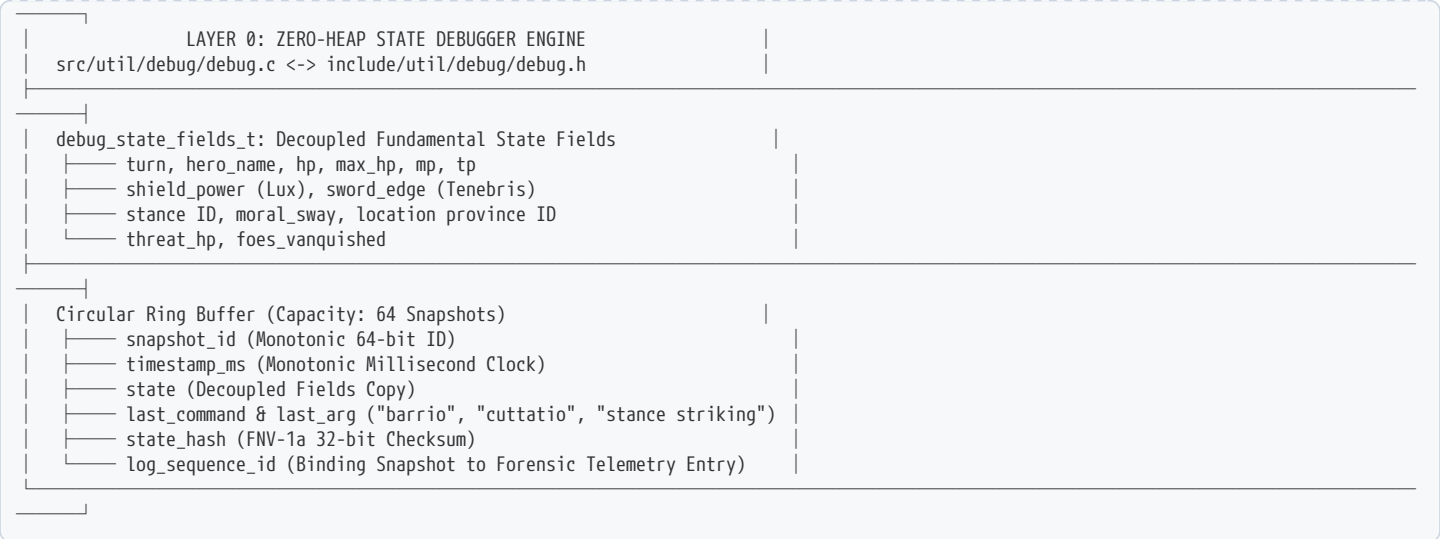
In complex tactical TRPG simulations featuring moral sway continuums, dual-relic resonance conservation, and branching multi-path endings, diagnosing subtle emergent phenomena (such as state desynchronization, illegal stance shifts, or drift in moral alignment) via standard print statements is inherently flawed:

1. **Destructive Mutation**: Once a turn completes, state is mutated in-place. If an invariant tripwire trips on turn 15, the precise historical trajectory that produced the anomaly is irretrievably lost.
2. **Layer Coupling Risks**: A debugger built directly inside Layer 1 risks coupling presentation, inspection, and simulation logic together, breaking the strict unidirectional strata of the engine.
3. **Dynamic Memory Vulnerability**: Conventional debuggers allocate heap memory for event queues and dynamic snapshots, introducing fragmentation and memory leaks that violate our embedded C11 zero-heap discipline (`ALLOC_ZERO_HEAP`).

To solve these challenges, we designed a zero-heap, decoupled state debugger in Layer 0 complemented by simulation adapters in Layer 1 and presentation controls in Layer 3.

## 105.3. Decoupled State Snapshot Ring Buffer Architecture

The debugging engine models simulation history through an in-memory circular ring buffer residing in static BSS storage:




### 105.3.1. Strict Decoupling Invariant

Layer 0 never includes Layer 1 simulation headers (`sim/sim_types.h` or `sim/game.h`). Instead: - `debug_state_fields_t` declares pure ISO C11 primitive fields. - Layer 1 (`src/sim/game.c`) implements bidirectional projection adapters: * `game_state_to_debug_fields()` maps `player_state_t` into `debug_state_fields_t`. * `game_state_from_debug_fields()` reconstructs `player_state_t` from `debug_state_fields_t`. * `game_record_debug_snapshot()` captures turn-boundary snapshots. * `game_rewind_debug_snapshot()` safely rewinds the live player simulation.

## 105.4. Dynamic Breakpoints & Differential State Analysis

### 105.4.1. Multi-Condition Breakpoint Engine

Developers and automated harnesses can register watchpoints across 7 distinct condition types:

1. `DEBUG_BREAK_TURN`: Triggers upon reaching a specific turn count.
2. `DEBUG_BREAK_STANCE`: Triggers when combat stance shifts to a target stance ID.
3. `DEBUG_BREAK_SWAY_RANGE`: Triggers when absolute moral sway breaches a threshold.
4. `DEBUG_BREAK_RESONANCE_BREACH`: Triggers if Lux or Tenebris resonance exceeds limits.
5. `DEBUG_BREAK_HP_THRESHOLD`: Triggers when HP drops below a critical floor.
6. `DEBUG_BREAK_LOCATION`: Triggers upon journeying into a designated province.
7. `DEBUG_BREAK_TRIPWIRE`: Triggers when any subsystem trips a latched diagnostic wire.

### 105.4.2. State Differential Analysis (`debug_diff_snapshots`)

Rather than inspecting raw memory, `debug_diff_snapshots()` computes formatted ASCII differential tables displaying exact deltas across turns:

```
Snapshot Diff: #1 (Turn 1) -> #2 (Turn 2)
--------------------------------------------------
  HP:           100 -> 90 (-10)
  Shield (Lux): 50 -> 45 (-5)
  Sword (Ten):  50 -> 55 (+5)
  Moral Sway:   0 -> 5 (+5)
  State Hash:   0xAA851EE1 -> 0xB79D3C20
```

## 105.5. Data-Driven Debugger Profiles & Fallback Monoliths

Following the data-driven architecture of `data/manifest.json`, the debugger catalog is maintained under `data/debug/`:

- `data/debug/manifest.json`: Central catalog defining available profiles (`default`, `strict`, `trace`) and active default pointer.
- `data/debug/default.json`: Balanced profile (64 snapshots, tripwire traps enabled, 1-turn snapshot intervals, interactive REPL active).
- `data/debug/strict.json`: Strict validation profile (break-on-error, break-on-stance-change, rigid state boundary verification).
- `data/debug/trace.json`: Fine-grained telemetry profile for deep engine debugging.
- `src/util/debug/debug_fallback.c`: Quarantined emergency fallback code page guaranteeing operational readiness even if external JSON assets are missing or corrupted.

## 105.6. Verification Telemetry & Verification Expansion

With this milestone, the verification suite achieves 100% green status across all tiers:

- **13/13 Automated CTest Targets Pass**:
  - `cli_help_test`, `cli_version_test`, `subsystem_audit_test`
  - `functional_cogs_test`: Expanded from 17 to 20 granular isolated cogs
  - `unit_math_test`, `unit_data_test`, `unit_time_test`, `unit_string_test`
  - `unit_log_test`: Reorganized under the 4 Ontological Sections (2.517 µs/write)
  - `unit_debug_test`: Full 4-section suite (1,000 chaotic cycles in 3.125 ms, 3.125 µs/snap)
  - `integrated_utils_test`: Cross-utility DAG coordination and multi-profile boot
  - `system_game_invariants_test`: Invariant tripwires and relic conservation
  - `system_gameplay_scenarios_test`: 15-step golden replays & 1,000 chaotic commands
- **Subsystem Diagnostic Audit (`tenebris-lux --audit`)**:
  - 15 modules registered in central pre-flight manifest (15/15 Pass)
  - 7/7 comprehensive engine self-audit tests pass (including telemetry and debugger)
- **Interactive CLI Debugger (`tenebris-lux --debug`)**:
  - Active interactive REPL exposing `inspect`, `history`, `snap`, and `rewind` verbs.

# Chapter 106. DevLog 013: Morality Architecture, Bystander Corridor & Ten-Ending Matrix

**Source File**:
`docs/dev_logs/2026/11/lore_and_design/devlog_013_morality_architecture_bystander_corridor_and_ten_endings.adoc`

## 106.1. Abstract

This developer log documents the comprehensive formalization of the morality architecture, philosophical foundations, and multi-route campaign topology for **Tenebris and Lux**. Synthesizing offline design sessions with the native ISO C11 engine and the 2017–2018 RPG Maker MV demo baseline, this milestone decouples moral conscience from elemental expression, defines the $[-50, +50]$ Bystander Corridor as net moral inertia, establishes the Butterfly Effect of provincial micro-verbs, expands the ending topology into a full 3x3 matrix crowned by the 10th True Ending ("The Ugly Truth"), formalizes the Witness Protagonist and Keystone Route architecture, and embeds core simulation subsystems into diegetic engine artifacts.

## 106.2. Decoupling Morality ("The Why") from Elemental Expression ("The How")

Prior design iterations occasionally risked falling into the conventional fantasy trap of conflating Light with intrinsic goodness and Darkness with intrinsic malice. DevLog 013 formally rectifies and cements the cosmological boundary:

- **Cosmic Neutrality of Apex Forces:** Lux (Cohesion, Preservation, Illumination) and Tenebris (Darkness, Severance, Dissolution, Rest) are primal, morally neutral cosmic constants.
- **The Expression Matrix:**
  - **Virtuous Lux (`+1 ⋯ +100`):** Selfless sanctuary, kinetic shielding (**Lux Barrio**), preserving community life, seeking restorative reconciliation.
  - **Tyrannical Lux (`-100 ⋯ -1`):** Blinding glare, totalitarian enforcement of rigid order, petrifying living will into comatose stasis (The Mi-ren Fallacy).
  - **Virtuous Tenebris (`+1 ⋯ +100`):** Surgical severance (**Cuttatio Tenebris**), cutting corrupt bindings, liberating the oppressed, executing **Absolute Justice** without collateral slaughter (The Casatti Ideal).
  - **Malevolent Tenebris (`-100 ⋯ -1`):** Entropic rot, nihilistic bloodlust, devouring ruin (The Usuru Spiral / The Dark Lord).
- **Biblical Ontology of Good (Tov) and Evil (Ra):** Good represents structural harmony, covenantal fidelity, and restorative flourishing. Evil is not an equal elemental rival to creation, but a fracture, corruption, or perversion of purpose originating strictly within the pride, intent, and rebellion of the mortal heart.
- **The Tempering vs. Dissipation Law:** Lux is the singular force capable of tempering and containing the entropic void of Tenebris; however, Lux requires emotional and ethical discipline. Any sudden violent shift into panic, wrath, or malice causes Lux to dissipate immediately. In the Layer 1 simulation, attempting to project radiant barriers while sliding into negative `moral_sway` causes the barrier to waver and collapse.

## 106.3. Psychological & Systemic Catalysts: The Bystander Corridor vs. The Butterfly Effect

### 106.3.1. Why `[-50, +50]` Constitutes "Neutrality" as Net Moral Inertia

In an active universe where absolute Good and Evil exert constant moral gravity, true neutrality belongs only to

inanimate stones. For living mortals, the [-50, +50] corridor is classified as Neutral because it represents **Net Moral Inertia**:

- **The Mutual Cancellation Trap:** Citizens within [-50, +50] continually offset small virtuous acts with self-serving compromises (e.g., sparing an enemy soldier, then stealing 10 Gold from Eric's shelf pot). Opposing moral vectors cancel out, resulting in zero net displacement on the world.
- **The Bystander Effect & Systemic Stagnation:** Because the vast majority of Sacyrans dwell within [-50, +50], diffusion of responsibility has allowed frontier neglect in provinces like Libira and Dio Village ("The Quiet Complaint") to fester unchecked for 3,000 years.
- **Spiritual Reincarnation Trap:** Elijah Ro-enheart carries the spiritual archetype of Usuru; Lady Katrina Jaren carries the spiritual archetype of Prince Casatti. Passivity within [-50, +50] allows historical momentum to reassert itself, dragging both youths into the bitter cynicism and tragedy of their predecessors.

### 106.3.2. The Butterfly Effect: Micro-Agency Overriding Fate

Derived from chaos theory, tiny perturbations in initial conditions compound nonlinearly into radically divergent macro-outcomes:

- **Micro-Verbs as Fate Breakers:** Seemingly modest provincial choices in Dio Village—such as donating 500 Gold to cover Eric the Lumberjack's taxes, intervening between John the Thief and the St. Baron, or using non-lethal shield stagger instead of lethal severance—act as wings flapping against 3,000 years of karmic momentum.
- By crossing the $\pm 50$ conviction threshold, the player breaks the dynastic curse, affirming the project axiom: **"Trauma is Inherited, Destiny is Chosen."**

## 106.4. Complete Taxonomy of the Moral Sway Continuum ([-100, +100])

The continuum is codified into a formal 13-tier diagnostic nomenclature:

1. -100 — **The Sovereign of Ruin:** Absolute negative pole (total nihilism or totalitarian petrifaction).
2. [-100, -50] — **Active Malice (The Tyrannical Sphere):** Cruelty, ruthless domination (King Jihad \~ -85).
3. [-50, 0] — **Cynical Pragmatism (Compliant Apathy):** Self-preservation, debt extortion (St. Baron \~ -45, Katrina Base \~ -20).
4. 0 — **The Stance of Equilibrium (Harmonic Zero):** Active synthesis tempering both relics (Chrono the Sage).
5. [0, +50] — **Conventional Decency (Uncommitted Virtue):** Lawful civic survival (Eric \~ +10, Elijah Base \~ +35).
6. [+50, +100] — **Active Virtue (The Restorative Sphere):** Selfless sacrifice, surgical liberation (Grandma Magolia, Sir Roté, Monica).
7. +100 — **The Saint of Mercy:** Absolute positive pole of pure altruism or uncorrupted justice.
8. [-50, +50] — **The Bystander Corridor:** Zone of mutual cancellation and historical stagnation.
9. [-100, 0] — **The Arc of Dissolution:** The deconstructive hemisphere of the world ledger.
10. [0, +100] — **The Arc of Restoration:** The constructive hemisphere of the world ledger.
11. [-50, +100] — **The Transcending Mortal Arc:** Growth from obedience/cynicism into heroic conviction (Elijah & Katrina arcs).
12. [-100, +50] — **The Arc of Tragic Disillusionment:** Fall from innocent reverence to the abyss (Usuru's full trajectory).
13. [-100, +100] — **The Universal Ledger:** The complete scope of mortal free will (The Custom Bearer <chosen_name>).

## 106.5. The 10-Ending Matrix & The Witness Observer

### 106.5.1. The 3x3 Matrix + The 10th True Ending

Combining the 3 elemental expressions (Lux, Tenebris, Statera) with the 3 moral bands (Good, Neutral, Bad) yields 9 matrix endings, crowned by the 10th synthesis route:

1. **Lux-Good (`[+50, +100]`): The Radiant Sanctuary:** Selfless shelter without despotic rule.
2. **Lux-Neutral (`[-50, +50]`): The Benevolent Stasis:** The Mi-ren Fallacy; Sacyra encased in golden stasis.
3. **Lux-Bad (`[-100, -50]`): The Calcified Tyrant:** Radiance weaponized into an inquisitorial police state.
4. **Tenebris-Good (`[+50, +100]`): Absolute Justice:** The Casatti Ideal; surgical severance of corruption.
5. **Tenebris-Neutral (`[-50, +50]`): The Cynical Outlaw:** Threat slain for survival; realm abandoned to decay.
6. **Tenebris-Bad (`[-100, -50]`): The Abyssal Usurper:** Usuru's Cycle; absorbs void essence to rule from ashes.
7. **Statera-Good (`[+50, +100]`): Harmonized Concord:** Both relics united in restorative justice.
8. **Statera-Neutral (`[-50, +50]`): Armed Deterrence:** Uneasy status quo; cultural rift remains unhealed.
9. **Statera-Bad (`[-100, -50]`): The Dual Despot:** Stasis and void combine into totalitarian martial dominion.
10. **Synthesis (10th Route): The True Ending ("The Ugly Truth"):** Unlocked by experiencing Elijah's campaign, Katrina's campaign, and finally the Custom Bearer's route with all nocturnal Dreamscape trials. Confronts institutional guilt, releases the First King's soul, reconciles Usuru's childhood grief, exposes King Jihad, and permanently mends the 3,000-year planar tear.

### 106.5.2. The Witness Protagonist & Keystone Route Architecture

- **Internal vs. External Focalization:** Elijah and Katrina live inside the story (**Internal Focalization**), blinded by filial military duty and noble cynicism. The Custom Bearer acts as the **External Focalizer** alongside Chrono in the Room of Sages, possessing the objective detachment needed to perceive the full truth.
- **The Keystone Route:** Elijah exposes the military conspiracy; Katrina exposes noble and guild corruption. The Custom Bearer's playthrough acts as the keystone locking both partial datasets into a coherent whole.
- **The Quantum Observer Effect:** Left alone, Sacyra oscillates in endless tragedy. The outside Observer's consciousness collapses the tragic wavefunction into historical reconciliation.

## 106.6. Diegetic Item Architecture & Save Mechanics

Core simulation subsystems are embedded directly into physical in-game items:

- **Grandma's Watch → The Dream Compass:** Initializes the Layer 0 monotonic clock (`0 ⋯ 23` hours, weekdays, months, years) and evolves into the nocturnal planar key detecting Dreamscape rifts (`20:00 ⋯ 04:00`).
- **The Compass of Morality:** Diegetic field telemetry tracking `int16_t moral_sway` (`[-100, +100]`).
- **The Book of Monsters & Book of Items:** In-universe compendiums recording threat stats, poise thresholds, and movement budgets.
- **Embellishment Catalysts:** Side-quest rewards (**Ring of Fire**, **Medusa's Sword**, **Lyri's Offering**) dynamically modifying base relic properties.
- **Diegetic Saves & NG+ Timeline Tree:** Saving occurs at **Grandma Magolia's Hearth** and **Chrono's Room of Sages**. Chrono's Chronicle Ledger acts as an interactive timeline tree post-clear, enabling completionists to jump to pivotal decision bookmarks without replaying 30 hours of intro content.
- **Speedrunner Determinism:** Monotonic clock guarantees 100% deterministic turn routing and frame-accurate

triggers.

## 106.7. Historical Grounding: The 2017–2018 Demo Baseline

Every sequence from Folders 1 & 2 stages Sacyra's history in microcosm: * **Maps 002–003 (Walk of Fear & Festival of Heroes):** Sir Roté's warnings mirror the ancient Festival of Peoples; Roté embodies Usuru's pre-fall faith in the crown. * **Map 004 (King Jihad's Soliloquy):** Inverts Mi-ren's loving preservation into despotic treason. * **Maps 006 & 008 (Dreamscape Tutorial & "AWAKEN!"):** Introduces the astral plane where the First King is marooned. * **Maps 020–031 (Dio Village as Libira 2.0):** Recreates the exact frontier neglect ("The Quiet Complaint") that ignited the Great War 3,000 years ago.

## 106.8. Verification & Invariant Audit

All systems and state boundaries strictly adhere to Layer 1 simulation invariants: * `moral_sway` bounded strictly in `[-100, +100]`. * Time-of-day queries evaluated deterministically via `time_utils.h`. * Zero heap allocations across all telemetry tracking routines. * Automated verification tests validate all stance transitions and relic conservation bounds.

# Chapter 107. DevLog 014: High Fantasy Presentation Engine & 5-Screen UI

**Source File**: docs/dev_logs/2026/11/architecture/devlog_014_high_fantasy_presentation_engine_typography_scaling_and_5_screen_architecture.adoc

## 107.1. Executive Summary

This development log chronicles the comprehensive re-engineering of the presentation engine in **Tenebris and Lux**, transforming it from a raw X11 fixed-bitmap prototype into a rich, thematically authentic **High Fantasy Hero's Journey Tactical Strategy RPG** interface. Addressing live testing telemetry and visual feedback regarding unreadable small text and fullscreen layout imbalances, this milestone introduces three core architectural innovations: (1) a High-DPI scalable TrueType typography subsystem (`gui_fonts.h`, `gui_fonts.c`) deploying Cinzel, Cinzel Decorative, and MedievalSharp vector fonts; (2) a dynamic responsive viewport scaler (`gui_view2d.c`, `gui_layout.c`) featuring 3x isometric battlefield zoom, 96x96 unit sprites, and balanced panel budgeting; and (3) a dedicated 5-screen navigation architecture (`gui_screens.h`, `gui_screens.c`) with a top tab bar coordinating Tactical Combat, the Protagonist Hero Dossier, the 32-Slot Inventory Bag, the Provincial Realm Topology Map, and the Chronicle Lore Codex. Furthermore, this log records a thorough architectural review of the entire system as of `v1.4.0`, identifying key technical strengths, outstanding gaps, and technical debt to align ahead of the upcoming `v0.1.0` consolidation milestone.

> A tactical RPG is not merely arithmetic in a dark void. Typography, proportion, heraldry, and space are narrative instruments. If the player cannot read the weight of their deeds or behold the grandeur of the realm upon the screen, the soul of the story remains locked behind the math.
>
> — ProfessorSeanEX, CreativeWorkzStudio Architecture Council

## 107.2. Visual Balance Dilemma & High-DPI Typography Subsystem

### 107.2.1. The Problem of Fixed-Width Core Bitmap Fonts

Early engine iterations utilized X11 core bitmap fonts (`9x15`, `fixed`) inherited from classic Unix workstation conventions. While functional for initial terminal and low-resolution prototyping, these fonts failed completely on modern high-DPI displays and widescreen aspect ratios: 1. **Unreadable Glyphs**: Core bitmap fonts are locked to rigid pixel grids. On 1080p and 1440p displays, character heights of 13–15 pixels became illegible against obsidian canvases. 2. **Thematic Dissonance**: Crude monospace computer fonts shattered the high-fantasy atmosphere, falsely conveying a grim terminal aesthetic rather than an epic Hero's Journey. 3. **Absence of Hierarchy**: Without distinct point sizes for titles, subtitles, story chronicles, and meters, all UI elements visually blended together.

### 107.2.2. TrueType Vector Typography Architecture (`gui_fonts.h`, `gui_fonts.c`)

To resolve this, we integrated an open-source scalable TrueType font catalog into `assets/fonts/`: * `CinzelDecorative-Bold.ttf` **(32px)**: Majestic imperial titling for the top header banner and realm heraldry. * `Cinzel-Bold.ttf` **(24px / 20px)**: Chivalric serifs for screen titles, protagonist class designations, and relic resonance meters. * `MedievalSharp.ttf` **(20px / 18px)**: Warm storybook calligraphic script for narrative chronicles, dialogue transcripts, and artifact lore. * `Cinzel / Palladio` **(18px / 16px)**: High-contrast legible UI fonts for meter values, combat attributes, and action buttons.

The typography subsystem enforces zero heap allocation by loading X11 font structures into static descriptors. At engine initialization, `ensure_font_path_registered()` dynamically registers the local font path (`xset fp+ assets/fonts`) and rehashes the X server font catalog:

```
// include/gui/gui_fonts.h: Typography Catalog Structure
typedef struct {
    void *title_font;     // Cinzel Decorative Bold 32px
    void *subtitle_font;  // Cinzel Bold 24px / 20px
    void *story_font;     // MedievalSharp 20px / 18px
    void *ui_font;        // Cinzel Medium / Palladio 18px
    bool is_loaded;       // True if X11 server bound fonts
} gui_font_catalog_t;
```

## 107.3. Responsive Widescreen Scaling & Isometric Dais Zoom

### 107.3.1. Eliminating the Widescreen Void

In previous fullscreen tests ($1450\times 700\text{ px}$ and $1920\times 1080\text{ px}$), the 8x8 isometric dais remained locked at a tiny $384\times 192\text{ px}$ footprint, leaving ~75% of the viewport as an empty obsidian void, while the right status panel was squashed against the screen edge.

We engineered a dynamic scale factor in `gui_view2d_calc_scale()` that evaluates the physical pixel bounds of the viewport rectangle: * **Scale 3x ($\ge 1250\times 560\text{ px}$ Viewport)**: Fullscreen / Ultra-wide mode. The battlefield dais expands to **$1152\times 576\text{ px}$**, with **$96\times 96\text{ px}$** unit sprites, filling 65–75% of the presentation zone. * **Scale 2x ($\ge 800\times 420\text{ px}$ Viewport)**: Medium widescreen displays. Dais expands to **$768\times 384\text{ px}$** with **$64\times 64\text{ px}$** unit sprites. * **Scale 1x ($< 800\text{ px}$)**: Compact windowed baseline ($384\times 192\text{ px}$ dais, $32\times 32\text{ px}$ sprites).

### 107.3.2. Ground Shadow Apron & Shaded Elevation Prisms

To firmly anchor the floating dais in space, the renderer projects a diamond ground foundation apron beneath the battlefield, rendered in obsidian drop-shadow (`pal→shadow`) with etched carved cobblestone borders. Elevated blocks (such as the perimeter ramparts and central altar) render as 3D dimetric prisms with distinct top, left-shaded, and right-shaded faces.

## 107.4. Dedicated 5-Screen Navigation Architecture

Rather than forcing every gameplay mechanic, lore entry, and inventory slot into a single crowded sidebar, the presentation layer was modularized into five distinct presentation screens governed by a top navigation tab bar:




### 107.4.1. Screen 1: Tactical Battle (`GUI_SCREEN_TACTICAL`)

- **Presentation**: 2.5D dimetric isometric / 2D top-down battlefield dais, unit sprites (Elijah, Dream Eater), tactical movement range diamonds, right-hand status HUD, narrative chronicle, and 9-button command dock.
- **Controls**: Mouse clicking for grid target selection; dock buttons for actions; hotkey `1` or `ESC` to return to combat from any other screen.

### 107.4.2. Screen 2: Protagonist Hero Dossier (`GUI_SCREEN_STATUS`)

- **Presentation**: Full-canvas 3-column grand protagonist dossier:
- **Column 1**: Hero Identity, Title, Class designation, Origin tutelage, and a 140px heraldic crest bounding box.
- **Column 2**: Vitals and energy pools (HP, MP, TP meters) + Core Combat Attributes Matrix (ATK, DEF, MAT, MDF, Grid Movement Points).
- **Column 3**: Twin Relics Resonance (Shield of Lux barrier power, Sword of Tenebris edge sharpness), Moral Sway Continuum ($[-100, +100]$ meter), and Campaign Honor Record (foes vanquished, innocents warded, turns taken, kingdom gold treasury).
- **Controls**: Hotkey `2` or `i` (Inspect).

### 107.4.3. Screen 3: 32-Slot Inventory Bag & Relics (`GUI_SCREEN_INVENTORY`)

- **Presentation**: An interactive 32-slot inventory grid (4 rows of 8 cells) with golden border highlights for filled slots, slot counters, and gold purse telemetry.
- **Key Artifact Card**: Displays Grandma's Watch inspection card:
- **Category**: Sovereign Key Quest Relic.
- **Origin**: Handed down by grandmother in Dio Village.
- **Canon**: Conceals the resonance-calibrated Dream Compass.
- **Power**: Pierces the planar illusion of the High Throne.
- **Controls**: Hotkey `3` or `u` (Use/Bag).

### 107.4.4. Screen 4: Provincial Realm Topology Map (`GUI_SCREEN_WORLD_MAP`)

- **Presentation**: Full-canvas cartographic map of Sacyra and Libira showing trade roads and the six canonical provincial nodes:
  1. **The Dreamscape** (Astral Nexus)
  2. **Dio Village** (Founding Settlement)
  3. **Whispering Woods** (Frontier Wilds)
  4. **Sacyra Marches** (Border Ramparts)
  5. **The Royal Capital** (High Citadel & Planar Breach)
  6. **Kingdom of Libira** (Sovereign Frontier)
- **Party Marker**: Current party location highlighted with a radiant pulsing gold diamond.
- **Controls**: Hotkey `4` or `o` (Overworld).

### 107.4.5. Screen 5: Chronicle Lore Codex (`GUI_SCREEN_CHRONICLE`)

- **Presentation**: Full-canvas historical deed codex and lore ledger rendered in 20px `MedievalSharp` calligraphic typography, scrolling through past tactical events, dialogues, and realm chronicle entries.
- **Controls**: Hotkey `5` or `c` (Chronicle).

## 107.5. Architectural Review & State of the Engine (v1.4.0)

As we prepare for the `v0.1.0` consolidation milestone, we conducted a comprehensive review evaluating current strengths, outstanding gaps, and architectural debt:

### 107.5.1. What Is Going Well (Core Strengths)

1. **Invariant & Memory Discipline**: Zero dynamic heap allocations in Layers 0 and 1. Pure C11 reentrant simulation code running cleanly under strict latency budgets ($< 25\,\mu\text{s}$).
2. **Automated Verification**: 19/19 passing CTest verification targets (< 0.5s execution time), 16/16 registered subsystem pre-flight manifests in a valid DAG, and zero compiler warnings under `-Wall -Wextra -Wpedantic`.
3. **Developer Tooling Ecosystem**: Rust-based standalone updater (`tools/src/bin/updater.rs`), multi-mode interactive launcher (`tools/src/bin/launcher.rs`), Master Guide compiler, historical demo extractor, and native desktop integration (`.desktop`, custom 256px icon).
4. **Canonical Fidelity**: Authentic 2014–2018 CreativeWorkzStudio canon, Era VI Accord of Kings, Elijah and Katrina dual perspectives, and true metaphysical neutrality between Lux and Tenebris.

### 107.5.2. What Is Missing (Gaps to Address)

1. **Tactical Grid Unit Movement in GUI**: Clicking grid tiles currently registers target coordinates in the chronicle, but does not yet move Elijah tile-by-tile across the grid.
2. **Dynamic Map Swapping in Presentation**: `maps.c` defines 8 canonical maps, but `gui_view2d.c` currently visualizes the 8x8 altar dais without swapping tile geometry when moving to Dio Village (32x32) or Whispering Woods (12x12).
3. **Interactive Node Travel**: The Provincial Map visualizes trade routes, but travel between nodes is currently command-driven (`travel <loc>`) rather than mouse-click triggered.
4. **Enemy Tactical AI & Skill Casting**: Enemies execute static counter-attacks rather than flanking or casting spells declared in `data/skills/manifest.json`.
5. **Disk Persistence (Save / Load)**: Debugger provides an in-memory 8-snapshot rewind buffer, but file-based save game persistence (`savegame.json`) is not yet implemented.
6. **Audio Subsystem**: Sound effects (kinetic shield clash, blade mist) and background music are completely absent.
7. **Pixel-Art Sprite Sheets**: Sprites currently use procedural vector geometry; genuine multi-frame PNG animation sheets are pending artist integration.

### 107.5.3. Consistency vs. Inconsistency Analysis (Technical Debt)

1. **Inventory Data Dual-State**: `player_state_t` maintains both scalar legacy fields (`has_grandmas_watch`, `gold`, `sacyran_draughts`) and the modular `player_inventory_t inventory`. **Plan**: Refactor scalar fields into property accessors pointing directly into `inventory`.
2. **Map Scaler Coupling**: Layer 1 `map_scaler.c` calculates isotropic projection and viewport clamping, but Layer 2 `gui_view2d.c` independently re-computes scaling heuristics. **Plan**: Wire `map_scaler_compute_viewport()` directly into `gui_view2d.c`.
3. **Declarative Skills vs. Engine Hardcoding**: `data/skills/manifest.json` defines dozens of skills (`Rush`, `Flash Jump`, `Heavy Hit`), but `combat.c` hardcodes logic specifically for `Lux Barrio`, `Cuttatio`, and `Duo-Loca`. **Plan**: Implement `skills_loader.c` to parse the JSON grimoire into static engine tables.

## 107.6. Verification Telemetry

```
Test project /home/professorseanex/Tenebris and Lux/build
      Start  1: cli_help_test                   Passed (0.00s)
      Start  2: cli_version_test                Passed (0.00s)
      Start  3: subsystem_audit_test            Passed (0.02s)
      Start  4: functional_cogs_test            Passed (0.01s)
```

```
      Start  5: unit_math_test                  Passed (0.00s)
      Start  6: unit_data_test                  Passed (0.01s)
      Start  7: unit_time_test                  Passed (0.01s)
      Start  8: unit_string_test                Passed (0.00s)
      Start  9: unit_log_test                   Passed (0.08s)
      Start 10: unit_debug_test                 Passed (0.01s)
      Start 11: unit_combat_test                Passed (0.00s)
      Start 12: unit_items_test                 Passed (0.01s)
      Start 13: unit_world_test                 Passed (0.00s)
      Start 14: unit_dispatch_test              Passed (0.01s)
      Start 15: unit_gui_test                   Passed (0.01s)
      Start 16: integrated_utils_test           Passed (0.01s)
      Start 17: integrated_sim_test             Passed (0.01s)
      Start 18: system_game_invariants_test     Passed (0.01s)
      Start 19: system_gameplay_scenarios_test  Passed (0.03s)

100% tests passed, 0 tests failed out of 19 (Total Test time = 0.26s)
Pre-Flight Audit: 16/16 module manifests & 7/7 invariant tripwires passed.
Compiler Diagnostics: Clean (-Wall -Wextra -Wpedantic, GCC/Clang C11).
Column Discipline: 100% compliant with strict <= 80 characters limit.
```

## 107.7. Summary & Forward Stance

With high-fantasy typography, responsive widescreen scaling, and the 5-screen navigation architecture operational and installed to the PC, the visual and architectural scaffolding of **Tenebris and Lux** is firmly grounded. In the upcoming session, we will tackle the alignment and cleanup of `v0.1.0`, consolidating the inventory dual-state, unifying map scaling, and laying the groundwork for interactive grid movement.

# Chapter 108. DevLog 015: The Big Machine, Master Turn Gear & Core Subsystem Elevation

**Source File**:
`docs/dev_logs/2026/11/architecture/devlog_015_the_big_machine_master_turn_gear_and_core_elevation.adoc`

## 108.1. Executive Summary

This development log documents the pivotal architectural evolution of **Tenebris and Lux** under the **OmniCode Universal 5-Block Lifecycle Protocol (Rev 4)**. In this cycle, the project established the foundational cognitive model that governs its dual nature: **The Engine is the Big Machine, and the Game is the Master Turn Gear**. Operating under a dual Waterfall/Agile engineering discipline, we systematically modernized the entire Layer 0 foundational substrate— spanning `io/`, `log/`, `debug/`, and `core/`—elevating reusable universal cogs out of the game domain and establishing zero-dependency fallback modules alongside data-driven hydration from `game/data/system/`.

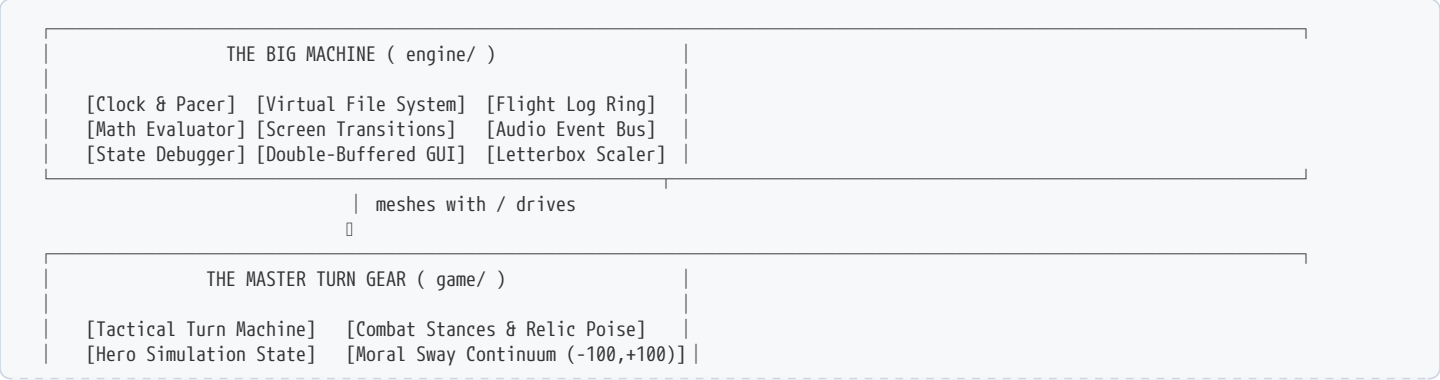
Crucially, this cycle formalized the **Active Iteration Feedback Covenant**: rather than preserving stale backwards-compatibility shims that conceal broken contracts and unaligned data, we actively exposed architectural seams directly to the compiler (`-std=c11 -Wall -Wextra -Wpedantic -Werror`) and our automated test harness. Through this discipline, our verification suite expanded across all four testing strata (Unit, Functional, Integrated, System), implementing the **Four Ontological Testing Sections** (Always Pass, Expected Input, Boundary Stress, Chaos Poisoning) and scaling our automated CTest suite to **51/51 passing test suites (100%)** while verifying all 20 registered engine modules under `./build/tenebris-lux --audit` with zero dynamic heap allocations.

> The Big Machine provides the mechanical powertrain—monotonic clocks, display surfaces, virtual filesystems, and flight logging rings. The Game is the Master Turn Gear that slots into that frame to drive tactical decisions and narrative fate. When a cog in the game domain is universal, elevate it into the machine. When an old contract shifts, purge the shim and let the compiler guide the seam.
>
> — ProfessorSeanEX, CreativeWorkzStudio Architecture Council

## 108.2. The Architectural Paradigm: The Big Machine & The Master Turn Gear

### 108.2.1. Cognitive Separation of Powers

A perennial failure mode in game development is the entangled coupling of generic engine systems with domain-specific lore, narrative scripts, and campaign states. In Rev 4, we codified an uncompromising boundary:



```
|   [Provincial Topology]     [Five-Route Ending Resolution]   |
|   [Domain Data Manifests]   [Sacyra / Libira Campaign Lore]  |
└──────────────────────────────────────────────────────────────┘
```

1. **The Big Machine (`engine/`)**: A universal, reusable, game-agnostic mechanical apparatus. Zero lore mentions. Operates strictly on generic IDs, bounded buffers, fixed-timestep ticks, and deterministic geometry.
2. **The Master Turn Gear (`game/`)**: The specific game implementation of **Tenebris and Lux**, bringing the tactical turn loop, the twin armaments (Shield of Lux, Sword of Tenebris), moral sway (`[-100, +100]`), and the continental chronicles of Sacyra and Libira.

### 108.2.2. Dual Waterfall & Agile Engineering Discipline

Game engine development demands two seemingly opposing disciplines at once: * **Waterfall Bedrock (The Cast-Iron Frame)**: Memory limits (`ALLOC_ZERO_HEAP` in Layer 0/1), strict ISO C11 adherence, strictly downward layer dependency flow, and universal 5-block structural envelopes are non-negotiable physical laws. * **Agile Living Iteration (Turning the Gears)**: Within that cast-iron frame, development proceeds empirically. We modernized concrete subsystem strata one by one (`io/` → `log/` → `debug/` → `core/`), using test failures and compiler diagnostics as precision feedback to guide refactoring.

## 108.3. Layer 0 Modernization: The Complete Four-Strata Engine Bedrock

Each foundational subsystem was refactored under the strict 5-block lifecycle protocol:

### 108.3.1. Virtual Filesystem & I/O (`io/`)

- **`path_finder`**: Zero-heap path resolver supporting multi-root searching across `game/data/`, `build/`, and system directories, eliminating buffer-truncation false positives.
- **`fs_utils`**: Bounded line and block file reading without dynamic heap allocations.
- **`json_utils`**: High-performance, streaming, zero-heap JSON tokenizer and recursive getter.
- **`master_loader`**: Schema-validated asset loader verifying `block_1_metadata` envelopes and requiring explicit `block_3_body` payloads (`tenlux.data.*`).

### 108.3.2. Telemetry & Ring Flight Logging (`log/`)

- **`log`**: Categorized flight recorder (`LOG_CAT_LOG`, `LOG_CAT_DEBUG`, `LOG_CAT_CHARACTER`) storing entries into a static ring buffer.
- Formatted with fixed-width column anchors, millisecond-precision ISO timestamps, and signed telemetry grades (`GRADE:+003`, `GRADE:-050`).
- Automatic flight log rotation (`tenebris_lux_dev.log.*`) quarantined into `build/logs/`.

### 108.3.3. Diagnostic Tripwires & State Rewind Debugger (`debug/`)

- **`debug`**: Non-crashing diagnostic tripwires (`debug_trap_tripwire`) logging violations with file and line number while allowing safe fallback recovery.
- Time-travel state debugger recording up to 64 frozen simulation snapshots.
- Bit-exact deterministic state validation using 32-bit and 64-bit FNV-1a hashing.

### 108.3.4. Core Utilities & Elevated Orchestrators (`core/`)

- `clock_pacer`: Fixed-timestep pacing orchestrator enforcing canonical 60Hz ($16.666\text{ ms}$) simulation ticks, rendering alpha calculation, and "spiral of death" lag clamping.
- `formula_eval`: Data-driven combat mechanics computing bounded damage mitigation ($10\%$ floor to $90\%$ ceiling), directional accuracy (Flank: $+10\%$, Rear: $+25\%$), critical strikes, and moral resonance.
- `geom_utils`: 2D AABB geometry, 2.5D dimetric/isometric grid coordinate math, and elevated letterbox/pillarbox viewport projection.
- `math_utils` / `string_utils` / `time_utils`: Safe bounded string manipulation, monotonic clocks, saturating integer math, and transition easing curves (`EASE_QUAD_IN`, `EASE_CUBIC_OUT`).

## 108.4. The Cog Elevation & Orchestrator Protocol

During this milestone, we practiced **Cog Elevation**: when a subsystem or helper inside `game/` is found to be universal, game-agnostic, and reusable, it must be elevated into `engine/`.

### 108.4.1. Elevation of Viewport Scalers

Previously, dynamic camera projection and letterbox scaling were embedded in `game/src/world/map_scaler.c`. We extracted the mathematical projection laws into `engine/src/core/geom_utils.c`: * `geom_viewport_init()`: Calculates uniform aspect-ratio scale factors and letterbox/pillarbox offsets. * `geom_world_to_screen()` & `geom_screen_to_world()`: Projects coordinates between simulation and display. * `geom_clamp_camera()` & `geom_is_in_viewport()`: Keeps camera views bounded within world limits.

`map_scaler.c` was updated to delegate its math directly to `geom_utils`, reducing domain code complexity while enriching the Big Machine.

### 108.4.2. Standalone Quarantined Fallback Modules

To ensure the engine can always compile and pass verification even when data files are missing, every Layer 0 cog features a quarantined fallback module (`*_fallback.[ch]`). Fallbacks provide hardcoded canonical profiles, zero dynamic allocations, and comprehensive pre-flight self-tests.

## 108.5. Four-Strata Verification Ontology & 51/51 CTest Pass Proof

We expanded our automated testing framework across the **Four Ontological Testing Sections**: 1. **Section 1: Always Pass (Nominal Golden Baseline)**: Validates standard functionality under ideal inputs. 2. **Section 2: Expected Input (Contracts & Permutations)**: Exercises valid domain edge cases and configurations. 3. **Section 3: Boundary & Edge Stress (Limits & Saturations)**: Tests clamping, maximum capacities, and extreme ranges. 4. **Section 4: Critical Failure & Chaos (Corrupt, Negative & Inverted)**: Injects poison inputs, NULL pointers, zero divisors, and lag spikes to verify tripwires and crash immunity.

```
============================================================================
CTest Automated Verification Suite Results
============================================================================
Test project /home/professorseanex/Tenebris and Lux/build
      Start  1: cli_help_test                   Passed    0.00 sec
      Start  2: cli_version_test                Passed    0.00 sec
      Start  3: subsystem_audit_test            Passed    0.01 sec
      Start  4: functional_cogs_test            Passed    0.00 sec
      Start  5: functional_io_test              Passed    0.00 sec
      Start  6: functional_log_test             Passed    0.00 sec
      Start  7: functional_debug_test           Passed    0.00 sec
      Start  8: functional_core_test            Passed    0.00 sec
```

```
      Start  9: unit_math_test                  Passed    0.00 sec
      Start 10: unit_data_test                  Passed    0.00 sec
      Start 11: unit_time_test                  Passed    0.00 sec
      Start 12: unit_string_test                Passed    0.00 sec
      Start 13: unit_log_test                   Passed    0.05 sec
      Start 14: unit_log_engine_test            Passed    0.01 sec
      Start 15: unit_log_fallback_test          Passed    0.00 sec
      Start 16: unit_debug_test                 Passed    0.00 sec
      Start 17: unit_debug_engine_test          Passed    0.00 sec
      Start 18: unit_debug_fallback_test        Passed    0.00 sec
      Start 19: unit_math_utils_test            Passed    0.00 sec
      Start 20: unit_math_utils_fallback_test   Passed    0.00 sec
      Start 21: unit_time_utils_test            Passed    0.00 sec
      Start 22: unit_time_utils_fallback_test   Passed    0.00 sec
      Start 23: unit_string_utils_test          Passed    0.00 sec
      Start 24: unit_string_utils_fallback_test  Passed    0.00 sec
      Start 25: unit_geom_utils_test            Passed    0.00 sec
      Start 26: unit_geom_utils_fallback_test   Passed    0.00 sec
      Start 27: unit_clock_pacer_test           Passed    0.00 sec
      Start 28: unit_clock_pacer_fallback_test  Passed    0.00 sec
      Start 29: unit_formula_eval_test          Passed    0.00 sec
      Start 30: unit_formula_eval_fallback_test  Passed    0.00 sec
      Start 31: unit_combat_test                Passed    0.00 sec
      Start 32: unit_items_test                 Passed    0.00 sec
      Start 33: unit_world_test                 Passed    0.00 sec
      Start 34: unit_dispatch_test              Passed    0.00 sec
      Start 35: unit_gui_test                   Passed    0.00 sec
      Start 36: unit_fs_utils_test              Passed    0.00 sec
      Start 37: unit_path_finder_test           Passed    0.00 sec
      Start 38: unit_json_utils_test            Passed    0.00 sec
      Start 39: unit_master_loader_test         Passed    0.00 sec
      Start 40: integrated_utils_test           Passed    0.00 sec
      Start 41: integrated_sim_test             Passed    0.01 sec
      Start 42: integrated_io_test              Passed    0.00 sec
      Start 43: integrated_log_test             Passed    0.00 sec
      Start 44: integrated_debug_test           Passed    0.00 sec
      Start 45: integrated_core_test            Passed    0.00 sec
      Start 46: system_game_invariants_test     Passed    0.00 sec
      Start 47: system_gameplay_scenarios_test  Passed    0.01 sec
      Start 48: system_io_test                  Passed    0.02 sec
      Start 49: system_log_test                 Passed    0.23 sec
      Start 50: system_debug_test               Passed    0.02 sec
      Start 51: system_core_test                Passed    0.00 sec

100% tests passed, 0 tests failed out of 51
Total Test time (real) = 0.50 sec
================================================================================
```

## 108.6. Telemetry & Pre-Flight Diagnostic Self-Audit

The diagnostic self-audit was expanded to 20 registered engine modules in `game/src/main.c`:

```
================================================================================
          Tenebris and Lux - Subsystem Invariant Pre-Flight Audit
================================================================================
Subsystem            Source File     Rev  Memory      Status  Description
--------------------------------------------------------------------------------
core.util.manifest   manifest.h      r2   Zero-Heap   [PASS]  Pre-flight manifest DAG orchestrator verified.
core.util.fs         fs_utils.h      r3   Zero-Heap   [PASS]  Zero-heap filesystem I/O verified.
core.util.json       json_utils.h    r3   Zero-Heap   [PASS]  JSON streaming parser verified.
core.util.time       time_utils.h    r3   Zero-Heap   [PASS]  Monotonic clock & ISO timestamps verified.
core.util.math       math_utils.h    r3   Zero-Heap   [PASS]  Clamping, saturating math & PRNG verified.
core.util.geom       geom_utils.h    r3   Zero-Heap   [PASS]  2D AABB & letterbox viewport scalers verified.
core.util.string     string_utils.h  r3   Zero-Heap   [PASS]  Buffer safety & path joining verified.
core.util.pacer      clock_pacer.h   r1   Zero-Heap   [PASS]  Fixed-timestep pacer & lag clamp verified.
core.util.formula    formula_eval.h  r1   Zero-Heap   [PASS]  Mitigation, hit chance & formulas verified.
core.util.log        log.h           r3   Zero-Heap   [PASS]  Static ring buffer & flight logs verified.
core.util.debug      debug.h         r3   Zero-Heap   [PASS]  State rewind & tripwires verified.
sim.engine.state     sim_state.h     r4   Zero-Heap   [PASS]  State initialization verified.
sim.engine.combat    combat.h        r5   Zero-Heap   [PASS]  Combat sub-assemblies satisfied pre-flight.
sim.engine.items     items.h         r2   Zero-Heap   [PASS]  Item catalog audited cleanly.
sim.engine.world     world.h         r5   Zero-Heap   [PASS]  World orchestrator & map scaler verified.
sim.engine.dispatcher dispatcher.h    r4   Zero-Heap   [PASS]  Command router & alias tables verified.
```

```
sim.engine.core      game.h          r4   Zero-Heap   [PASS]  Simulation core pre-flight satisfied.
gui.viewport.x11     gui.h           r2   Zero-Heap   [PASS]  X11 display connected & panels verified.
--------------------------------------------------------------------------------
Telemetry: 20 modules registered | 20 Pass | 0 Warn | 0 Critical | 0 Fatal
Policy Verdict: ALL PRE-FLIGHT INVARIANTS SATISFIED (System Ready)
Audit Invariant Checks: 7/7 Invariant Checks Passed (Relics, Stances, Tripwires)
================================================================================
```

## 108.7. The Fractal 5-Block Invariant: Code, Manifests, Tests & Git History

During this cycle, we discovered that the OmniCode Universal 5-Block Lifecycle Protocol is a **fractal design pattern** that governs all documentation, implementation, and history: * **Block 1 (Metadata)**: Identity, version, interface revision, stratum, author, invariants. * **Block 2 (Types & Framing)**: Ontology, data schemas, problem statement, domain types. * **Block 3 (Body Implementation)**: The executable work, architectural changes, sub-assemblies. * **Block 4 (Audit Procedures)**: Pre-flight checks, CTest verification proof, tripwire traps. * **Block 5 (Closing & Invariants)**: Diagnostic inspection cards, teardown, closing affirmations.

By extending this pattern directly into our **Git Commit Architecture** and **DevLog Chronicles**, the historical record of the repository mirrors the exact structural rigor of our source code.

## 108.8. Architectural Next Steps & Horizon Ahead

With Layer 0 of **The Big Machine** verified with 100% pass rates across 51 CTests and 20 registered modules, our engineering horizon points directly to: 1. **Screen Transition Orchestrator**: Generic scene dissolves, fades, and easing wipes in `engine/`. 2. **Audio Event Bus Orchestrator**: Zero-heap sound cue routing and channel volume mixing in `engine/`. 3. **Layer 2 Viewports Modernization**: Elevating native X11 rendering and High-DPI font blitting into the 5-block lifecycle architecture. 4. **Master Turn Gear Domain Modernization (`game/`)**: Aligning combat, stance state machines, and world topology with the curated data catalogs in `game/data/`.

# Chapter 109. DevLog 016: Native Git Commit Validation, Audio Elevation & Movie Reel

**Source File**: docs/dev_logs/2026/11/architecture/devlog_016_git_validator_movie_reel_and_audio_elevation.adoc



**Volume VI — Engineering Chronology (DevLog 016)**
**Milestone Dates:** 2026-11-07 – 2026-11-08 | **Baseline Commits:** 36427ba → 52149a3
**Series Navigation:** ← DevLog 015 | Nov 2026 Index | DevLog 017 →

Following DevLog 015's formalization of **The Big Machine** (engine/) and **The Master Turn Gear** ( game/), five rapid commits hardened the developer workflow and elevated audiovisual systems: integrating the native Rust git_validator hook, aligning High-Fantasy Hope presentation catalogs, folding Offline Work 4 into architecture docs, elevating audio/ cleanly into Layer 0, and building Layer 2's deterministic **Movie Reel** viewport architecture.

## 109.1. Milestone Scope & Commit Progression

| Commit | Date | Milestone Summary |
|---|---|---|
| 36427ba | 2026-11-07 | feat(ci): Integrated native Rust git_validator, .githooks/commit-msg hook, and modernized CI enforcement. |
| a138533 | 2026-11-07 | feat(presentation): Aligned presentation catalogs to 5-block JSON schemas and the High-Fantasy Hope color palette. |
| 22dea06 | 2026-11-07 | docs(architecture): Integrated Offline-Work-4, authored Layer 0 code tours, and compiled the expanded Master Guide. |
| 9c209ad | 2026-11-08 | feat(viewport): Elevated the audio mixing subsystem into Cornerstone Layer 0 and implemented the Movie Reel architecture. |
| 52149a3 | 2026-11-08 | feat(viewport): Completed presentation catalog alignment, audiovisual verification tiers, and grand orchestrators. |

## 109.2. Enforcing 5-Block Provenance at the Commit Boundary (36427ba)

As architectural rigor climbed inside engine/ and docs/, commit history itself risked drifting into informal one-line messages that hid invariants and test coverage. Rather than adding Python or shell lint scripts in violation of Cornerstone's **Native Compiled Toolchain Invariant**, commit enforcement was implemented inside the Rust workspace (tools/src/validator/git_commit.rs + git_validator binary + .githooks/commit-msg).

- **Strict Conventional Header Prefixes:** Only feat(, fix(, docs(, refactor(, test(, chore(, perf(, and build( headers are accepted.
- **Mandatory 5-Block Commit Body:** Every commit message must declare:
  1. BLOCK 1: METADATA & REVISION RECORD (including explicit Project:, Author:, and Date: tags),
  2. BLOCK 2: REASON FOR CHANGE & PROBLEM STATEMENT,
  3. BLOCK 3: ARCHITECTURAL MODIFICATIONS & ELEVATIONS,
  4. BLOCK 4: VERIFICATION, TRIPWIRES & PASS PROOF,
  5. BLOCK 5: INVARIANT COVENANTS & CLOSING AFFIRMATIONS.

- **Retroactive CI Sweep:** Running `tl_tools validate all` inspects the most recent 10 repository commits via `git log -n 10`, preventing discipline erosion after initial setup.

## 109.3. Elevating audio/ to Layer 0 & Building the Movie Reel (9c209ad, 52149a3)

In earlier builds, audio cue triggering was intertwined with presentation transitions. Guided by the **Cog Elevation & Orchestrator Protocol**, all general-purpose synthesis, channel routing, and fade pacing were cleanly elevated into `engine/{include,src}/audio/` as Layer 0 cogs:

```
Layer 2: Viewport (X11 / Headless Frame)
    └──── Movie Reel Orchestrator (scenes, wipes, typography scaling, Hope palette)
          │
          ▯ emits generic timeline triggers & cue IDs (zero lore bleed)
Layer 0: Cornerstone Audio Bus
    └──── Multi-bus mixer + cue queue + procedural BIOS synthesis fallback
```

- **5-Block Presentation Alignment (a138533):** Every viewport layout, typography rule, theme token, and audio channel catalog was migrated into strict `block_1_metadata` .. `block_5_closing` JSON envelopes.
- **Movie Reel Orchestrator (9c209ad):** Implemented deterministic scene reels, screen wipes, and frame pacing independent of story text so Layer 2 renders both graphical X11 (`640x360` letterboxed) and headless parity without touching heap memory (`ALLOC_ZERO_HEAP`).
- **Offline-Work-4 Convergence (22dea06):** Synchronized the statistical category sheet (Core 9 Combat Stats ▯ 8 Exploration Stats, Poise, Facing, Six-Power Factions) into `docs/architecture/` and regenerated the Master Guide volumes.

## 109.4. Verification & Cross-Tier Audiovisual Parity (52149a3)

- **4-Tier Test Expansion:** Added unit, functional, integrated, and system tests exercising audio bus mixing, procedural fallback waveforms, Movie Reel timeline progression, and Hope-palette theme loading under `-std=c11 -Wall -Wextra -Wpedantic`.
- **Headless Determinism:** Confirmed identical frame pacing, cue dispatch ordering, and state transitions whether running headless in automated CTest harnesses or rendering via native X11.
- **Commit Gate Proof:** `.githooks/commit-msg` verified every commit in the `36427ba`..`52149a3` chain cleanly under `cargo run --bin git_validator`.

## 109.5. Architectural Takeaway & Next Horizon

With commit hygiene guarded natively in Rust and audiovisual presentation elevated into clean Layer 0 / Layer 2 boundaries, attention turned back toward completing the game's deepest ludonarrative specifications (**Offline Work 5 & 6** → DevLog 017) and decoupling `engine/config/` cleanly from `game/data/` (→ DevLog 018).

**Navigation:** ← DevLog 015 | Nov 2026 DevLog Index | Next: DevLog 017 →

# Chapter 110. DevLog 017: Orthogonal 35 Classes, 28 Jobs, Generational War & 4D Dream Clock

**Source File**:
`docs/dev_logs/2026/11/lore_and_design/devlog_017_orthogonal_classes_jobs_generational_war_and_dream_clock.adoc`



**Volume VI — Engineering Chronology (DevLog 017)**
**Milestone Date:** 2026-11-09 | **Canonical Sources:** Offline Work 5 & Offline Work 6
**Series Navigation:** ← DevLog 016 | Nov 2026 Index | DevLog 018 →

This lore-and-systems milestone synthesizes **Offline Work 5** and **Offline Work 6**, resolving ambiguities inherited from the 2017–2018 RPG Maker MV archives (`Folder 1` and `Folder 2`), codifying the **35-Class Pascal Pyramid** alongside the **28-Job 5-Pillar Vocation Compendium**, framing the **10-to-20-year Generational War** under the **Two-Pass Grand Tour**, and specifying the zero-heap **4D Dream Clock** state contracts.

## 110.1. Resolving 2017–2018 Demo & Early Documentation Ambiguities

Cross-referencing the 2018 **classes_and_growth.adoc** tables against the full multi-route canon resolved four long-standing structural compressions:

- **Flattened "7 Combat Classes" → Orthogonal Class × Job:** Legacy RPG Maker MV tables packed **Warrior, Mage, Rogue (Thief), Priest, Summoner, Hero,** and **Guard** into one list. Canon separates them into: (a) 3 Basic + 2 Advanced Base Classes (`W, M, R, P, S`), (b) **The Hero** / **The Dark One** as Tier 5 Moral Apex Classes (`±50 .. ±100`), and (c) **Guard** (**Town Sentry** / **Royal Man-at-Arms**) as a Civic World Vocation (**Job**).
- **Protagonist Titles vs. Unofficial Classes: "Vanguard of Lux"**, **"Outrider of Tenebris"**, and **"Secret Chronicler"** are narrative focalizations. Mechanically in-story, Elijah and Katrina are multi-domain **Trainees** (`W+M+R`), while `<chosen_name>` is Chrono's initiate **Mystic** (`P+S`). Playthrough 1 grants single-domain "training wheels"; clearing the campaign unlocks official Hard-Mode **Trainee** (`2.2×` EXP) and **Mystic** (`1.8×` EXP) in New Game+.
- **Sir Roté's Dual Designation:** Recognized cleanly through Class vs. Job orthogonality—Roté's **Class** is **Priest**, while his Crown **Job** is **Border Outrider** / **Court Herald**.
- **Antagonist Identity & Class:** Confirms **Usuru** (personal name), **The Dark Lord** (title), and **The Dark One** (literal Tier 5 Moral Evil Apex Class at `-100 .. -50`).

## 110.2. Orthogonal Character Engine: 35 Classes & 28 World Jobs

| Progression Axis | Canonical Rules & Static Bitmask Representation |
|---|---|
| **Class (Combat Domain)** | 5-Tier combinatorial pyramid governed by Pascal's Triangle (`C(5,1)=5`, `C(5,2)=10`, `C(5,3)=10`, `C(5,4)=5` Paragons, plus 2 Special Initiates and 3 Cosmic Apex Classes = `35` total `class_id_t` values). Unlocks permanent muscle-memory skills and passive traits. Remaining inside `[-50, +50]` (**Bystander Corridor**) caps progression at Tier 4 Paragons (`High Justiciar`..`Ascendant Mystic`). |

| Progression Axis | Canonical Rules & Static Bitmask Representation |
|---|---|
| **Job (World Vocation)** | 28 vocations (job_id_t) across **5 Societal Pillars**: Frontier Labor (1–6), Civic/Military/Law (7–12), Scholarship/Trade/Underworld (13–18), Ecclesiastical Faith (19–23), and Ecology/Veil Seers (24–28). Enforces the **"No Starter Job in Dio Village"** rule (JOB_NONE_DIO_VILLAGE until crossing into Whispering Woods). Rank 1–4 tools swap with active station kits; Rank 5 banks permanent Mastery Perks into banked_job_perks_mask. |
| **NG+ Encore Mastery** | Re-selecting a previously mastered Base Class in NG+ starts mastery at 15%–25% (**Attunement Resonance**) and awards evolved capstone skills (**Sovereign Phantom Slash**, **Cataclysmic Convergence**, **Shadow Transposition**, **Chrono's Emergency Decree**, **Astral Triad Covenant**). |

## 110.3. Adolescent Crucible, Poisoned Mandate & Two-Pass Grand Tour

- **The Youth Tragedy:** Establishing that Elijah, Katrina, and <chosen_name> open at **15–18 years old** illuminates the genesis catastrophe: **Usuru** was an uncommissioned **11-year-old** nomad bound to Libira's **Three-Day Law**, **Prince Mi-ren** a terrified 15–18-year-old boy in oversized regent armor, and **Prince Casatti** a 14–17-year-old academy rebel.
- **The Poisoned Mandate & Midpoint Fork (Act I Climax):** Sending greenhorn protagonists on a sweeping Act I tour under King Jihad's royal writs is both a geopolitical smokescreen and a literal re-enactment of Usuru's **Three-Day Wanderer** pilgrimage (**Two-Pass Spiral World Design**). At the 50% Sacyra Castle Siege, **killing King Jihad** triggers a chaotic Neutral power vacuum, whereas **Uncorrupting Jihad** keeps him alive as a disgraced living indictment forcing institutional truth (**"The Ugly Truth"**).
- **10-to-20-Year Maturation:** Act II (**2nd Great War**, Years 4–12, Ages 19–27) and Act III (**Sovereign Eclipse**, Years 13–20, Ages 28–35+) visibly age Dio Village and its heroes so weathered 30-year-old adults finally confront an immortal warlord still frozen inside an 11-year-old binary wound.

## 110.4. The 4D "Dream Clock" Temporal Registration Engine

Turning the Dream Clock **OFF** does not freeze story time (protagonists still age across Acts I–III and reach all **9 Standard Matrix Endings** via 3D event vectors [Location, Action, Sway]). Enabling **Dream Clock ON** (DREAM_CLOCK_ON_REGISTERED) adds the 4th coordinate (sacyra_calendar_t timestamp) to story_point_receipt_t, dynamically alters what occurs at world story points (such as intercepting couriers across Libira's 7-day lag or acting during nocturnal 20:00–04:00 Veil windows), and is mathematically required to promote the campaign into the **10th True Ending ("The Ugly Truth")**.

## 110.5. Layer 1 Zero-Heap Contract Verification

- **Bitmask Density (ALLOC_ZERO_HEAP):** All 35 classes fit inside uint64_t mastered_classes_mask (0..34) and uint8_t class_mastery_pct[35]; all 28 jobs fit inside uint32_t banked_job_perks_mask and uint8_t job_rank[28] (actor_vocation_state_t).
- **Temporal Compactness:** sacyra_calendar_t, story_point_receipt_t, and campaign_chronicle_state_t require no heap pointers and map directly onto Layer 0 time_utils monotonic ticks scaled by tunable engine/config/runtime/time/ parameters.

**Archival Sources:** Offline Work 5 | Offline Work 6
**Navigation:** ← DevLog 016 | Nov 2026 DevLog Index | Next: DevLog 018 →

# Chapter 111. DevLog 018: Standalone Engine Config Decoupling & Rev 5 Block 1/2 Standards

**Source File:**
docs/dev_logs/2026/11/architecture/devlog_018_standalone_engine_config_decoupling_and_block1_block2_standards.adoc



**Volume VI — Engineering Chronology (DevLog 018)**
**Milestone Date:** 2026-11-10 | **Baseline Commits:** 489c3b6 (tag: v0.1.1) → d552752
**Series Navigation:** ← DevLog 017 | Nov 2026 Index | DevLog 019 →

Tagged as **v0.1.1**, this milestone physically severed engine baseline mechanics from RPG domain lore, codified the **Rev 5 Block 1 Growth Schema** and **Block 2 `extern "C"` Header Discipline**, modularized the Rust pre-flight validator, and resolved include-hierarchy discontinuities across every translation unit in engine/src/.

| Commit | Date | Scope & Architectural Deliverable |
|---|---|---|
| 489c3b6 (v0.1.1) | 2026-11-10 | feat(engine): Decoupled standalone engine/config/ from game/data/, modularized Rust validator/, formalized engine/config/standards/blocks/block_1..5.json, and rolled out Rev 5 Block 1 telemetry across headers. |
| d552752 | 2026-11-10 | refactor(engine): Standardized Block 2 opening blocks (extern "C" placement, 3-tier include hierarchy, C++-safe struct zero-initialization) and resolved module discontinuities across engine/src/. |

## 111.1. Separating Engine Mechanics (engine/config/) from Game Lore (game/data/)

Before v0.1.1, presentation profiles, viewport themes, audio channels, and core math configs lived alongside gameplay data. Because Cornerstone (engine/) must boot and power completely different games without game/ present, baseline engine configurations were extracted cleanly into engine/config/ while reserving game/data/ strictly for Sacyra lore and RPG content:

- **Standalone Mechanical Tree:** Organized engine/config/ across viewport/, audio/, input/, runtime/ (math/, time/, string/), diagnostics/ (telemetry/, debugger/, taxonomy/, formats/), and standards/ (manifest.json, subsystems.json, blocks/block_1..5.json).
- **Layered VFS Resolution (path_finder):** Established deterministic override search order—(1) game/config/<rel_path>, (2) engine/config/<rel_path>, and (3) game/data/presentation/config/<rel_path>—so game customizations override engine defaults without duplicating shared machinery.

## 111.2. Movement-Resilient Metadata & Unified Block 2 Opening Contracts

To turn file headers into real-time scaling radar rather than static boilerplate, engine/config/standards/blocks/block_1_metadata.json introduced the Rev 5 **Block 1 Growth Schema**:

- **Movement-Resilient @id vs. Volatile @file / @layer:** Stopped baking folder paths into @id (core.util.*), moving every cog toward invariant logical identifiers (cog.<module>) while tracking current paths and compilation strata separately.

- **Growth Telemetry Tags:** Every Block 1 header declares `@maturity`, `@chassis` (measured `.bss`/`.data`/`.rodata`/stack footprints), `@scale` (Big-O & hard config levers), and `@seams` (`in: cfg/rom | uses: deps | probes: ⋯ → DEBUG_STEP_AWAIT | dag: ⋯`).
- **Block 2 Strict Hygiene (`d552752`):** Placed `_POSIX_C_SOURCE 200809L` cleanly above Block 2 in POSIX sources, enforced the **3-Tier Downward Include Hierarchy** (ISO C stdlib → Tier 2 External [X11 in Layer 2 only] → Tier 3 Internal Engine), moved `extern "C" {` strictly **below** `#include` directives in headers, and replaced empty `{0}` aggregates with clean `.bss` zero-initialized singletons so C and C++ compilers agree identically.

## 111.3. Validator Modularization & Header Contract Verification

- **Modular Rust Validator (`tools/src/validator/`):** Split monolithic checks into domain modules (`blocks.rs`, `config.rs`, `headers.rs`, `git_commit.rs`), validating JSON 5-block schemas (`engine/config/standards/blocks/block_1..5.json`) alongside all `44` registered Cornerstone engine headers in `engine/config/standards/subsystems.json`.
- **Zero Discontinuities:** Swept `engine/src/` under strict `-Wall -Wextra -Wpedantic` to eliminate stray includes, broken guard naming, and `.h`/`.c` `@system` drift.

## 111.4. Readiness for Tier-0 BIOS & `.rodata` ROM Vaults

With `engine/config/` cleanly separated, validator contracts machine-checked, and Block 1/Block 2 interfaces unified, the engine was ready to compile `engine/config/` JSON payloads straight into `.rodata` Config ROM vaults and wire Safe-Mode downgrade probes across `log/`, `debug/`, and `io/` (DevLog 019).

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# Chapter 112. DevLog 019: BIOS/ROM Vaults, AWAIT Probes & The C23 Fall-Down Ladder

**Source File**:
docs/dev_logs/2026/11/architecture/devlog_019_bios_rom_vaults_await_probes_and_c23_falldown_ladder.adoc

> **Volume VI — Engineering Chronology (DevLog 019)**
> **Milestone Dates:** 2026-11-11 – 2026-11-12 | **Baseline:** Commit 76a443a + Root Stratum (log/, debug/, io/) & CMakeLists.txt Upgrade
> **Series Navigation:** ← DevLog 018 | Nov 2026 Index | DevLog 020 →
>
> This milestone fuses Cornerstone's Rev 5 autonomous root stratum (log/, debug/, and io/), replacing ad-hoc stubs with a unified .rodata Config ROM generator (cmake/GenerateConfigRom.cmake), quarantined BIOS fallback modules (*_fallback.[ch] + *_types.h), an autonomous **Probe → Safe-Mode Downgrade → DEBUG_STEP_AWAIT Tripwire Loop**, a Controlled Memory Chassis cleanup, and Cornerstone's **"Fall-Down Stairs" C23 → C17 → C11 (+ C++20/17)** portability ladder anchored in engine/include/core/manifest.h.

## 112.1. The Autonomous Root Spine: log/ + debug/ + io/

Rather than building independent utility islands, Rev 5 connects the first three subsystems into one self-healing boot tree ("roots feeding the rest of `engine/`"):

```
1. log/     (Boots Before Disk/VFS)
     |  - Quarantined log_types.h + log_fallback.[ch] + rom://telemetry
     v
2. debug/   (Captures Faults & Holds Execution)
     |  - Quarantined debug_types.h + debug_fallback.[ch] + rom://diagnostics
     |  - Registers probes -> triggers Safe-Mode downgrade -> DEBUG_STEP_AWAIT
     v
3. io/      (Hydrates Disk / VFS / rom:// Safely)
        - path_finder, fs_utils, json_utils, master_loader + io_fallback.[ch]
        - Hydrates only verified block_3_body spans; falls back to rom://io on fault
```

- **Unified .rodata Config ROM Vault (cmake/GenerateConfigRom.cmake):** Retired single-domain cmake/GenerateTelemetryRom.cmake in favor of a generalized build-time generator converting engine/config/*/.json catalogs into read-only ELF .rodata headers under build/generated/. Missing or corrupt disk files fall back directly to authentic structured engine defaults (rom://) rather than duplicated hardcoded literals.
- **Quarantined Leaf Headers & Fallbacks:** Extracted engine/include/log/log_types.h and engine/include/debug/debug_types.h to prevent circular includes, and introduced engine/include/{io/io_fallback.h} + engine/src/io/io_fallback.c alongside quarantined log_fallback.[ch] and debug_fallback.[ch].
- **Controlled Memory Chassis Diet:** Audited ELF .bss footprints (size -A), trimming idle static scratch allocations (such as oversized buffers in audio_bus.c) so chassis memory overhead remains minimal and honest.

## 112.2. 2. "Engine-as-a-Java" Portability: The C23 Fall-Down Ladder

Cornerstone targets forward-looking ISO **C23** (-std=c23) and **C20** (`-std=c20`) without dragging along deprecated legacy shims—yet steps gracefully down onto older compilers via CORNERSTONE_C_STANDARD=AUTO|23|17|11 and CORNERSTONE_CXX_STANDARD=AUTO|20|17 configured inside a 5-block refactored CMakeLists.txt.

*Pre-Flight Loose-Bolts Engine Walk (manifest_run_preflight) (engine/src/core/manifest.c#L412-L455 · Mode: Function)*

```
412 /**
413  * @brief Executes comprehensive pre-flight diagnostics across all registered modules.
414  */
415 preflight_status_t manifest_run_preflight(manifest_registry_t *registry)
416 {
417     if (registry == NULL) {
418         debug_trap_tripwire("manifest", "NULL registry passed to manifest_run_preflight", __FILE__, __LINE__);
419         return PREFLIGHT_STATUS_HALT_FATAL;
420     }
421 
422     registry->pass_count = 0;
423     registry->warn_count = 0;
424     registry->critical_count = 0;
425     registry->fatal_count = 0;
426 
427     /* Step 1: Active DAG Dependency Graph Verification */
428     for (size_t i = 0; i < registry->count; i++) {
429         const module_metadata_t *mod = registry->modules[i];
430         if (mod == NULL) continue;
431 
432         for (size_t d = 0; d < mod->dependency_count; d++) {
433             const char *req_dep = mod->dependencies[d];
434             if (req_dep == NULL) continue;
435 
436             bool found = false;
437             for (size_t j = 0; j < registry->count; j++) {
438                 if (registry->modules[j] != NULL &&
439                     strcmp(registry->modules[j]->module_id, req_dep) == 0) {
440                     found = true;
441                     break;
442                 }
443             }
444 
445             if (!found) {
446                 diag_result_t dep_res;
447                 dep_res.passed = false;
448                 dep_res.severity = SEVERITY_FATAL;
449                 dep_res.subsystem = mod->module_id;
450                 dep_res.check_name = "Dependency DAG";
451                 dep_res.details =
452                     "Fatal: Unregistered upstream dependency declared.";
453                 registry->results[i] = dep_res;
454                 registry->fatal_count++;
455                 registry->status = PREFLIGHT_STATUS_HALT_FATAL;
```

*Iterates every registered module sequentially, classifying OK, WARNINGS, DOWNGRADED, and HALT_FATAL states.*

Wholesale Listing: Complete verbatim code (#L412-L509) → Appendix: Pre-Flight Loose-Bolts Engine Walk (manifest_run_preflight).

- **Central Capability Macros (engine/include/core/manifest.h):** Exposes CS_C_STD_TIER (23/17/11/0) and CS_CXX_STD_TIER (20/17/11/0) plus CS_NODISCARD, CS_MAYBE_UNUSED, CS_FALLTHROUGH, CS_UNREACHABLE, CS_ALIGNOF, CS_NORETURN, CS_STATIC_ASSERT, and CS_HAS_EMBED, all compiling cleanly with zero GNU extension warnings under -Wall -Wextra -Wpedantic.
- **No Backward-Compatibility Crutches:** Instead of emulating removed historical idioms, building against the C17/C11 floor lets every module use modern attributes when compiled in C23 and fall safely down one step without losing deterministic correctness.

## 112.3. Crude-to-Fine Verification Across log/, debug/, and io/ (tests/)

Extracted shared deterministic harness macros into tests/common/ (cornerstone_test.h) and systematically upgraded every verification tier across the root subsystems (test_log_*, test_debug_*, test_io_*, plus new

tests/unit/io/test_io_fallback_unit.c):

1. **Layer 1 Unit & Fallback Isolation:** Exercises ROM parsing, ring wrap-around, watchpoints, and BIOS Safe-Mode boot with empty/broken VFS paths.
2. **Layer 2 Functional Contracts:** Validates 5-block JSON loading boundaries (block_3_body scoping) and categorized hex fault counters (0x0201..0x0207).
3. **Layer 3 Integrated Cross-Module Flows:** Tests io/ + log/ + debug/ probe escalation into DEBUG_STEP_AWAIT under live downgrades.
4. **Layer 4 System Suites:** Verifies deterministic full-boot parity and shutdown .bss zero-reset across repeated sessions (65 CTest targets in the suite).

## 112.4. Next Horizon: Documenting the Root Tree & Auditing core/

With the root spine (log/ → debug/ → io/), Config ROM Vaults, C23 Fall-Down macros, and tests/ fully aligned under Rev 5, the immediate priority became documenting this exact architecture through live code snapshots and clean multi-volume PDFs (DevLog 020) before advancing Block 3 audits into core/, audio/, and viewport/.

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# Chapter 113. DevLog 020: 5-Block AsciiDoc, Live Code Snapshots & Paired Textbooks

**Source File**:
docs/dev_logs/2026/11/architecture/devlog_020_five_block_asciidoc_snapshots_pdf_budgets_and_paired_docs.adoc

> **Volume VI — Engineering Chronology (DevLog 020)**
> **Milestone Date:** 2026-11-12 | **Scope:** tools/src/guide.rs, docs/adoc_pdf_layout_rules.json, docs/code/{log,debug,io}/, docs/architecture/chronicles/, docs/offline_work/, Volumes I–VI
> **Series Navigation:** ← DevLog 019 | Nov 2026 Index | **(Latest DevLog)**
>
> Inspecting generated PDF output—especially **Volume V** ( VOLUME_V_CORE_ENGINE_CODE_TOURS.pdf)—revealed awkward whitespace pages, broken [abstract] chapter numbering in Asciidoctor book builds, duplicate snapshot plate anchors, and stale manual prose drifting away from live C source files. To turn documentation into a true data-driven textbook system, master_guide gained live 5-mode code snapshots and PDF layout validation while docs/code/{log,debug,io}/, docs/architecture/chronicles/, and docs/offline_work/ were rebuilt onto the **Universal 5-Block AsciiDoc Standard**.

## 113.1. Textbook-Grade Live Code Snapshots (docs/code_snapshots_manifest.json)

Copying C functions by hand into documentation guarantees bit-rot. Extending tools/src/guide.rs introduced a declarative extraction manifest (docs/code_snapshots_manifest.json) supporting five surgical capture modes:

| Snapshot Mode | Extraction Behavior & Textbook Usage |
|---|---|
| whole_file | Extracts a clean compact file wholesale (such as cmake/GenerateConfigRom.cmake or core test headers). |
| block | Captures an entire OmniCode BLOCK 1..BLOCK 5 section from .h or .c. |
| sub_assembly | Extracts an exact Sub-Assembly N.M region (e.g., io/ 5-block getters or debug/ probe structs). |
| function | Isolates an individual C function body with its paired dual-audience docstring. |
| symbol_group | Collects related type declarations or helper families into one compact visual plate. |

- **Anchor Deduplication (deduplicate_snapshot_plate_anchors):** Eliminated duplicate [[snap_plate_*]] Asciidoctor anchor collisions whenever a snapshot appeared inline inside a chapter and again in a volume appendix.
- **Out-of-Box Theme & Title Formatting (docs/guide_theme_config.json, docs/themes/):** Cleaned Volume V synopsis whitespace so cover → TOC → overview → paired chapters flow cleanly across A4 pages without empty half-page breaks.

## 113.2. The 5-Block AsciiDoc Comment Standard & --validate-layout

Codified docs/adoc_5block_standard.json and docs/adoc_pdf_layout_rules.json so AsciiDoc manuscripts follow the exact same **Universal 5-Block Lifecycle Protocol** as .c, .h, and .json files:

- //// BLOCK 1..5 //// + // --- Sub-Assembly X.Y: AsciiDoc block/line comments frame Metadata, Opening, Body, Audit/Verification, and Closing/Navigation without rendering noise in HTML/PDF outputs.

- **PDF Hygiene Enforcement (`master_guide -- --validate-layout`):** Audits line-width budgets (`⇐ 104` cols with line numbers so Rouge blocks never soft-wrap in A4 PDFs), bans `[abstract]` above `==` headings, bans page-breaking `'''` horizontal rules in multi-chapter books, and enforces 6-hyphen `------` fences.

## 113.3. Paired Code Folders, Architectural Chronicles & Linked Offline Series

- `docs/code/{log,debug,io}/` **Paired Documentation:** Organized each root subsystem into its own folder pairing every `.h` and `.c` file into unified module manuals (plus a dedicated testing chapter per folder covering unit, functional, integrated, and system tiers).
- `docs/architecture/chronicles/`**:** Added accessible, mixed-audience deep dives explaining **why** the Rev 5 architecture was built for long-term game payoff (`01_root_stratum_story_and_game_payoff.adoc`) and how `AGENTS.md` evolved alongside AI-assisted engineering (`02_agents_md_evolution_and_ai_architecture.adoc`).
- `docs/offline_work/` **Series Unification (Parts 1–6):** Moved `Offline_Work_5.md` and `Offline_Work_6.md` out of repo root into `docs/offline_work/`, cleaned corrupted table math delimiters and citation tags in Part 6, authored `Offline-Work-5.adoc` + `Offline-Work-6.adoc`, upgraded `README.adoc` and `Offline-Work-1..4.{adoc,md}` into 5-block format with bidirectional navigation, and added `docs/offline_work/` to strict layout validation.

## 113.4. Verification Summary Across Code, Layout & Books

- **Native Rust Tooling (`RUSTFLAGS="-D warnings" cargo test`):** Verified snapshot extraction, anchor deduplication, layout rule checks, and validator modules cleanly with zero compiler warnings.
- **PDF Layout Validator (`master_guide -- --validate-layout`):** Enforced zero errors across strict targets (`docs/README.adoc`, `docs/architecture/`, `docs/code/{log,debug,io}/`, `docs/offline_work/`).
- **Full Book Compilation (`master_guide -- --all`):** Compiled standalone **Volumes I–VI** alongside the complete `MASTER_GUIDE.pdf` with DevLogs `001`–`020` indexed cleanly in Volume VI.

## 113.5. Synchronizing the Repository & Preparing for `core/` Audits

With code, tests, Config ROMs, snapshot books, pre-production chronicles, and DevLogs `016`–`020` all aligned under **Rev 5**, the workspace is ready for synchronized wave commits under `.githooks/commit-msg` before beginning the Block 3 audit across `core/`, `audio/`, `viewport/`, and `game/`.

**Navigation:** ← DevLog 019 | Nov 2026 DevLog Index | Documentation Root

# Chapter 114. Developer Log 002: The Sacyra Chronicles, Dual Protagonists & TRPG Evolution

**Source File**: docs/dev_logs/2026/10/lore_and_design/devlog_002_the_sacyra_chronicles_and_trpg_vision.adoc

## 114.1. Abstract

This developer log chronicles the foundational provenance and overarching creative vision of **Tenebris and Lux**, rooted in its 2014–2018 high school inception as the premier game project of CreativeWorkzStudio. It documents the elimination of placeholder factions in favor of the canonical **Kingdom of Sacyra**, the introduction of dual protagonists **Elijah Ro-enheart** and **Katrina Jaren**, and the technical specification of the game's true genre: a **Tactical Strategy RPG (TRPG)** fusing grid-based tactical maneuvering with cinematic traditional side-scroll / action-based combat resolution.

## 114.2. Historical Provenance: The 2014–2018 High School Genesis

Before CreativeWorkzStudio evolved into a multifaceted publishing house, **Tenebris and Lux** was its very first major game in active development.

- **The RPG Maker Era (2014–2018)**: Conceived while the author was in high school learning game design and programming. The initial iterations were built using the default offerings of RPG Maker, exploring community plugins, and crafting characters through its Character Generator.
- **The Lost Assets & The Blank Canvas**: With no legacy digital assets surviving from that era, the project stands not as a nostalgic port, but as a complete ground-up rebirth—realizing the original creative vision with the systems engineering maturity, mathematical rigor, and architectural discipline of modern custom engine design.

## 114.3. Narrative Realignment: The Kingdom of Sacyra

During early prototype scaffolding, generic placeholder factions ("The Solar Order" and "The Umbral Covenant") were inserted to test basic moral sway logic. On October 26, 2026, the creator firmly realigned the canon, stripping away these synthetic factions to reveal the true world:

### 114.3.1. The Kingdom of Sacyra (/s□□□ki□r□/)

- The setting is the historic **Kingdom of Sacyra** (pronounced **"saw-kee-ruh"**).
- Sacyra is a structured, chivalric realm anchored by three distinct socio-political pillars:
  1. **The Crown & Royal Guard**: Guardians of legal order, national defense, and the ancestral throne.
  2. **The Noble Houses & Royal Court**: High aristocracy governing regional fiefs and vying for political dominance on the royal council.
  3. **The Adventurers' Guild**: A pragmatic, frontier-based brotherhood operating beyond courtly vanity, answering commoner monster distress calls and exploring untamed ruins.
- **The Existential Threat**: The ancient sovereign **The Dark Lord** breaking through the Abyssal Vaults, threatening to extinguish all mortal agency.

### 114.3.2. The Dual Protagonists

Rather than a generic lone wanderer, the narrative is anchored by two contrasting protagonists whose fates, lineages, and martial philosophies intertwine:

| Character | Elijah Ro-enheart | Lady Katrina Jaren |
|---|---|---|
| **Social Origin** | Military Lineage (Common/Barracks Nobility) | High Aristocracy (House Jaren) |
| **Family Heritage** | Son of the former Head of the Royal Guard | Daughter of an influential noble house on the Royal Court |
| **Internal Desire** | Living up to his father's legendary military legacy while discovering what true honor demands | Escaping stifling aristocratic court expectations to join the local **Adventurers' Guild** |
| **Relic Synergy** | **The Shield of Lux** (Sanctuary, Warding, Formation Hold) | **The Sword of Tenebris** (Severance, Mist-Veil, Critical Strike) |
| **TRPG Doctrine** | Frontline heavy vanguard; zone-of-control denial; ally interception | Agile outrider skirmisher; bypassing enemy lines; flanking assassin |

## 114.4. Ludological Evolution: The Tactical Strategy RPG (TRPG)

The game's true mechanical identity is a **Tactical Strategy RPG (TRPG)**. This fundamentally informs how the simulation engine and presentation viewports are being constructed:

### 114.4.1. The Dual-Layer Combat Loop

The combat loop operates across two synchronized phases:

1. **Strategic Grid Phase**:
   - The battle takes place on a discrete 2D grid representing castles, ramparts, thickets, and void-blighted frontier passes.
   - Units position based on movement ranges, terrain defense bonuses, directional facing (Front, Flank, Rear), and Zone-of-Control (ZOC) restrictions.
2. **Cinematic Combat Resolution (Side-View Action)**:
   - When a unit initiates combat against an enemy target, the game transitions into a dynamic **2D traditional side-scroll / action-based clash**.
   - Attacks play out with timed player agency: precision blocks with the Shield of Lux, poise-breaking shield bashes, and devastating severance slashes with the Sword of Tenebris.
   - Upon duel conclusion, remaining unit health, stagger, and relic resonance seamlessly persist back to the tactical grid.

### 114.4.2. Architectural Implications for the C Engine

Knowing this was originally an RPG Maker project gives us clear architectural directives: * **Decoupled Data Architecture**: Static definitions (`Units`, `Skills`, `Equipment`, `TerrainTypes`) must exist as clean data tables distinct from session state (`player_state_t`, `grid_t`). * **Deterministic Event Pipeline**: Interactive map features (gates, switches, dialog triggers) will follow a clean, reentrant command interpreter model. * **Zero Engine Bloat**: By building this custom in C11 rather than relying on bloated runtimes, our TRPG will maintain sub-millisecond turn updates, instantaneous scene transitions, and absolute softlock immunity.

## 114.5. Next Milestone Roadmap

1. **Phase 1 (Complete)**: Established OmniCode 5-block layering, Layer 0 utilities, CTest verification suite, and canonical documentation.
2. **Phase 2 (Active)**: Expand simulation state to model dual protagonists (Elijah & Katrina), party data structures, and the foundational 2D tactical grid.
3. **Phase 3**: Implement the cinematic side-scroll combat transition in the presentation layer.
4. **Phase 4**: Expand the tactical campaign across the provinces of Sacyra to confront the vanguard of The Dark Lord.

# Appendix A: Wholesale Source Code Reference Plates

Complete verbatim reference plates for source files, blocks, sub-assemblies, and function groups excerpted in preceding chapters.

## A.1. Cornerstone Fall-Down Stairs Portability Ladder (C23/C17/C11 & C++20/17)

**Snapshot ID**: snap_manifest_ladder_sub2_1 | **Mode**: Subassembly | **File**: engine/include/core/manifest.h (L82–L194 of 707 lines) | Jump back to chapter text

```
 82 /*
 83 ------------------------------------------------------------------------------
 84 Sub-Assembly 2.1: Pre-Conditions, Progressive Standard Macros & Buffer Limits
 85 ------------------------------------------------------------------------------
 86 Compile-time sizing limits for registry module nodes, dependencies, rules, and
 87 Cornerstone's "Fall-Down" ISO C23 / C17 / C11 + C++20 / C++17 capability macros.
 88
 89 For Designers:
 90 Allows Cornerstone to take advantage of modern compiler safety checks on current
 91 hardware while stepping cleanly onto older systems without legacy clutter. Also
 92 bounds the engine registry to up to 32 major systems, 8 core rules, and 4 links.
 93
 94 For Programmers:
 95 - Guarantees 100% zero-heap allocation (ALLOC_ZERO_HEAP) in .rodata and .bss.
 96 - Exposes CS_C_STD_TIER (23/17/11/0) and CS_CXX_STD_TIER (20/17/11/0) plus
 97   graceful ISO attribute fall-down macros (CS_NODISCARD, CS_MAYBE_UNUSED,
 98   CS_FALLTHROUGH, CS_UNREACHABLE, CS_ALIGNOF, CS_NORETURN, CS_STATIC_ASSERT,
 99   CS_HAS_EMBED) compiled cleanly without extensions under -Wpedantic.
100 ------------------------------------------------------------------------------
101 */
102
103 /** Active ISO C standard tier compiled for this translation unit (23, 17, 11, or 0 in C++). */
104 #if defined(__STDC_VERSION__) && __STDC_VERSION__ >= 202311L
105 #    define CS_C_STD_TIER 23
106 #elif defined(__STDC_VERSION__) && __STDC_VERSION__ >= 201710L
107 #    define CS_C_STD_TIER 17
108 #elif defined(__STDC_VERSION__) && __STDC_VERSION__ >= 201112L
109 #    define CS_C_STD_TIER 11
110 #else
111 #    define CS_C_STD_TIER 0
112 #endif
113
114 /** Active ISO C++ standard tier when included from C++ (20, 17, 11, or 0 in C). */
115 #if defined(__cplusplus) && __cplusplus >= 202002L
116 #    define CS_CXX_STD_TIER 20
117 #elif defined(__cplusplus) && __cplusplus >= 201703L
118 #    define CS_CXX_STD_TIER 17
119 #elif defined(__cplusplus)
120 #    define CS_CXX_STD_TIER 11
121 #else
122 #    define CS_CXX_STD_TIER 0
123 #endif
124
125 /** Marks return values that callers MUST inspect (C23/C++17+ [[nodiscard]], C11/17 floor empty). */
126 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
127 #    define CS_NODISCARD [[nodiscard]]
128 #else
129 #    define CS_NODISCARD
130 #endif
131
132 /** Silences unused-entity diagnostics cleanly on intentional dormant/platform helpers. */
133 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
134 #    define CS_MAYBE_UNUSED [[maybe_unused]]
135 #else
136 #    define CS_MAYBE_UNUSED
137 #endif
138
139 /** Annotates explicit switch-case fall-through cleanly under -Wextra -Wpedantic. */
140 #if (CS_CXX_STD_TIER >= 17) || (CS_C_STD_TIER >= 23)
141 #    define CS_FALLTHROUGH [[fallthrough]]
```

```
#elif defined(__GNUC__) && (__GNUC__ >= 7)
#    define CS_FALLTHROUGH __attribute__((fallthrough))
#elif defined(__clang__) && defined(__has_attribute)
#    if __has_attribute(fallthrough)
#        define CS_FALLTHROUGH __attribute__((fallthrough))
#    else
#        define CS_FALLTHROUGH ((void)0)
#    endif
#else
#    define CS_FALLTHROUGH ((void)0)
#endif

/** Marks statically unreachable execution branches cleanly across C23/C17/C11/C++. */
#if (CS_C_STD_TIER >= 23) && defined(unreachable)
#    define CS_UNREACHABLE() unreachable()
#elif defined(__GNUC__) || defined(__clang__)
#    define CS_UNREACHABLE() __builtin_unreachable()
#else
#    define CS_UNREACHABLE() ((void)0)
#endif

/** Portable alignment query resolving to C23/C++ alignof vs C11/C17 _Alignof without <stdalign.h>. */
#if defined(__cplusplus) || (CS_C_STD_TIER >= 23)
#    define CS_ALIGNOF(type) alignof(type)
#else
#    define CS_ALIGNOF(type) _Alignof(type)
#endif

/** Portable non-returning function specifier (C23/C++ [[noreturn]] vs C11/C17 _Noreturn). */
#if defined(__cplusplus) || (CS_C_STD_TIER >= 23)
#    define CS_NORETURN [[noreturn]]
#else
#    define CS_NORETURN _Noreturn
#endif

/** Portable compile-time assertion (C23/C++ keyword vs C11/C17 <assert.h> macro). */
#define CS_STATIC_ASSERT(cond, msg) static_assert((cond), msg)

/** Non-zero when C23 #embed directive is natively available for binary/text ROM embedding. */
#ifdef __has_embed
#    define CS_HAS_EMBED 1
#else
#    define CS_HAS_EMBED 0
#endif

/** Maximum number of registered engine/game subsystem metadata nodes. */
#define MANIFEST_MAX_MODULES      32

/** Maximum documented invariant rules declared per module node. */
#define MANIFEST_MAX_INVARIANTS   8

/** Maximum upstream prerequisite dependencies declared per module node. */
#define MANIFEST_MAX_DEPENDENCIES 4
```

## A.2. Layer 3 Plug Entry & Manifest Wiring (init_engine_manifest + main)

**Snapshot ID**: snap_main_boot_group | **Mode**: Group | **File**: game/src/main.c (L71–L522 of 524 lines) | Jump back to chapter text

```
/**
 * @brief Registers all core engine subsystem manifests into central registry.
 *
 * ### Machine Role & Mission
 * Populates the central manifest registry with all Layer 0 utilities (manifest,
 * fs, json, time, math, geom, string, log, debug), Layer 1 simulation engine (state,
 * combat, world, dispatcher, game), and Layer 2 presentation viewport (gui).
 * Enables boot-time loose-bolts invariant checks and dependency graph resolution
 * before any game loop execution begins.
 *
 * ### Ontological Classification
 * - Stratum: Layer 3 (Application Entry & Presentation Router)
 * - Machine: app.entry.main (Executable Entry & Presentation Router)
```

```c
 * - Assembly: Pre-Flight Manifest Registry Initialization
 * - Cog: Module Manifest Registration Primitive
 *
 * ### Execution & Resource Contract
 * - Memory Model: ALLOC_ZERO_HEAP (Stack-allocated fixed manifest tables).
 * - Concurrency: CONCURRENCY_MAIN_THREAD (Called at startup on main thread).
 * - Latency Budget: < 50 microseconds.
 *
 * @param[out] registry Target manifest registry to initialize and populate.
 */
static void init_engine_manifest(manifest_registry_t *registry)
{
    /* Step 1: Zero-initialize registry table */
    manifest_registry_init(registry);

    /* Step 2: Register Layer 0 Core Utilities */
    manifest_register(registry, &manifest_utils_manifest);
    manifest_register(registry, &fs_utils_manifest);
    manifest_register(registry, &json_utils_manifest);
    manifest_register(registry, &path_finder_manifest);
    manifest_register(registry, &master_loader_manifest);
    manifest_register(registry, &time_utils_manifest);
    manifest_register(registry, &math_utils_manifest);
    manifest_register(registry, &geom_utils_manifest);
    manifest_register(registry, &string_utils_manifest);
    manifest_register(registry, &clock_pacer_manifest);
    manifest_register(registry, &formula_eval_manifest);
    manifest_register(registry, &log_manifest);
    manifest_register(registry, &debug_manifest);

    /* Step 3: Register Layer 1 Simulation Engine Subsystems */
    manifest_register(registry, &sim_state_manifest);
    manifest_register(registry, &combat_manifest);
    manifest_register(registry, &items_manifest);
    manifest_register(registry, &world_manifest);
    manifest_register(registry, &dispatcher_manifest);
    manifest_register(registry, &game_manifest);

    /* Step 4: Register Layer 2 Presentation & Audio Viewport */
    manifest_register(registry, &audio_bus_manifest);
    manifest_register(registry, &gui_manifest);
}
```

```c
/**
 * @brief Application entry point and presentation frontend router.
 *
 * ### Machine Role & Mission
 * Serves as the operating system process entry point. Parses command-line
 * arguments, executes pre-flight loose-bolts machine checks, initializes the
 * player simulation state, and dynamically routes to either native X11 GUI
 * or terminal CLI mode based on user flags and hardware display availability.
 *
 * ### Ontological Classification
 * - Stratum: Layer 3 (Application Entry & Presentation Router)
 * - Machine: app.entry.main (Executable Entry & Presentation Router)
 * - Assembly: Application Lifecycle & Frontend Router
 * - Cog: Process Entry Point & Presentation Mode Dispatcher
 *
 * ### Execution & Resource Contract
 * - Memory Model: ALLOC_ZERO_HEAP (Zero dynamic heap allocations in router).
 * - Concurrency: CONCURRENCY_MAIN_THREAD (Main thread process lifecycle).
 * - Latency Budget: Lifespan of game process.
 *
 * @param[in] argc Number of command-line arguments.
 * @param[in] argv Array of command-line argument strings.
 * @return Process exit code (0 on success, non-zero on failure).
 */
int main(int argc, char *argv[])
{
    /* Step 0: Arm Tier-0 BIOS Safe-Mode telemetry immediately (RAM Ring + stderr alerts) */
    log_fallback_boot(NULL);
    LOG_INFO(LOG_CAT_LOG, "BootBIOS", 0.0f, "Tier-0 BIOS Safe-Mode telemetry armed");

    /* Step 1: Initialize presentation mode flags */
```

```
    bool force_cli = false;
    bool force_gui = false;
    bool force_debug = false;

    /* Step 2: Parse command-line arguments */
    for (int i = 1; i < argc; i++) {
        if (strcmp(argv[i], "--help") == 0 || strcmp(argv[i], "-h") == 0) {
            print_usage();
            return 0;
        }
        if (strcmp(argv[i], "--version") == 0 || strcmp(argv[i], "-v") == 0) {
            print_version();
            return 0;
        }
        if (strcmp(argv[i], "--audit") == 0 || strcmp(argv[i], "-a") == 0) {
            return run_full_subsystem_audit();
        }
        if (strcmp(argv[i], "--debug") == 0 || strcmp(argv[i], "-d") == 0) {
            force_debug = true;
            force_cli = true;
        } else if (strcmp(argv[i], "--cli") == 0 || strcmp(argv[i], "-c") == 0) {
            force_cli = true;
        } else if (strcmp(argv[i], "--gui") == 0 ||
                   strcmp(argv[i], "-g") == 0) {
            force_gui = true;
        } else if (strcmp(argv[i], "--data-root") == 0 || strcmp(argv[i], "-r") == 0) {
            if (i + 1 < argc) {
                path_finder_set_data_root(argv[++i]);
            } else {
                log_fallback_record(
                    LOG_SEV_ERROR,
                    LOG_CAT_LOG,
                    "BootCLI",
                    -0.5f,
                    __FILE__,
                    __LINE__,
                    __func__,
                    "CLI argument --data-root missing directory parameter"
                );
                return 1;
            }
        } else if (strncmp(argv[i], "--data-root=", 12) == 0) {
            path_finder_set_data_root(argv[i] + 12);
        } else {
            log_fallback_record(
                LOG_SEV_ERROR,
                LOG_CAT_LOG,
                "BootCLI",
                -0.25f,
                __FILE__,
                __LINE__,
                __func__,
                "Unknown CLI option '%s'; use --help for usage",
                argv[i]
            );
            return 1;
        }
    }

    /* Step 3: Initialize forensic logging & debugging (auto-escalate to Safe Mode if needed) */
    if (!log_init(NULL)) {
        log_init_emergency_monolith();
    }
    debug_init(NULL);
    LOG_INFO(LOG_CAT_LOG, "Engine", 0.0f, "Tenebris and Lux boot initialized (mode: %s%s)",
             force_cli ? "CLI" : (force_gui ? "GUI" : "Auto"),
             force_debug ? " [DEBUG-ACTIVE]" : "");

    /* Step 4: Run Pre-Flight ("Don't start machine with loose bolts") */
    manifest_registry_t registry;
    init_engine_manifest(&registry);
    preflight_status_t preflight = manifest_run_preflight(&registry);

    if (preflight == PREFLIGHT_STATUS_HALT_FATAL) {
        LOG_FATAL(LOG_CAT_LOG, "Engine", -1.0f, "Engine startup halted due to loose bolts!");
```

```
        fprintf(stderr,
                "\n[FATAL ERROR] Engine startup halted due to loose bolts!\n");
        manifest_print_report(&registry);
        debug_shutdown();
        log_shutdown();
        return 1;
    }

    /* Step 5: Initialize simulation player state */
    player_state_t state;
    game_init(&state);

    /* Record initial genesis snapshot */
    game_record_debug_snapshot(&state, "init", "genesis");

    /* Step 6: Route explicit Debug / CLI launch request */
    if (force_debug) {
        printf("============================================================\n");
        printf("   TENEBRIS AND LUX - INTERACTIVE DEBUGGER ACTIVE\n");
        printf("   Type 'debug help' or 'inspect' during gameplay\n");
        printf("============================================================\n\n");
        LOG_INFO(LOG_CAT_DEBUG, "Engine", 0.0f, "Entering interactive debug mode in CLI");
        game_loop_cli(&state);
        debug_shutdown();
        log_shutdown();
        return 0;
    }

    if (force_cli) {
        LOG_INFO(LOG_CAT_LOG, "Engine", 0.0f, "Entering terminal CLI game loop");
        game_loop_cli(&state);
        debug_shutdown();
        log_shutdown();
        return 0;
    }

    /* Step 7: Route explicit GUI launch request */
    if (force_gui) {
        if (!gui_is_available()) {
            LOG_ERROR(LOG_CAT_LOG, "Engine", -0.5f, "No graphical X11 display available for --gui");
            fprintf(stderr,
                    "Error: No graphical X11 display available for --gui.\n");
            debug_shutdown();
            log_shutdown();
            return 1;
        }
        LOG_INFO(LOG_CAT_LOG, "Engine", 0.0f, "Launching native X11 GUI");
        int rc = gui_run(&state);
        debug_shutdown();
        log_shutdown();
        return rc;
    }

    /* Step 8: Graded policy fallback: downgrade to CLI if display missing */
    if (preflight == PREFLIGHT_STATUS_DOWNGRADED || !gui_is_available()) {
        LOG_WARN(LOG_CAT_LOG, "Engine", -0.2f, "Graphical display unavailable, downgrading to CLI");
        printf("[Notice] Graphical display unavailable. "
               "Downgrading to CLI safe mode.\n\n");
        game_loop_cli(&state);
        debug_shutdown();
        log_shutdown();
        return 0;
    }

    /* Step 9: Default presentation: launch native GUI */
    LOG_INFO(LOG_CAT_LOG, "Engine", 0.0f, "Launching native X11 GUI default presentation");
    int rc = gui_run(&state);
    debug_shutdown();
    log_shutdown();
    return rc;
}
```

# A.3. Build-Time Compiled .rodata Config ROM Generator

**Snapshot ID**: snap_cmake_rom_whole | **Mode**: Whole | **File**: cmake/GenerateConfigRom.cmake (L1–L302 of 302 lines) | Jump back to chapter text

```
# ============================================================================
# BLOCK 1: METADATA BLOCK
# ( Identity, Provenance, Scaling Telemetry & Architectural Boundary Contracts )
# ============================================================================
# @project       Tenebris and Lux
# @file          cmake/GenerateConfigRom.cmake
# @id            build.rom.config
# @layer         Layer 0 Infrastructure (Build Tooling)
# @archetype     operational | stateless
# @version       0.1.1-dev
# @interface_rev Rev 5
# @date          2026-11-11
# @author        CreativeWorkzStudio LLC / ProfessorSeanEX
# @repository    https://github.com/Novaking34/tenebris-lux
# @system        Universal Cornerstone Configuration ROM Embedder
# @role          Build-Time Script: Encodes canonical JSON manifests and presets
#                under engine/config/<subdir> into immutable ISO C11 .rodata
#                byte-array headers for zero-disk Safe-Mode BIOS boot.
# @maturity      Rev 5 Baseline | Generalized multi-subdir byte-array ROM compiler.
# @chassis       zero-heap | Emits pure static const .rodata C tables; zero disk I/O at runtime.
# @scale         O(F * B) compile-time hex conversion over F sorted JSON configs and B bytes.
# @seams         in: engine/config/**/*.json | uses: CMake file(READ HEX) | probes: Block 4 script guards
# @invariants    1. Pure ISO C11/C++17 byte arrays (avoids -Woverlength-strings).
#                2. Strictly ISO C-style /* ... */ comments only in emitted headers.
#                3. Deterministic alphabetical ordering across subdirectories and JSON stems.
#                4. Collision-free C symbol identifiers across multi-subdir vaults.
#                5. Content-preserving emission via cmake -E copy_if_different.
#
# PURPOSE & ARCHITECTURAL ROLE:
#   Generates compiled-in `.rodata` ROM vault headers directly from canonical
#   engine JSON configurations so `log_fallback`, `debug_fallback`, and
#   `io_fallback` retain structured Safe-Mode presets when disk files are absent.
#
#   ### For Game Designers & Non-Programmers:
#   Automatically bakes emergency read-only copies of JSON presets inside the
#   compiled engine binary every time `engine/config/` presets change, keeping
#   Safe-Mode fallbacks aligned without manual editing.
#
#   ### For Systems Programmers & Engine Architects:
#   Script-mode CMake generator (`cmake -P`) converting raw hex nibbles into
#   null-terminated 10-byte-per-line `static const unsigned char` arrays paired
#   with indexed lookup metadata structs.
# ============================================================================

# ============================================================================
# BLOCK 2: OPENING BLOCK
# ( Pre-Conditions, Default Contracts, Patterns & Script Accumulators )
# ============================================================================
# Engine:        Cornerstone Game Engine
# Stratum:       Layer 0 Infrastructure (Build Tooling)
# Module:        build.rom.config
# Mission:       Validate input arguments, establish symbol defaults, and
#                initialize deterministic ROM generation accumulators.
# ============================================================================

# ----------------------------------------------------------------------------
# Sub-Assembly 2.1: Parameter Pre-Conditions & Caller Contract Validation
# ----------------------------------------------------------------------------
if(NOT DEFINED CONFIG_ROOT OR NOT DEFINED SUBDIRS OR NOT DEFINED OUTPUT_FILE)
    message(FATAL_ERROR
        "GenerateConfigRom.cmake requires -DCONFIG_ROOT=<dir>, -DSUBDIRS=<list>, and -DOUTPUT_FILE=<file>")
endif()

if(NOT IS_DIRECTORY "${CONFIG_ROOT}")
    message(FATAL_ERROR
        "GenerateConfigRom.cmake: CONFIG_ROOT does not exist: '${CONFIG_ROOT}'")
endif()

```

```
# ------------------------------------------------------------------------------
# Sub-Assembly 2.2: Default Symbol Contract Bindings
# ------------------------------------------------------------------------------
if(NOT DEFINED HEADER_GUARD)
    set(HEADER_GUARD "CORNERSTONE_CONFIG_ROM_H")
endif()
if(NOT DEFINED SYMBOL_PREFIX)
    set(SYMBOL_PREFIX "CONFIG_ROM")
endif()
if(NOT DEFINED STRUCT_TAG)
    set(STRUCT_TAG "config_rom_asset")
endif()
if(NOT DEFINED TYPE_NAME)
    set(TYPE_NAME "config_rom_asset_t")
endif()
if(NOT DEFINED TABLE_NAME)
    set(TABLE_NAME "s_config_rom_table")
endif()
if(NOT DEFINED VAR_PREFIX)
    set(VAR_PREFIX "s_rom_")
endif()
if(NOT DEFINED BANNER_TITLE)
    set(BANNER_TITLE "AUTO-GENERATED ENGINE CONFIG ROM IMAGE (DO NOT EDIT MANUALLY)")
endif()
if(NOT DEFINED ID_MODE)
    set(ID_MODE "SUBDIR_STEM")
endif()

# ------------------------------------------------------------------------------
# Sub-Assembly 2.3: Hex Formatting Constants & Accumulation Buffers
# ------------------------------------------------------------------------------
set(_byte_pat "0x[0-9a-f][0-9a-f], ")
set(_ten_pat
    "(${_byte_pat}${_byte_pat}${_byte_pat}${_byte_pat}${_byte_pat}${_byte_pat}${_byte_pat}${_byte_pat}${_byte_pat}${_byte_pat})"
)

set(_arrays_code "")
set(_table_entries "")
set(_total_raw_bytes 0)
set(_rom_file_count 0)
set(_subdirs_doc "")
set(_seen_c_idents "")
set(_seen_asset_ids "")

# ==============================================================================
# BLOCK 3: BODY BLOCK
# ( Deterministic Discovery, Hex-to-C Encoding & Table Construction )
# ==============================================================================
# Engine:        Cornerstone Game Engine
# Stratum:       Layer 0 Infrastructure (Build Tooling)
# Module:        build.rom.config
# Mission:       Traverse sorted configuration domains and encode every JSON
#                asset into null-terminated .rodata C arrays and table rows.
# ==============================================================================

# ------------------------------------------------------------------------------
# Sub-Assembly 3.1: Sorted Domain Traversal & Identifier Mapping
# ------------------------------------------------------------------------------
set(_sorted_subdirs ${SUBDIRS})
list(SORT _sorted_subdirs)

foreach(_subdir IN LISTS _sorted_subdirs)
    set(_abs_dir "${CONFIG_ROOT}/${_subdir}")
    if(NOT IS_DIRECTORY "${_abs_dir}")
        message(FATAL_ERROR
            "GenerateConfigRom.cmake: Subdirectory does not exist: '${_abs_dir}'")
    endif()

    file(GLOB _dir_jsons "${_abs_dir}/*.json")
    list(SORT _dir_jsons)
    list(LENGTH _dir_jsons _dir_count)

    if(_dir_count EQUAL 0)
        message(FATAL_ERROR
            "GenerateConfigRom.cmake: No JSON files found in '${_abs_dir}'")
```

```
    endif()

    string(APPEND _subdirs_doc "  engine/config/${_subdir}/ (${_dir_count} assets)\n")
    get_filename_component(_subdir_leaf "${_subdir}" NAME)

    # ---------------------------------------------------------------------------
    # Sub-Assembly 3.2: Deterministic Byte-Array Encoding & Table Row Emission
    # ---------------------------------------------------------------------------
    foreach(_json_path IN LISTS _dir_jsons)
        get_filename_component(_stem "${_json_path}" NAME_WE)
        get_filename_component(_filename "${_json_path}" NAME)

        if(ID_MODE STREQUAL "STEM")
            set(_asset_id "${_stem}")
            string(REGEX REPLACE "[^A-Za-z0-9_]" "_" _c_ident "${_stem}")
        else()
            set(_asset_id "${_subdir_leaf}/${_stem}")
            string(REGEX REPLACE "[^A-Za-z0-9_]" "_" _c_ident "${_subdir_leaf}_${_stem}")
        endif()

        list(FIND _seen_c_idents "${_c_ident}" _dup_c_idx)
        if(NOT _dup_c_idx EQUAL -1)
            message(FATAL_ERROR
                "GenerateConfigRom.cmake: Duplicate C identifier '${_c_ident}' from '${_json_path}'")
        endif()
        list(APPEND _seen_c_idents "${_c_ident}")

        list(FIND _seen_asset_ids "${_asset_id}" _dup_id_idx)
        if(NOT _dup_id_idx EQUAL -1)
            message(FATAL_ERROR
                "GenerateConfigRom.cmake: Duplicate ROM asset ID '${_asset_id}' from '${_json_path}'")
        endif()
        list(APPEND _seen_asset_ids "${_asset_id}")

        file(READ "${_json_path}" _hex_content HEX)
        string(LENGTH "${_hex_content}" _hex_len)
        math(EXPR _hex_mod "${_hex_len} % 2")
        if(NOT _hex_mod EQUAL 0)
            message(FATAL_ERROR
                "GenerateConfigRom.cmake: Odd hex length (${_hex_len}) reading '${_json_path}'")
        endif()

        math(EXPR _byte_len "${_hex_len} / 2")
        math(EXPR _total_raw_bytes "${_total_raw_bytes} + ${_byte_len}")
        math(EXPR _rom_file_count "${_rom_file_count} + 1")

        if(_byte_len GREATER 0)
            string(REGEX REPLACE "([0-9a-f][0-9a-f])" "0x\\1, " _c_hex "${_hex_content}")
            set(_c_hex "${_c_hex}0x00")
            string(REGEX REPLACE "${_ten_pat}" "    \\1\n" _c_lines "${_c_hex}")
            string(REGEX REPLACE "\n(0x)" "\n    \\1" _c_lines "${_c_lines}")
            string(REGEX REPLACE " +\n" "\n" _c_lines "${_c_lines}")
            if(NOT "${_c_lines}" MATCHES "^    ")
                set(_c_lines "    ${_c_lines}")
            endif()
        else()
            set(_c_lines "    0x00")
        endif()

        string(APPEND _arrays_code
            "/* Embedded ROM Asset: ${_subdir}/${_filename} (${_byte_len} bytes) */\n"
            "static const unsigned char ${VAR_PREFIX}${_c_ident}[] = {\n"
            "${_c_lines}\n"
            "};\n\n"
        )

        string(APPEND _table_entries
            "    {\n"
            "        \"${_asset_id}\",\n"
            "        \"${_subdir}/${_filename}\",\n"
            "        ${VAR_PREFIX}${_c_ident},\n"
            "        ${_byte_len}U\n"
            "    },\n"
        )
    endforeach()
```

```
endforeach()

# ============================================================================
# BLOCK 4: AUDIT PROCEDURES BLOCK
# ( Build-Time Integrity Gates, Asset Parity & C++ Compatibility Synthesis )
# ============================================================================
# Engine:        Cornerstone Game Engine
# Stratum:       Layer 0 Infrastructure (Build Tooling)
# Module:        build.rom.config
# Mission:       Verify non-empty ROM image construction and assemble a clean
#                C11/C++17-compatible generated header unit.
# ============================================================================

# ----------------------------------------------------------------------------
# Sub-Assembly 4.1: Pre-Flight Asset Count & Header Synthesis Checks
# ----------------------------------------------------------------------------
if(_rom_file_count LESS_EQUAL 0)
    message(FATAL_ERROR
        "GenerateConfigRom.cmake: Produced 0 ROM assets across subdirs '${SUBDIRS}'")
endif()

set(_out_header "")
string(APPEND _out_header
    "/*\n"
    "================================================================================\n"
    "${BANNER_TITLE}\n"
    "Generated by cmake/GenerateConfigRom.cmake from:\n"
    "${_subdirs_doc}"
    "Assets: ${_rom_file_count} | Raw JSON Bytes: ${_total_raw_bytes}\n"
    "================================================================================\n"
    "*/\n"
    "#ifndef ${HEADER_GUARD}\n"
    "#define ${HEADER_GUARD}\n\n"
    "#include <stddef.h>\n\n"
    "#ifdef __cplusplus\n"
    "extern \"C\" {\n"
    "#endif\n\n"
    "#define ${SYMBOL_PREFIX}_ASSET_COUNT ${_rom_file_count}U\n"
    "#define ${SYMBOL_PREFIX}_TOTAL_BYTES ${_total_raw_bytes}U\n\n"
    "typedef struct ${STRUCT_TAG} {\n"
    "    const char          *id;\n"
    "    const char          *rel_path;\n"
    "    const unsigned char *data;\n"
    "    size_t               size;\n"
    "} ${TYPE_NAME};\n\n"
    "${_arrays_code}"
    "static const ${TYPE_NAME} ${TABLE_NAME}[${_rom_file_count}] = {\n"
    "${_table_entries}"
    "};\n\n"
    "#ifdef __cplusplus\n"
    "}\n"
    "#endif\n\n"
    "#endif /* ${HEADER_GUARD} */\n"
)

# ============================================================================
# BLOCK 5: CLOSING BLOCK
# ( Atomic Header Emission, Timestamp Preservation & Workspace Cleanup )
# ============================================================================
# Engine:        Cornerstone Game Engine
# Stratum:       Layer 0 Infrastructure (Build Tooling)
# Module:        build.rom.config
# Mission:       Write generated header atomically via copy_if_different to avoid
#                spurious incremental rebuilds when JSON bytes are unchanged.
# ============================================================================

# ----------------------------------------------------------------------------
# Sub-Assembly 5.1: Content-Preserving Atomic Write & Temporary File Cleanup
# ----------------------------------------------------------------------------
get_filename_component(_out_dir "${OUTPUT_FILE}" DIRECTORY)
file(MAKE_DIRECTORY "${_out_dir}")

set(_tmp_file "${OUTPUT_FILE}.tmp")
file(WRITE "${_tmp_file}" "${_out_header}")
execute_process(
```

```
    COMMAND "${CMAKE_COMMAND}" -E copy_if_different "${_tmp_file}" "${OUTPUT_FILE}"
    RESULT_VARIABLE _copy_rc
)
file(REMOVE "${_tmp_file}")

if(NOT _copy_rc EQUAL 0)
    message(FATAL_ERROR
        "GenerateConfigRom.cmake: Failed copying generated header to '${OUTPUT_FILE}' (rc=${_copy_rc})")
endif()
```

## A.4. Tier-0 I/O Safe-Mode BIOS Boot, Floppy Floor & Downgrade Latch

**Snapshot ID**: snap_io_fallback_bios_group | **Mode**: Group | **File**: engine/src/io/io_fallback.c (L2908–L3332 of 3355 lines) | Jump back to chapter text

```
/**
 * @brief Downgrade the BIOS I/O subsystem into Safe Mode (`FLOPPY_RO` or `ROM_ONLY`).
 *
 *   ### For Game Designers & Non-Programmers:
 *   Safely protects storage whenever external folders fail or policies break: locks out
 *   unsafe writes if requested, continues serving factory ROM data, and logs the event.
 *
 *   ### For Systems Programmers & Engine Architects:
 *   Arms BIOS, increments `s_io_fb.safe_downgrades`, switches `active_sink` to
 *   `IO_FB_SINK_FLOPPY_RO` or `IO_FB_SINK_ROM_ONLY`, and forwards telemetry into both
 *   `io_fallback_record_fault()` and `debug_fallback_record_fault()`.
 *
 * @param[in] reason         Concise Safe-Mode transition reason, or NULL for generic default.
 * @param[in] lock_read_only `true` to engage read-only floppy latch; `false` for `ROM_ONLY`.
 * @return `true` after Safe-Mode transition and telemetry recording complete.
 * @triggers Increments `safe_downgrades`, updates `active_sink`, and notifies debug BIOS.
 */
bool io_fallback_downgrade_safe_mode(const char *reason, bool lock_read_only)
{
    /* Step 1: Arm BIOS, count Safe-Mode downgrade, and transition active sink */
    io_fb_auto_ensure_booted();
    s_io_fb.safe_downgrades++;
    if (lock_read_only) {
        s_io_fb.config.read_only_mode = true;
        s_io_fb.active_sink = IO_FB_SINK_FLOPPY_RO;
    } else if (s_io_fb.active_sink == IO_FB_SINK_OFFLINE) {
        s_io_fb.active_sink = IO_FB_SINK_ROM_ONLY;
    }

    /* Step 2: Emit dual fault telemetry across I/O BIOS ring and debug BIOS ring */
    const char *msg = (reason != NULL) ? reason : "Engaged Safe-Mode I/O BIOS";
    io_fallback_record_fault("io_fallback", "downgrade_safe_mode", msg, IO_ERR_FILE_NOT_FOUND);
    (void)debug_fallback_record_fault("io_fallback", msg, __FILE__, __LINE__,
                                      "TW_ENG_IO_ROM_INTEGRITY");
    return true;
}
```

```
bool io_fallback_boot(const io_fallback_config_t *opt_cfg)
{
    io_fallback_config_t selected = s_io_fallback_template;
    bool hydrated = false;

    if (opt_cfg != NULL && io_fallback_validate_policy_contract(opt_cfg)) {
        selected = *opt_cfg;
        selected.loaded_from_data = false;
    } else if (io_fallback_load_rom_config("default", &selected)) {
        hydrated = true;
    }

    s_io_fb.config = selected;
    s_io_fb.rom_hydrated = hydrated;
    io_fb_hydrate_subpolicies();
    str_safe_copy(s_io_fb.active_root,
                  selected.default_root[0] ? selected.default_root : IO_FB_DEFAULT_ROOT,
                  sizeof(s_io_fb.active_root));

```

```
    if (selected.read_only_mode) {
        s_io_fb.active_sink = IO_FB_SINK_FLOPPY_RO;
    } else {
        s_io_fb.active_sink = IO_FB_SINK_DISK_ACTIVE;
    }

    s_io_fb.booted = true;
    io_fallback_register_debug_probes();
    return true;
}

bool io_fallback_boot_floppy(void)
{
    io_fallback_config_t floppy = s_io_floppy_template;
    if (io_fallback_load_rom_config("floppy_safe", &floppy) ||
        io_fallback_load_rom_config("default", &floppy)) {
        s_io_fb.rom_hydrated = true;
    } else {
        s_io_fb.rom_hydrated = false;
    }
    floppy.read_only_mode = true;
    floppy.loaded_from_data = false;
    if (floppy.profile_name[0] == '\0' || strcmp(floppy.profile_name, "default") == 0) {
        str_safe_copy(floppy.profile_name, IO_FALLBACK_FLOPPY_PROFILE_NAME,
                      sizeof(floppy.profile_name));
    }

    s_io_fb.config = floppy;
    io_fb_hydrate_subpolicies();
    s_io_fb.active_sink = IO_FB_SINK_FLOPPY_RO;
    str_safe_copy(s_io_fb.active_root, IO_FB_DEFAULT_ROOT, sizeof(s_io_fb.active_root));
    s_io_fb.booted = true;
    io_fallback_register_debug_probes();
    return true;
}
```

## A.5. Autonomous Validation Probes & Tripwire Trap Chain

**Snapshot ID**: snap_debug_probe_group | **Mode**: Group | **File**: engine/src/debug/debug.c (L1918–L2438 of 3221 lines) | Jump back to chapter text

```
/**
 * @brief Register or idempotently update a Block 4 subsystem validation probe.
 *
 *   ### For Game Designers & Non-Programmers:
 *   Plugs a subsystem's automated health monitor into the debugger so anomalies
 *   immediately report exact failing checks and test names.
 *
 *   ### For Systems Programmers & Engine Architects:
 *   Updates an existing `probe_id` slot in place while preserving its historical
 *   `trigger_count`, or claims the next free slot in `s_debug.probes[]`. Logs a
 *   guarded `LOG_WARN` if `DEBUG_MAX_VALIDATION_PROBES` capacity is exhausted.
 *
 * @param[in] probe Descriptor defining probe ID, subsystem tag, callback, and reaction.
 * @return Slot index `[0, DEBUG_MAX_VALIDATION_PROBES - 1]` on success; `-1` if null,
 *         empty `probe_id`, or probe registry is saturated.
 * @triggers Validates input guard and emits guarded `LOG_WARN` on registry saturation.
 */
int debug_register_validation_probe(const debug_validation_probe_t *probe)
{
    /* Step 1: Reject null descriptors or empty probe identifiers */
    if (probe == NULL || probe->probe_id[0] == '\0') {
        return -1;
    }

    /* Step 2: Update existing slot idempotently if probe_id already registered */
    for (size_t i = 0U; i < s_debug.probe_count; i++) {
        if (strcmp(s_debug.probes[i].probe_id, probe->probe_id) == 0) {
            uint32_t prev_count = s_debug.probes[i].trigger_count;
            s_debug.probes[i] = *probe;
            s_debug.probes[i].is_active = true;
            s_debug.probes[i].trigger_count = prev_count;
```

```
            return (int)i;
        }
    }

    /* Step 3: Guard static probe registry ceiling and log saturation telemetry */
    if (s_debug.probe_count >= DEBUG_MAX_VALIDATION_PROBES) {
        if (s_debug_active && !s_debug.in_audit && !s_debug.in_probe) {
            s_debug.in_probe = true;
            LOG_WARN(LOCAL_LOG_CAT, LOCAL_MODULE_TAG, -0.4f,
                     "Probe registry saturated (%zu/%d); rejected '%s'",
                     s_debug.probe_count, DEBUG_MAX_VALIDATION_PROBES, probe->probe_id);
            s_debug.in_probe = false;
        }
        return -1;
    }

    /* Step 4: Claim next free slot in `.bss` and activate probe */
    size_t slot = s_debug.probe_count++;
    s_debug.probes[slot] = *probe;
    s_debug.probes[slot].is_active = true;
    return (int)slot;
}
```

```
/**
 * @brief Trigger a validation probe tripwire, downgrade safely, and enter DEBUG_STEP_AWAIT.
 *
 *   ### For Game Designers & Non-Programmers:
 *   Handles a live safety tripwire cleanly: confirms which check failed, lets
 *   Safe Mode preserve critical data, freezes the offending frame, and prints an
 *   actionable diagnostic packet rather than crashing abruptly.
 *
 *   ### For Systems Programmers & Engine Architects:
 *   Non-reentrant (`in_probe`) pipeline: resolves matched probe defaults, runs optional
 *   `validate_fn()`, increments `traps_triggered`/`probe_failures`, notifies BIOS via
 *   `debug_fallback_record_fault()` and `debug_fallback_finish_with_fallback()`, locks the
 *   latest snapshot if present, arms `DEBUG_STEP_AWAIT` (when active and non-audit),
 *   populates `s_debug.last_report`, and emits a guarded `LOG_ERROR`.
 *
 * @param[in] probe_id  Registered probe identifier, or NULL for ad-hoc tripwire.
 * @param[in] subsystem Originating module identifier override, or NULL to inherit probe/module default.
 * @param[in] condition Human-readable invariant failure description, or NULL for callback/default message.
 * @param[in] file      Source file name captured at trap call site (`__FILE__`).
 * @param[in] line      Source line number captured at trap call site (`__LINE__`).
 * @param[in] func      Enclosing C function name (`__func__`).
 * @return `true` when latched and Safe-Mode downgrade completed; `false` if suppressed due
 *         to active `in_probe` recursion guard.
 * @triggers Enforces reentrancy guard, Safe-Mode downgrade, snapshot pinning, and AWAIT latch.
 */
bool debug_trigger_probe(
    const char *probe_id,
    const char *subsystem,
    const char *condition,
    const char *file,
    int         line,
    const char *func
) {
    /* Step 1: Prevent reentrant tripwire loops while handling an active probe */
    if (s_debug.in_probe) {
        return false;
    }

    s_debug.in_probe = true;

    /* Step 2: Locate registered probe metadata if present and merge fallback overrides */
    debug_validation_probe_t *matched = NULL;
    if (probe_id != NULL && probe_id[0] != '\0') {
        for (size_t i = 0U; i < s_debug.probe_count; i++) {
            if (s_debug.probes[i].is_active &&
                strcmp(s_debug.probes[i].probe_id, probe_id) == 0) {
                matched = &s_debug.probes[i];
                matched->trigger_count++;
                break;
            }
        }
```

```
    }

    const char *eff_sub = (subsystem != NULL && subsystem[0] != '\0')
                              ? subsystem
                              : ((matched != NULL && matched->subsystem[0] != '\0')
                                     ? matched->subsystem
                                     : MODULE_ID);
    const char *eff_cond = (condition != NULL && condition[0] != '\0')
                               ? condition
                               : "Validation tripwire raised";
    debug_tripwire_t eff_wire = (matched != NULL && matched->tripwire_code != DEBUG_TRIPWIRE_CLEAR)
                                    ? matched->tripwire_code
                                    : DEBUG_TRIPWIRE_TEST_FAILED;
    debug_tripwire_reaction_t eff_react = (matched != NULL)
                                              ? matched->reaction
                                              : dbg_default_reaction_for_wire(eff_wire);
    debug_test_tier_t eff_tier = (matched != NULL) ? matched->tier : DEBUG_TEST_TIER_AUDIT;
    const char *eff_sec = (matched != NULL && matched->source_section[0] != '\0')
                              ? matched->source_section
                              : ((func != NULL) ? func : "unknown_section");
    const char *eff_test = (matched != NULL && matched->test_suite_ref[0] != '\0')
                               ? matched->test_suite_ref
                               : "BLOCK4_PREFLIGHT";
    bool want_await = (matched != NULL) ? matched->auto_await : s_debug.config.auto_await_on_tripwire;

    /* Step 3: Invoke Block 4 validation callback when registered to confirm health */
    char probe_detail_buf[128];
    if (matched != NULL && matched->validate_fn != NULL) {
        diag_result_t probe_res = matched->validate_fn();
        if (!probe_res.passed && probe_res.details != NULL && condition == NULL) {
            eff_cond = probe_res.details;
        }
        snprintf(probe_detail_buf, sizeof(probe_detail_buf), "%s [probe=%s]",
                 eff_cond, probe_res.passed ? "SELF_OK_BY_FALLBACK" : "FAIL_CONFIRMED");
    } else {
        str_safe_copy(probe_detail_buf, eff_cond, sizeof(probe_detail_buf));
    }

    /* Step 4: Latch primary debugger tripwire status and forensic failure counters */
    s_debug.tripwire_status = eff_wire;
    s_debug.traps_triggered++;
    s_debug.probe_failures++;

    snprintf(s_debug.tripwire_details, sizeof(s_debug.tripwire_details),
             "[%s:%s] %s (%s:%d | %s)",
             eff_sub,
             debug_test_tier_name(eff_tier),
             probe_detail_buf,
             (file != NULL) ? file : "unknown",
             line,
             eff_test);

    /* Step 5: Record into Tier-0 BIOS & downgrade to complete safely in fallback */
    (void)debug_fallback_record_fault(eff_sub, probe_detail_buf, file, line, eff_test);

    char fb_action[96];
    snprintf(fb_action, sizeof(fb_action),
             "Reaction=%s | Safe-Mode completed execution -> AWAIT",
             debug_tripwire_reaction_name(eff_react));
    (void)debug_fallback_finish_with_fallback(eff_sub, fb_action);

    s_debug.safe_mode = true;
    s_debug.fallback_completed = true;

    /* Step 6: Pin critical forensic snapshot in circular ring if snapshots exist */
    uint64_t pinned_id = 0ULL;
    bool snap_locked = false;
    if (s_debug_active && s_debug.snapshot_count > 0U &&
        debug_lock_snapshot(0ULL, "TRIGGER_TRIPWIRE_LATCH")) {
        const debug_snapshot_t *latest = debug_get_latest_snapshot();
        if (latest != NULL) {
            pinned_id = latest->snapshot_id;
            snap_locked = true;
        }
    }
```

```
2258
2259     /* Step 7: Enter DEBUG_STEP_AWAIT when debugger active, non-audit, and armed */
2260     bool engaged_await = false;
2261     if (s_debug_active && !s_debug.in_audit &&
2262         s_debug.config.break_on_tripwire && want_await) {
2263         s_debug.step_mode = DEBUG_STEP_AWAIT;
2264         s_debug.await_active = true;
2265         engaged_await = true;
2266     }
2267
2268     /* Step 8: Populate structured forensic report and emit guarded error telemetry */
2269     memset(&s_debug.last_report, 0, sizeof(s_debug.last_report));
2270     s_debug.last_report.tripwire_status = eff_wire;
2271     s_debug.last_report.reaction = eff_react;
2272     s_debug.last_report.tier = eff_tier;
2273     s_debug.last_report.fallback_completed = true;
2274     s_debug.last_report.await_active = engaged_await;
2275     s_debug.last_report.snapshot_locked = snap_locked;
2276     s_debug.last_report.locked_snapshot_id = pinned_id;
2277     str_safe_copy(s_debug.last_report.probe_id,
2278                   (probe_id != NULL) ? probe_id : "ad_hoc.tripwire",
2279                   sizeof(s_debug.last_report.probe_id));
2280     str_safe_copy(s_debug.last_report.subsystem, eff_sub, sizeof(s_debug.last_report.subsystem));
2281     str_safe_copy(s_debug.last_report.condition, probe_detail_buf,
2282                   sizeof(s_debug.last_report.condition));
2283     snprintf(s_debug.last_report.source_location, sizeof(s_debug.last_report.source_location),
2284              "%s:%d", (file != NULL) ? file : "unknown", line);
2285     str_safe_copy(s_debug.last_report.source_section, eff_sec,
2286                   sizeof(s_debug.last_report.source_section));
2287     str_safe_copy(s_debug.last_report.test_suite_ref, eff_test,
2288                   sizeof(s_debug.last_report.test_suite_ref));
2289     str_safe_copy(s_debug.last_report.fallback_action, fb_action,
2290                   sizeof(s_debug.last_report.fallback_action));
2291
2292     if (s_debug_active && !s_debug.in_audit) {
2293         LOG_ERROR(LOCAL_LOG_CAT, LOCAL_MODULE_TAG, -0.75f,
2294                   "Probe tripwire latched: %s", s_debug.tripwire_details);
2295     }
2296
2297     s_debug.in_probe = false;
2298     return true;
2299 }
```

```
2367 /**
2368  * @brief Legacy compatibility helper that latches a state-corruption tripwire trap.
2369  *
2370  * @deprecated Prefer `debug_trigger_probe()` or `DEBUG_TRIGGER_PROBE()` so every trap
2371  *             carries a structured probe ID, tier classification, and forensic report.
2372  *
2373  *   ### For Game Designers & Non-Programmers:
2374  *   Preserved safety brake for older modules that report state errors without
2375  *   naming a specific modern probe identifier.
2376  *
2377  *   ### For Systems Programmers & Engine Architects:
2378  *   Latches `DEBUG_TRIPWIRE_STATE_CORRUPT`, increments `traps_triggered`, notifies
2379  *   BIOS via `debug_fallback_record_fault()`, performs guarded snapshot lock and
2380  *   `DEBUG_STEP_AWAIT` entry when active and non-audit, and emits `LOG_WARN`.
2381  *
2382  * @param[in] subsystem Subsystem tag reporting the fault, or NULL for `"unknown"`.
2383  * @param[in] condition Description of the violated condition, or NULL for `"anomaly"`.
2384  * @param[in] file      Originating source file (`__FILE__`).
2385  * @param[in] line      Originating source line (`__LINE__`).
2386  * @triggers Latches tripwire state, records BIOS fault, and warns via guarded log output.
2387  */
2388 void debug_trap_tripwire(const char *subsystem, const char *condition,
2389                          const char *file, int line)
2390 {
2391     /* Step 1: Latch classic STATE_CORRUPT code and forensic detail string verbatim */
2392     s_debug.tripwire_status = DEBUG_TRIPWIRE_STATE_CORRUPT;
2393     s_debug.traps_triggered++;
2394
2395     snprintf(s_debug.tripwire_details, sizeof(s_debug.tripwire_details),
2396              "[%s] %s (%s:%d)",
2397              (subsystem != NULL) ? subsystem : "unknown",
```

```
            (condition != NULL) ? condition : "anomaly",
            (file != NULL) ? file : "unknown", line);

    /* Step 2: Record fault in Tier-0 BIOS and complete via Safe-Mode fallback */
    (void)debug_fallback_record_fault(
        subsystem,
        (condition != NULL) ? condition : "anomaly",
        file,
        line,
        "debug_trap_tripwire"
    );

    if (s_debug_active && !s_debug.in_probe && !s_debug.in_audit) {
        (void)debug_fallback_finish_with_fallback(
            (subsystem != NULL) ? subsystem : MODULE_ID,
            "Completed trap safely via Safe-Mode fallback -> AWAIT"
        );
        s_debug.safe_mode = true;
        s_debug.fallback_completed = true;

        if (s_debug.snapshot_count > 0U) {
            (void)debug_lock_snapshot(0ULL, "TRIGGER_TRIPWIRE_LATCH");
        }
        if (s_debug.config.break_on_tripwire && s_debug.config.auto_await_on_tripwire) {
            s_debug.step_mode = DEBUG_STEP_AWAIT;
            s_debug.await_active = true;
        }
    }

    /* Step 3: Emit guarded telemetry warning without inducing recursive log traps */
    if (!s_debug.in_probe && !s_debug.in_audit) {
        s_debug.in_probe = true;
        LOG_WARN(LOCAL_LOG_CAT, LOCAL_MODULE_TAG, -0.5f,
                 "Tripwire trapped: %s", s_debug.tripwire_details);
        s_debug.in_probe = false;
    }

    if (s_debug_active && s_debug.config.auto_dump_on_tripwire && !s_debug.in_audit) {
        fprintf(stderr, "\n[DEBUG TRIPWIRE ALERT] %s\n", s_debug.tripwire_details);
    }
}
```

## A.6. Flight Log Subsystem Probe Registration & Validation Guards (Sub-Assembly 4.1)

**Snapshot ID**: snap_log_sub4_1_probes | **Mode**: Subassembly | **File**: engine/src/log/log.c (L1934–L2028 of 2833 lines) | Jump back to chapter text

```
/*
-------------------------------------------------------------------------------
Sub-Assembly 4.1: Contract, Input & Cog Verification Cogs
Mission: Verify received parameters, configuration contracts, and data provenance;
verify that declared subsystem cogs run properly as standalone units.
-------------------------------------------------------------------------------
*/

/**
 * @brief Returns whether the active logging subsystem was configured from data sources.
 *
 * ### For Game Designers & Non-Programmers:
 * Checks if settings were loaded from actual JSON files rather than emergency defaults.
 *
 * ### For Systems Programmers & Engine Architects:
 * Queries provenance boolean s_log.config.loaded_from_data.
 *
 * @return true if loaded from JSON data; false if using fallback code page.
 */
bool log_is_data_driven(void)
{
    return s_log.config.loaded_from_data;
}
```

```
1957
1958 /**
1959  * @brief Queries whether the logging engine is currently operating in Safe Mode.
1960  *
1961  * ### For Game Designers & Non-Programmers:
1962  * Returns true if the engine booted using compiled-in ROM backups or emergency
1963  * defaults instead of live external JSON directories on disk.
1964  *
1965  * ### For Systems Programmers & Engine Architects:
1966  * Checks initialization state alongside s_log.safe_mode (!loaded_from_data).
1967  *
1968  * @return true if active and running in Safe Mode; false otherwise.
1969  */
1970 bool log_is_safe_mode(void)
1971 {
1972     return s_log.initialized && s_log.safe_mode;
1973 }
1974
1975 /**
1976  * @brief Returns the current state of the logging data pipeline tripwire.
1977  *
1978  * ### For Game Designers & Non-Programmers:
1979  * Checks whether the data-driven logging pipeline encountered an error or is clean.
1980  *
1981  * ### For Systems Programmers & Engine Architects:
1982  * Queries s_log.tripwire_status latch.
1983  *
1984  * @return Active log_tripwire_t enum status code.
1985  */
1986 log_tripwire_t log_get_tripwire_status(void)
1987 {
1988     return s_log.tripwire_status;
1989 }
1990
1991 /**
1992  * @brief Returns a detailed message explaining why a tripwire was tripped.
1993  *
1994  * ### For Game Designers & Non-Programmers:
1995  * Gives a clear, readable explanation of what went wrong if a configuration fault occurred.
1996  *
1997  * ### For Systems Programmers & Engine Architects:
1998  * Returns pointer to internal diagnostic message buffer s_log.tripwire_details.
1999  *
2000  * @return Null-terminated string describing the latched tripwire condition.
2001  */
2002 const char *log_get_tripwire_details(void)
2003 {
2004     if (s_log.tripwire_status == LOG_TRIPWIRE_CLEAR &&
2005         s_log.tripwire_details[0] == '\0') {
2006         return "All data pipeline tripwires clear.";
2007     }
2008     return s_log.tripwire_details;
2009 }
2010
2011 /**
2012  * @brief Resets the tripwire state to CLEAR (used in diagnostic test suites).
2013  *
2014  * ### For Game Designers & Non-Programmers:
2015  * Clears any tripwire warning once the error has been noted or handled.
2016  *
2017  * ### For Systems Programmers & Engine Architects:
2018  * Restores s_log.tripwire_status to LOG_TRIPWIRE_CLEAR and reinitializes details.
2019  */
2020 void log_reset_tripwire(void)
2021 {
2022     s_log.tripwire_status = LOG_TRIPWIRE_CLEAR;
2023     str_safe_copy(
2024         s_log.tripwire_details,
2025         "All data pipeline tripwires clear.",
2026         sizeof(s_log.tripwire_details)
2027     );
2028 }
```

# A.7. Complete 5-Block VFS Precedence & Policy Configuration (vfs_routes.json)

**Snapshot ID**: snap_vfs_routes_cfg_whole | **Mode**: Whole | **File**: engine/config/io/vfs_routes.json (L1–L159 of 159 lines) |
Jump back to chapter text

```
{
  "block_1_metadata": {
    "project": "Tenebris and Lux",
    "file": "engine/config/io/vfs_routes.json",
    "id": "engine.config.io.vfs_routes",
    "layer": "Layer 0: Core Utilities / VFS Pathfinding",
    "version": "0.1.1-dev",
    "interface_rev": "Rev 5",
    "date": "2026-11-11",
    "author": "CreativeWorkzStudio LLC / ProfessorSeanEX",
    "repository": "https://github.com/Novaking34/tenebris-lux",
    "role": "Layered Virtual Filesystem routing specification defining 3-tier config precedence, candidate roots, and domain search directories.",
    "where_the_data_lives": {
      "topology": "Engine VFS Routing Table Specification",
      "path": "engine/config/io/vfs_routes.json",
      "header_binding": "engine/include/io/path_finder.h"
    },
    "what_it_enables": [
      "Deterministic 3-tier resolution: game overrides -> engine defaults -> presentation linkers",
      "Multi-root workspace probing (repo root, build/ subdir, or mounted mod directory)",
      "Zero-disk Safe-Mode virtual rom:// routing when disk files are absent",
      "Clean alignment to Rev 5 runtime/, diagnostics/, io/, and standards/ subtrees"
    ],
    "invariants": [
      "Zero dynamic heap allocation during VFS path probes (ALLOC_ZERO_HEAP)",
      "Resolved path output strictly < 512 bytes (PATH_FINDER_MAX_PATH)",
      "All resolved strings unconditionally null-terminated"
    ]
  },
  "block_2_opening": {
    "pre_conditions": {
      "path_finder_header": "io/path_finder.h",
      "bios_header": "io/io_fallback.h"
    },
    "static_declarations": {
      "subdomain_id": "io",
      "schema_id": "tenlux.data.io_vfs_routes.v1",
      "target_subsystem": "cog.path_finder",
      "max_root_bytes": 256,
      "max_path_bytes": 512
    },
    "generic_helper_contracts": {
      "mount_root": "path_finder_set_data_root",
      "load_routes": "path_finder_load_routes",
      "resolve": "path_finder_resolve",
      "resolve_asset": "path_finder_resolve_asset"
    },
    "capacities": {
      "config_tier_count": 3,
      "candidate_root_count": 6,
      "asset_root_count": 4,
      "domain_search_count": 36
    }
  },
  "block_3_body": {
    "default_data_root": "game/data",
    "env_override_var": "TL_DATA_ROOT",
    "block_path_traversal": true,
    "max_symlink_depth": 4,
    "allow_mod_mounts": true,
    "rom_fallback_enabled": true,
    "probe_await_on_corrupt": true,
    "config_precedence": [
      "game/config/",
      "engine/config/",
      "game/data/presentation/config/"
    ],
```

```
    "fallback_candidate_roots": [
      "game/data",
      "../game/data",
      "engine/config",
      "../engine/config",
      "data",
      "../data"
    ],
    "asset_roots": [
      "game/assets",
      "../game/assets",
      "assets",
      "../assets"
    ],
    "domain_search_paths": [
      "",
      "runtime/",
      "runtime/math/",
      "runtime/string/",
      "runtime/time/",
      "diagnostics/",
      "diagnostics/telemetry/",
      "diagnostics/debugger/",
      "diagnostics/taxonomy/",
      "diagnostics/formats/",
      "io/",
      "audio/",
      "viewport/",
      "input/",
      "standards/",
      "standards/blocks/",
      "mechanics/",
      "mechanics/items/",
      "mechanics/combat/",
      "mechanics/tactics/",
      "mechanics/campaign/",
      "mechanics/campaign/dispatch/",
      "world/",
      "world/geography/",
      "world/battlefields/",
      "world/terrains/",
      "entities/",
      "entities/classes/",
      "entities/roster/",
      "presentation/",
      "presentation/config/",
      "presentation/audio/",
      "presentation/ui/",
      "presentation/ui/fonts/",
      "presentation/visuals/",
      "presentation/visuals/palettes/"
    ],
    "diagnostic_aliases": {
      "log_prefix": "diagnostics/telemetry/",
      "debug_prefix": "diagnostics/debugger/",
      "io_prefix": "io/"
    }
  },
  "block_4_audit": {
    "audit_metrics": {
      "config_tier_count": 3,
      "candidate_root_count": 6,
      "asset_root_count": 4,
      "domain_search_count": 36,
      "zero_heap_compliant": true
    },
    "verification_tripwires": [
      {
        "name": "CONFIG_PRECEDENCE_3TIER",
        "rule": "config_precedence order preserves game overrides before engine defaults",
        "status": "VERIFIED_PASS"
      },
      {
        "name": "NO_STALE_SYSTEM_PREFIX",
        "rule": "diagnostic_aliases map cleanly to diagnostics/telemetry and diagnostics/debugger",
```

```
        "status": "VERIFIED_PASS"
      }
    ],
    "self_audit_target": "./build/tenebris-lux --audit"
  },
  "block_5_closing": {
    "execution_handshake": {
      "unit_test": "unit_path_finder_test",
      "rom_fallback_asset": "io/vfs_routes.json"
    },
    "navigation": {
      "parent_menu": "manifest.json",
      "root_manifest": "../manifest.json"
    }
  }
}
```

## A.8. JSON Block 3 Payload Isolation (block_3_body of vfs_routes.json)

**Snapshot ID**: snap_vfs_routes_body_block | **Mode**: Block | **File**: engine/config/io/vfs_routes.json (L55–L125 of 159 lines) | Jump back to chapter text

```
  "block_3_body": {
    "default_data_root": "game/data",
    "env_override_var": "TL_DATA_ROOT",
    "block_path_traversal": true,
    "max_symlink_depth": 4,
    "allow_mod_mounts": true,
    "rom_fallback_enabled": true,
    "probe_await_on_corrupt": true,
    "config_precedence": [
      "game/config/",
      "engine/config/",
      "game/data/presentation/config/"
    ],
    "fallback_candidate_roots": [
      "game/data",
      "../game/data",
      "engine/config",
      "../engine/config",
      "data",
      "../data"
    ],
    "asset_roots": [
      "game/assets",
      "../game/assets",
      "assets",
      "../assets"
    ],
    "domain_search_paths": [
      "",
      "runtime/",
      "runtime/math/",
      "runtime/string/",
      "runtime/time/",
      "diagnostics/",
      "diagnostics/telemetry/",
      "diagnostics/debugger/",
      "diagnostics/taxonomy/",
      "diagnostics/formats/",
      "io/",
      "audio/",
      "viewport/",
      "input/",
      "standards/",
      "standards/blocks/",
      "mechanics/",
      "mechanics/items/",
      "mechanics/combat/",
      "mechanics/tactics/",
      "mechanics/campaign/",
      "mechanics/campaign/dispatch/",
```

```
      "world/",
      "world/geography/",
      "world/battlefields/",
      "world/terrains/",
      "entities/",
      "entities/classes/",
      "entities/roster/",
      "presentation/",
      "presentation/config/",
      "presentation/audio/",
      "presentation/ui/",
      "presentation/ui/fonts/",
      "presentation/visuals/",
      "presentation/visuals/palettes/"
    ],
    "diagnostic_aliases": {
      "log_prefix": "diagnostics/telemetry/",
      "debug_prefix": "diagnostics/debugger/",
      "io_prefix": "io/"
    }
  },
```

## A.9. Scoped block_3_body JSON Accessor Contract (master_loader.h Sub-Assembly 3.2)

**Snapshot ID**: snap_master_loader_sub3_2 | **Mode**: Subassembly | **File**: engine/include/io/master_loader.h (L436–L608 of 987 lines) | Jump back to chapter text

```
/*
--------------------------------------------------------------------------------
Sub-Assembly 3.2: Scoped `block_3_body` Accessors
Mission: Restrict all payload getters and array locators strictly inside the
         verified `[body_start, body_start + body_len)` span (or named `sub_obj`).

  ### For Game Designers & Non-Programmers:
  Reads gameplay properties cleanly from inside `block_3_body` only, preventing
  file headers or audit sections from ever colliding with gameplay field names.

  ### For Systems Programmers & Engine Architects:
  Delegates scope isolation and typed extraction exclusively to span-bounded
  `json_*_span` APIs with unconditional NUL termination on extracted strings.
--------------------------------------------------------------------------------
*/

/**
 * @brief Extracts a string value strictly within `block_3_body` (or named sub-object).
 *
 * ### For Game Designers & Non-Programmers:
 * Safely reads text properties (character names, formulas, IDs) from the data body
 * without accidentally matching header text outside Block 3.
 *
 * ### For Systems Programmers & Engine Architects:
 * Uses span-bounded `json_find_object_span()` + `json_get_string_span()` bounded to
 * `loader->body_len`. Unconditionally NUL-terminates `out_val` when `out_size > 0`.
 * @triggers `io.json.depth_bounds` if scan breaches `max_json_depth`.
 *
 * @param[in]  loader   Ready `master_loader_t` instance.
 * @param[in]  sub_obj  Optional sub-object key inside `block_3_body` (or NULL for root body).
 * @param[in]  key      Target property key inside resolved body/sub-object span.
 * @param[out] out_val  Caller string buffer.
 * @param[in]  out_size Capacity of `out_val` in bytes.
 * @return true on clean extraction; false if missing, wrong type, or invalid params.
 */
bool master_loader_get_string(const master_loader_t *loader,
                              const char *sub_obj,
                              const char *key,
                              char *out_val,
                              size_t out_size);

/**
 * @brief Extracts a signed integer strictly within `block_3_body` (or named sub-object).
```

```
 *
 * ### For Game Designers & Non-Programmers:
 * Reads whole-number gameplay values (HP caps, percentages, cooldown turns) cleanly.
 *
 * ### For Systems Programmers & Engine Architects:
 * Span-bounded extraction via `json_get_int_span()` with overflow/syntax guards.
 * @triggers None on expected miss or parse rejection.
 *
 * @param[in]  loader  Ready `master_loader_t` instance.
 * @param[in]  sub_obj Optional sub-object key inside `block_3_body` (or NULL).
 * @param[in]  key     Target integer key.
 * @param[out] out_val Destination `int`.
 * @return true on valid integer parse; false otherwise.
 */
bool master_loader_get_int(const master_loader_t *loader,
                           const char *sub_obj,
                           const char *key,
                           int *out_val);

/**
 * @brief Extracts a non-negative `size_t` strictly within `block_3_body` (or `sub_obj`).
 *
 * ### For Game Designers & Non-Programmers:
 * Reads positive capacity counts (ring buffer slots, item stack limits) cleanly.
 *
 * ### For Systems Programmers & Engine Architects:
 * Rejects negative JSON integers safely via `json_get_size_t_span()`.
 * @triggers None on expected miss or parse rejection.
 *
 * @param[in]  loader  Ready `master_loader_t` instance.
 * @param[in]  sub_obj Optional sub-object key inside `block_3_body` (or NULL).
 * @param[in]  key     Target non-negative integer key.
 * @param[out] out_val Destination `size_t`.
 * @return true on valid unsigned parse; false otherwise.
 */
bool master_loader_get_size_t(const master_loader_t *loader,
                              const char *sub_obj,
                              const char *key,
                              size_t *out_val);

/**
 * @brief Extracts a floating-point number strictly within `block_3_body` (or `sub_obj`).
 *
 * ### For Game Designers & Non-Programmers:
 * Reads decimal multipliers (damage ratios, audio volume dB, difficulty scales).
 *
 * ### For Systems Programmers & Engine Architects:
 * Bounded extraction via `json_get_float_span()`, rejecting NaN/Inf tokens cleanly.
 * @triggers None on expected miss or parse rejection.
 *
 * @param[in]  loader  Ready `master_loader_t` instance.
 * @param[in]  sub_obj Optional sub-object key inside `block_3_body` (or NULL).
 * @param[in]  key     Target float key.
 * @param[out] out_val Destination `float`.
 * @return true on valid finite float parse; false otherwise.
 */
bool master_loader_get_float(const master_loader_t *loader,
                             const char *sub_obj,
                             const char *key,
                             float *out_val);

/**
 * @brief Extracts a JSON boolean strictly within `block_3_body` (or `sub_obj`).
 *
 * ### For Game Designers & Non-Programmers:
 * Reads true/false feature switches from data manifests cleanly.
 *
 * ### For Systems Programmers & Engine Architects:
 * Accepts strict JSON `true`/`false` literals only via `json_get_bool_span()`.
 * @triggers None on expected miss or parse rejection.
 *
 * @param[in]  loader  Ready `master_loader_t` instance.
 * @param[in]  sub_obj Optional sub-object key inside `block_3_body` (or NULL).
 * @param[in]  key     Target boolean key.
 * @param[out] out_val Destination `bool`.
```

```
 * @return true on valid boolean literal parse; false otherwise.
 */
bool master_loader_get_bool(const master_loader_t *loader,
                            const char *sub_obj,
                            const char *key,
                            bool *out_val);

/**
 * @brief Locates a JSON array span strictly within `block_3_body` (or `sub_obj`).
 *
 * ### For Game Designers & Non-Programmers:
 * Finds lists inside a file (such as enemy waves, settings menus, or item tables)
 * and counts how many entries exist.
 *
 * ### For Systems Programmers & Engine Architects:
 * Zero-copy slice via `json_find_array_span()` + `json_array_get_count()`.
 * @triggers `io.json.depth_bounds` if array bracket scan exceeds `max_json_depth`.
 *
 * @param[in]  loader    Ready `master_loader_t` instance.
 * @param[in]  sub_obj   Optional sub-object key inside `block_3_body` (or NULL).
 * @param[in]  array_key Target JSON array key.
 * @param[out] out_start Pointer to `[` delimiter inside loaded JSON buffer.
 * @param[out] out_len   Exact byte length from `[` through matching `]`.
 * @param[out] out_count Optional element count (may be NULL).
 * @return true if array found and well-formed; false otherwise.
 */
bool master_loader_find_array(const master_loader_t *loader,
                              const char *sub_obj,
                              const char *array_key,
                              const char **out_start,
                              size_t *out_len,
                              size_t *out_count);

/**
 * @brief Copies the `index`-th object element out of a located JSON array span.
 *
 * ### For Game Designers & Non-Programmers:
 * Steps through list entries one item at a time so loaders can import catalogs cleanly.
 *
 * ### For Systems Programmers & Engine Architects:
 * Bounded copy via `json_array_get_object()` guaranteeing NUL-termination.
 * @triggers `io.json.depth_bounds` if element object exceeds `max_json_depth`.
 *
 * @param[in]  arr_start Pointer to `[` array boundary from `master_loader_find_array()`.
 * @param[in]  arr_len   Exact byte span of the array.
 * @param[in]  index     Zero-based element index.
 * @param[out] out_obj   Caller destination buffer for the extracted JSON object slice.
 * @param[in]  out_size  Byte capacity of `out_obj`.
 * @return true on valid extraction; false on out-of-bounds index or overflow.
 */
bool master_loader_get_array_object(const char *arr_start,
                                    size_t arr_len,
                                    size_t index,
                                    char *out_obj,
                                    size_t out_size);
```

## A.10. Shared Zero-Heap JSON Block Resolver & VFS Parser Group

**Snapshot ID**: snap_io_fb_json_group | **Mode**: Group | **File**: engine/src/io/io_fallback.c (L779–L1194 of 3355 lines) | Jump back to chapter text

```
static bool io_fb_resolve_body_span(const char *json_text,
                                    size_t json_len,
                                    bool allow_strlen,
                                    const char **out_body_start,
                                    const char **out_body_end)
{
    if (json_text == NULL || out_body_start == NULL || out_body_end == NULL) {
        return false;
    }

    size_t effective_len = (json_len > 0U) ? json_len : (allow_strlen ? strlen(json_text) : 0U);
```

```
    if (effective_len == 0U || effective_len > IO_FB_JSON_MAX_BYTES) {
        return false;
    }

    const char *json_end = json_text + effective_len;
    const char *body_start = NULL;
    size_t body_len = 0U;
    if (!io_fb_find_block_span(json_text, json_end, "block_3_body", '{', '}',
                               &body_start, &body_len) ||
        body_start == NULL || body_len == 0U) {
        return false;
    }

    *out_body_start = body_start;
    *out_body_end = body_start + body_len;
    return true;
}
```

```
/**
 * @brief Hydrate VFS routing roots, 36+ domain paths, and symlink/traversal rules from JSON.
 *
 *   ### For Game Designers & Non-Programmers:
 *   Loads `vfs_routes.json` so asset folders, mod override precedence tiers, and
 *   diagnostic aliases stay configurable in data files rather than hardcoded C arrays.
 *
 *   ### For Systems Programmers & Engine Architects:
 *   Resolves `block_3_body` via `io_fb_resolve_body_span()`, seeds defaults with
 *   `io_fb_seed_default_vfs_routes()`, extracts precedence/root/domain arrays within
 *   fixed `.bss` caps, and forces `out_routes->loaded_from_data = false`.
 *
 * @param[in]  json_text  Raw `vfs_routes.json` text buffer.
 * @param[in]  json_len   Byte length of `json_text` (`0U` for `strlen()`).
 * @param[out] out_routes Hydrated output VFS routing descriptor.
 * @return `true` when body block parsed cleanly with non-empty root and domain lists.
 * @triggers Forces `loaded_from_data = false` so caller proves external disk provenance explicitly.
 */
bool io_fallback_parse_vfs_routes_json(const char *json_text,
                                       size_t json_len,
                                       io_fb_vfs_routes_t *out_routes)
{
    /* Step 1: Resolve `block_3_body` span and seed safe default routes */
    const char *body_start = NULL;
    const char *body_end = NULL;
    if (out_routes == NULL ||
        !io_fb_resolve_body_span(json_text, json_len, true, &body_start, &body_end)) {
        return false;
    }

    io_fb_seed_default_vfs_routes(out_routes);

    /* Step 2: Hydrate string identifiers and precedence/search directory arrays */
    char val_str[64];
    int val_int = 0;
    bool val_bool = false;

    if (io_fb_get_string_span(body_start, body_end, "default_data_root",
                              val_str, sizeof(val_str)) && val_str[0] != '\0') {
        str_safe_copy(out_routes->default_data_root, val_str,
                      sizeof(out_routes->default_data_root));
    }
    if (io_fb_get_string_span(body_start, body_end, "env_override_var",
                              val_str, sizeof(val_str)) && val_str[0] != '\0') {
        str_safe_copy(out_routes->env_override_var, val_str,
                      sizeof(out_routes->env_override_var));
    }

    (void)io_fb_get_string_array_span(body_start, body_end, "config_precedence",
                                      out_routes->config_precedence,
                                      IO_FB_MAX_PRECEDENCE_TIERS,
                                      &out_routes->config_precedence_count);
    (void)io_fb_get_string_array_span(body_start, body_end, "fallback_candidate_roots",
                                      out_routes->fallback_candidate_roots,
                                      IO_FB_MAX_VFS_CANDIDATES,
                                      &out_routes->candidate_root_count);
```

```
    (void)io_fb_get_string_array_span(body_start, body_end, "asset_roots",
                                      out_routes->asset_roots,
                                      IO_FB_MAX_ASSET_ROOTS,
                                      &out_routes->asset_root_count);
    (void)io_fb_get_string_array_span(body_start, body_end, "domain_search_paths",
                                      out_routes->domain_search_paths,
                                      IO_FB_MAX_VFS_DOMAIN_DIRS,
                                      &out_routes->domain_search_count);

    /* Step 3: Hydrate nested diagnostic domain alias prefixes if supplied */
    const char *alias_start = NULL;
    size_t alias_len = 0U;
    if (io_fb_find_block_span(body_start, body_end, "diagnostic_aliases", '{', '}',
                              &alias_start, &alias_len) &&
        alias_start != NULL && alias_len > 2U) {
        const char *alias_end = alias_start + alias_len;
        if (io_fb_get_string_span(alias_start, alias_end, "log_prefix",
                                  val_str, sizeof(val_str)) && val_str[0] != '\0') {
            str_safe_copy(out_routes->diag_log_prefix, val_str,
                          sizeof(out_routes->diag_log_prefix));
        }
        if (io_fb_get_string_span(alias_start, alias_end, "debug_prefix",
                                  val_str, sizeof(val_str)) && val_str[0] != '\0') {
            str_safe_copy(out_routes->diag_debug_prefix, val_str,
                          sizeof(out_routes->diag_debug_prefix));
        }
        if (io_fb_get_string_span(alias_start, alias_end, "io_prefix",
                                  val_str, sizeof(val_str)) && val_str[0] != '\0') {
            str_safe_copy(out_routes->diag_io_prefix, val_str,
                          sizeof(out_routes->diag_io_prefix));
        }
    }

    /* Step 4: Extract boolean security switches and verify non-empty route tables */
    if (io_fb_get_bool_span(body_start, body_end, "block_path_traversal", &val_bool)) {
        out_routes->block_path_traversal = val_bool;
    }
    if (io_fb_get_int_span(body_start, body_end, "max_symlink_depth", &val_int) &&
        val_int >= 0 && val_int <= 16) {
        out_routes->max_symlink_depth = val_int;
    }
    if (io_fb_get_bool_span(body_start, body_end, "allow_mod_mounts", &val_bool)) {
        out_routes->allow_mod_mounts = val_bool;
    }
    if (io_fb_get_bool_span(body_start, body_end, "rom_fallback_enabled", &val_bool)) {
        out_routes->rom_fallback_enabled = val_bool;
    }
    if (io_fb_get_bool_span(body_start, body_end, "probe_await_on_corrupt", &val_bool)) {
        out_routes->probe_await_on_corrupt = val_bool;
    }

    out_routes->loaded_from_data = false;
    return (out_routes->domain_search_count > 0U && out_routes->candidate_root_count > 0U);
}
```

# A.11. Central Manifest Metadata & Registry Ontology (manifest.h Sub-Assembly 2.3)

**Snapshot ID**: snap_manifest_types_sub2_3 | **Mode**: Subassembly | **File**: engine/include/core/manifest.h (L207–L370 of 707 lines) | Jump back to chapter text

```
/*
-------------------------------------------------------------------------------
Sub-Assembly 2.3: Structural Entity & Object Definitions
-------------------------------------------------------------------------------
Ontological types for architectural strata, memory models, and diagnostic audits.
-------------------------------------------------------------------------------
*/

/**
 * @enum module_layer
```

```
 * @brief Architectural layer strata within the hierarchical engine topology.
 *
 * For Designers: The 5 vertical tiers of the game engine, from deep bedrock
 * utilities (Layer 0) up to the game rules (Layer 1) and graphical display (Layer 2).
 *
 * For Programmers: Strict downward dependency hierarchy enforced at compile time.
 */
typedef enum module_layer {
    LAYER_0_CORE_UTIL = 0, /**< Layer 0: Foundational zero-dependency utils. */
    LAYER_1_SIMULATION,    /**< Layer 1: Headless game simulation engine. */
    LAYER_2_PRESENTATION,  /**< Layer 2: Graphical and text viewports. */
    LAYER_3_APPLICATION,   /**< Layer 3: Application entry and routing. */
    LAYER_4_TESTS,         /**< Layer 4: Verification test harness suites. */
    LAYER_COUNT
} module_layer_t;

/**
 * @enum alloc_policy
 * @brief Memory allocation contract governing a module's execution model.
 *
 * For Designers: Rules defining how a system manages memory, ensuring bedrock
 * systems never allocate heap memory to prevent crashes and stutter.
 *
 * For Programmers: Invariant memory policy verified during code audit.
 */
typedef enum alloc_policy {
    ALLOC_ZERO_HEAP = 0, /**< 100% stack or static layout; zero dynamic heap. */
    ALLOC_FIXED_POOL,    /**< Pre-allocated fixed-capacity buffer pools. */
    ALLOC_ARENA,         /**< Frame/scene temporary arena allocation. */
    ALLOC_DYNAMIC        /**< Managed dynamic heap allocation. */
} alloc_policy_t;

/**
 * @enum concurrency_policy
 * @brief Concurrency and reentrancy guarantees declared by a module.
 *
 * For Designers: Rules specifying whether a system can run simultaneously across
 * multiple CPU threads or must remain tied to the main display loop.
 *
 * For Programmers: Thread-safety model declared for scheduler evaluation.
 */
typedef enum concurrency_policy {
    CONCURRENCY_REENTRANT = 0, /**< Thread-safe, stateless pure functions. */
    CONCURRENCY_THREAD_LOCAL,  /**< Thread-local state isolation. */
    CONCURRENCY_MAIN_THREAD    /**< Main-thread execution affinity (GUI). */
} concurrency_policy_t;

/**
 * @enum diag_severity
 * @brief Five-tier diagnostic fault severity taxonomy for pre-flight machine audits.
 *
 * For Designers: A traffic light system for engine health:
 * PASS (green), INFO (blue notice), WARNING (yellow alert),
 * CRITICAL (orange, trigger safe fallback), FATAL (red, stop game safely).
 *
 * For Programmers: Strict escalation hierarchy: FATAL > CRITICAL > WARNING > PASS.
 */
typedef enum diag_severity {
    SEVERITY_PASS = 0, /**< Invariant holds cleanly; full readiness. */
    SEVERITY_INFO,     /**< Informational notice; no operational effect. */
    SEVERITY_WARNING,  /**< Non-fatal anomaly logged; execution proceeds. */
    SEVERITY_CRITICAL, /**< Subsystem offline; eligible for safe downgrade. */
    SEVERITY_FATAL     /**< Loose bolt detected; unrecoverable hard halt. */
} diag_severity_t;

/**
 * @enum preflight_status
 * @brief Aggregate pre-flight engine readiness state.
 *
 * For Designers: The overall health verdict after all engine systems test themselves.
 *
 * For Programmers: Composite state determined by highest severity finding across DAG.
 */
typedef enum preflight_status {
    PREFLIGHT_STATUS_OK = 0,    /**< Subsystems passed pre-flight audits. */
```

```
    PREFLIGHT_STATUS_WARNINGS,   /**< Non-fatal notices logged; system ready. */
    PREFLIGHT_STATUS_DOWNGRADED, /**< Critical fault downgraded to safe mode. */
    PREFLIGHT_STATUS_HALT_FATAL  /**< Loose bolt detected; engine halt. */
} preflight_status_t;

/**
 * @struct diag_result
 * @brief Individual diagnostic check outcome record.
 *
 * For Designers: A single test result card explaining if a test passed and any notes.
 *
 * For Programmers: Reentrant stack-allocated diagnostic result structure.
 */
typedef struct diag_result {
    bool            passed;     /**< Boolean pass/fail indicator. */
    diag_severity_t severity;   /**< Severity classification of the finding. */
    const char     *subsystem;  /**< Identifier of the audited subsystem. */
    const char     *check_name; /**< Descriptive name of the specific audit. */
    const char     *details;    /**< Human-readable diagnostic details. */
} diag_result_t;

/**
 * @struct module_metadata
 * @brief Compiled module metadata node binding identity, contracts, and tripwires.
 *
 * For Designers: The formal identity card of an engine subsystem, listing its name,
 * version, layer, who built it, what other systems it needs, and how to test it.
 *
 * For Programmers: Static immutable structure registered in central registry.
 */
typedef struct module_metadata {
    /* 1. Identity & Provenance */
    const char    *module_id;          /**< Canonical module ID. */
    const char    *file_name;          /**< Header file name. */
    module_layer_t layer;              /**< Architectural stratum. */
    struct {
        uint8_t major;
        uint8_t minor;
        uint8_t patch;
    } version;                         /**< Semantic version triplet. */
    uint8_t        interface_rev;      /**< Contract ABI revision counter. */
    const char    *last_updated;       /**< Full ISO 8601 date: YYYY-MM-DD. */
    const char    *description;        /**< Plain-English mission statement. */

    /* 2. Architectural Dependencies */
    size_t         dependency_count;   /**< Number of declared prerequisites. */
    const char    *dependencies[MANIFEST_MAX_DEPENDENCIES]; /**< Upstream IDs. */

    /* 3. Resource & Execution Contracts */
    alloc_policy_t       alloc_policy; /**< Memory allocation guarantee. */
    concurrency_policy_t concurrency;  /**< Concurrency / thread safety model. */
    uint32_t             max_budget_us;/**< Max latency budget (microseconds). */

    /* 4. Invariant Contracts */
    size_t         invariant_count;    /**< Number of documented invariants. */
    const char    *invariants[MANIFEST_MAX_INVARIANTS]; /**< Invariant rules. */

    /* 5. Pre-Flight Machine Self-Audit */
    diag_result_t (*preflight_check)(void); /**< Diagnostic callback. */
} module_metadata_t;

/**
 * @struct manifest_registry
 * @brief Central registry tracking registered modules, audit results, and status.
 *
 * For Designers: The master scoreboard storing every system's health report.
 *
 * For Programmers: Zero-heap fixed-capacity structure populated at engine boot.
 */
typedef struct manifest_registry {
    const module_metadata_t *modules[MANIFEST_MAX_MODULES]; /**< Pointers to registered module metadata nodes. */
    diag_result_t            results[MANIFEST_MAX_MODULES]; /**< Pre-flight check outcome records. */
    size_t                   count;                         /**< Number of currently registered modules. */
    size_t                   pass_count;                    /**< Total audits yielding SEVERITY_PASS. */
    size_t                   warn_count;                    /**< Total audits yielding SEVERITY_WARNING. */
```

```
    size_t                  critical_count;             /**< Total audits yielding SEVERITY_CRITICAL. */
    size_t                  fatal_count;                /**< Total audits yielding SEVERITY_FATAL. */
    preflight_status_t      status;                     /**< Aggregate pre-flight system readiness status. */
} manifest_registry_t;
```

## A.12. Pre-Flight Loose-Bolts Engine Walk (manifest_run_preflight)

**Snapshot ID**: snap_manifest_fn_preflight | **Mode**: Function | **File**: engine/src/core/manifest.c (L412–L509 of 669 lines) | Jump back to chapter text

```c
/**
 * @brief Executes comprehensive pre-flight diagnostics across all registered modules.
 */
preflight_status_t manifest_run_preflight(manifest_registry_t *registry)
{
    if (registry == NULL) {
        debug_trap_tripwire("manifest", "NULL registry passed to manifest_run_preflight", __FILE__, __LINE__);
        return PREFLIGHT_STATUS_HALT_FATAL;
    }

    registry->pass_count = 0;
    registry->warn_count = 0;
    registry->critical_count = 0;
    registry->fatal_count = 0;

    /* Step 1: Active DAG Dependency Graph Verification */
    for (size_t i = 0; i < registry->count; i++) {
        const module_metadata_t *mod = registry->modules[i];
        if (mod == NULL) continue;

        for (size_t d = 0; d < mod->dependency_count; d++) {
            const char *req_dep = mod->dependencies[d];
            if (req_dep == NULL) continue;

            bool found = false;
            for (size_t j = 0; j < registry->count; j++) {
                if (registry->modules[j] != NULL &&
                    strcmp(registry->modules[j]->module_id, req_dep) == 0) {
                    found = true;
                    break;
                }
            }

            if (!found) {
                diag_result_t dep_res;
                dep_res.passed = false;
                dep_res.severity = SEVERITY_FATAL;
                dep_res.subsystem = mod->module_id;
                dep_res.check_name = "Dependency DAG";
                dep_res.details =
                    "Fatal: Unregistered upstream dependency declared.";
                registry->results[i] = dep_res;
                registry->fatal_count++;
                registry->status = PREFLIGHT_STATUS_HALT_FATAL;
                debug_trap_tripwire("manifest", "Unregistered upstream DAG dependency detected", __FILE__, __LINE__);
                return registry->status;
            }
        }
    }

    /* Step 2: Individual Module Pre-Flight Verification */
    for (size_t i = 0; i < registry->count; i++) {
        const module_metadata_t *mod = registry->modules[i];
        if (mod == NULL) {
            continue;
        }

        diag_result_t res;
        if (mod->preflight_check != NULL) {
            res = mod->preflight_check();
        } else {
            res.passed = true;
```

```
            res.severity = SEVERITY_PASS;
            res.subsystem = mod->module_id;
            res.check_name = "Presence";
            res.details = "No active preflight check defined.";
        }

        registry->results[i] = res;

        bool is_fatal = (res.severity == SEVERITY_FATAL) ||
            (!res.passed && res.severity != SEVERITY_WARNING &&
             res.severity != SEVERITY_CRITICAL);

        if (is_fatal) {
            registry->fatal_count++;
            registry->status = PREFLIGHT_STATUS_HALT_FATAL;
            return registry->status;
        } else if (res.severity == SEVERITY_CRITICAL) {
            registry->critical_count++;
        } else if (res.severity == SEVERITY_WARNING) {
            registry->warn_count++;
        } else {
            registry->pass_count++;
        }
    }

    /* Step 3: Determine aggregate system status */
    if (registry->critical_count > 0) {
        registry->status = PREFLIGHT_STATUS_DOWNGRADED;
    } else if (registry->warn_count > 0) {
        registry->status = PREFLIGHT_STATUS_WARNINGS;
    } else {
        registry->status = PREFLIGHT_STATUS_OK;
    }

    return registry->status;
}
```

## A.13. Application Audit Integration (game/src/main.c BLOCK 4)

**Snapshot ID**: snap_main_preflight_block4 | **Mode**: Block | **File**: game/src/main.c (L193–L334 of 524 lines) | Jump back to chapter text

```
// ============================================================================
// BLOCK 4: AUDIT PROCEDURES BLOCK
// Stratum: Layer 3 (Application Entry) | Machine: Presentation Router
// ============================================================================

// ----------------------------------------------------------------------------
// [ SUB-ASSEMBLY 4.1: COMPREHENSIVE SUBSYSTEM SELF-AUDIT SUITE ]
// Mission: Execute pre-flight manifest checks, sim invariants, and GUI bounds.
// ----------------------------------------------------------------------------

/**
 * @brief Executes the comprehensive engine self-audit suite and manifests.
 *
 * ### Machine Role & Mission
 * Serves as the `--audit` diagnostic tripwire. Executes:
 * 1. Pre-flight verification across all registered modules with DAG checks.
 * 2. Layer 1 simulation invariant audit (`game_run_self_audit()`).
 * 3. Layer 2 GUI viewport layout and collision audit (`gui_audit_layout()`).
 * Returns non-zero if any invariant check or pre-flight test fails.
 *
 * ### Ontological Classification
 * - Stratum: Layer 3 (Application Entry & Presentation Router)
 * - Machine: app.entry.main (Executable Entry & Presentation Router)
 * - Assembly: Diagnostic Self-Audit & Pre-Flight Verification
 * - Cog: Full Engine Diagnostics & Invariant Self-Audit Suite
 *
 * ### Execution & Resource Contract
 * - Memory Model: ALLOC_ZERO_HEAP (Stack allocation only).
 * - Concurrency: CONCURRENCY_MAIN_THREAD (Synchronous diagnostic sweep).
 * - Latency Budget: < 5000 microseconds execution time.
```

```
 *
 * @return 0 if all invariant checks pass; non-zero if any check fails.
 */
static int run_full_subsystem_audit(void)
{
    log_init(NULL);
    LOG_INFO(LOG_CAT_LOG, "Audit", 0.0f, "Starting comprehensive diagnostic self-audit");

    /* Step 1: Initialize central manifest and run pre-flight diagnostics */
    manifest_registry_t registry;
    init_engine_manifest(&registry);
    manifest_run_preflight(&registry);
    manifest_print_report(&registry);

    /* Step 2: Evaluate pre-flight outcome for fatal loose-bolts */
    if (registry.status == PREFLIGHT_STATUS_HALT_FATAL) {
        LOG_FATAL(LOG_CAT_LOG, "Audit", -1.0f, "Pre-flight failed due to loose bolts!");
        fprintf(stderr, "[FATAL] Pre-flight failed due to loose bolts!\n");
        log_shutdown();
        return 1;
    }

    /* Step 2b: Run telemetry subsystem diagnostic audit */
    diag_result_t log_diag = log_audit(false);
    if (!log_diag.passed) {
        LOG_CRITICAL(LOG_CAT_LOG, "Audit", -0.8f, "Core telemetry audit failed: %s", log_diag.details);
        fprintf(stderr, "[FAIL] Core telemetry subsystem audit failed: %s\n", log_diag.details);
        log_shutdown();
        return 1;
    }
    printf("[PASS] Test 0a: Telemetry subsystem & grade invariants verified.\n");

    /* Step 2c: Run debugger subsystem diagnostic audit */
    diag_result_t debug_diag = debug_audit(false);
    if (!debug_diag.passed) {
        LOG_CRITICAL(LOG_CAT_LOG, "Audit", -0.8f, "Core debugger audit failed: %s", debug_diag.details);
        fprintf(stderr, "[FAIL] Core debugger subsystem audit failed: %s\n", debug_diag.details);
        debug_shutdown();
        log_shutdown();
        return 1;
    }
    printf("[PASS] Test 0b: Debugger subsystem & state rewind invariants verified.\n");

    /* Step 2d: Run frame pacing orchestrator diagnostic audit */
    if (!clock_pacer_audit()) {
        LOG_CRITICAL(LOG_CAT_LOG, "Audit", -0.8f, "Clock pacer orchestrator audit failed");
        fprintf(stderr, "[FAIL] Clock pacer orchestrator audit failed.\n");
        debug_shutdown();
        log_shutdown();
        return 1;
    }
    printf("[PASS] Test 0c: Frame pacing & accumulator invariants verified.\n");

    /* Step 2e: Run battle formula orchestrator diagnostic audit */
    if (!formula_eval_audit()) {
        LOG_CRITICAL(LOG_CAT_LOG, "Audit", -0.8f, "Formula evaluator orchestrator audit failed");
        fprintf(stderr, "[FAIL] Formula evaluator orchestrator audit failed.\n");
        debug_shutdown();
        log_shutdown();
        return 1;
    }
    printf("[PASS] Test 0d: Battle formulas & affinity invariants verified.\n");

    /* Step 2f: Run audio bus orchestrator diagnostic audit */
    if (!audio_bus_preflight_audit()) {
        LOG_CRITICAL(LOG_CAT_LOG, "Audit", -0.8f, "Audio event bus audit failed");
        fprintf(stderr, "[FAIL] Audio event bus orchestrator audit failed.\n");
        debug_shutdown();
        log_shutdown();
        return 1;
    }
    printf("[PASS] Test 0e: Audio event bus & multi-channel mixer invariants verified.\n");

    /* Step 2g: Run Movie Reel presentation orchestrator diagnostic audit */
    if (!reel_audit()) {
```

```
        LOG_CRITICAL(LOG_CAT_LOG, "Audit", -0.8f, "Movie Reel orchestrator audit failed");
        fprintf(stderr, "[FAIL] Movie Reel orchestrator audit failed.\n");
        debug_shutdown();
        log_shutdown();
        return 1;
    }
    printf("[PASS] Test 0f: Movie Reel scene strip & depth kinematics invariants verified.\n");

    /* Step 3: Run simulation engine invariant audit */
    if (game_run_self_audit() != 0) {
        LOG_CRITICAL(LOG_CAT_LOG, "Audit", -0.8f, "Core game simulation audit failed");
        fprintf(stderr, "[FAIL] Core game simulation audit failed.\n");
        debug_shutdown();
        log_shutdown();
        return 1;
    }

    /* Step 4: Run GUI layout bounds and collision audit */
    if (!gui_audit_layout()) {
        LOG_CRITICAL(LOG_CAT_LOG, "Audit", -0.8f, "GUI viewport layout bounds exceeded");
        fprintf(stderr, "[FAIL] GUI viewport layout bounds exceeded!\n");
        debug_shutdown();
        log_shutdown();
        return 1;
    }
    printf("[PASS] Test 5: GUI viewport layout invariants verified.\n");

    /* Step 5: Report comprehensive pass summary */
    printf("============================================================\n");
    printf("Audit Result: ALL INVARIANT CHECKS PASSED (9/9 + Pre-Flight)\n");
    printf("============================================================\n");

    LOG_INFO(LOG_CAT_LOG, "Audit", 1.0f, "All invariant checks passed successfully (9/9 + Pre-Flight)");
    debug_shutdown();
    log_shutdown();
    return 0;
}
```

## A.14. Layer 1 Deterministic Simulation State Audit (sim_state_audit)

**Snapshot ID**: snap_game_fn_audit | **Mode**: Function | **File**: game/src/hero/sim_state.c (L315–L376 of 420 lines) | Jump back to chapter text

```
game_audit_report_t sim_state_audit(const player_state_t *state)
{
    game_audit_report_t report;
    report.is_valid = true;
    report.error_count = 0;
    report.error_log[0] = '\0';

    if (state == NULL) {
        report.is_valid = false;
        report.error_count = 1;
        snprintf(report.error_log, sizeof(report.error_log),
                 "[ERROR] Player state pointer is NULL.");
        return report;
    }

    if (state->shield_power < 0 || state->shield_power > MAX_RESONANCE) {
        report.is_valid = false;
        report.error_count++;
        snprintf(report.error_log + strlen(report.error_log),
            sizeof(report.error_log) - strlen(report.error_log),
            "[ERROR] Shield power (%d) outside bounds [0, %d]. ",
            state->shield_power, MAX_RESONANCE);
    }

    if (state->sword_edge < 0 || state->sword_edge > MAX_RESONANCE) {
        report.is_valid = false;
        report.error_count++;
        snprintf(report.error_log + strlen(report.error_log),
            sizeof(report.error_log) - strlen(report.error_log),
```

```
            "[ERROR] Sword edge (%d) outside bounds [0, %d]. ",
            state->sword_edge, MAX_RESONANCE);
    }

    if (state->moral_sway < -100 || state->moral_sway > 100) {
        report.is_valid = false;
        report.error_count++;
        snprintf(report.error_log + strlen(report.error_log),
            sizeof(report.error_log) - strlen(report.error_log),
            "[ERROR] Moral sway (%d) outside bounds [-100, 100]. ",
            state->moral_sway);
    }

    if (state->stance < 0 || state->stance >= STANCE_COUNT) {
        report.is_valid = false;
        report.error_count++;
        snprintf(report.error_log + strlen(report.error_log),
            sizeof(report.error_log) - strlen(report.error_log),
            "[ERROR] Stance enum (%d) outside valid range [0, %d). ",
            state->stance, STANCE_COUNT);
    }

    if (state->location < 0 || state->location >= LOC_COUNT) {
        report.is_valid = false;
        report.error_count++;
        snprintf(report.error_log + strlen(report.error_log),
            sizeof(report.error_log) - strlen(report.error_log),
            "[ERROR] Location enum (%d) outside valid range [0, %d). ",
            state->location, LOC_COUNT);
    }

    return report;
}
```

## A.15. Layer 2 640x360 Viewport & 2.5D Movie Reel Data Structures (gui_types.h Sub-Assembly 2.3)

**Snapshot ID**: snap_gui_types_sub2_3 | **Mode**: Subassembly | **File**: engine/include/viewport/raster/gui_types.h (L138–L448 of 531 lines) | Jump back to chapter text

```
/*
--------------------------------------------------------------------------------
Sub-Assembly 2.3: Structural Entity & Object Definitions
--------------------------------------------------------------------------------
Core types, layout bounding rectangles, color models, view modes, and widget
structures for the desktop presentation engine.
--------------------------------------------------------------------------------
*/

/**
 * @enum gui_screen_mode
 * @brief Presentation screen views and modal navigation states.
 *
 * For Designers:
 * Represents the 5 primary game screens accessible via the top navigation tabs:
 * Tactical Grid, Hero Dossier, Inventory, Provincial Map, and Chronicle Codex.
 *
 * For Programmers:
 * Bounded enumerator indexed [0, 4] matching tab descriptors.
 */
typedef enum gui_screen_mode {
    GUI_SCREEN_TACTICAL = 0,    /**< 2.5D / 2D Tactical battlefield & combat */
    GUI_SCREEN_STATUS,          /**< Full protagonist dossier & attributes */
    GUI_SCREEN_INVENTORY,       /**< Bag, key items, Grandma's Watch */
    GUI_SCREEN_WORLD_MAP,       /**< Provincial realm topology & travel */
    GUI_SCREEN_CHRONICLE        /**< Complete lore codex & history of deeds */
} gui_screen_mode_t;

/**
 * @struct gui_tab
 * @brief Top navigation tab descriptor.
```

```
 *
 * For Designers:
 * A clickable header tab bar button linking directly to a primary screen.
 *
 * For Programmers:
 * Bounded rectangular button descriptor with single hotkey accelerator.
 */
typedef struct gui_tab {
    int x;
    int y;
    int width;
    int height;
    const char *label;
    gui_screen_mode_t screen;
    char hotkey;
} gui_tab_t;

/**
 * @enum gui_view_mode
 * @brief Display view mode for the tactical world viewport.
 *
 * For Designers:
 * Switches between classic top-down orthographic grid and dimetric isometric 2.5D.
 *
 * For Programmers:
 * Pure discrete rendering mode toggle.
 */
typedef enum gui_view_mode {
    GUI_VIEW_2D_FLAT = 0,         /**< Classic orthographic top-down grid */
    GUI_VIEW_25D_ISOMETRIC = 1    /**< Tactical dimetric grid with Z-height */
} gui_view_mode_t;

/**
 * @struct gui_palette
 * @brief Hardware and software color palette representation.
 *
 * For Designers:
 * The complete thematic color palette defining background, panel, text,
 * relic, morale, and tactical tile colors.
 *
 * For Programmers:
 * Pixel color values packed for X11 TrueColor (0xRRGGBB).
 */
typedef struct gui_palette {
    unsigned long bg;             /**< Obsidian canvas */
    unsigned long panel;          /**< Granite panel background */
    unsigned long border;         /**< Carved stone rim */
    unsigned long text_main;      /**< Ivory text */
    unsigned long text_muted;     /**< Silver secondary text */
    unsigned long shield_gold;    /**< Radiant Lux gold */
    unsigned long gold_bright;    /**< Lux edge highlight */
    unsigned long sword_violet;   /**< Tenebrous violet */
    unsigned long violet_bright;  /**< Tenebrous edge highlight */
    unsigned long virtue_green;   /**< Emerald virtue */
    unsigned long malice_red;     /**< Crimson malice */
    unsigned long btn_bg;         /**< Button slate */
    unsigned long btn_border;     /**< Button rim */
    unsigned long btn_hover;      /**< Button hover highlight */
    unsigned long tile_grid;      /**< Tactical grid lines */
    unsigned long tile_cobble;    /**< Cobblestone fill */
    unsigned long tile_grass;     /**< Meadow grass fill */
    unsigned long tile_water;     /**< Abyssal water fill */
    unsigned long tile_wall;      /**< Rampart stone wall */
    unsigned long tactical_move;  /**< Move range highlight */
    unsigned long tactical_atk;   /**< Attack range highlight */
    unsigned long shadow;         /**< Drop shadow tint */
} gui_palette_t;

/**
 * @struct gui_panel_rect
 * @brief Named bounding box representing a major desktop presentation zone.
 *
 * For Designers:
 * A positioned rectangular section of the screen with a label title.
 *
```

```
 * For Programmers:
 * Simple bounding rectangle with non-owning title string pointer.
 */
typedef struct gui_panel_rect {
    int x;
    int y;
    int width;
    int height;
    const char *title;
} gui_panel_rect_t;

/**
 * @struct gui_layout
 * @brief Unified desktop presentation layout geometry.
 *
 * For Designers:
 * Specifies the arrangement of header, 2D viewport, hero status, chronicle log,
 * and command dock across the window.
 *
 * For Programmers:
 * Fixed non-overlapping panel partitions filling window dimensions.
 */
typedef struct gui_layout {
    int window_width;
    int window_height;
    gui_panel_rect_t header;       /**< Top title banner and cycle indicator */
    gui_panel_rect_t viewport_2d;  /**< 2D / 2.5D tactical map viewport */
    gui_panel_rect_t status;       /**< Right hero vitals & relics panel */
    gui_panel_rect_t chronicle;    /**< Bottom left narrative log */
    gui_panel_rect_t command_dock; /**< Bottom right tactical action dock */
} gui_layout_t;

/**
 * @struct gui_button
 * @brief Interactive button widget definition in screen coordinates.
 *
 * For Designers:
 * A button in the tactical command dock with label, hotkey, and active/hover states.
 *
 * For Programmers:
 * Screen-coordinate widget descriptor with hover and active state flags.
 */
typedef struct gui_button {
    int x;
    int y;
    int width;
    int height;
    const char *label;
    const char *command;
    char hotkey;
    bool is_hovered;
    bool is_active;
} gui_button_t;

/**
 * @struct gui_log
 * @brief Scrollable ring-buffer storing recent chronicle narrative events.
 *
 * For Designers:
 * Stores the running history of recent battle logs and story narration.
 *
 * For Programmers:
 * Fixed 2D character matrix with count index, zero dynamic heap.
 */
typedef struct gui_log {
    char lines[GUI_MAX_LOG_LINES][GUI_LOG_ENTRY_LEN];
    int count;
} gui_log_t;

/**
 * @enum gui_asset_id
 * @brief Canonical asset identifier for characters, monsters, terrain, and relics.
 *
 * For Designers:
 * Defines generic engine visual slots (hero, minion, boss, tile, relic, vfx)
```

```
 * mapped cleanly to game domain narrative aliases (Elijah, Dream Eater, Shield of Lux).
 *
 * For Programmers:
 * Tagged enumeration representing integer asset slot keys in the asset registry.
 */
typedef enum gui_asset_id {
    ASSET_NONE = 0,
    ASSET_ENTITY_PROTAGONIST_1 = 1,  /**< Primary hero entity slot */
    ASSET_ENTITY_PROTAGONIST_2 = 2,  /**< Secondary companion entity slot */
    ASSET_ENTITY_PROTAGONIST_3 = 3,  /**< Tertiary companion entity slot */
    ASSET_ENTITY_ENEMY_MINION_1 = 10, /**< Standard enemy minion slot 1 */
    ASSET_ENTITY_ENEMY_MINION_2 = 11, /**< Standard enemy minion slot 2 */
    ASSET_ENTITY_BOSS_1        = 12, /**< Major boss entity slot */
    ASSET_TILE_PATH            = 20, /**< Standard walkable terrain tile */
    ASSET_TILE_PLAINS          = 21, /**< Vegetative / meadow terrain tile */
    ASSET_TILE_STRUCTURE       = 22, /**< Solid wall / obstacle terrain tile */
    ASSET_TILE_HAZARD          = 23, /**< Void / chasm terrain tile */
    ASSET_TILE_LIQUID          = 24, /**< Water / fluid terrain tile */
    ASSET_RELIC_DEFENSIVE      = 30, /**< Primary defensive ward / shield icon */
    ASSET_RELIC_OFFENSIVE      = 31, /**< Primary offensive blade icon */
    ASSET_VFX_BARRIER          = 40, /**< Radiant defensive arc visual effect */
    ASSET_VFX_SLASH            = 41, /**< Striking slash visual effect */

    /* Game Domain Aliases for Backward Compatibility during v0.1.1 iteration */
    ASSET_HERO_ELIJAH          = ASSET_ENTITY_PROTAGONIST_1,
    ASSET_HERO_KATRINA         = ASSET_ENTITY_PROTAGONIST_2,
    ASSET_HERO_CASATTI         = ASSET_ENTITY_PROTAGONIST_3,
    ASSET_ENEMY_DREAM_EATER    = ASSET_ENTITY_ENEMY_MINION_1,
    ASSET_ENEMY_SHADOW_STALKER = ASSET_ENTITY_ENEMY_MINION_2,
    ASSET_ENEMY_DARK_LORD      = ASSET_ENTITY_BOSS_1,
    ASSET_TILE_COBBLESTONE     = ASSET_TILE_PATH,
    ASSET_TILE_GRASS           = ASSET_TILE_PLAINS,
    ASSET_TILE_STONE_WALL      = ASSET_TILE_STRUCTURE,
    ASSET_TILE_VOID_CRAG       = ASSET_TILE_HAZARD,
    ASSET_TILE_WATER           = ASSET_TILE_LIQUID,
    ASSET_RELIC_SHIELD_LUX     = ASSET_RELIC_DEFENSIVE,
    ASSET_RELIC_SWORD_TENEBRIS = ASSET_RELIC_OFFENSIVE,
    ASSET_VFX_LUX_BARRIO       = ASSET_VFX_BARRIER,
    ASSET_VFX_CUTTATIO_SLASH   = ASSET_VFX_SLASH
} gui_asset_id_t;

/**
 * @enum gui_asset_kind
 * @brief Asset pipeline origin and representation kind.
 *
 * For Designers:
 * Distinguishes procedural vector art drawn directly by code from artist bitmaps
 * loaded from disk, ensuring the game is 100% playable with zero external art.
 *
 * For Programmers:
 * Discrete rendering mode indicator governing raster dispatch.
 */
typedef enum gui_asset_kind {
    ASSET_KIND_PROCEDURAL = 0,    /**< Synthesized geometric vector art */
    ASSET_KIND_BITMAP = 1         /**< External PNG/BMP artist spritesheet */
} gui_asset_kind_t;

/**
 * @struct gui_asset
 * @brief Asset specification in the Asset Abstraction Layer.
 *
 * For Designers:
 * Describes an individual graphic asset: its slot ID, display name, kind (procedural vs bitmap),
 * pixel dimensions, and external artist file path.
 *
 * For Programmers:
 * Lightweight asset metadata record with non-owning path pointers.
 */
typedef struct gui_asset {
    gui_asset_id_t id;
    const char *name;
    gui_asset_kind_t kind;
    int width;
    int height;
    const char *artist_file_path; /**< Future artist spritesheet path */
```

```
394 } gui_asset_t;
395
396 /**
397  * @struct gui_iso_point
398  * @brief Isometric screen point with elevation depth.
399  *
400  * For Designers:
401  * Represents a point in 2.5D space projected onto the screen with vertical elevation.
402  *
403  * For Programmers:
404  * Triplet of screen coordinates (screen_x, screen_y) and discrete elevation_z.
405  */
406 typedef struct gui_iso_point {
407     int screen_x;
408     int screen_y;
409     int elevation_z;
410 } gui_iso_point_t;
411
412 /**
413  * @struct gui_grid_coord
414  * @brief Discrete grid coordinate.
415  *
416  * For Designers:
417  * Specifies a specific square tile on the battlefield grid (gx, gy).
418  *
419  * For Programmers:
420  * Integer grid pair with zero floating point inaccuracy.
421  */
422 typedef struct gui_grid_coord {
423     int gx;
424     int gy;
425 } gui_grid_coord_t;
426
427 /**
428  * @struct gui_depth_entity
429  * @brief Depth-sorted entity render packet for the Painter's Algorithm.
430  *
431  * For Designers:
432  * An entity (hero, monster, or boss) queued for drawing with proper depth sorting
433  * so entities in front overlap entities behind them correctly.
434  *
435  * For Programmers:
436  * Lightweight packet containing calculated depth sorting key, screen coordinates,
437  * asset identifier, facing direction, and HP indicators.
438  */
439 typedef struct gui_depth_entity {
440     int depth_key;                /**< (gx + gy) * 1000 + gz * 10 */
441     int screen_x;
442     int screen_y;
443     gui_asset_id_t asset_id;
444     int facing;                   /**< 0: South, 1: West, 2: East, 3: North */
445     int hp;
446     int max_hp;
447     bool is_hero;
448 } gui_depth_entity_t;
```

## A.16. Flight Log Leaf Vocabulary & Entry Layout (log_types.h Sub-Assembly 2.3)

**Snapshot ID**: snap_log_types_sub2_3 | **Mode**: Subassembly | **File**: engine/include/log/log_types.h (L153–L303 of 421 lines) | Jump back to chapter text

```
153 /*
154 ------------------------------------------------------------------------------
155 Sub-Assembly 2.3: Shared Structural Entity & Telemetry Value Definitions
156 ------------------------------------------------------------------------------
157 Canonical non-anonymous enums and POD structures shared across operational and
158 quarantined BIOS telemetry cogs.
159 ------------------------------------------------------------------------------
160 */
161
162 /**
163  * @enum log_severity
```

```
164 * @brief Graded severity taxonomy for engine diagnostics and telemetry events.
165 *
166 * Ordered from highest severity (FATAL) down to finest detail (TRACE).
167 *
168 * ### For Game Designers & Non-Programmers:
169 * Defines how critical an event is:
170 *   - FATAL/CRITICAL: Showstoppers that threaten gameplay continuity.
171 *   - ERROR/WARNING: Problems handled gracefully with automatic fallbacks.
172 *   - INFO: Normal story, quest, and milestone announcements.
173 *   - DEBUG/TRACE: Granular technical values for combat balance and timing.
174 *
175 * ### For Systems Programmers & Engine Architects:
176 * Strict integer ordering enables threshold filtering and weight-table lookup.
177 */
178 typedef enum log_severity {
179     LOG_SEV_FATAL = 0,    /**< Unrecoverable failure or fatal state corruption. */
180     LOG_SEV_CRITICAL,     /**< Severe failure threatening gameplay loop. */
181     LOG_SEV_ERROR,        /**< Discrete operation failure with fallback. */
182     LOG_SEV_WARNING,      /**< Suspicious condition or degraded state. */
183     LOG_SEV_INFO,         /**< Major milestone or narrative state shift. */
184     LOG_SEV_DEBUG,        /**< Granular spatial, coordinate, or stance telemetry. */
185     LOG_SEV_TRACE,        /**< High-frequency per-tick calculation telemetry. */
186     LOG_SEV_COUNT         /**< Enumerator bound sentinel (not a valid severity). */
187 } log_severity_t;
188
189 /**
190  * @enum log_category
191  * @brief Functional categorization of logging events for focused channel filtering.
192  *
193  * ### For Game Designers & Non-Programmers:
194  * Divides events into distinct thematic channels (`CHARACTER`, `MESSAGE`,
195  * `EVENT`, `IO`, `LOG`, `DEBUG`) so designers can inspect one lane at a time.
196  *
197  * ### For Systems Programmers & Engine Architects:
198  * Fixed enum indexes into static descriptor rows and categorical archive streams.
199  */
200 typedef enum log_category {
201     LOG_CAT_DEBUG = 0,    /**< Low-level developer engine trace. */
202     LOG_CAT_LOG,          /**< Standard system lifecycle and startup logs. */
203     LOG_CAT_EVENT,        /**< Significant world, combat, or quest trigger. */
204     LOG_CAT_MESSAGE,      /**< Narrative dialogue or NPC communicative prompt. */
205     LOG_CAT_CHARACTER,    /**< Actor/enemy spatial and equipment stance action. */
206     LOG_CAT_IO,           /**< Filesystem, VFS, JSON & BIOS ROM I/O telemetry. */
207     LOG_CAT_COUNT         /**< Enumerator bound sentinel (not a valid category). */
208 } log_category_t;
209
210 /**
211  * @struct log_entry
212  * @brief Static telemetry record stored in memory ring buffer and emitted to disk.
213  *
214  * ### For Game Designers & Non-Programmers:
215  * A discrete "black box" flight recorder entry capturing when an event occurred,
216  * which actor or module triggered it, and the normalized scalar grade [-1.0..+1.0].
217  *
218  * ### For Systems Programmers & Engine Architects:
219  * 100% stack/static fixed-layout record. Source file/func pointers must point to
220  * static `.rodata` string literals (`__FILE__`, `__func__`) rather than stack storage.
221  */
222 typedef struct log_entry {
223     uint64_t       sequence_id;                 /**< Monotonic sequence identifier. */
224     uint64_t       monotonic_us;                /**< Monotonic clock delta in microseconds. */
225     char           iso_timestamp[32];           /**< UTC ISO 8601 timestamp string. */
226     log_severity_t severity;                    /**< Graded severity rank. */
227     log_category_t category;                    /**< Domain channel category. */
228     char           entity[LOG_MAX_ENTITY_LEN + 1];   /**< Actor or system entity identifier. */
229     float          grade;                       /**< Normalized continuum [-1.0f .. +1.0f]. */
230     int            grade_percent;               /**< Scaled integer readout [-100 .. +100]. */
231     const char    *file;                        /**< Source file anchor (__FILE__). */
232     int            line;                        /**< Source line anchor (__LINE__). */
233     const char    *func;                        /**< Source function anchor (__func__). */
234     char           message[LOG_MAX_MESSAGE_LEN + 1]; /**< NUL-terminated message payload. */
235 } log_entry_t;
236
237 /**
238  * @enum log_tripwire
```

```
 * @brief Diagnostic tripwire state for the data-driven logging pipeline.
 *
 * ### For Game Designers & Non-Programmers:
 * Circuit breakers catching corrupt telemetry JSON files so Safe Mode activates
 * automatically without crashing the session.
 *
 * ### For Systems Programmers & Engine Architects:
 * Latching status enum reported across shared `log_fallback_parse_config_json()`
 * and verified during Block 4 self-audits.
 */
typedef enum log_tripwire {
    LOG_TRIPWIRE_CLEAR = 0,        /**< No tripwires active; data pipeline healthy. */
    LOG_TRIPWIRE_MANIFEST_MISSING, /**< Telemetry manifest could not be found. */
    LOG_TRIPWIRE_MANIFEST_INVALID, /**< Manifest unparseable or lacks menu entries. */
    LOG_TRIPWIRE_SETTING_MISSING,  /**< Designated setting JSON profile not found. */
    LOG_TRIPWIRE_SETTING_INVALID,  /**< Setting file syntax or attributes corrupted. */
    LOG_TRIPWIRE_BOUNDS_VIOLATION  /**< Configuration violated byte/file limits. */
} log_tripwire_t;

/**
 * @struct log_config
 * @brief Runtime configuration for the logging subsystem.
 *
 * ### For Game Designers & Non-Programmers:
 * Configurable profile selecting log folder paths, max archive size, minimum
 * severity cutoff, and console color output.
 *
 * ### For Systems Programmers & Engine Architects:
 * Hydrated cleanly from `engine/config/diagnostics/telemetry/` or `.rodata` ROM;
 * `loaded_from_data` stays strictly `false` whenever hydrated from BIOS/ROM.
 */
typedef struct log_config {
    char           log_dir[64];            /**< Target directory for log files. */
    char           base_filename[64];      /**< Primary active log filename. */
    size_t         max_file_size_bytes;    /**< Byte threshold triggering file rotation. */
    size_t         max_rotated_files;      /**< Maximum archive files retained (1..25). */
    log_severity_t min_severity;           /**< Cutoff severity (looser levels filtered). */
    bool           save_by_category;       /**< Write per-category archives under log_dir. */
    bool           echo_to_console;        /**< Mirror formatted logs to console streams. */
    bool           use_colors;             /**< Emit ANSI color escapes on console. */
    char           profile_name[32];       /**< Active configuration preset identifier. */
    bool           loaded_from_data;       /**< True only when hydrated from live JSON VFS. */
} log_config_t;

/**
 * @struct log_stats
 * @brief Aggregate runtime telemetry report constructed by inspection procedures.
 *
 * ### For Game Designers & Non-Programmers:
 * Summary scorecard showing total logged events, active file byte count,
 * archive rotations, and whether disk or Safe-Mode presets are active.
 *
 * ### For Systems Programmers & Engine Architects:
 * Populated atomically by `log_execute_inspect()` with zero heap allocations.
 */
typedef struct log_stats {
    uint64_t       total_entries;       /**< Total log entries recorded since init. */
    size_t         ring_buffer_count;   /**< Active entries retained in circular ring. */
    size_t         rotation_count;      /**< Number of archive rotations performed. */
    size_t         active_file_bytes;   /**< Current size in bytes of primary log file. */
    log_severity_t min_severity;        /**< Active minimum severity cutoff. */
    bool           is_active;           /**< True when logging engine is initialized. */
    bool           loaded_from_data;    /**< True if active config originated from disk JSON. */
    char           active_profile[32];  /**< Active telemetry preset identifier. */
} log_stats_t;
```

## A.17. Flight Recorder Formatted Entry Dispatch API (log.h Sub-Assembly 3.3)

**Snapshot ID**: snap_log_ring_api_sub3_3 | **Mode**: Subassembly | **File**: engine/include/log/log.h (L406–L484 of 954 lines) | Jump back to chapter text

```
406 /*
```

```
/* ------------------------------------------------------------------------------
Sub-Assembly 3.3: Formatted Entry Dispatch Cogs
Mission: Format, serialize, and emit structured flight recorder entries.
@triggers
  - Tripwires      : LOG_TRIPWIRE_SETTING_INVALID / BOUNDS_VIOLATION -> probe "log.write_telemetry"
  - Block 4 Audits : log_validate_telemetry_integrity(), log_preflight()
  - Layer 4 Tests  :
      * tests/unit/log/test_log_unit.c
        [Section 3: INVARIANTS & CONSTRAINTS / test_log_unit_assertions_and_guarantees]
        [Section 4: RING BUFFER FIFO OVERWRITE & FILE ROTATION / test_log_unit_emergence]
      * tests/functional/test_log_functional.c
        [Section 3: INVARIANTS & SATURATION / test_log_functional_assertions_and_guarantees]
      * tests/integrated/test_log_integrated.c
        [Section 4: CATEGORICAL DISK ROUTING & HUD RING BUFFER / test_log_integrated_emergence]
      * tests/system/test_log_system.c
        [Section 1: NOMINAL HIGH-THROUGHPUT BURST / test_log_system_always_pass]
        [Section 3: FIXED-WIDTH REGEX DETERMINISM / test_log_system_assertions_and_guarantees]
------------------------------------------------------------------------------
*/

/**
 * @brief Emits a structured log entry into the ring buffer, file, and console.
 *
 * ### For Game Designers & Non-Programmers:
 * Writes an event to the game's flight recorder journal, capturing the character involved,
 * their current moral balance percentage, and what action took place.
 *
 * ### For Systems Programmers & Engine Architects:
 * Formats a fixed-column log record with microsecond monotonic timestamp and ISO 8601 UTC time,
 * pushes it to the static circular ring buffer, and writes to disk archives with zero heap allocation.
 *
 * @param[in] severity Graded severity level.
 * @param[in] category Target functional category.
 * @param[in] entity   Entity/Actor label (e.g. "Player", "Companion", "System").
 * @param[in] grade    Normalized grade [-1.0f .. +1.0f] (e.g. alignment sway or vitals).
 * @param[in] file     Source file name (__FILE__).
 * @param[in] line     Source line number (__LINE__).
 * @param[in] func     Source function name (__func__).
 * @param[in] fmt      Printf-style format string.
 * @return Monotonic sequence ID assigned to the entry, or 0 on drop/failure.
 */
uint64_t log_write_entry(log_severity_t severity,
                         log_category_t category,
                         const char *entity,
                         float grade,
                         const char *file,
                         int line,
                         const char *func,
                         const char *fmt,
                         ...);

/**
 * @brief Converts a normalized float grade [-1.0f, +1.0f] to integer percent [-100, +100].
 *
 * ### For Game Designers & Non-Programmers:
 * Converts an internal moral sway decimal into a clean -100% to +100% readout.
 *
 * ### For Systems Programmers & Engine Architects:
 * Clamps `grade` to [-1.0f, 1.0f] and scales symmetrically into a signed 32-bit integer percentage.
 *
 * @param[in] grade Normalized floating-point balance ratio in `[-1.0f .. +1.0f]`.
 * @return Integer percentage clamped to `[-100 .. +100]` (`0` on NaN).
 */
int log_grade_to_percent(float grade);

/**
 * @brief Converts an integer percent [-100, +100] to a normalized float grade [-1.0f, +1.0f].
 *
 * ### For Game Designers & Non-Programmers:
 * Converts a whole-number percentage into an internal decimal balance ratio.
 *
 * ### For Systems Programmers & Engine Architects:
 * Clamps `percent` to [-100, 100] and maps to floating-point representation in [-1.0f, 1.0f].
 *
 * @param[in] percent Integer percentage value (`-100 .. +100`).
```

```
482  * @return Normalized float clamped inside `[-1.0f .. +1.0f]`.
483  */
484 float log_percent_to_grade(int percent);
```

## A.18. Tier-0 Flight Log BIOS Boot & Early-Entry Replay Drain

**Snapshot ID**: snap_log_fb_bios_group | **Mode**: Group | **File**: engine/src/log/log_fallback.c (L1256–L1801 of 1865 lines) | Jump back to chapter text

```
1256 /**
1257  * @brief Drains buffered Tier-0 Safe-Mode entries in FIFO order for primary handoff.
1258  *
1259  * ### For Game Designers & Non-Programmers:
1260  * Hands off early boot messages stored in Safe Mode to the main log file cleanly.
1261  *
1262  * ### For Systems Programmers & Engine Architects:
1263  * Unrolls oldest-first entries via `log_fb_copy_ring_entries()` and decrements
1264  * `s_fb.ring_count` while incrementing `s_fb.total_replayed` for `log_init()` replay.
1265  *
1266  * @param[out] out_entries Caller array receiving entries (or NULL to discard cleanly).
1267  * @param[in]  max_entries Maximum capacity of `out_entries`.
1268  * @return Number of entries drained.
1269  */
1270 size_t log_fallback_drain_entries(log_entry_t *out_entries, size_t max_entries)
1271 {
1272     size_t count = log_fb_copy_ring_entries(out_entries, max_entries);
1273     if (count > 0U) {
1274         s_fb.ring_count -= count;
1275         s_fb.total_replayed += (uint64_t)count;
1276     }
1277
1278     return count;
1279 }
```

```
1701 /**
1702  * @brief Arms the Tier-0 BIOS Safe-Mode logger before CLI flags or disk VFS boot.
1703  *
1704  * ### For Game Designers & Non-Programmers:
1705  * Turns on emergency logging immediately when the program starts so early boot
1706  * bugs or missing folders are never lost, while leaving repo folders clean.
1707  *
1708  * ### For Systems Programmers & Engine Architects:
1709  * When opt_cfg is NULL, arms purely in .bss ring + stderr alert mode with zero
1710  * filesystem I/O. When opt_cfg is supplied, opens bounded Safe-Mode disk output
1711  * (applying build/logs sandbox protection) and degrades gracefully to RAM Ring +
1712  * stderr if mkdir/fopen fails on read-only or full media.
1713  *
1714  * @param[in] opt_cfg Optional Safe-Mode config (NULL for clean RAM BIOS boot).
1715  * @return true on clean BIOS Safe-Mode arming (survives disk failures).
1716  */
1717 bool log_fallback_boot(const log_config_t *opt_cfg)
1718 {
1719     if (s_fb.file != NULL) {
1720         fflush(s_fb.file);
1721         fclose(s_fb.file);
1722         s_fb.file = NULL;
1723     }
1724
1725     s_fb.disk_fault = false;
1726     s_fb.current_bytes = 0;
1727
1728     if (opt_cfg == NULL) {
1729         s_fb.config = s_log_floppy_template;
1730         s_fb.config.loaded_from_data = false;
1731         if (io_fallback_is_disk_read_only()) {
1732             s_fb.disk_fault = true;
1733         }
1734         s_fb.active_sink = LOG_FALLBACK_SINK_RING_AND_STDERR;
1735         s_fb.booted = true;
1736         return true;
1737     }
```

```
1738
1739     s_fb.config = *opt_cfg;
1740     s_fb.config.loaded_from_data = false;
1741
1742     if (s_fb.config.max_file_size_bytes == 0U) {
1743         s_fb.config.max_file_size_bytes = LOG_FALLBACK_FLOPPY_MAX_BYTES;
1744     }
1745     if (s_fb.config.max_rotated_files == 0U ||
1746         s_fb.config.max_rotated_files > LOG_FB_MAX_ROTATION_CAP) {
1747         s_fb.config.max_rotated_files = LOG_FALLBACK_FLOPPY_MAX_FILES;
1748     }
1749     if (s_fb.config.log_dir[0] == '\0' || s_fb.config.base_filename[0] == '\0') {
1750         s_fb.disk_fault = true;
1751         s_fb.active_sink = s_fb.config.echo_to_console
1752             ? LOG_FALLBACK_SINK_RING_AND_STDERR
1753             : LOG_FALLBACK_SINK_RING_ONLY;
1754         s_fb.booted = true;
1755         return true;
1756     }
1757
1758     if (io_fallback_is_disk_read_only()) {
1759         s_fb.disk_fault = true;
1760         s_fb.active_sink = s_fb.config.echo_to_console
1761             ? LOG_FALLBACK_SINK_RING_AND_STDERR
1762             : LOG_FALLBACK_SINK_RING_ONLY;
1763         s_fb.booted = true;
1764         return true;
1765     }
1766
1767     log_fb_resolve_safe_dir(&s_fb.config);
1768
1769     if (!fs_ensure_directory(s_fb.config.log_dir)) {
1770         s_fb.disk_fault = true;
1771         s_fb.active_sink = s_fb.config.echo_to_console
1772             ? LOG_FALLBACK_SINK_RING_AND_STDERR
1773             : LOG_FALLBACK_SINK_RING_ONLY;
1774         s_fb.booted = true;
1775         return true;
1776     }
1777
1778     char full_path[LOG_FB_PATH_CAP];
1779     str_path_join(full_path, sizeof(full_path), s_fb.config.log_dir, s_fb.config.base_filename);
1780
1781     s_fb.file = fopen(full_path, "a");
1782     if (s_fb.file == NULL) {
1783         s_fb.disk_fault = true;
1784         s_fb.active_sink = s_fb.config.echo_to_console
1785             ? LOG_FALLBACK_SINK_RING_AND_STDERR
1786             : LOG_FALLBACK_SINK_RING_ONLY;
1787         s_fb.booted = true;
1788         return true;
1789     }
1790
1791     struct stat st;
1792     if (stat(full_path, &st) == 0) {
1793         s_fb.current_bytes = (size_t)st.st_size;
1794     } else {
1795         s_fb.current_bytes = 0;
1796     }
1797
1798     s_fb.active_sink = LOG_FALLBACK_SINK_DISK_AND_RING;
1799     s_fb.booted = true;
1800     return true;
1801 }
```

## A.19. Flight Log System Emergence Verification (test_log_system_emergence)

**Snapshot ID**: snap_log_test_system | **Mode**: Function | **File**: tests/system/test_log_system.c (L283–L385 of 475 lines) | Jump back to chapter text

```
283 /**
284  * @brief Executes 10,000-entry chaos load, floppy BIOS endurance burst, and probe checks.
```

```
285 */
286 static void test_log_system_emergence(void)
287 {
288     printf("  [PASS-4] Running Section 4: Emergence (10,000-Entry Chaos, Floppy BIOS & Probes)...\n");
289 
290     TEST_ASSERT(debug_init(NULL), "Debugger online to monitor log system endurance");
291 
292     log_config_t cfg = log_get_default_config();
293     cfg.echo_to_console = false;
294     cfg.min_severity = LOG_SEV_TRACE;
295     cfg.save_by_category = true;
296     cfg.max_file_size_bytes = 16384U;
297     cfg.max_rotated_files = 10U;
298     str_safe_copy(cfg.log_dir, "build/logs/system_chaos", sizeof(cfg.log_dir));
299     str_safe_copy(cfg.base_filename, "chaos.log", sizeof(cfg.base_filename));
300 
301     TEST_ASSERT(log_init(&cfg), "Initialized for chaos stress");
302 
303     for (int i = 1; i <= TEST_LOG_SYS_CHAOS_COUNT; i++) {
304         log_severity_t sev = (log_severity_t)(i % LOG_SEV_COUNT);
305         log_category_t cat = (log_category_t)(i % LOG_CAT_COUNT);
306         float grade = ((float)(i % 200) - 100.0f) / 100.0f;
307 
308         LOG_RECORD(sev, cat, "ChaosAgent", grade, "High-stress event step #%06d", i);
309 
310         if (i % 2000 == 0) {
311             log_entry_t sample[10];
312             size_t fetched = log_get_top_entries(sample, 10U);
313             TEST_ASSERT_INT_EQ((int)fetched, 10,
314                                "Ring buffer top entries query succeeds during live write");
315         }
316     }
317 
318     log_stats_t final_stats;
319     TEST_ASSERT(log_execute_inspect(&final_stats), "Inspected final chaos stats");
320     TEST_ASSERT(final_stats.total_entries >= (uint64_t)TEST_LOG_SYS_CHAOS_COUNT,
321                 "Recorded at least 10,000 entries");
322     TEST_ASSERT(final_stats.rotation_count > 0U,
323                 "Multiple rotations executed under stress");
324 
325     /* Verify live telemetry probes remain healthy immediately after 10,000-entry stress */
326     cornerstone_test_probe_check_t p_rot = cornerstone_test_check_probe("log.rotation_io_bounds", LOG_MODULE_ID);
327     cornerstone_test_probe_check_t p_wri = cornerstone_test_check_probe("log.write_telemetry", LOG_MODULE_ID);
328     TEST_ASSERT(p_rot.found && p_rot.passed,
329                 "log.rotation_io_bounds probe remains PASS after 10,000-entry chaos load");
330     TEST_ASSERT(p_wri.found && p_wri.passed,
331                 "log.write_telemetry probe remains PASS after 10,000-entry chaos load");
332 
333     log_shutdown();
334 
335     /* Compact 1.44 MB floppy BIOS disk burst & tight 2-archive rotation cap */
336     remove("build/logs/system_floppy/floppy_stress.log");
337     remove("build/logs/system_floppy/floppy_stress.log.1");
338     remove("build/logs/system_floppy/floppy_stress.log.2");
339     remove("build/logs/system_floppy/floppy_stress.log.3");
340 
341     log_config_t floppy_cfg = log_fallback_get_floppy_config();
342     snprintf(floppy_cfg.log_dir, sizeof(floppy_cfg.log_dir), "build/logs/system_floppy");
343     snprintf(floppy_cfg.base_filename, sizeof(floppy_cfg.base_filename), "floppy_stress.log");
344     floppy_cfg.max_file_size_bytes = 2048U;
345     floppy_cfg.max_rotated_files = 2U;
346     floppy_cfg.echo_to_console = false;
347 
348     TEST_ASSERT(log_fallback_boot(&floppy_cfg),
349                 "Armed compact floppy BIOS disk sink for rotation stress");
350     for (unsigned int i = 0U; i < TEST_LOG_SYS_FLOPPY_FRAMES; ++i) {
351         uint64_t seq = log_fallback_record(
352             LOG_SEV_WARNING, LOG_CAT_LOG, "FloppyStress", 0.5f, __FILE__, __LINE__, __func__,
353             "Compact floppy rotation burst frame #%03u carrying fixed payload padding [0123456789ABCDEF0123456789ABCDEF]",
354             i);
355         TEST_ASSERT(seq > 0U, "Floppy BIOS stress entry recorded");
356     }
357 
358     log_fallback_stats_t fb_stats;
359     TEST_ASSERT(log_fallback_get_stats(&fb_stats),
```

```
                "Inspected floppy BIOS stats after 80-entry burst");
    TEST_ASSERT(!fb_stats.disk_fault && fb_stats.active_sink == LOG_FALLBACK_SINK_DISK_AND_RING,
                "Floppy BIOS sink sustained disk writes without fault");
    TEST_ASSERT(fb_stats.disk_bytes_written > 0U,
                "Floppy BIOS recorded non-zero disk writes");
    log_fallback_shutdown();
    debug_shutdown();

    FILE *f_rot1 = fopen("build/logs/system_floppy/floppy_stress.log.1", "r");
    FILE *f_rot2 = fopen("build/logs/system_floppy/floppy_stress.log.2", "r");
    FILE *f_rot3 = fopen("build/logs/system_floppy/floppy_stress.log.3", "r");
    TEST_ASSERT(f_rot1 != NULL, "Floppy BIOS rotation archive .1 exists");
    TEST_ASSERT(f_rot2 != NULL, "Floppy BIOS rotation archive .2 exists");
    TEST_ASSERT(f_rot3 == NULL, "Floppy BIOS rotation archive .3 suppressed by 2-file cap");
    if (f_rot1 != NULL) {
        fclose(f_rot1);
    }
    if (f_rot2 != NULL) {
        fclose(f_rot2);
    }
    if (f_rot3 != NULL) {
        fclose(f_rot3);
    }

    printf("  [PASS-4] Section 4 passed: 10,000-entry chaos, concurrent ring, probes & floppy BIOS verified.\n");
}
```

# A.20. State Debugger Leaf Vocabulary & Snapshot Structs (debug_types.h Sub-Assembly 2.3)

**Snapshot ID**: snap_debug_types_sub2_3 | **Mode**: Subassembly | **File**: engine/include/debug/debug_types.h (L147–L453 of 652 lines) | Jump back to chapter text

```
/*
-------------------------------------------------------------------------------
Sub-Assembly 2.3: Shared Structural Ontology & POD Debugger Types
-------------------------------------------------------------------------------
Canonical non-anonymous enums and structures shared identically between
`cog.debug` and quarantined `cog.debug.fallback`.
-------------------------------------------------------------------------------
*/

/**
 * @enum debug_tripwire
 * @brief Latched diagnostic tripwire states for the debugger and validation pipeline.
 *
 * ### For Game Designers & Non-Programmers:
 * Alarm switches indicating whether configuration data was missing or damaged,
 * an automated safety test fired, or Safe Mode engaged after a fault.
 *
 * ### For Systems Programmers & Engine Architects:
 * Latching error enum preserving `0..4` legacy discriminants and `5..7` active
 * validation probe, audit breach, and Safe-Mode downgrade discriminants.
 */
typedef enum debug_tripwire {
    DEBUG_TRIPWIRE_CLEAR = 0,         /**< All tripwires clear; simulation healthy. */
    DEBUG_TRIPWIRE_MANIFEST_MISSING,  /**< Debugger/subsystem manifest unavailable. */
    DEBUG_TRIPWIRE_SETTING_MISSING,   /**< Designated JSON setting profile missing. */
    DEBUG_TRIPWIRE_STATE_CORRUPT,     /**< Simulation state breached domain invariants. */
    DEBUG_TRIPWIRE_HASH_MISMATCH,     /**< Snapshot checksum diverged from recorded state. */
    DEBUG_TRIPWIRE_TEST_FAILED,       /**< Registered validation probe / fail-test fired. */
    DEBUG_TRIPWIRE_AUDIT_BREACH,      /**< Block 4 preflight or audit invariant failed. */
    DEBUG_TRIPWIRE_FALLBACK_ENGAGED   /**< Downgraded to Safe Mode to complete execution. */
} debug_tripwire_t;

/**
 * @enum debug_tripwire_reaction
 * @brief Recovery action taxonomy ingested from taxonomy/tripwire_matrix.json.
 *
 * ### For Game Designers & Non-Programmers:
 * Selects how the engine reacts when an alarm trips: logging warnings, clamping
```

```
 * bad numbers, dropping into Safe Mode, rewinding time, or halting in AWAIT.
 *
 * ### For Systems Programmers & Engine Architects:
 * Mapped 1:1 to `tripwire_matrix.json` canonical reaction identifier strings.
 */
typedef enum debug_tripwire_reaction {
    DEBUG_REACTION_LOG_AND_CONTINUE = 0, /**< Emit telemetry note and continue. */
    DEBUG_REACTION_SATURATE_CLAMP,       /**< Clamp out-of-range value to safe limits. */
    DEBUG_REACTION_FALLBACK_MODE,        /**< Downgrade step cleanly via fallback code. */
    DEBUG_REACTION_AUTO_REWIND,          /**< Restore last locked healthy snapshot. */
    DEBUG_REACTION_PANIC_HALT            /**< Complete via fallback then halt in AWAIT. */
} debug_tripwire_reaction_t;

/**
 * @enum debug_test_tier
 * @brief Ontological verification tier associated with a tripwire or probe.
 *
 * ### For Game Designers & Non-Programmers:
 * Identifies which inspection level (in-file self-check or one of the four
 * external verification suites) caught an anomaly.
 *
 * ### For Systems Programmers & Engine Architects:
 * Binds registered `debug_validation_probe_t` entries cleanly to Block 4 audits
 * and Layer 4 `Unit`, `Functional`, `Integrated`, and `System` suites.
 */
typedef enum debug_test_tier {
    DEBUG_TEST_TIER_AUDIT = 0,     /**< In-file Block 4 preflight / validation cog. */
    DEBUG_TEST_TIER_UNIT = 1,      /**< Layer 4 Unit isolation test suite. */
    DEBUG_TEST_TIER_FUNCTIONAL,    /**< Layer 4 Functional cog verification suite. */
    DEBUG_TEST_TIER_INTEGRATED,    /**< Layer 4 Cross-subsystem integrated suite. */
    DEBUG_TEST_TIER_SYSTEM         /**< Layer 4 End-to-end system stress & soak suite. */
} debug_test_tier_t;

/**
 * @enum debug_step_mode
 * @brief Execution stepping mode of the engine debugger.
 *
 * ### For Game Designers & Non-Programmers:
 * Controls how simulation steps advance (`RUN`, `OVER_TURN`, `INTO_COG`,
 * `PAUSED`, or emergency `AWAIT` wait state after Safe-Mode recovery).
 *
 * ### For Systems Programmers & Engine Architects:
 * Execution cadence discriminator driving loop gating and tripwire freezes.
 */
typedef enum debug_step_mode {
    DEBUG_STEP_RUN = 0,        /**< Continuous execution until breakpoint/tripwire. */
    DEBUG_STEP_OVER_TURN = 1,  /**< Pause cleanly at turn boundaries. */
    DEBUG_STEP_INTO_COG = 2,   /**< Micro-step individual cog sub-assemblies. */
    DEBUG_STEP_PAUSED = 3,     /**< Simulation paused for interactive inspection. */
    DEBUG_STEP_AWAIT = 4       /**< Halted idle awaiting repair and healthy restore. */
} debug_step_mode_t;

/**
 * @enum debug_breakpoint_type
 * @brief Watchpoint trigger condition taxonomy for simulation breaks.
 *
 * ### For Game Designers & Non-Programmers:
 * Chooses what pauses gameplay automatically: specific turns, stance shifts,
 * moral sway swings, weapon resonance limits, low HP, locations, or tripwires.
 *
 * ### For Systems Programmers & Engine Architects:
 * Evaluated in O(1) per breakpoint slot at each simulation step boundary.
 */
typedef enum debug_breakpoint_type {
    DEBUG_BREAK_NONE = 0,          /**< Disabled / unused breakpoint slot. */
    DEBUG_BREAK_TURN,              /**< Break when turn counter matches target value. */
    DEBUG_BREAK_STANCE,            /**< Break when actor enters target stance ID. */
    DEBUG_BREAK_SWAY_RANGE,        /**< Break when |moral_sway| exceeds threshold. */
    DEBUG_BREAK_RESONANCE_BREACH,  /**< Break when primary gauge exceeds threshold. */
    DEBUG_BREAK_HP_THRESHOLD,      /**< Break when HP falls at or below target. */
    DEBUG_BREAK_LOCATION,          /**< Break upon entering target zone/tile ID. */
    DEBUG_BREAK_TRIPWIRE           /**< Break whenever any diagnostic tripwire trips. */
} debug_breakpoint_type_t;

/**
```

```
 * @struct debug_state_fields
 * @brief Decoupled, generic state representation for Layer 0 snapshots.
 *
 * ### For Game Designers & Non-Programmers:
 * Captures an actor's generic vitals, posture, scalar balance, location index,
 * active turn timers, and threat readout inside the time-machine filmstrip.
 *
 * ### For Systems Programmers & Engine Architects:
 * Pure integer/char POD slot layout guaranteeing zero Layer 1 simulation headers
 * ever leak into `engine/`.
 */
typedef struct debug_state_fields {
    int  turn;                          /**< Active simulation turn index. */
    char hero_name[DEBUG_MAX_NAME_LEN];   /**< Generic active actor label slot. */
    int  hp;                            /**< Current health points slot. */
    int  max_hp;                        /**< Maximum health points capacity slot. */
    int  mp;                            /**< Secondary energy pool slot. */
    int  tp;                            /**< Tactical technique points slot. */
    int  shield_power;                  /**< Defensive artifact charge gauge slot. */
    int  sword_edge;                    /**< Offensive artifact charge gauge slot. */
    int  stance;                        /**< Active numerical posture/stance slot. */
    int  moral_sway;                    /**< Signed alignment/balance scalar slot. */
    int  location;                      /**< Discrete world zone/location identifier. */
    int  lux_barrio_turns;              /**< Protective ward duration counter slot. */
    int  charm_turns;                   /**< Status modifier duration counter slot. */
    int  threat_hp;                     /**< Opposing active target HP readout slot. */
    int  foes_vanquished;               /**< Cumulative defeated opposition counter. */
} debug_state_fields_t;

/**
 * @struct debug_snapshot
 * @brief Frozen state snapshot stored in the debugger circular ring buffer.
 *
 * ### For Game Designers & Non-Programmers:
 * One time-machine frame recording the exact turn values, command typed, log
 * line link, and whether the frame is pinned so later turns cannot overwrite it.
 *
 * ### For Systems Programmers & Engine Architects:
 * Fixed POD record verified by 32-bit FNV-1a checksum (`state_hash`) across
 * rewind and determinism comparisons.
 */
typedef struct debug_snapshot {
    uint64_t             snapshot_id;      /**< Monotonic snapshot sequence number. */
    uint64_t             timestamp_ms;     /**< Capture timestamp in milliseconds. */
    debug_state_fields_t state;            /**< Decoupled state scalar payload. */
    char                 last_command[64]; /**< Canonical command verb executed. */
    char                 last_arg[64];     /**< Optional command argument string. */
    uint32_t             state_hash;       /**< 32-bit FNV-1a deterministic hash. */
    uint64_t             log_sequence_id;  /**< Correlated flight recorder sequence. */
    bool                 locked;           /**< Eviction guard enforced up to quota. */
    char                 lock_reason[48];  /**< Human/probe lock tag (e.g. TRIPWIRE_LATCH). */
} debug_snapshot_t;

/**
 * @struct debug_breakpoint
 * @brief Dynamic watchpoint registration descriptor.
 *
 * ### For Game Designers & Non-Programmers:
 * Holds one active debug watchpoint rule with its trigger type, target value,
 * and readable designer label.
 *
 * ### For Systems Programmers & Engine Architects:
 * Fixed-width slot entry stored inside `debug_internal_state_t.breakpoints[]`.
 */
typedef struct debug_breakpoint {
    int                      id;           /**< Assigned positive breakpoint handle ID. */
    debug_breakpoint_type_t type;          /**< Watchpoint trigger discriminator. */
    int                      target_value; /**< Numeric threshold or target ID. */
    bool                     is_active;    /**< True while watchpoint is armed. */
    char                     label[64];    /**< Descriptive breakpoint tag. */
} debug_breakpoint_t;

/**
 * @struct debug_config
 * @brief Debugger configuration parameters loaded from data or BIOS fallback.
```

```
 *
 * ### For Game Designers & Non-Programmers:
 * Settings controlling rewind buffer length, watchpoint triggers, inspection
 * card auto-dumps, interactive console behavior, and Safe-Mode AWAIT pauses.
 *
 * ### For Systems Programmers & Engine Architects:
 * Shared cleanly between `cog.debug` and `cog.debug.fallback` with explicit
 * `loaded_from_data` provenance tracking (`false` for ROM/monolith defaults).
 */
typedef struct debug_config {
    bool   enabled;                         /**< Master debugger enable switch. */
    bool   break_on_error;                  /**< Break upon logged ERROR severity. */
    bool   break_on_tripwire;               /**< Break upon diagnostic tripwire latch. */
    bool   break_on_sway_breach;            /**< Break upon alignment bound breach. */
    bool   break_on_resonance_breach;       /**< Break upon artifact gauge overflow. */
    bool   break_on_hp_zero;                /**< Break when primary HP reaches zero. */
    bool   break_on_stance_change;          /**< Break upon actor posture transitions. */
    size_t max_snapshots;                   /**< Active rewind cap (<= DEBUG_MAX_SNAPSHOTS). */
    int    snapshot_interval_turns;         /**< Automatic snapshot cadence in turns. */
    bool   auto_dump_on_tripwire;           /**< Auto-format state card on tripwire. */
    bool   interactive_repl;                /**< Enable interactive terminal pause prompt. */
    bool   ansi_color;                      /**< Enable ANSI color styling on cards. */
    char   trace_channels[DEBUG_MAX_CHANNELS_LEN]; /**< Enabled channel filter CSV. */
    char   profile_name[32];                /**< Active configuration preset identifier. */
    bool   loaded_from_data;                /**< True only when loaded from disk VFS JSON. */
    bool   engine_traps_active;             /**< Gate for Layer 0 infrastructure traps. */
    bool   game_traps_active;               /**< Gate for Layer 1 simulation traps. */
    size_t locked_slot_quota;               /**< Reserved pinned critical slots in ring. */
    bool   verify_on_step;                  /**< Evaluate probes on each step check. */
    bool   auto_await_on_tripwire;          /**< Enter DEBUG_STEP_AWAIT after fallback step. */
} debug_config_t;

/**
 * @brief Validation probe callback signature returning a Block 4 `diag_result_t`.
 */
typedef diag_result_t (*debug_validation_probe_fn)(void);

/**
 * @struct debug_validation_probe
 * @brief Registered Block 4 / Layer 4 fail-test trigger linked into tripwire monitoring.
 *
 * ### For Game Designers & Non-Programmers:
 * Connects an automated safety check or verification rule directly to the
 * debugger so any violation stops execution in AWAIT instead of silently drifting.
 *
 * ### For Systems Programmers & Engine Architects:
 * Binds source location tags, optional callback function pointers, and Layer 4
 * test suite references inside a 32-slot zero-heap table.
 */
typedef struct debug_validation_probe {
    char                       probe_id[48];       /**< Unique probe ID (e.g. "io.loader.envelope"). */
    char                       subsystem[32];      /**< Owning subsystem ID (e.g. "core.io.master_loader"). */
    char                       source_section[64]; /**< File/block/function trigger anchor. */
    char                       test_suite_ref[64]; /**< Mapped Layer 4 test suite & section. */
    debug_test_tier_t          tier;               /**< Verification tier (AUDIT..SYSTEM). */
    debug_tripwire_t           tripwire_code;      /**< Categorized tripwire latched on fail. */
    debug_tripwire_reaction_t reaction;           /**< Configured recovery reaction policy. */
    debug_validation_probe_fn validate_fn;        /**< Optional Block 4 validation probe callback. */
    bool                       auto_await;         /**< True to pause in AWAIT after Safe-Mode step. */
    bool                       is_active;          /**< True if probe registration slot is live. */
    uint32_t                   trigger_count;      /**< Lifetime count of triggered firings. */
} debug_validation_probe_t;

/**
 * @struct debug_tripwire_report
 * @brief Forensic report capturing active tripwire, fallback completion, and AWAIT state.
 *
 * ### For Game Designers & Non-Programmers:
 * Full emergency incident ticket explaining which probe fired, which snapshot
 * got locked, what Safe Mode did, and whether the engine is paused in AWAIT.
 *
 * ### For Systems Programmers & Engine Architects:
 * Fixed-layout diagnostic report populated cleanly by `debug_trigger_probe()`.
 */
typedef struct debug_tripwire_report {
```

```
    debug_tripwire_t          tripwire_status;    /**< Categorized latched tripwire code. */
    debug_tripwire_reaction_t reaction;           /**< Selected reaction policy. */
    debug_test_tier_t         tier;               /**< Verification tier of firing probe. */
    bool                      fallback_completed; /**< True once Safe Mode finished step. */
    bool                      await_active;       /**< True while halted in DEBUG_STEP_AWAIT. */
    bool                      snapshot_locked;    /**< True if incident snapshot stayed pinned. */
    uint64_t                  locked_snapshot_id; /**< ID of pinned forensic snapshot. */
    char                      probe_id[48];       /**< Probe identifier that raised alarm. */
    char                      subsystem[32];      /**< Owning subsystem identifier. */
    char                      condition[128];     /**< Failure condition summary text. */
    char                      source_location[96];/**< Formatted `file:line` anchor. */
    char                      source_section[64]; /**< Source Block/function section anchor. */
    char                      test_suite_ref[64]; /**< Linked Layer 4 test reference. */
    char                      fallback_action[96];/**< Summary of Safe-Mode recovery action. */
} debug_tripwire_report_t;

/**
 * @struct debug_stats
 * @brief Aggregate runtime telemetry report constructed by Block 5 inspection procedures.
 *
 * ### For Game Designers & Non-Programmers:
 * Status scorecard showing rewind frame usage, locked emergency slots, probe
 * failure counts, Safe-Mode status, and whether playback is paused in AWAIT.
 *
 * ### For Systems Programmers & Engine Architects:
 * Zero-heap runtime snapshot copied from internal debugger state cleanly.
 */
typedef struct debug_stats {
    size_t            snapshot_count;         /**< Active snapshots stored in ring buffer. */
    size_t            max_snapshots;          /**< Configured snapshot ring capacity. */
    size_t            breakpoint_count;       /**< Count of armed watchpoint slots. */
    uint32_t          traps_triggered;        /**< Lifetime debugger trap activations. */
    debug_step_mode_t step_mode;              /**< Active stepping execution mode. */
    bool              is_active;              /**< True if primary debugger is initialized. */
    bool              loaded_from_data;       /**< True if active config came from disk JSON. */
    char              active_profile[32];     /**< Active configuration preset identifier. */
    size_t            locked_snapshot_count;  /**< Pinned critical snapshots in ring. */
    size_t            registered_probe_count; /**< Active registered validation probes. */
    uint32_t          probe_failures;         /**< Total validation probe fail triggers. */
    bool              safe_mode;              /**< True if operating via Safe-Mode fallback. */
    bool              await_active;           /**< True while halted in DEBUG_STEP_AWAIT. */
    bool              fallback_completed;     /**< True if Safe Mode completed halted step. */
    debug_tripwire_t  tripwire_status;        /**< Active debugger tripwire latch code. */
} debug_stats_t;
```

# A.21. Deterministic State Capture & Time-Travel Rewind (debug.c)

**Snapshot ID**: snap_debug_rewind_group | **Mode**: Group | **File**: engine/src/debug/debug.c (L1264–L1567 of 3221 lines) | Jump back to chapter text

```
/**
 * @brief Captures an actor/world snapshot into `.bss` circular ring storage with lock safety.
 *
 * ### For Game Designers & Non-Programmers:
 * Records the complete state of a simulation step in rewind memory while protecting pinned
 * forensic evidence slots from being erased when history wraps around.
 *
 * ### For Systems Programmers & Engine Architects:
 * Selects destination index under `CIRCULAR_FIFO_WITH_LOCKED_SLOTS` (invoking
 * `dbg_normalize_ring_for_locks()` and `memmove()` when preserving locked frames), hashes
 * `fields`, stamps monotonic `next_snapshot_id` + `time_get_monotonic_ms()`, and updates
 * watermark telemetry.
 *
 * @triggers
 * - Layer 4 Tests : `tests/unit/debug/test_debug_unit.c:Section1`,
 *                   `tests/functional/test_debug_functional.c`
 *
 * @param[in] fields Simulation state fields to capture.
 * @param[in] cmd    Associated command verb string (may be NULL).
 * @param[in] arg    Associated command argument string (may be NULL).
 * @return Assigned monotonic snapshot ID (`>= 1`), or `0ULL` if inactive or `fields == NULL`.
 */
```

```
uint64_t debug_record_snapshot(const debug_state_fields_t *fields,
                               const char *cmd, const char *arg)
{
    if (!s_debug_active || fields == NULL) {
        return 0ULL;
    }

    size_t cap = s_debug.config.max_snapshots;
    if (cap == 0U || cap > DEBUG_MAX_SNAPSHOTS) {
        cap = DEBUG_MAX_SNAPSHOTS;
    }

    size_t write_idx = 0U;

    /* Step 1: Select destination slot respecting CIRCULAR_FIFO_WITH_LOCKED_SLOTS */
    if (s_debug.snapshot_count < cap) {
        write_idx = s_debug.snapshot_head;
        s_debug.snapshot_head = (s_debug.snapshot_head + 1U) % cap;
        s_debug.snapshot_count++;
    } else if (s_debug.locked_count == 0U) {
        write_idx = s_debug.snapshot_head;
        s_debug.snapshot_head = (s_debug.snapshot_head + 1U) % cap;
    } else {
        /* Evict oldest UNLOCKED snapshot while preserving chronological order */
        dbg_normalize_ring_for_locks();
        size_t evict_idx = cap;
        for (size_t i = 0U; i < cap; i++) {
            if (!s_debug.snapshots[i].locked) {
                evict_idx = i;
                break;
            }
        }
        if (evict_idx == cap) {
            evict_idx = 0U;
            if (s_debug.locked_count > 0U) {
                s_debug.locked_count--;
            }
            if (!s_debug.in_audit) {
                LOG_WARN(LOCAL_LOG_CAT, LOCAL_MODULE_TAG, -0.25f,
                         "All snapshot ring slots locked; forcefully retired oldest slot.");
            }
        }
        if (evict_idx < cap - 1U) {
            memmove(
                &s_debug.snapshots[evict_idx],
                &s_debug.snapshots[evict_idx + 1U],
                (cap - 1U - evict_idx) * sizeof(debug_snapshot_t)
            );
        }
        write_idx = cap - 1U;
        s_debug.snapshot_head = 0U;
    }

    /* Step 2: Populate immutable snapshot frame */
    debug_snapshot_t *snap = &s_debug.snapshots[write_idx];
    memset(snap, 0, sizeof(*snap));
    snap->snapshot_id = s_debug.next_snapshot_id++;
    snap->timestamp_ms = time_get_monotonic_ms();
    snap->state = *fields;
    str_safe_copy(snap->last_command, (cmd != NULL) ? cmd : "", sizeof(snap->last_command));
    str_safe_copy(snap->last_arg, (arg != NULL) ? arg : "", sizeof(snap->last_arg));
    snap->state_hash = debug_compute_state_hash(fields);
    snap->log_sequence_id = log_get_total_entries();
    snap->locked = false;

    /* Step 3: Track ring buffer telemetry */
    dbg_update_snapshot_watermark(s_debug.snapshot_count);
    s_debug.total_snapshots_captured++;

    return snap->snapshot_id;
}
```

```
/**
 * @brief Rewinds state fields back to snapshot `index` after cryptographic checksum check.
```

```
 *
 * ### For Game Designers & Non-Programmers:
 * Restores actor stats from an earlier saved frame, pausing safely in AWAIT if memory
 * corruption altered the snapshot's stored fingerprint.
 *
 * ### For Systems Programmers & Engine Architects:
 * Fetches snapshot via `debug_get_snapshot()`, verifies `debug_compute_state_hash(&snap.state)`
 * matches `snap.state_hash` (engaging `DEBUG_TRIPWIRE_HASH_MISMATCH` + Safe-Mode AWAIT +
 * `LOG_ERROR` on mismatch), and emits `LOG_INFO` on verified restore.
 *
 * @triggers
 * - Layer 4 Tests : `tests/unit/debug/test_debug_unit.c:Section1`
 *
 * @param[in]  index      Chronological target snapshot offset.
 * @param[out] out_fields Restored simulation state fields on verified success.
 * @return true if checksum passed and `out_fields` populated; false on error or mismatch.
 */
bool debug_rewind_to_snapshot(size_t index, debug_state_fields_t *out_fields)
{
    /* Step 1: Validate preconditions and fetch chronological snapshot copy */
    if (!s_debug_active || out_fields == NULL) {
        return false;
    }

    debug_snapshot_t snap;
    if (!debug_get_snapshot(index, &snap)) {
        return false;
    }

    /* Step 2: Verify determinism checksum against tampering before restoring */
    uint32_t verify_hash = debug_compute_state_hash(&snap.state);
    if (verify_hash != snap.state_hash) {
        s_debug.tripwire_status = DEBUG_TRIPWIRE_HASH_MISMATCH;
        snprintf(s_debug.tripwire_details, sizeof(s_debug.tripwire_details),
                 "Rewind checksum mismatch on snap #%lu (0x%08X != 0x%08X)",
                 (unsigned long)snap.snapshot_id, verify_hash, snap.state_hash);
        s_debug.traps_triggered++;
        (void)debug_fallback_record_fault(
            MODULE_ID, s_debug.tripwire_details, __FILE__, __LINE__,
            "debug_rewind_to_snapshot"
        );
        (void)debug_fallback_finish_with_fallback(
            MODULE_ID, "Aborted corrupted rewind target and entered AWAIT"
        );
        if (!s_debug.in_audit) {
            LOG_ERROR(LOCAL_LOG_CAT, LOCAL_MODULE_TAG, -0.8f,
                      "%s", s_debug.tripwire_details);
        }
        s_debug.await_active = true;
        s_debug.step_mode = DEBUG_STEP_AWAIT;
        return false;
    }

    /* Step 3: Commit verified snapshot state into caller out_fields */
    *out_fields = snap.state;
    if (!s_debug.in_audit) {
        LOG_INFO(LOCAL_LOG_CAT, LOCAL_MODULE_TAG, 0.25f,
                 "Rewound to Snapshot #%lu (Turn %d, Hash 0x%08X%s)",
                 (unsigned long)snap.snapshot_id, snap.state.turn,
                 snap.state_hash, snap.locked ? ", LOCKED" : "");
    }
    return true;
}
```

## A.22. Tier-0 Debugger BIOS Boot & ROM Profile Hydration (debug_fallback.c)

**Snapshot ID**: `snap_debug_fb_bios_group` | **Mode**: `Group` | **File**: `engine/src/debug/debug_fallback.c` (`L1008`–`L1766` of `1852 lines`) | Jump back to chapter text

```
/**
 * @brief Hydrates a complete Safe-Mode debugger config from `.rodata` ROM JSON.
 *
```

```
 * ### For Game Designers & Non-Programmers:
 * Loads a factory debugger preset (`"default"`, `"strict"`, `"trace"`) plus snapshot locking
 * rules straight from executable ROM when disk folders are unreadable.
 *
 * ### For Systems Programmers & Engine Architects:
 * Parses `.rodata` 5-block envelope in memory via `master_loader_load_from_memory()` +
 * `debug_fallback_parse_config_json()`, stamps `loaded_from_data = false`, overlays ROM
 * `snapshot_policy`, and emits guarded BIOS telemetry on corruption or hydration.
 *
 * @triggers
 * - Primary Callers: `debug_get_default_config()`, `debug_init_emergency_monolith()`
 * - Layer 4 Tests  : `tests/unit/debug/test_debug_fallback_unit.c:Section3`,
 *                    `tests/integrated/test_debug_integrated.c:Section1`
 *
 * @param[in]  profile_id  Target profile stem (defaults cleanly to `"default"` if NULL).
 * @param[out] out_config  Caller `debug_config_t` destination buffer.
 * @param[in]  config_size Must equal `sizeof(debug_config_t)`.
 * @return true if ROM profile hydrated cleanly; false on missing/corrupt asset or bad args.
 */
bool debug_fallback_load_rom_profile(
    const char     *profile_id,
    debug_config_t *out_config,
    size_t          config_size
) {
    /* Step 1: Validate destination struct size and resolve target profile stem */
    if (out_config == NULL || config_size != sizeof(debug_config_t)) {
        return false;
    }

    const char *target = (profile_id != NULL && profile_id[0] != '\0')
                             ? profile_id
                             : "default";

    /* Step 2: Locate qualified debugger/<id> ROM payload or fallback leaf */
    char qualified_id[64];
    snprintf(qualified_id, sizeof(qualified_id), "debugger/%s", target);

    size_t raw_size = 0U;
    const char *raw_json = debug_fallback_find_rom_asset(qualified_id, &raw_size);
    if (raw_json == NULL) {
        raw_json = debug_fallback_find_rom_asset(target, &raw_size);
    }
    if (raw_json == NULL || raw_size == 0U) {
        return false;
    }

    /* Step 3: Unwrap 5-block envelope directly from immutable .rodata memory */
    master_loader_t loader;
    if (!master_loader_load_from_memory(&loader, raw_json, raw_size) ||
        loader.body_start == NULL) {
        fb_emit_bios_log(
            LOG_SEV_ERROR, __FILE__, __LINE__, __func__, MODULE_ID,
            "Corrupt 5-block envelope inside diagnostics ROM profile asset."
        );
        return false;
    }

    /* Step 4: Parse shared profile schema and overlay ROM snapshot retention quotas */
    debug_tripwire_t tw = DEBUG_TRIPWIRE_CLEAR;
    char err_msg[128];
    if (!debug_fallback_parse_config_json(
            loader.body_start,
            target,
            false,
            out_config,
            config_size,
            &tw,
            err_msg,
            sizeof(err_msg))) {
        fb_emit_bios_log(
            LOG_SEV_ERROR, __FILE__, __LINE__, __func__, MODULE_ID,
            err_msg
        );
        return false;
    }
```

```
1086
1087     (void)debug_fallback_load_rom_snapshot_policy(out_config, config_size);
1088
1089     /* Step 5: Guarantee non-disk provenance and log Safe-Mode ROM hydration */
1090     out_config->loaded_from_data = false;
1091
1092     s_fb.rom_hydrated = true;
1093     str_safe_copy(s_fb.active_rom_profile,
1094                   out_config->profile_name,
1095                   sizeof(s_fb.active_rom_profile));
1096
1097     fb_emit_bios_log(
1098         LOG_SEV_INFO,
1099         __FILE__,
1100         __LINE__,
1101         __func__,
1102         MODULE_ID,
1103         s_fb.active_rom_profile
1104     );
1105     return true;
1106 }
```

```
1715 /**
1716  * @brief Arms Tier-0 Safe-Mode BIOS state and seeds active ROM profile telemetry.
1717  *
1718  * ### For Game Designers & Non-Programmers:
1719  * Turns on emergency fault recording right at startup so any failure before external
1720  * configs finish loading is still captured and recoverable.
1721  *
1722  * ### For Systems Programmers & Engine Architects:
1723  * Copies caller profile name into `s_fb.active_rom_profile`, or hydrates `"default"`
1724  * via `debug_fallback_load_rom_profile()` with monolith fallback, sets `s_fb.booted = true`,
1725  * and logs an isolated `LOG_SEV_INFO` boot record into `cog.log.fallback`.
1726  *
1727  * @triggers
1728  * - Layer 4 Tests : `tests/unit/debug/test_debug_fallback_unit.c:Section4`
1729  *
1730  * @param[in] opt_config  Optional pre-populated `debug_config_t` (or NULL for ROM default).
1731  * @param[in] config_size Must equal `sizeof(debug_config_t)` when `opt_config != NULL`.
1732  * @return true once Tier-0 Safe-Mode debugger BIOS is booted.
1733  */
1734 bool debug_fallback_boot(const debug_config_t *opt_config, size_t config_size)
1735 {
1736     /* Step 1: Resolve active Safe-Mode profile string from caller, ROM, or monolith */
1737     debug_config_t scratch;
1738     if (opt_config != NULL && config_size == sizeof(debug_config_t)) {
1739         str_safe_copy(s_fb.active_rom_profile, opt_config->profile_name,
1740                       sizeof(s_fb.active_rom_profile));
1741     } else if (!debug_fallback_load_rom_profile("default", &scratch, sizeof(scratch))) {
1742         debug_get_monolith_fallback_config_raw(&scratch, sizeof(scratch));
1743         str_safe_copy(s_fb.active_rom_profile, scratch.profile_name,
1744                       sizeof(s_fb.active_rom_profile));
1745         fb_emit_bios_log(
1746             LOG_SEV_WARNING,
1747             __FILE__,
1748             __LINE__,
1749             __func__,
1750             MODULE_ID,
1751             "ROM default profile unavailable; seeded Safe-Mode BIOS from monolith template."
1752         );
1753     }
1754
1755     /* Step 2: Arm Tier-0 BIOS active flag and emit guarded boot log */
1756     s_fb.booted = true;
1757     fb_emit_bios_log(
1758         LOG_SEV_INFO,
1759         __FILE__,
1760         __LINE__,
1761         __func__,
1762         MODULE_ID,
1763         "Tier-0 Debugger Safe-Mode BIOS armed."
1764     );
1765     return true;
1766 }
```

# A.23. Debugger Unit Emergence & Tripwire Verification (test_debug_unit_emergence)

**Snapshot ID**: snap_debug_test_unit | **Mode**: Function | **File**: tests/unit/debug/test_debug_unit.c (L410–L510 of 563 lines)

| Jump back to chapter text

```
410 static void test_debug_unit_emergence(void)
411 {
412     cornerstone_test_section(4, "Emergence (1000-Turn Wrap, 4-Slot Lock Quota & AWAIT Recovery)");
413 
414     TEST_ASSERT(debug_init(NULL), "debug_init succeeds for emergence section");
415 
416     /* 4.1 1000-turn ring wrap-around */
417     for (int t = 1; t <= 1000; t++) {
418         debug_state_fields_t s = cornerstone_test_make_state(
419             t, "Elijah", 500 - (t % 200), t % 100, (t * 2) % 100, t % 4, (t % 200) - 100);
420         uint64_t sid = debug_record_snapshot(&s, "sim_turn", "step");
421         TEST_ASSERT_U64_EQ(sid, (uint64_t)t, "Monotonic ID strictly tracked across 1000 turns");
422     }
423 
424     TEST_ASSERT_U64_EQ(debug_get_snapshot_count(), (uint64_t)DEBUG_MAX_SNAPSHOTS,
425                        "Ring capacity capped at 64");
426 
427     debug_snapshot_t oldest;
428     TEST_ASSERT(debug_get_snapshot(0U, &oldest) && (oldest.state.turn == 937),
429                 "Oldest turn after 1000 pushes is 937");
430     const debug_snapshot_t *latest = debug_get_latest_snapshot();
431     TEST_ASSERT((latest != NULL) && (latest->state.turn == 1000), "Latest turn is 1000");
432 
433     debug_state_fields_t restored;
434     TEST_ASSERT(debug_rewind_to_snapshot(30U, &restored) &&
435                     (restored.turn == 967) &&
436                     (restored.hp == (500 - (967 % 200))),
437                 "Rewind index 30 restores turn 967 and exact HP formula");
438 
439     debug_shutdown();
440 
441     /* 4.2 Locked critical snapshot quota (4) + DEBUG_TRIGGER_PROBE -> AWAIT recovery */
442     TEST_ASSERT(debug_init(NULL), "Re-init debugger for lock quota & AWAIT check");
443     TEST_ASSERT(debug_can_step() && !debug_is_awaiting(), "Stepping armed initially");
444 
445     uint64_t lock_ids[5] = { 0U };
446     for (int t = 1; t <= 6; t++) {
447         debug_state_fields_t s = cornerstone_test_make_state(t, "Elijah", 450 + t, 60, 40, t % 4, t);
448         uint64_t sid = debug_record_snapshot(&s, "lock_setup", "critical");
449         if (t <= 5) {
450             lock_ids[t - 1] = sid;
451         }
452     }
453 
454     for (size_t i = 0U; i < DEBUG_FALLBACK_LOCKED_SLOT_QUOTA; i++) {
455         TEST_ASSERT(debug_lock_snapshot(lock_ids[i], "UNIT_CRITICAL_LOCK"),
456                     "Lock snapshot within 4-slot quota succeeds");
457     }
458     TEST_ASSERT(debug_lock_snapshot(lock_ids[4], "QUOTA_ROTATE"),
459                 "Locking 5th snapshot rotates oldest lock cleanly");
460 
461     debug_stats_t lock_stats;
462     TEST_ASSERT(debug_execute_inspect(&lock_stats) &&
463                     (lock_stats.locked_snapshot_count == DEBUG_FALLBACK_LOCKED_SLOT_QUOTA),
464                 "Active locked_snapshot_count clamped at quota == 4");
465 
466     for (int t = 7; t <= 126; t++) {
467         debug_state_fields_t s = cornerstone_test_make_state(
468             t, "Elijah", 400 + (t % 90), 50, 50, t % 4, 0);
469         (void)debug_record_snapshot(&s, "advance", "fifo");
470     }
471     TEST_ASSERT_U64_EQ(debug_get_snapshot_count(), (uint64_t)DEBUG_MAX_SNAPSHOTS,
472                        "Ring count remains clamped at 64 with locked slots");
473 
474     for (size_t idx = 0U; idx < DEBUG_FALLBACK_LOCKED_SLOT_QUOTA; idx++) {
475         debug_snapshot_t locked_snap;
476         TEST_ASSERT(debug_get_snapshot(idx, &locked_snap) &&
```

```
                        locked_snap.locked &&
                        (locked_snap.state.turn == (int)(idx + 2U)),
                    "Locked snapshots 2..5 survive 120-turn ring wrap");
    }
    debug_snapshot_t sliding_head;
    TEST_ASSERT(debug_get_snapshot(DEBUG_FALLBACK_LOCKED_SLOT_QUOTA, &sliding_head) &&
                    (sliding_head.state.turn == 67),
                "First evictable slot after 4 locks reflects turn 67");

    TEST_ASSERT(DEBUG_TRIGGER_PROBE("unit.state_guard", "unit.sim", "Synthetic fail-test trap"),
                "DEBUG_TRIGGER_PROBE latches unit validation probe");
    TEST_ASSERT(debug_is_awaiting() &&
                    (debug_get_step_mode() == DEBUG_STEP_AWAIT) &&
                    !debug_can_step(),
                "Debugger latches DEBUG_STEP_AWAIT and blocks stepping");

    debug_set_step_mode(DEBUG_STEP_RUN);
    TEST_ASSERT(debug_get_step_mode() == DEBUG_STEP_AWAIT,
                "debug_set_step_mode cannot bypass active DEBUG_STEP_AWAIT");

    debug_state_fields_t corrupt = cornerstone_test_make_state(126, "Elijah", -25, 50, 50, 0, 250);
    TEST_ASSERT(!debug_resolve_await(&corrupt, DEBUG_STEP_RUN, "Corrupt reject check") &&
                    debug_is_awaiting(),
                "debug_resolve_await rejects out-of-bounds HP/sway and holds AWAIT");

    debug_state_fields_t repaired = cornerstone_test_make_state(126, "Elijah", 450, 50, 50, 0, 10);
    TEST_ASSERT(debug_resolve_await(&repaired, DEBUG_STEP_RUN, "Unit state repaired cleanly") &&
                    !debug_is_awaiting() &&
                    debug_can_step() &&
                    (debug_get_step_mode() == DEBUG_STEP_RUN),
                "debug_resolve_await restores execution once valid state verified");

    debug_shutdown();
}
```

# A.24. Bounded POSIX Filesystem & Read-Only Guard API (fs_utils.h Sub-Assembly 3.2)

**Snapshot ID**: snap_io_fs_utils_sub3_2 | **Mode**: Subassembly | **File**: engine/include/io/fs_utils.h (L359–L455 of 838 lines) | Jump back to chapter text

```
/*
-------------------------------------------------------------------------------
Sub-Assembly 3.2: Bounded Buffered I/O & Directory Management Cogs
Mission: Bounded text reading, safe file writing, and parent directory synthesis.

  ### For Game Designers & Non-Programmers:
  Loads configuration and story files safely within memory caps and saves outputs
  atomically while blocking mutations during Read-Only Safe Mode.

  ### For Systems Programmers & Engine Architects:
  Enforces active policy ceilings (`max_config_bytes`, `path_max_bytes`), NUL
  termination, Safe-Mode RO protection, and shared Tier-0 BIOS atomic write/mkdir.
-------------------------------------------------------------------------------
*/

/**
 * @brief Reads entire text file contents into a bounded, caller-supplied buffer.
 *
 * Guarantees unconditional null termination and prevents buffer overruns.
 * If the file exceeds `buf_size - 1` or `max_config_bytes`, the read is safely rejected.
 *
 * ### For Game Designers & Non-Programmers:
 * Loads an entire text file safely into memory, guaranteeing it never overflows or crashes the engine.
 *
 * ### For Systems Programmers & Engine Architects:
 * Convenience wrapper delegating cleanly into `fs_read_file_bounded()` without length output.
 *
 * @param[in]  filepath Null-terminated path or `"rom://..."` URI to target file.
 * @param[out] buf      Caller-supplied memory destination buffer.
 * @param[in]  buf_size Byte capacity of destination buffer (including null terminator).
```

```
 * @return true if file opened and completely read within buffer bounds; false otherwise.
 * @triggers Inherits truncation fault and warning log behavior from `fs_read_file_bounded()`.
 */
bool fs_read_file_contents(const char *filepath, char *buf, size_t buf_size);

/**
 * @brief Reads file or `"rom://..."` text and reports exact count of bytes read.
 *
 * Similar to `fs_read_file_contents()`, and writes the exact number of bytes read
 * (excluding NUL terminator) into optional `out_bytes_read`.
 *
 * ### For Game Designers & Non-Programmers:
 * Reads a text file and tells callers the exact character byte length loaded.
 *
 * ### For Systems Programmers & Engine Architects:
 * Enforces `min(buf_size, max_config_bytes)` cap, increments `read_ops`/`truncation_rejections`,
 * and records BIOS faults (`IO_ERR_FILE_NOT_FOUND`/`IO_ERR_TRUNCATED_READ`).
 *
 * @param[in]  filepath       Null-terminated path or `"rom://..."` URI to target file.
 * @param[out] buf            Caller-supplied destination buffer.
 * @param[in]  buf_size       Capacity of destination buffer in bytes.
 * @param[out] out_bytes_read Optional pointer receiving bytes read (excluding null).
 * @return true if read completed without truncation; false otherwise.
 * @triggers Emits `LOG_WARN` and records BIOS fault on open error or policy truncation.
 */
bool fs_read_file_bounded(const char *filepath, char *buf, size_t buf_size,
                          size_t *out_bytes_read);

/**
 * @brief Safely writes or overwrites bounded data to a file on disk.
 *
 * Automatically creates parent directories recursively using configured permissions
 * and flushes/renames atomically when enabled in active policy.
 *
 * ### For Game Designers & Non-Programmers:
 * Saves files safely with automatic folder creation and read-only lock protection.
 *
 * ### For Systems Programmers & Engine Architects:
 * Validates path cap and Floppy RO lock (`io.fs.readonly_guard`), then delegates POSIX
 * write/flush/atomic rename to `io_fallback_fs_write_with_flags()`.
 *
 * @param[in] filepath Null-terminated target filepath.
 * @param[in] data     Buffer of data bytes to write.
 * @param[in] data_len Number of bytes to write.
 * @return true if all bytes were written successfully; false on guard block or error.
 * @triggers Triggers `io.fs.readonly_guard` when write attempted under read-only Safe Mode.
 */
bool fs_write_file_contents(const char *filepath, const char *data,
                            size_t data_len);

/**
 * @brief Ensures that a directory path exists, recursively creating missing parents.
 *
 * Mimics POSIX `mkdir -p` behavior using bounded stack buffers with zero heap allocations.
 *
 * ### For Game Designers & Non-Programmers:
 * Automatically creates folders needed for logs or outputs unless write-locked.
 *
 * ### For Systems Programmers & Engine Architects:
 * Validates `FS_PATH_MAX` cap and RO latch (`io.fs.readonly_guard`), then delegates to
 * `io_fallback_fs_ensure_dir_mode()`.
 *
 * @param[in] dir_path Null-terminated directory path to create.
 * @return true if directory exists or was created; false on RO lock, overflow, or error.
 * @triggers Triggers `io.fs.readonly_guard` when directory creation is blocked under RO latch.
 */
bool fs_ensure_directory(const char *dir_path);
```

## A.25. Multi-Tier VFS Path & Asset Resolution API (path_finder.h Sub-Assembly 3.2)

**Snapshot ID**: snap_io_path_finder_sub3_2 | **Mode**: Subassembly | **File**: engine/include/io/path_finder.h (L396–L510 of 719 lines) | Jump back to chapter text

```
/*
--------------------------------------------------------------------------------
Sub-Assembly 3.2: Multi-Tier VFS Path & Asset Resolution Cogs
Mission: Execute layered precedence search, diagnostic alias expansion, symlink
         depth bounds, traversal rejection, and `.rodata` ROM fallback cleanly.

  ### For Game Designers & Non-Programmers:
  Finds configuration manifests, game data files, fonts, and artwork across game
  overrides, engine baselines, asset folders, and zero-disk ROM fallbacks.

  ### For Systems Programmers & Engine Architects:
  Bounded-stack candidate resolution enforcing `max_symlink_depth`, `..` traversal
  containment, destination `out_size` overflow guards, and `io.vfs.traversal_guard`.
--------------------------------------------------------------------------------
*/

/**
 * @brief Resolves a logical domain and relative path to a concrete disk file or `rom://` URI.
 *
 * Search sequence (all tables driven by `io/vfs_routes.json`):
 *  1. `..` parent traversal sandbox guard (`block_path_traversal`) & explicit `rom://` check.
 *  2. Direct accessible file probe (enforcing `max_symlink_depth`).
 *  3. Mounted `active_root` (`<root>/<domain>/<clean>`, `<root>/<clean>`) and `env_override_var`.
 *  4. Layered `config_precedence` tiers (`game/config/`, `engine/config/`, linkers) + `diagnostic_aliases`.
 *  5. Hydrated `fallback_candidate_roots` x `domain_search_paths` matrix.
 *  6. Compiled `.rodata` ROM vault fallback (`rom://<clean>`) when disk targets are absent.
 *
 * ### For Game Designers & Non-Programmers:
 * Finds any configuration file, story file, or manifest automatically—checking game overrides
 * first, engine baselines second, and built-in ROM backups last.
 *
 * ### For Systems Programmers & Engine Architects:
 * Bounded stack probing (zero heap), strict `out_size` overflow rejection, `lstat`/`readlink`
 * symlink depth verification, and `LOG_CAT_IO` telemetry tracking.
 * @triggers `io.vfs.traversal_guard` (on `..`/symlink rejection), `LOG_CAT_IO`.
 *
 * @param[in]  domain   Optional logical domain hint (`"config"`, `"mechanics"`, `"io"`, etc.) or NULL.
 * @param[in]  rel_path Relative asset/config path or explicit `"rom://..."` URI.
 * @param[out] out_path Caller-supplied output path buffer (`out_path[0] = '\0'` on failure).
 * @param[in]  out_size Capacity of `out_path` in bytes.
 * @return true if resolved cleanly onto disk or ROM within `out_size`; false otherwise.
 */
bool path_finder_resolve(const char *domain,
                         const char *rel_path,
                         char *out_path,
                         size_t out_size);

/**
 * @brief Resolves a subsystem directory string to its `manifest.json` anchor file.
 *
 * ### For Game Designers & Non-Programmers:
 * Locates the master menu file (`manifest.json`) for any folder such as `"config"`,
 * `"config/diagnostics/telemetry"`, or `"mechanics"`.
 *
 * ### For Systems Programmers & Engine Architects:
 * Normalizes trailing slashes, enforces traversal guards before formatting
 * `<subsystem>/manifest.json`, and delegates to `path_finder_resolve()`. Zero heap.
 * @triggers `io.vfs.traversal_guard`, `LOG_CAT_IO`.
 *
 * @param[in]  subsystem_path Directory fragment naming the subsystem subtree.
 * @param[out] out_path       Caller output buffer receiving resolved manifest path.
 * @param[in]  out_size       Capacity of `out_path` in bytes.
 * @return true if a readable `manifest.json` was located; false otherwise.
 */
bool path_finder_find_manifest(const char *subsystem_path,
                               char *out_path,
                               size_t out_size);

/**
 * @brief Resolves a media or artwork asset under configured `asset_roots` and category folders.
 *
 * ### For Game Designers & Non-Programmers:
 * Finds fonts (`Cinzel-Bold.ttf`), sprites (`elijah_32x32.png`), audio tracks, and UI textures
 * across `game/assets/` or active mod asset folders.
 *
```

```
471  * ### For Systems Programmers & Engine Architects:
472  * Iterates hydrated `routes->asset_roots[]` (`game/assets`, `../game/assets`, etc.) plus active
473  * mount overlays with strict traversal and symlink checks. Zero heap.
474  * @triggers `io.vfs.traversal_guard`, `LOG_CAT_IO`.
475  *
476  * @param[in]  asset_category Category subfolder (`"fonts"`, `"sprites"`, `"audio"`, etc.).
477  * @param[in]  asset_name     Asset filename (`"Cinzel-Bold.ttf"`, etc.).
478  * @param[out] out_path       Caller destination path buffer.
479  * @param[in]  out_size       Capacity of `out_path` in bytes.
480  * @return true if asset file was located safely; false otherwise.
481  */
482 bool path_finder_resolve_asset(const char *asset_category,
483                                const char *asset_name,
484                                char *out_path,
485                                size_t out_size);
486
487 /**
488  * @brief Resolves an asset using an explicit artist file path hint before category fallbacks.
489  *
490  * ### For Game Designers & Non-Programmers:
491  * Checks an artist's exact custom artwork path first; if unavailable, seamlessly falls back to
492  * standard category folders (`sprites/`, `fonts/`) or data directories.
493  *
494  * ### For Systems Programmers & Engine Architects:
495  * Validates `direct_path_hint` against traversal guards, probes direct/VFS path first, and falls
496  * back to `path_finder_resolve_asset()` (when `asset_name != NULL`) or `path_finder_resolve()`.
497  * @triggers `io.vfs.traversal_guard`, `LOG_CAT_IO`.
498  *
499  * @param[in]  direct_path_hint Preferred direct or relative artist path (nullable).
500  * @param[in]  asset_category   Fallback category subfolder (`"sprites"`, `"fonts"`, etc., nullable).
501  * @param[in]  asset_name       Fallback filename (nullable when only probing hint).
502  * @param[out] out_path         Caller destination path buffer.
503  * @param[in]  out_size         Capacity of `out_path` in bytes.
504  * @return true if any valid target resolved safely; false otherwise.
505  */
506 bool path_finder_resolve_with_hint(const char *direct_path_hint,
507                                    const char *asset_category,
508                                    const char *asset_name,
509                                    char *out_path,
510                                    size_t out_size);
```

## A.26. Zero-Heap JSON Token Span Locator API (json_utils.h Sub-Assembly 3.1)

**Snapshot ID**: snap_io_json_utils_sub3_1 | **Mode**: Subassembly | **File**: engine/include/io/json_utils.h (L357–L461 of 1205 lines) | Jump back to chapter text

```
357 /*
358 ------------------------------------------------------------------------------
359 Sub-Assembly 3.1: JSON Token Locator Cogs
360 Mission: Scan flat or span-bounded JSON payloads for matching keys, objects, and
361          arrays while respecting escape sequences and nesting limits cleanly.
362
363   ### For Game Designers & Non-Programmers:
364   Finds named sections, tables, and lists inside JSON configuration files cleanly
365   without mistaking words inside text descriptions for structural headings.
366
367   ### For Systems Programmers & Engine Architects:
368   Exposes zero-copy, zero-heap token and balanced-container locators constrained
369   to `[json_start, json_start + json_len)` and active `max_json_depth` policy.
370 ------------------------------------------------------------------------------
371 */
372
373 /**
374  * @brief Scans a JSON buffer for a specific top-level or scoped key token.
375  *
376  * Core search primitive that locates occurrences of "key" followed by a colon
377  * (:), ignoring intervening whitespace. Prevents false positive matches on
378  * substring prefixes or keys embedded inside quoted strings.
379  *
380  * ### For Game Designers & Non-Programmers:
381  * Searches through a data file to locate where a property begins (e.g. finding "attack_power:").
382  *
```

```
 * ### For Systems Programmers & Engine Architects:
 * Reentrant zero-heap token locator respecting quote escapes and whitespace.
 * @triggers On nesting limit exceeded -> `IO_ERR_JSON_DEPTH_EXCEEDED` via `io.json.depth_bounds`.
 *
 * @param[in] json Null-terminated JSON buffer.
 * @param[in] key  Null-terminated key name to search for (without quotes).
 * @return Pointer to value position immediately following the colon (:),
 *         or NULL if key is not found or parameters are invalid.
 */
const char *json_find_key(const char *json, const char *key);

/**
 * @brief Span-bounded key locator preventing sibling overrun across sliced buffers.
 *
 * ### For Game Designers & Non-Programmers:
 * Searches strictly inside one specific section of a JSON file so properties from
 * neighboring sections can never bleed in accidentally.
 *
 * ### For Systems Programmers & Engine Architects:
 * Zero-copy locator bounded strictly to `[json_start, json_start + json_len)`.
 * Properly skips quoted string values (respecting `\\` and `\"` escapes) and
 * enforces active `max_json_depth`.
 *
 * @triggers On nesting overflow -> `IO_ERR_JSON_DEPTH_EXCEEDED` fault + `io.json.depth_bounds`.
 *
 * @param[in] json_start Pointer to JSON slice start.
 * @param[in] json_len   Exact byte length of slice (`0` measures NUL-terminated buffer).
 * @param[in] key        Key name to match.
 * @return Pointer to trimmed value token following ':', or NULL if absent/invalid.
 */
const char *json_find_key_span(const char *json_start, size_t json_len, const char *key);

/**
 * @brief Span-bounded nested JSON object (`{...}`) locator.
 *
 * ### For Game Designers & Non-Programmers:
 * Finds a sub-table inside a JSON file (such as `metrics` or `stats`) and
 * fences its boundaries cleanly so fields outside that table cannot leak in.
 *
 * ### For Systems Programmers & Engine Architects:
 * Delegates to `json_find_delimited_span_internal()` inside `[json_start, json_len)`,
 * matching balanced `{}` braces up to `max_json_depth` with zero allocation.
 *
 * @triggers On depth overflow -> `IO_ERR_JSON_DEPTH_EXCEEDED` fault + `io.json.depth_bounds`.
 *
 * @param[in]  json_start    Pointer to JSON slice start.
 * @param[in]  json_len      Exact byte length of slice (`0` measures NUL-terminated).
 * @param[in]  object_key    Target key whose value must be a JSON object.
 * @param[out] out_obj_start Receives pointer to opening ''{'`.
 * @param[out] out_obj_len   Receives byte length including closing ''}'`.
 * @return true if bounded object slice located cleanly; false otherwise.
 */
bool json_find_object_span(const char *json_start, size_t json_len,
                           const char *object_key,
                           const char **out_obj_start, size_t *out_obj_len);

/**
 * @brief Span-bounded nested JSON array (`[...]`) locator.
 *
 * ### For Game Designers & Non-Programmers:
 * Finds a JSON list (such as `"tags"` or `"skills"`) and fences its exact
 * start and end positions so adjacent lists are never read by mistake.
 *
 * ### For Systems Programmers & Engine Architects:
 * Bounded zero-copy bracket matcher operating strictly within `[json_start, json_len)`.
 * Enforces `max_json_depth` without touching caller memory beyond `json_len`.
 *
 * @triggers On depth overflow -> `IO_ERR_JSON_DEPTH_EXCEEDED` fault + `io.json.depth_bounds`.
 *
 * @param[in]  json_start    Pointer to JSON slice start.
 * @param[in]  json_len      Exact byte length of slice (`0` measures NUL-terminated).
 * @param[in]  array_key     Target key whose value must be a JSON array.
 * @param[out] out_arr_start Receives pointer to opening ''['`.
 * @param[out] out_arr_len   Receives byte length including closing '']'`.
 * @return true if bounded array slice located cleanly; false otherwise.
```

```
 */
bool json_find_array_span(const char *json_start, size_t json_len,
                          const char *array_key,
                          const char **out_arr_start, size_t *out_arr_len);
```

## A.27. I/O Pipeline System Emergence & ROM Downgrade Verification (test_io_system_emergence)

**Snapshot ID**: snap_io_test_system | **Mode**: Function | **File**: tests/system/test_io_system.c (L328–L399 of 477 lines) | Jump back to chapter text

```
/**
 * @brief Verifies sandboxed mount redirection and live v1 -> v2 JSON hot-patch reflection.
 */
static void test_io_system_emergence(void)
{
    printf("  [EMERGE-4] Running Section 4: Emergence (Dynamic Mount & Live Hot-Patching)...\n");

    const char *hotpatch_dir = "build/system_test_patch/system";
    const char *hotpatch_file = "build/system_test_patch/system/runtime_patch.json";

    TEST_ASSERT(fs_ensure_directory(hotpatch_dir), "Synthesized hot-patch directory");

    static const char patch_v1[] =
        "{\n"
        "  \"block_1_metadata\": {\"project\": \"Tenebris and Lux\", \"file\": \"runtime_patch.json\"},\n"
        "  \"block_2_types\": {\"schema_id\": \"tenlux.data.patch.v1\"},\n"
        "  \"block_3_body\": {\"difficulty_mult\": 1.0, \"active_boss\": \"Usuru\"}\n"
        "}\n";
    TEST_ASSERT(fs_write_file_contents(hotpatch_file, patch_v1, strlen(patch_v1)),
                "Wrote patch v1");

    path_finder_init();
    master_loader_init();
    path_finder_set_data_root("build/system_test_patch");

    master_loader_t loader;
    bool l1 = master_loader_execute_load_verified("system", "runtime_patch.json",
                                                  "tenlux.data.patch.v1",
                                                  s_io_sys_load_buf,
                                                  sizeof(s_io_sys_load_buf),
                                                  &loader);
    TEST_ASSERT(l1 && loader.status == LOADER_STATUS_OK, "Loaded patch v1");
    float diff1 = 0.0f;
    char boss1[64];
    TEST_ASSERT(master_loader_get_float(&loader, NULL, "difficulty_mult", &diff1) &&
                diff1 >= 0.999f && diff1 <= 1.001f,
                "Patch v1 difficulty_mult == 1.0");
    TEST_ASSERT(master_loader_get_string(&loader, NULL, "active_boss", boss1, sizeof(boss1)),
                "Patch v1 boss extracted");
    TEST_ASSERT_STR_EQ(boss1, "Usuru", "Boss is authentic Dark Lord Usuru");

    static const char patch_v2[] =
        "{\n"
        "  \"block_1_metadata\": {\"project\": \"Tenebris and Lux\", \"file\": \"runtime_patch.json\"},\n"
        "  \"block_2_types\": {\"schema_id\": \"tenlux.data.patch.v1\"},\n"
        "  \"block_3_body\": {\"difficulty_mult\": 2.5, \"active_boss\": \"The Dark Lord Usuru\"}\n"
        "}\n";
    TEST_ASSERT(fs_write_file_contents(hotpatch_file, patch_v2, strlen(patch_v2)),
                "Wrote patch v2");

    bool l2 = master_loader_execute_load_verified("system", "runtime_patch.json",
                                                  "tenlux.data.patch.v1",
                                                  s_io_sys_load_buf,
                                                  sizeof(s_io_sys_load_buf),
                                                  &loader);
    TEST_ASSERT(l2 && loader.status == LOADER_STATUS_OK, "Loaded patch v2");
    float diff2 = 0.0f;
    char boss2[64];
    TEST_ASSERT(master_loader_get_float(&loader, NULL, "difficulty_mult", &diff2) &&
                diff2 >= 2.499f && diff2 <= 2.501f,
```

```
388                 "Hot-patched difficulty_mult == 2.5");
389     TEST_ASSERT(master_loader_get_string(&loader, NULL, "active_boss", boss2, sizeof(boss2)),
390                 "Hot-patched boss extracted");
391     TEST_ASSERT_STR_EQ(boss2, "The Dark Lord Usuru",
392                        "Boss updated to The Dark Lord Usuru");
393 
394     path_finder_set_data_root(NULL);
395     master_loader_shutdown();
396     path_finder_shutdown();
397 
398     printf("  [EMERGE-4] Section 4 passed: Dynamic mount reconfiguration and hot-patching succeeded.\n");
399 }
```